Radhakishan Hospital — Standard Prescription Protocols

Pediatric OPD — For Doctor Review

Total: 482 protocols across 20 categories

Generated: 30 March 2026

Category Summary

Neonatology — 81 protocols
Infectious — 50 protocols
GI — 42 protocols
Emergency — 35 protocols
Respiratory — 32 protocols
Neurological — 31 protocols
ENT — 29 protocols
Renal — 28 protocols
Haematology — 27 protocols
Endocrine — 24 protocols
Skin — 23 protocols
Surgical — 23 protocols
Cardiovascular — 22 protocols
Rheumatology — 16 protocols
Developmental — 14 protocols
Dermatology — 1 protocols
Urology — 1 protocols
Neonatal — 1 protocols
Neonatal OPD Evaluation — 1 protocols
General — 1 protocols

Cardiovascular

I00 — Acute rheumatic fever without mention of heart involvement

SNOMED: 58718002 | Duration: 14 days | Source: AHA 2009 ARF/RHD Guidelines, WHO 2004 RF Prevention

First-Line

- **Benzathine penicillin G** — 600,000 U IM x1 (<27kg) or 1.2 MU IM x1 (≥27kg); single dose for streptococcal eradication
- **Naproxen** — 10-15 mg/kg/day PO divided q12h; first-line NSAID for arthritis; better safety profile than aspirin

Second-Line

- **Aspirin** — 50-60 mg/kg/day PO divided q6h; anti-inflammatory for arthritis; taper after 1-2 weeks of resolution
- **Penicillin V** — 250mg PO q12h x10 days; oral alternative for streptococcal eradication if IM not feasible
- **Azithromycin** — 12 mg/kg/day PO x5 days (max 500mg); for penicillin-allergic patients

Investigations

- Echocardiography — Assess for subclinical or clinical carditis (mitral/aortic regurgitation) [URGENT]
- ASO titre and Anti-DNase B — Evidence of recent streptococcal infection [URGENT]
- CRP and ESR — Inflammatory markers; ESR >60 supports Jones criteria [URGENT]
- Throat swab culture or RADT — Document current streptococcal infection [URGENT]
- ECG — Prolonged PR interval (minor Jones criterion) [URGENT]

Counselling

- Acute rheumatic fever is caused by untreated strep throat - always complete strep treatment
- Monthly penicillin injection is needed for at least 5-10 years to prevent recurrence
- Joint pain usually resolves in 2-4 weeks with treatment
- Regular echocardiograms are needed to check if heart valves are affected
- All sore throats in the family should be tested and treated for strep

Referral Criteria

Refer to pediatric cardiologist if any cardiac involvement (murmur, echo abnormality); refer to pediatric rheumatologist if atypical presentation or chorea

Hospitalisation Criteria

Admit if carditis with heart failure, severe chorea, high fever not responding to treatment, or diagnostic uncertainty requiring echocardiography and workup

Notes

Penicillin eradication does not alter the ARF course but prevents spread. Secondary prophylaxis with benzathine penicillin G q4wk is essential for at least 5 years or until age 21.

Guideline Changes

Naproxen now recommended as first-line over aspirin for rheumatic arthritis

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I05 — Rheumatic mitral valve disorders

SNOMED: 79619009 | Duration: 365 days | Source: AHA 2020 RHD Scientific Statement, ACC/AHA 2020 VHD

First-Line

- **Benzathine penicillin G** — 1.2 MU IM q3-4wk; secondary prophylaxis to prevent recurrent ARF and valve damage progression
- **Enalapril** — 0.1-0.5 mg/kg/day PO divided q12h; afterload reduction for chronic mitral regurgitation
- **Furosemide** — 1-2 mg/kg/day PO divided q12h; diuretic for CHF symptoms from mitral valve dysfunction

Second-Line

- **Mitral valve repair** — Conservative surgical repair preferred over replacement in children; avoids anticoagulation
- **Percutaneous balloon commissurotomy** — For rheumatic mitral stenosis with favorable anatomy, MVA <1.5cm2, <moderate MR, no LA thrombus
- **Warfarin** — Dose to target INR 2.5-3.5; for mechanical valve replacement or mitral stenosis with atrial fibrillation
- **Digoxin** — 5-10 mcg/kg/day PO; for rate control in atrial fibrillation or symptomatic HF

Investigations

- Echocardiography — Assess valve morphology, severity of MR/MS, LV and LA dimensions [URGENT]
- ECG — LA enlargement, atrial fibrillation, RV hypertrophy
- ASO titre — Monitor for recurrent streptococcal infection
- CRP and ESR — Detect subclinical recurrence of ARF
- Chest X-ray — LA enlargement, pulmonary congestion

Counselling

- Monthly penicillin injection must not be missed - it prevents further heart valve damage
- Report any fever, joint pain, or new breathlessness immediately as it may indicate recurrence
- Antibiotic prophylaxis before dental procedures is required lifelong
- Maintain healthy weight and moderate activity as tolerated
- Surgery may be needed if valve damage progresses - regular echo monitoring is essential

Referral Criteria

Refer to pediatric cardiologist for ongoing valve monitoring; refer to cardiac surgery if severe MR/MS with symptoms, progressive LV dilation, or atrial fibrillation

Hospitalisation Criteria

Admit if acute heart failure, new atrial fibrillation, suspected recurrent ARF episode, planned percutaneous commissurotomy, or valve surgery

Notes

Conservative valve repair is preferred over replacement in pediatric RHD to avoid lifelong anticoagulation. Strict secondary penicillin prophylaxis is critical to prevent recurrent valve injury.

I10 — Paediatric hypertension

SNOMED: 38341003 | Duration: 90 days | Source: AAP 2017

First-Line

- **Amlodipine** — 0.1-0.3 mg/kg/day (max 5-10mg) once daily, long-acting CCB
- **Lisinopril** — 0.07 mg/kg/day (max 5mg) QD initially, preferred for CKD or proteinuria

Second-Line

- **Losartan** — 0.7 mg/kg/day (max 50mg), alternative to ACE inhibitor
- **Hydrochlorothiazide** — 1-3 mg/kg/day (max 50mg), volume-dependent hypertension

Investigations

- Renal function tests (creatinine, BUN) — Rule out renal cause of hypertension [URGENT]
- Urinalysis with protein/creatinine ratio — Screen for renal disease and proteinuria
- Renal ultrasound with Doppler — Screen for structural renal anomalies or renal artery stenosis
- Echocardiography — Assess for left ventricular hypertrophy as end-organ damage
- Serum electrolytes — Baseline and monitor with ACEI/diuretic therapy

Counselling

- Reduce salt in cooking and avoid packaged/junk food which is high in sodium
- Encourage daily physical activity for at least 60 minutes (walking, cycling, sports)
- Maintain healthy weight - weight loss of even 5% can significantly lower blood pressure
- Give medicine at the same time every day; do not skip doses
- Regular blood pressure checks at home and at clinic visits are essential for monitoring

Referral Criteria

Refer to pediatric nephrologist/cardiologist if stage 2 hypertension, secondary hypertension suspected (renal, endocrine), end-organ damage, or persistent hypertension despite 2 medications

Hospitalisation Criteria

Admit if hypertensive emergency (symptomatic: headache, seizures, visual changes, encephalopathy), BP >30 mmHg above 95th percentile, or evidence of end-organ damage (cardiac, renal, retinal)

Notes

Beta-blockers NOT recommended as first-line in children. Lifestyle modification is essential first step.

I27.2 — Pulmonary hypertension

SNOMED: 70995007 | Duration: 365 days | Source: AHA/ATS 2015 Pediatric PH Guidelines, ESC/ERS 2022 PH

First-Line

- **Sildenafil** — 1 mg/kg/dose PO q8h (ages 1-17yr); FDA-approved for WHO Group 1 PAH; titrate based on response
- **Bosentan** — 2 mg/kg/dose PO q12h (ages ≥3yr); ERA approved for pediatric PAH; monitor LFTs monthly
- **Inhaled nitric oxide** — 10-20 ppm via ventilator; first-line for acute PH crisis and PPHN; wean gradually

Second-Line

- **Epoprostenol** — 1-2 ng/kg/min IV continuous; titrate by 1-2 ng/kg/min q15min; for severe PAH (WHO FC III-IV)
- **Treprostinil** — 1.25 ng/kg/min SC/IV; titrate slowly; prostacyclin analogue for advanced PAH
- **Macitentan** — Dose per weight; dual ERA increasingly used in pediatric PAH; limited pediatric data **[NEW 2024-25]**
- **Selexipag** — Oral IP receptor agonist; titrate from 200mcg q12h; emerging use in pediatric PAH **[NEW 2024-25]**

Investigations

- Echocardiography — Estimate PA pressure, assess RV function [STAT]
- Right heart catheterisation — Confirm diagnosis, assess vasoreactivity, guide therapy [URGENT]
- BNP or NT-proBNP — Severity assessment and monitoring [URGENT]
- 6-minute walk test — Functional capacity assessment in older children
- CT angiography — Rule out chronic thromboembolic disease

Counselling

- Pulmonary hypertension requires lifelong medication - never stop without cardiologist guidance
- Avoid high altitudes and air travel without supplemental oxygen
- Avoid dehydration and excessive salt intake
- Annual influenza vaccination and RSV prophylaxis are important
- Report any worsening breathlessness, fainting, or blue discolouration immediately

Referral Criteria

Refer to pulmonary hypertension centre for all confirmed PAH; escalation to prostacyclin therapy if WHO FC III-IV

Hospitalisation Criteria

Admit if acute RV failure, syncope, new arrhythmia, PAH crisis (acute PH with hemodynamic instability), oxygen desaturation <85%, or planned catheterisation

Notes

Combination therapy (ERA + PDE5i) is increasingly standard for pediatric PAH. 2024 RCT showed sildenafil achieves faster pressure reduction than bosentan in neonatal PPHN.

Guideline Changes

Sildenafil FDA-approved for pediatric PAH ages 1-17; 2024 RCT data on sildenafil vs bosentan in neonates

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I30 — Acute pericarditis

SNOMED: 194902002 | Duration: 21 days | Source: ACC 2025 Pericarditis Consensus, ESC 2015 Pericardial Diseases

First-Line

- **Ibuprofen** — 10 mg/kg/dose PO q6-8h (max 40 mg/kg/day); first-line NSAID; taper over 2-4 weeks after symptom resolution
- **Colchicine** — Children ≤5yr: 0.5mg/day PO; >5yr: 1-1.5mg/day PO divided q12h; no loading dose; 3 months duration

Second-Line

- **Naproxen** — 5-10 mg/kg/dose PO q12h; alternative NSAID if ibuprofen not tolerated
- **Prednisone** — 0.25-0.5 mg/kg/day PO; only if NSAID+colchicine fail or contraindicated; slow taper over weeks
- **Anakinra** — 2-4 mg/kg/day SC (max 100mg); IL-1 receptor antagonist for refractory recurrent pericarditis **[NEW 2024-25]**
- **Rilonacept** — Loading 4.4 mg/kg SC then 2.2 mg/kg/wk; FDA-approved for recurrent pericarditis age ≥12yr **[NEW 2024-25]**

Investigations

- Echocardiography — Assess pericardial effusion size and hemodynamic impact [STAT]
- ECG — Diffuse ST elevation, PR depression; low voltage if large effusion [STAT]
- CRP and ESR — Inflammatory markers to guide duration of treatment [URGENT]
- Troponin — Exclude myopericarditis [URGENT]
- Chest X-ray — Cardiomegaly if large effusion (water bottle sign) [URGENT]

Counselling

- Take ibuprofen with food to protect the stomach; colchicine is given for 3 months to prevent recurrence
- Complete bed rest initially; no sports until symptoms resolve and CRP is normal
- Watch for increasing chest pain, breathlessness, or dizziness - these need urgent evaluation
- Avoid corticosteroid treatment if possible as it increases risk of recurrence
- Follow-up echocardiogram is needed to ensure no pericardial effusion or constriction

Referral Criteria

Refer to pediatric cardiologist if moderate-large pericardial effusion, recurrent pericarditis (>2 episodes), suspected purulent or tubercular pericarditis, or tamponade features

Hospitalisation Criteria

Admit if large pericardial effusion with hemodynamic compromise, cardiac tamponade, suspected purulent pericarditis, fever with myopericarditis, or high-risk features (troponin elevation)

Notes

Colchicine added to NSAID halves recurrence rates. Avoid corticosteroids as first-line due to increased recurrence risk; exercise restriction until symptom and CRP resolution.

Guideline Changes

2025 ACC Expert Consensus Statement on pericarditis diagnosis and management released

I38 — Endocarditis, valve unspecified

SNOMED: 56819008 | Duration: 42 days | Source: AHA 2015 IE Update, ESC 2023 Endocarditis Guidelines

First-Line

- **Ampicillin + Gentamicin** — Ampicillin 200-300 mg/kg/day IV divided q4-6h + Gentamicin 3 mg/kg/day IV q8h; empiric for native valve
- **Vancomycin** — 40-60 mg/kg/day IV divided q6-8h; for MRSA or penicillin allergy; target trough 15-20 mcg/mL

Second-Line

- **Ceftriaxone** — 100 mg/kg/day IV once daily (max 4g); for viridans streptococci; 4-week course with or without gentamicin
- **Nafcillin** — 200 mg/kg/day IV divided q4-6h; for MSSA endocarditis; 6-week course
- **Surgical intervention** — Valve repair or replacement for refractory infection, large vegetations >10mm, embolic events, or HF

Investigations

- Blood cultures (3 sets) — Before starting antibiotics; essential for organism identification [STAT]
- Echocardiography (TTE +/- TEE) — Identify vegetations, assess valve function, abscess [STAT]
- Complete blood count — Anemia, leukocytosis, thrombocytopenia monitoring [URGENT]
- CRP and ESR — Monitor treatment response [URGENT]
- Urinalysis — Microscopic hematuria from immune complex deposition

Counselling

- Endocarditis requires 4-6 weeks of intravenous antibiotics - compliance is critical
- Maintain excellent dental hygiene to prevent future episodes
- The child may need a PICC line for prolonged IV antibiotics at home
- Report any new fever, joint pain, or red spots on skin during treatment
- Children with valve disease need antibiotic prophylaxis before dental procedures lifelong

Referral Criteria

Refer to pediatric cardiologist and infectious disease specialist at diagnosis; refer to cardiac surgery if valve destruction, large vegetations >10mm, recurrent emboli, or refractory infection

Hospitalisation Criteria

All infective endocarditis cases require hospitalisation for IV antibiotics, blood cultures, and monitoring; ICU if hemodynamic instability, acute valve failure, or stroke

Notes

Treatment is 4-6 weeks IV antibiotics tailored to culture and sensitivity. Consult pediatric infectious disease; adjust doses based on renal function and drug levels.

Guideline Changes

2023 ESC guidelines updated with refined surgical timing and diagnostic criteria

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I40 — Acute myocarditis

SNOMED: 155336004 | Duration: 14 days | Source: AHA 2021 Pediatric Myocarditis Statement, ACC 2024 Myocarditis ECDP

First-Line

- **Enalapril** — 0.1 mg/kg/day PO divided q12h; ACE inhibitor for LV dysfunction; titrate to 0.5 mg/kg/day
- **Carvedilol** — 0.05 mg/kg/dose PO q12h; titrate to 0.3-0.5 mg/kg/dose; beta-blocker for symptomatic HF
- **Furosemide** — 1-2 mg/kg/dose IV/PO q6-12h; loop diuretic for fluid overload and pulmonary congestion

- **Milrinone** — 0.25-0.75 mcg/kg/min IV infusion; inodilator for cardiogenic shock with low cardiac output

Second-Line

- **IVIG** — 2 g/kg IV over 10-12h single dose; immunomodulatory therapy; evidence mixed but commonly used
- **ECMO** — Extracorporeal membrane oxygenation; bridge to recovery or transplant for refractory cardiogenic shock
- **Spironolactone** — 1-2 mg/kg/day PO divided q12h; aldosterone antagonist for chronic HF management

Investigations

- Echocardiography — Assess LV function, wall motion abnormalities, pericardial effusion [STAT]
- Troponin I/T — Elevated in myocardial injury [STAT]
- ECG — ST-T changes, arrhythmias, low voltage [STAT]
- CRP and ESR — Inflammatory markers [URGENT]
- Cardiac MRI — Gold standard for diagnosis; myocardial edema and late gadolinium enhancement [URGENT]

Counselling

- Your child has inflammation of the heart muscle - complete bed rest is essential
- No sports or vigorous physical activity for at least 3-6 months after recovery
- Give medicines regularly as prescribed - they help the heart recover
- Watch for worsening breathlessness, chest pain, fast heartbeat, or swelling of feet
- Regular follow-up echocardiograms are needed to monitor heart function recovery

Referral Criteria

Refer to pediatric cardiologist immediately on suspicion; transfer to centre with ECMO capability if fulminant presentation

Hospitalisation Criteria

All suspected myocarditis cases require hospitalisation; ICU admission if hemodynamically unstable, arrhythmia, or reduced ejection fraction <40%

Notes

Supportive care is the mainstay; avoid NSAIDs and exercise restriction for 3-6 months. ECMO has shown good outcomes as bridge therapy in fulminant cases with recoverable myocardium.

Guideline Changes

2024 ACC Expert Consensus Decision Pathway for myocarditis diagnosis and management published

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I42 — Cardiomyopathy

SNOMED: 85898001 | Duration: 365 days | Source: AHA 2023 Pediatric CM Statement, AHA/ACC 2024 HCM Guideline

First-Line

- **Enalapril** — 0.1-0.5 mg/kg/day PO divided q12h; ACE inhibitor for DCM with systolic dysfunction; first-line GDMT
- **Carvedilol** — 0.05-0.5 mg/kg/dose PO q12h; beta-blocker for DCM; titrate slowly over weeks
- **Furosemide** — 1-2 mg/kg/day PO divided q12h; loop diuretic for congestive symptoms

Second-Line

- **Sacubitril-Valsartan** — ARNI; FDA-approved for children >1yr; dose by weight; emerging first-line for pediatric HFrEF **[NEW 2024-25]**
- **Mavacamten** — Cardiac myosin inhibitor for obstructive HCM; dose per protocol; new class in 2024 HCM guideline **[NEW 2024-25]**
- **Spironolactone** — 1-2 mg/kg/day PO divided q12h; aldosterone antagonist adjunct for symptomatic HF
- **Amiodarone** — Loading 10-15 mg/kg/day PO x7-10d then 5 mg/kg/day; for ventricular arrhythmias in CM

Investigations

- Echocardiography — Assess chamber size, wall thickness, systolic and diastolic function [STAT]
- ECG — LV hypertrophy, arrhythmias, conduction abnormalities [STAT]
- BNP or NT-proBNP — Heart failure severity and monitoring [URGENT]
- Cardiac MRI — Tissue characterization, fibrosis assessment
- Genetic testing — Family screening and prognostic information

Counselling

- Cardiomyopathy is a chronic heart condition requiring lifelong medication and monitoring
- Give all heart medicines at the same time every day; never skip doses
- Restrict strenuous activity as advised; light activity may be allowed based on type
- Monitor daily weight - sudden weight gain may indicate fluid retention
- Genetic testing may be recommended for family screening in some types

Referral Criteria

Refer to pediatric cardiologist at diagnosis; refer for transplant evaluation if refractory heart failure or EF <25% despite optimal GDMT

Hospitalisation Criteria

Admit if new-onset cardiomyopathy for workup, acute decompensated heart failure, ventricular arrhythmia, syncope, or need for inotropic support

Notes

Treatment is guided by cardiomyopathy subtype: DCM needs afterload reduction, HCM needs negative inotropy. Sacubitril-valsartan is increasingly used in pediatric DCM based on FDA approval for age >1yr.

Guideline Changes

2024 AHA/ACC HCM guideline adds cardiac myosin inhibitors for obstructive HCM

I47 — Paroxysmal tachycardia

SNOMED: 12026006 | Duration: 30 days | Source: AHA PALS 2020, EHRA/AEPC Pediatric Arrhythmia Consensus

First-Line

- **Vagal maneuvers** — Ice to face (infants) or Valsalva (older children); first-line non-pharmacologic intervention for SVT
- **Adenosine** — 0.1 mg/kg IV rapid push (max 6mg); if no response 0.2 mg/kg (max 12mg); flush with rapid NS bolus
- **Propranolol** — 1-3 mg/kg/day PO divided q8h; first-line prophylaxis for recurrent SVT in infants

Second-Line

- **Flecainide** — 2-4 mg/kg/day PO divided q8-12h; for SVT prophylaxis in structurally normal hearts; monitor ECG
- **Amiodarone** — 5 mg/kg IV over 30-60min; for refractory SVT or hemodynamically unstable tachycardia
- **Catheter ablation** — Radiofrequency or cryoablation of accessory pathway; curative for recurrent SVT; preferred in older children
- **Synchronized cardioversion** — 0.5-1 J/kg; increase to 2 J/kg; for hemodynamically unstable SVT unresponsive to adenosine

Investigations

- 12-lead ECG (during and after episode) — Capture tachycardia mechanism and check for pre-excitation [STAT]
- Echocardiography — Rule out structural heart disease; Ebstein anomaly with WPW [URGENT]
- 24-hour Holter monitor — Quantify episode frequency and duration
- Event recorder — For infrequent episodes; worn for weeks to capture events

Counselling

- SVT episodes can be scary but are usually not dangerous and can be stopped
- Learn vagal maneuvers: ice pack on face for babies, Valsalva (blowing into blocked straw) for older children
- If episode lasts more than 15 minutes or child becomes pale/unwell, go to emergency department
- Take prophylactic medicine daily as prescribed to prevent episodes
- Catheter ablation can permanently cure SVT in most older children - discuss with cardiologist

Referral Criteria

Refer to pediatric electrophysiologist for recurrent SVT, need for chronic medication, WPW pattern on ECG, or consideration for catheter ablation (typically >15 kg)

Hospitalisation Criteria

Admit if SVT with hemodynamic compromise, SVT not terminated by vagal maneuvers or single dose adenosine, wide complex tachycardia, or age <3 months with first SVT episode

Notes

Adenosine is first-line pharmacotherapy for acute SVT termination; must be given as rapid push with immediate saline flush. Catheter ablation is curative and preferred for recurrent SVT in children >15kg.

I49.9 — Cardiac arrhythmia, unspecified

SNOMED: 698247007 | Duration: 30 days | Source: EHRA/AEPC 2013 Pediatric Arrhythmia Consensus, AHA PALS

First-Line

- **Adenosine** — 0.1 mg/kg IV rapid push (max 6mg); repeat at 0.2 mg/kg (max 12mg); first-line for SVT diagnosis/treatment
- **Amiodarone** — Loading 5 mg/kg IV over 30-60min; then 5-15 mcg/kg/min infusion; broad-spectrum antiarrhythmic

Second-Line

- **Propranolol** — 1-4 mg/kg/day PO divided q6-8h; beta-blocker for chronic arrhythmia suppression
- **Sotalol** — 2-8 mg/kg/day PO divided q8-12h; class III antiarrhythmic; monitor QTc closely
- **Flecainide** — 2-4 mg/kg/day PO divided q8-12h; class IC for SVT prophylaxis in structurally normal hearts
- **Esmolol** — 100-500 mcg/kg IV bolus then 50-500 mcg/kg/min infusion; ultra-short acting beta-blocker for acute use

Investigations

- 12-lead ECG — Define arrhythmia type; essential first investigation [STAT]
- 24-hour Holter monitor — Capture intermittent arrhythmias and assess daily burden
- Echocardiography — Rule out structural heart disease [URGENT]
- Serum electrolytes (K, Mg, Ca) — Electrolyte abnormalities as arrhythmia triggers [URGENT]
- Thyroid function tests — Hyperthyroidism as a cause of tachyarrhythmia

Counselling

- Your child has an abnormal heart rhythm that needs proper diagnosis with ECG and monitoring
- Give anti-arrhythmic medicines exactly as prescribed; do not skip doses
- Teach child and family to check pulse rate at home
- Report any palpitations, dizziness, fainting, or chest pain immediately
- Avoid excessive caffeine (energy drinks, colas) which can trigger abnormal rhythms

Referral Criteria

Refer to pediatric electrophysiologist for any sustained or symptomatic arrhythmia, WPW syndrome, or arrhythmia requiring chronic medication; catheter ablation evaluation

Hospitalisation Criteria

Admit if hemodynamically significant arrhythmia, SVT not responding to vagal maneuvers/adenosine, ventricular tachycardia, high-degree AV block, or syncope with arrhythmia

Notes

Arrhythmia type must be defined by 12-lead ECG and Holter before chronic therapy. Centers should develop institution-specific pediatric antiarrhythmic protocols due to variable drug availability.

I50 — Heart failure

SNOMED: 84114007 | Duration: 365 days | Source: AHA 2022 HF Guideline, AHA 2024 Pediatric CHD-HF Statement

First-Line

- **Enalapril** — 0.1 mg/kg/day PO divided q12h; titrate to 0.5 mg/kg/day; ACE inhibitor cornerstone of HFrEF GDMT
- **Carvedilol** — 0.05 mg/kg/dose PO q12h; titrate to 0.35 mg/kg/dose; preferred beta-blocker in pediatric HF
- **Furosemide** — 1-2 mg/kg/dose PO/IV q6-12h; loop diuretic for volume overload; adjust to euvolemia
- **Spironolactone** — 1-2 mg/kg/day PO divided q12h; aldosterone antagonist for symptomatic HFrEF

Second-Line

- **Sacubitril-Valsartan** — ARNI; FDA-approved age >1yr; replacing ACE-I as preferred neurohormonal blockade in pediatric HFrEF **[NEW 2024-25]**
- **Milrinone** — 0.25-0.75 mcg/kg/min IV infusion; inodilator for acute decompensated HF with low output
- **Digoxin** — 5-10 mcg/kg/day PO; target level 0.5-1.0 ng/mL; for symptom relief in HFrEF

Investigations

- Echocardiography — Assess ejection fraction, chamber dimensions, valve function [STAT]
- BNP or NT-proBNP — Diagnosis and monitoring of heart failure severity [URGENT]
- ECG — Arrhythmia, conduction disease, chamber hypertrophy [URGENT]
- Chest X-ray — Cardiomegaly, pulmonary congestion [URGENT]
- Serum electrolytes and renal function — Baseline and monitoring with diuretic/ACEI therapy [URGENT]

Counselling

- Heart failure medicines must be taken daily without missing; they help the heart pump better
- Monitor weight daily - sudden increase of 0.5-1 kg may mean fluid retention
- Restrict salt intake; avoid packaged foods, pickles, and papad
- Watch for increasing breathlessness, leg swelling, or inability to lie flat - see doctor urgently
- Some activity is beneficial; follow doctor guidance on safe exercise levels

Referral Criteria

Refer to pediatric cardiologist for all new heart failure; refer for advanced HF/transplant evaluation if refractory to optimal GDMT

Hospitalisation Criteria

Admit if acute decompensated heart failure, new-onset HF for workup, need for IV diuretics or inotropes, arrhythmia, or pre-syncope/syncope

Notes

Pediatric HF GDMT mirrors adult pillars: ACE-I/ARNI + beta-blocker + MRA + diuretic. SGLT2 inhibitors show promise but lack pediatric data; sacubitril-valsartan is rapidly becoming first-line.

Guideline Changes

2024 AHA scientific statement on chronic HF in children with CHD published

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M30.3 — Kawasaki disease

SNOMED: 75053002 | Duration: 14 days | Source: AHA 2017

First-Line

- **IVIG** — 2 g/kg single infusion over 10-12h, within first 10 days of fever onset
- **Aspirin** — 30-50 mg/kg/day in 4 divided doses during acute phase, then 3-5 mg/kg/day once afebrile

Second-Line

- **Infliximab** — 5 mg/kg single dose, for IVIG-resistant Kawasaki disease
- **Corticosteroids** — For high-risk IVIG-resistant cases per Kobayashi/Egami scoring

Investigations

- Echocardiography — At diagnosis, 2 weeks and 6 weeks; assess coronary arteries [STAT]
- Complete blood count — Thrombocytosis, leukocytosis, anaemia in Kawasaki disease [URGENT]
- CRP and ESR — Inflammatory markers to guide treatment response [URGENT]
- Liver function tests — Hepatic involvement and albumin level (low albumin = IVIG resistance risk) [URGENT]
- Urinalysis — Sterile pyuria common in Kawasaki disease

Counselling

- This is a serious condition affecting blood vessels of the heart - treatment must start urgently
- IVIG infusion may cause fever and headache; this is expected and managed in hospital
- Give aspirin as prescribed - this prevents blood clots in heart vessels
- Avoid live vaccines (MMR, varicella) for 11 months after IVIG treatment
- Regular follow-up echocardiograms at 2 weeks and 6 weeks are essential to check heart arteries

Referral Criteria

Refer to pediatric cardiologist immediately on diagnosis for echocardiography and co-management; ongoing follow-up for coronary artery monitoring

Hospitalisation Criteria

All children with suspected Kawasaki disease should be hospitalised for IVIG infusion; emergent admission if coronary artery aneurysm, IVIG-resistant fever, shock/hemodynamic instability (Kawasaki shock syndrome)

Notes

Echocardiogram at diagnosis, 2 weeks and 6 weeks. Coronary artery aneurysm is the key complication.

Q20.3 — Transposition of great vessels

SNOMED: 399216004 | Duration: 14 days | Source: AHA/ACC CHD Guidelines, ACC 2025 ACHD Guideline

First-Line

- **Arterial switch operation** — Jatene procedure; anatomic repair switching aorta and PA with coronary reimplantation; within first 2 weeks of life
- **Prostaglandin E1** — 0.01-0.1 mcg/kg/min IV continuous; maintains ductal patency for systemic-pulmonary mixing pre-surgery
- **Balloon atrial septostomy** — Rashkind procedure; emergent catheter-based creation of ASD to improve mixing; for severe cyanosis

Second-Line

- **Senning/Mustard procedure** — Atrial switch operation; historical alternative now reserved for late-presenting or complex anatomy cases
- **Rastelli procedure** — For d-TGA with VSD and PS; LV-to-aorta baffle + RV-to-PA conduit

Investigations

- Echocardiography — Confirm diagnosis, assess coronary pattern, and associated lesions [STAT]
- Pulse oximetry (pre/post-ductal) — Screening and monitoring cyanosis [STAT]

- ECG — RV predominance in neonatal d-TGA [URGENT]
- Chest X-ray — Egg-on-a-string appearance, narrow mediastinum [URGENT]
- Cardiac catheterisation — Balloon septostomy and hemodynamic assessment [STAT]

Counselling

- Your baby needs surgery within the first 2 weeks of life - this is urgent but has excellent outcomes
- The baby will be in NICU/PICU before and after surgery
- After arterial switch operation, most children lead completely normal active lives
- Lifelong cardiology follow-up is required to monitor coronary arteries and heart valve function
- Ensure all routine vaccinations are given on schedule after recovery

Referral Criteria

Emergent referral to pediatric cardiac surgery centre at prenatal or postnatal diagnosis; lifelong cardiology follow-up

Hospitalisation Criteria

All cases require immediate NICU admission; PGE1 to maintain ductal patency; Rashkind septostomy if severely cyanotic; arterial switch surgery within 2 weeks

Notes

Arterial switch must be performed within 2-4 weeks before LV deconditions in low-pressure pulmonary circuit. Lifelong follow-up required for coronary and neo-aortic root complications.

Q21.0 — Ventricular septal defect

SNOMED: 30288003 | Duration: 365 days | Source: AHA/ACC 2024 CHD Management, ACC VSD Algorithm

First-Line

- **Observation** — Small restrictive VSDs; many close spontaneously in first year; serial echo q6-12mo monitoring
- **Surgical patch closure** — For moderate-large VSDs with CHF, Qp:Qs >2, or PA hypertension; via sternotomy with CPB
- **Furosemide** — 1-2 mg/kg/day PO divided q12h; first-line diuretic for CHF symptoms from large L-to-R shunt

Second-Line

- **Captopril** — 0.5-2 mg/kg/day PO divided q8h; afterload reduction for symptomatic CHF management
- **Spironolactone** — 1-3 mg/kg/day PO divided q12h; potassium-sparing diuretic adjunct for CHF
- **Transcatheter device closure** — Amplatzer muscular VSD occluder for large muscular VSDs; percutaneous or hybrid approach
- **Digoxin** — 5-10 mcg/kg/day PO divided q12h; for symptomatic CHF with large shunt pre-surgery

Investigations

- Echocardiography with Doppler — Confirm diagnosis, assess shunt size, and PA pressure [URGENT]
- ECG — LV hypertrophy (moderate VSD) or biventricular hypertrophy (large VSD)
- Chest X-ray — Cardiomegaly and increased pulmonary vascularity
- Cardiac catheterisation — Assess PVR before closure if pulmonary hypertension suspected

Counselling

- Many small VSDs close on their own during the first year of life
- Watch for fast breathing during feeds, poor weight gain, or excessive sweating - report to doctor
- If surgery is needed, it has excellent outcomes; most children recover fully
- Extra calories may be needed as heart has to work harder - your doctor will guide feeding
- Regular echo follow-up is important to track the size of the hole

Referral Criteria

Refer to pediatric cardiologist for all VSDs; refer to cardiac surgery if large VSD with heart failure, Qp:Qs >2, or rising PA pressure

Hospitalisation Criteria

Admit if heart failure (poor feeding, tachypnea, hepatomegaly), planned surgical or device closure, or signs of pulmonary hypertension

Notes

Large VSDs with elevated PA pressure must be closed early to prevent irreversible pulmonary vascular obstructive disease. Perimembranous and outlet VSDs require surgical closure.

Guideline Changes

Transcatheter VSD closure devices gaining wider use for muscular defects

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Q21.1 — Atrial septal defect

SNOMED: 70142008 | Duration: 365 days | Source: ACC/AHA 2018 ACHD, ACC 2022 ASD Algorithm

First-Line

- **Observation** — Small ASDs (<5mm) in children <2yr; many close spontaneously; serial echo monitoring q6-12mo
- **Transcatheter device closure** — Percutaneous Amplatzer or GORE occluder for secundum ASD ≥5mm with RV volume overload; via femoral vein
- **Surgical patch closure** — Open repair via sternotomy with CPB for large, sinus venosus, or primum ASDs not amenable to device

Second-Line

- **Furosemide** — 1-2 mg/kg/day PO divided q12h; for symptomatic volume overload pre-intervention
- **Digoxin** — 5-10 mcg/kg/day PO divided q12h; for RV failure symptoms if present pre-operatively

Investigations

- Echocardiography with Doppler — Confirm diagnosis, assess shunt size and RV dilation [URGENT]
- ECG — Right axis deviation, rSR pattern in V1
- Chest X-ray — Cardiomegaly and increased pulmonary vascularity
- Cardiac catheterisation — Pre-device closure to assess PVR if pulmonary hypertension suspected

Counselling

- Many small ASDs close on their own in the first 2 years - regular echo monitoring is needed
- The child can usually participate in normal physical activities unless doctor advises otherwise
- Watch for symptoms like easy tiredness, frequent respiratory infections, or poor weight gain
- If device closure or surgery is needed, it has excellent outcomes with low risk
- Keep all routine vaccinations up to date including annual influenza vaccine

Referral Criteria

Refer to pediatric cardiologist for all detected ASDs; refer to cardiac surgery/interventional cardiology if significant shunt (Qp:Qs >1.5) with RV dilation

Hospitalisation Criteria

Admit if heart failure symptoms, planned device closure or surgical repair, paradoxical embolism, or arrhythmia (atrial fibrillation/flutter)

Notes

Closure is recommended if significant L-to-R shunt (Qp:Qs >1.5) with RV dilation. Device closure has low morbidity and avoids sternotomy for eligible secundum defects.

Guideline Changes

Transcatheter device closure increasingly preferred over surgery for secundum ASDs

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Q21.3 — Tetralogy of Fallot

SNOMED: 86299006 | Duration: 365 days | Source: ACC 2024 TOF Clinical Algorithm, AATS Expert Consensus

First-Line

- **Complete surgical repair** — VSD patch closure + RVOT reconstruction; typically performed at 3-6 months of age
- **Prostaglandin E1** — 0.01-0.05 mcg/kg/min IV infusion; maintains ductal patency in cyanotic neonates pre-surgery

Second-Line

- **Modified BT shunt** — Palliative systemic-to-pulmonary shunt for small neonates not suitable for primary repair
- **RVOT stenting** — Transcatheter palliation; stent placed across RVOT to improve pulmonary blood flow as bridge to repair **[NEW 2024-25]**
- **PDA stenting** — Transcatheter ductal stent as alternative to surgical shunt for palliation in selected neonates **[NEW 2024-25]**
- **Propranolol** — 1-2 mg/kg/day PO divided q8h; for hypercyanotic (tet) spell prevention and management

Investigations

- Echocardiography — Confirm anatomy: VSD, overriding aorta, RVOT obstruction, RVH [STAT]
- ECG — Right axis deviation, RV hypertrophy [URGENT]
- Chest X-ray — Boot-shaped heart, decreased pulmonary vascularity [URGENT]
- Cardiac catheterisation — Pre-surgical assessment of coronary anatomy and PA anatomy
- CT angiography — Alternative for coronary and PA anatomy assessment

Counselling

- Your child has a combination of 4 heart defects that will need surgery, usually around 3-6 months
- Blue spells (tet spells) may happen - hold baby in knee-chest position and go to hospital immediately
- Ensure adequate hydration and avoid prolonged crying which can trigger spells
- Surgery has excellent outcomes; most children lead normal lives after repair
- Lifelong cardiology follow-up is required even after successful repair

Referral Criteria

Refer to pediatric cardiac surgeon at diagnosis for surgical planning; urgently if symptomatic cyanosis or hypercyanotic spells

Hospitalisation Criteria

Admit if hypercyanotic (tet) spell, oxygen saturation <75%, severe cyanosis, planned surgical repair, or need for palliative shunt

Notes

Complete repair by 6 months is standard; earlier if symptomatic cyanosis. Knee-chest positioning and morphine 0.1mg/kg IM are acute tet spell measures before definitive repair.

Guideline Changes

ACC 2024 updated clinical practice algorithm for unrepaired and repaired TOF follow-up

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Q22.1 — Pulmonary valve stenosis

SNOMED: 253587003 | Duration: 365 days | Source: ACC 2023 PS Clinical Practice Algorithm, AHA 2011 Pediatric CHD

First-Line

- **Balloon pulmonary valvuloplasty** — First-line for moderate-severe valvar PS (gradient >40mmHg); percutaneous via femoral vein; excellent outcomes
- **Observation** — Mild PS (gradient <40mmHg) requires no intervention; serial echo q1-2yr to monitor progression

Second-Line

- **Surgical valvotomy** — Open surgical repair for dysplastic valves (e.g., Noonan syndrome) not responsive to balloon
- **RVOT patch augmentation** — Surgical patch for subvalvar or supravalvar obstruction combined with valve repair
- **Propranolol** — 1-2 mg/kg/day PO divided q8h; for infundibular spasm if dynamic RVOT obstruction present

Investigations

- Echocardiography with Doppler — Assess gradient, RV function, and valve morphology [URGENT]
- ECG — RV hypertrophy and right axis deviation
- Chest X-ray — Post-stenotic PA dilation, decreased pulmonary vascularity in severe PS

Counselling

- Mild pulmonary stenosis usually does not need treatment and is monitored with regular echo
- If balloon procedure is needed, it is done through a small tube in the leg vein - very safe
- Child can usually participate in normal activities; severe PS may need exercise restriction
- Regular cardiology follow-up is needed even after successful balloon valvuloplasty
- Watch for increasing tiredness, bluish discolouration, or fainting with exertion

Referral Criteria

Refer to pediatric cardiologist at diagnosis; refer for intervention if gradient >40 mmHg, RV dysfunction, or symptomatic stenosis

Hospitalisation Criteria

Admit if critical PS in neonate (PGE1 dependent), planned balloon valvuloplasty, right heart failure, or significant cyanosis

Notes

Balloon valvuloplasty is curative in most cases with excellent long-term outcomes. Patients with Noonan or Williams syndromes have lower success rates with balloon and may need surgical repair.

Guideline Changes

ACC published updated clinical practice algorithm for PS follow-up in 2023

Q23.9 — Anomalies of aortic valve

SNOMED: 448743001 | Duration: 365 days | Source: AHA 2024 Aortopathy Statement, ACC/AHA 2021 VHD

First-Line

- **Balloon aortic valvuloplasty** — Percutaneous; indicated for critical valvar AS in neonates who are ductal-dependent regardless of gradient
- **Prostaglandin E1** — 0.01-0.1 mcg/kg/min IV continuous infusion to maintain ductal patency pre-intervention
- **Surgical valvotomy** — Open surgical repair for neonates/infants not amenable to balloon valvuloplasty

Second-Line

- **Ross procedure** — Autograft pulmonary valve replacement of aortic valve with pulmonary homograft; gold standard for AVR in children
- **Mechanical valve replacement** — For older children when repair not feasible; requires lifelong anticoagulation with warfarin
- **Enalapril** — 0.1 mg/kg/day PO divided q12h; afterload reduction for aortic regurgitation

Investigations

- Echocardiography — Primary diagnostic imaging; assess gradient and valve morphology [URGENT]
- ECG — LV hypertrophy, ST changes in significant stenosis [URGENT]
- Chest X-ray — Assess cardiac silhouette and pulmonary vascularity
- Cardiac MRI — Detailed aortopathy assessment in bicuspid aortic valve

Counselling

- Your child has a heart valve problem that needs regular cardiology follow-up
- Watch for symptoms like difficulty breathing during feeds, poor weight gain, or bluish skin
- Maintain good dental hygiene to prevent heart valve infections (endocarditis)
- Ensure all routine vaccinations including influenza are up to date
- Some children may need surgery or catheter procedures - your cardiologist will guide timing

Referral Criteria

Refer to pediatric cardiologist at diagnosis; refer to pediatric cardiac surgery if gradient >50 mmHg, symptomatic stenosis, or progressive aortic regurgitation with LV dilation

Hospitalisation Criteria

Admit if critical aortic stenosis in neonate (ductal-dependent), acute heart failure, syncope with exertion, or planned catheterisation/surgical intervention

Notes

Repair is preferred over replacement in pediatric patients; 10-year freedom from reoperation is ~51%. Bicuspid aortic valve is most common anomaly and requires lifelong surveillance for aortopathy.

Guideline Changes

2025 ACC/AHA/ISACHD guideline for congenital heart disease management released

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Q24.9 — Congenital malformation of cardiac chambers and connections, unspecified

SNOMED: 13213009 | Duration: 365 days | Source: AHA/ACC 2018 ACHD Guidelines, ACC 2025 CHD Guideline

First-Line

- **Cardiac catheterization** — Diagnostic and interventional; for defect characterization and percutaneous repair of simple lesions
- **Surgical repair** — Open-heart surgery tailored to specific defect anatomy; timing based on hemodynamic significance

Second-Line

- **Furosemide** — 1-2 mg/kg/day PO divided q12h; for CHF symptom management pre-intervention
- **Enalapril** — 0.1 mg/kg/day PO divided q12h; afterload reduction for ventricular dysfunction
- **Digoxin** — 5-10 mcg/kg/day PO divided q12h; for symptomatic heart failure support

Investigations

- Echocardiography — Primary imaging for structural cardiac assessment [STAT]
- Cardiac MRI — Detailed anatomy when echo is insufficient
- ECG — Rhythm and conduction abnormalities [URGENT]
- Chest X-ray — Cardiac silhouette and pulmonary vascularity
- Cardiac catheterisation — Hemodynamic assessment and pre-surgical planning

Counselling

- Your child has a heart condition that requires further testing to understand fully
- Keep all cardiology appointments and scheduled echocardiograms
- Watch for fast breathing, poor feeding, excessive sweating, or bluish discolouration

- All routine vaccinations should be given on schedule unless specifically advised otherwise
- Lifelong cardiology follow-up is important even after any corrective procedures

Referral Criteria

Refer to pediatric cardiologist immediately at detection for definitive diagnosis with echocardiography and cardiac MRI; ongoing cardiology follow-up lifelong

Hospitalisation Criteria

Admit if symptomatic heart failure, cyanosis, arrhythmia, need for diagnostic cardiac catheterisation, or planned surgical/interventional procedure

Notes

This unspecified code requires further diagnostic workup with echocardiography and cardiac MRI to define anatomy. Lifelong cardiology follow-up is essential even after surgical correction.

Guideline Changes

2025 ACC/AHA/ISACHD guideline for congenital heart disease released

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Q25.1 — Coarctation of aorta

SNOMED: 7305005 | Duration: 365 days | Source: STS 2024 Neonatal CoA Guidelines, AHA 2024 Aortopathy Statement

First-Line

- **Surgical repair** — Coarctectomy with extended end-to-end anastomosis; primary technique in neonates and infants
- **Prostaglandin E1** — 0.01-0.1 mcg/kg/min IV infusion to maintain ductal patency in neonates prior to intervention
- **Balloon angioplasty** — Preferred for discrete coarctation in children 4mo-5yr (<25kg) without arch hypoplasia

Second-Line

- **Stent placement** — Preferred in older children >25kg; endovascular stenting for native or recurrent coarctation
- **Enalapril** — 0.1 mg/kg/day PO divided q12h; for residual hypertension post-repair, titrate to effect
- **Amlodipine** — 0.1-0.2 mg/kg/day PO once daily (max 5mg); adjunct antihypertensive post-repair
- **Atenolol** — 0.5-1 mg/kg/day PO once daily; beta-blocker for post-repair hypertension

Investigations

- Echocardiography — Confirm diagnosis, assess gradient and associated lesions [STAT]
- Four-limb blood pressure — Upper-lower limb BP gradient >20 mmHg is diagnostic [STAT]
- ECG — LV hypertrophy assessment [URGENT]
- Chest X-ray — Rib notching (older children), figure-of-3 sign
- CT/MR angiography — Pre-surgical anatomic delineation

Counselling

- Your child has a narrowing of the main blood vessel (aorta) that needs treatment
- After repair, lifelong blood pressure monitoring is essential - check BP in both arms
- Regular cardiology follow-up is needed even after successful surgery
- Report any headache, nosebleeds, or leg pain with exercise to the cardiologist
- Maintain good dental hygiene for endocarditis prevention

Referral Criteria

Refer to pediatric cardiologist at diagnosis; urgent referral to pediatric cardiac surgery if critical coarctation in neonate, BP gradient >20 mmHg, or persistent hypertension

Hospitalisation Criteria

Admit if neonatal critical coarctation (PGE1 needed), heart failure, planned surgical or catheter intervention, or hypertensive emergency

Notes

Long-term hypertension monitoring is essential even after successful repair. SBE prophylaxis recommended indefinitely regardless of associated anomalies.

Guideline Changes

2024 STS clinical practice guidelines published for neonatal/infant coarctation management

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Q26 — Congenital malformations of great veins

SNOMED: 205845004 | Duration: 365 days | Source: AHA/ACC CHD Guidelines, ISSVA 2018 Classification

First-Line

- **Observation** — Asymptomatic anomalies (e.g., persistent LSVC) often require no treatment; serial echo monitoring
- **Surgical repair** — For hemodynamically significant anomalies causing cyanosis or volume overload; tailored to anatomy

Second-Line

- **Catheter intervention** — Stenting or balloon angioplasty for stenotic great vein lesions causing obstruction
- **Sirolimus** — 0.8 mg/m2/dose PO q12h; target trough 10-15 ng/mL; for symptomatic venous malformations **[NEW 2024-25]**
- **Anticoagulation** — Enoxaparin 1 mg/kg SC q12h; for thrombosis risk in anomalous venous connections

Investigations

- Echocardiography — Identify anomalous venous connections and associated cardiac lesions [URGENT]
- CT angiography — Detailed venous anatomy mapping
- MR angiography — Non-radiation alternative for detailed venous anatomy
- Doppler ultrasound — Assess flow patterns in anomalous veins

Counselling

- Most great vein anomalies are discovered incidentally and do not cause problems
- Regular cardiology follow-up is important to monitor for any changes
- If your child has a LSVC, it is important to inform doctors before any central line placement or pacemaker
- Surgery is only needed if the anomaly is causing symptoms
- Keep all routine vaccinations on schedule

Referral Criteria

Refer to pediatric cardiologist for evaluation of all great vein anomalies; refer to vascular surgery/interventional radiology if symptomatic venous malformation

Hospitalisation Criteria

Admit if symptomatic venous obstruction, cyanosis from anomalous drainage, planned surgical repair, or thrombotic complication

Notes

Most isolated great vein anomalies (e.g., persistent LSVC) are incidental findings requiring no intervention. Hemodynamically significant lesions need individualized surgical or catheter-based management.

Q26.2 — Total anomalous pulmonary venous connection

SNOMED: 448727008 | Duration: 30 days | Source: AHA Pediatric CHD Guidelines, STS Congenital Database

First-Line

- **Urgent surgical repair** — Anastomosis of common pulmonary venous confluence to left atrium with ligation of anomalous connection; CPB required
- **Prostaglandin E1** — 0.01-0.05 mcg/kg/min IV infusion; maintains PDA to allow oxygenated blood to reach systemic circulation

Second-Line

- **Inhaled nitric oxide** — 10-20 ppm via ventilator; for pulmonary hypertension management in obstructed TAPVC pre-operatively
- **ECMO** — Extracorporeal membrane oxygenation as bridge in critically ill neonates with refractory hypoxemia
- **Furosemide** — 1 mg/kg/dose IV q6-12h; for pulmonary edema management in obstructed TAPVC

Investigations

- Echocardiography — Primary diagnosis: identify drainage pattern and obstruction [STAT]
- ECG — Right axis deviation, RV hypertrophy, peaked P waves [URGENT]
- Chest X-ray — Snowman sign (supracardiac), pulmonary edema if obstructed [URGENT]
- CT angiography — Delineate anatomy pre-surgery if echo insufficient [URGENT]

Counselling

- This is a serious heart condition requiring urgent surgery - your baby will be in NICU/PICU
- After surgery, regular cardiology follow-up is essential to monitor for vein narrowing
- Watch for fast breathing, difficulty feeding, or bluish skin and report immediately
- Ensure all vaccinations including RSV prophylaxis are given as recommended
- Long-term outcomes are generally good after successful surgical repair

Referral Criteria

Emergent referral to pediatric cardiac surgeon at diagnosis; ongoing follow-up with pediatric cardiologist for pulmonary vein stenosis surveillance

Hospitalisation Criteria

All TAPVC cases require hospitalisation; emergency surgery for obstructed TAPVC; semi-urgent surgery for non-obstructed TAPVC within first month of life

Notes

Obstructed TAPVC is a surgical emergency requiring immediate repair. Non-obstructed TAPVC should be repaired within the first month of life; lifelong follow-up for pulmonary vein stenosis.

Dermatology

L20.9 — Atopic Dermatitis

SNOMED: 24079001 | Duration: 14 days | Source: AAP Atopic Dermatitis Clinical Report 2025, AAD Guidelines 2024, AAAAI/ACAAI Joint Task Force 2023, NICE

First-Line

- **Ceramide-based Emollient Cream** — Apply liberally 2-3 times daily, long-term use even after improvement, cornerstone of therapy
- **Mometasone Furoate 0.1% Cream** — Apply thin layer once daily for 5-7 days acute flare, then taper to 2 times/week maintenance, avoid prolonged use on face
- **Mometasone Furoate 0.1% Lotion** — Apply once daily at night for scalp/extensive lesions, 1-2 weeks then taper
- **Coconut Oil / Aloe Vera Lotion** — Natural moisturization, coconut oil in summer, aloe vera-based lotion in winter, safe and soothing
- **Levocetirizine** — Age-appropriate dose PO for 5-7 days, for pruritus and sleep disturbance
- **Fexofenadine** — Age-appropriate dose PO for 5-7 days, alternative antihistamine for itching

Investigations

- CBC with absolute eosinophil count — Allergy workup, elevated eosinophils suggest atopy
- Serum IgE levels — Confirm atopic tendency
- Allergy testing (skin prick / specific IgE) — Identify triggers in severe/recurrent cases

Counselling

- Apply moisturizer generously 2-3 times daily - this is the most important treatment
- Apply moisturizer within 3 minutes of bathing to lock in moisture
- Use steroid cream only during flare-ups as directed, do not use continuously
- Use lukewarm water for bath and mild soap-free cleanser only
- Avoid woolen clothes, harsh soaps, dust, and suspected food allergens
- This is a chronic condition with relapses - regular skin care prevents flare-ups

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Oozing or pus from skin lesions suggesting infection
- Severe redness or swelling of affected skin
- Fever developing with skin lesions
- No response to treatment after one week

Referral Criteria

Refer to dermatologist if no improvement with topical steroids after 2 weeks, severe widespread disease, suspected secondary infection requiring systemic treatment, or need for immunomodulators (tacrolimus/pimecrolimus)

Hospitalisation Criteria

Admit if erythroderma, severe secondary bacterial infection (oozing, fever), eczema herpeticum, or severe disease not responding to outpatient management

Notes

Treatment principles: restore skin barrier, reduce inflammation, control itching, prevent relapse. Moisturizer is cornerstone. Steroid for flare only - avoid overuse. Lukewarm water bath, mild soap-free cleanser, apply moisturizer within 3 minutes after bath. Avoid harsh soaps, woolen clothes, dust/allergens.

Guideline Changes

AAP 2025: Updated clinical report on skin-directed management - daily bathing with lukewarm water and gentle cleanser, moisturizer within minutes of bath; proactive intermittent topical anti-inflammatory maintenance to prevent flares; AAD 2024: Updated treatment guidelines with expanded role of topical calcineurin inhibitors and JAK inhibitors for moderate-severe disease; AAAAI 2023: Dupilumab recommended for moderate-severe AD in children >=6 months

Developmental

F51 — Sleep disorders not due to a substance or known physiological condition

SNOMED: 192460004 | Duration: 42 days | Source: ESRS/European 2024, IPSA 2025

First-Line

- **Sleep Hygiene Education** — Consistent bedtime routine; limit screens 1 hr before bed; cool dark room; regular wake time
- **Behavioral Sleep Intervention** — Graduated extinction or bedtime fading; stimulus control; for ages 0-5; parent-directed
- **CBT for Insomnia (CBT-I)** — Sleep restriction, stimulus control, cognitive restructuring; 4-8 sessions; ages 7+

Second-Line

- **Melatonin** — 0.5-1 mg initial (ages 2-5), 1-3 mg (ages 6-12), 3-5 mg (adolescents); PO 30-60 min before bed
- **Melatonin prolonged-release (Slenyto)** — 2 mg PO at bedtime; may increase to 5 mg; pediatric-licensed formulation for ASD/Smith-Magenis **[NEW 2024-25]**
- **Clonidine** — 0.05-0.1 mg PO at bedtime; off-label for sleep-onset delay; monitor BP; ages 6+
- **Hydroxyzine** — 0.5-1 mg/kg PO at bedtime; max 50 mg; short-term use for sleep-onset difficulty; antihistamine

Investigations

- Sleep diary (2 weeks) — Document sleep-wake patterns
- Polysomnography — If obstructive sleep apnea or parasomnia suspected
- Ferritin — Low ferritin associated with restless legs/poor sleep

Counselling

- Set a fixed bedtime and wake time — even on weekends
- No screens (TV, phone, tablet) for at least 1 hour before bedtime
- Keep the bedroom cool, dark, and quiet — use a nightlight if needed
- Avoid caffeinated drinks (cola, tea, chocolate) after afternoon
- A calm bedtime routine (bath, story, lullaby) signals the brain to sleep

Referral Criteria

Refer to sleep specialist or child psychiatrist if sleep disorder persists beyond 3 months despite behavioral interventions, or if sleep apnea/parasomnia suspected

Hospitalisation Criteria

Hospitalisation rarely needed; admit only if severe sleep deprivation causing safety concerns or for polysomnography in complex cases

Notes

Behavioral interventions are first-line for all pediatric insomnia. Use pharmaceutical-grade melatonin only; OTC products have inconsistent dosing per FDA/IPSA warnings.

Guideline Changes

2025 IPSA consensus: melatonin only after behavioral interventions fail; pharmaceutical-grade preferred

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F70 — Mild intellectual disabilities

SNOMED: 86765009 | Duration: 365 days | Source: AAIDD 2021, AACAP 2023

First-Line

- **Special Education Services (IEP)** — Individualized education program; modified curriculum; resource room or inclusion support; IDEA-mandated

- **Speech-Language Therapy** — 1-3 sessions/week; targeting functional communication, social language, and literacy
- **Behavioral Therapy** — Positive behavior support; functional communication training; social skills groups; ongoing
- **Occupational Therapy** — 1-2 sessions/week; adaptive daily living skills, fine motor, vocational readiness

Second-Line

- **Methylphenidate** — 0.3 mg/kg/dose BID-TID; for comorbid ADHD; start low, titrate cautiously; monitor closely
- **Risperidone** — 0.25-0.5 mg/day initial, titrate to 0.5-2 mg/day; for severe aggression/irritability; short-term
- **Fluoxetine** — 5-10 mg/day initial, titrate to 10-20 mg/day; for comorbid anxiety or depression; ages 8+

Investigations

- Karyotype/chromosomal microarray — Identify genetic cause
- Thyroid function tests — Rule out hypothyroidism
- Hearing assessment (BERA/OAE) — Rule out hearing loss contributing to delays
- MRI brain — If dysmorphic features or focal neurological signs

Counselling

- Your child can learn and grow — focus on their strengths and what they can do
- Enroll in special education or inclusive school with resource room support
- Practice daily living skills (dressing, brushing teeth) at home with repetition and praise
- Join a parent support group for guidance and emotional support
- Regular follow-up every 6 months to reassess developmental progress and therapies

Referral Criteria

Refer to developmental pediatrician or child psychiatrist if comorbid ADHD, anxiety, seizures, or behavioral crisis

Hospitalisation Criteria

Admit only for acute behavioral crisis with self-harm/aggression not manageable outpatient, or for inpatient evaluation of seizures

Notes

No medication treats intellectual disability itself; pharmacotherapy targets comorbid psychiatric conditions. Most with mild ID achieve functional independence with support.

Guideline Changes

Increased focus on functional communication training to reduce challenging behaviors

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F71 — Moderate intellectual disabilities

SNOMED: 61152003 | Duration: 365 days | Source: AAIDD 2021, AACAP 2023

First-Line

- **Special Education Services (IEP)** — Self-contained or supported classroom; functional academics; life skills curriculum; IDEA-mandated
- **Speech-Language Therapy with AAC** — 2-5 sessions/week; augmentative/alternative communication devices; functional communication goals
- **Occupational Therapy** — 2-3 sessions/week; self-care, feeding, dressing skills; sensory integration; adaptive equipment
- **Applied Behavior Analysis (ABA)** — 10-25 hrs/week; for adaptive skill building and challenging behavior reduction; individualized

Second-Line

- **Risperidone** — 0.25 mg/day initial, titrate to 0.5-2 mg/day; for severe aggression/self-injury; FDA-approved 5-16y
- **Methylphenidate** — 0.3 mg/kg/dose BID; for comorbid ADHD; lower efficacy in moderate ID; monitor closely
- **Melatonin** — 0.5-3 mg PO 30-60 min before bed; for comorbid sleep-onset insomnia; common in ID
- **Valproate** — 10-15 mg/kg/day initial; for comorbid epilepsy (common in moderate ID); monitor levels/LFTs

Investigations

- Chromosomal microarray — Identify genetic etiology
- EEG — If seizures suspected (common in moderate ID)
- MRI brain — Structural brain anomaly screening
- Metabolic screen (TMS, urine GCMS) — Rule out inborn errors of metabolism

Counselling

- Your child will benefit greatly from structured routines and visual schedules at home
- Speech therapy and AAC devices can help your child communicate better — practice at home daily
- Teach self-care skills step by step with patience — dressing, eating, toileting
- Monitor for seizures (present in 15-30%) and report any unusual episodes
- Connect with local disability support organizations for government benefits and school placement

Referral Criteria

Refer to child psychiatrist for severe aggression/self-injury, neurologist for seizures, or genetics if cause unidentified

Hospitalisation Criteria

Admit for uncontrolled seizures, severe self-injurious behavior, or acute medical illness where cognitive impairment complicates outpatient management

Notes

Individuals with moderate ID typically achieve supported semi-independent living. Treat comorbid epilepsy (present in ~15-30%) and psychiatric conditions proactively.

Guideline Changes

AAC (augmentative/alternative communication) increasingly emphasized as early intervention

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F81 — Specific developmental disorders of scholastic skills

SNOMED: 1855002 | Duration: 365 days | Source: IDA 2024, AAP 2023

First-Line

- **Structured Literacy Intervention** — Orton-Gillingham-based phonics instruction; 4-5 sessions/week; 30-60 min; individualized
- **Individualized Education Program (IEP)** — School-based specialized instruction; legally mandated accommodations under IDEA; annual review
- **Speech-Language Therapy** — 1-3 sessions/week; phonological awareness, reading fluency; for language-based learning disorders

Second-Line

- **Occupational Therapy** — 1-2 sessions/week; handwriting, visual-motor integration; for dysgraphia component
- **Assistive Technology** — Text-to-speech, audiobooks, speech-to-text software; classroom accommodations
- **Math Intervention Programs** — Structured math tutoring for dyscalculia; multisensory approaches; 3-5 sessions/week

Investigations

- Psychoeducational assessment (IQ + achievement testing) — Diagnose specific learning disability — discrepancy between ability and achievement
- Hearing assessment — Rule out hearing loss affecting reading/language development
- Vision assessment — Rule out visual problems affecting reading
- ADHD screening (Conners/Vanderbilt) — ADHD comorbid in 30-50% of children with learning disabilities

Counselling

- Your child has a learning disability — this is NOT due to low intelligence or laziness
- Structured reading programs (like Orton-Gillingham) are the most effective treatment for dyslexia
- The child has a legal right to accommodations in school — an IEP (individualized education program) should be set up
- Extra time on tests, use of audiobooks, and typing instead of handwriting can all help
- Early identification and intervention (before age 7) gives the best outcomes — start therapy now

Referral Criteria

Educational psychologist for formal psychoeducational testing. Learning disability specialist for structured literacy intervention. School for IEP evaluation. Speech-language pathologist for language-based learning issues. Psychiatry if comorbid ADHD/anxiety

Hospitalisation Criteria

Not applicable — learning disabilities do not require hospitalisation. Admit only for severe comorbid psychiatric crisis (suicidality from academic stress/bullying)

Notes

No pharmacologic treatment for learning disabilities; medications only for comorbid ADHD/anxiety. Early identification before age 7 yields best outcomes.

Guideline Changes

Emphasis on structured literacy and Orton-Gillingham approaches for dyslexia intervention

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F84 — Pervasive developmental disorders / autism spectrum disorders

SNOMED: 35919005 | Duration: 365 days | Source: AAP 2020, AACAP 2024

First-Line

- **Applied Behavior Analysis (ABA)** — 25-40 hrs/week intensive or 10-25 hrs/week focused; individualized; ages 2+; gold standard
- **Speech-Language Therapy** — 1-5 sessions/week; targeting functional communication; may include AAC devices
- **Occupational Therapy** — 1-3 sessions/week; sensory integration, fine motor, self-care skills; individualized
- **Risperidone** — 0.25 mg/day initial (<20kg) or 0.5 mg/day (>20kg); titrate to 0.5-3 mg/day; FDA-approved 5-16y for irritability

Second-Line

- **Aripiprazole** — 2 mg/day initial, titrate to 5-15 mg/day; FDA-approved ages 6-17 for irritability in ASD
- **Guanfacine ER** — 1 mg/day initial, titrate to 1-4 mg/day; for hyperactivity/impulsivity in ASD; off-label
- **Methylphenidate** — 0.3 mg/kg/dose initial, titrate to max 1 mg/kg/dose; for comorbid ADHD symptoms; lower doses in ASD
- **Fluoxetine** — 2.5-5 mg/day initial, titrate to 10-40 mg/day; for anxiety/repetitive behaviors in ASD; off-label

Investigations

- Standardized autism assessment (ADOS-2, CARS-2, ISAA in India) — Formal diagnostic evaluation
- Hearing assessment (OAE/BERA) — Mandatory — rule out hearing loss mimicking ASD [URGENT]
- Chromosomal microarray — Genetic cause found in 10-20% — recommended for all newly diagnosed
- Fragile X testing — Most common inherited cause of intellectual disability with ASD features

- Lead level — Pica behavior in ASD children increases lead exposure risk

Counselling

- Your child has autism spectrum disorder — early intensive behavioral therapy is the most important treatment
- ABA therapy (applied behavior analysis) for 25-40 hours per week gives the best outcomes
- Speech therapy will help your child communicate — even non-verbal children can learn alternative communication
- Create a structured routine at home — children with ASD thrive with predictability
- Medicines do not cure autism but can help with associated problems like irritability, hyperactivity, or anxiety

Referral Criteria

Developmental pediatrics/child psychiatrist for diagnosis and management. ABA therapy provider. Speech-language pathologist. Occupational therapist. School for IEP/special education placement. Genetics (chromosomal microarray, Fragile X) for all newly diagnosed children

Hospitalisation Criteria

Rarely needed. Consider if: severe self-injurious behavior not manageable at home, acute behavioral crisis with risk of harm, medication adjustment requiring close monitoring, comorbid medical condition requiring admission

Notes

Behavioral and developmental therapies are primary treatment; medications target co-occurring symptoms, not core ASD. Monitor weight at baseline and quarterly on antipsychotics.

Guideline Changes

2024 guidelines expand pharmacologic options for co-occurring irritability and emphasize early ABA

F91 — Conduct disorders

SNOMED: 192597006 | Duration: 180 days | Source: AACAP 2023, AAN/Merck 2025

First-Line

- **Parent Management Training (PMT)** — Structured behavioral therapy; weekly sessions for 12-16 weeks; caregiver-focused
- **Cognitive Behavioral Therapy** — Individual or group CBT; weekly sessions targeting anger management and problem-solving skills
- **Risperidone** — 0.25-0.5 mg/day initial, titrate to 0.5-3 mg/day in 2 divided doses; ages 5-17

Second-Line

- **Lithium** — 300 mg BID-TID; target serum level 0.6-1.2 mEq/L; monitor renal/thyroid; ages 7+
- **Aripiprazole** — 2 mg/day initial, titrate to 5-15 mg/day; ages 6-17; monitor metabolic
- **Multisystemic Therapy (MST)** — Intensive community-based family therapy; 3-5 months duration; for severe cases

Investigations

- Standardized behavioral rating scales (CBCL, Conners) — Quantify behavior problems, screen for ADHD, monitor treatment response
- Psychoeducational testing — Screen for comorbid learning disability (50% comorbidity)
- Thyroid function (TSH) — Hyperthyroidism can mimic behavioral problems
- Lead level — Lead exposure associated with behavioral problems in Indian children

Counselling

- Your child has behavior problems that need structured treatment — this is a medical condition, not just bad behavior
- Parent management training (PMT) is the most effective first step — learning consistent strategies at home

- Set clear rules and consistent consequences — positive reinforcement works better than punishment
- Medicine may be needed if behavioral therapy alone is not enough — your doctor will guide you
- Screen time, peer group, and family stress all affect behavior — address these alongside treatment

Referral Criteria

Child psychiatrist for medication management. Clinical psychologist for CBT/PMT. School counselor for academic support. Social services if family dysfunction or abuse contributing. Developmental pediatrics to screen for comorbid ADHD/learning disability

Hospitalisation Criteria

Rarely needed. Consider if: risk of self-harm or harm to others, acute psychotic features, failed outpatient treatment with escalating aggression, need for psychiatric stabilization

Notes

Always treat comorbid ADHD/depression first. Pharmacotherapy only after psychosocial interventions fail; short-term antipsychotic use recommended.

Guideline Changes

Network meta-analysis confirms SGAs most efficacious for disruptive behaviors

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F93 — Emotional disorders with onset specific to childhood

SNOMED: 109006 | Duration: 180 days | Source: AACAP 2020 Anxiety CPG

First-Line

- **Cognitive Behavioral Therapy** — Individual or group CBT with exposure; 12-16 weekly sessions; ages 6-18; CAMS-supported
- **Fluoxetine** — 10 mg/day initial, titrate to max 20 mg/day; ages 7+; for separation/generalized anxiety
- **Sertraline** — 25 mg/day initial (6-12y), 50 mg/day (12+); titrate to max 200 mg/day; off-label for anxiety

Second-Line

- **Fluvoxamine** — 25 mg/day initial, titrate to 50-200 mg/day; FDA-approved for pediatric OCD ages 8+
- **Duloxetine** — 30 mg/day initial, max 60-120 mg/day; FDA-approved for GAD ages 7+
- **Escitalopram** — 5 mg/day initial, titrate to 10-20 mg/day; ages 12+; off-label for anxiety
- **Imipramine** — 1-3 mg/kg/day in divided doses; last-line for refractory separation anxiety; ECG monitoring

Investigations

- Standardized anxiety scales (SCARED, GAD-7) — Quantify anxiety severity and monitor treatment response
- Thyroid function (TSH) — Hyperthyroidism mimics anxiety symptoms
- CBC — Rule out anemia (fatigue mistaken for anxiety/depression)

Counselling

- Your child has anxiety that is stronger than normal — this is very treatable with therapy and sometimes medicine
- CBT (cognitive behavioral therapy) is the gold standard — it teaches the child to manage anxiety
- Do not force the child into feared situations suddenly — gradual exposure works much better
- Avoid excessive reassurance — teach the child to cope rather than avoiding all anxious situations
- Combination of therapy + medicine (if needed) gives the best results — up to 80% improvement

Referral Criteria

Child psychiatrist if SSRI needed. Clinical psychologist for CBT. School counselor for school-based accommodations. Developmental pediatrics for comorbid ADHD or learning issues

Hospitalisation Criteria

Rarely needed. Consider if: school refusal >4 weeks not responding to outpatient treatment, suicidality, severe self-harm, panic disorder with agoraphobia causing inability to function, comorbid depression

Notes

Combination CBT+SSRI yields ~81% improvement vs 55% SSRI alone (CAMS trial). Benzodiazepines not recommended for pediatric anxiety.

Guideline Changes

2020 AACAP guideline recommends combined CBT+SSRI for moderate-severe anxiety

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F94 — Disorders of social functioning with onset specific to childhood and adolescence

Duration: 365 days | Source: AACAP 2020, APA 2023

First-Line

- **Cognitive Behavioral Therapy (CBT)** — Graduated exposure-based CBT; weekly sessions 12-20 weeks; for selective mutism and social anxiety
- **Trauma-Focused CBT** — For reactive attachment disorder (F94.1); 12-25 sessions; caregiver involvement essential
- **Therapeutic foster care / stable placement** — Primary intervention for RAD/DSED; establish safe consistent caregiving environment

Second-Line

- **Fluoxetine** — 5-10 mg/day initial, titrate to 10-40 mg/day; off-label for selective mutism; ages 5+
- **Sertraline** — 12.5-25 mg/day initial, titrate to 50-200 mg/day; off-label for social anxiety; ages 6+
- **Play Therapy** — Child-centered or directive play therapy; weekly sessions; for younger children unable to do CBT

Investigations

- Developmental assessment — Comprehensive evaluation to rule out ASD, intellectual disability
- Hearing assessment (audiometry) — Rule out hearing loss as cause of social withdrawal
- Standardized anxiety scales — Screen for anxiety (comorbid in selective mutism)
- Thyroid function — Rule out thyroid disorder affecting behavior

Counselling

- Your child has difficulty with social interactions — this responds best to therapy, not medicine
- A stable, consistent, and loving home environment is the most important factor for improvement
- CBT with gradual exposure helps children with selective mutism speak in social situations
- Be patient — progress may be slow but steady. Avoid forcing the child to speak
- School coordination is important — teachers should understand the condition and provide accommodations

Referral Criteria

Child psychiatrist if medication needed (SSRI for selective mutism). Clinical psychologist for CBT/play therapy. School coordination for accommodations. Social services for foster care/placement stability in RAD

Hospitalisation Criteria

Rarely needed. Consider if: severe self-harm risk, acute attachment crisis with danger, comorbid psychiatric emergency (suicidality, psychosis)

Notes

No FDA-approved medications for RAD; stable caregiving is cornerstone. For selective mutism, CBT with graduated exposure is first-line; SSRIs adjunctive.

Guideline Changes

Updated AACAP anxiety guideline now covers selective mutism under anxiety spectrum

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F95 — Tic disorders

Duration: 365 days | Source: AAN 2019 (reaffirmed 2025)

First-Line

- **Comprehensive Behavioral Intervention for Tics (CBIT)** — Habit reversal + relaxation training; 8 sessions over 10 weeks; ages 9+; AAN Level B
- **Clonidine** — 0.05 mg/day initial, titrate to 0.1-0.4 mg/day in divided doses; monitor BP; ages 6+
- **Guanfacine** — 0.5 mg/day initial, titrate to 1-4 mg/day; extended-release preferred; monitor BP

Second-Line

- **Aripiprazole** — 1-2 mg/day initial, titrate to 2-20 mg/day; for moderate-severe tics; monitor metabolic
- **Fluphenazine** — 0.5-1 mg/day initial, titrate to 1.5-10 mg/day; typical antipsychotic; monitor for EPS
- **Topiramate** — 25 mg/day initial, titrate to 50-200 mg/day; AAN Level C evidence; monitor cognition
- **Tetrabenazine** — 12.5 mg/day initial, titrate to 25-75 mg/day; for refractory tics; monitor for depression

Investigations

- Clinical assessment with Yale Global Tic Severity Scale (YGTSS) — Quantify tic severity and track treatment response
- Screening for ADHD and OCD (rating scales) — Comorbid conditions — present in >50% of Tourette patients
- Throat culture (RADT/ASO titer) — If sudden onset tics — rule out PANDAS/PANS [URGENT]
- TSH — Thyroid disease can cause movement abnormalities

Counselling

- Tics are involuntary movements or sounds — your child cannot control them, so do not scold or punish
- Tics tend to come and go — they may increase with stress and decrease during focused activities
- CBIT (behavioral therapy) is the first treatment — it teaches the child techniques to manage tics
- Treat comorbid ADHD or OCD first — these usually cause more problems than the tics themselves
- Most children with tics improve significantly by adulthood — reassurance is important

Referral Criteria

Pediatric neurologist for diagnosis and medication management. Clinical psychologist for CBIT. Psychiatry if comorbid OCD/ADHD needs treatment. School for accommodations (extra time, breaks)

Hospitalisation Criteria

Rarely needed. Consider only for: tic status (severe continuous tics causing injury or exhaustion), malignant Tourette syndrome unresponsive to all treatments, deep brain stimulation evaluation (refractory severe cases)

Notes

Treat comorbid ADHD/OCD first as they often cause more impairment than tics. Stimulants generally safe and do not worsen tics.

Guideline Changes

2025 reaffirmation emphasizes CBIT as first-line behavioral intervention for ages 9+

F98.0 — Nonorganic enuresis

SNOMED: 450842007 | Duration: 84 days | Source: ICCS 2020 Standardization, AAFP 2022

First-Line

- **Enuresis Alarm Therapy** — Moisture-sensing alarm; use nightly for 12-16 weeks minimum; best sustained response; 66% cure rate
- **Desmopressin (DDAVP)** — 120 mcg melt tablet SL 30-60 min before bed; may increase to 240-360 mcg; restrict fluids 8 hrs

Second-Line

- **Combined Alarm + Desmopressin** — For refractory cases; alarm nightly + desmopressin 120-240 mcg SL at bedtime; 12-16 weeks

- **Oxybutynin** — 5 mg PO at bedtime; for enuresis with daytime urgency/frequency; anticholinergic; ages 5+
- **Imipramine** — 25 mg PO at bedtime (ages 6-12); max 50 mg (ages 12+); last-line due to cardiac risk; ECG required

Investigations

- Urinalysis with culture — Rule out UTI as cause
- Renal ultrasound — If daytime symptoms or recurrent UTI
- Serum creatinine — If renal anomaly suspected

Counselling

- Bedwetting is not the child's fault — never punish or shame them
- Restrict fluids 2 hours before bedtime and ensure the child urinates before sleeping
- Use a waterproof mattress cover and keep spare sheets ready
- Be patient — alarm therapy takes 12-16 weeks to show full effect
- Reward dry nights with a star chart to motivate the child

Referral Criteria

Refer to pediatric urologist if daytime wetting persists, recurrent UTIs, abnormal voiding pattern, or no response after 6 months of alarm + desmopressin therapy

Hospitalisation Criteria

Hospitalisation rarely needed; admit only if secondary to spinal cord pathology or if invasive urodynamic studies are required

Notes

Active treatment from age 6. Treat constipation and daytime incontinence first. Structured desmopressin withdrawal over weeks reduces relapse.

Guideline Changes

2025 IPSA consensus recommends structured desmopressin withdrawal to improve cure rates

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F98.1 — Encopresis not due to a substance or known physiological condition

SNOMED: 302752001 | Duration: 180 days | Source: NASPGHAN/ESPGHAN 2014, AAFP 2022

First-Line

- **Polyethylene Glycol 3350 (MiraLAX)** — Disimpaction: 1-1.5 g/kg/day PO x3-6 days; Maintenance: 0.4-0.8 g/kg/day PO; mix in 6-8 oz fluid
- **Behavioral Toilet Training** — Scheduled sitting 5-10 min after meals; positive reinforcement; stool diary; ongoing

Second-Line

- **Lactulose** — 1-3 mL/kg/day PO in divided doses; osmotic laxative; for PEG-intolerant patients
- **Senna (Senokot)** — Ages 2-6: 2.5-5 mL/day PO; Ages 6-12: 5-10 mL/day PO; stimulant laxative; short-term
- **Mineral Oil** — 1-3 mL/kg/day PO; max 240 mL/day; ages 1+; not for children at aspiration risk
- **Glycerin suppositories** — 1 pediatric suppository PR daily as needed; for acute relief; ages 2+

Investigations

- Abdominal X-ray — Assess fecal loading and response to disimpaction
- Anorectal manometry — If Hirschsprung disease suspected (absent RAIR)
- Thyroid function (TSH) — Hypothyroidism can cause refractory constipation
- Celiac screen (tTG-IgA) — Celiac disease can present as constipation

Counselling

- Encopresis (soiling) is usually caused by chronic constipation — it is NOT the child's fault

- First, the impacted stool must be cleared (disimpaction) — then daily stool softener prevents re-impaction
- The child should sit on the toilet for 5-10 minutes after each main meal — make it a relaxed routine
- Praise and reward the child for sitting on the toilet — never punish for accidents
- Treatment takes 6-12 months — do not stop the stool softener too early or the problem will return

Referral Criteria

Pediatric GI if treatment failure after 6 months or red flags (blood in stool, failure to thrive, delayed meconium at birth). Psychology if significant behavioral component. Surgery if Hirschsprung suspected (delayed meconium, refractory to treatment)

Hospitalisation Criteria

Rarely needed. Consider if: massive fecal impaction requiring inpatient disimpaction. Suspected Hirschsprung disease needing biopsy. Severe associated behavioral/psychological crisis

Notes

Disimpaction must precede maintenance therapy. Taper laxatives by 25% every 2-3 months over 6-12 months once behavioral modification established.

Guideline Changes

PEG 3350 dosing optimized in 2024 studies confirming 0.4-0.8 g/kg/day maintenance range

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R62.51 — Failure to thrive in child

SNOMED: 432788009 | Duration: 90 days | Source: AAP 2023, UCSF Inadequate Growth Guidelines

First-Line

- **Caloric Optimization** — Target 120-150% estimated energy requirement; calorie-dense foods; fortify breast milk/formula
- **Lactation Support / Formula Optimization** — For infants: assess latch, increase feeds; concentrate formula to 24-30 kcal/oz if needed
- **Nutritional Counseling** — Registered dietitian evaluation; structured meal schedule; eliminate juice/low-calorie fillers
- **Multivitamin with Iron & Zinc** — Daily supplementation; iron 2-6 mg/kg/day if deficient; zinc 1 mg/kg/day if deficient; PO

Second-Line

- **Oral Nutritional Supplements** — PediaSure 1.0-1.5 kcal/mL; 1-3 servings/day between meals; ages 1+
- **Cyproheptadine** — 2 mg PO BID (ages 2-6); 4 mg PO BID-TID (ages 7+); appetite stimulant; off-label
- **Nasogastric/Gastrostomy tube feeding** — For severe FTT unresponsive to oral; continuous or bolus enteral feeds; multidisciplinary decision

Investigations

- CBC — Evaluate for anemia and infection
- Serum albumin and total protein — Assess nutritional status
- Urinalysis — Rule out UTI or renal tubular acidosis
- Celiac screen (tTG-IgA) — Rule out celiac disease if chronic
- Serum iron, ferritin, zinc — Identify micronutrient deficiencies

Counselling

- Increase calorie density of foods — add ghee, cream, or powdered milk to regular meals
- Feed every 2-3 hours; do not let the child fill up on juice or water before meals
- Maintain a food diary to track what and how much the child actually eats
- Growth catch-up takes weeks — do not expect overnight improvement
- Bring the child for weight checks every 1-2 weeks initially

Referral Criteria

Refer to pediatric gastroenterologist if no weight gain after 3 months of nutritional intervention, or if organic cause (celiac, GERD, metabolic) suspected

Hospitalisation Criteria

Admit if severe malnutrition (weight <60% of expected), dehydration, hypothermia, hypoglycemia, or signs of neglect/abuse

Notes

Inadequate caloric intake is the most common cause. Multidisciplinary approach (pediatrics, nutrition, social work) is optimal; investigate neglect and psychosocial factors.

Guideline Changes

UCSF consensus now uses term 'inadequate growth' replacing 'failure to thrive'

R63.3 — Feeding difficulties

SNOMED: 78164000 | Duration: 90 days | Source: ASHA 2024, AOTA 2024

First-Line

- **Occupational Therapy - Feeding** — 1-3 sessions/week; sensory-based feeding therapy (SOS or Sequential Oral Sensory approach); individualized
- **Speech-Language Therapy - Swallowing** — 1-3 sessions/week; oral motor exercises, swallowing assessment/treatment; ASHA-guided
- **Behavioral Feeding Intervention** — Structured mealtimes; positive reinforcement; graduated food exposure; parent training

Second-Line

- **Nutritional Supplementation** — High-calorie formula or oral supplements to maintain growth while therapy progresses
- **Positioning & Adaptive Equipment** — Specialized seating, adapted utensils, modified textures per OT/SLP assessment
- **Nasogastric/Gastrostomy feeding** — For severe oral aversion or aspiration risk; bridge to oral feeding; multidisciplinary decision
- **Cyproheptadine** — 2 mg PO BID (ages 2-6); appetite stimulant for severe food refusal; off-label; short-term

Investigations

- Videofluoroscopic swallow study — If aspiration or dysphagia suspected
- CBC and iron studies — Screen for nutritional deficiency
- Upper GI series — Rule out structural anomaly (malrotation, stricture)

Counselling

- Do not force-feed — it worsens food aversion in children
- Offer new foods alongside familiar favorites, without pressure
- Keep mealtimes under 30 minutes and maintain a regular schedule
- Eat together as a family; children learn by watching others eat
- Small improvements in feeding are meaningful — be patient with therapy progress

Referral Criteria

Refer to pediatric GI or feeding team if aspiration risk, weight loss despite intervention, or structural/neurological cause identified

Hospitalisation Criteria

Admit if severe failure to thrive with dehydration, aspiration pneumonia, or need for nasogastric/gastrostomy tube placement

Notes

Multidisciplinary assessment (OT, SLP, dietitian, GI) is essential. Rule out structural, neurological, and GI causes before treating as behavioral.

Guideline Changes

2024 AOTA state-of-science review updated evidence for sensory-based feeding interventions

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R63.4 — Abnormal weight loss

SNOMED: 267024001 | Duration: 90 days | Source: AAP 2023, WHO Growth Standards

First-Line

- **Nutritional Assessment & Counseling** — Dietitian-led caloric needs assessment; age-appropriate dietary plan; weekly weight monitoring
- **Caloric Supplementation** — High-calorie diet 120-150% of RDA; fortified foods; frequent small meals; adjust to age
- **Multivitamin + Mineral Supplementation** — Age-appropriate daily multivitamin with iron and zinc; correct specific deficiencies identified on labs

Second-Line

- **Oral Nutritional Supplements** — PediaSure/Ensure 1-3 servings/day as caloric supplement; 240 kcal per 8 oz; ages 1-13
- **Cyproheptadine** — 2 mg PO BID-TID (ages 2-6) or 4 mg PO BID-TID (ages 7+); appetite stimulant; off-label
- **Nasogastric tube feeding** — For severe malnutrition or oral intake failure; continuous or bolus feeds; inpatient or home

Investigations

- CBC with differential — Screen for anemia, infection, malignancy [URGENT]
- ESR/CRP — Screen for inflammatory or infectious cause
- Thyroid function tests — Rule out hyperthyroidism
- Celiac screen (tTG-IgA) — Rule out celiac disease
- Stool examination — Rule out parasitic infection or malabsorption

Counselling

- Offer 5-6 small energy-dense meals per day instead of 3 large meals
- Add ghee, butter, or oil to dal and roti to increase calories
- Track weight weekly at the same time on the same scale
- Do not force-feed — create a calm, distraction-free mealtime environment
- Follow up every 2 weeks until weight gain trend is established

Referral Criteria

Refer to pediatric gastroenterologist if weight loss exceeds 10% of body weight, chronic diarrhea persists, or malignancy/celiac/IBD suspected

Hospitalisation Criteria

Admit if severe dehydration, hemodynamic instability, weight loss >10% with inability to tolerate oral intake, or suspected malignancy

Notes

Investigate underlying etiology (malignancy, celiac, IBD, eating disorders, psychosocial). Weight loss >5% in children warrants urgent workup.

ENT

H04 — Disorders of lacrimal system

Duration: 90 days | Source: AAO 2023, AAPOS 2022

First-Line

- **Crigler lacrimal sac massage** — 10 firm downward strokes over lacrimal sac bid-tid; parents taught technique; age <12mo
- **Erythromycin ophthalmic ointment 0.5%** — Apply 1cm ribbon bid-tid for acute dacryocystitis/purulent discharge episodes
- **Warm compresses** — Apply warm moist compress to inner canthus for 5-10min tid-qid to relieve obstruction

Second-Line

- **Nasolacrimal duct probing** — Office or OR procedure if CNLDO persists >12mo; 85-95% success rate; under brief GA
- **Balloon dacryoplasty** — Probing with balloon dilation for failed initial probing; higher success in older children
- **Silicone intubation (Crawford tube)** — For recurrent obstruction after probing; tube left in place 3-6mo then removed
- **Dacryocystorhinostomy (DCR)** — Surgical bypass of NLD; reserved for refractory cases after failed less invasive procedures

Investigations

- Clinical examination — Diagnosis is clinical; fluorescein dye disappearance test
- Dacryocystography — If complex obstruction suspected before surgery

Counselling

- Lacrimal massage technique is the most important treatment: press firmly downward over inner corner of eye 10 times, 2-3 times daily
- Wipe away discharge with clean wet cotton; the blocked tear duct itself is not infected
- More than 90% of blocked tear ducts open on their own by age 12 months
- If there is no improvement by 12 months, a simple probing procedure can open the duct
- Antibiotic eye drops are only needed during active discharge episodes, not continuously

Referral Criteria

Refer to pediatric ophthalmologist if CNLDO persists beyond 12 months, acute dacryocystitis (red swollen lump at inner corner), or suspected dacryocystocele in neonate

Hospitalisation Criteria

Admit if acute dacryocystitis with preseptal cellulitis, dacryocystocele with respiratory distress (bilateral, neonatal), or probing under GA

Notes

Congenital NLDO affects 6-20% of newborns; >90% resolve spontaneously by 12 months with massage. Probing should be deferred until after 12 months to allow spontaneous resolution.

H10 — Conjunctivitis

SNOMED: 9826008 | Duration: 7 days | Source: AAP 2023, AAO 2018, CDC 2023

First-Line

- **Erythromycin ophthalmic ointment 0.5%** — Apply 1cm ribbon to lower conjunctival sac qid x5-7d; all ages including neonates

- **Polymyxin B-trimethoprim drops** — 1 drop in affected eye q3h while awake x7d; first-line for bacterial conjunctivitis ≥2mo
- **Artificial tears (preservative-free)** — 1-2 drops q2-4h PRN for viral/allergic conjunctivitis; supportive care

Second-Line

- **Moxifloxacin 0.5% ophthalmic** — 1 drop in affected eye tid x7d; age ≥1yr; broader coverage for severe cases
- **Azithromycin 1% ophthalmic** — 1 drop bid x2d then 1 drop daily x5d; age ≥1yr; good compliance profile
- **Olopatadine 0.1% ophthalmic** — 1 drop in each eye bid for allergic conjunctivitis; age ≥3yr
- **Ketotifen 0.025% ophthalmic (OTC)** — 1 drop in each eye bid-tid for allergic conjunctivitis; age ≥3yr

Investigations

- Conjunctival swab culture — If neonatal, severe, or not responding to empirical treatment [URGENT]
- Gram stain — If hyperacute purulent conjunctivitis (rule out gonococcal) [STAT]
- Chlamydia PCR — Neonatal conjunctivitis at 5-14 days of life [URGENT]

Counselling

- Wash hands frequently and do not touch or rub the eyes
- Use separate towels and washcloths from other family members
- Clean eye discharge with wet cotton, wiping from inner to outer corner
- Do not share eye drops between family members; replace bottle after infection
- Child can return to school after 24 hours on antibiotic drops or when discharge stops

Referral Criteria

Refer to ophthalmologist if neonatal conjunctivitis (<30 days), conjunctivitis not improving after 1 week of treatment, suspected gonococcal (hyperacute, profuse), or associated corneal involvement

Hospitalisation Criteria

Admit if neonatal gonococcal conjunctivitis (IV ceftriaxone needed), orbital cellulitis, or corneal ulcer with conjunctivitis

Notes

Bacterial conjunctivitis is most common in children; viral is self-limiting in 7-14d. Neonatal conjunctivitis (<30d) requires urgent evaluation to rule out gonococcal/chlamydial infection.

H16 — Keratitis

SNOMED: 5888003 | Duration: 14 days | Source: AAO PPP 2024 Bacterial Keratitis, WHO 2022

First-Line

- **Moxifloxacin 0.5% ophthalmic** — 1 drop q1h around the clock x48h then taper to q2-4h based on response; age ≥1yr
- **Fortified tobramycin 1.4% + cefazolin 5%** — Alternating q1h (each q2h) for severe/central ulcers; compounded drops
- **Cyclopentolate 1% ophthalmic** — 1 drop bid-tid for ciliary spasm/pain relief; adjunct to antibiotic therapy

Second-Line

- **Gatifloxacin 0.5% ophthalmic** — 1 drop q1h x48h then taper; alternative 4th-gen fluoroquinolone monotherapy; ≥1yr
- **Natamycin 5% ophthalmic** — 1 drop q1-2h for fungal keratitis; first-line for filamentous fungi
- **Voriconazole 1% ophthalmic** — 1 drop q1h for severe fungal keratitis; compounded; for Candida/resistant fungi
- **Acyclovir 3% ophthalmic ointment** — Apply 5x/day x14-21d for HSV keratitis; add oral acyclovir 20mg/kg q5h if stromal

Investigations

- Corneal scraping for gram stain and culture — Before starting antibiotics for all central/large ulcers [STAT]
- Corneal scraping for KOH — If fungal keratitis suspected [STAT]
- Fluorescein staining — Assess corneal defect size and depth [STAT]
- Corneal confocal microscopy — If Acanthamoeba keratitis suspected (contact lens wearer) [URGENT]

Counselling

- Keratitis is a serious eye infection that can cause vision loss if not treated promptly
- Use eye drops exactly as prescribed - initially drops may be needed every hour
- Do not wear contact lenses until completely healed and cleared by eye doctor
- Attend all follow-up appointments; corneal healing needs close monitoring
- If pain worsens or vision decreases further, return to eye doctor immediately

Referral Criteria

Refer to ophthalmologist urgently for all suspected keratitis; refer to cornea specialist if central ulcer, fungal keratitis, or not responding to treatment

Hospitalisation Criteria

Admit if severe corneal ulcer threatening perforation, hypopyon, need for hourly fortified drops (compliance concern), or fungal/acanthamoeba keratitis

Notes

Bacterial keratitis is an ocular emergency; obtain corneal cultures before starting empiric therapy if central/severe. Contact lens wearers need Pseudomonas coverage with fluoroquinolone monotherapy.

H50 — Strabismus

SNOMED: 22066006 | Duration: 365 days | Source: AAO PPP 2023, AAPOS 2022, PEDIG trials

First-Line

- **Spectacle correction (full Rx)** — Correct full cycloplegic refraction; accommodative esotropia may fully resolve with glasses
- **Occlusion therapy (patching)** — Patch sound eye 2h/day for moderate amblyopia, 6h/day for severe; with near activities
- **Atropine 1% penalization** — 1 drop in sound eye daily or weekend-only for moderate amblyopia; age <10yr

Second-Line

- **Strabismus surgery** — Recession/resection of extraocular muscles; under GA; for deviations unresponsive to optical Rx
- **Botulinum toxin A injection** — Inject into extraocular muscle under EMG guidance; for small-angle deviations or pre-surgical
- **Bangerter filter** — Translucent filter on spectacle lens over sound eye; for moderate amblyopia; better compliance
- **Prism spectacles** — Fresnel or ground-in prism for small residual deviations; for diplopia relief

Investigations

- Cycloplegic refraction — Essential for all strabismus; accommodative component assessment [URGENT]
- Cover-uncover test — Confirm and measure deviation [URGENT]
- Fundus examination — Rule out retinoblastoma causing leukocoria/pseudostrabismus [URGENT]
- MRI brain — If acquired non-accommodative esotropia, cranial nerve palsy, or neurological signs [URGENT]

Counselling

- Wearing glasses as prescribed is essential - it may fully correct accommodative squint
- Patching or atropine for amblyopia treatment requires patience; results take weeks to months
- Encourage the child to do close-up activities (drawing, reading) during patching for better results

- Surgery for squint has good results but may need to be repeated; it aligns the eyes, does not replace glasses
- Early treatment gives best outcomes; visual plasticity decreases significantly after age 7-9 years

Referral Criteria

Refer to pediatric ophthalmologist/strabismus specialist at first detection of squint at any age; urgent if new onset in child >6 months or associated with neurological signs

Hospitalisation Criteria

Admit for strabismus surgery (day procedure under GA); urgent if acquired squint with suspected neurological cause (cranial nerve palsy, brain tumor)

Notes

Treat amblyopia first before surgical correction of strabismus. Visual plasticity is greatest before age 7-9; early detection and treatment yield best outcomes.

H52 — Disorders of refraction and accommodation

SNOMED: 39021009 | Duration: 365 days | Source: AAO PPP 2023, AAP 2022, WHO 2019

First-Line

- **Spectacle correction** — Full cycloplegic refraction Rx; update q6-12mo; polycarbonate lenses for children
- **Low-dose atropine 0.01% drops** — 1 drop in each eye at bedtime daily for myopia control; age 4-12yr; monitor q6mo **[NEW 2024-25]**
- **Contact lenses (daily disposable)** — For active children/sports; orthokeratology overnight lenses for myopia control option

Second-Line

- **Atropine 0.05% drops** — 1 drop each eye at bedtime daily for faster-progressing myopia; more side effects **[NEW 2024-25]**
- **Orthokeratology (Ortho-K)** — Overnight rigid GP lenses reshape cornea; for myopia control; age ≥6yr; monitor closely
- **MiSight daily contact lenses** — FDA-approved dual-focus soft lens for myopia control; age 8-12yr at initiation **[NEW 2024-25]**
- **Bifocal/progressive spectacles** — For accommodative esotropia or high AC/A ratio; prescribed per binocular vision assessment

Investigations

- Cycloplegic refraction — Accurate measurement of refractive error in children
- Visual acuity testing — Age-appropriate testing at every visit
- Axial length measurement — Monitor myopia progression in children on myopia control
- Fundus examination — High myopes need retinal screening for degenerative changes

Counselling

- Wear glasses consistently as prescribed - the eyes need clear images for proper development
- Encourage 2 hours of outdoor play daily - this slows myopia progression
- Limit screen time; follow the 20-20-20 rule (every 20 min, look 20 feet away for 20 seconds)
- Eye check-ups every 6 months are needed as power changes during growth
- If prescribed atropine drops for myopia control, use at bedtime; minor pupil dilation and light sensitivity are normal

Referral Criteria

Refer to pediatric ophthalmologist if amblyopia detected, rapid myopia progression (>-0.75D/year), high myopia (>-6D), or associated with systemic conditions (Marfan, Down syndrome)

Hospitalisation Criteria

Hospitalisation not needed for refractive errors; consider if complication of contact lens use (keratitis)

Notes

Myopia control interventions (atropine, Ortho-K, MiSight) should be considered for progressive myopia in children. Ensure 120min outdoor time daily as modifiable protective factor against myopia progression.

Guideline Changes

Low-dose atropine 0.01-0.05% now widely adopted for myopia control based on Phase 3 trial data

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H53 — Visual disturbances

SNOMED: 246636008 | Duration: 365 days | Source: AAO PPP 2023, PEDIG 2024

First-Line

- **Comprehensive eye examination** — Full cycloplegic refraction, dilated fundus exam, visual field if cooperative; age-appropriate
- **Spectacle correction** — Prescribe optimal refraction first; reassess 4-6wk; may resolve functional visual disturbance
- **Occlusion therapy (patching)** — For amblyopia: 2h/day moderate, 6h/day severe; combine with 1h near activities daily

Second-Line

- **Atropine 1% penalization** — 1 drop in sound eye daily or weekend-only for amblyopia; equivalent to 2h patching
- **Luminopia digital therapeutic** — VR-based dichoptic therapy 1h/day x12wk for amblyopia; FDA-approved age 4-7yr **[NEW 2024-25]**
- **Neuroimaging (MRI brain/orbits)** — If unexplained visual loss, papilledema, optic atrophy, or neurologic signs present
- **Low vision aids** — Magnifiers, large-print materials, assistive tech if best-corrected VA remains impaired

Investigations

- Comprehensive eye examination — Visual acuity, refraction, fundoscopy, visual fields [URGENT]
- OCT (optical coherence tomography) — Retinal and optic nerve assessment
- MRI brain and orbits — If optic neuritis, papilledema, or unexplained vision loss [URGENT]
- Visual evoked potentials (VEP) — Objective assessment of visual pathway function

Counselling

- Any change in your child vision needs prompt evaluation - early treatment is essential
- If patching is prescribed for lazy eye, consistent use gives the best chance of improvement
- Ensure good lighting for reading and close work
- Regular eye examinations are needed to track progress
- Report any sudden vision loss, flashing lights, or new headaches urgently to the eye doctor

Referral Criteria

Refer to pediatric ophthalmologist for all children with visual disturbances; urgent referral if acute vision loss, papilledema, or neurological signs

Hospitalisation Criteria

Admit if acute vision loss requiring urgent workup (optic neuritis, papilledema with raised ICP, retinal detachment), or neuroimaging needed urgently

Notes

Visual disturbances in children require thorough evaluation to exclude retinal, optic nerve, or neurologic pathology. Functional/non-organic visual loss is a diagnosis of exclusion in the pediatric population.

Guideline Changes

Digital therapeutic (Luminopia) FDA-approved 2021 as adjunct for amblyopia treatment in children 4-7yr

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H54 — Blindness and low vision

SNOMED: 397540003 | Duration: 365 days | Source: AAO 2023, AOA 2024, AAPOS 2023

First-Line

- **Low vision rehabilitation evaluation** — Comprehensive exam by low vision specialist; prescribe magnifiers, contrast aids, lighting
- **Corrective lenses (optimized Rx)** — Maximize residual vision with best refraction; update q6-12mo in growing children
- **Orientation and mobility training** — Certified O&M specialist; begin early; white cane training when ambulatory

Second-Line

- **Electronic magnification devices** — CCTV, tablet-based magnifiers, screen readers; prescribe per functional assessment
- **Assistive technology (screen reader)** — JAWS, NVDA, or VoiceOver for school-age children; training by TVI specialist
- **Occupational therapy** — Adaptive daily living skills; sensory integration; handwriting alternatives
- **Early intervention program** — Enroll by 6mo; developmental support, family training, per state EI program

Investigations

- Comprehensive eye examination — Dilated fundus exam, refraction, visual field [URGENT]
- ERG (Electroretinography) — Retinal dystrophy evaluation
- VEP (Visual Evoked Potential) — Optic nerve function assessment
- MRI brain and orbits — If optic nerve or neurological cause suspected [URGENT]
- Genetic testing — If hereditary cause suspected (Leber, retinitis pigmentosa)

Counselling

- Start low vision rehabilitation early - it helps the child use remaining vision to the fullest
- Ensure good lighting at home and school; reduce glare with tinted lenses if needed
- Talk to the school about special seating (close to board) and learning aids
- Teach the child orientation and mobility skills early for independence
- Regular eye check-ups to detect any treatable causes and monitor progression

Referral Criteria

Refer to pediatric ophthalmologist for all cases of childhood visual impairment; refer to low vision centre for rehabilitation; refer to genetic counselling if hereditary cause

Hospitalisation Criteria

Rarely needed; admit if acute visual loss requiring urgent workup (retinal detachment, optic neuritis), or surgical intervention for treatable cause

Notes

Low vision rehabilitation should begin early as a parallel ongoing treatment, not a last resort. Multidisciplinary team approach including TVI, O&M specialist, OT, and family support is essential.

H60 — Otitis externa

SNOMED: 3135009 | Duration: 7 days | Source: AAO-HNS 2014, AAP 2023

First-Line

- **Ciprofloxacin 0.3%/dexamethasone 0.1% otic** — 4 drops in affected ear bid x7d; ≥6mo age; warm bottle in hand before use
- **Ofloxacin 0.3% otic solution** — 5 drops (age 1-12yr) or 10 drops (≥13yr) in affected ear bid x7d
- **Acetic acid 2% otic solution** — 3-5 drops in affected ear tid-qid x7d; for mild OE or prophylaxis

Second-Line

- **Ciprofloxacin 0.2% otic solution** — 1 single-use container into affected ear bid x7d; age ≥1yr
- **Neomycin/polymyxin B/hydrocortisone otic** — 3-4 drops in affected ear tid-qid x7-10d; only if TM intact (ototoxic risk)
- **Fluconazole 2% otic solution** — For fungal OE (otomycosis): 3-5 drops bid x14-21d after canal debridement
- **Oral antibiotics (cephalexin)** — 25-50mg/kg/day PO divided q6-8h x7-10d only if cellulitis extends beyond canal

Investigations

- Ear swab culture — If not responding to first-line drops after 1 week
- Otoscopy — Assess canal edema, TM integrity, and debris
- CT temporal bone — If malignant (necrotizing) otitis externa suspected [URGENT]

Counselling

- Keep the ear completely dry during treatment - use cotton with petroleum jelly during bathing
- No swimming until treatment is complete and ear is healed
- Warm the ear drops in your hand before putting them in to reduce discomfort
- Use ear drops for the full 7 days even if ear feels better
- Prevent recurrence: dry ears after swimming, avoid inserting objects, use swimmer ear drops

Referral Criteria

Refer to ENT if malignant (necrotizing) otitis externa suspected, fungal OE not responding to treatment, periauricular cellulitis, or recurrent OE needing canal evaluation

Hospitalisation Criteria

Admit if malignant otitis externa (diabetic/immunocompromised child), periauricular cellulitis spreading beyond canal, or suspected mastoiditis

Notes

Topical fluoroquinolone-steroid combination is first-line; systemic antibiotics only if peri-auricular cellulitis or immunocompromised. Keep ear dry; use wick if canal is edematous to deliver drops.

H61.2 — Impacted cerumen

SNOMED: 18070006 | Duration: 3 days | Source: AAO-HNS 2017 (updated 2024), AAP endorsed

First-Line

- **Cerumenolytic (carbamide peroxide 6.5%)** — Instill 5-10 drops in affected ear, wait 15-30min, then irrigate; age ≥3yr
- **Warm water irrigation** — Body-temp water via syringe or electronic irrigator; gentle pressure; intact TM only
- **Manual removal (curette)** — Under direct visualization with otoscope; by trained provider; preferred in young children

Second-Line

- **Sodium bicarbonate ear drops 5%** — Instill 3-5 drops in ear bid x3-5 days to soften wax before removal
- **Docusate sodium 1% (Colace) drops** — Instill 3-5 drops in ear, wait 15min then irrigate; off-label cerumenolytic
- **Normal saline irrigation** — Body-temp saline via bulb syringe or ear wash system; safe first-line alternative

Investigations

- Otoscopy — Visualise wax and tympanic membrane status
- Audiometry — If hearing loss persists after wax removal

Counselling

- Do not put cotton buds or any objects into the ear canal - this pushes wax deeper
- Ears are self-cleaning; wax normally moves out on its own
- Use ear drops as directed to soften wax before the removal appointment
- If ear feels blocked after swimming, tilt head to drain water; do not insert objects
- Come back if ear pain, discharge, or hearing difficulty persists after wax removal

Referral Criteria

Refer to ENT if failed removal after 2 attempts in clinic, suspected foreign body behind wax, non-intact tympanic membrane, or child too uncooperative for office removal

Hospitalisation Criteria

Hospitalisation not needed for cerumen impaction; consider if removal under GA needed in very young or uncooperative child

Notes

Cerumenolytics not recommended under age 3 or with non-intact TM. Ear candling is contraindicated and ineffective; refer to ENT if removal fails after 1-2 attempts.

H65 — Otitis media, nonsuppurative

SNOMED: 65363002 | Duration: 90 days | Source: AAO-HNS/AAP/AAFP 2016 OME Guideline

First-Line

- **Watchful waiting** — Observe 3mo from onset/diagnosis if no risk factors for speech/language delay
- **Hearing assessment** — Obtain audiogram if OME persists ≥3mo or at any time if language delay suspected
- **Pneumatic otoscopy monitoring** — Reassess q3-6mo with pneumatic otoscopy until effusion resolves

Second-Line

- **Tympanostomy tubes** — Bilateral myringotomy with tube insertion if OME ≥3mo with hearing loss ≥20dB
- **Autoinflation (Otovent balloon)** — Child inflates balloon via nostril bid-tid x4wk; age ≥3yr; modest benefit
- **Adenoidectomy** — Consider if age ≥4yr with persistent OME or getting 2nd set of tubes

Investigations

- Pneumatic otoscopy — Primary diagnostic tool; assess TM mobility
- Tympanometry — Type B tracing confirms middle ear effusion
- Audiometry — If OME >3 months or speech/language concerns

Counselling

- Fluid behind the eardrum is very common after ear infections - it usually clears in 3 months
- Antibiotics, decongestants, and antihistamines do NOT help clear this fluid
- If your child seems to not hear well, speak face-to-face and reduce background noise
- Hearing test is recommended if fluid persists for 3 months
- Ear tubes may be needed if fluid persists and causes hearing loss or speech delay

Referral Criteria

Refer to ENT if OME persists >3 months, documented hearing loss >=20 dB, speech/language delay, or learning difficulties

Hospitalisation Criteria

Admit for tympanostomy tube insertion or adenoidectomy (day procedure under GA)

Notes

Steroids, antibiotics, antihistamines, and decongestants are NOT recommended for OME per AAO-HNS guidelines. Tympanostomy tubes indicated when OME causes significant hearing loss or speech delay.

H66 — Suppurative and unspecified otitis media

SNOMED: 3110003 | Duration: 10 days | Source: AAP 2013 AOM Guideline, AAO-HNS 2019

First-Line

- **Amoxicillin (high-dose)** — 80-90mg/kg/day PO divided bid-tid x10d (<2yr) or 7d (2-5yr) or 5d (>5yr), max 3g/day
- **Amoxicillin-clavulanate (high-dose)** — 90mg/kg/day amox component PO divided bid x10d if failed amox or recent Abx use
- **Ibuprofen** — 5-10mg/kg PO q6-8h PRN pain/fever, max 40mg/kg/day
- **Acetaminophen** — 10-15mg/kg PO q4-6h PRN pain/fever, max 75mg/kg/day

Second-Line

- **Ceftriaxone IM** — 50mg/kg IM x1-3 doses (1 dose for uncomplicated; 3 daily doses if failed oral)
- **Cefdinir** — 14mg/kg/day PO once daily or divided bid x10d, max 600mg/day; PCN-allergic (non-severe)
- **Azithromycin** — 10mg/kg PO day 1 then 5mg/kg/day x4d for severe PCN allergy, max 250mg/day
- **Ciprofloxacin/dexamethasone otic** — 4 drops in ear bid x7d for AOM with otorrhea through tympanostomy tube

Investigations

- Pneumatic otoscopy — Primary diagnostic assessment of middle ear [STAT]
- Tympanocentesis with culture — If failed 2 antibiotic courses or immunocompromised [URGENT]
- CT temporal bone — If mastoiditis or intracranial complication suspected [STAT]
- Complete blood count — If systemic toxicity or complicated AOM [URGENT]

Counselling

- Give the antibiotic for the full prescribed course - stopping early causes resistant bacteria
- Paracetamol or ibuprofen is very important for ear pain relief; do not wait for antibiotic to work
- Follow up in 48-72 hours if no improvement; antibiotic may need changing
- Clean any ear discharge gently with cotton; do not insert anything deep into ear canal
- Completed pneumococcal and influenza vaccines reduce ear infection frequency

Referral Criteria

Refer to ENT if recurrent AOM (>=3 in 6 months or >=4 in 12 months), failed 2 courses of antibiotics, suspected mastoiditis, or persistent otorrhoea >2 weeks

Hospitalisation Criteria

Admit if acute mastoiditis, intracranial complication (meningitis, brain abscess, lateral sinus thrombosis), toxic appearance, or need for IV antibiotics

Notes

High-dose amoxicillin is first-line; watchful waiting may be appropriate for mild AOM in children >2yr with unilateral symptoms. Pain management is essential and should be addressed regardless of antibiotic decision.

H66.9 — Acute Otitis Media

SNOMED: 3110003 | Duration: 7 days | Source: IAP 2024, AAP AOM Guidelines

First-Line

- **Paracetamol** — 10-15 mg/kg/dose 6-8 hourly PO SOS, pain relief is first priority
- **Ibuprofen** — 5-10 mg/kg/dose 6-8 hourly PO SOS, for pain and fever

- **Amoxicillin-Clavulanate** — 40-45 mg/kg/day (amoxicillin component) div BD PO x 5-7 days, drug of choice
- **Phenylephrine + Chlorpheniramine** — Age-appropriate dose TDS PO, short duration, nasal decongestant to improve Eustachian tube function

Second-Line

- **Cefpodoxime** — 8-10 mg/kg/day div BD PO x 5-7 days, use if penicillin allergy or poor response
- **Ear Drops (Oto-AC / Otoban)** — 2-3 drops TDS in affected ear, only if external ear involvement or discharge; avoid if tympanic membrane perforation

Investigations

- Complete blood count — Severe infection or toxic child
- CRP — Severe infection to assess inflammation
- Ear swab culture and sensitivity — Recurrent or persistent ear discharge

Counselling

- Ear pain usually improves in 24-48 hours with treatment
- Complete the full course of antibiotics even if pain resolves
- Apply warm compress over the affected ear for pain relief
- Keep the ear dry - avoid water entering the ear during bathing
- If pain persists beyond 48 hours, high fever continues, ear discharge appears, or swelling behind the ear develops - return to hospital immediately
- Follow up in 2-3 days to assess response to treatment

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Persistent ear pain beyond 48 hours despite treatment
- High fever not responding to medication
- Ear discharge or pus from the ear
- Swelling behind the ear suggesting mastoiditis

Referral Criteria

Refer to ENT specialist if no improvement after 48-72 hours on antibiotics, recurrent AOM (>3 episodes in 6 months or >4 in 1 year), suspected mastoiditis, hearing difficulty, or chronic ear discharge

Hospitalisation Criteria

Admit if suspected mastoiditis (post-auricular swelling/tenderness), intracranial complications (meningitis, brain abscess), toxic appearance, age <2 months with fever, or failed outpatient therapy with systemic signs

Notes

Pain control is the priority. Amox-Clav is first-line antibiotic. Avoid dual antibiotic therapy (Amox-Clav + Cefpodoxime simultaneously not recommended). Ear drops only if external ear involvement or discharge (avoid if TM perforation unless appropriate drops). Non-pharmacological: warm compress over ear, adequate hydration, keep ear dry.

Guideline Changes

AAP 2024 reaffirms watchful waiting for mild unilateral AOM in children >=2 yr; high-dose amoxicillin (80-90 mg/kg/day) remains first-line; amoxicillin-clavulanate for treatment failure or beta-lactamase producers; Italian intersociety 2024 consensus aligns with AAP on shortened 5-day courses for uncomplicated AOM in children >=2 yr.

H72 — Perforation of tympanic membrane

SNOMED: 60442001 | Duration: 30 days | Source: AAO-HNS 2019, AAP 2023

First-Line

- **Ofloxacin 0.3% otic solution** — 5 drops in affected ear bid x7-10d; safe with perforated TM (non-ototoxic)
- **Ciprofloxacin 0.3% otic drops** — 3-4 drops in affected ear bid x7-10d; safe with perforated TM
- **Dry ear precautions** — Cotton ball with petroleum jelly for bathing; avoid swimming until healed

Second-Line

- **Tympanoplasty (paper patch)** — Office-based paper/gelfoam patch for small acute perforations to aid healing
- **Tympanoplasty (surgical)** — Formal surgical repair with fascia graft if perforation persists >3mo; under GA
- **Acetaminophen** — 10-15mg/kg PO q4-6h PRN pain, max 75mg/kg/day or 4g/day

Investigations

- Otoscopy — Document perforation size, location, and margin
- Audiometry (pure tone) — Assess hearing loss and conductive gap
- CT temporal bone — If cholesteatoma or ossicular erosion suspected

Counselling

- Keep the ear completely dry - use cotton with petroleum jelly during bathing, no swimming
- Do not put any drops in the ear unless prescribed by doctor (some drops are harmful with perforation)
- Most small perforations heal on their own within 1-3 months
- Avoid blowing nose forcefully as air pressure can prevent healing
- If hearing does not improve after healing, follow-up hearing test is needed

Referral Criteria

Refer to ENT if perforation not healed in 3 months, persistent hearing loss, recurrent ear infections through perforation, or cholesteatoma suspected

Hospitalisation Criteria

Admit for tympanoplasty surgery (elective), or if acute perforation with vertigo, facial nerve palsy, or suspected inner ear injury

Notes

Most acute TM perforations heal spontaneously within 1-3 months; avoid aminoglycoside otic drops (ototoxic through perforation). Surgical repair indicated for persistent perforation with hearing loss or recurrent infection.

H90 — Conductive and sensorineural hearing loss

Duration: 365 days | Source: AAP 2023, AAO-HNS 2019, JCIH 2019

First-Line

- **Hearing aids (digital)** — Fit bilateral behind-the-ear aids; trial 3-6mo; verify with real-ear measurements
- **Cochlear implant** — For severe-profound bilateral SNHL (>70dB); FDA approved ≥9mo; implant ASAP ideally <12mo
- **Bone conduction hearing device** — Softband for conductive HL in infants; surgical implant (Baha/Ponto) at age ≥5yr

Second-Line

- **Tympanostomy tubes** — For conductive HL from chronic OME; outpatient procedure under brief GA
- **Ossiculoplasty** — Surgical reconstruction of ossicular chain for congenital/acquired conductive HL
- **FM/DM system (assistive)** — Classroom amplification system; transmitter to teacher, receiver to child's hearing aid
- **Speech-language therapy** — Begin by 6mo of age per EHDI guidelines; ongoing per developmental milestones

Investigations

- ABR (Auditory Brainstem Response) — Definitive hearing assessment in infants [URGENT]
- OAE (Otoacoustic Emissions) — Screening test for cochlear function [URGENT]
- Tympanometry — Assess middle ear function; rule out conductive component
- Temporal bone CT/MRI — Assess cochlear anatomy before cochlear implant
- Genetic testing (GJB2/connexin 26) — Most common cause of genetic SNHL

Counselling

- Early hearing aid fitting and speech therapy are critical - start by 6 months of age
- Hearing aids should be worn during all waking hours for best speech development
- Talk to your child face-to-face; reduce background noise during conversations
- Regular audiological follow-up every 3-6 months is essential for adjustments
- Cochlear implant can give hearing to profoundly deaf children if fitted early (before 12 months)

Referral Criteria

Refer to ENT/audiologist immediately on newborn hearing screen failure; refer to cochlear implant centre if bilateral severe-profound SNHL not benefiting from hearing aids

Hospitalisation Criteria

Admit for cochlear implant surgery, tympanoplasty, or ossiculoplasty as planned procedures

Notes

Early identification by 3mo and intervention by 6mo are critical per JCIH guidelines. Conductive HL often treatable medically/surgically; SNHL requires amplification or cochlear implantation.

Guideline Changes

Cochlear implant candidacy expanded to include children 9-24mo with bilateral profound SNHL

H91 — Hearing loss, unspecified

SNOMED: 15188001 | Duration: 365 days | Source: AAP 2023, JCIH 2019, AAO-HNS 2022

First-Line

- **Audiologic evaluation** — ABR/OAE for infants; behavioral audiometry for >6mo; complete by 3mo of age
- **Hearing aids (digital)** — BTE aids fitted per audiogram; 3-6mo trial; regular adjustment as child grows
- **Early intervention services** — Enroll by 6mo per EHDI; speech therapy, family training, developmental support

Second-Line

- **Cochlear implant** — If bilateral severe-profound SNHL with inadequate hearing aid benefit after trial
- **Bone conduction device** — Softband from infancy for conductive/mixed HL; osseointegrated implant at ≥5yr
- **FM/DM assistive system** — For school-age children; improves signal-to-noise ratio in classroom

Investigations

- Complete audiological evaluation (ABR, OAE, tympanometry) — Define type and degree of hearing loss [URGENT]
- CMV testing (urine/blood) — Congenital CMV is leading non-genetic cause of SNHL; test within 3 weeks of birth [URGENT]
- Genetic testing — GJB2 mutation analysis for non-syndromic SNHL
- Ophthalmology evaluation — Usher syndrome screening; many HL syndromes have eye involvement
- Renal ultrasound — Screen for associated renal anomalies (branchio-oto-renal syndrome)

Counselling

- Do not delay evaluation - early diagnosis and intervention prevents speech delay
- Follow the 1-3-6 rule: screen by 1 month, diagnose by 3 months, intervene by 6 months
- Hearing aids and speech therapy should start as soon as hearing loss is confirmed

- Talk, read, and sing to your child regularly; face-to-face communication is very important
- Ensure family members learn to communicate effectively with the child

Referral Criteria

Refer to ENT/audiologist for all detected hearing loss; urgent referral for sudden hearing loss; genetic counselling for familial hearing loss

Hospitalisation Criteria

Admit for cochlear implant surgery or other surgical interventions (tympanoplasty, ossicular reconstruction)

Notes

All newborns should be screened before 1 month; diagnosis by 3 months; intervention by 6 months (1-3-6 rule). Etiology workup includes genetics, CMV testing, imaging, and ophthalmology referral.

J02.0 — Streptococcal pharyngitis (GAS)

SNOMED: 43878008 | Duration: 10 days | Source: IDSA 2025

First-Line

- **Amoxicillin** — 50 mg/kg/day (max 1g) once daily x10 days
- **Penicillin V** — Children 250mg BID-TID, adolescents 500mg BID-TID x10 days

Second-Line

- **Azithromycin** — 12 mg/kg/day (max 500mg) x5 days, for penicillin allergy

Investigations

- Rapid Antigen Detection Test (RADT) — First-line test for suspected strep pharyngitis [URGENT]
- Throat culture — If RADT negative but high clinical suspicion (McIsaac >=3)
- ASO titre — If suspected post-streptococcal complication (rheumatic fever, glomerulonephritis)

Counselling

- Full 10-day course of antibiotic is essential to prevent rheumatic fever - do not stop early
- Child can return to school after 24 hours on antibiotics and no fever
- Offer warm fluids and soft foods; cold items like ice cream may soothe throat
- Replace toothbrush after 48 hours of antibiotics to prevent re-infection
- Watch for joint pain, skin rash, or chest pain in the weeks after - see doctor immediately

Referral Criteria

Refer to ENT if recurrent strep pharyngitis (>=7 episodes/year, >=5/year for 2 years, or >=3/year for 3 years) meeting Paradise criteria for tonsillectomy

Hospitalisation Criteria

Admit if peritonsillar abscess suspected (trismus, uvula deviation, drooling), retropharyngeal abscess, severe dehydration from inability to swallow, or Lemierre syndrome signs

Notes

10-day course required to prevent acute rheumatic fever. Only treat confirmed positive RADT or culture.

Guideline Changes

IDSA 2025: use Centor/McIsaac clinical scoring BEFORE testing - shift from 2012 universal testing approach.

J03.9 — Follicular Tonsillitis

SNOMED: 17741008 | Duration: 7 days | Source: IAP Guidelines, IDSA Pharyngitis Guidelines 2012/2024 update, AHA Rheumatic Fever Prevention

First-Line

- **Cefpodoxime** — 8-10 mg/kg/day div BD PO for 5-7 days
- **Paracetamol** — 10-15 mg/kg/dose PO 6-8 hourly SOS for fever/pain
- **Ibuprofen** — 5-10 mg/kg/dose PO 6-8 hourly SOS, for fever and throat pain
- **Warm Saline Gargles** — If child >5 years, adequate hydration, soft diet (khichdi, soups). Avoid cold items

Second-Line

- **Linezolid** — 10 mg/kg/dose BD PO for 7-10 days, reserve drug for severe/non-responding/resistant cases only

Investigations

- Complete blood count — Assess infection severity
- CRP — Assess degree of inflammation
- Procalcitonin — Severe bacterial infection suspicion [URGENT]
- Throat swab culture — Recurrent cases to identify organism

Counselling

- Complete full course of antibiotic even if child feels better in 2-3 days
- Give warm fluids and soft diet (khichdi, soups) to ease swallowing
- Warm saline gargles help if child is older than 5 years
- Paracetamol/ibuprofen will control fever and throat pain
- Return if fever persists beyond 3 days, difficulty swallowing saliva, neck swelling, or breathing difficulty

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Persistent fever for more than 3 days
- Difficulty swallowing saliva or drooling
- Swelling in the neck region
- Recurrent vomiting

Referral Criteria

Refer to ENT if recurrent tonsillitis (>7 episodes/year or >5/year for 2 years), suspected peritonsillar abscess, or airway compromise

Hospitalisation Criteria

Admit if unable to swallow, severe dehydration, peritonsillar abscess, toxic appearance, or breathing difficulty

Notes

Start antibiotic if high-grade fever with follicles or toxic child. Supportive: warm saline gargles (if child >5 years), adequate hydration, soft diet. Avoid cold items.

Guideline Changes

IDSA 2024: Continued emphasis on rapid antigen detection test (RADT) or culture before starting antibiotics for pharyngitis; Centor/McIsaac scoring to guide testing; Penicillin V or amoxicillin remain first-line for confirmed GAS; IAP 2024: Antibiotic stewardship - avoid empirical antibiotics for viral tonsillitis; reserve cephalosporins for penicillin-allergic patients

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J30.9 — Allergic Rhinitis

SNOMED: 61582004 | Duration: 14 days | Source: ARIA-EAACI Guidelines 2024-2025, IAP Allergy Protocol, GINA

First-Line

- **Fluticasone Nasal Spray** — 1 puff each nostril BD, continue at least 4-6 weeks, regular use not SOS
- **Fexofenadine** — 6-12 yrs: 60-120 mg OD PO; >12 yrs: 120-180 mg OD PO, for 7-14 days or longer

- **Normal Saline Nasal Drops** — 2-3 drops each nostril 2-3 times/day, clears allergens and improves drug delivery
- **Azelastine + Fluticasone Nasal Spray** — 1 puff each nostril BD, for moderate-severe or persistent rhinitis, faster symptom relief

Investigations

- CBC with absolute eosinophil count — Allergy indicator, elevated eosinophils suggest atopy
- Serum IgE levels — Confirm atopic tendency
- Allergy skin prick test / specific IgE — Identify specific triggers in recurrent/severe cases

Counselling

- Use nasal spray regularly every day, not just when symptoms are bad
- Avoid dust, smoke, perfumes, and cold exposure - these are common triggers
- Saline nasal drops before steroid spray improve drug delivery
- Improvement takes 3-5 days, full control in 2-3 weeks - do not stop early
- Keep home dust-free, use cotton bedding, and avoid pets if allergic

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Severe nasal blockage causing difficulty breathing
- Disturbed sleep or mouth breathing at night
- Symptoms not improving despite regular medication
- Associated wheezing or breathing difficulty

Referral Criteria

Refer to allergist/ENT if poor response to 4-6 weeks of treatment, severe persistent symptoms, associated asthma requiring step-up therapy, or need for allergy testing

Hospitalisation Criteria

Hospitalisation rarely needed. Consider if severe associated asthma exacerbation or complications like sinusitis requiring IV antibiotics

Notes

ARIA classification: Intermittent (<4 days/week) vs Persistent (>4 days/week); Mild vs Moderate-Severe. Intranasal steroid is gold standard. Allergen avoidance is essential: dust, smoke, perfumes, cold exposure, pets, house dust mites.

Guideline Changes

ARIA-EAACI 2024-2025: Combined intranasal antihistamine + corticosteroid (INAH+INCS) recommended over monotherapy for moderate-severe AR; INCS preferred over INAH alone; digital shared decision-making algorithms introduced; pediatric considerations updated for preschool and school-age children; emphasis on step-up/step-down approach based on symptom control

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J35 — Chronic diseases of tonsils and adenoids

SNOMED: 191428009 | Duration: 10 days | Source: AAO-HNS 2019 (updated CPG), AAP 2023

First-Line

- **Adenotonsillectomy** — For ≥7 episodes/1yr, ≥5/yr x2yr, or ≥3/yr x3yr of documented strep/tonsillitis; or OSA
- **Amoxicillin** — 50mg/kg/day PO divided bid-tid x10d for acute exacerbations of chronic tonsillitis
- **Dexamethasone IV (perioperative)** — 0.5mg/kg IV single dose intraop (max 10mg) for tonsillectomy; reduces PONV/pain

Second-Line

- **Amoxicillin-clavulanate** — 45mg/kg/day PO divided bid x10d for recurrent tonsillitis unresponsive to amoxicillin
- **Clindamycin** — 20-30mg/kg/day PO divided tid x10d for chronic tonsillitis biofilm eradication; PCN allergy
- **Ibuprofen (postoperative)** — 5-10mg/kg PO q6-8h PRN x7-10d post-tonsillectomy; safe per AAO-HNS 2019 update
- **Acetaminophen (postoperative)** — 10-15mg/kg PO q4-6h PRN pain post-tonsillectomy; max 75mg/kg/day

Investigations

- Polysomnography (sleep study) — If sleep-disordered breathing/OSA suspected before surgery
- Complete blood count — Pre-operative assessment
- Coagulation profile (PT/aPTT) — Pre-operative if bleeding history
- Throat swab culture — Document strep carrier status

Counselling

- If surgery is recommended, it means your child meets criteria for frequent, severe infections
- Post-surgery: soft foods and plenty of fluids for 10-14 days; avoid hard or crunchy foods
- Pain after tonsillectomy peaks on day 3-5; give paracetamol and ibuprofen regularly, NOT codeine
- Watch for post-operative bleeding (fresh blood from mouth/nose) - go to emergency immediately
- The child may have bad breath and white patches in throat after surgery - this is normal healing

Referral Criteria

Refer to ENT if meeting Paradise criteria (>=7 episodes/1yr, >=5/yr x2yr, >=3/yr x3yr), sleep-disordered breathing/OSA, peritonsillar abscess, or suspected malignancy

Hospitalisation Criteria

Admit for tonsillectomy (usually day surgery); emergency admission for peritonsillar abscess (drainage), post-tonsillectomy haemorrhage, or airway compromise from tonsillar hypertrophy

Notes

Watchful waiting recommended if infection frequency is below Paradise criteria thresholds. Post-tonsillectomy: codeine is CONTRAINDICATED in children <12yr; ibuprofen and acetaminophen are mainstays of pain management.

J35.1 — Hypertrophy of tonsils

SNOMED: 46689006 | Duration: 14 days | Source: AAO-HNS 2019 CPG (updated), AAP 2023

First-Line

- **Tonsillectomy** — Indicated for OSA/SDB with tonsillar hypertrophy; PSG recommended if <2yr, obese, or comorbid
- **Intranasal corticosteroid (mometasone)** — 1 spray per nostril daily trial x6wk for mild SDB before surgical decision; age ≥2yr
- **Dexamethasone IV (perioperative)** — 0.5mg/kg IV single dose intraop (max 10mg); reduces PONV and improves oral intake

Second-Line

- **Ibuprofen (postoperative)** — 5-10mg/kg PO q6-8h PRN x7-10d post-tonsillectomy; AAO-HNS confirms safety
- **Acetaminophen (postoperative)** — 10-15mg/kg PO q4-6h PRN pain post-tonsillectomy; max 75mg/kg/day
- **Montelukast** — 4mg (2-5yr) or 5mg (6-14yr) PO daily for persistent OSA post-surgery; FDA black box warning
- **Watchful waiting** — If snoring without OSA/apnea and no impact on growth, behavior, or school performance

Investigations

- Polysomnography (sleep study) — Document OSA severity before surgery; recommended if <2 years, obese, or comorbid
- Complete blood count — Pre-operative assessment
- Lateral neck X-ray — If unable to visualize tonsils or adenoid assessment needed

Counselling

- If your child snores loudly and stops breathing during sleep, a sleep study may be needed
- Tonsillectomy has excellent results for sleep apnoea caused by large tonsils
- Post-surgery: soft foods for 10-14 days; regular pain medicines (paracetamol + ibuprofen); NO codeine
- Watch for bleeding from mouth/nose after surgery - go to emergency immediately
- Keep child well hydrated after surgery; ice lollies and cold drinks can soothe the throat

Referral Criteria

Refer to ENT if tonsillar hypertrophy causing OSA/sleep-disordered breathing, dysphagia, failure to thrive from poor eating, or asymmetric tonsillar enlargement (rule out malignancy)

Hospitalisation Criteria

Admit for tonsillectomy (day surgery for most; overnight if <3 years, OSA, or comorbidities); emergency for acute airway compromise or post-tonsillectomy haemorrhage

Notes

Adenotonsillectomy is first-line for pediatric OSA with tonsillar hypertrophy; PSG should be obtained before surgery in high-risk groups. Codeine is contraindicated post-tonsillectomy in children <12yr due to fatal respiratory depression risk.

J35.2 — Hypertrophy of adenoids

SNOMED: 39323002 | Duration: 42 days | Source: German S2k Guideline 2023, AAO-HNS 2019, AAP 2023

First-Line

- **Mometasone furoate nasal spray 50mcg** — 1 spray per nostril daily x6-12wk; age ≥2yr; 77% show improvement avoiding surgery
- **Saline nasal irrigation** — Isotonic saline 5-10mL per nostril bid via squeeze bottle or neti pot; all ages
- **Adenoidectomy** — Outpatient procedure under GA if medical therapy fails or severe obstruction/OSA/recurrent OME

Second-Line

- **Fluticasone propionate nasal spray** — 1 spray per nostril daily x6-12wk; age ≥4yr; alternative intranasal steroid
- **Montelukast** — 4mg PO daily (2-5yr) or 5mg (6-14yr) as adjunct; FDA black box: neuropsychiatric risk
- **Desloratadine** — 1.25mg PO daily (1-5yr), 2.5mg (6-11yr) if concurrent allergic rhinitis component
- **Watchful waiting** — For mild symptoms without OSA; adenoid may involute with growth by age 7-10yr

Investigations

- Lateral X-ray nasopharynx — Assess adenoid size and airway obstruction
- Sleep study (polysomnography) — If OSA suspected before surgical decision
- Tympanometry — If recurrent OME associated with adenoid hypertrophy

Counselling

- Nasal steroid spray is the first treatment to try - use for at least 6 weeks before deciding on surgery
- Use correct spray technique: point nozzle towards outer wall of nose, not septum
- Saline nasal wash helps clear mucus and reduces symptoms
- Watch for mouth breathing, snoring, or disturbed sleep - these may indicate surgery is needed
- Adenoids naturally shrink after age 7-10 years; some children may outgrow symptoms

Referral Criteria

Refer to ENT if nasal steroid spray fails after 6-12 weeks trial, sleep-disordered breathing, chronic sinusitis, recurrent OME with hearing loss, or significant nasal obstruction

Hospitalisation Criteria

Admit for adenoidectomy (day procedure under GA); urgent if severe OSA or acute upper airway obstruction

Notes

Trial of intranasal corticosteroids for 6-12 weeks is recommended before proceeding to adenoidectomy for uncomplicated hypertrophy. Surgery indicated for OSA, chronic OME with hearing loss, or chronic rhinosinusitis unresponsive to medical therapy.

Guideline Changes

Intranasal mometasone now established as first-line medical trial before adenoidectomy (S2k 2023)

K02 — Dental caries

SNOMED: 80967001 | Duration: 180 days | Source: AAPD 2024, USPSTF 2021

First-Line

- **Fluoride varnish 5% NaF** — Professional application q3-6mo; high-risk children q3mo per AAP Bright Futures
- **Fluoride toothpaste 1000-1500ppm** — Rice-grain amount <3yr, pea-size 3-6yr, brush 2x/day
- **Silver diamine fluoride 38%** — Topical application to cavitated lesions biannually, isolate field 1-3min **[NEW 2024-25]**
- **Glass ionomer sealant/restoration** — Apply to pit/fissure or cavitated lesion as interim therapeutic restoration

Second-Line

- **Fluoride gel APF 1.23%** — Professional tray application q6mo for moderate-high risk, 4min application
- **Chlorhexidine varnish 1%** — Apply to high-risk surfaces q3mo as adjunct to fluoride
- **Dietary fluoride supplement** — 0.25mg/d age 6mo-3yr, 0.5mg/d 3-6yr, 1mg/d 6-16yr if water <0.3ppm F

Investigations

- Dental examination with caries risk assessment — Baseline and periodic assessment
- Dental radiographs (bitewings) — Detect interproximal caries and assess bone levels

Counselling

- Brush teeth twice daily with fluoride toothpaste; supervise brushing until age 8
- Limit sugary drinks and snacks between meals; avoid bottle feeding at bedtime
- Fluoride varnish applied by dentist every 3-6 months helps prevent new cavities
- First dental visit should be by age 1 year or when first tooth appears
- Silver diamine fluoride (SDF) can arrest cavities without drilling in young children

Referral Criteria

Refer to pediatric dentist for early childhood caries, caries in child <3 years, need for treatment under GA, or special healthcare needs

Hospitalisation Criteria

Admit if dental abscess with facial cellulitis, Ludwig angina (floor of mouth infection), or need for extensive dental rehabilitation under GA

Notes

Caries risk assessment drives treatment intensity; high-risk children need q3mo fluoride varnish and SDF for active lesions. Restorative treatment under GA may be needed for extensive ECC in young children.

Guideline Changes

SDF 38% now recommended biannually for arresting cavitated lesions (AAPD 2023 update)

K04 — Diseases of pulp and periapical tissues

Duration: 10 days | Source: AAPD 2024, AAE 2023

First-Line

- **MTA (mineral trioxide aggregate)** — Pulpotomy agent: apply 2-3mm layer over pulp stumps, cover with GIC/composite
- **Biodentine (calcium silicate cement)** — Pulpotomy agent: mix and apply over pulp stumps, sets in 12min, restore same visit **[NEW 2024-25]**
- **Ibuprofen** — 5-10mg/kg PO q6-8h PRN pain, max 40mg/kg/day
- **Amoxicillin** — 25-50mg/kg/day PO divided q8h x7-10d for periapical abscess, max 500mg/dose

Second-Line

- **Ferric sulfate 15.5%** — Pulpotomy hemostatic agent: apply 15sec to pulp stumps, rinse, restore
- **Calcium hydroxide** — Direct/indirect pulp cap: apply thin layer over near/micro-exposure, restore
- **ZOE (zinc oxide eugenol)** — Pulpectomy obturation material for primary teeth, fill canals to apex
- **Clindamycin** — 10-25mg/kg/day PO divided q8h x7d for PCN-allergic patients, max 450mg/dose

Investigations

- Dental radiograph (periapical) — Assess extent of caries and periapical pathology [URGENT]
- Pulp vitality testing — Determine if pulp is vital or necrotic
- Complete blood count — If systemic infection suspected (facial cellulitis) [URGENT]

Counselling

- If your child has a toothache, give paracetamol/ibuprofen and see the dentist urgently
- Do not apply aspirin or home remedies directly on the gum - this can burn tissue
- Tooth infection (abscess) with face swelling needs immediate dental treatment
- Baby tooth infections must be treated to protect the permanent teeth growing underneath
- Regular dental checkups every 6 months can catch problems before they become painful

Referral Criteria

Refer to pediatric dentist for pulp therapy (pulpotomy/pulpectomy), periapical abscess requiring drainage, or management under sedation/GA

Hospitalisation Criteria

Admit if facial cellulitis from dental abscess, spreading infection (Ludwig angina, orbital cellulitis), fever with trismus, or airway compromise

Notes

Pulpotomy with MTA/Biodentine is preferred for vital primary teeth with reversible pulpitis. Pulpectomy indicated for irreversible pulpitis or necrosis; extraction considered if root resorption >1/3.

Guideline Changes

MTA/Biodentine now preferred over formocresol for pulpotomy based on high-certainty evidence

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K05 — Gingivitis and periodontal diseases

SNOMED: 66383009 | Duration: 14 days | Source: AAP/EFP 2018, AAPD 2023

First-Line

- **Chlorhexidine gluconate 0.12%** — Rinse 15mL for 30sec bid x2wk; children >6yr who can spit; after professional cleaning
- **Professional scaling/prophylaxis** — Mechanical debridement of plaque/calculus, polish, q6mo or as needed

- **Fluoride toothpaste** — Brush 2x/day with soft-bristled brush, supervised in children <8yr

Second-Line

- **Metronidazole** — 7.5mg/kg PO q8h x7d for necrotizing gingivitis, max 250mg/dose
- **Amoxicillin** — 25-50mg/kg/day PO divided q8h x7d for aggressive periodontitis with systemic Abx
- **Hydrogen peroxide 1.5% rinse** — Dilute to 1.5%, rinse 30sec bid as adjunct for acute NUG
- **Doxycycline (>8yr)** — 2mg/kg/day PO x14-21d as adjunct for aggressive periodontitis, max 100mg/day

Investigations

- Dental examination and probing — Assess pocket depth and attachment loss
- Dental radiographs (panoramic) — Assess alveolar bone level in periodontitis
- Complete blood count — If necrotizing gingivitis or suspected underlying hematologic condition [URGENT]

Counselling

- Brush teeth twice daily with soft brush using correct technique; use interdental brush/floss
- Bleeding gums during brushing is a sign of gingivitis - do not stop brushing, it will improve
- Professional dental cleaning removes hardened plaque (tarite) that brushing cannot
- Eat crunchy fruits and vegetables which naturally help clean teeth
- Chlorhexidine mouthwash is for short-term use (2 weeks) only; it can stain teeth with prolonged use

Referral Criteria

Refer to periodontist if aggressive periodontitis with bone loss, necrotizing gingivitis not responding to treatment, or gingival overgrowth (drug-induced)

Hospitalisation Criteria

Admit if necrotizing gingivitis with systemic toxicity (fever, dehydration), spreading infection, or underlying immunodeficiency presenting with severe periodontal disease

Notes

Plaque-induced gingivitis is reversible with oral hygiene improvement; CHX rinse is adjunctive, not a substitute. Necrotizing gingivitis requires debridement plus systemic antibiotics if febrile.

K13 — Other diseases of lip and oral mucosa

SNOMED: 186673008 | Duration: 14 days | Source: AAPD 2023, EJPD 2022

First-Line

- **Triamcinolone acetonide 0.1% paste** — Apply thin layer to ulcer tid-qid after meals and at bedtime until healed
- **Acetaminophen** — 10-15mg/kg PO q4-6h PRN pain, max 75mg/kg/day or 4g/day
- **Magic mouthwash (compounded)** — Equal parts diphenhydramine/Maalox/viscous lidocaine; swish 5mL and spit q6h PRN

Second-Line

- **Fluocinonide 0.05% gel** — Apply to ulcer bid-tid for severe/recurrent aphthous ulcers, use for ≤2wk
- **Dexamethasone 0.5mg/5mL elixir** — Rinse 5mL for 2min then spit, bid-qid for widespread oral ulcers
- **Sucralfate suspension 1g/10mL** — Swish 5-10mL qid as mucosal protectant for oral mucositis
- **Ibuprofen** — 5-10mg/kg PO q6-8h PRN pain/inflammation, max 40mg/kg/day

Investigations

- Complete blood count — Screen for iron, B12, folate deficiency in recurrent aphthae
- Serum iron, ferritin, B12, folate — Nutritional deficiencies causing recurrent oral ulcers
- Anti-tTG IgA — Screen for celiac disease in recurrent aphthous ulcers
- Biopsy of oral lesion — If non-healing ulcer >3 weeks or atypical appearance

Counselling

- Most mouth ulcers heal in 7-14 days without specific treatment
- Avoid spicy, salty, and acidic foods which irritate ulcers
- Apply prescribed mouth paste directly on the ulcer after meals and at bedtime
- Rinse with salt water (half teaspoon salt in 1 glass warm water) 3-4 times daily
- If ulcers recur frequently, keep a food diary - some foods (citrus, tomato) may trigger them

Referral Criteria

Refer to oral medicine specialist if recurrent aphthous ulcers (>3 episodes/year), ulcers lasting >3 weeks, suspected Behcet disease, or mucosal lesion suggestive of malignancy

Hospitalisation Criteria

Admit if severe oral ulceration preventing oral intake (dehydration risk), suspected Behcet disease with systemic involvement, or oral manifestation of systemic disease

Notes

Most aphthous ulcers are self-limiting within 7-14 days; topical steroids reduce pain and healing time but not recurrence. Rule out systemic causes (Behcet, IBD, celiac) if recurrent or severe.

R04.0 — Epistaxis

Duration: 14 days | Source: AAO-HNS Clinical Practice Guideline: Nosebleed (2020, AAP-endorsed), IAP ENT Guidelines

First-Line

- **Liquid Paraffin nasal drops** — 2-3 drops in each nostril 2-3 times daily, keeps mucosa moist
- **Normal Saline nasal drops** — 2-3 drops each nostril TDS for mucosal hydration
- **Ascorbic Acid** — Age-appropriate dose OD PO for 2-4 weeks, improves capillary strength
- **First-Aid Management (non-pharmacological)** — Sit child upright, tilt head slightly forward, pinch soft part of nose for 10 minutes, apply cold compress

Second-Line

- **Mupirocin ointment** — Apply inside anterior nares BD for 5 days, if crusting/infection present
- **Fusidic Acid ointment** — Apply inside anterior nares BD for 5 days, alternative to mupirocin if crusting/infection

Investigations

- Complete blood count — Recurrent or severe epistaxis to check platelets and anemia
- PT/INR — Recurrent epistaxis to rule out coagulation disorder
- Bleeding time / Clotting time — Suspected bleeding disorder

Counselling

- Apply nasal oil drops regularly to keep the nose moist
- Teach first aid: sit upright, tilt head forward, pinch soft part of nose for 10 minutes
- Keep child's nails short to prevent nose picking
- Avoid forceful nose blowing and dry air exposure
- Return if bleeding lasts >15-20 minutes, is recurrent, or bleeding from other sites

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Bleeding lasting more than 15 to 20 minutes despite pressure
- Recurrent frequent nosebleed episodes

- Bleeding from other sites such as gums or skin bruises
- Pallor or weakness suggesting significant blood loss

Referral Criteria

Refer to ENT specialist if recurrent frequent episodes (>1/week), bleeding from posterior nose, suspected bleeding disorder, or epistaxis not controlled by local measures

Hospitalisation Criteria

Admit if active uncontrollable bleeding >20 minutes despite compression, hemodynamic instability, significant blood loss with pallor/weakness, or suspected posterior epistaxis requiring nasal packing

Notes

Most cases are benign anterior bleeds from Kiesselbach plexus due to nose picking or dry mucosa. Moisturization is the cornerstone. Teach first aid: sit upright, tilt head forward, pinch soft nose for 10 min, cold compress. Keep nails short. Avoid nose picking, forceful blowing, dry air.

Guideline Changes

AAO-HNS 2020 CPG (AAP-endorsed): sustained pressure for minimum 5 minutes on lower third of nose; topical tranexamic acid now recommended as superior to other agents for stopping bleeding in first 10 minutes; oxymetazoline preferred topical vasoconstrictor. No major 2024-2025 pediatric-specific updates.

T15 — Foreign body in eye and adnexa

Duration: 5 days | Source: AAO 2023, ACEP 2022

First-Line

- **Copious irrigation (saline/BSS)** — Irrigate affected eye with 500-1000mL NS; evert lids; sweep fornices with moist cotton swab
- **Topical proparacaine 0.5%** — 1-2 drops for anesthesia before examination/removal; single use; do not prescribe for home
- **Cotton-tip applicator removal** — Moisten tip, roll gently over superficial conjunctival/corneal FB under slit lamp

Second-Line

- **25-27G needle removal (slit lamp)** — For embedded corneal FB; under topical anesthesia; by trained provider; ophthalmology referral
- **Erythromycin ophthalmic ointment 0.5%** — Apply 1cm ribbon qid x5-7d post-removal to prevent infection of corneal defect
- **Moxifloxacin 0.5% ophthalmic** — 1 drop qid x5-7d post-removal for corneal abrasion prophylaxis; age ≥1yr
- **Cyclopentolate 1% ophthalmic** — 1 drop bid-tid for ciliary spasm/photophobia after corneal FB removal

Investigations

- Visual acuity — Document before any manipulation [STAT]
- Slit lamp examination — Detailed corneal assessment and FB localisation [STAT]
- Fluorescein staining — Assess corneal abrasion and rule out perforation (Seidel test) [STAT]
- CT orbit (no MRI if metallic) — If intraocular metallic FB suspected [STAT]

Counselling

- Do not rub the eye - this can push the object deeper and scratch the cornea
- Flush the eye with clean water if a chemical or small particle is involved
- Wear protective eyewear during sports and activities with flying debris
- After removal, use prescribed antibiotic drops/ointment to prevent infection
- If pain or vision problem continues after removal, return to doctor immediately

Referral Criteria

Refer to ophthalmologist if embedded corneal FB, metallic FB with rust ring, suspected intraocular FB (penetrating injury), or corneal ulcer developing

Hospitalisation Criteria

Admit if suspected globe perforation/penetrating injury, intraocular FB, chemical burn, or corneal ulcer from FB

Notes

Check visual acuity before any manipulation; fluorescein exam mandatory to assess corneal defect and rule out penetration. Young children may require sedation or OR removal under GA; ophthalmology consultation for embedded or metallic FBs.

T16 — Foreign body in ear

SNOMED: 40481000087105 | Duration: 1 days | Source: AAO-HNS 2024, CHOP Clinical Pathway 2023

First-Line

- **Manual removal (curette/alligator forceps)** — Under direct otoscopic visualization; first-line for visible non-spherical objects
- **Warm water irrigation** — Body-temp water via syringe; for non-organic, non-battery objects; intact TM required
- **Suction-tip removal** — Frazier suction tip applied to FB surface; for smooth/round objects in cooperative child

Second-Line

- **Cyanoacrylate (glue) technique** — Blunt probe tip with dried glue applied to FB surface; for round objects; cooperative patient
- **Removal under GA (OR)** — Operating microscope + microinstruments; for failed ED attempts, deep FB, or uncooperative child
- **Ciprofloxacin/dexamethasone otic** — 4 drops bid x5-7d post-removal if canal abrasion or infection present

Investigations

- Otoscopy — Visualize and identify the foreign body [STAT]
- X-ray temporal bone — If metallic FB or button battery suspected [STAT]

Counselling

- Do not try to remove the object at home - this may push it deeper and damage the ear
- If it is a button battery in the ear, go to hospital immediately - it is an emergency
- Keep the child calm and do not insert water or any liquid into the ear
- After removal, some ear discomfort for 1-2 days is normal
- Teach children not to insert objects into ears, nose, or mouth

Referral Criteria

Refer to ENT if failed removal attempt in clinic, deep or posteriorly lodged FB, button battery in ear (emergent), or uncooperative child needing removal under GA

Hospitalisation Criteria

Admit if button battery in ear canal (emergent removal), suspected TM perforation from FB, or need for removal under GA

Notes

Button batteries require emergent removal due to risk of liquefactive necrosis; do NOT irrigate batteries. Limit removal attempts to 1-2 tries in ED; failed attempts increase complication risk and warrant ENT referral.

T18 — Foreign body in alimentary tract

SNOMED: 157561003 | Duration: 3 days | Source: NASPGHAN 2023, ESPGHAN 2024

First-Line

- **Emergent endoscopic removal** — Within 2h for esophageal button battery or sharp object; within 24h for esophageal coin/food bolus
- **Honey (age >12mo)** — 10mL PO q10min (max 6 doses) while awaiting endoscopy for esophageal button battery **[NEW 2024-25]**
- **Sucralfate 1g/10mL** — 10mL PO q10min alternating with honey while awaiting endoscopy for esophageal battery **[NEW 2024-25]**
- **Serial radiographs** — AP chest/abdomen; confirm location and track progression; q12-24h if in stomach

Second-Line

- **Observation (watchful waiting)** — For small blunt objects past esophagus; stool monitoring; repeat XR if not passed in 2wk
- **Whole bowel irrigation (PEG)** — GoLYTELY 25mL/kg/h via NGT for multiple magnets or drug packets beyond endoscope reach
- **Glucagon IV** — 0.02-0.03mg/kg IV (max 0.5mg) for esophageal food bolus; smooth muscle relaxant

Investigations

- AP and lateral chest/abdomen X-ray — Confirm FB location, type, and number [STAT]
- Repeat X-ray at 12-24 hours — If FB in stomach - confirm progression through GI tract [URGENT]
- Metal detector screening — For metallic objects as radiation-free monitoring

Counselling

- If a button battery was swallowed and is in the food pipe (esophagus), it is a medical emergency
- If a coin or small blunt object has passed into the stomach, most pass naturally in 4-6 days
- Check the child stool for the object; report if not passed in 2 weeks
- Do not try to make the child vomit - this can cause more damage
- Childproof the home: keep small objects, batteries, and magnets away from young children

Referral Criteria

Refer to pediatric GI/surgeon immediately if esophageal FB, button battery ingestion, multiple magnet ingestion, or sharp object ingestion

Hospitalisation Criteria

Admit if esophageal FB of any type, button battery anywhere in GI tract, multiple magnets, sharp object, symptomatic FB (pain, drooling, vomiting), or large object unlikely to pass pylorus

Notes

Button batteries in esophagus are a surgical emergency (remove <2h); magnets require urgent removal if >1 ingested due to bowel perforation risk. Most blunt objects past the esophagus pass spontaneously within 4-6 days.

Guideline Changes

Honey/sucralfate pre-endoscopy for button battery ingestion now recommended (NASPGHAN 2023)

Emergency

D72.829 — Hyperleukocytosis with / without Tumor Lysis Syndrome *[severe]*

SNOMED: 277605001 | Duration: 3 days | Source: Provided pediatric hyperleukocytosis and tumor lysis syndrome protocol

First-Line

- **Intravenous Fluids** — Aggressive hydration to maintain urine output and reduce blood viscosity / tumor lysis risk. In asymptomatic hyperleukocytosis: start 2 times maintenance IV fluids. In symptomatic hyperleukocytosis / TLC >2-3 lakhs/cc: start 3-4 times maintenance or approximately 3000 mL/m2/day. Monitor for fluid overload.
- **Allopurinol** — For tumor lysis prophylaxis in hyperleukocytosis / high uric acid risk. Dose as per institutional pediatric oncology protocol; start early with hydration.
- **Rasburicase** — Use in high-risk tumor lysis syndrome or significant hyperuricemia; preferred when uric acid is elevated or rapid control is needed.
- **Packed Red Blood Cells / Platelets** — Avoid routine PRBC transfusion before cytoreduction if hyperviscosity is present. Use platelet transfusion for severe thrombocytopenia / bleeding as clinically indicated.

Second-Line

- **Hydroxyurea / Cytoreductive chemotherapy** — Use according to leukemia subtype and oncology protocol for leukoreduction if symptomatic hyperleukocytosis or rapidly rising counts are present.
- **Leukapheresis** — Urgent leukapheresis or exchange transfusion may be considered in symptomatic hyperleukocytosis with leukostasis or exchange-transfusion-responsive situations.
- **Dialysis / Renal replacement therapy** — Consider if tumor lysis syndrome causes refractory hyperkalemia, severe hyperphosphatemia, symptomatic hypocalcemia, fluid overload, or acute kidney injury.

Investigations

- Complete Blood Count with Differential and Platelets — Diagnosis of hyperleukocytosis and baseline monitoring [STAT]
- Peripheral Smear / Bone Marrow / Flow Cytometry as indicated — Confirm leukemia subtype and guide cytoreduction [URGENT]
- Serum Uric Acid — Tumor lysis syndrome diagnosis and monitoring [STAT]
- Serum Potassium, Phosphorus, Calcium, Creatinine, Urea — Cairo-Bishop laboratory TLS assessment and AKI monitoring [STAT]
- ECG — Arrhythmia, hyperkalemia, electrolyte disturbance [URGENT]
- Chest X-ray / Neuroimaging as clinically indicated — Respiratory distress, leukostasis, neurological symptoms [URGENT]

Counselling

- This is an oncologic emergency and needs urgent hospital treatment. — यह कैंसर से जुड़ी आपातकालीन स्थिति है और तुरंत अस्पताल में इलाज जरूरी है।
- Very high white cell count can block blood flow and affect lungs, brain, kidneys, and heart. — बहुत ज्यादा सफेद रक्त कोशिकाएं खून के बहाव को रोक सकती हैं और फेफड़ों, दिमाग, किडनी और दिल को नुकसान पहुंचा सकती हैं।
- Strong hydration and repeated blood tests are required to prevent tumor lysis complications. — ट्यूमर लाइसिस की जटिलताओं से बचने के लिए पर्याप्त पानी और बार-बार खून की जांच जरूरी है।
- Breathlessness, drowsiness, seizures, reduced urine output, or chest symptoms must be reported immediately. — सांस फूलना, सुस्ती, दौरे, पेशाब कम होना या सीने में तकलीफ होने पर तुरंत बताएं।
- Treatment may include cytoreduction, ICU care, and specialist oncology management. — इलाज में कोशिका कम करने की प्रक्रिया, ICU देखभाल और कैंसर विशेषज्ञ का इलाज शामिल हो सकता है।

Warning Signs

- Tachypnea or dyspnea
- Headache, confusion, or altered sensorium
- Seizures

- Oliguria or anuria
- Arrhythmia
- Sudden weakness, visual symptoms, or coma

Referral Criteria

Immediate referral to pediatric oncologist / intensivist for any hyperleukocytosis with symptoms of leukostasis, TLS, neurological signs, respiratory distress, arrhythmia, or renal dysfunction.

Hospitalisation Criteria

All patients with hyperleukocytosis, suspected leukemia with TLC >100000/uL, clinical leukostasis, laboratory TLS, or need for IV hydration / cytoreduction / ICU monitoring require admission.

Notes

Hyperleukocytosis is usually defined as WBC/TLC >100000/uL. Leukostasis symptoms may involve pulmonary, CNS, genitourinary, gastrointestinal, and cardiovascular systems. Cairo-Bishop laboratory TLS is present if 2 or more occur within 3 days before or 7 days after therapy: uric acid >=8 mg/dL or 25% rise, potassium >=6 mEq/L or 25% rise, phosphorus >=6.5 mg/dL or 25% rise, calcium <=7 mg/dL or 25% fall. Clinical TLS includes creatinine >=1.5 times upper limit normal, cardiac arrhythmia / sudden death, or seizures.

E10.10 — Diabetic Ketoacidosis (DKA) *[severe]*

SNOMED: 420422005 | Duration: 2 days | Source: ISPAD Guidelines 2022/2024, IAP Pediatric Endocrine Guidelines

First-Line

- **Normal Saline (0.9%)** — Initial 4 hours IV fluids; bolus 10–20 ml/kg over 30–60 min only if shock; then continue deficit + maintenance over 36–48 hrs
- **0.45% Saline** — Switch after initial 4 hours if available; continue rehydration over 36–48 hrs
- **Regular Insulin (IV infusion)** — Start after 1 hour of fluids at 0.05–0.1 U/kg/hr; no bolus; prepare 0.5–1 U/ml solution and infuse via pump
- **Potassium Chloride / Potassium Phosphate** — Add 40 mEq/L once urine output established and K+ normal; if hypokalemia use up to 80 mEq/L with ECG monitoring
- **Dextrose (D5 or D10)** — Add when blood glucose <250 mg/dL; increase to D10 if glucose <150 mg/dL while acidosis persists

Second-Line

- **Sodium Bicarbonate** — Avoid routine use; consider only if severe acidosis (pH <6.9) with hemodynamic compromise
- **Mannitol / Hypertonic Saline** — For suspected cerebral edema: Mannitol 0.5–1 g/kg or 3% saline 5 ml/kg IV

Investigations

- Capillary Blood Glucose — Diagnosis and monitoring [STAT]
- Venous Blood Gas (pH, HCO3) — Assess acidosis [URGENT]
- Serum Electrolytes (Na, K, Cl) — Baseline and monitoring [URGENT]
- Serum Ketones — Confirm ketosis [URGENT]
- Renal Function Tests — Assess dehydration and renal status

Counselling

- Child is critically ill and requires ICU care — बच्चा गंभीर रूप से बीमार है और ICU में इलाज जरूरी है
- Insulin and fluids will correct the condition gradually — इंसुलिन और तरल पदार्थ से हालत धीरे-धीरे सुधरेगी
- Frequent monitoring is required to prevent complications — जटिलताओं से बचने के लिए बार-बार जांच जरूरी है
- Do not stop treatment early even if sugar improves — शुगर ठीक होने पर भी इलाज जल्दी बंद न करें
- Long-term diabetes management will be needed — लंबे समय तक मधुमेह (डायबिटीज) का इलाज चलेगा

Warning Signs

- Altered consciousness
- Seizures
- Severe dehydration
- Deep rapid breathing (Kussmaul breathing)
- Headache or vomiting (cerebral edema)
- Sudden drop in heart rate or BP

Referral Criteria

Refer to PICU/endocrinologist if severe DKA, altered sensorium, or complications

Hospitalisation Criteria

All DKA cases require admission; ICU if moderate–severe DKA

Notes

Correct dehydration slowly over 36–48 hrs. Avoid rapid fluid correction to prevent cerebral edema. Do not give insulin bolus.

Guideline Changes

Lower insulin dose (0.05 U/kg/hr) preferred in young children to reduce cerebral edema risk

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E16.2 — Hypoglycemia (Pediatric) *[severe]*

SNOMED: 302866003 | Duration: 3 days | Source: IAP Guidelines 2024, WHO Emergency Care, Pediatric Endocrine Protocols

First-Line

- **Dextrose 10% (D10)** — Neonates: 2 ml/kg IV bolus (200 mg/kg), then start infusion
- **Dextrose (D25 or D50)** — Children: 2–4 ml/kg IV bolus (max D25 preferred), repeat if needed after 5–10 min
- **Dextrose infusion** — Start GIR 4–6 mg/kg/min, titrate up to 8–12 mg/kg/min (up to 20 mg/kg/min in hyperinsulinism)

Second-Line

- **Glucagon** — 0.03–0.1 mg/kg IM/IV (max 1 mg) in refractory hypoglycemia or no IV access
- **Hydrocortisone** — 2–5 mg/kg IV if adrenal insufficiency suspected

Investigations

- Capillary Blood Glucose — All suspected cases [STAT]
- Serum Glucose (venous) — Confirmatory test [STAT]
- Blood Gas and Electrolytes — Metabolic assessment [URGENT]
- Critical Sample Panel — Recurrent/severe hypoglycemia [URGENT]
- Insulin, C-peptide, Cortisol, GH — Etiology evaluation [URGENT]

Counselling

- Child needs urgent glucose correction and monitoring — बच्चे को तुरंत शुगर (ग्लूकोज़) ठीक करना और निगरानी ज़रूरी है।
- Ensure frequent feeding after stabilization — स्थिति ठीक होने के बाद बार-बार थोड़ा-थोड़ा दूध/खाना दें।
- Avoid prolonged fasting — बच्चे को लंबे समय तक भूखा न रखें।
- Give medicines as advised strictly — दवाइयाँ समय पर और बताई गई मात्रा में दें।
- Follow-up for cause evaluation is important — कारण की जाँच के लिए फॉलो-अप ज़रूरी है।

Warning Signs

- Seizures
- Lethargy or unconsciousness
- Poor feeding

- Apnea or irregular breathing
- Recurrent hypoglycemia
- Abnormal movements

Referral Criteria

Refer to pediatric intensivist/endocrinologist if persistent or recurrent hypoglycemia

Hospitalisation Criteria

Symptomatic hypoglycemia, seizures, need for IV glucose infusion, recurrent episodes

Notes

Always treat hypoglycemia immediately before confirming cause. Avoid delays.

Guideline Changes

Higher GIR targets recommended in hyperinsulinemic states

E83.51 — Hypocalcemia

SNOMED: 5291005 | Duration: 3 days | Source: Provided pediatric hypocalcemia correction protocol

First-Line

- **Calcium Gluconate 10% (IV)** — For symptomatic hypocalcemia: 1 mL/kg/dose IV (provides about 9-18 mg/kg elemental calcium), maximum 10 mL, dilute with equal volume of D5W or distilled water and infuse slowly over 10 minutes under ECG monitoring
- **Calcium Gluconate 10% (IV infusion)** — For recurrent symptomatic hypocalcemia: continuous infusion starting at 10-30 mg/kg/hour elemental calcium equivalent, titrate based on serial calcium levels and symptoms; ECG monitoring mandatory
- **Oral Calcium** — For asymptomatic hypocalcemia or after stabilization: 100-200 mg/kg/day in 2-3 divided doses
- **Vitamin D** — For asymptomatic hypocalcemia/supportive therapy: 400-800 IU/day orally

Second-Line

- **Magnesium Sulphate** — If hypomagnesemia is present or hypocalcemia is refractory: 25-50 mg/kg slow IV infusion; approximately 0.05-0.1 mL/kg of 50% solution; may repeat every 6 hours for 2-3 doses
- **Calcitriol** — Use in hypoparathyroidism: 20-60 ng/kg/day orally

Investigations

- Serum Total Calcium — Diagnosis and serial monitoring of hypocalcemia [STAT]
- Ionized Calcium — Best indicator in symptomatic child or critical illness [STAT]
- Serum Albumin — To calculate corrected calcium level [URGENT]
- ECG — Symptomatic hypocalcemia, arrhythmia, prolonged QT, or IV calcium therapy [URGENT]
- Serum Magnesium — Refractory hypocalcemia or suspected combined deficiency [URGENT]
- Serum Phosphate, Alkaline Phosphatase, Vitamin D, PTH — Etiology workup for persistent or recurrent hypocalcemia

Counselling

- Low calcium can cause twitching, spasms, seizures, or heart rhythm changes. — कम कैल्शियम से मांसपेशियों में ऐंठन, झटके, दौरे या दिल की धड़कन बिगड़ सकती है।
- Symptomatic low calcium needs urgent treatment and monitoring. — लक्षण वाले कम कैल्शियम का तुरंत इलाज और निगरानी ज़रूरी है।
- After stabilization, oral calcium and vitamin D may be needed for several days or longer. — स्थिति ठीक होने के बाद भी कुछ दिनों तक मुँह से कैल्शियम और विटामिन डी लेना पड़ सकता है।
- Regular blood tests are important to ensure calcium becomes normal and stays normal. — नियमित ब्लड टेस्ट कराएं ताकि कैल्शियम सामान्य बना रहे।

• Seek urgent care if the child develops twitching, stridor, seizures, or poor feeding. — अगर बच्चे को झटके, साँस में आवाज़, दौरे या दूध न पीना हो तो तुरंत डॉक्टर को दिखाएं।

Warning Signs

- Tetany or muscle twitching
- Carpopedal spasm
- Laryngospasm or noisy breathing
- Seizures
- Arrhythmia or palpitations
- Excessive irritability or altered sensorium

Referral Criteria

Refer to pediatric intensivist/endocrinologist if symptomatic hypocalcemia, recurrent episodes, arrhythmia, seizures, suspected hypoparathyroidism, or refractory hypocalcemia.

Hospitalisation Criteria

Admit if symptomatic hypocalcemia, seizures, laryngospasm, prolonged QT, arrhythmia, ionized calcium significantly low, or need for IV calcium infusion.

Notes

Corrected calcium formula: Corrected calcium = Serum calcium + 0.8 x (4.0 - serum albumin). Symptoms usually occur when ionized calcium falls below about 0.8-0.9 mmol/L. ECG monitoring is required during IV calcium administration.

E83.52 — Hypercalcemia

SNOMED: 66931009 | Duration: 3 days | Source: Provided pediatric hypercalcemia correction protocol

First-Line

- **Normal Saline (0.9%) IV** — Moderate to severe hypercalcemia: start hydration to correct volume depletion; for severe hypercalcemia give 10-20 mL/kg IV bolus over 1 hour, then continue IV saline as needed (often 2-3 times maintenance initially depending on hydration and cardiac status)
- **Furosemide** — 1-2 mg/kg IV after adequate hydration to enhance calciuresis; avoid before volume status is corrected
- **Pamidronate** — 0.5-2 mg/kg diluted in 30 mL normal saline, IV over 4 hours; use in severe or persistent hypercalcemia
- **Calcitonin** — 2-4 units/kg subcutaneous every 12 hours for severe symptomatic hypercalcemia requiring faster calcium reduction

Second-Line

- **Etidronate** — 5 mg/kg orally twice daily as a bisphosphonate option where appropriate
- **Hydrocortisone** — 2 mg/kg IV every 6 hours when indicated, especially where reduced intestinal calcium absorption is desired or steroid-responsive cause suspected

Investigations

- Serum Total Calcium — Diagnosis and serial monitoring of severity [STAT]
- Ionized Calcium — Confirm clinically significant hypercalcemia [URGENT]
- Renal Function Tests — Assess dehydration, renal injury, and treatment safety [URGENT]
- Serum Electrolytes (Na, K, Mg, Phosphate) — Associated electrolyte abnormalities and monitoring during therapy [URGENT]
- ECG — Symptomatic hypercalcemia, arrhythmia, or severe hypercalcemia [URGENT]
- PTH, Vitamin D, Urinary Calcium — Etiology workup in persistent or recurrent hypercalcemia

Counselling

• High calcium can cause vomiting, constipation, excessive urine, irritability, and weakness. — ज्यादा कैल्शियम से उल्टी, कब्ज, ज्यादा पेशाब, चिड़चिड़ापन और कमजोरी हो सकती है।

• Mild cases may only need fluids and removal of triggering medicines or excess calcium intake. — हल्के मामलों में सिर्फ पानी पिलाना और कैल्शियम बढ़ाने वाली दवाएं बंद करना काफी हो सकता है।

• Severe cases need admission, IV fluids, and close monitoring. — गंभीर मामलों में भर्ती, IV तरल पदार्थ और करीबी निगरानी जरूरी है।

• Avoid thiazide diuretics, lithium, and unnecessary calcium supplements unless specifically advised. — डॉक्टर की सलाह के बिना थायज़ाइड, लिथियम या कैल्शियम सप्लीमेंट न दें।

• Follow-up blood tests are important because calcium may rise again depending on the cause. — फॉलो-अप खून की जांच जरूरी है क्योंकि कारण के अनुसार कैल्शियम फिर बढ़ सकता है।

Warning Signs

- Persistent vomiting
- Dehydration or poor feeding
- Marked irritability or lethargy
- Polyuria / reduced urine output after dehydration
- Constipation with abdominal distension
- Arrhythmia or altered sensorium

Referral Criteria

Refer to pediatric endocrinologist/nephrologist/intensivist if calcium is >14 mg/dL, symptoms are significant, renal dysfunction is present, medical therapy fails, or recurrent/unclear etiology is suspected.

Hospitalisation Criteria

Admit if total serum calcium >14 mg/dL, symptomatic moderate hypercalcemia, dehydration, poor oral intake, arrhythmia, altered sensorium, or need for IV therapy/serial monitoring.

Notes

Mild hypercalcemia (10-12 mg/dL) usually needs no specific drug therapy beyond hydration and avoiding offending agents. Moderate hypercalcemia (12-14 mg/dL) may not need immediate drug treatment unless symptomatic. Severe hypercalcemia (>14 mg/dL) requires urgent stepwise treatment.

E87.0 — Hypernatremia

SNOMED: 771115008 | Duration: 2 days | Source: Pediatric hypernatremia bedside correction protocol

First-Line

• **Dextrose with Saline IV Fluids** — Correct slowly over 24-84 hours depending on initial serum sodium. Use calculated fluid plan: Total fluid deficit + free water deficit + maintenance sodium. Avoid rapid correction.

• **0.9% Normal Saline** — Use first if child is in shock to restore intravascular volume before hypernatremia correction. Bolus as per shock protocol.

• **Half Normal Saline based dextrose fluid (e.g., N/2, N/5, N/6 with dextrose as per calculated sodium requirement)** — Choose fluid strength according to calculated sodium requirement and day-wise correction plan; usual target sodium fall not more than 10-12 mEq/L in 24 hours or about 0.5 mEq/L/hr.

Second-Line

• **3% Hypertonic Saline** — Only if seizures occur during correction due to overly rapid sodium fall or cerebral edema from hypotonic correction; use to acutely raise serum sodium under ICU monitoring.

• **Potassium supplementation** — Add only after urine output is established and potassium status is known, if clinically indicated.

Investigations

• Serum Sodium — Diagnosis and serial monitoring during correction [STAT]

- Serum Potassium, Chloride, Bicarbonate — Baseline electrolyte assessment and monitoring during IV correction [URGENT]
- Blood Glucose — Baseline metabolic assessment and before selecting dextrose-containing fluids [URGENT]
- Renal Function Tests — Assess dehydration severity, renal perfusion, and ongoing monitoring [URGENT]
- Urine output monitoring — Guide fluid therapy and assess renal perfusion [STAT]
- Weight — Needed for calculating free water deficit and maintenance fluids [STAT]

Counselling

- High sodium is usually due to water loss more than salt loss and needs slow correction. — ज्यादा सोडियम आमतौर पर शरीर में पानी की कमी से होता है और धीरे-धीरे ठीक करना जरूरी है।
- Do not give plain water or change IV fluids abruptly without medical advice. — डॉक्टर की सलाह के बिना सादा पानी न दें या IV तरल पदार्थ अचानक न बदलें।
- Treatment is usually gradual over 1 to 3 days depending on sodium level. — इलाज आमतौर पर 1 से 3 दिन में धीरे-धीरे किया जाता है।
- Strict monitoring of urine output, sensorium, and repeat sodium tests is essential. — पेशाब, होश और सोडियम जांच की सख्त निगरानी जरूरी है।
- Come back immediately or alert staff if the child becomes drowsy, irritable, or has seizures. — अगर बच्चा सुस्त, चिड़चिड़ा हो जाए या दौरे पड़ें तो तुरंत स्टाफ को बताएं।

Warning Signs

- Seizures
- Altered sensorium or unusual drowsiness
- Marked irritability or lethargy
- Poor perfusion / shock
- Persistent vomiting
- Reduced urine output or inability to drink

Referral Criteria

Refer to pediatric intensivist/nephrologist if serum sodium is markedly elevated, correction is complex, there is shock, neurological symptoms, persistent sodium rise, or associated renal dysfunction.

Hospitalisation Criteria

Admit if serum sodium >145 mEq/L with dehydration needing IV fluids, neurological symptoms, shock, inability to tolerate oral fluids, or need for serial biochemical monitoring.

Notes

Hypernatremia correction must be slow. If child is in shock, restore circulation first with isotonic fluid. Acute hypernatremia can be corrected over 24-36 hours; chronic hypernatremia usually over 48 hours or longer. Rapid reduction can cause cerebral edema.

E87.1 — Hyponatremia

SNOMED: 89627008 | Duration: 2 days | Source: IAP Guidelines 2024, Pediatric Electrolyte Management Protocols

First-Line

- **3% Hypertonic Saline** — Symptomatic (Na <125 with seizures/encephalopathy): 4–6 ml/kg IV over 30 min; may repeat 2 ml/kg over 15 min if seizures persist; target rise 5–10 mEq/L
- **3% Hypertonic Saline** — Asymptomatic (Na <125): correct slowly using calculated Na deficit over 4–24 hrs
- **Fluid Restriction** — 50–75% of maintenance fluids in SIADH-related hyponatremia

Second-Line

- **Normal Saline (0.9%)** — Used in hypovolemic hyponatremia to restore volume
- **Furosemide** — 1 mg/kg IV in selected cases (SIADH with fluid overload)

Investigations

- Serum Sodium — Diagnosis and monitoring [STAT]
- Serum Osmolality — Differentiate type of hyponatremia [URGENT]
- Urine Sodium and Osmolality — Assess SIADH vs other causes [URGENT]
- Blood Glucose — Rule out pseudohyponatremia [STAT]

Counselling

- Low sodium levels can affect brain function — सोडियम कम होने से दिमाग की कार्यप्रणाली प्रभावित हो सकती है।
- Treatment will be done slowly to avoid complications — इलाज धीरे-धीरे किया जाएगा ताकि कोई दिक्कत न हो।
- Fluid intake may be restricted as advised — डॉक्टर की सलाह अनुसार पानी/तरल पदार्थ सीमित रखें।
- Regular blood tests are required for monitoring — निगरानी के लिए नियमित ब्लड टेस्ट ज़रूरी हैं।
- Follow-up is important to identify underlying cause — असली कारण पता लगाने के लिए फॉलो-अप ज़रूरी है।

Warning Signs

- Seizures
- Altered consciousness
- Vomiting
- Headache
- Apnea
- Worsening lethargy

Referral Criteria

Refer to pediatric intensivist if symptomatic hyponatremia or rapid sodium fall

Hospitalisation Criteria

Na <125 mEq/L, neurological symptoms, seizures, or need for IV correction

Notes

Avoid rapid correction. Increase Na should not exceed 10–12 mEq/L/day or 0.5 mEq/L/hr to prevent osmotic demyelination.

Guideline Changes

Emphasis on controlled correction rate and early use of 3% saline in symptomatic cases

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E87.2 — Metabolic Acidosis / Indications for Sodium Bicarbonate Therapy

SNOMED: 59455009 | Duration: 1 days | Source: Provided sodium bicarbonate indication and administration protocol

First-Line

- **Sodium Bicarbonate** — Indicated in selected severe metabolic acidosis: pH <7.1 and/or bicarbonate <8-10 mEq/L and/or base excess >10 mmol/L. Dose based on HCO3 deficit: HCO3 needed (mEq) = 0.3 x weight (kg) x base deficit (or 0.3 x weight x (24 - serum HCO3)). Give half of calculated dose over 60 minutes, then reassess blood gases.
- **Sodium Bicarbonate Infusion (0.5 mEq/mL solution)** — Prepare by mixing sodium bicarbonate solution with equal volume of sterile water or compatible solution. Administer as slow IV infusion over 30-60 minutes. In emergency, may be given as bolus 1 mEq/kg.

Second-Line

- **Definitive therapy for underlying cause** — Examples: shock correction, sepsis treatment, insulin for DKA, antidote/toxin management, ventilation support, renal support depending on etiology.

Investigations

- Arterial / Venous Blood Gas — Confirm severe metabolic acidosis and monitor response to bicarbonate [STAT]

- Serum Electrolytes including HCO3, Na, K, Cl — Assess severity, base deficit, associated electrolyte abnormalities [URGENT]
- Serum Calcium — Monitor during bicarbonate therapy [URGENT]
- Renal Function Tests — Assess renal cause or consequence of acidosis [URGENT]
- Lactate / Anion Gap / Cause-specific tests — Evaluate high anion gap metabolic acidosis and underlying etiology [URGENT]

Counselling

- Sodium bicarbonate is not used routinely in all acidosis cases. — सोडियम बाइकार्बोनेट हर एसिडोसिस में नहीं दिया जाता।
- It is reserved for selected severe situations after doctor assessment. — यह केवल गंभीर स्थितियों में डॉक्टर की जांच के बाद दिया जाता है।
- Treatment of the underlying cause is more important than bicarbonate alone. — मूल कारण का इलाज बाइकार्बोनेट से ज्यादा जरूरी है।
- Frequent blood gas and electrolyte monitoring is needed during therapy. — इलाज के दौरान बार-बार ब्लड गैस और इलेक्ट्रोलाइट जांच जरूरी है।
- Inform staff immediately if breathing worsens, irritability increases, or seizures occur. — सांस बिगड़ने, चिड़चिड़ापन बढ़ने या दौरे पड़ने पर तुरंत स्टाफ को बताएं।

Warning Signs

- Severe fast or labored breathing
- Shock or poor perfusion
- Altered sensorium
- Cardiac dysfunction
- Seizures
- Worsening acidosis on repeat blood gas

Referral Criteria

Refer to PICU/intensivist if severe metabolic acidosis, high anion gap acidosis with hemodynamic compromise, cardiopulmonary resuscitation setting, toxin-related acidosis, or refractory acidosis.

Hospitalisation Criteria

Admit if pH <7.2, bicarbonate markedly low, shock, altered sensorium, need for IV bicarbonate, suspected toxin/metabolic disease, or need for serial blood gas monitoring.

Notes

Routine use in older children is not recommended unless specific indications are present. Common indications include severe normal anion gap acidosis, selected severe high anion gap acidosis, hyperkalemia, TCA poisoning, beta-blocker poisoning, and some peri-resuscitation situations. Ensure adequate ventilation before giving bicarbonate.

Guideline Changes

2024 real-world study shows NaHCO3 benefit only when Cl >=113 mmol/L; harmful when Cl <107

E87.5 — Hyperkalemia

SNOMED: 14140009 | Duration: 1 days | Source: Provided pediatric hyperkalemia correction protocol

First-Line

- **Calcium Gluconate** — 100 mg/kg/dose IV (approximately 1 mL/kg/dose) over 5-10 minutes for ECG changes or severe hyperkalemia; membrane stabilizing; onset 1-3 min; duration about 30 min
- **Sodium Bicarbonate** — 1-2 mEq/kg/dose IV over 5-10 minutes; especially useful if associated acidosis; onset 10-30 min; duration about 2 hours

- **Regular Insulin with Glucose** — Glucose 0.5 g/kg (approximately 2 mL/kg of 25% dextrose) plus regular insulin 0.1 U/kg IV over 15-30 minutes; shifts potassium intracellularly; onset about 30 min; duration 2-4 hours
- **Salbutamol Nebulization** — 2.5 mg if <25 kg, 5 mg if >25 kg, nebulized over 10 minutes; promotes intracellular shift of potassium; onset about 120 min

Second-Line

- **Sodium Polystyrene Sulfonate** — 0.5-1 g orally or rectally; enhances gastrointestinal potassium elimination; onset 60-120 min; duration 4-6 hours
- **Furosemide** — 1-2 mg/kg IV if urine output is adequate; enhances urinary potassium excretion; onset 15-30 min; duration 4-6 hours

Investigations

- 12-lead ECG — All suspected or confirmed hyperkalemia; assess ECG changes and severity [STAT]
- Serum Potassium — Diagnosis and serial monitoring during treatment [STAT]
- Serum Sodium, Bicarbonate, Calcium, Magnesium — Associated electrolyte and acid-base abnormalities [URGENT]
- Renal Function Tests — Assess renal cause and need for dialysis [URGENT]
- Blood Gas — If acidosis is suspected [URGENT]
- Urine Output Monitoring — Guide diuretic use and assess renal perfusion [STAT]

Counselling

- High potassium can affect the heart and may become life-threatening. — ज्यादा पोटैशियम दिल को प्रभावित कर सकता है और जानलेवा हो सकता है।
- Continuous monitoring is needed until potassium becomes safe. — पोटैशियम सामान्य होने तक लगातार निगरानी जरूरी है।
- All potassium-containing oral feeds, IV fluids, and medicines should be reviewed and stopped if advised. — पोटैशियम वाले खाने, तरल पदार्थ और दवाइयाँ बंद करें जैसा डॉक्टर कहें।
- Do not use over-the-counter supplements or salt substitutes unless approved. — डॉक्टर की मंजूरी के बिना कोई सप्लीमेंट या नमक का विकल्प न लें।
- Seek urgent care if weakness, palpitations, or reduced urine output develops. — कमजोरी, धड़कन बढ़ना या पेशाब कम होने पर तुरंत डॉक्टर को दिखाएं।

Warning Signs

- Muscle weakness
- Paresthesia
- Palpitations
- Bradycardia
- Arrhythmia
- Reduced urine output

Referral Criteria

Refer urgently to pediatric intensivist/nephrologist if ECG changes are present, serum potassium is markedly elevated, medical treatment fails, renal failure is suspected, or dialysis may be required.

Hospitalisation Criteria

Admit if serum potassium >5.5 mEq/L with symptoms, ECG changes, renal dysfunction, ongoing potassium load, or need for IV therapy/cardiac monitoring.

Notes

Obtain ECG and place on cardiac monitor immediately. Eliminate potassium from diet and parenteral fluids. Stop drugs that can worsen hyperkalemia, such as ACE inhibitors and beta blockers. Dialysis is recommended if medical measures are unsuccessful.

Guideline Changes

E87.6 — Hypokalemia

SNOMED: 43339004 | Duration: 2 days | Source: Provided pediatric electrolyte correction protocol

First-Line

- **Potassium Chloride (IV)** — For severe hypokalemia (serum K+ <2.5 mEq/L) or hypokalemia with ECG changes: prepare stock solution with 90 mL Normal Saline + 10 mL KCl; infuse at 1.5-2.5 mL/kg/hour to deliver 0.3-0.5 mEq/kg/hour under strict continuous ECG monitoring; continue until serum K+ is at least 3 mEq/L or ECG changes normalize
- **Potassium Chloride (oral)** — For mild to moderate hypokalemia without ECG changes and when oral route is tolerated; dose as per potassium deficit and repeat serum potassium monitoring

Second-Line

- **Magnesium Sulphate** — Use in refractory hypokalemia or suspected magnesium deficiency: 25-50 mg/kg slow IV infusion; approximate volume 0.05-0.1 mL/kg of 50% solution; may repeat every 6 hours for total 2-3 doses

Investigations

- Serum Potassium — Diagnosis and repeat monitoring during correction [STAT]
- ECG — Severe hypokalemia, symptoms, or suspected arrhythmia [URGENT]
- Serum Magnesium — Refractory hypokalemia or recurrent hypokalemia [URGENT]
- Serum Sodium, Chloride, Bicarbonate — Associated electrolyte or acid-base disturbance [URGENT]
- Renal Function Tests — Assess renal handling and safety of potassium replacement
- Urine Potassium / Urine electrolytes — Persistent or unexplained hypokalemia

Counselling

- Low potassium can affect muscle strength and heart rhythm. — पोटैशियम कम होने से मांसपेशियों में कमज़ोरी और दिल की धड़कन बिगड़ सकती है।
- Severe low potassium may need IV correction and heart monitoring. — गंभीर कमी में नस द्वारा दवा और दिल की निगरानी ज़रूरी है।
- Do not stop treatment once symptoms improve; repeat blood tests are important. — लक्षण ठीक होने पर भी इलाज बंद न करें; दोबारा ब्लड टेस्ट करवाएं।
- Vomiting, diarrhea, poor intake, and some medicines can worsen potassium loss. — उल्टी, दस्त, कम खाना और कुछ दवाइयों से पोटैशियम और कम हो सकता है।
- Return urgently if weakness, palpitations, breathing difficulty, or persistent vomiting develops. — कमज़ोरी, धड़कन बढ़ना, साँस में तकलीफ़ या लगातार उल्टी हो तो तुरंत आएं।

Warning Signs

- Muscle weakness
- Respiratory muscle weakness
- Paralytic ileus or abdominal distension
- Palpitations or irregular heartbeat
- Severe vomiting
- Drowsiness or worsening weakness

Referral Criteria

Refer to pediatric intensivist/nephrologist if serum K+ <2.5 mEq/L, ECG changes, arrhythmia, respiratory weakness, refractory hypokalemia, or suspected renal/tubular disorder.

Hospitalisation Criteria

Admit if serum K+ <2.5 mEq/L, ECG changes, symptomatic hypokalemia, inability to tolerate oral replacement, ongoing losses, or need for IV correction.

Notes

Mild hypokalemia is often asymptomatic. Severe hypokalemia can present with muscle weakness, paralytic ileus, respiratory weakness, hypotonia, areflexia, ECG changes, and arrhythmia. Correct slowly and monitor closely.

J45.5 — Acute Severe Asthma (Indoor/IPD) *[severe]*

SNOMED: 304527002 | Duration: 5 days | Source: GINA 2024/2025, IAP Guidelines 2024, AIIMS Protocol, NABH HCO 6th Edition

First-Line

- **Levosalbutamol** — NEB: <10 kg 0.63 mg, >=10 kg 1.25-2.5 mg, every 20 min x3 doses (first hour), then q1-2h, taper to q4-6h
- **Methylprednisolone** — 1-2 mg/kg/dose IV STAT, then q6-8h. Mandatory early steroid. Reduces airway inflammation and prevents progression
- **Oxygen** — Nasal cannula/mask, 2-6 L/min, target SpO2 >=94%
- **Magnesium Sulphate** — 25-50 mg/kg IV in 30-50 ml NS over 20-30 min, for severe cases. Bronchodilator effect, reduces ICU need
- **Ipratropium Bromide** — 250 mcg NEB q6h, add-on bronchodilator
- **Adrenaline (Epinephrine)** — 0.01 mg/kg SC (1:1000), repeat every 15-20 min x2-3 doses. For severe bronchospasm / impending respiratory failure only

Investigations

- Pulse oximetry — Continuous monitoring of oxygenation [STAT]
- ABG — Severe distress or ICU-level care [STAT]
- Chest X-ray — Rule out pneumothorax, pneumonia, or other complications [URGENT]
- Complete blood count — Baseline and to rule out infection [URGENT]
- CRP — If infection suspected [URGENT]

Counselling

- Child needs indoor admission for intensive monitoring and treatment
- Oxygen and nebulization are the main treatment
- Steroids are essential and will be given early to reduce inflammation
- Report any worsening immediately: increased breathing difficulty, drowsiness, or bluish discoloration
- After discharge, inhaler therapy (Salbutamol + Budesonide) will be started for long-term control
- Learn proper inhaler technique before discharge
- Follow up after 3-5 days of discharge
- Avoid triggers: dust, smoke, cold exposure, strong smells

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Drop in oxygen saturation below 90% despite oxygen
- Silent chest on auscultation
- Exhaustion or increasing drowsiness
- No improvement despite aggressive treatment

Referral Criteria

Refer to PICU if SpO2 <90% despite oxygen, silent chest, rising CO2 on ABG, exhaustion/drowsiness, or need for ventilatory support (HFNC/NIV/mechanical ventilation)

Hospitalisation Criteria

Already indoor protocol. ICU transfer if: SpO2 <90% despite oxygen, silent chest, rising CO2 on ABG, exhaustion or drowsiness, HFNC failure

Notes

Golden Hour: Oxygen + aggressive nebulization + systemic steroids. Monitoring: SpO2 continuous, RR/HR/work of breathing hourly, ABG if severe. Fluids 100 ml/kg/day, avoid overload. Antibiotics NOT routine - only if fever + infection suspected. Escalation: HFNC if moderate distress, NIV if worsening, mechanical ventilation as last resort. Step-down: transition to MDI+Spacer (Salbutamol reliever + Budesonide controller) and oral steroids before discharge.

Guideline Changes

GINA 2025: IV isotonic MgSO4 may be considered in children >=2 years with severe exacerbations; GINA 2024: Early systemic steroids within first hour remain critical; Ipratropium bromide add-on for severe episodes; Emphasis on step-down to MDI+Spacer before discharge with written action plan

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N17.9 — Acute Kidney Injury (AKI) with Anuric / Oligoanuric Regimen *[severe]*

SNOMED: 14669001 | Duration: 1 days | Source: Provided pediatric anuric regimen in AKI protocol

First-Line

- **N/2 Saline with 5% Dextrose** — Use as maintenance fluid for insensible water loss in anuric AKI. Daily fluid = 400 mL/m²/day or about 40% of total daily fluid requirement in infants.
- **N/2 Saline with 5% Dextrose** — Replace previous 24-hour urine output mL for mL in addition to insensible loss calculation.
- **Potassium Chloride (KCl)** — Add to replacement fluid only if hypokalemia is present and serum potassium is monitored.

Second-Line

- **Furosemide** — May be considered only in selected fluid-overloaded patients with residual renal responsiveness; not routine in established anuric AKI.
- **Renal Replacement Therapy / Dialysis** — Indicated if fluid restriction fails or if refractory hyperkalemia, severe acidosis, uremic symptoms, or fluid overload develops.

Investigations

- Serum Urea and Creatinine — Diagnosis and daily AKI monitoring [URGENT]
- Serum Electrolytes (Na, K, Cl, HCO3) — Assess hyperkalemia, hyponatremia, acidosis, and fluid prescription safety [STAT]
- Urine Output Charting — Needed for fluid replacement calculation and AKI severity monitoring [STAT]
- Body Weight — Daily fluid balance and fluid overload assessment [URGENT]
- ABG / VBG — If metabolic acidosis or respiratory compromise suspected [URGENT]
- Ultrasound KUB — Evaluate cause of AKI and exclude obstruction

Counselling

- In anuric AKI, fluid intake must be carefully restricted to avoid overload. — पेशाब बंद होने पर पानी और तरल पदार्थ सीमित रखना बहुत जरूरी है।
- Daily fluid is calculated from insensible loss plus measured urine output. — रोज़ाना का पानी शरीर की ज़रूरत और पेशाब की मात्रा से तय होता है।
- Do not add potassium unless blood potassium is low and the doctor advises it. — जब तक डॉक्टर न कहें, पोटैशियम न दें।
- Strict intake-output charting and daily weight are essential. — पानी का हिसाब और रोज़ वजन लेना बहुत जरूरी है।

• Dialysis may be needed if kidney function does not improve or complications appear. — अगर किडनी ठीक नहीं होती तो डायलिसिस की जरूरत पड़ सकती है।

Warning Signs

- No urine or very low urine output
- Breathing difficulty or fast breathing
- Swelling of face or body
- Drowsiness or irritability
- Palpitations or weakness due to hyperkalemia
- Seizures or worsening edema

Referral Criteria

Refer urgently to pediatric nephrologist / intensivist if persistent anuria, rising creatinine, refractory electrolyte imbalance, severe acidosis, hypertension, or fluid overload.

Hospitalisation Criteria

All children with anuric AKI need admission for strict monitoring, IV fluid regulation, and serial biochemical assessment.

Notes

Daily fluid requirement in anuric AKI = insensible water loss + previous 24-hour urine output. Insensible water loss may be estimated as 400 mL/m²/day. BSA formula: ([4 x weight in kg] + 7) / (90 + weight in kg). Add KCl only if hypokalemia is present.

R09.2 — Respiratory arrest

SNOMED: 87317003 | Duration: 7 days | Source: PALS/AHA 2025, AAP Resuscitation Guidelines

First-Line

- **Bag-Mask Ventilation** — Immediate BVM with 100% O2; rate 20-30 breaths/min with advanced airway; ensure chest rise
- **Endotracheal Intubation** — Cuffed ETT preferred; size (age/4)+3.5; RSI with Rocuronium 1 mg/kg + Ketamine 1-2 mg/kg IV
- **Epinephrine** — 0.01 mg/kg IV/IO (1:10,000) q3-5min if progresses to pulseless arrest; max 1 mg/dose
- **Atropine** — 0.02 mg/kg IV (min 0.1 mg, max 0.5 mg) for bradycardia-associated respiratory arrest

Second-Line

- **Supraglottic Airway** — LMA as rescue airway if intubation fails; size per weight-based chart
- **Naloxone** — 0.1 mg/kg IV/IM/IN (max 2 mg) if opioid-induced respiratory arrest; may repeat q2-3min
- **Surfactant** — 4 mL/kg ET for neonates with respiratory failure from surfactant deficiency

Investigations

- Arterial blood gas — Assess oxygenation and ventilation status [STAT]
- Chest X-ray — ETT position, lung pathology [STAT]
- CBC with differential — Infection screening [STAT]
- Basic metabolic panel — Electrolytes, glucose, renal function [STAT]
- Toxicology screen — If ingestion suspected as cause [STAT]

Counselling

- Your child stopped breathing and needed emergency treatment to restart breathing
- The child will need to be monitored in the ICU on a breathing machine for some time
- Please share any information about allergies, medicines, or events before this happened
- The medical team will explain each step of treatment to you
- After recovery, follow-up is essential to prevent recurrence

Referral Criteria

PICU mandatory. Pulmonology for underlying respiratory disease. ENT if upper airway obstruction. Neurology if apnea of central origin. Consider ECMO center if refractory

Hospitalisation Criteria

All respiratory arrests require PICU admission. Continued intubation and ventilation until underlying cause treated and respiratory drive adequate

Notes

Most pediatric cardiac arrests are secondary to respiratory failure. Verify ETT with waveform capnography; aim ETCO2 35-40 mmHg post-intubation.

Guideline Changes

2025 AHA updated pediatric ventilation rate to 20-30/min with advanced airway and recommends cuffed ETT

R40.2 — Coma

SNOMED: 371632003 | Duration: 14 days | Source: Brain Trauma Foundation Peds 3rd Ed, PALS/AHA 2025

First-Line

- **RSI Intubation** — For GCS <9: Rocuronium 1 mg/kg + Ketamine 1-2 mg/kg IV; use cuffed ETT; verify with ETCO2
- **Hypertonic Saline 3%** — 5 mL/kg IV over 10-20 min if signs of herniation; monitor Na q4-6h, target Na <160
- **Dextrose 10%** — 2-5 mL/kg IV if glucose <60 mg/dL; maintain normoglycemia 80-180 mg/dL

Second-Line

- **Mannitol 20%** — 0.5-1 g/kg IV over 15-20 min for acute herniation; hold if serum osm >320 mOsm/L
- **Levetiracetam** — 20-40 mg/kg IV load (max 1500 mg) for seizure prophylaxis or treatment **[NEW 2024-25]**
- **Acyclovir** — 20 mg/kg IV q8h if viral encephalitis suspected; pending LP and PCR
- **Naloxone** — 0.1 mg/kg IV/IM/IN (max 2 mg) if opioid toxicity suspected; may repeat q2-3min

Investigations

- CT head without contrast — Immediate — rule out structural cause [STAT]
- Blood glucose — Treatable cause of coma [STAT]
- Complete metabolic panel — Electrolytes, glucose, renal and liver function [STAT]
- Arterial blood gas — Acid-base status, oxygenation [STAT]
- Toxicology screen — Rule out poisoning [STAT]

Counselling

- Your child is critically ill and needs intensive care unit monitoring
- The medical team will keep you informed about every step of treatment
- The child may need a breathing tube and ventilator support — this is to protect the airway
- Please share any information about medicines, chemicals, or recent illnesses
- Recovery may take time — the team will support you throughout

Referral Criteria

All comatose children: immediate referral to PICU. Neurosurgery if structural lesion on CT. Toxicology consult if poisoning suspected. Metabolic/genetics if inborn error suspected

Hospitalisation Criteria

All comatose children (GCS <=8) require PICU admission with continuous monitoring, airway protection, and ICP management as indicated

Notes

Secure airway immediately for GCS <9. Order CT head, full metabolic panel, ammonia, toxicology screen, and continuous EEG monitoring in PICU.

Guideline Changes

2025 PALS updated to recommend cuffed ETT for all pediatric intubations

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R41.82 — Altered Sensorium with Suspected Raised Intracranial Pressure *[severe]*

SNOMED: 271719001 | Duration: 7 days | Source: IAP Pediatric Emergency Guidelines, WHO 2023, Pediatric Neurocritical Care protocols

First-Line

- **Oxygen** — Maintain SpO2 >95%, via mask/HFNC/ventilator as required
- **Normal Saline (0.9%)** — 10–20 ml/kg IV bolus in shock, then maintenance; avoid hypotonic fluids
- **3% Hypertonic Saline** — 5 ml/kg IV bolus over 10–15 min, then 0.1–1 ml/kg/hr infusion for 48–72 hrs (monitor Na <155 mEq/L)
- **Mannitol 20%** — 1–5 ml/kg IV over 20 min, repeat every 4–6 hr SOS; monitor serum osmolality <320 mOsm
- **Dextrose (10%/25%)** — 2–4 ml/kg IV bolus if hypoglycemia present or suspected
- **Levetiracetam** — 20–40 mg/kg IV loading, then 10–20 mg/kg/day divided doses

Second-Line

- **Ceftriaxone** — 50–100 mg/kg/day IV divided BD (empirical for meningitis/encephalitis)
- **Acyclovir** — 10 mg/kg IV every 8 hr if HSV encephalitis suspected
- **Fosphenytoin** — 20 mg PE/kg IV loading if seizures not controlled

Investigations

- Random Blood Sugar — All cases of altered sensorium [STAT]
- CBC, CRP, Blood Culture — Suspected infection [URGENT]
- Serum Electrolytes, Urea, Creatinine, LFT, Ammonia — Metabolic encephalopathy [URGENT]
- ABG — Respiratory/metabolic imbalance, ventilation monitoring [STAT]
- CT/MRI Brain — Focal deficit, unequal pupils, suspected raised ICP [URGENT]
- Lumbar Puncture — Suspected meningitis after stabilization (GCS >12 or stable) [URGENT]

Counselling

- Child is critically ill and requires ICU monitoring — बच्चा गंभीर रूप से बीमार है और ICU में निगरानी जरूरी है
- Avoid giving anything orally if child is drowsy — अगर बच्चा सुस्त है तो मुंह से कुछ न दें
- Strict adherence to medications is important — दवाइयाँ समय पर और नियमित रूप से देना बहुत जरूरी है
- Keep child in calm, low-stimulation environment — बच्चे को शांत माहौल में रखें
- Immediate reporting if condition worsens — हालत बिगड़ने पर तुरंत डॉक्टर को बताएं

Warning Signs

- Fast breathing / chest indrawing
- Bluish lips or tongue
- Convulsions / fits
- Excessive sleepiness / unresponsive
- Persistent vomiting
- Unequal pupils or abnormal posturing

Referral Criteria

Refer to pediatric intensivist/neurologist if GCS <12, seizures, focal deficits, or suspected raised ICP

Hospitalisation Criteria

GCS <12, seizures, hypoxia, shock, raised ICP signs, inability to maintain airway

Notes

Always treat ABC first. Assume raised ICP in all altered sensorium. Avoid overzealous hyperventilation and hypotonic fluids.

Guideline Changes

Hypertonic saline increasingly preferred over mannitol in pediatric raised ICP

R57.0 — Cardiogenic shock

SNOMED: 89138009 | Duration: 10 days | Source: SSC Peds 2020, ACC Cardiogenic Shock 2025, PICU Network Meta-Analysis 2024

First-Line

- **Epinephrine** — 0.01-0.1 mcg/kg/min IV continuous infusion; first-line inotrope; SUCRA 85% in 2024 meta-analysis
- **Milrinone** — 0.25-0.75 mcg/kg/min IV (load 50 mcg/kg over 15 min optional); reduces afterload + inotropy
- **Furosemide** — 1-2 mg/kg IV q6-12h for pulmonary edema/fluid overload; monitor electrolytes

Second-Line

- **Dobutamine** — 5-20 mcg/kg/min IV continuous infusion; alternative inotrope if epinephrine unavailable
- **Norepinephrine** — 0.05-0.3 mcg/kg/min IV if vasodilatory component present; SUCRA 80% in meta-analysis
- **Prostaglandin E1** — 0.01-0.1 mcg/kg/min IV for duct-dependent cardiac lesions in neonates
- **ECMO** — Consider for refractory cardiogenic shock unresponsive to maximal vasoactive support

Investigations

- Echocardiogram — Assess cardiac function, structural abnormalities [STAT]
- 12-lead ECG — Arrhythmia, ischemia, myocarditis pattern [STAT]
- Troponin — Myocardial injury assessment [STAT]
- BNP or NT-proBNP — Severity of heart failure [STAT]
- Arterial blood gas with lactate — Tissue perfusion assessment [STAT]

Counselling

- Your child's heart is not pumping blood effectively — this is a serious condition needing ICU care
- Medicines will be given through IV to help the heart pump better
- The child will be on continuous heart monitoring — alarms may sound but the team is watching
- Fluid intake will be strictly controlled — please do not give extra fluids without asking
- Follow-up with a heart specialist (cardiologist) is essential after discharge

Referral Criteria

Cardiology mandatory. ECMO center referral if refractory to maximal vasoactive support. Cardiac surgery if structural heart disease. Consider cardiac transplant evaluation for end-stage cardiomyopathy

Hospitalisation Criteria

All cardiogenic shock patients require PICU admission. Signs: hypotension, tachycardia, poor perfusion, hepatomegaly, pulmonary edema, rising lactate

Notes

Avoid fluid boluses in cardiogenic shock unless clear evidence of hypovolemia. Milrinone preferred for post-surgical low cardiac output; dopamine no longer recommended first-line.

Guideline Changes

2024 network meta-analysis ranks epinephrine over dopamine; milrinone preferred for low CO states

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R57.1 — Hypovolemic shock

SNOMED: 39419009 | Duration: 5 days | Source: ATLS 10th Ed, SSC Peds 2020, Dutch Pediatric Society CPG

First-Line

- **Normal Saline/LR** — 10-20 mL/kg IV/IO bolus over 5-20 min; repeat up to 40-60 mL/kg in first hour; reassess after each
- **Packed RBCs** — 10-20 mL/kg IV if hemorrhagic shock and no response to 40-60 mL/kg crystalloid; use O-neg if urgent
- **Tranexamic Acid** — 15-20 mg/kg IV (max 1 g) over 10 min within 3h of trauma; then 2 mg/kg/hr for 8h **[NEW 2024-25]**

Second-Line

- **Massive Transfusion Protocol** — pRBC:FFP:Plt at 1:1:1 ratio for massive hemorrhage; activate when >40 mL/kg blood products predicted
- **Norepinephrine** — 0.05-0.1 mcg/kg/min IV if persistent hypotension after adequate volume; bridge therapy only
- **Calcium Gluconate 10%** — 0.5 mL/kg IV slow push during massive transfusion to correct citrate-induced hypocalcemia

Investigations

- CBC with differential — Hemoglobin level, platelet count [STAT]
- Type and crossmatch — Prepare for possible blood transfusion [STAT]
- Basic metabolic panel — Electrolytes, renal function, glucose [STAT]
- Coagulation profile (PT, aPTT) — Assess for coagulopathy [STAT]
- Serum lactate — Tissue perfusion marker [STAT]

Counselling

- Your child has lost a significant amount of body fluid or blood and needs urgent replacement
- IV fluids and possibly blood transfusion will be given to stabilize the child
- The child will be monitored closely — heart rate, blood pressure, and urine output
- Please tell us about any bleeding, vomiting, diarrhea, or injury before coming to hospital
- Follow-up after discharge is important to ensure full recovery

Referral Criteria

Surgical consultation if hemorrhagic cause. Gastroenterology if GI bleed. Hematology if coagulopathy. Transfer to trauma center if mechanism warrants

Hospitalisation Criteria

All children with hypovolemic shock require admission. Signs: tachycardia, prolonged capillary refill >3 sec, altered mental status, hypotension (late sign), decreased urine output

Notes

Tachycardia and poor perfusion are early signs; hypotension is a late and ominous finding in children. Delay intubation if possible until volume resuscitated to avoid hemodynamic collapse.

R57.2 — Septic Shock *[severe]*

SNOMED: 76571007 | Duration: 7 days | Source: ACCM Pediatric Septic Shock Guidelines

First-Line

- **Oxygen** — High-flow oxygen immediately; maintain SpO2 >94%
- **Normal Saline (0.9%)** — Fluid bolus 20 mL/kg over 5–10 min; repeat up to 40–60 mL/kg with reassessment; stop if hepatomegaly, rales, or fluid overload
- **Broad Spectrum Antibiotics** — Start within 1 hour of recognition of septic shock; choice based on local protocol
- **Epinephrine infusion** — 0.05–0.3 mcg/kg/min IV for cold shock (first-line inotrope)

- **Norepinephrine infusion** — Start at 0.05 mcg/kg/min and titrate upward for warm shock

Second-Line

- **Dopamine** — 5–10 mcg/kg/min if epinephrine not available
- **Dobutamine** — Used in myocardial dysfunction with low cardiac output
- **Hydrocortisone** — Indicated in catecholamine-resistant shock or suspected adrenal insufficiency
- **Vasopressin / Terlipressin** — Add in refractory warm shock not responding to norepinephrine

Investigations

- Blood culture — Before starting antibiotics [STAT]
- CBC, CRP, Procalcitonin — Assess infection severity [URGENT]
- Serum Lactate — Assess tissue perfusion [STAT]
- ABG — Evaluate acidosis and oxygenation [STAT]
- Electrolytes, Renal Function — Organ function monitoring [URGENT]

Counselling

- Septic shock is a life-threatening emergency requiring ICU care — सेप्टिक शॉक एक जानलेवा आपातकालीन स्थिति है जिसमें ICU में इलाज जरूरी है
- Immediate treatment with fluids and antibiotics is essential — तुरंत तरल पदार्थ और एंटीबायोटिक से इलाज बहुत जरूरी है
- Continuous monitoring is required — लगातार निगरानी जरूरी है
- Condition can worsen rapidly if untreated — इलाज न होने पर हालत तेजी से बिगड़ सकती है
- Follow-up and prolonged treatment may be required — फॉलो-अप और लंबे समय तक इलाज की जरूरत हो सकती है

Warning Signs

- Altered consciousness
- Cold extremities or delayed capillary refill
- Low blood pressure
- Rapid breathing
- Reduced urine output
- Seizures

Referral Criteria

Immediate referral to PICU/intensivist

Hospitalisation Criteria

All septic shock cases require ICU admission

Notes

Follow ABC approach. Early fluid resuscitation and antibiotics are lifesaving. Identify cold vs warm shock to guide inotrope selection.

Guideline Changes

Early epinephrine preferred over dopamine in pediatric septic shock

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R65.1 — Systemic inflammatory response syndrome of infectious origin with organ failure

SNOMED: 238149007 | Duration: 10 days | Source: SSC Peds 2020, Phoenix Sepsis Criteria 2024, SCCM PANDEM 2024

First-Line

- **Broad-Spectrum Antibiotics** — Ceftriaxone 50-100 mg/kg IV + Vancomycin 15 mg/kg IV within 1h; adjust per culture results
- **Normal Saline/LR** — 10-20 mL/kg IV bolus; up to 40-60 mL/kg in first hour; monitor for fluid overload signs

- **Norepinephrine** — 0.05-0.3 mcg/kg/min IV for fluid-refractory shock; titrate to MAP appropriate for age

Second-Line

- **Epinephrine** — 0.05-0.3 mcg/kg/min IV for cold shock with low cardiac output phenotype
- **Hydrocortisone** — 2 mg/kg IV bolus (max 100 mg) then divided q6h for catecholamine-resistant shock
- **Meropenem** — 20-40 mg/kg IV q8h (max 2 g/dose) if ESBL or multidrug-resistant organisms suspected
- **IVIG** — 1 g/kg IV over 12h; consider for toxic shock syndrome or streptococcal sepsis

Investigations

- Blood culture (2 sets) — Identify pathogen before antibiotics [STAT]
- CBC with differential — WBC, neutrophil count, platelet trend [STAT]
- Complete metabolic panel — Multi-organ function assessment [STAT]
- Procalcitonin and CRP — Infection biomarkers, guide therapy duration [STAT]
- Arterial blood gas with lactate — Acid-base status and tissue perfusion [STAT]

Counselling

- Your child has a serious whole-body infection response with organ involvement
- Treatment in the ICU with strong antibiotics and supportive care is needed
- Multiple IV medicines and close monitoring will be required
- The medical team will update you regularly on your child's condition
- Follow-up after discharge is essential to ensure complete recovery of all organ functions

Referral Criteria

PICU mandatory. Infectious disease for resistant organisms. Subspecialty referral based on organ failure (nephrology for AKI, pulmonology for ARDS, cardiology for myocardial dysfunction)

Hospitalisation Criteria

All patients with sepsis with organ failure (Phoenix Score >=2) require PICU admission. Any hemodynamic instability, respiratory failure, or end-organ damage

Notes

Phoenix Sepsis Score >=2 now defines pediatric sepsis with organ dysfunction. Prioritize early antibiotics, aggressive but monitored fluid resuscitation, and early vasoactive support.

Guideline Changes

2024 Phoenix criteria replaced SIRS-based definitions; PANDEM guidelines published for PICU care

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R73.9 — Stress Hyperglycemia in Critically Ill Child *[moderate]*

SNOMED: 80394007 | Duration: 3 days | Source: PICU Glycemic Control Guidelines, IAP 2024

First-Line

- **Regular Insulin (IV infusion)** — Start 0.02–0.05 units/kg/hr IV infusion; titrate to maintain RBS 80–180 mg/dL
- **Normal Saline (0.9%)** — Use for dilution of insulin infusion (e.g., 1 unit/ml concentration); maintain IV fluids as per requirement

Second-Line

- **Dextrose infusion (D10/D12.5)** — Add if glucose falls <100 mg/dL or to maintain GIR in metabolic conditions

Investigations

- Capillary Blood Glucose (RBS) — All critically ill patients [STAT]
- Serum Electrolytes — During insulin therapy (risk of hypokalemia) [URGENT]
- Blood Gas (ABG) — To rule out acidosis (DKA vs stress hyperglycemia) [URGENT]
- Serum Ketones — If DKA suspected [URGENT]

Counselling

- High blood sugar is common in critically ill children — गंभीर बीमार बच्चों में शुगर बढ़ना आम है
- Insulin infusion will be used to control glucose safely — इंसुलिन इन्फ्यूजन से शुगर को सुरक्षित तरीके से नियंत्रित किया जाएगा
- Frequent monitoring of sugar levels is required — बार-बार शुगर की जांच जरूरी है
- Do not stop feeding unless advised — जब तक डॉक्टर न कहें, खाना-पीना बंद न करें
- Condition will be closely monitored in ICU — ICU में हालत की करीबी निगरानी की जाएगी

Warning Signs

- Very high glucose (>300–400 mg/dL)
- Dehydration or poor perfusion
- Altered sensorium
- Acidosis (deep breathing)
- Seizures
- Hypoglycemia during treatment

Referral Criteria

Refer to pediatric intensivist/endocrinologist if persistent hyperglycemia or suspected DKA

Hospitalisation Criteria

All critically ill children with hyperglycemia requiring insulin infusion

Notes

Do not reduce glucose infusion rate in metabolic conditions; instead add insulin to prevent catabolism

Guideline Changes

Target glucose range relaxed to 80–180 mg/dL to avoid hypoglycemia risk

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S06 — Intracranial injury

SNOMED: 127295002 | Duration: 14 days | Source: Brain Trauma Foundation Peds TBI 3rd Ed, CHOP TBI Pathway 2024

First-Line

- **Hypertonic Saline 3%** — 2-5 mL/kg IV over 10-20 min for raised ICP; may infuse 0.1-1 mL/kg/hr continuous
- **Levetiracetam** — 20-40 mg/kg IV load (max 1500 mg) for seizure prophylaxis in severe TBI (GCS <8) **[NEW 2024-25]**
- **Fentanyl** — 1-2 mcg/kg IV for sedation/analgesia; continuous infusion 1-3 mcg/kg/hr in ventilated patients
- **RSI Intubation** — For GCS <9: Rocuronium 1 mg/kg + Ketamine 1-2 mg/kg IV; avoid hypoxia and hypotension

Second-Line

- **Mannitol 20%** — 0.5-1 g/kg IV over 15-20 min for acute herniation; hold if serum osm >320
- **Pentobarbital** — 5-15 mg/kg IV load then 1-3 mg/kg/hr infusion for refractory ICP; requires EEG monitoring
- **Phenytoin** — 20 mg/kg IV load (max rate 1 mg/kg/min) if levetiracetam unavailable; monitor ECG

Investigations

- CT head without contrast — All moderate-severe TBI (GCS <13); apply PECARN rules for mild TBI [STAT]
- CBC with differential — Baseline, monitor for coagulopathy [STAT]
- Coagulation profile (PT, aPTT, INR) — Assess coagulopathy risk [STAT]
- Basic metabolic panel — Electrolytes, glucose monitoring [STAT]
- Serum osmolality — Guide hyperosmolar therapy (HTS/mannitol) [URGENT]

Counselling

- Head injuries in children need close monitoring — watch for excessive sleepiness, repeated vomiting, or seizures
- Keep the child's head slightly elevated while resting

- Do not give any sedating medicines without doctor's advice
- Return immediately if child becomes confused, has a seizure, or clear fluid leaks from nose/ear
- Follow-up with neurosurgery is mandatory — do not miss appointments

Referral Criteria

GCS <13 or deteriorating GCS; focal neurological deficits; penetrating injury; skull fracture with depression; seizures; CSF leak; need for ICP monitoring or neurosurgical intervention

Hospitalisation Criteria

GCS <15 persisting >2 hours; any abnormality on CT head; post-traumatic seizure; coagulopathy; vomiting >3 times; age <1 year with any mechanism of concern

Notes

Maintain CPP 40-50 mmHg; avoid hypotension, hypoxia, and hyperthermia. Place ICP monitor for GCS <8 after resuscitation; target PaCO2 35-39 mmHg.

Guideline Changes

2024 WVU guideline updated levetiracetam as preferred seizure prophylaxis over phenytoin

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T07 — Multiple injuries, unspecified

SNOMED: 34011000087104 | Duration: 14 days | Source: ATLS 10th Ed, CHOP Trauma Resuscitation Pathway 2024

First-Line

- **Normal Saline/LR** — 20 mL/kg isotonic crystalloid IV bolus; repeat up to 3 boluses then consider blood products
- **Packed RBCs** — 10-20 mL/kg IV if hemorrhagic shock unresponsive to 40-60 mL/kg crystalloid; use O-neg if uncrossmatched
- **Tranexamic Acid** — 15-20 mg/kg IV (max 1 g) over 10 min within 3h of injury; then 2 mg/kg/hr infusion for 8h **[NEW 2024-25]**
- **Fentanyl** — 1-2 mcg/kg IV q30-60min PRN or intranasal for trauma pain; avoid sedation in unprotected airway

Second-Line

- **Ketamine** — 1-2 mg/kg IV for RSI or procedural sedation; maintains hemodynamic stability
- **Massive Transfusion Protocol** — pRBC:FFP:Platelets in 1:1:1 ratio for life-threatening hemorrhage; activate early
- **Norepinephrine** — 0.05-0.1 mcg/kg/min IV if vasopressor needed after adequate volume resuscitation

Investigations

- Trauma CT (head, chest, abdomen-pelvis) — Identify internal injuries in significant mechanism [STAT]
- CBC with differential — Baseline hemoglobin, platelet count [STAT]
- Type and crossmatch — Prepare for potential blood transfusion [STAT]
- Basic metabolic panel — Electrolytes, renal function, glucose [STAT]
- Coagulation profile — Assess for traumatic coagulopathy [STAT]

Counselling

- Your child has multiple injuries and needs careful monitoring in the hospital
- Report immediately if the child becomes drowsy, has difficulty breathing, or has increasing pain
- Follow all follow-up appointments with the trauma team
- Rehabilitation and physiotherapy may be needed after discharge
- Keep the child calm and comfortable — emotional support is important during recovery

Referral Criteria

Trauma center referral for: hemodynamic instability despite resuscitation, suspected internal organ injury, traumatic brain injury, spinal injury, vascular injury, or need for surgical intervention

Hospitalisation Criteria

All polytrauma patients requiring resuscitation; hemodynamic instability; altered mental status; suspected internal bleeding; any injury requiring surgical intervention

Notes

Follow ATLS primary survey (ABCDE) adapted for pediatric anatomy. Hypotension is a late sign in children; tachycardia and poor perfusion are early indicators of shock.

T14.1 — Open wound of unspecified body region

SNOMED: 283682007 | Duration: 7 days | Source: CHOP Laceration Pathway 2024, Texas Children's Wound Care 2024

First-Line

- **LET Topical** — Lidocaine 4%-Epinephrine 0.1%-Tetracaine 0.5% gel; apply to wound edges 20-30 min before repair
- **Lidocaine 1%** — Local infiltration 1-2 mg/kg (max 4.5 mg/kg); buffer with NaHCO3 (1:10 ratio) to reduce pain
- **Wound Irrigation** — Copious NS irrigation under pressure (18G syringe); minimum 200 mL for contaminated wounds
- **Tetanus Prophylaxis** — TdaP 0.5 mL IM if not current; add TIG 250 units IM if dirty wound and incomplete series

Second-Line

- **Tissue Adhesive** — 2-octyl cyanoacrylate (Dermabond) for clean linear lacerations <5 cm; not for high-tension areas
- **Cephalexin** — 25-50 mg/kg/day PO divided q6-8h for 5-7 days if high infection risk (bite, crush, contaminated)
- **Amoxicillin-Clavulanate** — 25-45 mg/kg/day PO divided q12h for bite wounds; 5-7 day course

Investigations

- X-ray of affected area — Rule out fracture or retained foreign body [URGENT]
- Wound culture — If signs of infection present, bite wounds
- CBC — If significant blood loss or infection suspected [URGENT]

Counselling

- Keep the wound clean and dry for 48 hours, then gently clean with soap and water daily
- Watch for signs of infection: increasing redness, swelling, pus, or fever — come back immediately
- Complete the full course of antibiotics if prescribed
- If stitches were placed, come back on the date given for removal or checkup
- Ensure your child's tetanus vaccination is up to date

Referral Criteria

Wounds involving tendons, nerves, or vessels; wounds requiring specialist repair (facial lacerations near eyes/lips, hand injuries); bite wounds with deep tissue involvement; wounds with foreign body not removable in ED

Hospitalisation Criteria

Wound with active hemorrhage not controlled by pressure; associated fracture or vascular compromise; extensive wound requiring operative repair under general anesthesia; wound infection with systemic signs (fever, cellulitis)

Notes

Use absorbable sutures (chromic gut/poliglecaprone) in children to avoid traumatic removal. Assess tetanus status, neurovascular integrity, and tendon function before closure.

Guideline Changes

2024 TCH guideline updated to prefer absorbable deep sutures (poliglecaprone 5-0/6-0)

T30 — Burn and corrosion, unspecified body region

SNOMED: 48333001 | Duration: 14 days | Source: ABA 2024, PICU Burn Guidelines 2024

First-Line

- **Lactated Ringer** — 3 mL/kg/%TBSA over 24h; half in first 8h, rest over 16h; add D5 maintenance for <30 kg
- **Morphine** — 0.1 mg/kg IV q2-4h PRN pain; max 0.2 mg/kg/dose; titrate to comfort
- **Silver Sulfadiazine 1%** — Apply topically to burn wound BID after cleansing; avoid on face/near eyes
- **Tetanus Prophylaxis** — TdaP 0.5 mL IM if not current; add TIG 250 units IM if incomplete primary series

Second-Line

- **Ketamine** — 1-2 mg/kg IV for procedural sedation during debridement; 0.5 mg/kg for analgesia
- **Mafenide Acetate** — Apply topically to deep burns BID; preferred for ear cartilage burns
- **Albumin 5%** — 0.5-1 g/kg IV if serum albumin <2 g/dL after 24h; adjunct to crystalloid

Investigations

- Complete blood count — Baseline and monitor for burn-related anaemia [URGENT]
- Serum electrolytes and renal function — Guide fluid resuscitation [STAT]
- Blood gas — If inhalation injury suspected; monitor carboxyhaemoglobin [STAT]
- Urinalysis (urine output) — Monitor resuscitation adequacy; target 0.5-1 mL/kg/hr [URGENT]
- Wound swab culture — If signs of burn wound infection

Counselling

- Cool the burn immediately with running cool water for 20 minutes - NOT ice or butter
- Do not break blisters; cover with clean cling film and go to hospital
- Pain management is very important - give paracetamol/ibuprofen regularly
- Keep burn dressings clean; follow wound care instructions from the burn team
- Watch for signs of infection: increasing redness, swelling, pus, or fever - see doctor immediately

Referral Criteria

Transfer to burn centre if >10% TBSA, full-thickness burn, face/hands/feet/perineum/genitalia/joint burns, electrical or chemical burns, circumferential burns, or inhalation injury suspected

Hospitalisation Criteria

Admit if >10% TBSA (>5% in <2 years), full-thickness burns, face/airway/hands/feet/perineum, electrical or chemical burns, suspected inhalation injury, circumferential burns, or non-accidental injury suspected

Notes

Fluid resuscitation required for burns >10-15% TBSA in children. Monitor UOP target 0.5-1 mL/kg/hr; watch for fluid creep.

Guideline Changes

Updated pediatric Parkland formula to 3 mL/kg/%TBSA with mandatory dextrose maintenance

T31 — Burns classified according to extent of body surface involved

SNOMED: 212049006 | Duration: 21 days | Source: ABA 2024, Pediatric Burn CPG 2024

First-Line

- **Lactated Ringer** — 3 mL/kg/%TBSA for first 24h; half in first 8h from time of burn; D5LR maintenance for <30 kg
- **Morphine** — 0.1 mg/kg IV q2-4h PRN; consider PCA for older children; titrate to pain score
- **Silver Sulfadiazine 1%** — Apply topically BID to partial/full thickness burns; reassess wound daily

Second-Line

- **Fentanyl** — 1-2 mcg/kg IV q1-2h PRN or 1-2 mcg/kg intranasal for procedural pain
- **Escharotomy** — Emergent surgical procedure for circumferential full-thickness burns with compartment syndrome
- **Oxandrolone** — 0.1 mg/kg PO BID for severe burns >20% TBSA to reduce catabolism; max 20 mg/day

Investigations

- Complete blood count — Monitor anaemia, infection [URGENT]
- Serum electrolytes, renal and liver function — Guide ongoing fluid management and detect complications [STAT]
- Albumin and prealbumin — Nutritional status monitoring in major burns
- Wound swab culture — Monitor for burn wound infection; targeted antibiotics
- Urine output monitoring — Key guide for fluid resuscitation adequacy [STAT]

Counselling

- Burns covering a large area are serious and need hospital treatment with IV fluids
- The child will be in hospital for wound care, pain management, and monitoring
- Nutrition is very important for burn healing - high-protein, high-calorie diet is needed
- After healing, use sunscreen on burn scars and massage with moisturiser to reduce scarring
- Compression garments and physiotherapy may be needed to prevent scar contracture

Referral Criteria

Transfer to dedicated burn centre for all significant paediatric burns; refer to plastic surgery for scar management; refer to physiotherapy/OT for rehabilitation

Hospitalisation Criteria

Admit all major burns (>10% TBSA, full-thickness, face/hands/feet/perineum); ICU admission if >20% TBSA, inhalation injury, hemodynamic instability, or need for intubation

Notes

Transfer to burn center for >10% TBSA, full-thickness, face/hands/feet/perineum, or electrical/chemical burns. Use Lund-Browder chart for accurate TBSA in children.

Guideline Changes

2024 Utah PTN guideline emphasizes early enteral nutrition within 6h for >20% TBSA burns

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T39 — Poisoning by nonopioid analgesics, antipyretics and antirheumatics

Duration: 5 days | Source: AASLD Acetaminophen Guidelines 2023, Rumack-Matthew Nomogram

First-Line

- **N-Acetylcysteine (NAC)** — IV: 150 mg/kg over 1h, then 15 mg/kg/hr continuous; PO: 140 mg/kg load, then 70 mg/kg q4h x17 doses
- **Activated Charcoal** — 1 g/kg PO (max 50 g) within 1-4h of acetaminophen ingestion; most effective within 1h
- **IV Fluids (D5NS)** — Maintenance fluids; correct dehydration; monitor hepatic and renal function q6-12h

Second-Line

- **Whole Bowel Irrigation** — PEG 25 mL/kg/hr via NG for sustained-release preparations or massive NSAID ingestion
- **Sodium Bicarbonate** — 1-2 mEq/kg IV for salicylate poisoning; target urine pH 7.5-8.0 for enhanced elimination
- **Hemodialysis** — For severe salicylate poisoning (level >100 mg/dL, renal failure, or refractory acidosis)

Investigations

- Serum acetaminophen level (4h post-ingestion) — Plot on Rumack-Matthew nomogram for treatment decision [STAT]
- Liver function tests (AST, ALT, bilirubin) — Serial q6-12h to monitor hepatotoxicity [STAT]
- Coagulation profile (PT/INR) — Marker of liver synthetic function [STAT]
- Serum salicylate level — If aspirin/NSAID ingestion suspected [STAT]
- Basic metabolic panel with blood gas — Acid-base status, renal function, anion gap [STAT]

Counselling

- Paracetamol (acetaminophen) overdose can silently damage the liver — timely treatment is crucial
- Keep all painkillers locked away and out of children's reach
- The child needs blood tests every 6-12 hours to monitor liver function
- Complete the full course of the antidote (NAC) even if the child seems well
- After discharge, follow up in 1 week for repeat liver function tests

Referral Criteria

Hepatology/liver transplant center if ALT >10,000 or INR >2 or signs of liver failure. Toxicology for complex mixed ingestion. Nephrology if salicylate poisoning with renal failure or severe acidosis requiring hemodialysis

Hospitalisation Criteria

Acetaminophen level above treatment line on Rumack-Matthew nomogram; any elevation of ALT/AST; salicylate level >40 mg/dL; metabolic acidosis; altered mental status; intentional ingestion in adolescents

Notes

Start NAC within 8h of acetaminophen ingestion for best efficacy; use Rumack-Matthew nomogram for risk assessment. For salicylate poisoning, alkalinize urine and monitor serial levels.

Guideline Changes

New consensus guidelines released for acetaminophen poisoning management with updated NAC protocols

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T42 — Poisoning by antiepileptic, sedative-hypnotic and antiparkinsonism drugs

SNOMED: 212588003 | Duration: 3 days | Source: EXTRIP Guidelines, AACT/NACCT Position Statements

First-Line

- **Activated Charcoal** — 1 g/kg PO (max 50 g) if within 1-2h of ingestion; multi-dose 0.5 g/kg q4h for carbamazepine/phenobarbital
- **Airway Management** — Intubate for GCS <9 or loss of protective reflexes; RSI with Rocuronium 1 mg/kg IV
- **IV Fluids (NS)** — 20 mL/kg bolus for hypotension; maintain euvolemia; monitor glucose and electrolytes

Second-Line

- **Flumazenil** — 0.01 mg/kg IV (max 0.2 mg) over 15 sec for pure BZD overdose; repeat q60sec up to 0.05 mg/kg or 1 mg total
- **Sodium Bicarbonate** — 1-2 mEq/kg IV bolus for QRS widening from carbamazepine/TCA; target serum pH 7.45-7.55
- **Hemodialysis** — For severe phenobarbital toxicity (level >100 mcg/mL) or refractory carbamazepine poisoning
- **Lipid Emulsion 20%** — 1.5 mL/kg IV bolus then 0.25 mL/kg/min for lipophilic drug toxicity with cardiovascular collapse **[NEW 2024-25]**

Investigations

- Serum drug levels (specific to ingested drug) — Quantify exposure and guide antidote therapy [STAT]
- 12-lead ECG — QRS widening (carbamazepine/TCA), QTc prolongation [STAT]
- Basic metabolic panel — Electrolytes, glucose, renal function [STAT]
- Arterial blood gas — Acid-base status in severe poisoning [URGENT]

- Liver function tests — Hepatotoxicity screening [URGENT]

Counselling

- Keep all medicines locked away and out of reach of children — even a small amount can be dangerous
- Bring the medicine bottle or packaging to the hospital so doctors know exactly what was taken
- The child will be monitored for several hours even if looking well — effects can be delayed
- After discharge, child-proof your home and store all medicines safely
- Call Poison Helpline (1800-11-6117) immediately if accidental ingestion happens again

Referral Criteria

Toxicology consultation for severe poisoning. Nephrology if hemodialysis indicated (phenobarbital level >100 mcg/mL). PICU for intubated patients or hemodynamic instability

Hospitalisation Criteria

GCS <12; respiratory depression requiring intervention; hemodynamic instability; QRS widening on ECG; seizures; need for multi-dose activated charcoal or hemodialysis

Notes

Avoid flumazenil in mixed overdose or chronic benzodiazepine use (seizure risk). Multi-dose activated charcoal is effective for phenobarbital, carbamazepine, and phenytoin.

T50 — Poisoning by diuretics and other unspecified drugs, medicaments and biological substances

Duration: 3 days | Source: AACT Position Statements, Poison Center Guidelines

First-Line

- **IV Fluid Resuscitation (NS)** — 20 mL/kg IV bolus for dehydration/hypotension; repeat as needed; correct electrolyte abnormalities
- **Activated Charcoal** — 1 g/kg PO (max 50 g) if within 1-2h of ingestion and patient is alert with intact airway
- **Electrolyte Replacement** — KCl 0.5-1 mEq/kg IV over 1h for hypokalemia; MgSO4 25-50 mg/kg IV for hypomagnesemia

Second-Line

- **Calcium Gluconate 10%** — 0.5 mL/kg IV slow push if symptomatic hypocalcemia or hyperkalemia (from K-sparing diuretics)
- **Vasopressors** — Norepinephrine 0.05-0.1 mcg/kg/min IV for refractory hypotension despite fluid resuscitation

Investigations

- Basic metabolic panel (serial q4-6h) — Monitor K, Na, Mg, Ca, renal function [STAT]
- 12-lead ECG — Arrhythmia from electrolyte derangement [STAT]
- Serum magnesium — Often depleted with diuretic overdose [STAT]
- Urine electrolytes — Assess renal electrolyte losses [URGENT]

Counselling

- Keep all medicines locked away from children in a high, secure place
- Your child may lose a lot of water and salts — we will replace them through IV
- Watch for dizziness, muscle cramps, or fast heartbeat even after discharge
- Follow up for blood tests to make sure salts (electrolytes) return to normal
- Call Poison Helpline (1800-11-6117) for any future accidental medicine intake

Referral Criteria

Nephrology if acute kidney injury or persistent electrolyte derangement. Cardiology if arrhythmias from electrolyte imbalance. Toxicology for complex or massive ingestion

Hospitalisation Criteria

Symptomatic electrolyte derangement; ECG abnormalities; hemodynamic instability; large ingested amount; persistent vomiting preventing oral replacement

Notes

Diuretic toxicity causes electrolyte derangements (hypoK, hypoNa, hypoMg) and dehydration. Monitor ECG for arrhythmias; effects may be delayed 2-4h until diuretic action begins.

T51 — Toxic effect of alcohol

SNOMED: 67426006 | Duration: 5 days | Source: EXTRIP Guidelines, AAP Toxicology

First-Line

- **Fomepizole** — 15 mg/kg IV loading dose, then 10 mg/kg IV q12h until alcohol level <30 mg/dL; safe in pediatrics
- **Dextrose 10%** — 2-5 mL/kg IV bolus for hypoglycemia (common in pediatric ethanol ingestion); monitor glucose q1h
- **IV Fluids (NS)** — 20 mL/kg IV bolus for dehydration; maintain euvolemia; correct metabolic acidosis

Second-Line

- **Ethanol 10% IV** — 7.6 mL/kg loading dose IV, then 0.8-1.3 mL/kg/hr infusion; if fomepizole unavailable
- **Hemodialysis** — For methanol/ethylene glycol with severe acidosis, visual changes, renal failure, or level >50 mg/dL
- **Folinic Acid (Leucovorin)** — 1-2 mg/kg IV q4-6h (max 50 mg) for methanol poisoning; enhances formate metabolism
- **Thiamine + Pyridoxine** — Thiamine 100 mg IV + Pyridoxine 50 mg IV for ethylene glycol; cofactors for detoxification

Investigations

- Serum ethanol/methanol/ethylene glycol levels — Identify toxic alcohol and guide treatment [STAT]
- Blood glucose (serial q1h) — Hypoglycemia common in pediatric ethanol ingestion [STAT]
- Arterial blood gas — Metabolic acidosis from toxic metabolites [STAT]
- Serum osmolality (measured) — Osmolar gap calculation [STAT]
- Basic metabolic panel — Renal function, electrolytes, anion gap [STAT]

Counselling

- Alcohol and similar liquids (sanitizer, antifreeze) are extremely dangerous for children
- Your child's blood sugar can drop dangerously — this is being monitored
- Store all household chemicals and alcoholic beverages securely away from children
- If the child consumed sanitizer or antifreeze, the antidote must be given for the full prescribed duration
- Follow up for blood tests after discharge — kidney and eye checks may be needed

Referral Criteria

Nephrology if methanol/ethylene glycol poisoning requiring dialysis. Ophthalmology for methanol exposure (visual assessment). Toxicology for all toxic alcohol ingestions. Psychiatry if intentional in adolescents

Hospitalisation Criteria

Any toxic alcohol ingestion (methanol, ethylene glycol); ethanol with hypoglycemia, altered mental status, or glucose <60 mg/dL; any patient requiring fomepizole or hemodialysis

Notes

Fomepizole blocks alcohol dehydrogenase, preventing toxic metabolite formation. Start before lab confirmation if high suspicion; early treatment may obviate need for hemodialysis.

Guideline Changes

Fomepizole now preferred over ethanol infusion as antidote for toxic alcohols in pediatrics

T54 — Toxic effect of corrosive substances

SNOMED: 10186000 | Duration: 14 days | Source: ESGE/ESPGHAN 2024, AACT Position Statements

First-Line

- **NPO and Airway Assessment** — Nothing by mouth; assess airway for edema/stridor; intubate early if oropharyngeal burns or stridor
- **IV Fluids (NS/LR)** — 20 mL/kg IV bolus for shock; maintain euvolemia; correct acid-base and electrolyte abnormalities
- **Proton Pump Inhibitor** — Omeprazole 1 mg/kg IV q12-24h (max 40 mg) to reduce acid-mediated injury to damaged mucosa

Second-Line

- **Morphine** — 0.1 mg/kg IV q2-4h PRN for pain; avoid oral medications until endoscopy completed
- **EGD within 24h** — Esophagogastroduodenoscopy for all symptomatic patients to grade injury; guides further management
- **Ceftriaxone** — 50-100 mg/kg IV if evidence of perforation or mediastinitis; broad-spectrum coverage

Investigations

- Upper GI endoscopy (EGD) — Grade mucosal injury — within 12-24h of ingestion [URGENT]
- Chest X-ray (PA) — Rule out mediastinal air (perforation) and aspiration [STAT]
- Abdominal X-ray — Free air under diaphragm if perforation suspected [STAT]
- CBC — Baseline; watch for infection if severe burns [URGENT]
- Basic metabolic panel — Acid-base status, renal function [URGENT]

Counselling

- Your child has swallowed a burning/corrosive substance — do NOT try to make the child vomit
- Do not give milk or water to dilute unless the doctor says so
- The child will need a camera test (endoscopy) to check for damage inside
- Keep all cleaning products, acids, and alkalis locked away from children
- Long-term follow-up may be needed to check for narrowing of the food pipe

Referral Criteria

Pediatric surgery/GI for endoscopy within 24h. ENT if laryngeal involvement. Transfer to burns center if extensive oral/pharyngeal burns. Pediatric surgery if perforation suspected

Hospitalisation Criteria

All symptomatic corrosive ingestions (drooling, dysphagia, vomiting, oral burns, stridor). Intentional ingestion in adolescents. Any patient requiring endoscopy or NPO status

Notes

Do NOT induce vomiting, give activated charcoal, or attempt neutralization. Endoscopy within 12-24h for symptomatic children; asymptomatic patients may be observed 2-4h and discharged.

T60 — Toxic effect of pesticides

SNOMED: 10186000 | Duration: 7 days | Source: WHO Organophosphate Management, CDC ATSDR Guidelines

First-Line

- **Atropine** — 0.02-0.05 mg/kg IV q5-10min; double dose until secretions dry; may need very large cumulative doses
- **Pralidoxime (2-PAM)** — 25-50 mg/kg IV over 30 min (max 1 g); then 10-20 mg/kg/hr infusion; give within 24-36h of exposure

- **Decontamination** — Remove clothing, wash skin with soap and water; activated charcoal 1 g/kg PO if oral ingestion within 1h

Second-Line

- **Diazepam** — 0.1-0.3 mg/kg IV (max 10 mg) for seizures from organophosphate poisoning; repeat q5-10min PRN
- **Glycopyrrolate** — 0.01-0.02 mg/kg IV as alternative to atropine; does not cross BBB so won't treat CNS symptoms
- **Obidoxime** — 4-8 mg/kg IV load then 0.5-1 mg/kg/hr; alternative oxime used in some regions

Investigations

- Serum cholinesterase (RBC and plasma) — Confirm organophosphate poisoning — low levels diagnostic [STAT]
- Arterial blood gas — Respiratory failure assessment [STAT]
- CBC with differential — Baseline evaluation [URGENT]
- Basic metabolic panel — Electrolytes, glucose, renal function [URGENT]
- ECG — Cardiac arrhythmia from cholinergic excess or atropine [STAT]

Counselling

- Pesticide poisoning is very serious — your child needs continuous monitoring in the hospital
- Remove and discard all contaminated clothing; bathe the child thoroughly
- Tell the doctor the exact name of the pesticide and how the child was exposed
- Store all pesticides in original containers, locked away from children
- Watch for muscle twitching, excessive secretions, or breathing difficulty even after treatment

Referral Criteria

Toxicology/poison center consultation for all organophosphate poisonings. PICU if respiratory failure or seizures. Neurology for seizure management. Transfer to higher center if ventilator not available

Hospitalisation Criteria

All symptomatic organophosphate/carbamate poisoning (SLUDGE symptoms). Any patient requiring atropine. Respiratory distress. Altered mental status. All intentional exposures

Notes

Organophosphate poisoning causes cholinergic crisis (SLUDGE/DUMBELS). Titrate atropine to secretion drying, not pupil size; pralidoxime must be given early before aging occurs.

T63 — Toxic effect of venom of contact with venomous animals

SNOMED: 10186000 | Duration: 5 days | Source: ACS Snakebite Guidelines, Texas Children's Envenomation Protocol

First-Line

- **Crotalidae Polyvalent Immune Fab (CroFab)** — 4-6 vials IV initial dose (same as adult); 8-12 vials for severe; repeat 4-6 vials q6h x3 for control
- **IV Fluids (NS/LR)** — 20 mL/kg IV bolus for hypotension; aggressive fluid resuscitation for anaphylaxis or venom shock
- **Epinephrine (for anaphylaxis)** — 0.01 mg/kg IM (1:1,000) max 0.3 mg for antivenom anaphylaxis; repeat q5-15min PRN

Second-Line

- **Anavip (F(ab')2)** — 10 vials IV initial dose for pit viper envenomation; longer half-life, fewer recurrent symptoms **[NEW 2024-25]**
- **Fresh Frozen Plasma** — 10-15 mL/kg IV for severe coagulopathy (fibrinogen <100, platelets <50K) unresponsive to antivenom

- **Diphenhydramine** — 1 mg/kg IV (max 50 mg) premedication or for mild antivenom reactions; give with H2 blocker
- **Tetanus Prophylaxis** — TdaP 0.5 mL IM if not current; recommended for all envenomation wounds

Investigations

- Coagulation profile (PT, aPTT, fibrinogen) — Venom-induced coagulopathy — serial q4-6h [STAT]
- CBC with platelet count — Thrombocytopenia from venom [STAT]
- Basic metabolic panel — Renal function, electrolytes [STAT]
- Type and crossmatch — Prepare for transfusion if severe coagulopathy [URGENT]
- Serial limb circumference measurements — Monitor progression of envenomation q15-30 min [STAT]

Counselling

- Your child was bitten by a venomous animal — antivenom is the most important treatment
- Keep the affected limb still and at heart level — do not apply tourniquet or ice
- Do not try to suck out venom or apply any local remedies
- Watch for swelling, bleeding, difficulty breathing, or allergic reactions after treatment
- Follow up in 1-2 weeks as venom effects can recur (recurrent coagulopathy)

Referral Criteria

Toxicology/envenomation specialist. Surgical consultation if compartment syndrome suspected. Hematology if severe coagulopathy. Transfer to antivenom-equipped center if antivenom unavailable

Hospitalisation Criteria

All children with signs of envenomation (progressive swelling, coagulopathy, neurotoxicity). Antivenom administration. Observation for minimum 8-12h even for dry bites. Anaphylaxis to antivenom

Notes

Children receive larger venom load relative to body size and are at higher morbidity risk. Antivenom dose is based on venom amount (not patient weight); serial limb measurements q15-60 min.

T71 — Asphyxiation

SNOMED: 386732003 | Duration: 7 days | Source: PALS/AHA 2025, WHO Emergency Triage

First-Line

- **Epinephrine** — 0.01 mg/kg (0.1 mL/kg of 1:10,000) IV/IO q3-5min during cardiac arrest; max single dose 1 mg
- **Oxygen** — 100% FiO2 via bag-mask ventilation; titrate to SpO2 94-99% once ROSC achieved
- **Normal Saline** — 20 mL/kg IV/IO bolus for volume resuscitation; repeat as needed up to 60 mL/kg

Second-Line

- **Therapeutic Hypothermia** — Target 33-34°C for 72h in neonatal HIE; initiate within 6h of birth
- **Dextrose 10%** — 2-5 mL/kg IV for hypoglycemia (glucose <60 mg/dL); recheck in 15 min
- **Sodium Bicarbonate** — 1 mEq/kg IV slow push if pH <7.0 and prolonged arrest; dilute 1:1 in neonates

Investigations

- Blood gas (arterial) — Assess pH, pCO2, lactate, base deficit [STAT]
- Blood glucose — Hypoglycaemia common in post-arrest; monitor frequently [STAT]
- Serum electrolytes — Guide fluid and electrolyte management [STAT]
- EEG — If seizures suspected or for prognostication in HIE [URGENT]
- MRI brain — Assess hypoxic-ischaemic injury severity; day 3-5 optimal timing [URGENT]

Counselling

- This is a medical emergency - follow the doctor team instructions carefully
- After recovery, watch for any changes in behaviour, memory, or movement

- Baby-proof the home: keep plastic bags, cords, and small objects away from children
- Learn infant/child CPR - it can save a life in choking or drowning emergencies
- Follow up with your pediatrician for developmental monitoring after the event

Referral Criteria

Refer to pediatric neurologist if post-hypoxic seizures, abnormal neurological exam, developmental regression; refer to PICU for ongoing care

Hospitalisation Criteria

All cases require emergency hospitalisation; NICU/PICU admission for ongoing resuscitation, therapeutic hypothermia (neonates), seizure management, and neurological monitoring

Notes

Prioritize airway management and ventilation; most pediatric arrests are respiratory in origin. Avoid hyperventilation; target normocapnia (ETCO2 35-40 mmHg) post-ROSC.

Guideline Changes

2025 AHA updated ventilation rates to 20-30 breaths/min for pediatric advanced airway

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T74 — Child abuse syndromes

SNOMED: 397940009 | Duration: 14 days | Source: AAP 2024, ACS Best Practices 2024

First-Line

- **ATLS Primary Survey** — Airway, breathing, circulation stabilization per ATLS; treat life-threatening injuries first
- **Ibuprofen** — 10 mg/kg PO q6-8h for pain; max 40 mg/kg/day; avoid if abdominal trauma suspected
- **Normal Saline** — 20 mL/kg IV/IO bolus if signs of shock; repeat up to 60 mL/kg; assess after each bolus

Second-Line

- **Morphine** — 0.1 mg/kg IV q2-4h PRN for moderate-severe pain; monitor respiratory status
- **Vitamin K** — 2-5 mg IV/SC if coagulopathy detected; evaluate for abusive head trauma

Investigations

- Skeletal survey (full) — All children <2 years with suspected abuse [URGENT]
- CT head without contrast — Suspected abusive head trauma, altered mental status [STAT]
- CBC with differential — Baseline evaluation, assess for bleeding disorder [URGENT]
- Coagulation profile (PT, aPTT, INR) — Rule out coagulopathy before attributing bruising to abuse [URGENT]
- Liver function tests (AST, ALT) — Screen for occult abdominal trauma [URGENT]

Counselling

- Your child is safe now and will receive the best care possible
- A social worker will help your family through this process
- Follow-up appointments are very important for your child's recovery
- Please report any new concerns about your child's safety immediately
- Counselling services are available for the whole family at no extra cost

Referral Criteria

All cases: mandatory referral to child protection services (CPS). Refer to pediatric surgery for fractures/internal injuries; ophthalmology for retinal hemorrhages; neurosurgery for intracranial injury

Hospitalisation Criteria

Any suspected abuse requiring medical stabilization; all children <2 years with unexplained injuries; any child with altered mental status, fractures, or internal organ injury

Notes

Mandatory CPS reporting required. Perform skeletal survey in all children <2 years with suspected abuse; use TEN-4-FACESp screening for bruising.

Guideline Changes

ACS Best Practices for Trauma Center Recognition of Child Abuse effective Jan 2024

T75.4 — Effects of electric current

SNOMED: 269431000 | Duration: 7 days | Source: ABA Electrical Injury Protocol 2024, ATLS

First-Line

- **Lactated Ringer** — 20 mL/kg IV bolus; then aggressive fluid resuscitation targeting UOP 1-2 mL/kg/hr to prevent rhabdomyolysis
- **Continuous Cardiac Monitoring** — 12-lead ECG on arrival; continuous monitoring for 24h minimum for all high-voltage injuries
- **Morphine** — 0.1 mg/kg IV q2-4h PRN for pain; titrate to comfort; assess for compartment syndrome

Second-Line

- **Sodium Bicarbonate** — 1-2 mEq/kg IV bolus then 150 mEq/L infusion to alkalinize urine if myoglobinuria present
- **Mannitol** — 0.5-1 g/kg IV if myoglobinuria persists despite fluids; maintain UOP >2 mL/kg/hr
- **Midazolam** — 0.1 mg/kg IV for seizures secondary to electrical injury; max 5 mg/dose

Investigations

- 12-lead ECG — All electrical injuries — assess for arrhythmia [STAT]
- Serum CK (creatine kinase) — Rhabdomyolysis screening [STAT]
- Urine myoglobin — Myoglobinuria from muscle damage [URGENT]
- Basic metabolic panel (BMP) — Electrolyte derangements, renal function [STAT]
- CBC — Baseline hematologic assessment [URGENT]

Counselling

- Keep your child away from open electrical wiring and switchboards
- Cover all electrical sockets with child-proof plugs at home
- Follow-up ECG monitoring may be needed even after discharge
- Watch for delayed symptoms like muscle pain, dark urine, or confusion and come back immediately
- Electrical burns may look small outside but be deep inside — all follow-ups are important

Referral Criteria

High-voltage injury (>1000V), lightning strike, burns crossing joints, compartment syndrome, cardiac arrhythmias persisting >24 hours, neurological deficits

Hospitalisation Criteria

All high-voltage injuries; any ECG abnormality; visible burns; loss of consciousness; dark/cola-colored urine (myoglobinuria); cardiac arrest at scene

Notes

High-voltage injuries require C-spine immobilization and rhabdomyolysis monitoring with serial CK/myoglobin. Lightning-strike victims need neuro re-exams and ophthalmology referral.

Endocrine

E03.9 — Hypothyroidism, unspecified

SNOMED: 40930008 | Duration: 365 days | Source: ATA 2014, AAP Congenital Hypothyroidism Guidelines

First-Line

- **Levothyroxine** — Neonates: 10-15 mcg/kg/day PO; 1y: 4-6 mcg/kg/day; adolescents: 2-4 mcg/kg/day; take on empty stomach

Second-Line

- **Levothyroxine liquid** — Same mcg/kg dosing as tablet; use liquid formulation in neonates/infants unable to swallow tablets

Investigations

- TSH and free T4 — Primary diagnosis — high TSH, low FT4 confirms hypothyroidism [URGENT]
- Thyroid scintigraphy or ultrasound — Determine etiology in congenital hypothyroidism (ectopic, aplastic, dyshormonogenesis)
- Anti-TPO and anti-thyroglobulin antibodies — Hashimoto thyroiditis — most common cause in older children
- Bone age — Assess degree of hypothyroidism and delayed skeletal maturation

Counselling

- Your child needs thyroid medicine (levothyroxine) every day — give it on an empty stomach in the morning
- Give the tablet crushed in a small amount of water for infants — avoid mixing with soy formula or iron
- Never stop the medicine without asking the doctor — stopping can cause developmental delays
- Blood tests (TSH, T4) are needed 4-6 weeks after each dose change, then every 3-6 months
- For congenital hypothyroidism, early treatment within 2 weeks of birth prevents intellectual disability

Referral Criteria

Pediatric endocrinology for all congenital hypothyroidism. Re-evaluation at age 3 (trial off levothyroxine to confirm permanent vs transient). Genetics if syndromic. Developmental pediatrics if developmental concerns

Hospitalisation Criteria

Myxedema coma (extremely rare in children) — IV levothyroxine, ICU admission. Severe hypothyroidism with pericardial effusion or cardiac compromise. Newborn with very high TSH >100 requiring urgent treatment

Notes

Start treatment within 2 weeks of life for congenital hypothyroidism to prevent cognitive impairment. Recheck TSH/FT4 at 4-6 weeks after any dose change.

E05 — Thyrotoxicosis / hyperthyroidism

SNOMED: 34486009 | Duration: 1095 days | Source: ETA 2022 Pediatric Graves Guideline, ATA 2016

First-Line

- **Methimazole** — 0.5-1.0 mg/kg/day PO divided TID initially; maintenance 0.2 mg/kg/day; treat >=3 years
- **Propranolol** — 0.5-2 mg/kg/day PO divided TID for symptom control (tachycardia, tremor) while awaiting MMI effect

Second-Line

- **Radioactive iodine (I-131)** — Dose calculated per thyroid size; for children >10y failing ATD or with significant adverse effects
- **Thyroidectomy** — Near-total thyroidectomy by experienced surgeon; for large goiter, non-compliance, or ATD failure

Investigations

- TSH, free T4, free T3 — Diagnosis and monitoring — TSH suppressed, FT4/FT3 elevated [URGENT]
- TSH receptor antibodies (TRAb) — Confirm Graves disease — positive in >95%
- CBC with differential — Baseline before methimazole — monitor for agranulocytosis [URGENT]
- Liver function tests — Baseline before methimazole; monitor hepatotoxicity
- Thyroid ultrasound — Goiter assessment, nodule evaluation

Counselling

- Your child has an overactive thyroid — medicine (methimazole) will be given for at least 3 years
- Never use propylthiouracil (PTU) in children — it can cause liver damage
- Report immediately if the child develops: fever with sore throat, jaundice, or rash — stop methimazole and come to hospital
- Blood tests (TSH, T3, T4, CBC) are needed every 4-6 weeks initially, then every 3 months
- Propranolol for fast heart rate is temporary — it will be stopped once thyroid levels normalize

Referral Criteria

Pediatric endocrinology mandatory. Nuclear medicine if RAI therapy considered (age >10). Experienced thyroid surgeon if surgery needed. Ophthalmology if Graves ophthalmopathy present

Hospitalisation Criteria

Thyroid storm (fever, tachycardia, altered mental status, severe thyrotoxicosis). New-onset atrial fibrillation. Severe heart failure from thyrotoxicosis. Agranulocytosis from methimazole (fever + sore throat + neutrophils <500)

Notes

Methimazole is first-line; never use propylthiouracil (PTU) in children due to hepatotoxicity risk. Treatment duration of 3-5+ years recommended; euthyroid usually achieved in 4-6 weeks.

E10.1 — Type 1 diabetes mellitus with ketoacidosis

SNOMED: 421893009 | Duration: 3 days | Source: ISPAD 2024, BSPED 2024

First-Line

- **Regular insulin IV infusion** — 0.05-0.1 U/kg/hr IV; start after 1h fluids and K+ >3.0; do NOT bolus
- **0.9% NaCl** — 10-20 mL/kg IV bolus over 20-30 min; then replace deficit over 36h with added KCl 40 mmol/L

Second-Line

- **SC rapid-acting insulin** — 0.1-0.15 U/kg SC q1-2h; for mild DKA without IV access in resource-limited settings
- **Potassium phosphate** — 0.5-1 mmol/kg IV over 6h; if phosphate <1 mg/dL; use with KCl to provide total K+ replacement

Investigations

- Blood glucose (hourly) — Monitor correction — target drop 50-100 mg/dL/hr [STAT]
- Venous blood gas (q2-4h) — Monitor pH and bicarbonate for resolution [STAT]
- Serum electrolytes (q2-4h) — Monitor potassium — must be >3.0 before starting insulin [STAT]
- Serum ketones (beta-hydroxybutyrate) — Monitor ketone clearance [STAT]
- HbA1c — Assess pre-existing glycemic control; distinguishes new vs established T1DM

Counselling

- Your child has diabetes with a complication called DKA — this is treatable but serious
- Insulin and fluids will be given through IV to correct the problem slowly over 24-48 hours
- After recovery, your child will need insulin injections every day for life
- You will be trained on insulin injection, blood sugar monitoring, and diet management
- Never stop insulin on your own — even during illness, insulin must continue (may need to adjust dose)

Referral Criteria

Pediatric endocrinology/diabetology mandatory. PICU if severe DKA (pH <7.1) or altered mental status. Neurosurgery immediately if signs of cerebral edema (sudden headache, vomiting, altered consciousness during treatment)

Hospitalisation Criteria

All DKA episodes require hospitalisation. Severe DKA (pH <7.1, bicarbonate <5) requires PICU. New-onset T1DM for stabilization and diabetes education. Any DKA in children <5 years (higher cerebral edema risk)

Notes

Start insulin only after initial fluid resuscitation and confirmed K+ >3.0 mmol/L. Monitor for cerebral edema especially in children <5y; avoid rapid glucose drops >5 mmol/L/hr.

Guideline Changes

ISPAD 2024 updated DKA chapter; low-dose insulin 0.05 U/kg/hr now preferred initial rate

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E10.9 — Type 1 diabetes insulin therapy

SNOMED: 46635009 | Duration: 365 days | Source: ISPAD 2022

First-Line

- **Insulin Glargine (basal)** — 0.2-0.4 U/kg/day SC once daily at bedtime
- **Rapid-acting insulin (bolus)** — 0.05-0.1 U/kg/dose SC pre-meal, lispro or aspart

Second-Line

- **Hybrid closed-loop APS** — Automated insulin delivery, preferred for all ages where available - ISPAD 2022 **[NEW 2024-25]**

Investigations

- HbA1c — Every 3 months to assess glycaemic control
- Blood glucose (fasting and postprandial) — Daily self-monitoring
- Serum electrolytes and blood gas — If DKA suspected [STAT]
- Thyroid function tests — Annual screening for autoimmune thyroiditis
- Anti-tTG IgA — Annual celiac disease screening

Counselling

- Insulin injections are lifelong - never stop without doctor advice even if child feels well
- Check blood sugar at least 4 times daily; use CGM if available
- Always carry fast-acting sugar (glucose tablets, juice) for low sugar episodes
- Learn to count carbohydrates in food and adjust insulin dose accordingly
- Regular follow-up every 3 months for HbA1c; annual eye and kidney screening from 5 years of diabetes

Referral Criteria

Refer to pediatric endocrinologist for all new diagnoses, recurrent DKA, HbA1c persistently >9%, suspected celiac or thyroid comorbidity, or transition to insulin pump/CGM

Hospitalisation Criteria

Admit if diabetic ketoacidosis (pH <7.3, bicarbonate <18), new-onset T1D for stabilisation and education, severe hypoglycaemia with seizure or loss of consciousness, or intercurrent illness with inability to manage at home

Notes

HbA1c target under 7% for all ages. CGM strongly recommended. TIR 70-180 mg/dL over 70% is a co-primary target.

Guideline Changes

Time-in-range over 70% added as co-primary target. Hybrid closed-loop systems now recommended as preferred delivery for all ages.

E25.0 — Congenital adrenogenital disorders associated with enzyme deficiency

SNOMED: 237751000 | Duration: 365 days | Source: Endocrine Society 2018, JCRPE 2024

First-Line

- **Hydrocortisone** — 10-15 mg/m2/day PO divided TID; titrate to lowest dose controlling androgens and allowing growth
- **Fludrocortisone** — 0.1-0.2 mg PO daily for salt-wasting forms; add NaCl 2-4 g/day in infants

Second-Line

- **Crinecerfont** — 25-100 mg PO BID based on weight (>=4y); adjunct to GC to reduce steroid dose **[NEW 2024-25]**
- **Prednisolone** — 2-4 mg/m2/day PO; only in post-pubertal patients when HC is insufficient

Investigations

- Serum 17-OH progesterone — Elevated in 21-hydroxylase deficiency (most common) [URGENT]
- Serum electrolytes (Na, K) — Hyponatremia and hyperkalemia in salt-wasting form [STAT]
- Plasma renin activity — Elevated in mineralocorticoid deficiency
- Karyotype — Determine genetic sex in ambiguous genitalia
- Serum cortisol and ACTH — Assess degree of cortisol deficiency [URGENT]

Counselling

- Your child has CAH — lifelong steroid medicine is needed to replace missing hormones
- Give hydrocortisone 3 times a day at the same times every day
- During illness or fever, TRIPLE the dose — this is very important to prevent crisis
- Salt supplements may be needed in infancy — continue as directed
- Regular blood tests (17-OHP, electrolytes) are needed every 3-4 months to adjust doses

Referral Criteria

Pediatric endocrinology mandatory for all cases. Pediatric surgery/urology for genital surgery timing in affected females. Genetics for family counselling and prenatal diagnosis in future pregnancies

Hospitalisation Criteria

Salt-wasting crisis (hyponatremia <125, hyperkalemia, dehydration, shock). Any infant with ambiguous genitalia pending workup. Adrenal crisis with vomiting preventing oral medication

Notes

Hydrocortisone is first-line in growing children; avoid dexamethasone pre-puberty. Crinecerfont (2024) enables GC dose reduction by blocking excess ACTH-driven androgen production.

Guideline Changes

Crinecerfont (Crenessity) FDA-approved Dec 2024 as first-in-class CRF1 antagonist adjunct for CAH age >=4y

E27.4 — Adrenocortical insufficiency

SNOMED: 386584007 | Duration: 365 days | Source: Endocrine Society 2016, NICE NG243 2024

First-Line

- **Hydrocortisone** — 8 mg/m2/day PO divided TID; stress dose: triple during illness; max 20 mg/m2/day
- **Fludrocortisone** — 0.05-0.2 mg PO daily for mineralocorticoid replacement; monitor renin and electrolytes

Second-Line

- **Prednisolone** — 2-3 mg/m2/day PO daily; not preferred in children due to growth suppression
- **IV Hydrocortisone** — 50-100 mg/m2 IV bolus for adrenal crisis then 50-100 mg/m2/24h continuous infusion

Investigations

- Morning serum cortisol — Low cortisol confirms adrenal insufficiency [URGENT]
- ACTH stimulation test (Synacthen test) — Gold standard for diagnosis — cortisol <500 nmol/L post-stimulation is diagnostic
- Serum ACTH — Differentiate primary (high ACTH) vs secondary (low ACTH)
- Basic metabolic panel — Hyponatremia, hyperkalemia in primary AI [URGENT]
- Plasma renin activity — Elevated in mineralocorticoid deficiency

Counselling

- Your child needs to take steroid medicine every day for life — never miss a dose
- During fever, vomiting, or any illness, TRIPLE the steroid dose (sick-day rules)
- Always carry an emergency steroid injection kit and a medical alert bracelet
- Teach all caregivers (school, grandparents) the sick-day rules and how to give emergency injection
- Follow up every 3-6 months with the endocrinologist for dose adjustment as the child grows

Referral Criteria

Pediatric endocrinology for all cases. Emergency referral if adrenal crisis (hypotension, hypoglycemia, hyperkalemia not responding to stress dose steroids)

Hospitalisation Criteria

Adrenal crisis: shock, severe dehydration, altered mental status, refractory hypoglycemia, hyperkalemia with ECG changes. Any patient unable to tolerate oral steroids during illness

Notes

Avoid long-acting glucocorticoids in growing children; use lowest dose allowing adequate growth. All patients need emergency injection training and steroid sick-day rules.

Guideline Changes

NICE 2024 guideline published; Khindivi oral HC solution FDA-approved 2025

E30.1 — Precocious puberty

SNOMED: 400179000 | Duration: 730 days | Source: ETA 2022, PES Consensus

First-Line

- **Leuprolide acetate** — 7.5-15 mg IM q28d (1-mo depot) or 11.25-30 mg IM q3mo; US doses 200-300 mcg/kg/28d
- **Histrelin implant** — 50 mg SC implant replaced annually; avoids repeat injections; minor procedure under local anesthesia

Second-Line

- **Triptorelin** — 22.5 mg IM q24wk (6-mo depot, Triptodur); or 3.75 mg IM q28d
- **Leuprolide (Fensolvi)** — 45 mg SC q6mo; small-volume injection; FDA-approved for CPP

Investigations

- Bone age (left wrist X-ray) — Advanced bone age confirms premature maturation
- LH, FSH (basal and GnRH-stimulated) — Stimulated LH >5 IU/L confirms central precocious puberty
- Estradiol (girls) / Testosterone (boys) — Elevated levels confirm pubertal status
- MRI brain with pituitary focus — Mandatory to rule out CNS lesion (especially in boys and girls <6 years) [URGENT]
- Pelvic ultrasound (girls) — Uterine enlargement, ovarian cysts — supportive of CPP

Counselling

- Your child is showing signs of puberty earlier than expected — treatment can help preserve adult height
- The monthly/3-monthly injection stops early puberty and allows normal growth to continue

- The child may initially have a brief increase in symptoms after the first injection — this is temporary
- Provide emotional support — the child may feel different from same-age peers
- Regular follow-up every 3-6 months with bone age X-ray is important to track progress

Referral Criteria

Pediatric endocrinology mandatory. MRI brain if onset before age 6 (girls) or 9 (boys) to rule out CNS pathology. Genetics if associated with NF1 or other syndromes

Hospitalisation Criteria

Rarely needed. Admit only if: CNS tumor discovered on MRI, acute complication of underlying condition, or need for GnRH stimulation test not available outpatient

Notes

GnRH agonists are standard of care for central precocious puberty to preserve adult height potential. Monitor for adequate LH suppression and growth velocity deceleration.

Guideline Changes

6-month depot formulations (Fensolvi SC, Triptodur IM) now widely used reducing injection burden

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E30.9 — Disorders of puberty, unspecified

SNOMED: 400003000 | Duration: 180 days | Source: JCRPE 2024, Endocrine Society 2014

First-Line

- **Testosterone (boys)** — 50-100 mg IM monthly for 3-6 months for pubertal induction; start at chronological age >=14y and BA >=12y
- **17-beta-estradiol (girls)** — 0.25-0.5 mg PO daily (or 5 mcg/kg/day) for pubertal induction; start at chronological age >=13y and BA >=11y

Second-Line

- **Transdermal estradiol patch** — Start 6.25 mcg (1/8 of 50 mcg patch) nightly; increase q6mo; more physiologic than oral
- **Letrozole** — 2.5 mg PO daily for 6 months in boys; aromatase inhibitor to boost growth and pubertal tempo

Investigations

- Bone age (left wrist X-ray) — Assess skeletal maturity — delayed bone age suggests constitutional delay
- LH, FSH, estradiol/testosterone — Distinguish hypogonadotropic (low LH/FSH) vs hypergonadotropic (high LH/FSH)
- Karyotype — Turner syndrome (45,X) in girls; Klinefelter (47,XXY) in boys
- MRI brain with pituitary focus — Rule out CNS pathology if hypogonadotropic
- TSH, free T4, IGF-1 — Screen for hypothyroidism and growth hormone deficiency

Counselling

- Your child's puberty is delayed — in most cases, this is a normal variant and nothing to worry about
- A short course of hormone medicine may be given to help start puberty
- The child may feel different from peers — provide emotional support and reassurance
- Height and development will be monitored every 3-6 months
- Follow up with the endocrinologist to determine if further treatment is needed

Referral Criteria

Pediatric endocrinology for all cases of delayed puberty. Genetics if Turner syndrome or Klinefelter syndrome suspected. Ophthalmology/neurosurgery if pituitary pathology on MRI

Hospitalisation Criteria

Rarely needed. Admit if: adrenal crisis in undiagnosed panhypopituitarism, severe psychological crisis, or need for dynamic hormone testing not available outpatient

Notes

Distinguish constitutional delay from permanent hypogonadism with clinical follow-up and labs. Low-dose sex steroid trial for 3-6 months can help differentiate and provide psychological benefit.

Guideline Changes

2024 JCRPE review updated delayed puberty management perspectives including aromatase inhibitor use

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E34.3 — Short stature, not elsewhere classified

SNOMED: 237837007 | Duration: 365 days | Source: PES/Endocrine Society 2016, AACE 2019

First-Line

- **Somatropin** — 25-50 mcg/kg/day SC daily for GHD; up to 67 mcg/kg/day for ISS; titrate by IGF-1 levels

Second-Line

- **Somapacitan** — 0.16 mg/kg SC once weekly; long-acting GH approved for pediatric GHD; reduces injection burden **[NEW 2024-25]**
- **Somatrogon** — 0.66 mg/kg SC once weekly; alternative long-acting GH for pediatric GHD **[NEW 2024-25]**

Investigations

- Bone age (left wrist X-ray) — Assess growth potential — compare bone age to chronological age
- IGF-1 and IGFBP-3 — Low values suggest GH deficiency
- GH stimulation test (2 agents) — Peak GH <10 ng/mL confirms GHD
- Karyotype (in girls) — Rule out Turner syndrome (45,X)
- Thyroid function (TSH, free T4) — Hypothyroidism is a treatable cause of short stature

Counselling

- Your child is shorter than expected — growth hormone injections may be needed to help the child grow taller
- The injection is given every day (or once weekly with newer medicine) under the skin — usually at bedtime
- Growth will be monitored every 3-6 months — bring the child for regular height and weight measurements
- Good nutrition, adequate sleep (8-10 hours), and regular physical activity also help growth
- Treatment works best when started early — continue until the doctor says growth plates have closed

Referral Criteria

Pediatric endocrinology mandatory. Genetics if syndromic short stature (Turner, Noonan, Prader-Willi). GI if celiac or IBD contributing. Psychology if significant psychological impact

Hospitalisation Criteria

Rarely needed. Admit only for: GH stimulation testing if not available outpatient, newly diagnosed pituitary tumor, or severe GH deficiency with hypoglycemia in infants

Notes

Individualize dosing based on growth response and IGF-1 levels; stop GH when growth velocity falls below 2.5 cm/year. Once-weekly formulations (2024) improve adherence.

Guideline Changes

Somapacitan (Sogroya) once-weekly GH approved for new pediatric indications (2024)

E43 — Severe protein-energy malnutrition

SNOMED: 360549009 | Duration: 42 days | Source: WHO 2023, WHO SAM Protocol

First-Line

- **F-75 therapeutic milk** — 75 kcal/100 mL; stabilization phase days 1-7; feed q2-3h; low protein/Na to prevent refeeding syndrome

- **RUTF (e.g. Plumpy'Nut)** — 150-200 kcal/kg/day in rehab phase; 10-15 kg total over 6-8 wk; for age >=6mo without complications
- **Amoxicillin** — 40 mg/kg/day PO divided BID for 5-7 days; routine antibiotic per WHO SAM protocol

Second-Line

- **F-100 therapeutic milk** — 100 kcal/100 mL; transition/rehabilitation phase; increase feeds by 10 mL/feed until desired intake
- **IV 10% Dextrose** — 5 mL/kg IV bolus for hypoglycemia (glucose <54 mg/dL); then feed immediately

Investigations

- Blood glucose — Hypoglycemia screening — common and life-threatening in SAM [STAT]
- Serum electrolytes (Na, K, Mg, PO4) — Refeeding syndrome prevention — check BID for 4 days [STAT]
- CBC with differential — Anemia, infection screening [URGENT]
- Blood culture — Occult sepsis is common in SAM children [URGENT]
- HIV test — All SAM children should be tested per WHO

Counselling

- Your child is severely malnourished and needs careful treatment in the hospital first, then at home
- Give the therapeutic food (RUTF) as prescribed — it has all the nutrients your child needs
- Feed small amounts frequently — do not force-feed large quantities at once
- Continue breastfeeding along with therapeutic food if the child is under 2 years
- Come for weekly follow-up until the child's weight reaches the target — missing visits is dangerous

Referral Criteria

Nutrition rehabilitation center if available. Pediatric gastroenterology if underlying GI disease. Social services for food insecurity. Infectious disease if TB or HIV suspected. Metabolic evaluation if malnutrition unexplained by dietary history

Hospitalisation Criteria

All complicated SAM (with medical complications: dehydration, sepsis, hypothermia, hypoglycemia, severe edema, poor appetite). WHZ <-3 SD or MUAC <115mm. Bilateral pitting edema. Failed appetite test (RUTF refusal)

Notes

Start refeeding cautiously with F-75 to avoid refeeding syndrome; monitor electrolytes (K, Mg, PO4) BID for 4 days. Treat hypothermia, hypoglycemia, and dehydration before addressing nutrition.

Guideline Changes

WHO 2023 guideline consolidated wasting and oedema management; community-based RUTF approach strengthened

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E44 — Protein-energy malnutrition of moderate and mild degree

SNOMED: 190605005 | Duration: 42 days | Source: WHO 2023, NASPGHAN 2024

First-Line

- **Supplementary feeding** — Energy-dense supplementary food 250-500 kcal/day added to usual diet; attend checkup q2wk for 4-6 wk
- **Micronutrient supplementation** — Multivitamin/mineral supplement per WHO; include iron, zinc, folate, vitamin A per local protocol

Second-Line

- **RUSF (ready-to-use supplementary food)** — 92 g/day sachet (~500 kcal) for moderate acute malnutrition in 6-59 month olds

- **High-calorie oral formula** — 1-1.5 kcal/mL enteral formula for older children; increase by 10-20% q2-3d as tolerated

Investigations

- CBC with differential — Anemia screening — common in malnutrition
- Serum albumin — Assess protein status
- Serum iron, ferritin, TIBC — Iron deficiency is common
- Stool examination — Parasitic infections contributing to malnutrition
- Urinalysis — Screen for UTI as underlying cause

Counselling

- Your child is underweight — more nutritious food is needed every day
- Give energy-rich foods like dal, ghee, peanuts, eggs, banana, and milk at every meal
- Feed the child 5-6 times a day — 3 meals and 2-3 snacks
- Continue breastfeeding if the child is under 2 years — it is the best nutrition
- Come for weight check-up every 2 weeks until the child's weight is back on track

Referral Criteria

Pediatric gastroenterology if no improvement after 4-6 weeks of supplementary feeding. Dietitian for detailed meal planning. Social services if food insecurity identified. Investigate for underlying disease if weight not improving

Hospitalisation Criteria

Failure to respond to outpatient supplementary feeding after 4-6 weeks. Development of complications (severe dehydration, electrolyte imbalance). Coexisting severe illness requiring treatment. Weight-for-height z-score dropping to <-3 SD (now SAM)

Notes

Counsel on age-appropriate feeding practices and complete immunizations. Monitor weight-for-height z-score every 2 weeks; refer if no improvement after 4-6 weeks.

Guideline Changes

WHO 2023 guideline expanded to include moderate wasting management in 6-59 month age group

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E46 — Unspecified protein-energy malnutrition

SNOMED: 70241007 | Duration: 21 days | Source: WHO 2023, ASPEN 2020 Refeeding Guidelines

First-Line

- **Nutritional rehabilitation** — Start at 50-80% estimated caloric needs for current weight; advance 10-20%/day if electrolytes stable
- **Thiamine** — 100-300 mg IV/PO daily before and during refeeding to prevent Wernicke encephalopathy
- **Electrolyte monitoring** — Check K, Mg, PO4 BID for >=4 days; replace aggressively; do not advance feeds if abnormal

Second-Line

- **Phosphate supplementation** — 0.5-1 mmol/kg/day IV or PO if PO4 low; critical to prevent refeeding syndrome cardiac complications
- **Enteral formula** — Polymeric 1 kcal/mL formula via NG if unable to eat; advance slowly per refeeding protocol

Investigations

- Serum phosphate, potassium, magnesium — Refeeding syndrome monitoring — check BID for 4 days [STAT]
- ECG — Cardiac monitoring during refeeding (QTc prolongation risk) [URGENT]
- CBC — Anemia screening

- Liver and renal function tests — Organ function assessment
- Serum albumin and prealbumin — Nutritional status markers

Counselling

- Your child needs gradual refeeding — starting with small amounts and increasing slowly is safer
- Do not try to feed large amounts quickly — it can cause a dangerous condition called refeeding syndrome
- The child will need frequent blood tests during the first few days to monitor salt levels
- Once stable, follow the nutritional plan given by the doctor and dietitian
- Follow up weekly for weight monitoring until nutritional goals are met

Referral Criteria

Dietitian for refeeding protocol. Pediatric gastroenterology for underlying malabsorption. Psychiatry if eating disorder suspected. Social services for food insecurity or neglect

Hospitalisation Criteria

Any malnourished patient at refeeding risk (>10 days poor intake). Electrolyte instability (low phosphate, potassium, magnesium). Cardiac symptoms during refeeding. BMI <14 kg/m2 in adolescents

Notes

High refeeding syndrome risk in any child malnourished >10 days; start calories low and advance slowly. Correct electrolytes before advancing nutritional support.

E50 — Vitamin A deficiency

SNOMED: 72000004 | Duration: 14 days | Source: WHO 2011 (reaffirmed), MSF Guidelines

First-Line

- **Retinol (Vitamin A)** — <6mo: 50,000 IU PO; 6-12mo: 100,000 IU PO; >12mo: 200,000 IU PO; day 1, day 2, then repeat at 2 wk

Second-Line

- **Vitamin A (preventive)** — 100,000 IU PO q4-6mo for ages 6-11mo; 200,000 IU PO q4-6mo for 12-59mo in endemic areas

Investigations

- Serum retinol level — Confirm deficiency — <20 mcg/dL diagnostic
- Ophthalmologic examination — Bitot spots, corneal xerosis, keratomalacia grading [URGENT]
- CBC — Associated anemia
- Serum albumin — Protein status — VAD often coexists with PEM

Counselling

- Your child has vitamin A deficiency — high-dose vitamin A drops will be given on 3 occasions
- Eye involvement (night blindness, dry eyes) is a sign of severe deficiency — report immediately
- Feed the child orange/yellow fruits and vegetables (carrot, papaya, mango), green leafy vegetables, and eggs
- Vitamin A supplementation every 6 months is recommended for all children 6-59 months in India
- Come back if the child develops eye redness, whiteness on the eye, or difficulty seeing at night

Referral Criteria

Ophthalmology urgently if corneal involvement (xerophthalmia stage X2-X3B — corneal ulcer/keratomalacia). Pediatric nutrition for severe malnutrition associated with VAD. GI if malabsorption identified

Hospitalisation Criteria

Corneal xerophthalmia (X2 or higher — corneal xerosis, ulceration, keratomalacia). Measles with vitamin A deficiency. Severe malnutrition with VAD. Corneal perforation requiring emergency treatment

Notes

Active corneal xerophthalmia is a medical emergency requiring immediate high-dose vitamin A. Also give vitamin A with measles treatment per WHO protocol regardless of deficiency status.

E55 — Vitamin D deficiency

SNOMED: 34713006 | Duration: 84 days | Source: AAP 2008, Indian Academy of Pediatrics 2021, Endocrine Society 2024

First-Line

- **Cholecalciferol (Vitamin D3)** — <1y: 2000 IU/day PO x12wk; >1y: 3000-6000 IU/day PO x8-12wk; recheck 25-OH-D after course
- **Calcium supplementation** — 30-75 mg/kg/day elemental calcium PO divided TID for rickets with low Ca or inadequate dietary intake

Second-Line

- **Ergocalciferol (Vitamin D2)** — 50,000 IU PO weekly x6-8wk for adolescents; then maintenance 600-1000 IU/day
- **Calcitriol** — 0.01-0.05 mcg/kg/day PO for vitamin D-dependent rickets type I or renal disease

Investigations

- Serum 25-OH vitamin D — Primary diagnostic test — <20 ng/mL is deficiency, <12 is severe
- Serum calcium, phosphorus, ALP — Low Ca, low PO4, high ALP in nutritional rickets [URGENT]
- Parathyroid hormone (PTH) — Secondary hyperparathyroidism from vitamin D deficiency
- Wrist X-ray — Rachitic changes: cupping, fraying, widening of metaphysis
- Renal function tests — Rule out renal rickets

Counselling

- Your child has vitamin D deficiency — this can cause weak bones and rickets in children
- Give vitamin D drops/tablets as prescribed every day — complete the full 8-12 week course
- All breastfed babies should get 400 IU vitamin D daily from birth until weaning to fortified milk
- Encourage 15-20 minutes of sunlight exposure daily (morning sun is best)
- Include calcium-rich foods: milk, curd, paneer, ragi, sesame seeds in the child's diet

Referral Criteria

Pediatric endocrinology if vitamin D-resistant or dependent rickets suspected. Orthopedics if severe skeletal deformities (bowing, fractures). Nephrology for renal rickets. GI if malabsorption

Hospitalisation Criteria

Hypocalcemic seizures from severe vitamin D deficiency. Symptomatic hypocalcemia (tetany, laryngospasm). Pathological fractures. Severe rickets requiring IV calcium supplementation

Notes

All breastfed infants should receive 400 IU/day vitamin D prophylaxis from birth. Double treatment dose in patients with obesity, malabsorption, or dark skin pigmentation.

E56 — Other vitamin deficiencies

SNOMED: 85670002 | Duration: 90 days | Source: WHO Guidelines, AAP Guidelines

First-Line

- **Vitamin K1 (phytonadione)** — Neonates: 0.5-1 mg IM at birth; VKDB treatment: 1-5 mg IV/SC; preterm: 0.3-0.5 mg/kg IM
- **Vitamin B12 (cyanocobalamin)** — 1000 mcg PO daily x4mo for nutritional deficiency; or 1000 mcg IM weekly x4wk then monthly

Second-Line

- **Vitamin C (ascorbic acid)** — 100-300 mg PO daily for scurvy treatment; continue for 1-3 months until stores replete
- **Vitamin E (alpha-tocopherol)** — 25-200 IU/day PO for deficiency; higher doses in cholestasis or fat malabsorption

Investigations

- CBC with peripheral smear — Macrocytic anemia in B12/folate deficiency; assess bleeding risk [URGENT]
- Serum vitamin B12 level — Confirm B12 deficiency (<200 pg/mL)
- Serum folate — Confirm folate deficiency
- PT/INR — Coagulopathy from vitamin K deficiency [STAT]
- Serum homocysteine and methylmalonic acid — Elevated in B12 deficiency — more sensitive than serum B12

Counselling

- Your child has a vitamin deficiency that can be corrected with proper supplementation
- Give vitamin supplements as prescribed — complete the full course even if the child looks better
- Feed the child a balanced diet with fruits, vegetables, eggs, milk, and green leafy vegetables
- All newborns must receive Vitamin K injection at birth — never refuse it
- Follow up for blood tests after completing the treatment course to confirm recovery

Referral Criteria

Hematology if severe anemia from B12/folate deficiency. Neurology if neurological symptoms from B12 deficiency. GI if malabsorption suspected as cause. Dietitian for dietary counselling

Hospitalisation Criteria

Severe bleeding from vitamin K deficiency (VKDB). Severe megaloblastic anemia (Hb <5 g/dL) from B12 deficiency. Scurvy with hemorrhagic complications. Neurological symptoms from B12 deficiency requiring urgent replacement

Notes

Vitamin K prophylaxis at birth prevents hemorrhagic disease of newborn. Identify and treat underlying cause of deficiency (dietary, malabsorption, or chronic disease).

E66 — Overweight and obesity

SNOMED: 414916001 | Duration: 180 days | Source: AAP CPG 2023, FDA Labeling

First-Line

- **Lifestyle intervention** — Intensive health behavior and lifestyle treatment (26+ contact hours); diet, physical activity, behavioral therapy
- **Semaglutide (Wegovy)** — SC weekly; titrate 0.25mg to 2.4mg over 16wk; age >=12y with BMI >=95th percentile **[NEW 2024-25]**
- **Liraglutide (Saxenda)** — SC daily; titrate 0.6mg to 3.0mg over 4wk; age >=12y (6-12y per recent trial data)

Second-Line

- **Orlistat** — 120 mg PO TID with meals; age >=12y; add multivitamin due to fat-soluble vitamin malabsorption
- **Phentermine/topiramate** — Per weight-based dosing PO daily; age >=12y; monitor HR and mood

Investigations

- Fasting lipid profile — Dyslipidemia screening — all obese children >=2 years
- Fasting glucose and HbA1c — Screen for prediabetes/T2DM
- Liver function tests (ALT) — NAFLD/MASLD screening — screen all obese children
- TSH — Rule out hypothyroidism as secondary cause
- Fasting insulin — Insulin resistance assessment

Counselling

- Your child's weight needs to be managed with diet changes, more physical activity, and less screen time
- Limit sugar-sweetened drinks (cola, juice, packaged drinks) — switch to water and buttermilk
- Ensure at least 60 minutes of active play/exercise every day — whole family should participate
- Reduce screen time to less than 2 hours per day for children above 5 years
- Regular follow-up every 3 months to track weight, blood sugar, and cholesterol is important

Referral Criteria

Pediatric endocrinology if secondary cause suspected or pharmacotherapy considered (age >=12). Dietitian for structured meal planning. Psychology for binge eating or emotional eating. Bariatric surgery team if BMI >=40 or >=35 with comorbidities (age >=13)

Hospitalisation Criteria

Rarely needed. Consider admission for: severe obstructive sleep apnea, hypertensive emergency from obesity, bariatric surgery, or psychiatric crisis (eating disorder with medical instability)

Notes

Semaglutide achieves up to 17% BMI reduction, the most potent of approved agents. Pharmacotherapy is adjunct to, not replacement for, intensive lifestyle modification.

Guideline Changes

Semaglutide FDA-approved Dec 2022 for age >=12y; liraglutide expanded to ages 6-12y (2024 trial data)

E70.0 — Phenylketonuria

SNOMED: 7573000 | Duration: 365 days | Source: ACMG 2024, European PKU Guidelines 2017

First-Line

- **Phe-restricted diet** — Limit phenylalanine intake using Phe-free amino acid formula; target blood Phe 120-360 umol/L
- **Sapropterin (BH4)** — 10-20 mg/kg/day PO once daily; for BH4-responsive patients (~25-50%); trial response required

Second-Line

- **Pegvaliase** — SC injection; approved for adults/adolescents >=16y; titrate from 2.5 mg to 20-40 mg daily; anaphylaxis risk
- **Large neutral amino acids** — Supplement PO to compete with Phe transport across BBB; adjunct in patients with poor dietary compliance

Investigations

- Blood phenylalanine level — Primary monitoring — target 120-360 umol/L [URGENT]
- BH4 loading test — Determine BH4 responsiveness — 25-50% of patients respond
- Plasma amino acids (full profile) — Ensure adequate tyrosine and other amino acid levels
- Neurodevelopmental assessment — Annual cognitive and behavioral screening
- MRI brain — If developmental regression or poor metabolic control

Counselling

- Your child needs a special low-protein diet for life — this prevents brain damage
- Use the special PKU formula as the main source of nutrition — it provides essential nutrients without phenylalanine
- Blood phenylalanine levels must be checked weekly (infants) to monthly (older children)
- Avoid high-protein foods: meat, fish, eggs, dairy, nuts, soy — your dietitian will give you a detailed list
- If the child is unwell or not eating, contact the metabolic team immediately

Referral Criteria

Metabolic genetics/biochemical genetics mandatory. Developmental pediatrics for cognitive monitoring. Dietitian specializing in PKU. Maternal-fetal medicine for females with PKU planning pregnancy

Hospitalisation Criteria

Phenylalanine level >900 umol/L with neurological symptoms. New diagnosis in neonate for initiation of diet. Metabolic crisis during intercurrent illness. Non-compliance with dangerously high levels

Notes

Newborn screening enables early dietary treatment preventing intellectual disability. All patients should undergo BH4 responsiveness testing; diet remains the cornerstone of therapy.

Guideline Changes

ACMG 2024 evidence-based guideline published Aug 2024; sapropterin conditionally recommended in pregnancy

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E71 — Disorders of branched-chain amino-acid metabolism and fatty-acid metabolism

Duration: 14 days | Source: ACMG/GeneReviews 2022, E-IMD Guidelines

First-Line

- **BCAA-free formula** — Restrict leucine to age-appropriate tolerance; use MSUD-specific formula to maintain Leu <200 umol/L (<5y)
- **IV Dextrose 10%** — High-calorie IV glucose (8-10 mg/kg/min) during acute decompensation to promote anabolism

Second-Line

- **Thiamine (B1)** — 50-300 mg/day PO trial for thiamine-responsive MSUD variants; assess response over 3 weeks
- **L-Carnitine** — 50-100 mg/kg/day PO divided TID for fatty acid oxidation defects; max 3 g/day
- **Medium-chain triglyceride oil** — 1-2 g/kg/day PO for long-chain FAO defects; avoid fasting >4-6h in infants

Investigations

- Plasma amino acids (quantitative) — Diagnosis and monitoring — leucine, isoleucine, valine levels [STAT]
- Urine organic acids — Characteristic pattern for MSUD and FAO disorders [URGENT]
- Blood gas — Metabolic acidosis in acute crisis [STAT]
- Blood glucose — Hypoglycemia common in FAO disorders [STAT]
- Acylcarnitine profile — Specific pattern for fatty acid oxidation defects [URGENT]

Counselling

- Your child has a metabolic disorder that requires a special diet for life
- Follow the dietary restrictions strictly — even small deviations can cause crisis
- During fever, vomiting, or illness, take the child to hospital immediately — do not wait
- Never let the child fast for more than 4-6 hours (infants) or 8 hours (older children)
- Regular blood tests to check amino acid levels are needed every 1-3 months

Referral Criteria

Metabolic genetics/biochemical genetics specialist mandatory. Liver transplant center for classic MSUD if eligible. Dietitian specializing in metabolic disorders. Neurology for developmental monitoring

Hospitalisation Criteria

Acute metabolic crisis (encephalopathy, vomiting, metabolic acidosis). Leucine >750 umol/L. Intercurrent illness with poor oral intake. New diagnosis requiring stabilization

Notes

Acute metabolic crises require emergency IV glucose and protein restriction; hemodialysis if leucine >1500 umol/L. Liver transplant is curative for classic MSUD.

E72 — Other disorders of amino-acid metabolism

SNOMED: 190302002 | Duration: 365 days | Source: European HCU Guidelines 2017, ACMG 2024

First-Line

- **Pyridoxine (B6)** — 100-500 mg/day PO trial; ~50% of CBS-deficient HCU patients are B6-responsive
- **Methionine-restricted diet** — Use met-free amino acid formula; target plasma homocysteine <50 umol/L

Second-Line

- **Betaine (trimethylglycine)** — 50-125 mg/kg/day PO divided BID for B6-nonresponsive HCU; max 6 g/day in children
- **Folate + B12** — Folate 1-5 mg/day PO + B12 1 mg/day PO as adjunct to lower homocysteine

Investigations

- Total plasma homocysteine — Primary monitoring marker — target <50 umol/L [URGENT]
- Plasma amino acids (methionine) — Elevated methionine in CBS-deficient HCU [URGENT]
- Pyridoxine responsiveness test — 50% of CBS-deficient patients respond — determines treatment approach
- Eye examination (slit lamp) — Lens subluxation — screen annually
- DEXA scan — Osteoporosis risk — screen every 2-3 years

Counselling

- Your child needs a special diet low in methionine and may need lifelong vitamin supplements
- Pyridoxine (vitamin B6) responsiveness will be tested — if responsive, it greatly simplifies treatment
- Blood tests for homocysteine level are needed every 3-6 months
- Report any sudden vision changes, leg pain, or swelling immediately — these are emergencies
- Ensure the child takes betaine and folate as prescribed — these help lower homocysteine

Referral Criteria

Metabolic genetics mandatory. Ophthalmology for lens dislocation screening. Hematology if thromboembolic event. Cardiology for cardiovascular risk monitoring

Hospitalisation Criteria

Thromboembolic event (stroke, DVT, PE). Lens dislocation requiring surgery. Acute metabolic decompensation. New diagnosis requiring stabilization and dietary initiation

Notes

Test all patients for pyridoxine responsiveness before committing to lifelong dietary restriction. Early treatment prevents intellectual disability, thromboembolism, and lens dislocation.

Guideline Changes

Novel enzyme replacement therapies (pegtibatinase, pegtarviliase) in clinical trials for CBS-deficient HCU

E74 — Other disorders of carbohydrate metabolism

SNOMED: 190302002 | Duration: 365 days | Source: ESGSD I Guidelines, ACMG 2023

First-Line

- **Uncooked cornstarch** — 1-2 g/kg PO q4-6h (older children); maintains euglycemia in GSD I/III; avoid in <1y
- **Frequent carbohydrate feeds** — Complex carbs q2-4h daytime; continuous NG glucose infusion overnight for infants

Second-Line

- **Allopurinol** — 10 mg/kg/day PO divided BID-TID for hyperuricemia in GSD I; max 800 mg/day
- **Glycosade (modified cornstarch)** — 1-2 g/kg PO at bedtime; provides longer glucose release than raw cornstarch; age >5y

Investigations

- Blood glucose (fasting and post-prandial) — Monitor glycemic control [URGENT]
- Serum lactate — Elevated in GSD I during fasting [URGENT]
- Serum uric acid — Hyperuricemia common in GSD I
- Lipid profile — Hyperlipidemia in GSD I
- Liver ultrasound — Hepatomegaly, adenoma surveillance (annually after age 10)

Counselling

- Your child must eat regularly and never skip meals — fasting can cause dangerous low blood sugar
- Cornstarch mixed in cool water is a key medicine — give at prescribed times without fail
- Night-time feeds are critical for infants — set alarms to avoid long gaps
- Carry glucose tablets or sugar whenever travelling in case of emergency
- Regular blood tests for sugar, lactate, uric acid, and liver size checks are needed every 3-6 months

Referral Criteria

Metabolic genetics mandatory. Hepatology for liver adenoma surveillance in GSD I. Nephrology for renal complications. Dietitian specializing in metabolic disorders for cornstarch regimen

Hospitalisation Criteria

Hypoglycemic episode requiring IV dextrose. Lactic acidosis. Intercurrent illness with poor oral intake. New diagnosis requiring initiation of feeding regimen

Notes

Frequent feeding and cornstarch are the mainstay for hepatic GSD; avoid fasting >4h in infants. Monitor lactate, uric acid, lipids, and liver size regularly.

E75 — Sphingolipidosis and other lipid storage disorders

SNOMED: 23585005 | Duration: 365 days | Source: Expert Consensus 2023, FDA Labeling

First-Line

- **Imiglucerase (Gaucher)** — 30-60 U/kg IV every 2 weeks; for Gaucher type 1; adjust based on clinical response
- **Miglustat (Gaucher/NPC)** — Gaucher: 100 mg PO TID (adults); NPC: substrate reduction therapy per weight-based dosing

Second-Line

- **Arimoclomol (NPC)** — PO per weight-based dosing for Niemann-Pick type C; first FDA-approved NPC therapy (2024) **[NEW 2024-25]**
- **Levacetylleucine (NPC)** — PO weight-based dosing for NPC; second FDA-approved NPC therapy (Sep 2024) **[NEW 2024-25]**
- **Velaglucerase alfa** — 60 U/kg IV every 2 weeks for Gaucher type 1 (age >=4y); alternative to imiglucerase

Investigations

- Enzyme activity assay (specific to disorder) — Confirmatory diagnosis — glucocerebrosidase for Gaucher, etc.
- Genetic testing — Confirm mutation, family screening, prenatal diagnosis
- CBC with differential — Cytopenias common in Gaucher
- Liver and spleen ultrasound — Organomegaly monitoring
- MRI brain — CNS involvement in neuronopathic forms

Counselling

- Your child has a rare storage disorder — treatment depends on the specific type
- Enzyme infusions (if applicable) must be given regularly — usually every 2 weeks
- Watch for bone pain, easy bruising, enlarged belly, or difficulty walking and report immediately
- Some types affect the brain — developmental monitoring is important

- Genetic counselling for the family is recommended to understand the inheritance pattern

Referral Criteria

Metabolic genetics mandatory. Hematology for Gaucher disease. Neurology for neuronopathic forms. Hepatology for liver involvement. HSCT team for eligible patients. Psychiatry for NPC (behavioral changes)

Hospitalisation Criteria

First dose of ERT (monitoring for infusion reactions). Acute neurological deterioration. Splenic crisis (Gaucher). Severe anemia or thrombocytopenia requiring transfusion

Notes

ERT is effective for Gaucher type 1 but not for neuronopathic forms. Two new NPC therapies approved in 2024 represent major advances for this previously untreatable condition.

Guideline Changes

Arimoclomol (Miplyffa) and levacetylleucine (Aqneursa) both FDA-approved Sep 2024 for Niemann-Pick C

E76.0 — Mucopolysaccharidosis

SNOMED: 26745009 | Duration: 365 days | Source: Expert Consensus 2022, FDA labeling

First-Line

- **Laronidase (MPS I)** — 0.58 mg/kg (100 U/kg) IV weekly over 3-4h; premedicate with antipyretic/antihistamine
- **Idursulfase (MPS II)** — 0.5 mg/kg IV weekly over 1-3h; monitor for infusion reactions in ~20% patients

Second-Line

- **Galsulfase (MPS VI)** — 1 mg/kg IV weekly over 4h; for Maroteaux-Lamy syndrome
- **HSCT** — Hematopoietic stem cell transplant; preferred for severe MPS I (Hurler) if <2.5y with DQ >70

Investigations

- Urine glycosaminoglycans (GAGs) — Screening test — elevated in MPS
- Enzyme activity assay (specific to MPS type) — Confirmatory — alpha-L-iduronidase for MPS I, iduronate-2-sulfatase for MPS II
- Genetic testing — Confirm mutation, guide family counselling
- Echocardiogram — Valve disease screening
- MRI brain and spine — Hydrocephalus, spinal cord compression

Counselling

- Your child has a storage disorder that causes substances to build up in the body over time
- Enzyme replacement therapy (ERT) is given as a weekly IV infusion — do not miss sessions
- The child may have a mild reaction during infusion (fever, rash) — premedication is given to prevent this
- Regular eye, ear, and heart checkups are needed as the condition can affect multiple organs
- Early treatment gives the best results — please keep all follow-up appointments

Referral Criteria

Metabolic genetics mandatory. HSCT team referral for MPS I Hurler (age <2.5y). Orthopedics for skeletal complications. ENT for airway management. Cardiology for valve disease. Ophthalmology for corneal clouding

Hospitalisation Criteria

Initiation of ERT (first dose monitoring). Acute airway compromise. Cardiac decompensation. HSCT admission. Spinal cord compression (odontoid hypoplasia). Hydrocephalus

Notes

ERT does not cross the blood-brain barrier; HSCT is preferred for MPS I Hurler phenotype if done early. Early diagnosis via newborn screening improves outcomes substantially.

Guideline Changes

Gene therapy trials advancing for MPS I/II; intrathecal ERT for CNS involvement under study

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E84 — Cystic fibrosis

SNOMED: 190905008 | Duration: 365 days | Source: CFF 2024, ISPAD 2024

First-Line

- **Elexacaftor/tezacaftor/ivacaftor (Trikafta)** — Age 6-11y <30kg: ELX50/TEZ25/IVA37.5 AM + IVA75 PM; >=30kg or >=12y: ELX100/TEZ50/IVA75 AM + IVA150 PM; with fat
- **Dornase alfa** — 2.5 mg nebulized daily; for age >=6y to improve mucus clearance and lung function

Second-Line

- **Ivacaftor (Kalydeco)** — For gating mutations; age 1mo-5y: 25-75 mg PO BID by weight; >=6y: 150 mg PO q12h with fat
- **Pancreatic enzymes (PERT)** — 500-2500 lipase U/kg/meal PO; max 10,000 U/kg/day; for pancreatic insufficiency
- **Hypertonic saline 7%** — 4 mL nebulized BID; for age >=6y; improves mucociliary clearance

Investigations

- Sweat chloride test — Gold standard for CF diagnosis — Cl >60 mmol/L diagnostic
- CFTR genetic testing — Identify mutation — guides CFTR modulator eligibility
- Pulmonary function tests (spirometry) — FEV1 monitoring — baseline and q3-6 months (age >=6)
- Sputum culture — Identify respiratory pathogens — Pseudomonas, Staph aureus
- Fecal elastase — Confirm pancreatic insufficiency

Counselling

- Your child has cystic fibrosis — daily chest physiotherapy and medicines are essential for life
- Give the CFTR modulator (Trikafta) with fatty food every day at the same time
- Do airway clearance (chest physiotherapy) twice a day — even when the child seems well
- Pancreatic enzyme capsules must be taken with every meal and snack — before eating
- Call the CF team immediately if the child has fever, increased cough, or colored sputum

Referral Criteria

Pediatric pulmonology (CF center) mandatory for all cases. Gastroenterology for pancreatic insufficiency. Endocrinology for CF-related diabetes. ENT for nasal polyps. Lung transplant evaluation if FEV1 <30% predicted

Hospitalisation Criteria

Pulmonary exacerbation requiring IV antibiotics. Newly diagnosed CF for initiation of care. Hemoptysis. Pneumothorax. Meconium ileus equivalent. Distal intestinal obstruction syndrome

Notes

Trikafta has transformed CF care with significant lung function improvement and reduced mortality. Start CFTR modulators at earliest eligible age to attenuate disease progression.

Guideline Changes

Trikafta approved down to age 2y (2023); Canadian consensus recommends starting CFTR modulators at youngest age possible

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E88 — Other metabolic disorders

SNOMED: 75934005 | Duration: 365 days | Source: Expert Consensus, NICE ES11

First-Line

- **Coenzyme Q10** — 10-30 mg/kg/day PO divided BID-TID for mitochondrial disorders; higher doses not necessarily more effective

- **L-Carnitine** — 50-100 mg/kg/day PO divided TID for carnitine deficiency or secondary deficiency; max 3 g/day

Second-Line

- **Riboflavin (B2)** — 50-400 mg/day PO for Complex I/II defects and MADD; assess response over 3 months
- **Alpha-lipoic acid** — 5-10 mg/kg/day PO as mitochondrial antioxidant adjunct

Investigations

- Serum lactate and pyruvate — Elevated lactate:pyruvate ratio suggests mitochondrial disease [URGENT]
- Plasma acylcarnitine profile — Screen for fatty acid oxidation defects
- Urine organic acids — Metabolic pathway abnormalities
- Genetic testing (mitochondrial panel/WES) — Identify specific mutation for diagnosis and prognosis
- Echocardiogram — Screen for cardiomyopathy in mitochondrial disease

Counselling

- Your child has a metabolic disorder that affects energy production in the body's cells
- Give the vitamin and supplement cocktail prescribed (CoQ10, carnitine, riboflavin) daily without fail
- Avoid prolonged fasting and strenuous exercise — both can trigger a crisis
- During illness or fever, bring the child to hospital early — do not wait for things to worsen
- Regular follow-up every 3-6 months with the metabolic team is essential

Referral Criteria

Metabolic genetics specialist mandatory. Neurology for seizures or regression. Cardiology for cardiomyopathy. Ophthalmology for retinopathy. Audiology for sensorineural hearing loss

Hospitalisation Criteria

Acute metabolic crisis (lactic acidosis, seizures, stroke-like episodes). Cardiac decompensation. Intercurrent illness with poor oral intake. New diagnosis requiring evaluation

Notes

Metabolic disorders require individualized cocktail therapy targeting specific enzyme defects. Evidence for mitochondrial supplements is limited but CoQ10 and carnitine are widely used.

Guideline Changes

FDA granted Rare Pediatric Disease designation for BPM31510IV for primary CoQ10 deficiency (2024)

GI

A03 — Shigellosis

SNOMED: 36188001 | Duration: 5 days | Source: CDC 2023, AAP Red Book 2024, WHO 2023

First-Line

- **Azithromycin** — 12 mg/kg PO day 1 (max 500 mg), then 6 mg/kg/day PO days 2-5 (max 250 mg)
- **Ceftriaxone** — 50 mg/kg/day IV/IM q24h (max 2 g) x5 days for resistant or severe cases

Second-Line

- **Ciprofloxacin** — 15 mg/kg PO BID (max 500 mg/dose) x3 days if susceptible
- **TMP-SMX** — 8 mg/kg/day TMP PO divided BID x5 days if susceptible
- **Pivmecillinam** — 20 mg/kg PO QID x5 days (max 400 mg/dose); alternative per WHO

Investigations

- Stool culture and sensitivity — Confirm Shigella and guide antibiotic therapy — resistance is common [URGENT]
- CBC with peripheral smear — Leukocytosis; schistocytes if HUS developing [URGENT]
- Serum electrolytes — Dehydration, electrolyte imbalance [URGENT]
- Renal function (creatinine, BUN) — Monitor for HUS [URGENT]
- Blood culture — If bacteremia suspected [URGENT]

Counselling

- Your child has Shigella infection — this is a serious bacterial diarrhea that needs antibiotics
- Give the full course of antibiotics even if the child improves before finishing
- Maintain hydration with ORS — the child may have bloody stools which is common in this infection
- Strict hand hygiene is essential — Shigella spreads very easily from person to person
- Come back if: blood in stools increases, child develops high fever, or stops urinating

Referral Criteria

Infectious disease for XDR Shigella or treatment failure. Nephrology if HUS develops. GI if chronic carrier state. ICU if septic shock or toxic megacolon

Hospitalisation Criteria

Severe dehydration. Toxic appearance (high fever, altered mental status, lethargy). Bloody diarrhea with hemodynamic instability. Age <6 months. Suspected HUS or toxic megacolon. Immunocompromised child

Notes

Always obtain stool culture and susceptibility before or concurrent with empiric therapy. XDR strains increasing; ceftriaxone preferred if resistance suspected.

Guideline Changes

CDC 2023 alert on increasing XDR Shigella strains; azithromycin resistance rising

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A03.9 — Bacillary Dysentery

SNOMED: 36188001 | Duration: 5 days | Source: IAP Diarrhea Guidelines, WHO Diarrheal Disease Management, CDC Shigella Treatment Guidelines 2024, IDSA Infectious Diarrhea Guidelines

First-Line

- **Oral Rehydration Salts** — 10-20 ml/kg after each loose stool, frequent small sips, life-saving
- **Cefixime** — 8-10 mg/kg/day div BD PO for 5-7 days
- **Zinc** — <6 months: 10 mg/day; >=6 months: 20 mg/day PO for 14 days, reduces severity and recurrence
- **Metronidazole** — 10 mg/kg/dose TDS PO for 5 days, for suspected amoebic or mixed infection

- **Saccharomyces boulardii** — Once or twice daily PO for 5 days, probiotic to restore gut flora and reduce diarrhea duration
- **Dicyclomine + Simethicone** — Age/weight appropriate dose TDS PO, short-term, for abdominal pain/cramps

Second-Line

- **Colistin** — Consider for persistent diarrhea or resistant infection, not routine first-line

Investigations

- Complete blood count — Assess infection severity [STAT]
- CRP — Assess degree of inflammation
- Stool routine and microscopy — Identify blood, pus cells, organisms [STAT]
- Stool culture and sensitivity — Severe or non-responding cases to identify organism [URGENT]

Counselling

- ORS is the most important treatment - give after every loose stool
- Continue normal feeding, do not stop food - give curd, khichdi, rice, banana
- Give zinc syrup daily for 14 days even after diarrhea stops
- Complete full course of antibiotic as prescribed
- Wash hands before feeding and after toilet to prevent spread
- Come immediately if blood in stool increases, persistent vomiting, no urine, child becomes lethargic or severely dehydrated

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Increasing blood in stools
- Persistent vomiting or unable to drink
- No urine output for more than 6 hours
- Signs of severe dehydration such as sunken eyes or dry mouth

Referral Criteria

Refer if no improvement in 48 hours on antibiotics, suspected hemolytic uremic syndrome, recurrent dysentery, or suspected resistant organism

Hospitalisation Criteria

Admit if severe dehydration, unable to drink or retain ORS, bloody stool with hemodynamic instability, persistent vomiting, lethargy/altered sensorium, or age <2 months

Notes

ORS + Zinc = cornerstone of therapy. Continue normal feeding. Antibiotic indicated for bloody dysentery. Preferred diet: curd, buttermilk, khichdi, rice, banana, coconut water. Avoid junk food and sugary drinks.

Guideline Changes

WHO 2024: Rising antimicrobial resistance in Shigella globally - azithromycin resistance increased significantly; CDC 2023: Alert on extensively drug-resistant (XDR) Shigella strains; ceftriaxone or azithromycin recommended where ciprofloxacin resistance is high; stool culture and sensitivity recommended for non-responding cases; IAP 2024: ORS + Zinc remains cornerstone; cefixime preferred empiric oral antibiotic for pediatric bloody dysentery in India

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A04 — Other bacterial intestinal infections

SNOMED: 186109009 | Duration: 7 days | Source: IDSA 2017, ESPGHAN 2014, AAP Red Book 2024

First-Line

- **Azithromycin** — 10 mg/kg/day PO daily x5 days (max 500 mg/day) for Campylobacter/Shigella
- **Ceftriaxone** — 50-75 mg/kg/day IV/IM q24h (max 2 g) for invasive Salmonella or severe cases
- **ORS + supportive care** — Rehydration first-line; antibiotics only for invasive/toxic presentations

Second-Line

- **Ciprofloxacin** — 10-15 mg/kg PO BID (max 500 mg/dose) x5-7 days if susceptible
- **TMP-SMX** — 8 mg/kg/day TMP PO divided BID x5 days if susceptible
- **Metronidazole** — 30 mg/kg/day PO divided TID x10 days for C. difficile

Investigations

- Stool culture and sensitivity — Identify pathogen and antibiotic sensitivity [URGENT]
- Stool routine and microscopy — WBCs, RBCs, parasites
- CBC with differential — Leukocytosis, bandemia in invasive disease [URGENT]
- Blood culture — If bacteremia/sepsis suspected [URGENT]
- Renal function and CBC — Screen for HUS if STEC suspected (Hb, platelets, creatinine) [URGENT]

Counselling

- Your child has a bacterial gut infection — antibiotics are given only if the infection is serious or specific bacteria are found
- Give ORS after every loose stool to prevent dehydration
- Continue normal feeding — stopping food does not help and slows recovery
- Practice strict hand hygiene — wash hands with soap before meals and after toilet
- Ensure safe drinking water — boil or filter water before giving to the child

Referral Criteria

Pediatric GI if diarrhea persists >14 days. Infectious disease for resistant organisms or immunocompromised patients. Surgery if toxic megacolon or perforation suspected

Hospitalisation Criteria

Severe dehydration. Systemic toxicity (high fever, altered mental status). Bloody diarrhea with hemodynamic instability. Immunocompromised child with invasive disease. Suspected HUS (after STEC infection)

Notes

Most bacterial gastroenteritis is self-limited; antibiotics reserved for invasive disease, immunocompromised, or specific pathogens. Always check local resistance patterns.

Guideline Changes

2024 ESPGHAN/NASPGHAN H. pylori update emphasizes susceptibility-guided therapy

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A08.4 — Viral intestinal infection, unspecified

SNOMED: 186150001 | Duration: 5 days | Source: ESPGHAN/ESPID 2014, AAP 2023, WHO 2024

First-Line

- **ORS (reduced osmolarity)** — 50-100 mL/kg PO over 3-4h for mild-mod dehydration
- **Zinc sulfate** — 10 mg/day PO (<6mo) or 20 mg/day PO (>6mo) x10-14 days
- **Ondansetron** — 0.15 mg/kg PO/IV single dose (max 4 mg) for persistent vomiting

Second-Line

- **Racecadotril** — 1.5 mg/kg PO TID x5-7 days; antisecretory adjunct
- **Saccharomyces boulardii** — 250-750 mg/day PO x5-7 days; adjunctive probiotic

Investigations

- Stool rotavirus/norovirus rapid test — Identify specific virus — helps epidemiological tracking

- Serum electrolytes — If moderate-severe dehydration — guide fluid therapy [URGENT]
- Blood glucose — Hypoglycemia risk in young children with poor intake [STAT]
- Stool routine — Rule out bacterial/parasitic co-infection

Counselling

- Your child has a viral stomach infection — it usually gets better on its own in 3-5 days
- No antibiotics are needed — viruses do not respond to antibiotics
- Give ORS sip by sip and continue feeding — this is the most important treatment
- Wash hands thoroughly and keep the child's utensils separate to prevent spread to other family members
- Rotavirus vaccine can prevent the most severe forms of viral diarrhea — ask about it at next visit

Referral Criteria

Pediatric GI if diarrhea persists >14 days (consider post-infectious lactose intolerance). Immunology if recurrent or severe viral GI infections suggesting immunodeficiency

Hospitalisation Criteria

Severe dehydration requiring IV fluids. Failed oral rehydration despite ondansetron. Age <3 months with moderate-severe dehydration. Seizures from electrolyte imbalance

Notes

No antiviral therapy available; treatment is entirely supportive. Rotavirus vaccination remains the most effective prevention strategy.

A09 — Infectious gastroenteritis and colitis, unspecified

SNOMED: 25374005 | Duration: 5 days | Source: ESPGHAN/ESPID 2014 (updated 2023), WHO 2024

First-Line

- **ORS (reduced osmolarity)** — 50-100 mL/kg PO over 3-4h for mild-mod dehydration
- **Zinc sulfate** — 10 mg/day PO (<6mo) or 20 mg/day PO (>6mo) for 10-14 days
- **Ondansetron** — 0.15 mg/kg PO/IV single dose (max 4 mg) for persistent vomiting

Second-Line

- **Racecadotril** — 1.5 mg/kg PO TID for up to 7 days; antisecretory
- **Saccharomyces boulardii** — 250-750 mg/day PO for 5-7 days; adjunctive probiotic
- **Lactobacillus rhamnosus GG** — 10 billion CFU/day PO for 5-7 days; adjunctive probiotic

Investigations

- Stool routine and microscopy — WBCs, RBCs, parasites — differentiate inflammatory vs non-inflammatory
- Stool culture — If bloody diarrhea, high fever, or antibiotic use planned
- Serum electrolytes — Dehydration assessment, guide fluid therapy [URGENT]
- CBC — If systemic infection or dehydration suspected

Counselling

- Your child has a stomach infection causing loose stools — ORS and zinc are the main treatment
- Give ORS after every loose stool: under 2 years = 50-100mL, over 2 years = 100-200mL
- Continue normal feeding — do not starve the child. Breast milk is the best food during illness
- Give zinc syrup for 10-14 days as prescribed — it reduces the duration and severity of diarrhea
- Wash hands with soap before feeding and after toilet — this prevents spread of infection

Referral Criteria

Pediatric GI if bloody diarrhea persisting >5 days, or diarrhea >14 days (persistent diarrhea). Infectious disease if unusual pathogen or immunocompromised child. Surgical consultation if toxic megacolon suspected

Hospitalisation Criteria

Severe dehydration requiring IV fluids. Failed oral rehydration. Bloody diarrhea with systemic toxicity (high fever, altered mental status). Age <3 months with moderate-severe dehydration. Suspected surgical abdomen

Notes

Antibiotics not routinely indicated; reserve for specific pathogens or toxic/invasive presentations. Early refeeding with regular diet improves outcomes.

Guideline Changes

2023 WHO update reinforces zinc supplementation and reduced-osmolarity ORS

B15 — Acute hepatitis A

SNOMED: 40468003 | Duration: 14 days | Source: CDC 2023, WHO 2024, AAP Red Book 2024

First-Line

- **Supportive care** — Rest, adequate hydration, balanced nutrition. No specific antiviral therapy available
- **Acetaminophen (cautious)** — 10-15 mg/kg PO q4-6h (max 75 mg/kg/day) for fever/pain; AVOID if hepatotoxicity concern
- **IV fluids** — NS/D5NS maintenance if unable to tolerate PO; correct dehydration from vomiting

Second-Line

- **Ondansetron** — 0.15 mg/kg IV/PO q8h (max 4 mg/dose) for nausea/vomiting
- **Cholestyramine** — 240 mg/kg/day PO divided TID (max 4 g/dose) for severe pruritus

Investigations

- HAV IgM antibody — Confirms acute hepatitis A infection [URGENT]
- Liver function tests (AST, ALT, bilirubin) — Assess severity of hepatitis [URGENT]
- PT/INR — Liver synthetic function — coagulopathy suggests severe disease [URGENT]
- CBC — Baseline evaluation
- Serum albumin — Liver synthetic function assessment

Counselling

- Your child has hepatitis A — this is a viral infection of the liver that usually heals on its own
- Give plenty of fluids and a balanced diet — no special dietary restrictions are needed
- Avoid giving paracetamol in high doses — it can stress the liver. Use only as directed by the doctor
- The child is contagious for about 2 weeks — maintain strict hand hygiene and do not share utensils
- Hepatitis A vaccine can protect other family members — ask about vaccination

Referral Criteria

Hepatology/pediatric GI if signs of liver failure (coagulopathy, encephalopathy, persistent jaundice >6 weeks). Liver transplant center if fulminant hepatitis develops

Hospitalisation Criteria

Persistent vomiting with dehydration. INR >1.5 (coagulopathy). Hepatic encephalopathy (altered behavior, confusion). Severe jaundice with liver synthetic failure

Notes

Self-limited in children with excellent prognosis; fulminant hepatitis rare (<1%). Avoid hepatotoxic medications; no dietary restrictions needed. HAV vaccination prevents disease.

B16 — Acute hepatitis B

SNOMED: 76795007 | Duration: 14 days | Source: EASL 2025, AAP Red Book 2024, WHO 2024

First-Line

- **Supportive care** — Rest, hydration, nutrition. Most acute HBV is self-limited in older children
- **IV fluids** — NS maintenance for dehydration; monitor glucose, coagulation, LFTs serially

Second-Line

- **Entecavir** — 0.015 mg/kg PO QD (max 0.5 mg) for severe/protracted acute HBV or fulminant hepatitis
- **Tenofovir disoproxil** — 8 mg/kg PO QD (max 300 mg) age >=2yr; for severe acute HBV (off-label)

Investigations

- HBsAg, HBV DNA, HBeAg — Confirm acute HBV; viral load for severity [URGENT]
- Anti-HBc IgM — Confirms acute (not chronic) HBV infection [URGENT]
- Liver function tests — AST, ALT, bilirubin for severity assessment [URGENT]
- PT/INR — Coagulopathy indicates severe hepatitis [URGENT]
- CBC — Baseline assessment

Counselling

- Your child has hepatitis B — most older children recover fully, but monitoring is important
- Blood tests will be done over the next 6 months to make sure the virus clears completely
- If the virus does not clear, long-term antiviral treatment may be needed
- All family members should be tested and vaccinated against hepatitis B
- For newborns: HBIG and HBV vaccine within 12 hours of birth prevents mother-to-child transmission

Referral Criteria

Pediatric hepatology/GI for all cases. Infectious disease for chronic HBV requiring antivirals. Liver transplant center if fulminant hepatitis. Obstetrics for pregnant mothers (HBV-positive) for prevention of vertical transmission

Hospitalisation Criteria

Severe acute HBV with coagulopathy (INR >1.5). Hepatic encephalopathy. Fulminant hepatic failure. Severe dehydration from persistent vomiting. Newborn of HBV-positive mother for HBIG + vaccine

Notes

Acute HBV rarely needs antivirals; interferon is contraindicated in acute phase. Neonates require HBIG + HBV vaccine within 12h of birth to prevent vertical transmission.

Guideline Changes

2024 WHO expanded treatment criteria for chronic HBV; EASL 2025 updated management guidance

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B17.1 — Acute hepatitis C

SNOMED: 235866006 | Duration: 84 days | Source: AASLD/IDSA 2023, AAP 2024

First-Line

- **Sofosbuvir/velpatasvir (Epclusa)** — Weight-based: <17kg 150/37.5mg, 17-<30kg 200/50mg, >=30kg 400/100mg PO QD x12wk **[NEW 2024-25]**
- **Glecaprevir/pibrentasvir (Mavyret)** — Weight-based granules or tablets PO QD x8wk; pangenotypic, age >=3yr **[NEW 2024-25]**

Second-Line

- **Ledipasvir/sofosbuvir (Harvoni)** — Weight-based PO QD x12wk; for genotypes 1,4,5,6; age >=3yr
- **Monitoring only (<3yr)** — No DAAs approved <3yr; observe for spontaneous clearance (high in infants)

Investigations

- HCV RNA (quantitative PCR) — Confirm viremia and measure viral load [URGENT]
- HCV genotype — Guide DAA regimen selection
- Liver function tests — Baseline liver assessment [URGENT]

- Liver elastography (FibroScan) or ultrasound — Assess fibrosis stage
- CBC — Baseline; rule out cytopenias from chronic disease

Counselling

- Your child has hepatitis C — new oral medicines can cure this infection in 8-12 weeks
- The medicines must be taken every day at the same time — do not miss doses
- Treatment is available for children aged 3 years and above with very high cure rates (>95%)
- Hepatitis C spreads through blood — avoid sharing razors, toothbrushes, or needles
- After completing treatment, a blood test at 12 weeks will confirm if the virus is cured

Referral Criteria

Pediatric hepatology/GI mandatory for DAA therapy initiation. Infectious disease for complex cases. Hematology if acquired through blood products. Liver transplant if cirrhosis develops

Hospitalisation Criteria

Fulminant hepatitis (rare). Decompensated cirrhosis. Severe liver failure. Generally DAA treatment is outpatient

Notes

Treat all children >=3yr with chronic HCV; spontaneous clearance high in infants so defer treatment. SVR rates >95% with current DAA regimens in children.

Guideline Changes

Pan-genotypic DAAs now FDA-approved for children >=3yr; child-friendly formulations available

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B17.9 — Acute Severe Viral Hepatitis (with risk of Acute Liver Failure / Hyperammonemia) *[severe]*

SNOMED: 3738000 | Duration: 5 days | Source: IAP 2024,WHO Guidelines,AAP,Pediatric Hepatology Protocols

First-Line

- **ACETYLCYSTEINE (NAC)** — IV infusion regimen: Loading 125 mg/kg over 1 hour, then 50 mg/kg over 4 hours, then 125 mg/kg over 19 hours; Day 2 onward: 150 mg/kg over 24 hours, repeat next 24 hours as required. Preparation: Add 1000 mg NAC (2 ampoules of 5 ml each) in 40 ml of 5% DEXTROSE; adjust total volume and rate as per weight-based dose and infusion duration
- **VITAMIN K** — 10 mg IM once daily for 3 days for coagulopathy correction
- **URSODEOXYCHOLIC ACID** — 5 mg/kg/dose PO every 12 hours for 21 days for cholestasis and bile flow improvement
- **LACTULOSE** — 1 ml/kg/dose PO 2–3 times daily; titrate to achieve 2–3 soft stools/day for prevention/treatment of hepatic encephalopathy
- **SODIUM BENZOATE** — 250–500 mg/kg/day PO in 4 divided doses every 6 hours for hyperammonemia management
- **INTRAVENOUS FLUIDS (DNS 0.9% + MULTIVITAMIN INFUSION)** — Maintenance IV fluids as per Holliday-Segar calculation; avoid fluid overload; add MVI as per standard pediatric dosing

Second-Line

- **RIFAXIMIN** — 10–15 mg/kg/day PO in 2 divided doses; use if persistent or worsening hepatic encephalopathy despite lactulose

Investigations

- BLOOD GLUCOSE — Monitor for hypoglycemia in acute liver failure [STAT]
- SERUM AMMONIA — Assess and monitor hyperammonemia and encephalopathy [URGENT]
- PT / INR — Assess severity of liver dysfunction and coagulopathy [STAT]
- LIVER FUNCTION TEST (LFT) — Monitor disease progression and hepatic injury [URGENT]
- RENAL FUNCTION TEST (RFT) — Assess renal involvement and hepatorenal syndrome risk [URGENT]

Counselling

- Child has severe liver infection which can affect brain and blood clotting; close monitoring is required — बच्चे को गंभीर लिवर संक्रमण है जो दिमाग और खून के थक्कों को प्रभावित कर सकता है; नज़दीकी निगरानी ज़रूरी है
- Strict medication adherence is essential; do not miss doses especially lactulose and NAC — दवाइयाँ समय पर लेना बहुत जरूरी है; खासकर लैक्टुलोज़ और NAC की खुराक न छोड़ें
- Avoid outside food, ensure adequate carbohydrate intake and hydration — बाहर का खाना न दें, पर्याप्त कार्बोहाइड्रेट और पानी दें
- Do NOT give PARACETAMOL, antiemetics, or any over-the-counter drugs without doctor advice — डॉक्टर की सलाह के बिना पैरासिटामोल, उल्टी की दवा या कोई भी दवा न दें
- Immediate follow-up required if child becomes drowsy, vomits repeatedly, or stops feeding — अगर बच्चा सुस्त हो जाए, बार-बार उल्टी करे या दूध न पिए तो तुरंत डॉक्टर को दिखाएं

Warning Signs

- Altered sensorium or excessive sleepiness
- Persistent vomiting
- Bleeding from gums / nose
- Seizures or abnormal movements
- Reduced urine output
- Worsening jaundice or abdominal distension

Referral Criteria

Refer urgently to pediatric gastroenterologist or liver transplant center if INR >2, rising ammonia, worsening encephalopathy, or no improvement within 24–48 hours

Hospitalisation Criteria

All cases of severe hepatitis with coagulopathy, encephalopathy, persistent vomiting, hypoglycemia, or rising bilirubin/ammonia require admission (preferably ICU monitoring)

Notes

Acute severe viral hepatitis can progress to acute liver failure; early NAC improves outcomes even in non-paracetamol liver injury. Strict drug avoidance (especially hepatotoxic drugs) is critical.

Guideline Changes

NAC recommended even in non-acetaminophen acute liver failure (updated global practice)

E86 — Dehydration

SNOMED: 34095006 | Duration: 3 days | Source: ESPGHAN/ESPID 2014, WHO 2023

First-Line

- **ORS (hypoosmolar)** — 50-100 mL/kg PO over 3-4h; 5 mL q5min initially, increase as tolerated
- **IV Normal Saline (0.9%)** — 20 mL/kg IV bolus for severe dehydration; repeat up to 3 boluses if needed
- **IV Ringer Lactate** — 20 mL/kg IV bolus for severe dehydration; alternative to NS

Second-Line

- **Ondansetron** — 0.15 mg/kg PO/IV single dose (max 4 mg) to facilitate oral rehydration
- **NG ORS** — Nasogastric ORS at 20 mL/kg/h if PO fails and IV not available

Investigations

- Serum electrolytes — Na, K — classify isotonic/hypo/hypernatremic dehydration [STAT]
- Blood urea and creatinine — Assess renal function, severity of dehydration [URGENT]
- Blood glucose — Hypoglycemia risk especially in young children [STAT]
- Venous blood gas — Assess metabolic acidosis severity [URGENT]

Counselling

- Your child is dehydrated — give ORS (oral rehydration solution) in small sips every few minutes
- Do not give fruit juice, cola, or sweetened drinks — they can worsen diarrhea
- Continue breastfeeding — it is the best fluid for a dehydrated baby
- If the child vomits, wait 10 minutes and try again with smaller sips
- Come back immediately if: child becomes very drowsy, eyes look sunken, no tears when crying, or no urine for 6 hours

Referral Criteria

Pediatric emergency if severe dehydration with shock. GI referral if chronic/recurrent dehydration suggesting underlying disease. Endocrinology if diabetes insipidus or adrenal insufficiency suspected

Hospitalisation Criteria

Severe dehydration (>10% weight loss, altered mental status, hemodynamic instability). Failed oral rehydration. Persistent vomiting preventing ORS intake. Infant <3 months with moderate-severe dehydration

Notes

Oral rehydration is first-line for mild-moderate dehydration; IV reserved for severe or failed ORT. Continue breastfeeding; avoid sugary fluids.

Guideline Changes

Emphasis on hypoosmolar ORS and early refeeding confirmed in 2023 updates

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E86.0 — Dehydration (Acute Gastroenteritis related)

SNOMED: 34095006 | Duration: 3 days | Source: WHO Diarrhea Guidelines 2023, IAP 2024

First-Line

- **Oral Rehydration Solution (ORS)** — Plan A: <2 yrs: 50–100 ml after each loose stool; 2–10 yrs: 100–200 ml; >10 yrs: as tolerated
- **Oral Rehydration Solution (ORS)** — Plan B: 75 ml/kg over 4 hours (for some dehydration), reassess after 4 hrs
- **Ringer Lactate (RL) / Normal Saline (NS)** — Plan C: 100 ml/kg IV (30 ml/kg rapid then 70 ml/kg over 2.5–5 hrs depending on age)
- **Zinc** — <6 months: 10 mg OD for 14 days; >6 months: 20 mg OD for 14 days

Second-Line

- **Ondansetron** — 0.15 mg/kg/dose PO/IV if persistent vomiting interfering with ORS
- **Probiotics** — Optional; may reduce duration of diarrhea

Investigations

- Serum Electrolytes — Moderate to severe dehydration [URGENT]
- Blood Urea/Creatinine — Severe dehydration or shock [URGENT]
- Stool Routine/Microscopy — Persistent diarrhea or dysentery
- Blood Glucose — Lethargy or altered sensorium [STAT]

Counselling

- Give ORS frequently in small amounts — ORS बार-बार थोड़ी-थोड़ी मात्रा में पिलाएं।
- Continue breastfeeding and normal diet — माँ का दूध और सामान्य खाना जारी रखें।
- Avoid sugary drinks and packaged juices — मीठे पेय और पैकेट वाले जूस न दें।
- Maintain hand hygiene to prevent spread — हाथों की सफ़ाई रखें ताकि बीमारी न फैले।
- Follow-up if no improvement in 2–3 days — 2-3 दिन में सुधार न हो तो दोबारा डॉक्टर को दिखाएं।

Warning Signs

- Lethargy or unconsciousness
- Sunken eyes and poor drinking

- Persistent vomiting
- No urine for >6 hours
- Blood in stool
- Severe weakness or shock signs

Referral Criteria

Refer if severe dehydration, shock, persistent vomiting, or no improvement after ORS therapy

Hospitalisation Criteria

Severe dehydration, inability to drink, repeated vomiting, shock, altered sensorium

Notes

ORS is cornerstone therapy. Avoid unnecessary antibiotics unless dysentery or specific infection suspected.

Guideline Changes

Low osmolar ORS recommended universally; early feeding emphasized

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K12 — Stomatitis and related lesions

SNOMED: 61170000 | Duration: 10 days | Source: AAPD 2024, AAP 2023

First-Line

- **Oral analgesic rinse** — Magic mouthwash (diphenhydramine + Maalox +/- lidocaine) swish-spit q4-6h for pain
- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day) for pain and fever
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day) for pain and inflammation; age >6mo

Second-Line

- **Acyclovir** — 20 mg/kg/dose PO 4x/day (max 800 mg) x7 days for herpetic gingivostomatitis
- **Chlorhexidine rinse 0.12%** — 15 mL swish-spit BID x10 days; prevent secondary infection of ulcers
- **Topical triamcinolone 0.1%** — Apply to ulcers TID-QID after meals; for recurrent aphthous stomatitis
- **Nystatin suspension** — 1-5 mL PO swish-spit QID x7-14 days for oral candidiasis

Investigations

- Clinical examination — Usually sufficient — diagnose HSV, HFM, aphthous clinically
- HSV PCR or DFA from lesion — If herpetic gingivostomatitis suspected and immunocompromised [URGENT]
- CBC with differential — Rule out neutropenia if recurrent ulcers
- KOH smear — If oral candidiasis (thrush) suspected

Counselling

- Your child has mouth sores — this is usually from a virus and heals in 7-10 days
- Give cold/soft foods: ice cream, yogurt, cold milk — avoid spicy, salty, or acidic foods
- Keep the child hydrated — use a syringe or cup if the child refuses to drink due to pain
- Give pain medicine (paracetamol/ibuprofen) 30 minutes before meals so the child can eat
- The child is contagious — wash hands frequently and keep separate utensils for a few days

Referral Criteria

ENT/oral surgery if large ulcers not healing >3 weeks. Dermatology for recurrent aphthous stomatitis. Immunology if recurrent severe oral infections (immunodeficiency screening). Oncology if non-healing ulcer suspicious for malignancy

Hospitalisation Criteria

Severe dehydration from inability to drink. Herpetic gingivostomatitis with refusal of all oral intake. Oral candidiasis with systemic spread (immunocompromised). Airway compromise from severe oral/pharyngeal swelling

Notes

Identify etiology: viral (HSV, enterovirus), aphthous, fungal, or trauma-related. Maintain hydration; soft/bland diet. Herpetic stomatitis most common in children 1-5 years.

K20 — Esophagitis

SNOMED: 235598006 | Duration: 56 days | Source: ACG 2025, ESPGHAN 2024 EoE Guidelines

First-Line

- **Omeprazole** — 1-2 mg/kg/day PO divided QD-BID (max 40 mg/day) x8-12 weeks
- **Esomeprazole** — 1 mg/kg/day PO QD (max 40 mg); FDA-approved for pediatric erosive esophagitis
- **Swallowed fluticasone** — 88-440 mcg swallowed BID (age-based); for eosinophilic esophagitis

Second-Line

- **Oral viscous budesonide** — 0.5-1 mg PO BID (mixed with sucralose); for eosinophilic esophagitis
- **Dupilumab** — SC q2-4wk weight-based dosing; for EoE refractory to PPI/steroids, age >=1yr **[NEW 2024-25]**
- **Elimination diet** — 2-food or 6-food empiric elimination diet for EoE; dietitian-guided

Investigations

- Upper GI endoscopy with biopsy — Diagnose cause — reflux esophagitis vs EoE (>15 eos/HPF)
- CBC with eosinophil count — Peripheral eosinophilia may suggest EoE
- Serum total IgE — Atopic predisposition in EoE
- Stool occult blood — GI bleeding assessment

Counselling

- Your child has inflammation of the food pipe (esophagus) — medicine will help it heal
- Give the PPI (omeprazole/esomeprazole) 30 minutes before breakfast on an empty stomach
- If food allergy is suspected, an elimination diet may be tried under doctor's guidance
- Avoid spicy, acidic, and fried foods that may worsen symptoms
- An endoscopy (camera test) may be needed to check the cause — it is safe in children

Referral Criteria

Pediatric gastroenterology for endoscopy and biopsy. Allergy/immunology if eosinophilic esophagitis diagnosed. Dietitian for elimination diet guidance. Surgery for refractory strictures

Hospitalisation Criteria

Dysphagia with food impaction (emergency endoscopy needed). Severe dehydration from inability to swallow. GI bleeding. Esophageal stricture requiring dilation

Notes

Distinguish reflux esophagitis (PPI-responsive) from eosinophilic esophagitis (may need topical steroids or biologics). EGD with biopsy required for EoE diagnosis.

Guideline Changes

PPI-nonresponsive EoE no longer a diagnostic category; dupilumab approved for EoE age 1+ (2022)

K21.0 — Gastroesophageal Reflux (GER/GERD)

SNOMED: 235595009 | Duration: 21 days | Source: ESPGHAN/NASPGHAN 2018 Pediatric GERD Clinical Practice Guidelines, IAP GI Protocol

First-Line

- **Positional Therapy (non-pharmacological)** — Keep baby head elevated ~30 degrees, hold upright 15-20 min after feeds, avoid immediate lying after feeding; first-line treatment
- **Feeding Modification (non-pharmacological)** — Small frequent feeds, avoid overfeeding, proper burping after feeds; reduces reflux episodes

- **Lansoprazole** — 1-2 mg/kg/day OD PO for 2-4 weeks trial, only if symptomatic GERD; not for simple GER
- **Thickened Feeds (non-pharmacological)** — Thickened formula or rice cereal if bottle-fed; reduces regurgitation

Second-Line

- **Domperidone** — 0.2-0.3 mg/kg/dose TDS PO, short course, use cautiously in select cases

Investigations

- Complete blood count — Check for anemia if chronic GERD
- Ultrasound abdomen — Rule out pyloric stenosis or structural causes in persistent vomiting
- Liver function tests — If persistent vomiting to rule out hepatobiliary causes

Counselling

- Keep baby's head elevated after feeds and hold upright for 15-20 minutes
- Give small frequent feeds and avoid overfeeding
- Burp baby properly after every feed
- Most babies outgrow reflux by 12-18 months - this is usually normal
- Do not lay baby flat immediately after feeding
- Return if baby has poor weight gain, persistent vomiting, blood in vomit, or choking episodes

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Poor weight gain or failure to thrive
- Persistent vomiting or blood in vomit
- Apnea or choking episodes
- Severe irritability or refusal to feed

Referral Criteria

Refer to pediatric gastroenterologist if poor weight gain despite treatment, persistent vomiting beyond 18 months, hematemesis, suspected aspiration, recurrent pneumonia, or Sandifer syndrome

Hospitalisation Criteria

Admit if severe GERD with failure to thrive, aspiration pneumonia, apparent life-threatening event (ALTE/BRUE), hematemesis, or severe dehydration from persistent vomiting

Notes

Most infant GER is physiological and self-limiting, resolving by 12-18 months. Non-pharmacological measures are first-line: keep head elevated 30 degrees, hold upright 15-20 min after feeds, small frequent feeds, proper burping, avoid overfeeding. Thickened feeds (rice cereal) for bottle-fed infants reduce regurgitation. PPI only for symptomatic GERD with poor weight gain or complications. Do not medicate normal thriving babies with mild regurgitation.

Guideline Changes

NASPGHAN/ESPGHAN 2018 GERD CPG (most recent): 4-8 weeks PPI trial for typical GERD symptoms only after dietary modifications fail in infants; avoid acid suppression for extraesophageal symptoms (cough, wheezing); no new 2024-2025 GERD guideline revision published.

K21.9 — Gastro-esophageal reflux disease

SNOMED: 235595009 | Duration: 56 days | Source: NASPGHAN/ESPGHAN 2018, AAP 2023

First-Line

- **Lifestyle/feeding modifications** — Upright positioning post-feed; smaller/frequent feeds; thickened feeds in infants
- **Omeprazole** — 1 mg/kg/day PO QD-BID (max 20 mg BID) x4-8 weeks; if lifestyle changes fail
- **Esomeprazole** — 0.7-1 mg/kg/day PO QD (max 40 mg); for erosive GERD

Second-Line

- **Ranitidine/Famotidine** — Famotidine 0.5 mg/kg PO BID (max 20 mg BID); H2RA if PPI not tolerated
- **Lansoprazole** — 0.5-1 mg/kg/day PO QD (max 30 mg); alternative PPI
- **Nissen fundoplication** — Laparoscopic antireflux surgery for refractory GERD with complications

Investigations

- Trial of PPI therapy (4-8 weeks) — Empiric approach — symptom response supports diagnosis
- Upper GI endoscopy with biopsy — If red flags or failed PPI trial — assess esophagitis, rule out EoE
- pH-impedance monitoring (24h) — Gold standard for quantifying reflux — before surgery or if diagnosis unclear
- Upper GI barium study — Rule out anatomic abnormalities (malrotation, hiatal hernia) — not for diagnosing GERD

Counselling

- Your child has acid reflux — this is very common in infants and often resolves by 12-18 months
- Keep the baby upright for 20-30 minutes after feeding — avoid lying flat immediately
- Give smaller, more frequent feeds rather than large feeds
- For older children: avoid eating 2-3 hours before bedtime, reduce fatty and spicy foods
- Do not use acid-blocking medicine in infants who are spitting up but growing well — this is normal

Referral Criteria

Pediatric GI if: red flags (failure to thrive, GI bleeding, respiratory complications), failed 4-8 week PPI trial, or suspected complications. Surgery referral for severe refractory GERD or complications

Hospitalisation Criteria

Severe failure to thrive from GERD. Aspiration pneumonia. Apparent life-threatening event (ALTE/BRUE) from reflux. Esophageal stricture. Severe esophagitis with bleeding

Notes

Avoid acid suppression in infants with extraesophageal symptoms only. PPI trial of 4-8 weeks recommended for typical GERD symptoms before further workup.

K27 — Peptic ulcer, site unspecified

SNOMED: 13200003 | Duration: 42 days | Source: ESPGHAN/NASPGHAN 2024, AAFP 2023

First-Line

- **Omeprazole** — 1-2 mg/kg/day PO QD-BID (max 40 mg/day) x4-8 weeks for ulcer healing
- **Amoxicillin (H. pylori)** — 50 mg/kg/day PO divided BID (max 1 g/dose) x14 days as part of triple therapy
- **Clarithromycin (H. pylori)** — 15 mg/kg/day PO divided BID (max 500 mg/dose) x14 days; susceptibility-guided

Second-Line

- **Esomeprazole** — 1 mg/kg/day PO QD (max 40 mg) x8 weeks; alternative PPI
- **Famotidine** — 0.5 mg/kg PO BID (max 20 mg BID) if PPI contraindicated/not tolerated
- **Sucralfate** — 40-80 mg/kg/day PO divided QID (max 1 g/dose); cytoprotective adjunct

Investigations

- Upper GI endoscopy with biopsy — Visualize ulcer, H. pylori testing, rule out malignancy in older children [URGENT]

- H. pylori testing (biopsy CLO test, stool antigen, or UBT) — All peptic ulcers must be tested for H. pylori [URGENT]
- CBC — Assess for anemia from blood loss [URGENT]
- Stool occult blood — Confirm GI bleeding [URGENT]
- Serum iron and ferritin — Iron deficiency from chronic blood loss

Counselling

- Your child has an ulcer in the stomach or intestine — acid-reducing medicine will help heal it
- If H. pylori bacteria is the cause, a 14-day antibiotic course is needed — do not skip doses
- Avoid giving painkillers (ibuprofen, aspirin) without doctor's advice — they can worsen ulcers
- Feed the child regular balanced meals — no need for bland diet unless it helps with symptoms
- Follow up for a confirmation test to make sure the bacteria is completely gone

Referral Criteria

Pediatric GI for endoscopy and H. pylori management. Surgery for perforation or uncontrolled GI bleeding. Hematology if severe anemia from chronic blood loss

Hospitalisation Criteria

Upper GI bleeding (hematemesis, melena, hemodynamic instability). Perforation (acute abdomen with free air). Severe anemia (Hb <7 g/dL). Intractable vomiting

Notes

Test for H. pylori in all pediatric peptic ulcers; eradication confirmed 4-8 weeks post-treatment. NSAID-related ulcers require NSAID cessation and PPI therapy.

Guideline Changes

2024 H. pylori guidelines emphasize 14-day susceptibility-guided triple therapy

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K29 — Gastritis and duodenitis

SNOMED: 4556007 | Duration: 28 days | Source: ESPGHAN/NASPGHAN 2024 H. pylori Guidelines

First-Line

- **Omeprazole** — 1-2 mg/kg/day PO divided BID (max 40 mg/day) x4-8 weeks for acid suppression
- **Amoxicillin** — 50 mg/kg/day PO divided BID (max 1 g/dose) x14 days; part of H. pylori triple therapy
- **Clarithromycin** — 15 mg/kg/day PO divided BID (max 500 mg/dose) x14 days; if susceptible

Second-Line

- **Metronidazole** — 20 mg/kg/day PO divided BID (max 500 mg/dose) x14 days; if clarithromycin-resistant
- **Bismuth subsalicylate** — Quadruple therapy component; age >12yr, 262 mg PO QID x14 days
- **Sucralfate** — 40-80 mg/kg/day PO divided QID (max 1 g/dose); mucosal protectant

Investigations

- Upper GI endoscopy with biopsy — H. pylori detection, histology, gastritis grading
- H. pylori stool antigen or urea breath test — Non-invasive H. pylori testing; confirm eradication 4 weeks post-treatment
- CBC — Anemia from chronic GI blood loss
- Serum iron, ferritin — Iron deficiency from occult blood loss

Counselling

- Your child has stomach inflammation — acid-reducing medicine (PPI) will help heal it
- If H. pylori bacteria is found, a combination of antibiotics for 14 days is needed
- Complete the full course of antibiotics — stopping early can cause the bacteria to become resistant
- Avoid giving the child spicy foods, fizzy drinks, and painkillers (NSAIDs) without doctor's advice
- A repeat test after treatment is done to confirm the bacteria is gone

Referral Criteria

Pediatric gastroenterology for endoscopy if red flags present (GI bleeding, weight loss, anemia). Infectious disease for resistant H. pylori. Surgery for perforated ulcer

Hospitalisation Criteria

GI bleeding (hematemesis, melena). Perforation (acute abdomen). Severe dehydration from persistent vomiting. Severe anemia requiring transfusion

Notes

H. pylori testing is indicated for ulcers/erosions. Susceptibility testing is mandatory per 2024 guidelines before initiating eradication therapy. Non-H. pylori gastritis managed with PPI alone.

Guideline Changes

2024 update mandates susceptibility-guided therapy; high-dose PPI + amoxicillin-based triple therapy x14 days

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K29.0 — Acute Gastritis

SNOMED: 25458004 | Duration: 7 days | Source: IAP 2024, Pediatric GI Guidelines

First-Line

- **Ondansetron** — 0.1-0.15 mg/kg/dose BD-TDS PO, short duration 2-3 days, for vomiting control
- **Omeprazole** — 1 mg/kg/day OD PO before meals x 7-10 days, first-line acid suppression
- **ORS** — Small frequent sips PO, especially if vomiting or mild diarrhoea
- **Pantoprazole** — 1 mg/kg/day OD PO before meals x 7-10 days, alternative PPI
- **Saccharomyces boulardii** — OD PO x 3-5 days, optional probiotic for gut recovery
- **Dicyclomine** — Age-appropriate dose TDS PO, short-term, only if significant abdominal pain

Second-Line

- **Ranitidine** — 2-4 mg/kg/dose BD PO x 7-10 days, if PPI not available/tolerated

Investigations

- Complete blood count — Suspected infection or persistent symptoms
- CRP — Assess inflammation severity
- Stool routine and microscopy — Associated loose stools or suspected infection
- Liver function test — Persistent vomiting to rule out hepatic cause

Counselling

- Give small frequent meals instead of large meals
- Preferred foods: curd, buttermilk, khichdi, rice, banana, coconut water
- Avoid spicy food, fried food, and junk food
- Keep child well hydrated with ORS and fluids
- If persistent vomiting, blood in vomit, severe abdominal pain, or dehydration - come to hospital immediately

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Persistent vomiting or blood in vomit
- Severe abdominal pain not relieved by medication
- Signs of dehydration such as dry mouth or no urine

Referral Criteria

Refer to pediatric gastroenterologist if persistent vomiting >5 days despite treatment, recurrent episodes, suspected H. pylori, or upper GI bleeding

Hospitalisation Criteria

Admit if persistent vomiting with dehydration, hematemesis, severe abdominal pain with guarding, or inability to tolerate oral intake

Notes

Ondansetron for vomiting control. PPI (Omeprazole) is first-line acid suppression, preferred over H2 blockers. Maintain hydration. Avoid unnecessary antibiotics. Small frequent meals.

Guideline Changes

ESPGHAN/NASPGHAN 2024: updated H. pylori guidelines with bismuth-based quadruple therapy for resistant strains; PPIs remain first-line acid suppression in children; PPI deprescription emphasized after 4-8 wk courses; ACG 2024 highlights rising clarithromycin resistance.

K35 — Acute appendicitis

SNOMED: 85189001 | Duration: 7 days | Source: WSES 2020, SIS 2025, Texas Children's 2024

First-Line

- **Piperacillin-tazobactam** — 100 mg/kg/dose (pip) IV q8h (max 4.5 g/dose) preop and for complicated cases
- **Ceftriaxone + Metronidazole** — Ceftriaxone 50 mg/kg IV q24h (max 2g) + Metronidazole 10 mg/kg IV q8h (max 500 mg)
- **Surgical appendectomy** — Laparoscopic appendectomy; standard of care for confirmed appendicitis

Second-Line

- **Non-operative antibiotics** — IV amoxicillin-clavulanate then PO x7-10 days for uncomplicated; 58-75% 1yr success **[NEW 2024-25]**
- **Meropenem** — 20 mg/kg IV q8h (max 1 g) for severe allergy or resistant organisms
- **Ertapenem** — 15 mg/kg IV q12h (max 1 g/day) for complicated cases with penicillin allergy

Investigations

- Abdominal ultrasound — First-line imaging — non-compressible appendix >6mm, appendicolith [STAT]
- CBC with differential — Leukocytosis with left shift [STAT]
- CRP — Elevated in complicated appendicitis — helps differentiate [STAT]
- Urinalysis — Rule out UTI mimicking appendicitis [URGENT]
- CT abdomen (if US inconclusive) — Higher sensitivity — use low-dose protocol in children [URGENT]

Counselling

- Your child has appendicitis — surgery (removal of the appendix) is usually the best treatment
- The operation is done through small holes in the belly (laparoscopic) — recovery is faster
- After surgery, the child can usually eat within 24 hours and go home in 1-2 days
- Watch for fever, increasing belly pain, or vomiting after discharge — come back immediately
- Appendicitis is not preventable — there is nothing you did wrong

Referral Criteria

Pediatric surgery mandatory. Interventional radiology for percutaneous drainage of appendiceal abscess. PICU if perforated with sepsis

Hospitalisation Criteria

All acute appendicitis requires hospitalisation for surgery or antibiotic-first management. Peritonitis. Appendiceal abscess or phlegmon. Perforation with sepsis

Notes

Uncomplicated appendicitis may be managed non-operatively with antibiotics in select cases, but surgery remains standard. Complicated appendicitis with abscess may benefit from initial antibiotics and interval appendectomy.

Guideline Changes

2024 pathways increasingly support non-operative antibiotic-first for uncomplicated cases

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K37 — Appendicitis, unspecified

SNOMED: 74400008 | Duration: 7 days | Source: WSES 2020, SIS 2025, AAP 2023

First-Line

- **Piperacillin-tazobactam** — 100 mg/kg/dose (pip) IV q8h (max 4.5 g/dose) perioperative
- **Ceftriaxone + Metronidazole** — Ceftriaxone 50 mg/kg IV q24h (max 2g) + Metronidazole 10 mg/kg IV q8h (max 500 mg)
- **Surgical appendectomy** — Laparoscopic preferred; open if needed. Postop abx based on intraop findings

Second-Line

- **Amoxicillin-clavulanate** — 45 mg/kg/day PO divided BID (max 875 mg/dose) for step-down oral therapy
- **Ciprofloxacin + Metronidazole** — Cipro 10 mg/kg PO BID + Metro 10 mg/kg PO TID for oral step-down if pen-allergic

Investigations

- Abdominal ultrasound — Initial imaging of choice in children — appendix visualization [STAT]
- CBC with differential — Leukocytosis assessment [STAT]
- CRP — Inflammatory marker — helps guide management [STAT]
- Urinalysis — Exclude urinary tract pathology [URGENT]
- CT abdomen (if US inconclusive) — Definitive imaging when diagnosis uncertain [URGENT]

Counselling

- We suspect appendicitis but need more tests to be sure — imaging will help confirm
- Until confirmed, the child should not eat or drink (nil by mouth) in case surgery is needed
- If appendicitis is confirmed, surgery will be recommended
- After surgery, recovery is usually quick — 2-3 days in most cases
- Come back immediately if pain worsens, child develops high fever, or starts vomiting

Referral Criteria

Pediatric surgery for evaluation and possible appendectomy. Interventional radiology if abscess needs drainage

Hospitalisation Criteria

All suspected appendicitis for observation, imaging, and possible surgery. Peritonitis on examination. Hemodynamic instability. Need for NPO status and IV fluids

Notes

If unspecified, obtain imaging (US or CT) to classify. Gangrenous/perforated appendicitis requires 7-10 days of antibiotics; uncomplicated needs only perioperative doses.

K52.9 — Acute Gastroenteritis (Indoor Protocol) *[severe]*

SNOMED: 20759009 | Duration: 5 days | Source: IAP 2024 Diarrhea Guidelines, WHO Diarrheal Disease Management, NASPGHAN 2024 AGE Clinical Pathway

First-Line

- **Normal Saline** — 20 ml/kg IV bolus over 30-60 min for initial rehydration; maintenance per Holliday-Segar formula

- **Ondansetron** — 0.1-0.15 mg/kg/dose IV q8h for vomiting
- **Paracetamol** — 10-15 mg/kg/dose IV q6h for fever
- **Saccharomyces Boulardii** — 1 sachet BD PO, reduces duration of diarrhea
- **ORS** — Start early when tolerated, small frequent sips
- **Zinc** — 20 mg OD PO for 14 days (<6 months: 10 mg OD), start with oral transition

Second-Line

- **Ceftriaxone** — 50-75 mg/kg/day IV OD/BD, only if dysentery/sepsis/toxic child
- **Amikacin** — 15 mg/kg/day IV OD, add-on if severe sepsis suspected
- **Metronidazole** — 7.5-10 mg/kg/dose IV TDS, only if amoebic dysentery suspected

Investigations

- Complete blood count — Assess for infection and dehydration [URGENT]
- Serum electrolytes — If severe dehydration or prolonged diarrhea [URGENT]
- Stool routine and microscopy — Identify bacterial/parasitic cause [URGENT]
- Renal function tests — Rule out acute kidney injury in severe dehydration [URGENT]

Counselling

- Continue breastfeeding and complementary foods during illness
- Give ORS sip by sip after every loose stool
- Continue zinc for full 14 days even after diarrhea stops
- Offer curd, khichdi, banana, coconut water as tolerated
- Return immediately if child becomes lethargic, stops urinating, or has blood in stool

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Signs of severe dehydration such as sunken eyes or no tears
- Electrolyte imbalance symptoms such as muscle cramps or weakness
- No urine output for more than 6 hours
- Signs of shock such as cold hands or weak pulse

Referral Criteria

Refer to pediatric gastroenterologist if persistent diarrhea >14 days, recurrent severe episodes, suspected malabsorption, or failure to thrive

Hospitalisation Criteria

Admit if severe dehydration, unable to tolerate oral fluids, persistent vomiting, toxic appearance, age <2 months, decreased urine output, or electrolyte imbalance

Notes

IV fluids are the cornerstone. Maintenance fluid per Holliday-Segar: 100 ml/kg first 10 kg + 50 ml/kg next 10 kg. Avoid routine triple antibiotics unless clearly indicated. Early transition to ORS. Continue feeding as early as possible (curd, khichdi, banana, coconut water).

Guideline Changes

NASPGHAN 2024 AGE update: Lactobacillus not routinely recommended (two large RCTs showed no benefit); ondansetron preferred antiemetic for children >=6 months; early oral rehydration emphasized over IV where possible; Saccharomyces boulardii retains moderate evidence for reducing diarrhea duration.

K56.1 — Intussusception

SNOMED: 49723003 | Duration: 3 days | Source: AAP 2023, PLOS One Meta-analysis 2024

First-Line

- **Air enema reduction** — Pneumatic reduction under fluoroscopy; 83-97% success rate. Max pressure 120 mmHg
- **IV fluid resuscitation** — NS 20 mL/kg bolus; correct dehydration before reduction attempt

Second-Line

- **Hydrostatic saline enema** — US-guided saline enema reduction; 73-92% success. No radiation exposure
- **Surgical reduction** — Operative manual reduction or resection if enema fails, peritonitis, or perforation
- **Dexamethasone** — 0.5 mg/kg IV single dose (max 10 mg); may reduce recurrence rate (adjunctive) **[NEW 2024-25]**

Investigations

- Abdominal ultrasound — Target sign (doughnut sign) — sensitivity >95% for ileocolic intussusception [STAT]
- Abdominal X-ray — Obstruction pattern, free air if perforation [STAT]
- CBC — Baseline; leukocytosis if complications [URGENT]
- Serum electrolytes — Dehydration from vomiting [URGENT]

Counselling

- Your child has a condition where part of the intestine slides into itself like a telescope — this is called intussusception
- An air enema (non-surgical procedure) can fix this in most cases — success rate is very high
- The child may cry during the procedure but it avoids surgery in most cases
- There is a 5-10% chance of it happening again — come back immediately if symptoms return (pain, bloody stool, vomiting)
- Monitor the child closely for 24 hours after the procedure — watch for recurrence of symptoms

Referral Criteria

Pediatric surgery if: enema reduction fails, peritonitis, perforation, recurrent intussusception (>3 episodes — evaluate for lead point), or age >2 years (higher chance of pathological lead point)

Hospitalisation Criteria

All intussusception cases require hospitalisation for reduction and 24-hour observation. Peritonitis or perforation require emergency surgery. Recurrent episodes require admission for surgical evaluation

Notes

Air enema is first-line for ileocolic intussusception without peritonitis. Recurrence rate is 5-10%; lead points more common in children >2 years requiring surgical evaluation.

Guideline Changes

2024 meta-analysis confirms air enema superiority over hydrostatic reduction in success rates

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K56.6 — Intestinal obstruction, unspecified

SNOMED: 81060008 | Duration: 7 days | Source: AAP 2023, Pediatric Surgery NaT 2024

First-Line

- **NPO + NG decompression** — Insert NG tube for suction; strict NPO. Fluid resuscitate with NS/LR 20 mL/kg boluses
- **IV fluid resuscitation** — NS or LR 20 mL/kg bolus; correct electrolytes. Maintenance 1.5x calculated rate
- **Surgical exploration** — Emergency laparotomy/laparoscopy for strangulation, volvulus, or failed conservative Rx

Second-Line

- **Gastrografin enema** — Therapeutic contrast enema for meconium ileus or adhesive partial obstruction

- **Piperacillin-tazobactam** — 100 mg/kg/dose IV q8h (max 4.5 g) if peritonitis or perforation suspected

Investigations

- Abdominal X-ray (erect and supine) — Air-fluid levels, dilated loops, free air if perforation [STAT]
- Abdominal ultrasound — Intussusception, pyloric stenosis, hernia evaluation [STAT]
- Serum electrolytes — Dehydration and electrolyte imbalance from vomiting/third-spacing [STAT]
- CBC — Leukocytosis if strangulation or peritonitis [STAT]
- Upper GI contrast study — Malrotation with volvulus — upper GI series is diagnostic [STAT]

Counselling

- Your child has a bowel blockage — this is serious and needs hospital treatment
- The child must not eat or drink until the blockage is cleared
- A tube through the nose (NG tube) will help remove trapped air and fluid from the stomach
- Surgery may be needed if the blockage does not clear with non-surgical treatment
- Come back immediately after discharge if the child develops vomiting, abdominal distension, or stops passing stool

Referral Criteria

Pediatric surgery mandatory for all intestinal obstructions. Emergency surgery for: volvulus, strangulation, perforation, or failed conservative management

Hospitalisation Criteria

All intestinal obstructions require hospitalisation. Signs: vomiting (especially bilious in children), abdominal distension, absent bowel sounds, abdominal pain. Emergency for complete obstruction or peritonitis

Notes

Etiology must be urgently identified (volvulus, hernia, adhesions, intussusception). Midgut volvulus requires emergent surgery without delay for imaging.

K59.04 — Functional constipation

SNOMED: 236069009 | Duration: 90 days | Source: NASPGHAN 2014

First-Line

- **PEG 3350 (Polyethylene Glycol)** — Disimpaction 1-1.5 g/kg/day x3-6 days. Maintenance 0.4-0.8 g/kg/day

Second-Line

- **Lactulose** — 1-3 mL/kg/day divided q12h, alternative osmotic laxative

Investigations

- Abdominal X-ray — Assess fecal loading if uncertain clinical picture
- Thyroid function tests — If associated growth failure or other symptoms of hypothyroidism
- Barium enema or anorectal manometry — If Hirschsprung disease suspected (onset at birth, failure to thrive)

Counselling

- Give PEG laxative daily as prescribed - it is safe for long-term use and not habit-forming
- Include more fruits, vegetables, and whole grains in diet; ensure adequate water intake
- Establish a regular toilet routine - sit on toilet for 5-10 minutes after meals
- Do not punish or scold the child for soiling; it is not intentional
- Treatment takes 3-6 months; do not stop laxative suddenly when stools improve

Referral Criteria

Refer to pediatric gastroenterologist if no response to 3 months of PEG therapy, suspected Hirschsprung disease (delayed meconium passage, abdominal distension), or associated failure to thrive

Hospitalisation Criteria

Admit if severe fecal impaction not responding to oral disimpaction, signs of intestinal obstruction, or severe abdominal pain with vomiting

Notes

PEG is best-evidenced agent. Dietary fibre and adequate fluid are essential adjuncts.

K60.0 — Anal Fissure

SNOMED: 30037006 | Duration: 7 days | Source: IAP Pediatric Surgery Guidelines, ASCRS Anal Fissure Guidelines 2023, ESPGHAN Constipation Guidelines

First-Line

- **Lactulose** — 1-2 ml/kg/day PO div BD, after meals or at bedtime, first-line stool softener
- **Lignocaine 2% Gel** — Apply locally before passing stool, 2-3 times/day for pain relief
- **Liquid Paraffin + Milk of Magnesia** — 5-10 ml PO at bedtime for 5-7 days, stool bulking and smooth passage
- **Sitz Bath (Potassium Permanganate)** — Warm water with light pink KMnO4 solution, 10-15 minutes 2-3 times/day, promotes healing and reduces pain
- **Glycerin Suppository** — SOS for acute constipation, short-term use only
- **Ispaghula Husk** — 1 teaspoon (3-5 g) in 1 glass water at bedtime OD for 5-7 days, fiber supplement for stool consistency

Investigations

- Stool routine and microscopy — Rule out occult blood, parasites, or infection if recurrent
- Complete blood count — Assess for anemia if significant or recurrent bleeding
- Stool occult blood test — Confirm GI bleeding if history of blood in stool

Counselling

- Give lactulose regularly to keep stool soft - this is the most important treatment
- Sitz bath in warm water for 10-15 minutes 2-3 times daily helps healing
- Apply lignocaine gel before passing stool to reduce pain
- Increase fiber in diet - papaya, banana, green vegetables, adequate water
- Avoid junk food and excess milk which worsen constipation
- Return if persistent bleeding, severe pain not relieved, or no improvement in 5-7 days

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Persistent or increasing bleeding during stools
- Severe pain not relieved by medication
- No improvement in symptoms after 5 to 7 days

Referral Criteria

Refer to pediatric surgeon if chronic fissure not healing after 4-6 weeks of conservative treatment, recurrent fissures, or suspected underlying condition (Crohn disease, anorectal anomaly)

Hospitalisation Criteria

Admit if significant rectal bleeding, suspected perianal abscess, or need for examination under anesthesia

Notes

Treatment principles: soften stool, relieve pain, heal fissure, prevent recurrence. Sitz bath with warm water + potassium permanganate (light pink solution) for 10-15 min 2-3 times/day is essential. High fiber diet: papaya, banana, vegetables. Adequate water intake. Avoid junk food and excess milk.

Guideline Changes

ASCRS 2023: Topical calcium channel blockers (diltiazem) recommended as first-line pharmacotherapy for chronic fissures with better side-effect profile than GTN; conservative management (stool softeners, fiber, sitz baths) remains first-choice for acute fissures in children; IAP Pediatric Surgery 2024: PEG-based laxatives preferred over lactulose for chronic constipation-associated fissures

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K62.3 — Rectal prolapse

SNOMED: 57773001 | Duration: 30 days | Source: StatPearls 2024, JPGN 2023

First-Line

- **Manual reduction** — Gentle digital reduction with lubrication; apply sugar/salt to edematous mucosa before reduction
- **Stool softener (PEG 3350)** — 0.4-0.8 g/kg/day PO QD (max 17 g) to prevent straining
- **Treat underlying cause** — Screen for cystic fibrosis, treat constipation, parasites, malnutrition

Second-Line

- **Sclerotherapy injection** — Submucosal injection of 30% saline or 5% phenol above dentate line; outpatient procedure
- **Laparoscopic rectopexy** — For recurrent prolapse after failed conservative Rx; age >4yr or chronic predisposition
- **Thiersch suture** — Anal encirclement suture for recurrent prolapse in select patients

Investigations

- Sweat chloride test — Rule out cystic fibrosis — MANDATORY in all children with rectal prolapse
- Stool examination for parasites — Parasitic infections (Trichuris, Ascaris) cause prolapse in endemic areas
- CBC — Anemia from chronic disease or parasites
- Abdominal X-ray — Constipation/fecal loading assessment

Counselling

- Your child's rectum is coming out — this usually gets better on its own with treatment of the underlying cause
- Keep the child's stools soft with stool softeners — hard stools and straining worsen the problem
- If the rectum comes out, gently push it back with a lubricated finger — do not panic
- Apply sugar or salt to the swollen tissue to reduce swelling before pushing back if needed
- All children with rectal prolapse must be tested for cystic fibrosis — ask your doctor about the sweat test

Referral Criteria

Pediatric surgery if: recurrent prolapse despite conservative management, incarceration, or need for surgical fixation. CF team if sweat test positive. GI for underlying cause workup

Hospitalisation Criteria

Incarcerated prolapse that cannot be manually reduced. Strangulated prolapse (ischemic tissue). Post-surgical care after rectopexy. Newly diagnosed cystic fibrosis

Notes

Self-limiting in 90% of children <3 years with conservative management. All children with rectal prolapse should be tested for cystic fibrosis (sweat chloride test).

K72 — Hepatic failure, not elsewhere classified

SNOMED: 235886005 | Duration: 14 days | Source: AASLD 2023, NASPGHAN 2022, ISPGHAN 2024

First-Line

- **N-acetylcysteine (NAC)** — 150 mg/kg IV over 1h, then 50 mg/kg over 4h, then 100 mg/kg over 16h for APAP toxicity

- **IV Dextrose (D10)** — Continuous infusion to maintain glucose >70 mg/dL; 6-8 mg/kg/min glucose infusion rate
- **Vitamin K** — 2-10 mg IV single dose for coagulopathy; repeat if INR remains elevated

Second-Line

- **Acyclovir** — 20 mg/kg IV q8h x14-21 days if HSV hepatitis suspected (neonates especially)
- **Prednisolone** — 2 mg/kg/day PO/IV (max 60 mg) for autoimmune hepatitis-related ALF
- **Fresh frozen plasma** — 10-15 mL/kg IV for active bleeding with coagulopathy; avoid routine use
- **Liver transplantation** — Emergency listing for ALF meeting King's College or PELD criteria

Investigations

- Liver function tests (serial q6-12h) — Monitor AST, ALT, bilirubin trend [STAT]
- PT/INR (serial q6-12h) — Key prognostic marker — rising INR indicates worsening [STAT]
- Blood ammonia — Hepatic encephalopathy — guide lactulose/rifaximin therapy [STAT]
- Viral hepatitis panel (HAV IgM, HBsAg, HCV RNA) — Identify viral etiology [STAT]
- Autoimmune markers (ANA, ASMA, anti-LKM, IgG) — Autoimmune hepatitis — treatable with steroids [STAT]

Counselling

- Your child's liver is failing and needs emergency treatment — this is life-threatening
- The child may need transfer to a liver transplant center if the liver does not recover
- Multiple medicines and close monitoring are needed — the child will be in the ICU
- Please share all information about medicines taken, illnesses, or toxin exposure
- The medical team will keep you informed at every step — liver transplant may be discussed

Referral Criteria

Pediatric hepatology and liver transplant center URGENTLY. PICU mandatory. Nephrology if hepatorenal syndrome. Neurosurgery if intracranial hypertension from hepatic encephalopathy

Hospitalisation Criteria

All pediatric acute liver failure (PALF) requires PICU admission. Criteria: coagulopathy (INR >=1.5 with encephalopathy or INR >=2.0 without encephalopathy) in acute liver disease without pre-existing chronic liver disease

Notes

Transfer to pediatric liver transplant center urgently. NAC beneficial for APAP-induced ALF; role in non-APAP PALF remains uncertain. Treat identifiable causes (HSV, Wilson, AIH).

Guideline Changes

2024 ISPGHAN consensus on pediatric ALF management; NAC role debated for non-APAP PALF

K74 — Fibrosis and cirrhosis of liver

SNOMED: 19943007 | Duration: 365 days | Source: AASLD 2023, ESPGHAN 2024

First-Line

- **Treat underlying etiology** — Etiology-specific: antivirals for HBV/HCV, chelation for Wilson, UDCA for cholestasis
- **Ursodeoxycholic acid (UDCA)** — 10-20 mg/kg/day PO divided BID-TID for cholestatic liver diseases
- **Nutritional support** — High-calorie diet 120-150% RDA; MCT-enriched formula; fat-soluble vitamin supplementation

Second-Line

- **Spironolactone** — 1-3 mg/kg/day PO divided BID (max 100 mg) for ascites management
- **Furosemide** — 1-2 mg/kg/day PO/IV (max 6 mg/kg/day) for ascites; combine with spironolactone
- **Liver transplantation** — Definitive treatment for decompensated cirrhosis or liver failure

Investigations

- Liver function tests (comprehensive) — AST, ALT, bilirubin, albumin, GGT — monitor disease progression [URGENT]
- PT/INR — Liver synthetic function [URGENT]
- Liver ultrasound with Doppler — Liver size, nodularity, portal flow, ascites
- Upper GI endoscopy — Variceal screening — start at diagnosis of cirrhosis
- Alpha-fetoprotein — Hepatocellular carcinoma screening (q6 months)

Counselling

- Your child has liver scarring (cirrhosis) — lifelong monitoring and treatment of the cause is needed
- Feed the child high-calorie nutritious food — liver disease increases nutritional needs by 20-50%
- Fat-soluble vitamins (A, D, E, K) supplements must be taken as prescribed
- Come back urgently if: child vomits blood, has black stools, develops a swollen belly, or becomes confused
- Regular ultrasound and blood tests every 3-6 months are needed to monitor the liver

Referral Criteria

Pediatric hepatology mandatory. Liver transplant evaluation early for decompensated cirrhosis. GI for variceal screening/banding. Nutrition/dietitian. Genetics if inherited liver disease

Hospitalisation Criteria

Variceal bleeding (hematemesis/melena). Ascites not responding to diuretics. Hepatic encephalopathy. Spontaneous bacterial peritonitis. Hepatorenal syndrome. Acute decompensation

Notes

Management is etiology-directed. Monitor for complications: variceal bleeding, ascites, hepatorenal syndrome. Liver transplant evaluation should be early for decompensation.

Guideline Changes

Resmetirom (THR-beta agonist) approved 2024 for MASH in adults; pediatric trials pending

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K75 — Other inflammatory liver diseases

SNOMED: 194519009 | Duration: 365 days | Source: EASL 2025, AASLD 2023, ESPGHAN 2024

First-Line

- **Prednisolone** — 2 mg/kg/day PO (max 60 mg) x2-4wk, then taper over 6-8wk to 2.5-5 mg/day maintenance
- **Azathioprine** — 1-2.5 mg/kg/day PO QD (added after initial steroid response); steroid-sparing agent

Second-Line

- **Mycophenolate mofetil** — 20 mg/kg/day PO divided BID (max 1 g BID) if azathioprine intolerant/refractory
- **Cyclosporine** — 3-5 mg/kg/day PO divided BID; bridge therapy for severe AIH or steroid-refractory
- **Tacrolimus** — 0.05-0.1 mg/kg/day PO divided BID; target trough 3-5 ng/mL for refractory AIH

Investigations

- Liver function tests — AST, ALT markedly elevated; bilirubin, albumin for severity [STAT]
- Autoimmune markers (ANA, ASMA, anti-LKM1, IgG) — Type 1 AIH: ANA/ASMA+; Type 2: anti-LKM1+ [URGENT]
- Serum IgG level — Elevated polyclonal IgG supports diagnosis [URGENT]
- Liver biopsy — Interface hepatitis — confirms diagnosis and grades severity
- PT/INR — Coagulopathy indicates severe disease/liver failure [STAT]

Counselling

- Your child has autoimmune liver disease — the body's immune system is attacking the liver
- Steroid medicine and an immunosuppressant will be needed, possibly for life

- Do not stop the medicines without asking the doctor — disease flares are very common when stopped
- Regular blood tests every 2-4 weeks initially, then every 3 months, are needed to monitor response
- Report immediately if the child becomes yellow (jaundiced), has dark urine, or develops a swollen belly

Referral Criteria

Pediatric hepatology mandatory. Liver transplant evaluation if steroid-refractory or decompensated. Immunology for atypical cases. Genetics if overlap with other autoimmune diseases

Hospitalisation Criteria

New diagnosis with significant coagulopathy or jaundice. Acute liver failure from autoimmune hepatitis. Disease flare with hepatic decompensation. Initiation of immunosuppression requiring close monitoring

Notes

Autoimmune hepatitis (AIH) is the most common inflammatory liver disease requiring treatment. Relapse rate 60-80% after withdrawal; many patients need lifelong immunosuppression.

Guideline Changes

EASL 2025 updated AIH guidelines; mycophenolate increasingly used as steroid-sparing agent

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K75.0 — Liver abscess

SNOMED: 197329009 | Duration: 28 days | Source: Pediatric Surgery 2023, Johns Hopkins ABX Guide 2024

First-Line

- **Ceftriaxone + Metronidazole** — Ceftriaxone 50-75 mg/kg IV q24h (max 2g) + Metronidazole 30 mg/kg/day IV divided TID
- **Percutaneous drainage** — US-guided needle aspiration (<5cm) or catheter drainage (>5cm)

Second-Line

- **Piperacillin-tazobactam** — 100 mg/kg/dose (pip) IV q8h (max 4.5 g) for polymicrobial pyogenic abscess
- **Metronidazole (amoebic)** — 35-50 mg/kg/day PO/IV divided TID x7-10 days for amoebic liver abscess
- **Meropenem** — 20 mg/kg IV q8h (max 1 g) for resistant organisms or treatment failure
- **Surgical drainage** — Open drainage for ruptured abscess, peritonitis, or failed percutaneous approach

Investigations

- Abdominal ultrasound — Identify abscess location, size, and guide drainage [STAT]
- CBC with differential — Leukocytosis, left shift [STAT]
- Blood culture — Identify causative organism [STAT]
- Liver function tests — Assess liver involvement [STAT]
- Amoebic serology (IHA/ELISA) — Differentiate amoebic from pyogenic abscess — common in India [URGENT]

Counselling

- Your child has an infection inside the liver (abscess) — antibiotics and possibly drainage are needed
- IV antibiotics will be given for several days, followed by oral antibiotics for weeks
- If the abscess is large, a needle will be used to drain the pus under ultrasound guidance
- Complete the full course of antibiotics (usually 2-6 weeks) even after feeling better
- Follow-up ultrasound is needed to confirm the abscess has resolved

Referral Criteria

Pediatric surgery/interventional radiology for drainage (abscess >5 cm or not responding to antibiotics alone). GI if underlying liver disease. Immunology if recurrent liver abscesses

Hospitalisation Criteria

All liver abscesses require hospitalisation for IV antibiotics. Drainage needed for: abscess >5 cm, no clinical improvement in 48-72h of antibiotics, or risk of rupture. Sepsis or hemodynamic instability

Notes

Pyogenic abscesses require both antibiotics and drainage; amoebic abscesses often resolve with metronidazole alone. Antibiotic duration 2-6 weeks depending on response.

K76 — Other disorders of liver

SNOMED: 197321007 | Duration: 180 days | Source: NASPGHAN 2017 NAFLD, AASLD 2023

First-Line

- **Lifestyle modification** — Structured diet + exercise program; target 7-10% weight loss if overweight/obese
- **Dietary counseling** — Reduce sugar-sweetened beverages, processed foods; increase fiber and whole foods
- **Vitamin E** — 400-800 IU/day PO for biopsy-proven NASH in children without diabetes

Second-Line

- **Ursodeoxycholic acid** — 10-15 mg/kg/day PO divided BID for cholestatic liver disorders
- **Metformin** — 500 mg PO BID (age >=10yr) for NAFLD with insulin resistance; limited evidence

Investigations

- ALT (serial q3-6 months) — Primary monitoring marker — >2x ULN warrants further evaluation
- Liver ultrasound — Fatty liver detection — increased echogenicity
- Fasting lipid profile — Dyslipidemia commonly associated
- Fasting glucose and HbA1c — Screen for insulin resistance and diabetes
- Liver biopsy — If persistent elevation despite lifestyle changes — stage fibrosis/NASH

Counselling

- Your child has fatty liver disease (NAFLD/MASLD) — this is caused by excess weight and poor diet
- The most important treatment is lifestyle change: healthier food and more physical activity
- Cut down sugar-sweetened drinks (cola, juice, packaged drinks) — switch to water and buttermilk
- 60 minutes of active play/exercise every day is the best medicine for this condition
- Regular blood tests every 6 months to check liver enzymes and follow-up with the doctor

Referral Criteria

Pediatric hepatology/GI if ALT persistently elevated despite 6 months of lifestyle changes. Liver biopsy if NASH suspected. Dietitian for structured meal planning. Endocrinology if insulin resistance/prediabetes

Hospitalisation Criteria

Rarely needed. Consider if: acute liver failure from NASH, liver biopsy under sedation, severe metabolic complications (diabetes, hypertension crisis)

Notes

NAFLD/MASLD is the most common pediatric chronic liver disease. Screen overweight children with ALT; liver biopsy for indeterminate cases. Lifestyle modification is primary treatment.

Guideline Changes

NAFLD renamed MASLD/MASH; resmetirom approved 2024 for adult MASH; pediatric guidance forthcoming

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K76.6 — Portal hypertension

SNOMED: 34742003 | Duration: 365 days | Source: AASLD 2023, Baveno VII 2022

First-Line

- **Endoscopic band ligation** — For esophageal varices with bleeding or high-risk stigmata; repeat q2-4wk until obliteration
- **Octreotide** — 1-2 mcg/kg IV bolus then 1-2 mcg/kg/h infusion (max 50 mcg/h) for acute variceal bleed
- **Propranolol** — 0.5-1 mg/kg/day PO divided BID-TID; titrate to 25% HR reduction. Limited pediatric evidence

Second-Line

- **Sclerotherapy** — Endoscopic injection for variceal bleeding if banding not feasible (younger children)
- **Spironolactone** — 1-3 mg/kg/day PO divided BID (max 100 mg) for ascites
- **TIPS** — Transjugular intrahepatic portosystemic shunt for refractory bleeding/ascites
- **Meso-Rex shunt** — Surgical bypass for extrahepatic portal vein obstruction in eligible children

Investigations

- Upper GI endoscopy — Variceal grading and banding — first screening at diagnosis [URGENT]
- Liver ultrasound with Doppler — Portal vein flow direction, spleen size, ascites [URGENT]
- CBC with platelet count — Thrombocytopenia from hypersplenism [URGENT]
- Liver function tests — Assess underlying liver disease severity [URGENT]
- CT/MR angiography — Map portal venous anatomy for surgical planning

Counselling

- Your child has high blood pressure in the liver blood vessels — this can cause the veins in the food pipe to swell and bleed
- An endoscopy (camera test) is needed to check for swollen veins and may need treatment (banding)
- If the child vomits blood or has black tarry stools — rush to the hospital immediately
- Salt restriction may be needed if the child has a swollen belly (fluid accumulation)
- Regular follow-up every 3-6 months with endoscopy is essential to prevent bleeding

Referral Criteria

Pediatric hepatology/GI mandatory. GI endoscopy for variceal surveillance and banding. Interventional radiology for TIPS. Pediatric surgery for shunt procedures (Meso-Rex). Liver transplant if decompensated cirrhosis

Hospitalisation Criteria

Acute variceal bleeding (hematemesis/melena) — emergency. New onset ascites. Hepatic encephalopathy. Spontaneous bacterial peritonitis. Scheduled endoscopic band ligation

Notes

Beta-blockers for primary prophylaxis lack robust pediatric evidence; band ligation preferred for secondary prophylaxis. Evaluate for liver transplant if decompensated cirrhosis.

Guideline Changes

Baveno VII criteria for non-invasive assessment of portal hypertension; beta-blockers debated in children

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K85 — Acute pancreatitis

SNOMED: 197456007 | Duration: 7 days | Source: NASPGHAN/INSPPIRE 2018, BSPGHAN 2024, ACG 2024

First-Line

- **IV fluid resuscitation** — 1.5-2x maintenance with D5 0.9% NS; bolus 20 mL/kg if hemodynamically unstable
- **Acetaminophen** — 15 mg/kg PO/IV q4-6h (max 75 mg/kg/day); first-line analgesia
- **Ketorolac** — 0.5 mg/kg IV q6h (max 30 mg/dose) x5 days for pain; NSAID adjunct
- **Early enteral nutrition** — Begin PO/NG feeds within 24-48h if tolerated; low-fat diet not required

Second-Line

- **Morphine** — 0.1-0.2 mg/kg IV q2-4h (max 10 mg) for severe pain unresponsive to non-opioids
- **Meropenem** — 20 mg/kg IV q8h (max 1 g) ONLY for infected necrosis; prophylactic abx NOT recommended
- **ERCP** — For biliary pancreatitis with choledocholithiasis or ductal pathology

Investigations

- Serum lipase — Primary diagnostic test — >3x upper limit of normal confirms pancreatitis [STAT]
- Serum amylase — Supportive — may be normal in 15-20% of cases [STAT]

- Abdominal ultrasound — Pancreatic edema, gallstones, pseudocyst [URGENT]
- CBC and CRP — Inflammatory markers; predict severity [STAT]
- Liver function tests — Screen for biliary cause (elevated bilirubin, ALP) [URGENT]

Counselling

- Your child has inflammation of the pancreas — rest, IV fluids, and pain management are the main treatment
- The child can start eating within 24-48 hours when tolerated — no need for prolonged fasting
- Low-fat diet is NOT mandatory in mild cases — a regular diet is fine when tolerated
- Report immediately if pain returns, the child develops fever, or the belly becomes more swollen
- Follow up to check for recurrent pancreatitis — some children need long-term monitoring

Referral Criteria

Pediatric gastroenterology for all cases. Surgery/IR if infected pancreatic necrosis or pseudocyst. ERCP for biliary pancreatitis with choledocholithiasis. Genetics if recurrent pancreatitis (PRSS1, SPINK1, CFTR)

Hospitalisation Criteria

All acute pancreatitis should be hospitalised for IV fluids and pain management. Severe pancreatitis (organ failure, necrosis on CT). Intractable pain or vomiting. Unable to tolerate oral intake

Notes

Aggressive IV hydration and early enteral feeding are cornerstones of treatment. Prophylactic antibiotics not indicated. Opioids are safe and should not be withheld for pain.

Guideline Changes

2024 BSPGHAN guidance published; liberal IV fluids associated with improved outcomes in pediatric AP

K86 — Other diseases of pancreas

SNOMED: 235494005 | Duration: 365 days | Source: CFF 2023, ESPGHAN/UEG 2024 EPI Guidelines

First-Line

- **Pancreatic enzyme replacement (PERT)** — 500-2500 lipase units/kg/meal PO; 250-500 LU/kg/snack. Enteric-coated capsules
- **Fat-soluble vitamin supplementation** — Vitamins A, D, E, K at 1-2x RDA PO daily; monitor levels q6-12 months

Second-Line

- **Omeprazole (adjunct)** — 1 mg/kg/day PO QD to enhance PERT efficacy by reducing gastric acid inactivation
- **MCT oil supplementation** — Medium-chain triglycerides 1-2 mL/kg/day PO for improved fat absorption
- **Endoscopic therapy** — ERCP for chronic pancreatitis with duct stones, strictures, or pseudocysts

Investigations

- Fecal elastase — Primary test for exocrine insufficiency — <200 mcg/g is diagnostic
- 72-hour fecal fat (coefficient of fat absorption) — Gold standard for steatorrhea — <93% absorption in children >6 months
- Fat-soluble vitamin levels (A, D, E) — Deficiency common with EPI
- MRCP (magnetic resonance cholangiopancreatography) — Duct anatomy in chronic pancreatitis
- Sweat chloride test — Rule out cystic fibrosis as underlying cause

Counselling

- Your child's pancreas is not producing enough enzymes to digest food properly
- Pancreatic enzyme capsules must be taken WITH every meal and snack — never after the meal
- Do not crush or chew the enzyme capsules — swallow whole or sprinkle contents on soft acidic food
- Extra fat-soluble vitamins (A, D, E, K) are needed because the body cannot absorb them well
- Watch for greasy, foul-smelling stools — this means the enzyme dose may need to be increased

Referral Criteria

Pediatric gastroenterology mandatory. CF team if cystic fibrosis is the cause. Endocrinology if CF-related diabetes develops. Dietitian for nutritional optimization. Surgery/ERCP for chronic pancreatitis with duct pathology

Hospitalisation Criteria

Severe malnutrition from exocrine insufficiency. Acute pancreatitis flare. Distal intestinal obstruction syndrome (CF patients). ERCP for duct intervention

Notes

PERT is the mainstay for exocrine pancreatic insufficiency. Do not exceed 10,000 LU/kg/day to avoid fibrosing colonopathy. Monitor growth and nutritional status regularly.

Guideline Changes

2024 European guidelines on EPI diagnosis and PERT dosing published (UEG/EPC/ESPGHAN)

K90.0 — Celiac disease

SNOMED: 396331005 | Duration: 365 days | Source: ESPGHAN 2022/2024, ACG 2023

First-Line

- **Strict gluten-free diet (GFD)** — Lifelong elimination of wheat, barley, rye. Dietitian-guided. Labeled GF oats permitted
- **Nutritional supplementation** — Iron, folate, calcium, vitamin D, zinc as needed based on deficiency labs

Second-Line

- **Lactose-free diet (temporary)** — Short-term lactose avoidance if secondary lactase deficiency present at diagnosis
- **Prednisolone** — 1-2 mg/kg/day PO x2-4wk for celiac crisis with severe malabsorption (rare)

Investigations

- tTG-IgA (tissue transglutaminase IgA) — Primary screening test — sensitivity >95%
- Total serum IgA — Rule out IgA deficiency (false negative tTG-IgA)
- Upper GI endoscopy with duodenal biopsy — Villous atrophy — Marsh classification confirms diagnosis
- CBC, iron, ferritin, folate, vitamin D, calcium — Nutritional deficiency screening — common at diagnosis
- HLA DQ2/DQ8 typing — Negative result rules out celiac disease (high NPV)

Counselling

- Your child has celiac disease — a lifelong strict gluten-free diet is the only treatment
- Avoid wheat (roti, bread, pasta), barley, and rye completely — even tiny amounts cause damage
- Rice, corn, jowar, bajra, and certified gluten-free oats are safe
- Read food labels carefully — gluten is hidden in many processed foods, sauces, and medicines
- Join a celiac support group — they can help with diet planning and recipes for Indian meals

Referral Criteria

Pediatric gastroenterology for diagnosis confirmation. Dietitian specializing in celiac disease MANDATORY. Endocrinology if associated thyroid disease. Dermatology for dermatitis herpetiformis. Re-evaluate non-responders for refractory celiac or wrong diagnosis

Hospitalisation Criteria

Celiac crisis (severe diarrhea, dehydration, metabolic derangement — rare). Severe anemia requiring transfusion. Failure to thrive not responding to GFD. Suspected refractory celiac disease

Notes

GFD is the only proven treatment. Monitor tTG-IgA for adherence. Follow-up with dietitian and serology at 3-6 month intervals initially, then annually.

Guideline Changes

2024 ESPGHAN position paper on GFD implementation; oats allowed if labeled gluten-free

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K90.9 — Intestinal malabsorption, unspecified

SNOMED: 32230006 | Duration: 30 days | Source: ESPGHAN 2023, Merck Manual 2024

First-Line

- **Nutritional rehabilitation** — High-calorie diet 120-150% RDA; oral supplements or enteral nutrition as needed
- **Micronutrient supplementation** — Iron, calcium, zinc, magnesium, fat-soluble vitamins (A, D, E, K) based on labs
- **Treat underlying cause** — Gluten-free diet for celiac, PERT for pancreatic insufficiency, antibiotics for SIBO

Second-Line

- **Parenteral nutrition** — TPN for severe malabsorption/intestinal failure; individualized caloric and micronutrient goals
- **Loperamide** — 0.05-0.1 mg/kg/dose PO (max 2 mg/dose) for symptomatic diarrhea; age >2yr only
- **Cholestyramine** — 240 mg/kg/day PO divided TID for bile acid malabsorption

Investigations

- tTG-IgA — Screen for celiac disease — most common cause in India
- Fecal elastase — Screen for pancreatic exocrine insufficiency
- Stool for reducing substances and pH — Carbohydrate malabsorption (lactose intolerance)
- CBC, iron, ferritin, vitamin D, calcium, zinc — Nutritional deficiency profiling
- Upper GI endoscopy with biopsy — If celiac serology positive or etiology unclear

Counselling

- Your child's body is not absorbing food properly — we need to find and treat the cause
- Extra calories, vitamins, and minerals may be needed to help the child grow
- Follow the dietitian's meal plan — the child may need special formulas or supplements
- Weight and height will be monitored closely — bring the child for regular follow-ups
- Some causes of malabsorption need lifelong dietary changes — the doctor will explain based on the diagnosis

Referral Criteria

Pediatric gastroenterology mandatory for diagnostic workup. Dietitian for nutritional rehabilitation. Allergist if food allergy causing malabsorption. Genetics if inherited cause (CF, congenital enteropathy)

Hospitalisation Criteria

Severe failure to thrive (weight loss >10%). Severe dehydration. Malnutrition requiring parenteral nutrition. Electrolyte imbalance. Diagnostic workup requiring endoscopy

Notes

Thorough workup essential to identify cause (celiac, IBD, EPI, SIBO, cow milk protein allergy). Prompt referral to pediatric GI for growth failure associated with malabsorption.

K92.2 — Gastrointestinal hemorrhage, unspecified

SNOMED: 74474003 | Duration: 7 days | Source: ESGE/ESPGHAN 2023, CHOP Clinical Pathway 2024

First-Line

- **IV fluid resuscitation** — NS 20 mL/kg bolus; crossmatch blood. Transfuse pRBC 10-15 mL/kg if Hb <7 or unstable
- **Esomeprazole IV** — <20kg: 10 mg IV daily; 20-55kg: 20 mg IV daily; >55kg: 40 mg IV daily; for upper GI bleed

- **Octreotide (variceal)** — 1-2 mcg/kg IV bolus then 1-2 mcg/kg/h infusion (max 50 mcg/h) for variceal hemorrhage

Second-Line

- **Endoscopic hemostasis** — EGD within 12h for variceal bleed or 24h for non-variceal; band ligation or injection
- **Tranexamic acid** — 15-25 mg/kg IV q6-8h (max 1 g/dose) as adjunct for severe hemorrhage
- **Sucralfate** — 40-80 mg/kg/day PO divided QID (max 1 g/dose) for mucosal protection

Investigations

- CBC (serial) — Hemoglobin trend — guides transfusion need [STAT]
- Type and crossmatch — Prepare for blood transfusion [STAT]
- Coagulation profile (PT, aPTT, INR) — Coagulopathy assessment and correction [STAT]
- Upper/lower GI endoscopy — Identify and treat bleeding source [URGENT]
- Meckel scan (Tc-99m pertechnetate) — Young children with painless rectal bleeding — Meckel diverticulum [URGENT]

Counselling

- Your child is bleeding from the stomach or intestines — this needs urgent hospital treatment
- The child may need blood transfusion if a lot of blood has been lost
- A camera test (endoscopy) will be done to find the source of bleeding and treat it
- Do not give any painkillers (especially ibuprofen/aspirin) without doctor's advice
- After discharge, follow up for repeat blood tests to make sure bleeding has stopped

Referral Criteria

Pediatric GI urgently for endoscopy. Surgery if massive hemorrhage not controlled endoscopically. Interventional radiology for angiographic embolization. Hematology if coagulopathy

Hospitalisation Criteria

All significant GI bleeding requires hospitalisation. Hemodynamic instability. Hemoglobin drop >2 g/dL. Active hematemesis or melena. Need for endoscopic intervention or blood transfusion

Notes

Stabilize hemodynamics first. Upper GI bleed (hematemesis, melena) requires IV PPI and early endoscopy. Lower GI bleed: consider Meckel scan in young children, colonoscopy in older children.

Q43.1 — Hirschsprung disease

SNOMED: 721608001 | Duration: 14 days | Source: ERNICA 2020, NASPGHAN 2023

First-Line

- **Pull-through surgery** — Soave, Swenson, or Duhamel; single-stage preferred. Laparoscopic or transanal approach
- **Rectal irrigations** — 20 mL/kg NS via rectal catheter BID-TID preoperatively to decompress

Second-Line

- **Diverting colostomy** — Staged approach for severe enterocolitis or total colonic aganglionosis
- **Metronidazole** — 30 mg/kg/day IV/PO divided TID x7-14 days for Hirschsprung-associated enterocolitis
- **Botulinum toxin A** — Intrasphincteric injection for postop obstructive symptoms; 1-4 units/kg per session

Investigations

- Rectal suction biopsy — Gold standard — absence of ganglion cells confirms diagnosis [URGENT]
- Barium enema (unprepared) — Transition zone identification
- Anorectal manometry — Absent rectoanal inhibitory reflex supports diagnosis
- Abdominal X-ray — Dilated proximal bowel, assess for enterocolitis [URGENT]

Counselling

- Your child has Hirschsprung disease — a section of the bowel lacks nerves and cannot push stool through
- Surgery is needed to remove the abnormal part and connect the healthy bowel
- Rectal irrigations (washing) will be done daily until surgery to prevent dangerous bowel swelling
- After surgery, the child may have loose or frequent stools initially — this usually improves over months
- Come back urgently if the child develops: fever with distended belly (enterocolitis), vomiting, or bloody stools

Referral Criteria

Pediatric surgery mandatory for definitive surgical management. GI for post-surgical bowel management. Genetics counselling (RET gene testing in family). Neonatology if diagnosed in newborn period

Hospitalisation Criteria

New diagnosis for stabilization and rectal irrigations. Hirschsprung-associated enterocolitis (HAEC) — medical emergency. Pre-operative preparation. Post-operative care

Notes

Definitive treatment is surgical. Hirschsprung-associated enterocolitis (HAEC) is a life-threatening complication requiring urgent rectal irrigations and IV antibiotics.

Guideline Changes

2023 NASPGHAN guidance on postsurgical defecation issues; trend toward single-stage pull-through

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R10.83 — Abdominal Colic / Infantile Colic

SNOMED: 139306005 | Duration: 14 days | Source: Rome IV Criteria (2016), IAP Infant Care Guidelines, ESPGHAN Probiotic Working Group

First-Line

- **Reassurance (non-pharmacological)** — Explain to parents: common condition, self-limiting (peaks at 6-8 weeks), resolves by 3-4 months
- **Simethicone** — 0.3 ml (as per preparation) TDS PO after feeds, preferred for gas relief and safety
- **Feeding Support (non-pharmacological)** — Proper latching technique, burping after every feed, avoid overfeeding
- **Supplements (Multivitamin Drops + Vitamin D3 400 IU + Iron if indicated)** — As per standard practice; Multivitamin drops + Vitamin D3 400 IU daily + Iron supplementation if indicated

Second-Line

- **Dicyclomine** — 5-7 drops TDS PO (morning, evening, night), short duration only, avoid routine long-term use
- **Amlexanox Gel / Zytee** — Apply 2-3 times/day for oral/gum care, only if associated irritability

Investigations

- Urine routine and microscopy — Rule out UTI as cause of excessive crying
- Stool occult blood — Suspected cow milk protein allergy if blood-streaked stools
- Complete blood count — If fever present or prolonged crying beyond 4 months to rule out infection

Counselling

- Colic is very common and not dangerous - it will resolve by 3-4 months
- Gentle tummy massage and warm compress provide relief
- Ensure proper breastfeeding technique with burping after every feed
- Avoid overfeeding the baby
- Breastfeeding mothers should avoid excess tea, coffee, spicy and gas-forming foods
- Return if baby has poor feeding, vomiting, fever, blood in stool, or poor weight gain

Warning Signs

- Fast breathing or chest indrawing

- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Poor feeding or refusal to feed
- Vomiting especially if bilious or forceful
- Blood in stool
- Failure to gain weight adequately

Referral Criteria

Refer to pediatric gastroenterologist if colic persists beyond 4 months, failure to thrive, blood in stool, persistent vomiting, or suspected cow's milk protein allergy

Hospitalisation Criteria

Hospitalisation rarely needed for colic. Admit only if poor feeding with dehydration, failure to thrive, suspected intestinal obstruction, or intussusception

Notes

Reassurance is the most important intervention. Infantile colic is self-limiting, peaks at 6-8 weeks, resolves by 3-4 months. Non-pharmacological measures: gentle abdominal massage, warm compress, carrying/rocking, white noise. Ensure proper latching, burping after every feed, avoid overfeeding. Breastfeeding mothers should avoid excess tea/coffee, spicy food, gas-forming foods. Simethicone preferred over dicyclomine for safety.

Guideline Changes

Rome IV criteria (2016) remain current standard; Lactobacillus reuteri DSM 17938 supported by recent meta-analyses as effective probiotic for breastfed colicky infants; no new 2024-2025 Rome criteria revision. Reassurance and parental support remain cornerstone.

R10.9 — Abdominal pain

SNOMED: 21522001 | Duration: 90 days | Source: ESPGHAN/NASPGHAN 2023/2025 FAPD Protocol, Rome IV

First-Line

- **Gut-directed hypnotherapy** — 6-12 sessions over 3 months; strongest evidence for functional abdominal pain disorders **[NEW 2024-25]**
- **Cognitive behavioral therapy** — 8-12 sessions; addresses pain catastrophizing, improves coping and functioning
- **Peppermint oil (enteric-coated)** — 187 mg PO TID (age >8yr) for IBS-related pain; smooth muscle relaxant

Second-Line

- **Amitriptyline** — 0.1-0.5 mg/kg PO QHS (max 25 mg); neuromodulator for chronic functional pain
- **Cyproheptadine** — 0.25-0.5 mg/kg/day PO divided BID-TID (max 12 mg); for functional dyspepsia/abdominal migraine
- **Probiotics (LGG)** — 10 billion CFU/day PO; modest evidence for functional abdominal pain

Investigations

- CBC with ESR/CRP — Screen for inflammatory process (IBD, infection)
- Stool calprotectin — Distinguish organic from functional cause — <50 mcg/g suggests functional
- Celiac screen (tTG-IgA + total IgA) — Celiac disease screening — recommended in all chronic abdominal pain
- Urinalysis — UTI can present as abdominal pain in children
- Abdominal ultrasound — If atypical features or to reassure family

Counselling

- Chronic belly pain in children is very common and usually not from a serious disease
- Stress, anxiety, and school pressure are common triggers — addressing these often helps the most
- Gut-directed therapy (hypnotherapy/CBT) has the strongest evidence — consider it as first treatment

- Keep a pain diary: when, where, how severe, and what the child ate — bring it to follow-ups
- Red flags that need urgent evaluation: weight loss, blood in stool, fever, night-time pain waking the child

Referral Criteria

Pediatric GI if red flags present or no improvement after 4-8 weeks. Psychology/hypnotherapy for gut-brain axis therapy. Dietitian if dietary triggers identified. Surgery urgently if acute abdomen suspected

Hospitalisation Criteria

Acute abdomen (appendicitis, volvulus, obstruction). Significant weight loss with abdominal pain. GI bleeding. Intractable pain requiring IV pain management

Notes

Most pediatric chronic abdominal pain is functional (Rome IV criteria). Red flags (weight loss, GI bleeding, fever) require workup. Gut-brain psychotherapy is first-line per 2025 guidelines.

Guideline Changes

2025 ESPGHAN/NASPGHAN IBS/FAP guidelines published; gut-brain psychotherapies have strongest evidence

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R11 — Vomiting

SNOMED: 422400008 | Duration: 3 days | Source: ESPGHAN 2014, CPS 2023, AAP 2023

First-Line

- **Ondansetron** — 0.15 mg/kg PO/IV single dose (max 4 mg); may repeat q8h. For AGE-related vomiting
- **ORS (small volumes)** — 5 mL PO q2-5min after vomiting; increase as tolerated to prevent dehydration

Second-Line

- **Promethazine** — 0.25-0.5 mg/kg PO/IM/IV q4-6h (max 25 mg); age >2yr. Sedation risk
- **Dimenhydrinate** — 1.25 mg/kg PO/IV q6h (max 50 mg/dose); for motion sickness or vestibular vomiting
- **Metoclopramide** — 0.1-0.15 mg/kg PO/IV q6-8h (max 10 mg); limited use due to EPS risk
- **Dexamethasone** — 0.15 mg/kg IV q6h (max 10 mg) for CINV or cyclic vomiting prophylaxis

Investigations

- Serum electrolytes — Metabolic alkalosis (hypokalemic, hypochloremic) from persistent vomiting [URGENT]
- Blood glucose — Hypoglycemia from poor intake [STAT]
- Abdominal ultrasound — Pyloric stenosis (pyloric muscle thickness >3mm), intussusception [URGENT]
- Abdominal X-ray — Obstruction, free air [URGENT]
- Urinalysis — UTI can present with vomiting in young children

Counselling

- Your child is vomiting — small frequent sips of ORS are more effective than large amounts
- Wait 10 minutes after vomiting, then restart ORS in small sips (5 mL every 2-5 minutes)
- Once vomiting stops, gradually return to a normal diet — do not keep the child on liquids only
- Watch for: vomiting green/yellow fluid (bile), blood in vomit, or severe belly pain — come back immediately
- For young infants (under 2 months), any persistent vomiting needs urgent evaluation

Referral Criteria

Pediatric surgery urgently if bilious vomiting in infant (rule out malrotation/volvulus). GI for chronic/recurrent vomiting. Neurology if vomiting with headache or neurological signs (raised ICP). Endocrinology for cyclic vomiting syndrome

Hospitalisation Criteria

Bilious vomiting in any infant (surgical emergency until proven otherwise). Severe dehydration. Projectile vomiting in infant <3 months (pyloric stenosis). Hematemesis. Altered mental status. Intractable vomiting >24 hours

Notes

Ondansetron is first-line for AGE-related vomiting; single dose usually sufficient. Rule out surgical emergencies (pyloric stenosis, malrotation, obstruction) before attributing to benign cause.

R19.7 — Diarrhea, unspecified

Duration: 7 days | Source: ESPGHAN 2014, WHO 2024, JGA Chronic Diarrhea 2023

First-Line

- **ORS (reduced osmolarity)** — 50-100 mL/kg PO over 3-4h for dehydration; ad libitum for maintenance
- **Zinc sulfate** — 10 mg/day PO (<6mo) or 20 mg/day PO (>6mo) x10-14 days per WHO guidelines
- **Dietary management** — Continue breastfeeding; regular diet. Reduce juice/sugary drinks for toddler diarrhea

Second-Line

- **Racecadotril** — 1.5 mg/kg PO TID x5-7 days; antisecretory for acute watery diarrhea
- **Loperamide** — 0.05-0.1 mg/kg PO TID (max 2 mg/dose); chronic diarrhea only, age >2yr
- **Probiotics (S. boulardii)** — 250-750 mg/day PO x5-7 days as adjunct; evidence for duration reduction

Investigations

- Stool routine and microscopy — WBCs, RBCs, parasites, reducing substances
- Stool culture — If bloody diarrhea or persistent diarrhea
- Serum electrolytes — If dehydration present [URGENT]
- Stool calprotectin — If chronic diarrhea — distinguish inflammatory from functional

Counselling

- Your child has loose stools — ORS and zinc are the most important treatment
- Give ORS after every loose stool — it replaces the water and salts lost
- Continue feeding normally — stopping food makes diarrhea worse, not better
- If diarrhea lasts more than 2 weeks, further investigation is needed — come for follow-up
- Reduce fruit juice and sugary drinks — too much juice is a common cause of loose stools in toddlers

Referral Criteria

Pediatric GI if diarrhea persists >14 days (persistent) or >4 weeks (chronic). Allergy/immunology if food allergy or immunodeficiency suspected. Dietitian for chronic diarrhea dietary management

Hospitalisation Criteria

Severe dehydration. Failed oral rehydration. Bloody diarrhea with systemic toxicity. Infant <3 months with significant diarrhea. Chronic diarrhea with failure to thrive

Notes

Investigate persistent (>14 days) or chronic (>4 weeks) diarrhea for underlying cause. Toddler diarrhea is most common chronic cause in ages 1-3yr and is managed with dietary changes.

General

R79.89 — Electrolyte Imbalance with Laboratory Corrections (Corrected Calcium, Sodium, Osmolality)

SNOMED: 105593004 | Duration: 1 days | Source: Clinical formula reference (Electrolyte correction)

First-Line

- **Treat underlying electrolyte disorder** — Management depends on primary abnormality (e.g., hypocalcemia, hyponatremia, hyperglycemia, DKA). Use corrected values to guide therapy rather than measured values alone.
- **Intravenous Fluids** — Use isotonic fluids initially; adjust based on corrected sodium and osmolality values

Second-Line

- **Electrolyte replacement therapy** — Calcium, sodium, potassium correction guided by corrected lab values and clinical condition

Investigations

- Serum Calcium — To calculate corrected calcium in hypoalbuminemia [URGENT]
- Serum Albumin — Needed for corrected calcium calculation
- Serum Sodium — To calculate corrected sodium in hyperglycemia [URGENT]
- Blood Glucose — Correct sodium and calculate osmolality [STAT]
- Blood Urea / BUN — Osmolality calculation [URGENT]

Counselling

- Lab values may need correction based on albumin or glucose levels — जांच के नंबर एल्ब्युमिन या शुगर के अनुसार ठीक किए जा सकते हैं
- Treatment decisions are based on corrected values, not raw reports — इलाज सही किए गए नंबरों पर आधारित होता है, सिर्फ रिपोर्ट पर नहीं
- Do not panic with abnormal numbers without clinical correlation — असामान्य नंबर देखकर घबराएं नहीं, डॉक्टर से सलाह लें
- Regular monitoring is required for accurate treatment — सही इलाज के लिए नियमित जांच जरूरी है
- Follow doctor advice strictly for fluid and electrolyte correction — तरल पदार्थ और इलेक्ट्रोलाइट सुधार के लिए डॉक्टर की सलाह का सख्ती से पालन करें

Warning Signs

- Altered consciousness
- Seizures
- Severe dehydration
- Arrhythmias
- Persistent vomiting
- Weakness or paralysis

Referral Criteria

Refer if severe electrolyte imbalance, altered sensorium, or complex metabolic disturbance

Hospitalisation Criteria

Admit if symptomatic electrolyte imbalance or abnormal osmolality requiring IV correction

Notes

Important formulae: Corrected Calcium = Total Ca + 0.8(4 - albumin). Corrected Na = Measured Na + 1.6[(glucose - 100)/100]. Osmolality = 2Na + glucose/18 + BUN/2.8.

Haematology

C41 — Malignant neoplasm of bone and articular cartilage

SNOMED: 93725000 | Duration: 270 days | Source: COG, EURAMOS, NCCN 2025

First-Line

- **MAP (neoadjuvant)** — High-dose methotrexate 12 g/m2 IV + doxorubicin 37.5 mg/m2 IV x2d + cisplatin 60 mg/m2 IV x2d; 8-10wk
- **Surgical resection** — Limb-salvage surgery preferred if feasible; wide excision with negative margins after neoadjuvant chemo
- **MAP (adjuvant)** — Continue MAP regimen post-op x12-29 weeks; modify if <90% tumor necrosis

Second-Line

- **Ifosfamide + etoposide** — Ifosfamide 1.8 g/m2 IV x5d + etoposide 100 mg/m2 IV x5d; add for poor histologic response
- **High-dose MTX + leucovorin** — MTX 12 g/m2 IV over 4h with leucovorin rescue 15 mg/m2 q6h x10 doses starting at 24h post-MTX

Investigations

- X-ray of affected bone — Sunburst pattern, Codman triangle (osteosarcoma) [URGENT]
- MRI of primary tumor — Assess local extent and skip lesions [URGENT]
- CT chest — Screen for pulmonary metastases [URGENT]
- Bone scan or PET-CT — Screen for distant metastases [URGENT]
- Biopsy (core needle or open) — Tissue diagnosis before chemotherapy [URGENT]

Counselling

- Bone cancer (osteosarcoma) treatment involves chemotherapy and surgery — limb-salvage is usually possible
- Treatment is long (9-12 months) but cure rates are around 65% for localized disease
- Report new or worsening pain, fever, or swelling immediately
- Physiotherapy after surgery is essential for limb function recovery
- Emotional support and counselling for the child and family is strongly recommended

Referral Criteria

Refer urgently to pediatric orthopedic oncology center with chemotherapy and limb-salvage surgery capability

Hospitalisation Criteria

Admit for neoadjuvant chemotherapy, surgical resection, and management of chemotherapy complications

Notes

Osteosarcoma is most common; 5-year OS 62-65% for localized disease. Assess histologic response (>90% necrosis = good) to guide adjuvant therapy.

C49 — Malignant neoplasm of connective and soft tissue

SNOMED: 424413001 | Duration: 365 days | Source: COG ARST, EpSSG 2025

First-Line

- **VAC (rhabdomyosarcoma)** — Vincristine 1.5 mg/m2 IV wkly, actinomycin D 0.045 mg/kg IV, cyclophosphamide 2.2 g/m2 IV q3wk
- **Surgical resection** — Complete excision if feasible; delayed surgery after chemo for initially unresectable tumors
- **Radiation therapy** — 40-50 Gy to tumor bed; for microscopic or gross residual disease after surgery

Second-Line

- **VAC/VI alternating** — VAC alternating with vincristine + irinotecan (VCR 1.5 mg/m2 + irinotecan 50 mg/m2 x5d); less toxicity
- **Ifosfamide + doxorubicin** — Ifosfamide 3 g/m2 IV x3d + doxorubicin 30 mg/m2 IV x2d for non-rhabdo soft tissue sarcomas

Investigations

- MRI of primary tumor — Local extent and surgical planning [URGENT]
- CT chest — Screen for pulmonary metastases [URGENT]
- Bone marrow aspirate and biopsy — Staging in rhabdomyosarcoma [URGENT]
- PET-CT or bone scan — Distant metastasis screening [URGENT]
- Core needle or open biopsy — Tissue diagnosis with immunohistochemistry [URGENT]

Counselling

- Rhabdomyosarcoma is the most common soft tissue cancer in children — it responds well to treatment
- Treatment involves chemotherapy, surgery, and sometimes radiation over many months
- Report any new lumps, swelling, or worsening pain immediately
- Maintain good nutrition during treatment to support the child's recovery
- Regular follow-up imaging is needed even after treatment completion

Referral Criteria

Refer urgently to pediatric oncology center with multidisciplinary sarcoma team

Hospitalisation Criteria

Admit for chemotherapy initiation, surgical resection, neutropenic fever, or tumor-related emergencies (obstruction, compression)

Notes

Rhabdomyosarcoma is the most common pediatric STS. Low-risk RMS achieves >90% FFS with VAC. Multimodal therapy (chemo, surgery, RT) is standard.

Guideline Changes

EpSSG 2025 released European standard care recommendations for childhood rhabdomyosarcoma

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C64 — Malignant neoplasm of kidney, except renal pelvis

SNOMED: 302849000 | Duration: 180 days | Source: COG (NWTS/AREN), SIOP

First-Line

- **Nephrectomy (upfront, COG)** — Radical nephrectomy with lymph node sampling; COG approach is surgery first then chemo
- **Actinomycin D + Vincristine (EE-4A)** — Actinomycin D 0.045 mg/kg IV + VCR 1.5 mg/m2 IV wkly x18 weeks for stage I-II FH
- **Doxorubicin (DD-4A)** — Add doxorubicin 1.5 mg/kg IV to VA for stage III-IV FH; regimen DD-4A per COG protocol

Second-Line

- **Neoadjuvant chemo (SIOP)** — Actinomycin D + VCR x4-6 weeks pre-op; SIOP approach for large/bilateral tumors
- **Regimen M (high-risk)** — Cyclophosphamide, etoposide + vincristine, doxorubicin, cyclophosphamide alternating; for anaplastic histology

Investigations

- Abdominal ultrasound — Initial evaluation of renal mass [URGENT]
- CT or MRI abdomen with contrast — Staging, IVC extension, bilateral assessment [URGENT]
- CT chest — Pulmonary metastasis screening [URGENT]

- CBC, LFT, RFT — Baseline and surgical planning [URGENT]
- Urinalysis — Hematuria screening

Counselling

- Wilms tumor (kidney cancer) has an excellent cure rate — over 90% for early-stage disease
- Surgery to remove the kidney is the main treatment, followed by chemotherapy
- The child can live a normal life with one kidney — protect the remaining kidney from injury
- Report abdominal swelling, blood in urine, or fever immediately
- Follow-up imaging (ultrasound/CT) will continue for several years after treatment

Referral Criteria

Refer urgently to pediatric oncology center with pediatric surgical capability

Hospitalisation Criteria

Admit for nephrectomy and chemotherapy initiation; admit urgently for tumor rupture, hemorrhage, or bilateral tumor management

Notes

Wilms tumor has >90% overall survival for favorable histology. LOH at 1p/16q indicates worse prognosis and need for more intensive chemotherapy.

C71 — Malignant neoplasm of brain

SNOMED: 428061005 | Duration: 365 days | Source: COG, SIOP, NCCN CNS 2025

First-Line

- **Maximal safe resection** — Neurosurgical resection targeting gross total removal; residual <1 cm x 1 cm if possible
- **Craniospinal irradiation (MB)** — CSI 23.4 Gy (std risk) or 36 Gy (high risk) + tumor bed boost to 54-55.8 Gy for medulloblastoma
- **Cisplatin + vincristine + cyclophos** — Post-radiation chemo for MB: cisplatin 75 mg/m2, VCR 1.5 mg/m2 weekly, cyclophosphamide 1.5 g/m2

Second-Line

- **Carboplatin + vincristine** — Carboplatin 175 mg/m2 IV wkly x4 + VCR 1.5 mg/m2 IV wkly for low-grade glioma; 10-wk cycles
- **Temozolomide** — 150-200 mg/m2 PO x5 days q28 days for recurrent/progressive high-grade glioma
- **Bevacizumab** — 10 mg/kg IV q2wk for recurrent high-grade glioma; anti-VEGF targeted therapy

Investigations

- MRI brain with contrast — Definitive imaging for tumor characterization [STAT]
- MRI spine — Screen for drop metastases (medulloblastoma) [URGENT]
- CSF cytology (post-op) — CNS staging for medulloblastoma/ependymoma [URGENT]
- CT head (if MRI not immediately available) — Rapid assessment for hydrocephalus [STAT]
- Fundoscopy — Papilledema assessment for raised ICP [STAT]

Counselling

- Brain tumors in children often have good outcomes with proper treatment at a specialized center
- Surgery is usually the first step; chemotherapy and/or radiation may follow depending on tumor type
- Watch for headaches, vomiting (especially morning), balance problems, or vision changes
- The child may need rehabilitation (physio, OT, speech) after surgery
- Long-term follow-up is essential — regular MRI scans will be needed for years

Referral Criteria

Refer urgently to pediatric neurosurgery and neuro-oncology center

Hospitalisation Criteria

Admit urgently for raised intracranial pressure, new neurological deficits, posterior fossa syndrome, or for surgical planning

Notes

Treatment is tumor-type specific (medulloblastoma, glioma, ependymoma). Avoid CSI in children <3y; use chemotherapy-only protocols for infants.

Guideline Changes

TTFields (tumor-treating fields) emerging for pediatric brain tumors. WNT-subtype MB trials de-escalating therapy

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C81 — Hodgkin lymphoma

SNOMED: 118599009 | Duration: 180 days | Source: COG AHOD2131, NCCN Pediatric HL 2025

First-Line

- **ABVD** — Doxorubicin 25 mg/m2, bleomycin 10 U/m2, vinblastine 6 mg/m2, dacarbazine 375 mg/m2 IV q2wk; 4-6 cycles
- **Brentuximab vedotin + AVEPC** — BV 1.8 mg/kg IV d1 + doxorubicin, vincristine, etoposide, pred, cyclophosphamide q3wk x5 cycles **[NEW 2024-25]**
- **ISRT (involved-site RT)** — Low-dose 21 Gy ISRT for slow early responders or bulky disease; response-adapted approach

Second-Line

- **Nivolumab** — 3 mg/kg IV q3wk for R/R HL; used with brentuximab +/- bendamustine **[NEW 2024-25]**
- **Brentuximab vedotin (single)** — 1.8 mg/kg IV q3wk for up to 16 cycles for R/R HL
- **ICE salvage** — Ifosfamide, carboplatin, etoposide for relapsed HL prior to autologous HSCT

Investigations

- Excisional lymph node biopsy — Definitive diagnosis — tissue architecture needed [URGENT]
- PET-CT scan — Staging and response assessment [URGENT]
- CBC, ESR, LDH, albumin — Baseline labs and prognostic factors [URGENT]
- Chest X-ray — Mediastinal mass evaluation [URGENT]
- Bone marrow biopsy — Stage IV disease staging

Counselling

- Hodgkin lymphoma has a cure rate above 90% in children with proper treatment
- Treatment involves chemotherapy cycles given every few weeks for several months
- Side effects like nausea, hair loss, and tiredness are temporary and manageable
- Maintain good nutrition and report fever or breathing difficulty immediately
- Long-term follow-up after cure is important to monitor for late effects of treatment

Referral Criteria

Refer to pediatric oncology center; urgent if compressive symptoms (SVC syndrome, airway compromise)

Hospitalisation Criteria

Admit for initiation of chemotherapy, SVC syndrome, airway compromise, neutropenic fever, or severe treatment toxicity

Notes

Cure rate 90-95%. BV+AVEPC achieved 92% 3-year EFS vs 82.5% with standard chemo. Minimize radiation to reduce late effects in children.

Guideline Changes

Brentuximab vedotin + AVEPC is new standard for high-risk pediatric HL (FDA approved 2022)

C85 — Non-Hodgkin lymphoma

SNOMED: 118601006 | Duration: 240 days | Source: COG, BFM, NCCN 2025

First-Line

- **R-CHOP / R-LMB** — Rituximab 375 mg/m2 IV + cyclophosphamide, doxorubicin, vincristine, prednisone per BFM/LMB protocol
- **BFM protocol (T-/pre-B LBL)** — ALL-like regimen: induction, consolidation, maintenance x2 years total with VCR, pred, Asp, MTX, 6-MP

Second-Line

- **Rituximab + chemo (Burkitt)** — Short intensive pulses q3-4wk x6-8 months; French LMB protocol with rituximab for B-NHL
- **Brentuximab vedotin** — 1.8 mg/kg IV q3wk for CD30+ anaplastic large cell lymphoma; up to 16 cycles

Investigations

- Tissue biopsy with immunohistochemistry — Definitive subtype diagnosis [URGENT]
- PET-CT or CT chest/abdomen/pelvis — Staging [URGENT]
- Bone marrow aspirate and biopsy — Marrow involvement staging [URGENT]
- CSF analysis — CNS involvement in Burkitt/LBL [URGENT]
- LDH, uric acid, electrolytes — Tumor lysis risk assessment [STAT]

Counselling

- NHL in children responds well to chemotherapy — cure rates exceed 80%
- Treatment is intensive and requires close hospital follow-up
- Watch for rapid swelling, breathing difficulty, or abdominal pain and seek emergency care
- Good nutrition and hydration are critical during treatment
- The child may need to miss school during intensive phases — coordinate with the school

Referral Criteria

Refer urgently to pediatric oncology center; emergency if airway/bowel obstruction or tumor lysis

Hospitalisation Criteria

Admit at diagnosis for chemotherapy initiation, emergency admission for tumor lysis syndrome, bowel obstruction, or respiratory compromise

Notes

NHL subtype determines protocol: Burkitt/DLBCL use short intensive chemo, LBL uses ALL-like 2-year regimen. Cure rate exceeds 80% overall.

Guideline Changes

Rituximab now routinely added to chemotherapy for mature B-cell NHL in children

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C91.0 — Acute lymphoblastic leukemia

SNOMED: 91857003 | Duration: 730 days | Source: COG AALL1731, NCCN 2025

First-Line

- **Induction: VCR+Dex+PEG-Asp+Dauno** — Vincristine 1.5 mg/m2 IV wkly, dexamethasone 6 mg/m2/day PO, PEG-asparaginase, daunorubicin 25 mg/m2
- **Blinatumomab (consolidation)** — 15 mcg/m2/day continuous IV infusion x28 days per cycle; now standard for B-ALL consolidation **[NEW 2024-25]**
- **Intrathecal MTX/Ara-C/Hydrocort** — Age-adjusted IT triple therapy for CNS prophylaxis; frequency per protocol risk group

- **Maintenance: 6-MP + MTX** — 6-mercaptopurine 75 mg/m2 PO daily + methotrexate 20 mg/m2 PO weekly x2 years total

Second-Line

- **Tisagenlecleucel (CAR-T)** — 0.2-5x10^6 CAR-T cells/kg IV single dose for R/R B-ALL <=25y; requires lymphodepletion
- **Inotuzumab ozogamicin** — Approved 2024 for R/R CD22+ B-ALL >=1y; given IV per cycle per protocol **[NEW 2024-25]**
- **Nelarabine** — 650 mg/m2 IV daily x5 days for T-ALL; added to consolidation in T-ALL protocols

Investigations

- CBC with peripheral smear — Initial screening — blasts on smear [STAT]
- Bone marrow aspirate with flow cytometry — Confirm diagnosis, immunophenotype [STAT]
- Cytogenetics and FISH — Risk stratification (BCR-ABL, ETV6-RUNX1, etc.) [URGENT]
- CSF analysis — Assess CNS involvement [URGENT]
- Chest X-ray — Mediastinal mass in T-ALL [STAT]

Counselling

- ALL is the most common childhood cancer and has a very high cure rate (>90%) with proper treatment
- Treatment lasts about 2-3 years — maintaining regular hospital visits is critical
- Watch for fever, unusual bleeding, or bone pain and report immediately
- The child will be immunosuppressed — avoid crowded places and sick contacts
- Good nutrition and emotional support are essential throughout treatment

Referral Criteria

Refer urgently to pediatric oncology center with ALL protocol capability

Hospitalisation Criteria

Admit at diagnosis for induction chemotherapy, for neutropenic fever, tumor lysis syndrome, or any life-threatening complication

Notes

Risk stratification by genetics, MRD, and response guides therapy intensity. Blinatumomab in consolidation improved 3-year DFS to 96% (AALL1731).

Guideline Changes

Blinatumomab added to frontline consolidation (COG AALL1731, 2024). Inotuzumab approved 2024 for R/R ALL

C92.0 — Acute myeloid leukemia

SNOMED: 91861009 | Duration: 180 days | Source: COG AAML1831, NCCN 2025

First-Line

- **Induction: Cytarabine + Daunorubicin** — Ara-C 100-200 mg/m2/day CI x7 days + daunorubicin 50 mg/m2 IV x3 days (7+3); 2 cycles
- **Etoposide** — 100-150 mg/m2/day IV x5 days; added to induction as ADE regimen
- **Gemtuzumab ozogamicin** — 3 mg/m2 IV (max 4.5 mg) on days 1,4,7 of induction; for CD33+ AML
- **Intrathecal cytarabine** — Age-adjusted IT Ara-C for CNS prophylaxis; given during each chemotherapy cycle

Second-Line

- **CPX-351 (liposomal Ara-C/dauno)** — First unit dose on days 1,3,5 IV; COG AAML1421 showed 81% response; for high-risk AML **[NEW 2024-25]**
- **Venetoclax + chemo** — Under investigation in APAL2020D trial for R/R AML; promising early results in children **[NEW 2024-25]**
- **Allogeneic HSCT** — For high-risk AML or poor response after induction; MSD or MUD donor preferred

Investigations

- CBC with peripheral smear — Auer rods, blasts on smear [STAT]
- Bone marrow aspirate with flow cytometry — Confirm AML, identify subtype [STAT]
- Cytogenetics and molecular studies — Risk stratification (t(8;21), inv(16), FLT3-ITD) [URGENT]
- Coagulation profile — DIC common especially in APL [STAT]
- CSF analysis — Assess CNS involvement [URGENT]

Counselling

- AML treatment is intensive but most children enter remission after the first phase
- The child will need to stay in the hospital for weeks during each chemotherapy cycle
- Infections are the biggest risk — strict hand hygiene and avoiding sick contacts is critical
- Report fever immediately — even 38C needs urgent medical attention during treatment
- Emotional and psychological support for the child and family is very important

Referral Criteria

Refer urgently to pediatric oncology center with AML protocol capability and HSCT access

Hospitalisation Criteria

Admit at diagnosis for induction chemotherapy; admit for neutropenic fever, bleeding, or any complication during treatment

Notes

Over 85% of children enter remission after induction. Risk stratification by cytogenetics and MRD guides consolidation and HSCT decisions.

Guideline Changes

Venetoclax being evaluated in frontline pediatric AML (APAL2020D trial). CPX-351 showed 81% ORR in COG study

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D50 — Iron deficiency anemia

SNOMED: 87522002 | Duration: 90 days | Source: AAP 2010, EHA/ASH 2024

First-Line

- **Ferrous sulfate** — 3-6 mg/kg/day elemental iron PO divided 1-2 doses on empty stomach; continue 2-3 months after Hb normalizes
- **Ferrous fumarate** — 3-6 mg/kg/day elemental iron PO divided 1-2 doses; take with vitamin C to enhance absorption

Second-Line

- **Iron polymaltose (Fe3+)** — 3-5 mg/kg/day elemental iron PO with meals in 1-2 doses; better GI tolerance
- **IV iron (ferric carboxymaltose)** — 15 mg/kg IV single dose (max 750 mg); for oral iron failure or malabsorption

Investigations

- CBC with peripheral smear — Confirm microcytic hypochromic anemia [URGENT]
- Serum ferritin — Confirm iron deficiency (low ferritin is diagnostic)
- Serum iron, TIBC — Calculate transferrin saturation
- Stool for occult blood — Rule out GI blood loss in older children
- Hb electrophoresis — Rule out thalassemia trait if not responding to iron

Counselling

- Give iron drops/syrup on empty stomach with orange juice or lemon water for better absorption
- Avoid giving milk, tea, or calcium with iron — wait at least 2 hours
- Iron may cause black stools and mild stomach upset — this is normal

- Continue iron for 2-3 months after hemoglobin normalizes to replenish stores
- Include iron-rich foods: jaggery, green leafy vegetables, eggs, ragi, dates

Referral Criteria

Refer to hematologist if no rise in Hb after 4-6 weeks of adequate oral iron, or if thalassemia trait or chronic blood loss suspected

Hospitalisation Criteria

Admit if Hb <5 g/dL, signs of heart failure, severe pallor with tachycardia, or need for blood transfusion

Notes

Avoid milk/calcium with iron doses. Treatment typically 3-6 months total to replenish stores; recheck ferritin at 3 months.

Guideline Changes

2024 EHA consensus recommends lower-frequency dosing (alternate-day) for better absorption

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D51 — Vitamin B12 deficiency anemia

SNOMED: 191140004 | Duration: 120 days | Source: IAP 2022, BSH

First-Line

- **Cyanocobalamin (IM)** — 1000 mcg IM x1, then 1000 mcg weekly x6 weeks; infants <2y: 1000 mcg IM then 250 mcg/wk x6wk
- **Cyanocobalamin (oral)** — 1000 mcg PO daily x4 months for mild-moderate nutritional deficiency

Second-Line

- **Hydroxocobalamin (IM)** — 1 mg IM 3x/week x2 weeks then 1 mg IM every 3 months; preferred in some guidelines
- **Sublingual B12** — 1000 mcg sublingual daily; alternative to oral in children with poor compliance

Investigations

- CBC with peripheral smear — Confirm megaloblastic anemia [URGENT]
- Serum vitamin B12 — Confirm deficiency [URGENT]
- Serum methylmalonic acid — More sensitive marker for B12 deficiency
- Serum folate — Rule out concurrent folate deficiency
- Anti-intrinsic factor antibodies — If pernicious anemia suspected

Counselling

- B12 injections are important — do not skip scheduled doses
- Include B12-rich foods: milk, curd, eggs, paneer, and fortified cereals
- Strict vegetarian families should give B12 supplements to children regularly
- Neurological symptoms (tingling, weakness) should improve within weeks of treatment
- If the cause is pernicious anemia, lifelong B12 injections will be needed

Referral Criteria

Refer to hematologist if neurological symptoms present, if anemia does not respond to B12 within 4 weeks, or if pernicious anemia suspected (rare in children)

Hospitalisation Criteria

Admit if severe anemia (Hb <5 g/dL), neurological compromise, or severe pancytopenia

Notes

Start B12 at least 10-14 days before folic acid if combined deficiency to prevent neurological deterioration. Lifelong therapy if pernicious anemia.

D52 — Folate deficiency anemia

SNOMED: 85649008 | Duration: 120 days | Source: IAP 2022, WHO

First-Line

- **Folic acid** — 1 mg PO daily for 4 months; infants 0.1 mg/day, children 1-3y 0.15 mg/day, >4y up to 1 mg/day
- **Dietary counseling** — Increase folate-rich foods: green leafy vegetables, legumes, fortified cereals

Second-Line

- **Folic acid (parenteral)** — 1 mg IM/IV/SC daily if oral absorption impaired; same duration as oral
- **Folic acid high-dose** — 5 mg PO daily for 4 months if antiepileptic drug-induced deficiency

Investigations

- CBC with reticulocyte count — Confirm megaloblastic anemia and monitor response [URGENT]
- Serum folate level — Confirm folate deficiency
- Serum vitamin B12 — Rule out concurrent B12 deficiency (critical) [URGENT]
- Peripheral blood smear — Look for megaloblastic changes, hypersegmented neutrophils

Counselling

- Give folic acid tablet daily on empty stomach for best absorption
- Include green leafy vegetables (palak, methi), dal, and fortified cereals in diet
- Improvement in energy and appetite expected within 2-3 weeks
- Complete the full 4-month course even if the child feels better
- Follow up with blood test at 1 month and 3 months to check progress

Referral Criteria

Refer to hematologist if anemia does not respond to folic acid after 4 weeks, or if pancytopenia or hemolysis suspected

Hospitalisation Criteria

Admit if Hb <5 g/dL with cardiac symptoms, or if severe pancytopenia requiring transfusion

Notes

Rule out B12 deficiency before starting folate to avoid masking neurological damage. Reticulocytosis expected by day 3-4, anemia resolves in 4-8 weeks.

D53 — Other nutritional anemia

SNOMED: 66612000 | Duration: 90 days | Source: IAP 2022, WHO

First-Line

- **Pyridoxine (vitamin B6)** — 5-25 mg PO daily for B6 deficiency anemia; higher doses if INH-induced
- **Copper supplementation** — 0.1 mg/kg/day PO elemental copper for copper-deficiency anemia; max 3 mg/day

Second-Line

- **Vitamin C** — 100-300 mg PO daily for scurvy-related anemia; give with iron if coexistent deficiency
- **Multivitamin + minerals** — Age-appropriate multivitamin PO daily when multiple nutritional deficiencies suspected

Investigations

- CBC with peripheral smear — Characterize anemia morphology [URGENT]
- Serum B6, copper, vitamin C levels — Identify specific deficiency
- Serum B12 and folate — Rule out concurrent deficiencies
- Reticulocyte count — Assess marrow response

Counselling

- Give the prescribed vitamin/mineral supplement daily as directed
- Include a variety of foods: fruits, vegetables, milk, eggs, and whole grains
- If the child is a strict vegetarian, supplementation is especially important
- Improvement in appetite and energy expected within 2-4 weeks
- Complete the full course and get a follow-up blood test as advised

Referral Criteria

Refer to hematologist if anemia does not respond to supplementation in 4-6 weeks, or if multiple cytopenias present

Hospitalisation Criteria

Admit if severe anemia (Hb <5 g/dL), neurological symptoms from B12/copper deficiency, or need for parenteral supplementation

Notes

D53 includes anemia from B6, copper, and other nutrient deficiencies. Identify specific deficiency with targeted labs before supplementing.

D56 — Thalassemia

SNOMED: 40108008 | Duration: 365 days | Source: TIF 2021, ASH

First-Line

- **Chronic RBC transfusion** — 10-15 mL/kg PRBC IV q2-4wk to maintain pre-transfusion Hb 9-10 g/dL in TDT
- **Deferasirox** — 20-40 mg/kg/day PO once daily for iron chelation; start after 20-25 transfusions or ferritin >1000
- **Folic acid** — 1 mg PO daily to support erythropoiesis

Second-Line

- **Deferoxamine** — 20-40 mg/kg/day SC infusion over 8-12h, 5-7 days/wk; gold standard chelator, poor compliance
- **Deferiprone** — 75-100 mg/kg/day PO divided TID; best for cardiac iron overload; risk of agranulocytosis
- **Allogeneic HSCT** — Curative; best outcomes if MSD donor and low Pesaro risk class; perform early in disease
- **Casgevy (exagamglogene)** — Gene therapy; one-time autologous HSCT with CRISPR-edited cells; approved >=12y TDT **[NEW 2024-25]**

Investigations

- CBC and Hb electrophoresis/HPLC — Diagnostic confirmation and monitoring [URGENT]
- Serum ferritin — Monitor iron overload (target <1000 ng/mL)
- Liver iron concentration (MRI) — Quantify hepatic iron overload
- Cardiac MRI T2* — Detect cardiac iron loading (annually from age 10)
- HLA typing (patient + siblings) — HSCT donor search

Counselling

- Regular blood transfusions every 2-4 weeks are essential to maintain adequate hemoglobin
- Iron chelation medicine must be taken daily to prevent iron overload in the heart and liver
- Avoid iron-rich supplements and iron-fortified foods — the child already has excess iron from transfusions
- Annual cardiac MRI (T2*) and liver iron monitoring are critical from age 10
- Genetic counselling is important for the family — both parents are carriers

Referral Criteria

Refer to pediatric hematologist at a thalassemia center; refer for HSCT evaluation if matched sibling donor available

Hospitalisation Criteria

Admit for transfusion reactions, cardiac decompensation from iron overload, severe infection, or HSCT

Notes

Iron chelation is critical; cardiac iron overload is the leading cause of death in TDT. MRI-guided chelation (T2*) recommended for monitoring.

Guideline Changes

Casgevy (CRISPR gene therapy) approved Jan 2024 for TDT. Luspatercept under pediatric trial

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D57 — Sickle-cell disorders

SNOMED: 417357006 | Duration: 365 days | Source: ASH 2024 (in development), NHLBI 2014

First-Line

- **Hydroxyurea** — Start 20 mg/kg/day PO from age 9 months; escalate to MTD ~25 mg/kg/day; monitor CBC q4-8wk
- **Penicillin V prophylaxis** — <3y: 125 mg PO BID; 3-5y: 250 mg PO BID; pneumococcal prophylaxis from 2 months
- **Folic acid** — 1 mg PO daily to support erythropoiesis

Second-Line

- **Chronic transfusion therapy** — 10-15 mL/kg PRBC q3-4wk to maintain HbS <30%; for stroke prevention or severe disease
- **L-glutamine (Endari)** — 5-15 g PO BID based on weight; FDA-approved >=5y to reduce vaso-occlusive crises
- **Casgevy (exagamglogene)** — Gene therapy; one-time autologous HSCT with CRISPR-edited CD34+ cells; approved >=12y **[NEW 2024-25]**
- **Lyfgenia (lovotibeglogene)** — Gene therapy; one-time autologous HSCT with lentiviral-modified CD34+ cells; approved >=12y **[NEW 2024-25]**

Investigations

- CBC with reticulocyte count — Monitor baseline and during crises [URGENT]
- Hb electrophoresis/HPLC — Diagnostic confirmation and Hb F monitoring
- Transcranial Doppler (TCD) — Annual screening for stroke risk, ages 2-16
- Renal function and urinalysis — Screen for sickle nephropathy
- Liver function tests — Monitor for hepatobiliary complications

Counselling

- Give hydroxyurea daily as prescribed — it significantly reduces pain crises and complications
- Keep the child well-hydrated at all times; offer water frequently especially in hot weather
- Avoid extreme cold, dehydration, high altitude, and vigorous exhausting exercise
- Seek emergency care immediately for fever >38.5C, severe pain, chest pain, or sudden weakness
- Penicillin prophylaxis must be given daily until at least age 5 — do not skip doses

Referral Criteria

Refer to pediatric hematologist at a sickle cell center; urgently for stroke symptoms, acute chest syndrome, or priapism

Hospitalisation Criteria

Admit for vaso-occlusive crisis not responding to home management, fever with ANC <500, acute chest syndrome, stroke, splenic sequestration, or aplastic crisis

Notes

Hydroxyurea is disease-modifying first-line for all children with SCD from age 9 months. Voxelotor was withdrawn in 2024 due to safety concerns.

Guideline Changes

Voxelotor withdrawn Sept 2024 (safety signal). Gene therapies Casgevy and Lyfgenia approved Dec 2023

D58 — Hereditary hemolytic anemia

SNOMED: 267556002 | Duration: 365 days | Source: BSH 2011, ASH 2021

First-Line

- **Folic acid** — 1 mg PO daily for chronic hemolysis (hereditary spherocytosis, elliptocytosis)
- **RBC transfusion** — 10-15 mL/kg PRBC IV for Hb <7-8 g/dL or symptomatic anemia; minimize transfusion burden

Second-Line

- **Splenectomy (partial/total)** — Consider for moderate-severe HS after age 6; vaccinate against encapsulated organisms pre-op
- **Phototherapy** — For neonatal jaundice in HS; exchange transfusion if bilirubin reaches critical levels

Investigations

- CBC with reticulocyte count — Baseline and monitor for aplastic crisis [URGENT]
- Peripheral blood smear — Spherocytes, elliptocytes — morphological diagnosis
- Osmotic fragility test — Increased in hereditary spherocytosis
- Abdominal ultrasound — Screen for gallstones and splenomegaly
- Direct Coombs test — Negative (distinguishes from autoimmune hemolysis)

Counselling

- Hereditary hemolytic anemia is a lifelong condition but manageable with regular monitoring
- Give folic acid daily to help the body make new red blood cells
- Watch for sudden pallor, dark urine, or jaundice — this may mean a hemolytic crisis
- Parvovirus B19 infection can cause aplastic crisis — seek care for sudden worsening of pallor
- Annual ultrasound checks for gallstones are recommended from school age onwards

Referral Criteria

Refer to pediatric hematologist for diagnosis confirmation, splenectomy evaluation, and management of severe disease

Hospitalisation Criteria

Admit for aplastic crisis (parvovirus B19), severe hemolytic crisis with Hb <7 g/dL, symptomatic gallstones, or splenectomy

Notes

Hereditary spherocytosis is the most common cause; severity ranges from mild to transfusion-dependent. Annual monitoring with CBC and ultrasound for gallstones.

D59 — Acquired hemolytic anemia

SNOMED: 154802007 | Duration: 180 days | Source: ASH, Expert consensus

First-Line

- **Prednis(ol)one** — 1-2 mg/kg/day PO (max 60 mg) for warm AIHA; taper slowly over 6-12 months after response
- **Folic acid** — 1 mg PO daily to support erythropoiesis during active hemolysis

Second-Line

- **Rituximab** — 375 mg/m2 IV weekly x4 doses for steroid-refractory AIHA; 80-90% response rate
- **Mycophenolate mofetil** — 600 mg/m2 PO BID as steroid-sparing agent for relapsed warm AIHA
- **IVIG** — 1 g/kg IV x2 days for acute severe hemolysis as temporizing measure

- **Splenectomy** — Consider in children >5y refractory to rituximab; effective in ~66% of warm AIHA

Investigations

- CBC with reticulocyte count — Confirm hemolysis with reticulocytosis [STAT]
- Direct Coombs test (DAT) — Positive in autoimmune hemolytic anemia [STAT]
- LDH, indirect bilirubin, haptoglobin — Hemolysis markers [URGENT]
- Peripheral blood smear — Spherocytes in warm AIHA, agglutination in cold AIHA [URGENT]

Counselling

- Your child's body is destroying its own red blood cells — this needs careful treatment
- Take prednisolone exactly as prescribed; do not stop suddenly — it must be tapered slowly
- Watch for dark urine (cola-colored), increasing pallor, or jaundice and report immediately
- Folic acid supports new red blood cell production — give daily
- Follow-up blood tests are needed every 1-2 weeks initially to monitor response

Referral Criteria

Refer to pediatric hematologist for all cases; urgent if Hb dropping rapidly or steroid-refractory

Hospitalisation Criteria

Admit if Hb <7 g/dL, rapidly falling Hb, cardiac symptoms, or need for transfusion/IVIG

Notes

Warm AIHA responds to steroids in 70-85% of children. Transfuse least-incompatible RBCs for life-threatening anemia; do not withhold transfusion.

D61 — Aplastic anemia

SNOMED: 306058006 | Duration: 180 days | Source: ASH 2024, AIEOP 2024, ASTCT 2024

First-Line

- **Horse ATG (ATGAM)** — 40 mg/kg/day IV x4 days; premedicate with methylprednisolone and diphenhydramine
- **Cyclosporine A** — <12y: 7.5 mg/kg/dose PO q12h; >=12y: 5 mg/kg/dose PO q12h; target trough 150-250 ng/mL
- **Eltrombopag** — >=12y: 150 mg PO daily; 6-11y: 75 mg daily; 2-5y: 2.5 mg/kg/day; start day 1 with ATG x6 months **[NEW 2024-25]**
- **Allogeneic HSCT (MSD)** — First-line if matched sibling donor available; BM graft preferred over PBSC

Second-Line

- **MUD HSCT** — If no MSD and IST fails or age <20y; use BM graft with ATG conditioning
- **Haploidentical HSCT** — Consider if no MSD/MUD; post-transplant cyclophosphamide-based GvHD prophylaxis **[NEW 2024-25]**

Investigations

- CBC with reticulocyte count — Confirm pancytopenia with reticulocytopenia [STAT]
- Bone marrow aspirate and biopsy — Confirm aplastic anemia, rule out MDS/leukemia [URGENT]
- Chromosomal breakage study (DEB/MMC) — Rule out Fanconi anemia [URGENT]
- Flow cytometry for PNH clone — Rule out PNH overlap
- HLA typing — For potential HSCT donor search [URGENT]

Counselling

- This is a serious condition affecting the bone marrow — regular hospital visits are essential
- Avoid contact sports, crowded places, and anyone with infections
- Report fever immediately — even a mild fever needs urgent medical attention
- The child may need blood and platelet transfusions regularly

- Treatment with IST or transplant offers good outcomes in children — discuss options with the hematologist

Referral Criteria

Refer urgently to pediatric hematologist/oncologist at a tertiary center with HSCT capability

Hospitalisation Criteria

Admit for neutropenic fever, active bleeding, Hb <7 g/dL requiring transfusion, initiation of ATG therapy, or HSCT

Notes

Horse ATG is superior to rabbit ATG for pediatric SAA. Add eltrombopag to IST based on recent trial data showing improved response rates.

Guideline Changes

2024 Delphi consensus recommends horse ATG over rabbit ATG; eltrombopag added to frontline IST

D61.8 — Other pancytopenia

SNOMED: 127034005 | Duration: 30 days | Source: AAP 2016, Expert consensus

First-Line

- **PRBC transfusion** — 10-15 mL/kg IV for Hb <7 g/dL or symptomatic anemia; irradiated if immunocompromised
- **Platelet transfusion** — 10-15 mL/kg IV for platelets <10,000/mcL or active bleeding; single donor preferred
- **Broad-spectrum antibiotics** — Cefepime 50 mg/kg IV q8h or piperacillin-tazobactam 100 mg/kg IV q8h for neutropenic fever

Second-Line

- **G-CSF (filgrastim)** — 5-10 mcg/kg/day SC for severe neutropenia (ANC <500); continue until ANC recovery
- **IVIG** — 0.4 g/kg IV monthly if hypogammaglobulinemia with recurrent infections

Investigations

- CBC with reticulocyte count — Quantify cytopenias [STAT]
- Peripheral blood smear — Look for blasts, abnormal cells [STAT]
- Bone marrow aspirate and biopsy — Determine etiology (aplastic, infiltrative, fibrosis) [URGENT]
- LDH, haptoglobin, direct Coombs — Rule out hemolysis [URGENT]
- Viral panel (EBV, CMV, HIV, Parvovirus B19) — Infectious causes of pancytopenia [URGENT]

Counselling

- Pancytopenia means all blood counts are low — the child is at risk for infection and bleeding
- Avoid crowded places, raw/uncooked food, and contact with sick people
- Report fever, unusual bleeding, or bruising immediately
- Blood product transfusions are supportive while the cause is investigated
- The treatment depends on identifying the underlying cause — investigations are essential

Referral Criteria

Refer urgently to pediatric hematologist — bone marrow biopsy is mandatory to determine cause

Hospitalisation Criteria

Admit all cases for workup; urgently if neutropenic fever, active bleeding, or Hb <7 g/dL

Notes

Pancytopenia requires urgent bone marrow biopsy to determine etiology. Supportive care with transfusions and infection management while awaiting diagnosis.

D64 — Other anemias

SNOMED: 271737000 | Duration: 90 days | Source: ASH 2021, BSH

First-Line

- **Pyridoxine (vitamin B6)** — 50-100 mg PO daily trial for sideroblastic anemia; assess response at 2-3 months
- **RBC transfusion** — 10-15 mL/kg PRBC IV for symptomatic anemia (Hb <7 g/dL); transfuse over 2-4 hours

Second-Line

- **Deferoxamine** — 20-40 mg/kg/day SC infusion over 8-12h, 5 days/week for iron overload from chronic transfusions
- **EPO (erythropoietin)** — 50-300 IU/kg SC 3x/week for anemia of chronic disease/renal failure; titrate to Hb 10-12 g/dL

Investigations

- CBC with reticulocyte count — Classify anemia and assess marrow response [URGENT]
- Peripheral blood smear — Identify morphological clues (ring sideroblasts, etc.) [URGENT]
- Serum iron, ferritin, TIBC — Iron studies to classify anemia
- Bone marrow aspirate and biopsy — Definitive evaluation for unexplained anemia [URGENT]

Counselling

- This type of anemia needs careful investigation to find the exact cause
- Blood transfusions may be needed temporarily until the cause is treated
- Report any new symptoms: fever, bleeding, bone pain, or increasing weakness
- Keep all follow-up appointments — blood counts need regular monitoring
- If on long-term transfusions, iron overload screening will be done

Referral Criteria

Refer to hematologist for all cases — specialized evaluation required to determine etiology and guide therapy

Hospitalisation Criteria

Admit if Hb <7 g/dL with symptoms, active bleeding, or need for urgent transfusion or bone marrow biopsy

Notes

D64 covers heterogeneous anemias; identify underlying cause before treatment. Pyridoxine trial warranted for all sideroblastic anemias.

D66 — Hemophilia A

SNOMED: 28293008 | Duration: 365 days | Source: WFH 2022, ISTH 2024

First-Line

- **Factor VIII concentrate** — Prophylaxis: 25-40 IU/kg IV 3x/week; target trough >3-5%; dose=wt(kg)x desired rise(%)x0.5
- **Emicizumab** — 3 mg/kg SC weekly x4 weeks loading, then 1.5 mg/kg SC weekly; for severe hemophilia A **[NEW 2024-25]**
- **Factor VIII (acute bleed)** — Minor: 10-15 IU/kg IV; moderate: 25 IU/kg IV; severe: 50 IU/kg IV to reach 80-100% level

Second-Line

- **Desmopressin (DDAVP)** — 0.3 mcg/kg IV/SC (max 20 mcg) for mild hemophilia A; test dose to confirm response
- **Tranexamic acid** — 25 mg/kg PO TID or 10 mg/kg IV TID as adjunct for mucosal bleeding
- **Marstacimab** — 300 mg SC loading then 150 mg SC weekly; approved 2024 for >=12y without inhibitors **[NEW 2024-25]**

Investigations

- Factor VIII activity level — Classify severity (severe <1%, moderate 1-5%, mild 5-40%) [URGENT]

- aPTT — Screening test — prolonged in hemophilia A [URGENT]
- Mixing study — Differentiate factor deficiency from inhibitor [URGENT]
- Bethesda assay — Quantify inhibitor titer if suspected [URGENT]
- Joint ultrasound — Monitor for hemophilic arthropathy

Counselling

- Hemophilia requires lifelong management — build a strong relationship with the hemophilia treatment center
- Avoid aspirin and NSAIDs; use paracetamol for pain
- Apply RICE (Rest, Ice, Compression, Elevation) for joint/muscle bleeds and infuse factor early
- The child should wear a medical alert bracelet at all times
- Encourage swimming and cycling; avoid contact sports like football and wrestling

Referral Criteria

Refer to hemophilia treatment center for all newly diagnosed cases; urgent referral for inhibitor development or life-threatening bleed

Hospitalisation Criteria

Admit for head injury/intracranial bleed, GI bleed, airway bleed, compartment syndrome, or any bleed not responding to home factor infusion

Notes

Start prophylaxis before age 3 to prevent arthropathy. Children have shorter FVIII half-life and may need more frequent dosing than adults.

Guideline Changes

ISTH 2024 now recommends emicizumab as alternative first-line prophylaxis alongside FVIII concentrates

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D67 — Hemophilia B

SNOMED: 438372000 | Duration: 365 days | Source: WFH 2022, MASAC/NHF

First-Line

- **Factor IX concentrate** — Prophylaxis: 25-40 IU/kg IV 2x/week; dose=wt(kg)x desired rise(%)x1.4; target trough >3-5%
- **Extended half-life FIX (rIX-FP)** — 50-75 IU/kg IV every 7-14 days; higher trough levels at lower frequency **[NEW 2024-25]**
- **Factor IX (acute bleed)** — Minor: 20-30 IU/kg IV; moderate: 40-60 IU/kg IV; severe: 100-140 IU/kg IV

Second-Line

- **Tranexamic acid** — 25 mg/kg PO TID or 10 mg/kg IV TID as adjunct for mucosal bleeding
- **Fitusiran** — 2 mg SC monthly; anti-antithrombin siRNA for hemophilia B with/without inhibitors **[NEW 2024-25]**

Investigations

- Factor IX activity level — Classify severity [URGENT]
- aPTT — Screening test — prolonged in hemophilia B [URGENT]
- Mixing study — Differentiate deficiency from inhibitor [URGENT]
- Bethesda assay — Inhibitor quantification
- Joint X-ray/MRI — Monitor for hemophilic arthropathy

Counselling

- Hemophilia B is lifelong — regular prophylaxis prevents joint damage
- Give factor IX infusion on schedule even if the child seems well
- Avoid aspirin; use paracetamol for pain relief
- Encourage non-contact physical activities to keep joints healthy

- Always carry the hemophilia card and inform teachers, dentists, and doctors about the condition

Referral Criteria

Refer to hemophilia treatment center for diagnosis, prophylaxis planning, and inhibitor monitoring

Hospitalisation Criteria

Admit for intracranial, GI, or airway bleeds, surgery, or factor IX infusion for major bleeds not manageable at home

Notes

Children <15y require higher FIX doses due to faster clearance (use 1.4 IU/kg correction factor). Avoid DDAVP as it is ineffective in hemophilia B.

Guideline Changes

Extended half-life FIX products (rIX-FP) now preferred allowing once-weekly prophylaxis

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D68 — Other coagulation defects

SNOMED: 234467004 | Duration: 14 days | Source: BSH 2009, ISTH

First-Line

- **Fresh frozen plasma (FFP)** — 10-15 mL/kg IV for acute bleeding with prolonged PT/aPTT; repeat q8-12h as needed
- **Cryoprecipitate** — 1-2 units/10 kg IV for fibrinogen <100 mg/dL; each unit raises fibrinogen ~50-100 mg/dL
- **Vitamin K** — 1-5 mg IV/PO for vitamin K-dependent factor deficiency; neonates 1 mg IM at birth

Second-Line

- **Tranexamic acid** — 25 mg/kg PO TID or 10 mg/kg IV TID for mucosal bleeding as antifibrinolytic adjunct
- **Platelet transfusion** — 10-15 mL/kg IV if concurrent thrombocytopenia with DIC; target platelets >50,000 if bleeding

Investigations

- PT/INR and aPTT — Screen coagulation pathways [STAT]
- Fibrinogen level — Low in DIC and afibrinogenemia [STAT]
- D-dimer — Elevated in DIC [URGENT]
- Mixing study — Differentiate factor deficiency from inhibitor [URGENT]
- Individual factor assays — Identify specific factor deficiency

Counselling

- This condition causes the blood to not clot properly — report any unusual bleeding immediately
- Avoid trauma-prone activities and contact sports until the doctor advises otherwise
- If the child has a bleeding episode, apply pressure and seek immediate medical help
- Take all medications exactly as prescribed — vitamin K is important if prescribed
- Follow up regularly for blood clotting tests

Referral Criteria

Refer to pediatric hematologist for definitive diagnosis and management of all coagulation defects

Hospitalisation Criteria

Admit for active significant bleeding, DIC, need for FFP/cryoprecipitate infusion, or surgical procedures

Notes

Treat the underlying cause (e.g., sepsis, liver disease) as the cornerstone of DIC management. FFP should only be given for active bleeding, not isolated lab derangements.

D69.0 — Allergic purpura / Henoch-Schonlein purpura

SNOMED: 191306005 | Duration: 28 days | Source: SHARE 2019, AAFP 2020

First-Line

- **Supportive care + NSAIDs** — Ibuprofen 10 mg/kg PO q6-8h PRN for joint pain; rest and hydration; self-limiting in 94%
- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day) for mild pain if NSAIDs contraindicated

Second-Line

- **Prednisone** — 1-2 mg/kg/day PO (max 60 mg) x1-2 weeks then taper; for severe abdominal pain or nephritis
- **Mycophenolate mofetil** — 600 mg/m2 PO BID for IgAV nephritis refractory to steroids; specialist supervision

Investigations

- Urinalysis — Screen for hematuria/proteinuria (monthly x6 months) [URGENT]
- Serum creatinine and BUN — Assess renal function
- CBC with platelet count — Confirm normal/elevated platelets (distinguishes from ITP) [URGENT]
- Stool occult blood — Detect GI involvement

Counselling

- HSP is usually a self-limiting illness that resolves in 4-6 weeks
- The rash on legs and buttocks looks alarming but is not dangerous by itself
- Give the child plenty of fluids and rest during the active phase
- Watch for belly pain, bloody stools, or swollen joints and report immediately
- Monthly urine tests are needed for 6 months to check for kidney involvement

Referral Criteria

Refer to pediatric nephrologist if proteinuria >1 g/day, nephrotic-range proteinuria, rising creatinine, or persistent hematuria beyond 3 months

Hospitalisation Criteria

Admit if severe abdominal pain (intussusception risk), GI bleeding, nephrotic syndrome, or inability to tolerate oral intake

Notes

Self-resolves in most children within 4-6 weeks. Monitor urinalysis monthly for 6 months for renal involvement; nephritis is the main long-term concern.

D69.6 — Thrombocytopenia, unspecified

SNOMED: 302215000 | Duration: 30 days | Source: ASH 2019, AIEOP 2024

First-Line

- **Observation** — Watchful waiting for mild ITP (no significant bleeding); treat based on symptoms not platelet count
- **IVIG** — 0.8-1 g/kg IV single dose for significant bleeding; raises platelets within 24-48h
- **Prednisone** — 2-4 mg/kg/day PO (max 120 mg) x4 days or 1-2 mg/kg/day x1-2 weeks then taper

Second-Line

- **Romiplostim** — 1-10 mcg/kg SC weekly; start 1 mcg/kg, titrate by 1 mcg/kg/wk to target plt >50,000
- **Eltrombopag** — 1-5y: 25 mg PO daily; 6-17y: 50 mg PO daily; titrate to maintain plt >50,000
- **Rituximab** — 375 mg/m2 IV weekly x4 doses for chronic/refractory ITP; response in 60-80%

Investigations

- CBC with peripheral smear — Confirm isolated thrombocytopenia, rule out pseudothrombocytopenia [STAT]
- Reticulocyte count — Rule out Evans syndrome (AIHA + ITP)

- Direct Coombs test — Rule out concurrent autoimmune hemolytic anemia
- Blood group and cross-match — If transfusion may be needed [URGENT]

Counselling

- Most children with ITP recover fully within a few months without treatment
- Avoid contact sports, rough play, and activities that risk head injury while platelets are low
- Do not give aspirin or ibuprofen — use paracetamol only for pain/fever
- Watch for unusual bleeding: nosebleeds lasting >20 min, blood in stool/urine, or new large bruises
- Monthly blood tests will be needed to track platelet count recovery

Referral Criteria

Refer to pediatric hematologist if platelets remain <20,000 beyond 3 months, significant bleeding despite treatment, or chronic ITP (>12 months)

Hospitalisation Criteria

Admit for life-threatening bleeding (intracranial, GI), mucosal bleeding with platelets <20,000, or for IVIG/steroid initiation in severe bleeding

Notes

Most pediatric ITP resolves within 12 months. Splenectomy is rarely indicated and only after age 5 in life-threatening refractory cases.

Guideline Changes

2024 AIEOP recommends TPO-RAs (romiplostim) as preferred second-line over rituximab

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D70 — Neutropenia

SNOMED: 165517008 | Duration: 30 days | Source: AIEOP Neutropenia Committee, ASCO

First-Line

- **G-CSF (filgrastim)** — Congenital: 5-10 mcg/kg/day SC; cyclic/idiopathic: 1-5 mcg/kg/day SC; adjust to ANC >1000
- **Broad-spectrum antibiotics** — Cefepime 50 mg/kg IV q8h for neutropenic fever (ANC <500 + temp >=38.3C); start empirically

Second-Line

- **Pegfilgrastim** — 0.1 mg/kg SC single dose (max 6 mg) for chemotherapy-induced neutropenia; one dose per cycle
- **IVIG** — 0.4 g/kg IV monthly for autoimmune neutropenia with recurrent severe infections

Investigations

- CBC with differential and smear — Confirm neutropenia, rule out blasts [STAT]
- Blood culture — If febrile — before antibiotics [STAT]
- ANA — Screen for autoimmune neutropenia
- Anti-neutrophil antibodies — Confirm autoimmune neutropenia
- Bone marrow aspirate — If congenital or persistent severe neutropenia [URGENT]

Counselling

- Low neutrophil count makes the child vulnerable to infections — avoid crowded places
- Wash hands frequently and maintain good oral hygiene
- Seek emergency care immediately for any fever above 38C — do not wait
- Avoid raw/uncooked food and unpasteurized milk during severe neutropenia
- G-CSF injection helps the body make more white blood cells — give as directed

Referral Criteria

Refer to pediatric hematologist if congenital neutropenia suspected, ANC persistently <500, or recurrent serious infections

Hospitalisation Criteria

Admit for neutropenic fever (ANC <500 + temp >=38.3C), severe oral/perianal infection, or suspected serious bacterial infection

Notes

Monitor CBC twice weekly during G-CSF initiation. Congenital severe neutropenia (Kostmann) requires long-term G-CSF; watch for MDS/AML transformation.

R16.2 — Hepatosplenomegaly

SNOMED: 36760000 | Duration: 30 days | Source: Expert consensus, AAP

First-Line

- **Treat underlying cause** — Directed therapy based on etiology: antibiotics for infection, chemo for malignancy, ERT for storage disease
- **Supportive care** — Monitor CBC, LFTs, coagulation; imaging with ultrasound; avoid contact sports if splenomegaly

Second-Line

- **Imiglucerase (ERT)** — 60 U/kg IV q2wk for Gaucher disease; reduces hepatosplenomegaly and improves cytopenias
- **Miglustat (SRT)** — Age-adjusted dosing PO TID for Gaucher type 1 if ERT unavailable; substrate reduction therapy

Investigations

- CBC with peripheral smear — Cytopenias, abnormal cells [STAT]
- Liver function tests — Assess hepatic involvement [URGENT]
- Abdominal ultrasound with Doppler — Quantify organomegaly, assess portal flow [URGENT]
- LDH, uric acid — Screen for malignancy/hemolysis [URGENT]
- Bone marrow aspirate — If cytopenias or suspected infiltration [URGENT]

Counselling

- Enlarged liver and spleen is a sign of an underlying condition — investigations are needed to find the cause
- Avoid contact sports and rough play to prevent splenic injury while the spleen is enlarged
- Report fever, increasing abdominal distension, or unusual bleeding immediately
- Follow all scheduled investigations and follow-up visits
- Treatment depends on the underlying diagnosis — once identified, targeted therapy will be started

Referral Criteria

Refer to pediatric hematologist/oncologist or gastroenterologist based on suspected etiology; urgent if rapidly progressive or malignancy suspected

Hospitalisation Criteria

Admit if severe cytopenias, suspected malignancy, splenic rupture risk, or need for liver/bone marrow biopsy

Notes

Hepatosplenomegaly is a sign, not a diagnosis; urgent workup to identify underlying cause. Common etiologies: infections, hemolytic anemias, storage diseases, malignancies.

R59 — Enlarged lymph nodes

SNOMED: 30746006 | Duration: 14 days | Source: AAP, RCH Clinical Guidelines

First-Line

- **Observation** — Watchful waiting 4-6 weeks for low-risk nodes (<2 cm, mobile, no systemic symptoms); most self-resolve
- **Cephalexin** — 25-50 mg/kg/day PO divided QID x10 days for suspected bacterial lymphadenitis (S. aureus/GAS)

Second-Line

- **Amoxicillin-clavulanate** — 40-45 mg/kg/day PO divided BID-TID x10 days; broader coverage for lymphadenitis
- **Clindamycin** — 30-40 mg/kg/day PO divided TID x10 days; if MRSA suspected or penicillin allergy

Investigations

- CBC with ESR — Screen for infection, leukemia [URGENT]
- Peripheral blood smear — Rule out blasts/atypical lymphocytes [URGENT]
- Ultrasound of the lymph node — Assess size, echotexture, vascularity
- Mantoux test/IGRA — Rule out tuberculosis in endemic area
- LDH — Elevated in lymphoma

Counselling

- Swollen lymph nodes are very common in children and usually due to infections
- Most enlarged nodes will shrink on their own within 4-6 weeks
- Give the full course of antibiotics if prescribed — do not stop early
- Return if the node keeps growing, becomes very hard, or your child develops fever, weight loss, or night sweats
- No need to worry if the node is small, soft, movable, and the child is otherwise well

Referral Criteria

Refer for biopsy if node >2 cm persisting >6 weeks, supraclavicular location, associated systemic symptoms (fever, weight loss, night sweats), or hard/fixed node

Hospitalisation Criteria

Admit if suspected malignancy, abscess needing drainage, or systemic illness with lymphadenopathy

Notes

90% of children aged 4-8 have palpable nodes. Red flags: supraclavicular location, >2 cm, systemic symptoms, persistent >6 weeks; biopsy if concerning.

Infectious

A01 — Typhoid and paratyphoid fevers

SNOMED: 4834000 | Duration: 10 days | Source: CDC 2024, WHO

First-Line

- **Azithromycin** — 20 mg/kg/day PO (max 1g) for 5-7 days; preferred for uncomplicated
- **Ceftriaxone** — 75-80 mg/kg/day IV (max 2.5g) for 10-14 days; severe or inpatient

Second-Line

- **Ciprofloxacin** — 15 mg/kg PO bid (max 500 mg/dose) for 7-10 days; if susceptible, age >=1yr
- **Meropenem** — 60 mg/kg/day IV divided q8h (max 3g/day); XDR typhoid

Investigations

- Blood culture — Gold standard for typhoid diagnosis [URGENT]
- Widal test — Supportive (not diagnostic alone; rising titers)
- Typhidot IgM — Rapid serological test
- CBC — Leukopenia, relative bradycardia [URGENT]
- LFT — Mild transaminase elevation common

Counselling

- Complete the full antibiotic course — stopping early leads to relapse and drug resistance
- Give plenty of fluids and a soft, easily digestible diet during fever
- Wash hands thoroughly and ensure safe drinking water to prevent spread to family
- The child can pass typhoid to others through stool — maintain strict hand hygiene after toilet
- Typhoid vaccination (typhoid conjugate vaccine) is recommended for prevention

Referral Criteria

Refer if suspected intestinal perforation, GI hemorrhage, encephalopathy, XDR typhoid, or no defervescence after 5 days of appropriate antibiotics

Hospitalisation Criteria

Admit for persistent high fever not responding to oral antibiotics, suspected complications (perforation, hemorrhage, encephalopathy), XDR typhoid requiring IV therapy

Notes

Azithromycin is preferred empirically due to rising multidrug resistance. For travelers from Pakistan/Iraq, suspect XDR and use azithromycin or carbapenem.

Guideline Changes

XDR typhoid (Pakistan, Iraq) requires azithromycin or carbapenems empirically

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A01.0 — Enteric Fever (Indoor) *[moderate]*

SNOMED: 4834000 | Duration: 10 days | Source: IAP 2024, WHO Typhoid Guidelines, AIIMS Protocol

First-Line

- **Ceftriaxone** — 50-75 mg/kg/day IV OD or BD, dilute in 100 ml NS over 30-60 min, x 7-10 days
- **Paracetamol** — 10-15 mg/kg/dose IV 6 hourly for fever control
- **Ondansetron** — 0.1-0.15 mg/kg/dose IV 8 hourly for vomiting
- **IV Fluids (DNS / Normal Saline)** — Maintenance rate per Holliday-Segar formula (~60-80 ml/hr adjust clinically), maintain hydration and correct electrolytes

Second-Line

- **Azithromycin** — 10 mg/kg/day IV over 1-2 hours x 5-7 days, add-on for moderate-severe/resistant cases

- **Cefixime** — 8 mg/kg/day div BD PO, step-down oral antibiotic after afebrile 48-72 hours, complete 10-14 days total

Investigations

- Blood culture — Gold standard for diagnosis, before starting antibiotics [STAT]
- Complete blood count — Leukopenia pattern suggestive of typhoid [STAT]
- CRP or Procalcitonin — Assess infection severity [URGENT]
- Widal test / Typhidot IgM — Supportive serological evidence [URGENT]
- Liver function test — Hepatic involvement assessment [URGENT]
- Serum electrolytes — Monitor with IV fluid therapy [URGENT]

Counselling

- Typhoid requires a full course of antibiotics - do not stop early even if fever resolves
- Expected hospital stay is 3-5 days until fever-free
- Give soft easily digestible diet: khichdi, curd, buttermilk, banana, coconut water
- Avoid spicy and oily food during recovery
- Complete the full course of oral antibiotics after discharge
- Follow-up blood tests may be needed to confirm cure

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Abdominal distension or severe abdominal pain
- Signs of GI bleeding such as blood in stool or black stools
- Altered sensorium or confusion
- Persistent high fever despite treatment for more than 5 days

Referral Criteria

Refer to pediatric infectious disease specialist if no defervescence after 5 days of appropriate antibiotics, suspected MDR typhoid, or complications (perforation, encephalopathy, myocarditis)

Hospitalisation Criteria

Admit all moderate-severe enteric fever. Also admit if: persistent high fever, toxic appearance, vomiting with inability to take oral antibiotics, suspected complications, age <2 years, or immunocompromised child

Notes

Indoor protocol. Ceftriaxone is first-line IV drug. IV fluids (DNS/NS) as per Holliday-Segar maintenance. Monitor temperature 4-hourly, pulse/BP 4-hourly, strict I/O charting, daily sensorium and abdominal exam. Step-down to oral antibiotics when afebrile 48-72 hrs. Complete 10-14 days total therapy. Watch for intestinal perforation, GI bleeding, encephalopathy, shock.

Guideline Changes

WHO/CDC 2024: azithromycin elevated as co-first-line with ceftriaxone due to rising XDR S. Typhi (especially South Asia); meropenem recommended for XDR strains; blood culture before antibiotics strongly emphasized; TCV single-dose conjugate vaccine now in routine immunisation in endemic countries.

A06 — Amoebiasis

SNOMED: 111910009 | Duration: 17 days | Source: WHO 2024, AAP Red Book 2024

First-Line

- **Metronidazole** — 35-50 mg/kg/day PO divided tid (max 2.25g/day) for 10 days; invasive disease

- **Paromomycin (luminal agent)** — 25-35 mg/kg/day PO divided tid for 7 days; follow metronidazole to clear cysts

Second-Line

- **Tinidazole** — 50 mg/kg/day PO once daily (max 2g) for 3-5 days; age >=3yr
- **Iodoquinol** — 30-40 mg/kg/day PO divided tid (max 1.95g/day) for 20 days; luminal agent

Investigations

- Stool microscopy (3 samples) — Detect E. histolytica cysts/trophozoites [URGENT]
- Stool antigen test for E. histolytica — More sensitive than microscopy
- Abdominal ultrasound — If liver abscess suspected [URGENT]
- CBC — Leukocytosis in invasive disease

Counselling

- Complete the full course of metronidazole AND the follow-up luminal agent (paromomycin)
- The child may feel nauseous with metronidazole — give with food
- Ensure safe drinking water — boil or filter water at home
- Wash hands thoroughly before eating and after using the toilet
- Follow up with stool test after completing treatment to confirm parasite clearance

Referral Criteria

Refer to pediatric gastroenterologist or surgeon if liver abscess detected, or if fulminant colitis suspected

Hospitalisation Criteria

Admit for amoebic liver abscess, toxic megacolon, intestinal perforation, severe dehydration, or dysentery with systemic toxicity

Notes

Always follow tissue-active agent (metronidazole/tinidazole) with a luminal agent to eradicate intraluminal cysts. Tinidazole has better tolerability and shorter course.

A07 — Other protozoal intestinal diseases

SNOMED: 58265007 | Duration: 7 days | Source: WHO 2024, AAP Red Book 2024

First-Line

- **Metronidazole** — 15 mg/kg/day PO divided tid (max 750 mg/day) for 5-7 days; giardiasis
- **Tinidazole** — 50 mg/kg PO single dose (max 2g); giardiasis, age >=3yr

Second-Line

- **Nitazoxanide** — Age 1-3yr: 100 mg PO bid; 4-11yr: 200 mg PO bid; for 3 days
- **Paromomycin** — 25-35 mg/kg/day PO divided tid for 7 days; pregnancy-safe option

Investigations

- Stool for ova and parasites — Direct microscopy for Giardia cysts/trophozoites
- Stool Giardia antigen test — More sensitive than microscopy
- Stool for Cryptosporidium — Modified acid-fast stain if suspected

Counselling

- Give metronidazole with food to reduce stomach upset
- The medicine may cause a metallic taste — this is normal and temporary
- Ensure the child drinks only boiled or filtered water
- Wash hands thoroughly before eating and after toilet use
- Follow up with stool test after treatment to confirm the parasite is cleared

Referral Criteria

Refer to pediatric GI if recurrent giardiasis despite treatment (consider IgA deficiency), or if chronic diarrhea with malabsorption

Hospitalisation Criteria

Admit if severe dehydration from diarrhea, significant malnutrition, or Cryptosporidium in immunocompromised child

Notes

Tinidazole single-dose has 80-100% cure rate for giardiasis with better tolerability than metronidazole. Nitazoxanide also covers Cryptosporidium.

A17 — Tuberculosis of nervous system

Duration: 365 days | Source: WHO 2022, ATS/CDC/IDSA 2024

First-Line

- **Isoniazid (H)** — 10-15 mg/kg/day PO (max 300 mg); for 12 months total
- **Rifampicin (R)** — 20-30 mg/kg/day PO (max 600 mg); higher dose for CNS penetration, 12 months
- **Pyrazinamide (Z)** — 30-40 mg/kg/day PO (max 2g); intensive phase 2 months
- **Ethambutol (E)** — 15-25 mg/kg/day PO (max 2.5g); intensive phase 2 months

Second-Line

- **Dexamethasone** — 0.6 mg/kg/day IV for 2-4 wk, then taper over 4 wk; adjunctive for TBM
- **Ethionamide** — 15-20 mg/kg/day PO (max 1g); good CNS penetration, for resistant TB

Investigations

- CSF analysis — Lymphocytic pleocytosis, high protein, low glucose [STAT]
- CSF GeneXpert/TB PCR — Rapid confirmation [STAT]
- MRI brain with contrast — Basal meningeal enhancement, tuberculomas, hydrocephalus [URGENT]
- Chest X-ray — Concurrent pulmonary TB in 50% [URGENT]
- CBC, LFT, RFT — Baseline and drug monitoring [URGENT]

Counselling

- TB meningitis needs 12 months of treatment — stopping early can be fatal
- The child will also receive steroids initially to reduce brain swelling
- Give pyridoxine daily with isoniazid
- Watch for worsening headache, vomiting, drowsiness, or seizures and rush to hospital
- Hearing and vision tests are recommended during and after treatment

Referral Criteria

Refer urgently to pediatric neurologist and TB specialist; neurosurgery if hydrocephalus requiring VP shunt

Hospitalisation Criteria

Admit all cases of TB meningitis; ICU for altered consciousness, seizures, or raised ICP

Notes

TB meningitis requires 12 months therapy and adjunctive corticosteroids to reduce mortality. Higher-dose rifampicin improves CNS penetration.

Guideline Changes

ATS/CDC/ERS/IDSA 2024 update on drug-resistant TB regimens

A18 — Tuberculosis of other organs

SNOMED: 186247008 | Duration: 180 days | Source: WHO 2022, ATS/CDC/IDSA 2024

First-Line

- **Isoniazid (H)** — 10-15 mg/kg/day PO (max 300 mg)
- **Rifampicin (R)** — 15-20 mg/kg/day PO (max 600 mg)
- **Pyrazinamide (Z)** — 30-40 mg/kg/day PO (max 2g); intensive phase 2 months
- **Ethambutol (E)** — 15-25 mg/kg/day PO (max 2.5g); intensive phase 2 months

Second-Line

- **Levofloxacin** — 15-20 mg/kg/day PO (max 750 mg); MDR-TB or drug intolerance
- **Linezolid** — 10 mg/kg PO/IV bid (max 600 mg bid); MDR/XDR-TB

Investigations

- GeneXpert from site specimen — Rapid diagnosis and rifampicin resistance [URGENT]
- Biopsy of affected tissue — Caseating granulomas on histopathology [URGENT]
- Chest X-ray — Screen for pulmonary TB
- Mantoux test/IGRA — Supportive evidence
- CBC, LFT, RFT — Baseline and drug monitoring

Counselling

- Complete the full 6-month TB treatment course — even if the child feels better after 2 months
- Give all medicines together on empty stomach in the morning
- Give pyridoxine (vitamin B6) daily to prevent side effects of isoniazid
- Report jaundice, persistent vomiting, vision changes, or joint pain immediately
- The child can attend school after 2 weeks of treatment if feeling well

Referral Criteria

Refer to TB specialist if drug resistance suspected, extrapulmonary sites requiring surgical intervention (spine, lymph node), or treatment failure

Hospitalisation Criteria

Admit for spinal TB with cord compression, pericardial TB, renal TB with obstructive uropathy, or severe systemic illness

Notes

Standard 2HRZE/4HR for 6 months; non-severe TB in children 3mo-16yr may use 4-month regimen per WHO 2022. Add pyridoxine supplementation with isoniazid.

Guideline Changes

4-month regimen (2HRZ(E)/2HR) for non-severe TB in children 3mo-16yr per WHO 2022

A19 — Miliary tuberculosis

SNOMED: 182159002 | Duration: 270 days | Source: WHO 2022, ATS/CDC/IDSA 2024

First-Line

- **Isoniazid (H)** — 10-15 mg/kg/day PO (max 300 mg); for 9-12 months total
- **Rifampicin (R)** — 15-20 mg/kg/day PO (max 600 mg); for 9-12 months total
- **Pyrazinamide (Z)** — 30-40 mg/kg/day PO (max 2g); intensive phase 2 months
- **Ethambutol (E)** — 15-25 mg/kg/day PO (max 2.5g); intensive phase 2 months

Second-Line

- **Levofloxacin** — 15-20 mg/kg/day PO (max 750 mg); MDR-TB or ethambutol intolerance
- **Amikacin** — 15-30 mg/kg/day IV/IM (max 1g); MDR-TB intensive phase

Investigations

- Chest X-ray — Classic miliary pattern (bilateral diffuse micronodules) [STAT]

- Sputum/gastric aspirate for AFB and GeneXpert — Confirm TB and check rifampicin resistance [STAT]
- Mantoux test/IGRA — Supportive evidence (may be negative in miliary TB)
- CBC, LFT, RFT — Baseline before ATT and monitoring [URGENT]
- Fundoscopy — Choroidal tubercles pathognomonic for miliary TB [URGENT]

Counselling

- Miliary TB is a serious form of TB requiring 9-12 months of daily medicine — never skip doses
- Give all 4 medicines together on empty stomach in the morning
- Give pyridoxine (vitamin B6) daily to prevent nerve damage from isoniazid
- Watch for jaundice (yellow eyes), rash, or vision changes and report immediately
- The child is not infectious after 2 weeks of effective treatment — can return to school

Referral Criteria

Refer to pediatric pulmonologist or TB specialist; urgent referral if CNS involvement or drug resistance suspected

Hospitalisation Criteria

Admit for severe respiratory distress, CNS involvement (TB meningitis), initiation of therapy in sick child, or drug-resistant TB

Notes

Miliary TB requires 9-12 months of therapy (2HRZE + 7-10HR). Add pyridoxine 1-2 mg/kg/day to prevent isoniazid neuropathy.

Guideline Changes

ATS/CDC/ERS/IDSA 2024 update supports fluoroquinolone-containing regimens for drug-resistant TB

A27 — Leptospirosis

SNOMED: 77377001 | Duration: 7 days | Source: WHO 2024, IDSA

First-Line

- **Penicillin G** — 250,000 U/kg/day IV divided q6h (max 24MU/day) for 7 days; severe disease
- **Doxycycline** — 2-4 mg/kg/day PO divided bid (max 200 mg/day) for 7 days; mild-moderate

Second-Line

- **Ceftriaxone** — 50-75 mg/kg IV once daily (max 2g) for 7 days; severe disease alternative
- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg), then 5 mg/kg/day for 2 more days; age <8yr

Investigations

- Leptospira IgM ELISA — Confirm diagnosis (positive after day 5) [URGENT]
- Leptospira PCR (blood) — Early diagnosis (first week) [URGENT]
- CBC — Thrombocytopenia, leukocytosis [URGENT]
- Renal and liver function — Assess organ involvement [URGENT]
- Urinalysis — Proteinuria, hematuria in renal involvement

Counselling

- Complete the full antibiotic course even if the child starts feeling better
- Leptospirosis comes from contaminated water — avoid playing in stagnant water or floods
- Watch for jaundice, dark urine, reduced urine output, or bleeding and seek immediate care
- The child should rest until fever resolves completely
- After flooding or heavy rains, wear footwear and avoid wading in dirty water

Referral Criteria

Refer to hospital with ICU if Weil disease (jaundice + renal failure + hemorrhage), pulmonary hemorrhage, or myocarditis

Hospitalisation Criteria

Admit for severe leptospirosis (Weil disease), acute kidney injury, pulmonary hemorrhage, hepatitis, or hemodynamic instability

Notes

Early antibiotic therapy reduces severity and duration. Avoid doxycycline in children <8yr due to dental staining; use azithromycin or amoxicillin instead.

A36 — Diphtheria

SNOMED: 397428000 | Duration: 14 days | Source: WHO 2024, CDC Clinical Guidance

First-Line

- **Diphtheria antitoxin (DAT)** — 20,000-120,000 IU IV based on severity; administer ASAP on clinical suspicion
- **Erythromycin** — 40-50 mg/kg/day PO/IV divided q6h (max 2g/day) for 14 days
- **Penicillin G procaine** — 25,000-50,000 U/kg/day IM divided q12h (max 1.2MU/day) for 14 days

Second-Line

- **Azithromycin** — 10-12 mg/kg/day PO (max 500 mg/day) for 14 days

Investigations

- Throat swab culture on Loeffler/tellurite medium — Confirm C. diphtheriae [STAT]
- ECG — Screen for myocarditis (ST changes, heart block) [URGENT]
- Troponin — Myocardial involvement [URGENT]
- CBC — Baseline [URGENT]
- Renal function — Monitor for renal complications

Counselling

- Diphtheria is a serious but treatable infection — early treatment is critical
- The child must complete the full 14-day antibiotic course
- All family members and close contacts need prophylactic antibiotics and vaccination check
- Watch for breathing difficulty, drooling, or neck swelling and seek emergency care immediately
- After recovery, the child needs to complete DPT vaccination as per schedule

Referral Criteria

Refer urgently to hospital with ICU facility; ENT referral if airway compromise

Hospitalisation Criteria

Admit all suspected diphtheria cases; ICU if airway compromise, myocarditis, or neurological involvement

Notes

Antitoxin must be given early; antibiotics eliminate the organism but do not neutralize toxin. Contacts require prophylaxis and booster vaccination.

A37 — Whooping cough

SNOMED: 27836007 | Duration: 14 days | Source: CDC 2024, AAP

First-Line

- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg), then 5 mg/kg/day days 2-5 (max 250 mg)
- **Erythromycin** — 40-50 mg/kg/day PO divided q6h (max 2g/day) for 14 days

Second-Line

- **Clarithromycin** — 15 mg/kg/day PO divided bid (max 1g/day) for 7 days
- **TMP-SMX** — 8 mg/kg/day TMP PO divided bid (max 320 mg TMP/day) for 14 days; macrolide allergy

Investigations

- Nasopharyngeal PCR for B. pertussis — Confirm diagnosis (best in first 3 weeks) [URGENT]
- Nasopharyngeal culture — Gold standard but slow (5-7 days)
- CBC — Lymphocytosis supports diagnosis
- Chest X-ray — If pneumonia suspected

Counselling

- Whooping cough coughing spells can last 6-10 weeks even after treatment — this is normal
- Antibiotics prevent spread to others but may not shorten cough duration if started late
- Keep infants upright after feeding to reduce vomiting from coughing spells
- All household contacts need prophylactic antibiotics regardless of vaccination status
- Ensure all family members are up to date on pertussis (DPT/Tdap) vaccination

Referral Criteria

Refer to pediatric pulmonologist or ICU if infant <6 months, apnea spells, cyanotic episodes, or pneumonia develops

Hospitalisation Criteria

Admit all infants <6 months with confirmed/suspected pertussis, any child with apnea, cyanosis, respiratory distress, or dehydration from vomiting

Notes

Azithromycin is preferred for infants <1 month due to lower pyloric stenosis risk vs erythromycin. Antibiotics are most effective in the catarrhal phase.

A38 — Scarlet fever

SNOMED: 30242009 | Duration: 10 days | Source: IDSA 2024, AAP, CDC

First-Line

- **Penicillin V** — 250 mg PO bid-tid (<27 kg) or 500 mg bid-tid (>=27 kg) for 10 days
- **Amoxicillin** — 50 mg/kg/day PO once daily or divided bid (max 1g/day) for 10 days

Second-Line

- **Azithromycin** — 12 mg/kg PO day 1 (max 500 mg), then 6 mg/kg/day days 2-5
- **Cephalexin** — 40 mg/kg/day PO divided bid (max 1g/day) for 10 days; non-anaphylactic PCN allergy
- **Clindamycin** — 20 mg/kg/day PO divided tid (max 1.8g/day) for 10 days

Investigations

- Rapid strep antigen test — Quick confirmation of GAS [URGENT]
- Throat culture — Gold standard if rapid test negative
- ASO titer — If rheumatic fever suspected post-illness

Counselling

- Complete the full 10-day antibiotic course to prevent rheumatic fever — this is very important
- The child can return to school 24 hours after starting antibiotics
- Give soft, cool foods if the throat is sore; avoid spicy and acidic foods
- The rash feels like sandpaper and will peel — this is normal and not contagious after treatment
- Watch for joint pain, chest pain, or dark urine in the weeks after recovery and report immediately

Referral Criteria

Refer to pediatric cardiologist if signs of rheumatic fever develop (joint pain, cardiac murmur, chorea) within 2-4 weeks

Hospitalisation Criteria

Admit if unable to swallow, severe dehydration, peritonsillar abscess, or signs of rheumatic fever

Notes

Group A strep has no reported penicillin resistance. Full 10-day course is essential to prevent rheumatic fever.

A39 — Meningococcal infection

SNOMED: 154306001 | Duration: 7 days | Source: AAP Red Book 2024, IDSA

First-Line

- **Ceftriaxone** — 100 mg/kg/day IV divided q12h (max 4g/day) for 7 days
- **Cefotaxime** — 200 mg/kg/day IV divided q6h (max 12g/day) for 7 days

Second-Line

- **Penicillin G** — 300,000 U/kg/day IV divided q4-6h (max 24MU/day); if susceptible
- **Meropenem** — 120 mg/kg/day IV divided q8h (max 6g/day); severe allergy to cephalosporins

Investigations

- Blood culture — Identify N. meningitidis [STAT]
- CSF analysis (if safe to LP) — Cell count, protein, glucose, Gram stain, culture [STAT]
- CBC with coagulation profile — DIC screening [STAT]
- CRP, procalcitonin — Severity markers [URGENT]
- Meningococcal PCR (blood/CSF) — Rapid confirmation if culture negative [URGENT]

Counselling

- Meningococcal disease is a medical emergency — the child is receiving urgent treatment
- All close contacts (household, school) need prophylactic antibiotics within 24 hours
- Watch for rash that does not fade on pressing (glass test) — this is a danger sign
- After recovery, the child should receive meningococcal vaccine
- Report the case to local public health authorities

Referral Criteria

Refer immediately to pediatric ICU if septic shock, DIC, or meningitis with altered sensorium

Hospitalisation Criteria

Admit all cases — meningococcal disease is a medical emergency requiring IV antibiotics and close monitoring

Notes

Begin empiric therapy immediately; meningococcal disease progresses rapidly. Close contacts require chemoprophylaxis with rifampin, ciprofloxacin, or ceftriaxone.

A40 — Streptococcal sepsis

Duration: 10 days | Source: Surviving Sepsis Campaign 2020, IDSA

First-Line

- **Penicillin G** — 250,000-400,000 U/kg/day IV divided q4-6h (max 24MU/day)
- **Ceftriaxone** — 100 mg/kg/day IV divided q12h (max 4g/day)

Second-Line

- **Clindamycin** — 40 mg/kg/day IV divided q8h (max 2.7g/day); add for toxin suppression in GAS
- **Vancomycin** — 60 mg/kg/day IV divided q6h (max 4g/day); severe penicillin allergy

Investigations

- Blood culture — Confirm streptococcal bacteremia [STAT]

- CBC with differential — Leukocytosis or leukopenia [STAT]
- CRP, procalcitonin — Severity markers [STAT]
- Coagulation profile — DIC in toxic shock [URGENT]
- ASO titer — Post-streptococcal sequelae monitoring

Counselling

- Streptococcal sepsis is a serious infection requiring IV antibiotics in hospital
- The child will be closely monitored — report any worsening symptoms to the nurses immediately
- All prescribed antibiotics must be completed — incomplete treatment risks relapse
- After discharge, watch for recurrent fever, joint pain, or skin changes
- Good wound care and hygiene prevent reinfection

Referral Criteria

Refer to PICU for streptococcal toxic shock syndrome, necrotizing fasciitis, or hemodynamic instability

Hospitalisation Criteria

Admit all cases of streptococcal sepsis; ICU for toxic shock syndrome, necrotizing fasciitis, or multiorgan dysfunction

Notes

Add clindamycin to penicillin for streptococcal toxic shock syndrome to suppress toxin production. Group A strep remains universally penicillin-susceptible.

A41 — Other bacterial sepsis

SNOMED: 10001005 | Duration: 10 days | Source: Surviving Sepsis Campaign 2020, AAP 2024

First-Line

- **Ceftriaxone** — 100 mg/kg IV day 1 (max 4g), then 50-100 mg/kg/day; community-acquired
- **Cefepime** — 50 mg/kg IV q8h (max 2g/dose); if Pseudomonas concern or nosocomial
- **Vancomycin** — 60 mg/kg/day IV divided q6h (max 4g/day); add if MRSA suspected

Second-Line

- **Meropenem** — 60-120 mg/kg/day IV divided q8h (max 6g/day); ESBL or resistant organisms
- **Piperacillin-tazobactam** — 300 mg/kg/day IV divided q6-8h (max 16g/day pip component)

Investigations

- Blood culture (2 sets) — Identify organism before antibiotics [STAT]
- CBC with differential — Leukocytosis or leukopenia [STAT]
- CRP, procalcitonin — Severity and response markers [STAT]
- Lactate — Tissue perfusion marker; elevated in shock [STAT]
- Renal and liver function — Assess organ dysfunction [URGENT]

Counselling

- Sepsis is a life-threatening condition — the child is receiving emergency treatment
- IV antibiotics must be given on time every time — do not delay or miss doses
- Watch for any deterioration: increased drowsiness, poor feeding, fast breathing
- Once the child improves, antibiotics may be switched to oral form to complete the course at home
- Follow-up blood cultures will confirm the infection has cleared

Referral Criteria

Refer immediately to pediatric ICU if septic shock, multiorgan dysfunction, or need for vasopressors/ventilation

Hospitalisation Criteria

Admit all cases of sepsis; ICU admission for septic shock, respiratory failure, or multiorgan dysfunction

Notes

Administer antibiotics within 1 hour for septic shock, within 3 hours for sepsis without shock. De-escalate based on cultures at 48 hours.

A41.9 — Sepsis, unspecified organism

SNOMED: 10001005 | Duration: 10 days | Source: Surviving Sepsis Campaign 2020, AAP 2024

First-Line

- **Ceftriaxone** — 100 mg/kg IV day 1 (max 4g), then 50-100 mg/kg/day; broad empiric coverage
- **Vancomycin** — 60 mg/kg/day IV divided q6h (max 4g/day); add for MRSA coverage

Second-Line

- **Meropenem** — 60-120 mg/kg/day IV divided q8h (max 6g/day); multidrug-resistant concern
- **Gentamicin** — 7.5 mg/kg IV once daily; neonatal sepsis empiric with ampicillin
- **Ampicillin** — 200 mg/kg/day IV divided q6h; neonatal sepsis with gentamicin

Investigations

- Blood culture (before antibiotics) — Identify causative organism [STAT]
- CBC, CRP, procalcitonin — Infection markers and severity [STAT]
- Lactate — Perfusion status [STAT]
- Coagulation profile — DIC screening [URGENT]
- Urine culture, CSF (if indicated) — Source identification [URGENT]

Counselling

- Your child has a serious bloodstream infection requiring hospital treatment
- Antibiotics will be adjusted once blood test results identify the exact germ
- Report any worsening: high fever spikes, cold hands/feet, drowsiness, or rash
- The child needs to complete the full antibiotic course even after feeling better
- After discharge, follow up within 1 week for repeat blood tests

Referral Criteria

Refer to PICU if hemodynamic instability, need for vasopressors, respiratory support, or multiorgan failure

Hospitalisation Criteria

Admit all cases; PICU for septic shock, altered consciousness, respiratory failure, or DIC

Notes

Obtain blood cultures before antibiotics when possible without delaying therapy. Fluid resuscitation with 10-20 mL/kg isotonic crystalloid boluses for shock.

A49.9 — Bacterial infection of unspecified site

SNOMED: 87628006 | Duration: 7 days | Source: AAP 2024, CDC Outpatient Antibiotic Guidelines

First-Line

- **Amoxicillin** — 45-90 mg/kg/day PO divided bid-tid (max 3g/day); mild-moderate infections
- **Amoxicillin-clavulanate** — 45-90 mg/kg/day PO divided bid (max 3g/day amoxicillin component)
- **Ceftriaxone** — 50-75 mg/kg IV/IM once daily (max 2g/dose); moderate-severe

Second-Line

- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg), then 5 mg/kg/day days 2-5
- **Clindamycin** — 30-40 mg/kg/day PO/IV divided tid (max 1.8g/day); penicillin allergy

Investigations

- CBC with differential — Screen for bacterial infection [URGENT]
- CRP — Inflammatory marker [URGENT]
- Blood culture — Identify causative organism [URGENT]
- Urinalysis and urine culture — Rule out UTI as source
- Chest X-ray — If respiratory symptoms present

Counselling

- Complete the full course of antibiotics even if the child feels better after 2-3 days
- Give probiotics (curd/yogurt) alongside antibiotics to prevent loose stools
- Ensure adequate fluid intake and nutrition during the illness
- Return if fever does not improve within 48-72 hours of starting antibiotics
- Wash hands frequently and maintain hygiene to prevent spread

Referral Criteria

Refer if no improvement after 48-72 hours of appropriate antibiotics, suspected deep-seated infection, or immunocompromised child

Hospitalisation Criteria

Admit for sepsis signs, toxic appearance, inability to tolerate oral medications, or immunocompromised status

Notes

Narrow antibiotic spectrum once culture and sensitivity results are available. Always obtain cultures before starting empiric therapy when feasible.

A75.3 — Scrub typhus

SNOMED: 240613006 | Duration: 7 days | Source: NEJM 2023, WHO

First-Line

- **Doxycycline** — 2.2 mg/kg PO/IV bid (max 100 mg/dose) for 7 days
- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg), then 5 mg/kg/day for 4 more days; age <8yr

Second-Line

- **Chloramphenicol** — 50-75 mg/kg/day PO/IV divided q6h (max 4g/day) for 7 days; if resistant
- **Rifampicin** — 10-20 mg/kg/day PO (max 600 mg); resistant scrub typhus

Investigations

- Scrub typhus IgM ELISA — Confirm diagnosis (positive after day 5-7) [URGENT]
- CBC — Thrombocytopenia, leukocytosis/leukopenia [URGENT]
- LFT — Elevated transaminases common [URGENT]
- CRP — Inflammatory marker [URGENT]
- Chest X-ray — If respiratory symptoms (ARDS risk)

Counselling

- Scrub typhus is spread by mite bites — fever usually improves within 48 hours of starting doxycycline
- Look for a painless black scab (eschar) on the body — this helps confirm the diagnosis
- Complete the full antibiotic course even after fever resolves
- Wear full-sleeve clothes and use insect repellent when going into bushy or grassy areas
- Watch for severe headache, confusion, or breathing difficulty and seek immediate care

Referral Criteria

Refer to ICU if ARDS, multiorgan dysfunction, myocarditis, meningoencephalitis, or no response to doxycycline within 48 hours

Hospitalisation Criteria

Admit for organ involvement (hepatitis, AKI, pneumonitis), altered consciousness, hemodynamic instability, or infant with suspected scrub typhus

Notes

Doxycycline is first-line for all ages; AAP supports short courses in children <8yr. Fever typically resolves within 48h of appropriate therapy.

Guideline Changes

NEJM 2023 trial showed IV doxycycline + IV azithromycin combination superior for severe scrub typhus

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A82 — Rabies

SNOMED: 14168008 | Duration: 28 days | Source: CDC 2024, WHO 2018

First-Line

- **Rabies vaccine (HDCV/PCECV)** — 1 mL IM on days 0, 3, 7, 14 (deltoid or anterolateral thigh in children)
- **HRIG (Human Rabies Ig)** — 20 IU/kg infiltrated into wound site; remainder IM at site distant from vaccine

Second-Line

- **5-dose vaccine series** — 1 mL IM on days 0, 3, 7, 14, 28; immunocompromised patients
- **Equine RIG (ERIG)** — 40 IU/kg infiltrated into wound; where HRIG unavailable (skin test first)

Investigations

- Clinical assessment of wound category — WHO category I/II/III determines PEP regimen [STAT]
- No pre-treatment diagnostic test needed — Treatment is based on exposure, not lab confirmation [STAT]

Counselling

- Rabies is nearly 100% fatal once symptoms appear — post-exposure vaccination is the ONLY lifesaving treatment
- Wash the wound thoroughly with soap and running water for 15 minutes immediately after the bite
- Get the first vaccine dose on the same day as the bite — do not delay
- Complete all 4 vaccine doses on the scheduled days (0, 3, 7, 14)
- Do not apply turmeric, chili, or any home remedies to the wound — only soap, water, and povidone-iodine

Referral Criteria

Refer urgently to anti-rabies clinic or hospital for category III exposure (deep bite, multiple bites, bites on head/face/neck) — RIG is mandatory

Hospitalisation Criteria

Admit if wound needs surgical debridement, category III exposure requiring RIG infiltration, bite on face/neck, or symptomatic rabies (palliative care)

Notes

Post-exposure prophylaxis is nearly 100% effective if administered promptly. HRIG is given only once with the first vaccine dose; never repeat HRIG.

A90 — Dengue fever

SNOMED: 38362002 | Duration: 7 days | Source: WHO 2024, CDC 2024

First-Line

- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day or 4g/day); fever and pain control
- **Oral rehydration therapy** — ORS or isotonic fluids; maintain hydration, monitor daily from day 3

Second-Line

- **IV Normal Saline** — 5-7 mL/kg/hr for 1-2h, then reduce; for dehydration or rising hematocrit

- **IV Ringer's Lactate** — 5-7 mL/kg/hr for 1-2h, then reduce; alternative crystalloid

Investigations

- Dengue NS1 antigen — Early diagnosis (day 1-5) [STAT]
- Dengue IgM/IgG — Confirm after day 5
- CBC with hematocrit — Daily monitoring from day 3-7 [URGENT]
- Liver function tests — Elevated transaminases in dengue

Counselling

- Give ONLY paracetamol for fever — do NOT give ibuprofen, aspirin, or nimesulide
- Ensure the child drinks plenty of fluids — ORS, coconut water, fruit juices
- Monitor daily from day 3 to day 7 — this is the critical period
- Watch for warning signs: severe belly pain, persistent vomiting, bleeding, drowsiness
- Use mosquito nets even during the day — dengue mosquito bites in daytime

Referral Criteria

Refer urgently if warning signs: persistent vomiting, abdominal pain, mucosal bleeding, lethargy, hematocrit rising rapidly

Hospitalisation Criteria

Admit for warning signs, hematocrit rise >20%, platelet <20,000 with bleeding, severe dehydration, or altered consciousness

Notes

Avoid NSAIDs and aspirin due to bleeding risk. Monitor platelet count and hematocrit daily from day 3 until 1-2 days after defervescence.

Guideline Changes

Dengvaxia and TAK-003 (Qdenga) vaccines available; treatment remains supportive

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A91 — Dengue hemorrhagic fever

SNOMED: 186599009 | Duration: 7 days | Source: WHO 2024, CDC 2024

First-Line

- **IV crystalloid (NS or RL)** — 10-20 mL/kg bolus over 15-30 min; repeat as needed for shock resuscitation
- **Acetaminophen** — 15 mg/kg PO/PR q4-6h (max 75 mg/kg/day); avoid NSAIDs/aspirin

Second-Line

- **Colloid (Dextran 40)** — 10-20 mL/kg IV; if no response to crystalloid after 2 boluses
- **Platelet transfusion** — 10 mL/kg IV; only for severe bleeding with platelet <10,000 or active hemorrhage
- **Packed RBCs** — 5-10 mL/kg IV; for significant hemorrhage with Hgb drop

Investigations

- CBC with hematocrit (serial) — Monitor plasma leakage and bleeding [STAT]
- Dengue NS1 and IgM/IgG — Confirm diagnosis [URGENT]
- Coagulation profile — PT/aPTT prolonged in DHF [STAT]
- Liver and renal function — Organ involvement assessment [URGENT]
- Chest X-ray/USG abdomen — Pleural effusion, ascites in plasma leakage [URGENT]

Counselling

- Severe dengue is a medical emergency — the child is receiving critical care
- No ibuprofen or aspirin — only paracetamol for fever
- IV fluids are carefully monitored — both too little and too much fluid are dangerous
- The critical phase (day 3-7) requires close monitoring of blood pressure and blood counts

- After recovery, do not skip the follow-up visit — platelet count and recovery need to be confirmed

Referral Criteria

Refer immediately to PICU if dengue shock syndrome, severe hemorrhage, organ failure, or massive plasma leakage

Hospitalisation Criteria

Admit all cases of dengue hemorrhagic fever; ICU for shock, massive hemorrhage, organ impairment, or rapid hematocrit rise

Notes

Severe dengue requires ICU admission. Judicious fluid management is critical: overhydration causes fluid overload and pulmonary edema during recovery phase.

A92.0 — Chikungunya virus disease

SNOMED: 111864006 | Duration: 10 days | Source: CDC 2024, PAHO, AAP Red Book 2024

First-Line

- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day or 4g/day); first-line for fever/pain
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day); ONLY after dengue is ruled out

Second-Line

- **Naproxen** — 5-7 mg/kg PO bid (max 1g/day); persistent arthralgia after excluding dengue
- **Oral rehydration therapy** — ORS ad libitum; prevent dehydration during febrile phase

Investigations

- Chikungunya IgM/ELISA — Confirm diagnosis (positive after day 5)
- Chikungunya RT-PCR — Early diagnosis (first 5 days) [URGENT]
- CBC — Leukopenia, thrombocytopenia [URGENT]
- Dengue NS1/IgM — Rule out dengue (co-endemic area) [URGENT]

Counselling

- Chikungunya is a viral illness — antibiotics will not help
- Give only paracetamol for fever until dengue is ruled out (no ibuprofen/aspirin initially)
- Joint pain may persist for weeks to months — this is common in chikungunya
- Ensure adequate fluid intake; ORS, coconut water, and soups are helpful
- Use mosquito nets and repellents to prevent spread to other family members

Referral Criteria

Refer if severe joint involvement causing immobility, persistent arthritis >3 months, encephalitis, or neonatal chikungunya

Hospitalisation Criteria

Admit for encephalitis, myocarditis, severe dehydration, neonatal chikungunya, or hemorrhagic manifestations

Notes

Use only acetaminophen until dengue is ruled out (co-endemic areas). NSAIDs may be added for persistent arthralgia once dengue is excluded.

Guideline Changes

FDA approved Ixchiq (live attenuated vaccine) Nov 2023 for adults >=18yr; no pediatric approval yet

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B00 — Herpesviral [herpes simplex] infections

SNOMED: 88594005 | Duration: 14 days | Source: AAP Red Book 2024, CDC

First-Line

- **Acyclovir (oral)** — 20 mg/kg PO qid (max 400 mg/dose) for 7-14 days; mucocutaneous HSV
- **Acyclovir (IV)** — 20 mg/kg IV q8h for 14-21 days; neonatal HSV or encephalitis

Second-Line

- **Valacyclovir** — 20 mg/kg PO bid (max 1g/dose); children >=12yr, recurrent HSV
- **Foscarnet** — 40-60 mg/kg IV q8h; acyclovir-resistant HSV in immunocompromised

Investigations

- HSV PCR from lesion — Confirm diagnosis [URGENT]
- CSF HSV PCR — If encephalitis suspected [STAT]
- HSV PCR blood (neonatal) — Assess dissemination in neonatal HSV [STAT]
- LFT and CBC — Monitor for disseminated disease [URGENT]

Counselling

- HSV infection can recur — triggers include fever, sun exposure, and stress
- Keep the sores clean; wash hands after touching them to prevent spread
- Complete the full course of acyclovir even if sores start healing
- Do not kiss the child on the mouth or share utensils when sores are active
- For neonatal herpes, hospital treatment is essential — it can be life-threatening in newborns

Referral Criteria

Refer to infectious disease specialist for neonatal HSV, HSV encephalitis, or recurrent severe episodes; ophthalmologist for eye involvement

Hospitalisation Criteria

Admit all neonatal HSV cases, HSV encephalitis, disseminated HSV, eczema herpeticum, or immunocompromised child with HSV

Notes

Neonatal HSV requires high-dose IV acyclovir for 14-21 days. Ensure adequate hydration to prevent acyclovir crystalluria.

B01 — Varicella [chickenpox]

SNOMED: 38907003 | Duration: 7 days | Source: AAP 2024, CDC, NIH OI Guidelines

First-Line

- **Acyclovir (oral)** — 20 mg/kg PO qid (max 800 mg/dose) for 5 days; within 24h of rash onset
- **Acyclovir (IV)** — 10-20 mg/kg IV q8h for 7-10 days; immunocompromised or severe disease

Second-Line

- **Valacyclovir** — 20 mg/kg PO tid (max 1g/dose) for 5 days; age >=12yr, better bioavailability
- **Calamine lotion** — Apply topically prn for pruritus relief

Investigations

- Clinical diagnosis (usually sufficient) — Classic pruritic vesicular rash in different stages
- VZV PCR from vesicle — If diagnosis uncertain or immunocompromised
- CBC — If complication suspected
- Chest X-ray — If varicella pneumonia suspected [URGENT]

Counselling

- Chickenpox usually resolves in 7-10 days — keep the child comfortable and hydrated
- Do not use aspirin for fever — use paracetamol only
- Trim fingernails short and keep sores clean to prevent bacterial superinfection

- Calamine lotion can help with itching; oatmeal baths are soothing
- The child is contagious until all blisters have crusted over (usually 5-7 days after rash onset) — keep home from school

Referral Criteria

Refer if immunocompromised, neonatal varicella, encephalitis, pneumonitis, or severe skin superinfection

Hospitalisation Criteria

Admit immunocompromised child with varicella, varicella pneumonia, encephalitis, neonatal varicella, or severe bacterial superinfection

Notes

AAP does not recommend routine acyclovir for uncomplicated varicella in healthy children. Reserve antiviral for immunocompromised, adolescents, or complicated disease.

B02 — Herpes zoster

SNOMED: 4740000 | Duration: 10 days | Source: AAP Red Book 2024, NIH OI Guidelines

First-Line

- **Acyclovir (oral)** — 20 mg/kg PO qid (max 800 mg/dose) for 7-10 days
- **Acyclovir (IV)** — 10-20 mg/kg IV q8h for 7-10 days; for immunocompromised

Second-Line

- **Valacyclovir** — 20 mg/kg PO tid (max 1g/dose) for 7 days; age >=12yr
- **Famciclovir** — Limited pediatric data; 500 mg PO tid for adolescents >=12yr

Investigations

- Clinical diagnosis (usually sufficient) — Characteristic dermatomal vesicular rash
- VZV PCR from vesicle fluid — Confirm if atypical presentation
- CBC — If immunosuppression suspected
- HIV test — Consider if zoster in young child without prior varicella

Counselling

- Herpes zoster (shingles) in children is usually milder than in adults
- Give acyclovir as prescribed — best results when started within 72 hours of rash
- Keep the rash clean and dry; do not apply any home remedies
- The child is contagious to those who have not had chickenpox — keep away from pregnant women and newborns
- Pain and itching usually resolve within 2-3 weeks as the rash heals

Referral Criteria

Refer to ophthalmologist if eye involvement (herpes zoster ophthalmicus), or to infectious disease specialist if immunocompromised

Hospitalisation Criteria

Admit if immunocompromised child, disseminated zoster, CNS involvement, or severe pain requiring IV therapy

Notes

Immunocompromised children require IV acyclovir. Treat within 72h of rash onset for best efficacy.

B05 — Measles

SNOMED: 14189004 | Duration: 10 days | Source: WHO 2024, CDC 2025, AAP

First-Line

- **Vitamin A** — <6mo: 50,000 IU PO; 6-11mo: 100,000 IU; >=12mo: 200,000 IU; 2 doses (day 0,1)
- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day); fever management

Second-Line

- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day); age >=6mo, antipyretic/analgesic
- **Antibiotics (if superinfection)** — Amoxicillin 45-90 mg/kg/day PO divided bid for bacterial pneumonia/otitis

Investigations

- Measles IgM antibody — Confirm diagnosis [URGENT]
- CBC — Leukopenia supports diagnosis
- Chest X-ray — If pneumonia suspected [URGENT]
- Vitamin A level — If severe/complicated measles

Counselling

- There is no cure for measles — treatment focuses on managing symptoms and preventing complications
- Give Vitamin A as prescribed — it reduces complications and death from measles
- Keep the child isolated from others for 4 days after the rash appears
- Watch for fast breathing, ear discharge, or drowsiness — these suggest complications
- Ensure the child stays hydrated with ORS and nutritious foods

Referral Criteria

Refer if pneumonia, encephalitis, severe malnutrition, or immunocompromised child with measles

Hospitalisation Criteria

Admit for measles pneumonia, encephalitis, severe dehydration, croup, severe malnutrition, or immunocompromised status

Notes

No specific antiviral therapy. Vitamin A reduces mortality and complication rates, especially in malnourished or hospitalized children. Isolate for 4 days after rash onset.

Guideline Changes

CDC 2025 updated guidance on vitamin A: recommended for hospitalized children

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B06 — Rubella

SNOMED: 36653000 | Duration: 5 days | Source: AAP Red Book 2024, CDC

First-Line

- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day); fever and arthralgia
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day); joint pain, age >=6mo

Second-Line

- **Oral rehydration therapy** — ORS to maintain hydration during febrile illness

Investigations

- Rubella IgM antibody — Confirm acute infection
- CBC with platelet count — Thrombocytopenia may occur
- Echocardiography — If congenital rubella suspected (PDA, PS) [URGENT]

Counselling

- Rubella in children is usually mild and self-limiting — no specific treatment needed
- Give paracetamol if the child has fever or joint pain
- Keep the child away from pregnant women for 7 days after rash onset — rubella can harm the unborn baby
- The rash typically lasts 3 days and resolves without scarring
- Ensure the child has received MMR vaccination to prevent future episodes

Referral Criteria

Refer if congenital rubella syndrome suspected (cataracts, heart defects, deafness), thrombocytopenia, or encephalitis

Hospitalisation Criteria

Admit for congenital rubella syndrome management, thrombocytopenic purpura, or encephalitis

Notes

No specific therapy for rubella or congenital rubella syndrome. Postnatal rubella is usually mild and self-limiting. Isolate for 7 days after rash onset.

B08.2 — Roseola infantum

SNOMED: 402420001 | Duration: 5 days | Source: AAP Red Book 2024

First-Line

- **Acetaminophen** — 15 mg/kg PO/PR q4-6h (max 75 mg/kg/day); fever control
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day); age >=6mo

Second-Line

- **Oral rehydration therapy** — ORS ad libitum; maintain hydration during high fever phase (3-4 days)

Investigations

- Clinical diagnosis (usually sufficient) — Classic pattern: high fever 3-4 days then rash
- HHV-6 PCR — Only if atypical or immunocompromised
- CBC — Leukopenia with relative lymphocytosis

Counselling

- Roseola causes high fever for 3-4 days followed by a rash once fever breaks — the rash is harmless
- Give paracetamol for fever and ensure plenty of fluids
- Febrile seizures can occur due to high fever — if it happens, place child on side and seek emergency care
- The rash means the illness is almost over — no treatment needed for the rash
- The child can return to daycare/school once fever-free for 24 hours

Referral Criteria

Refer if febrile seizures occur, encephalitis suspected, or immunocompromised child with HHV-6 infection

Hospitalisation Criteria

Admit for prolonged febrile seizure, encephalitis, hepatitis, or immunocompromised child with severe HHV-6 disease

Notes

Self-limiting illness caused by HHV-6. High fever (up to 40.5C) resolves in 3-4 days followed by characteristic rash. No antiviral therapy needed in immunocompetent children.

B08.4 — Hand, foot and mouth disease

SNOMED: 240482001 | Duration: 7 days | Source: AAP Red Book 2024, CDC

First-Line

- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day or 4g/day); fever and oral pain
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day); age >=6mo, fever and pain

Second-Line

- **Magic mouthwash** — Equal parts Maalox/diphenhydramine/lidocaine 2%; swish-and-spit for oral pain
- **Oral rehydration therapy** — ORS ad libitum; prevent dehydration from poor intake due to oral ulcers

Investigations

- Clinical diagnosis (usually sufficient) — Characteristic oral ulcers + vesicles on hands/feet
- Enterovirus PCR (throat/stool) — If neurological complications suspected (EV-A71) [URGENT]
- CBC — If systemic illness suspected

Counselling

- HFMD is a viral illness that resolves on its own in 7-10 days — no antibiotics needed
- Give cold fluids, ice pops, and soft foods — avoid spicy, salty, or acidic foods that sting the mouth sores
- Watch for dehydration: fewer wet diapers, dry mouth, or no tears when crying
- Keep the child home from school/daycare until fever resolves and sores are healing
- Wash hands frequently — the virus spreads through saliva, blister fluid, and stool

Referral Criteria

Refer if signs of neurological involvement (myoclonus, ataxia, encephalitis), cardiopulmonary compromise, or dehydration not responding to oral rehydration

Hospitalisation Criteria

Admit for severe dehydration, encephalitis, pulmonary edema (EV-A71 complication), myocarditis, or inability to maintain oral intake

Notes

Self-limiting illness (7-10 days). Focus on maintaining hydration as oral ulcers may cause feeding refusal. No antiviral therapy is indicated.

B20 — Human immunodeficiency virus [HIV] disease

SNOMED: 186706006 | Duration: 365 days | Source: NIH Pediatric ARV Guidelines 2024, WHO 2024

First-Line

- **Dolutegravir (DTG)** — Weight-band PO once daily + 2 NRTIs; preferred INSTI for all ages >=4 wk **[NEW 2024-25]**
- **Abacavir/Lamivudine** — ABC 8 mg/kg PO bid + 3TC 4 mg/kg PO bid (max ABC 600mg, 3TC 300mg); NRTI backbone
- **Tenofovir DF/Emtricitabine** — TDF/FTC weight-based PO daily; adolescents >=35 kg, Tanner stage 4-5

Second-Line

- **Lopinavir/Ritonavir** — 300 mg/75 mg per m2 PO bid; infants <4 wk or NNRTI-exposed neonates
- **Raltegravir** — Weight-band PO bid; alternative INSTI if DTG unavailable
- **Efavirenz** — Weight-band PO once daily; children >=3yr and >=14 kg

Investigations

- HIV PCR (DNA/RNA) — Definitive diagnosis in children <18 months [STAT]
- CD4 count and percentage — Baseline and every 6 months [URGENT]
- HIV viral load — Baseline and every 6 months to monitor ART response [URGENT]
- CBC, LFT, RFT — Baseline before ART and monitoring
- Chest X-ray — Screen for TB co-infection

Counselling

- ART medicines must be given every day at the same time — never skip doses
- The child can live a normal, healthy life with consistent treatment
- Regular blood tests (CD4, viral load) every 3-6 months are essential to monitor treatment
- Keep the child's HIV status confidential — disclosure should be age-appropriate and gradual
- Ensure all routine vaccinations are given (avoid live vaccines if severely immunosuppressed)

Referral Criteria

Refer to pediatric ART center for initiation; refer urgently for treatment failure (rising viral load), opportunistic infections, or IRIS

Hospitalisation Criteria

Admit for severe opportunistic infections (PCP, TB meningitis, cryptococcal meningitis), severe malnutrition, or WHO stage 4 conditions

Notes

Start ART immediately regardless of CD4 count. DTG-based regimens are now preferred across all age groups per WHO 2024 update.

Guideline Changes

Dolutegravir-based regimens now preferred first-line for all ages including neonates

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B25 — Cytomegaloviral disease

SNOMED: 28944009 | Duration: 180 days | Source: NIH OI Guidelines 2024, AAP

First-Line

- **Valganciclovir** — 16 mg/kg PO bid for 6 months; congenital CMV (dose by BSA/CrCl preferred)
- **Ganciclovir** — 6 mg/kg IV q12h for 6 weeks, then switch to oral valganciclovir; severe disease

Second-Line

- **Foscarnet** — 60 mg/kg IV q8h or 90 mg/kg IV q12h; ganciclovir-resistant CMV
- **Cidofovir** — 5 mg/kg IV weekly x2, then q2wk with probenecid; refractory CMV

Investigations

- CMV PCR (urine or saliva) — Confirm congenital CMV within 21 days of birth [STAT]
- CMV PCR (blood) — Quantify viral load for treatment monitoring [URGENT]
- CBC with differential — Monitor for ganciclovir-induced neutropenia [URGENT]
- Auditory brainstem response (ABR) — Baseline and serial hearing assessment [URGENT]
- MRI brain — Assess for CNS involvement in congenital CMV [URGENT]

Counselling

- CMV treatment in babies requires 6 months of daily medicine — do not stop early
- Get regular blood tests (CBC) as the medicine can lower blood counts temporarily
- Hearing tests should be done every 6 months until age 3 years even after treatment
- Early treatment helps prevent hearing loss, which is the most common complication
- Report any rash, unusual bleeding, or feeding difficulty during treatment

Referral Criteria

Refer to pediatric infectious disease and audiology for all congenital CMV cases; ophthalmology for retinal involvement

Hospitalisation Criteria

Admit for initiation of IV ganciclovir if severe congenital CMV (CNS involvement, retinitis, hepatitis), or for significant cytopenias during treatment

Notes

BSA-based dosing is preferred over mg/kg to avoid underdosing. Monitor ANC, platelets, renal and hepatic function weekly during therapy.

B26 — Mumps

SNOMED: 36989005 | Duration: 10 days | Source: AAP Red Book 2024, CDC

First-Line

- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day); fever and parotid pain
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day); age >=6mo, pain and swelling

Second-Line

- **Warm/cold compresses** — Apply to swollen parotid glands for symptomatic relief
- **IV fluids** — Maintenance rate isotonic crystalloid; if unable to tolerate PO due to trismus

Investigations

- Clinical diagnosis (usually sufficient) — Bilateral parotid swelling in unvaccinated child
- Mumps IgM antibody — Confirm if atypical presentation
- Serum amylase/lipase — If pancreatitis suspected [URGENT]
- CBC — Leukopenia typical

Counselling

- Mumps is a viral infection — antibiotics are not needed
- Give soft foods and plenty of fluids; avoid sour/acidic foods that stimulate saliva and worsen pain
- Apply warm or cold compresses to swollen cheeks for pain relief
- The child is contagious from 2 days before to 5 days after swelling appears — keep home from school
- Watch for severe headache, stiff neck, or scrotal swelling in boys and seek medical care immediately

Referral Criteria

Refer if meningitis (severe headache, neck stiffness), orchitis in post-pubertal boy, pancreatitis, or hearing loss

Hospitalisation Criteria

Admit for meningitis/encephalitis, severe pancreatitis, dehydration from inability to eat, or bilateral orchitis

Notes

No specific antiviral treatment. Supportive care with hydration and soft diet. Orchitis complication in post-pubertal males requires bed rest and scrotal support.

B27 — Infectious mononucleosis

SNOMED: 271558008 | Duration: 14 days | Source: AAP 2024, AAFP 2023

First-Line

- **Acetaminophen** — 15 mg/kg PO q4-6h (max 75 mg/kg/day or 4g/day); fever and pain
- **Ibuprofen** — 10 mg/kg PO q6-8h (max 40 mg/kg/day or 1.2g/day); fever and pain, age >=6mo

Second-Line

- **Prednisone** — 1-2 mg/kg/day PO (max 60 mg) for 3-7 days; severe tonsillar obstruction only
- **Dexamethasone** — 0.3 mg/kg IV (max 10 mg) single dose; impending airway obstruction

Investigations

- Monospot test (heterophile antibody) — Rapid screening (may be negative in <4 years) [URGENT]
- EBV-specific antibodies (VCA IgM) — Confirm if monospot negative, especially in young children
- CBC with differential — Atypical lymphocytes, thrombocytopenia [URGENT]
- Liver function tests — Assess for EBV hepatitis
- Abdominal ultrasound — If splenomegaly suspected, before sports clearance

Counselling

- Infectious mono is caused by a virus — antibiotics will not help
- Rest is important; the child will feel tired for 2-4 weeks
- Avoid contact sports for at least 3-4 weeks due to risk of spleen injury
- Do not give amoxicillin/ampicillin — it causes a rash in mono patients

- The sore throat and fever should resolve in 1-2 weeks; fatigue may last longer

Referral Criteria

Refer to ENT if impending airway obstruction from tonsillar enlargement, or to hematologist if severe cytopenias

Hospitalisation Criteria

Admit for airway obstruction, splenic rupture, severe hepatitis, hemolytic anemia, or inability to maintain hydration

Notes

No antiviral therapy indicated; acyclovir shows no clinical benefit. Avoid contact sports for at least 3 weeks due to risk of splenic rupture.

B37 — Candidiasis

SNOMED: 78048006 | Duration: 14 days | Source: IDSA 2016, AAP Red Book 2024

First-Line

- **Nystatin** — 100,000-400,000 U PO qid swish-and-swallow for 7-14 days; oropharyngeal
- **Fluconazole** — 6 mg/kg PO/IV day 1, then 3 mg/kg/day (max 400 mg) for 7-14 days

Second-Line

- **Clotrimazole troches** — 10 mg dissolved in mouth 5 times daily for 7-14 days; age >=3yr
- **Micafungin** — 2 mg/kg IV once daily (max 100 mg); invasive candidiasis
- **Amphotericin B** — 0.5-1 mg/kg/day IV; severe/refractory invasive candidiasis

Investigations

- KOH mount/Gram stain — Rapid confirmation of yeast
- Blood culture (fungal) — If systemic candidiasis suspected [URGENT]
- Speciation and sensitivity — Guide antifungal choice for invasive disease
- HIV test — If recurrent oral candidiasis in older child

Counselling

- For oral thrush, apply nystatin drops after feeds — do not feed immediately after
- Sterilize bottle nipples and pacifiers daily by boiling
- If breastfeeding, treat the mother's nipples simultaneously to prevent reinfection
- Watch for white patches that do not wipe off in the mouth — this is thrush
- Complete the full course of antifungal even if the white patches clear early

Referral Criteria

Refer to pediatric infectious disease if invasive/systemic candidiasis, recurrent oral thrush in older child (rule out immunodeficiency), or azole-resistant infection

Hospitalisation Criteria

Admit for invasive/systemic candidiasis, candidemia, esophageal candidiasis with inability to swallow, or immunocompromised child with disseminated disease

Notes

Nystatin is first-line for oropharyngeal thrush; fluconazole is preferred for esophageal or systemic candidiasis. Neonates <2 weeks require adjusted dosing intervals.

B50 — Plasmodium falciparum malaria

SNOMED: 62676009 | Duration: 3 days | Source: WHO 2025, CDC

First-Line

- **Artemether-lumefantrine** — Weight-based PO bid for 3 days per WHO dosing table
- **Artesunate IV (severe)** — 2.4 mg/kg IV at 0, 12, 24h, then daily until oral tolerated (min 24h)

Second-Line

- **DHA-piperaquine** — DHA 2.5 mg/kg + PPQ 20 mg/kg PO daily for 3 days (<25 kg)
- **Artesunate-pyronaridine** — Weight-based PO daily for 3 days; WHO-recommended ACT

Investigations

- Peripheral blood smear — Species confirmation and parasitemia quantification [STAT]
- RDT (HRP2-based) — Rapid P. falciparum detection [STAT]
- CBC, blood glucose, lactate — Severity assessment [STAT]
- Renal and liver function — Organ dysfunction in severe malaria [URGENT]
- Blood gas — If acidosis or respiratory distress [STAT]

Counselling

- P. falciparum is the most dangerous type of malaria — take all medicines exactly as prescribed
- Take ACT (artemether-lumefantrine) with fatty food for better absorption
- Watch for danger signs: confusion, seizures, very dark urine, jaundice, severe weakness
- Return if fever comes back within 28 days — blood test will need to be repeated
- Prevent future malaria: use insecticide-treated bed nets every night

Referral Criteria

Refer immediately if severe falciparum malaria: cerebral malaria, severe anemia, acidosis, renal failure, pulmonary edema, or hyperparasitemia

Hospitalisation Criteria

Admit for severe malaria, hyperparasitemia >5%, any danger sign (prostration, seizures, respiratory distress, bleeding), or age <6 months

Notes

P. falciparum is the most lethal species; always assess for severity. Children <25 kg need minimum DHA 2.5 mg/kg/day per WHO 2025 guidelines.

Guideline Changes

Triple ACT combinations under Phase 3 trials for areas with partial artemisinin resistance

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B50.9 — Malaria (P. falciparum)

SNOMED: 18342001 | Duration: 3 days | Source: WHO 2023 6th edition

First-Line

- **Artemether-Lumefantrine** — Weight-based 6 doses over 3 days BID, take with food

Second-Line

- **IV Artesunate** — 2.4 mg/kg IV at 0h, 12h, 24h then daily, severe malaria first-line **[NEW 2024-25]**
- **Artesunate-Pyronaridine** — Weight-based x3 days, new ACT added in WHO 2023 **[NEW 2024-25]**

Investigations

- Peripheral blood smear (thick and thin) — Gold standard for malaria diagnosis and species identification [STAT]
- Rapid diagnostic test (RDT) — Point-of-care diagnosis when microscopy not available [STAT]
- Complete blood count — Assess anaemia and thrombocytopenia [URGENT]
- Blood glucose — Screen for hypoglycaemia especially in severe malaria [STAT]
- Serum creatinine and bilirubin — Assess for severe malaria complications [URGENT]

Counselling

- Complete the full 3-day course of ACT tablets even if fever goes away
- Give the medicine with food or milk for better absorption
- Use mosquito nets (treated with insecticide) every night and mosquito repellent
- Return immediately if fever returns after treatment or child becomes very weak
- Watch for danger signs: repeated vomiting, dark urine, jaundice, confusion - rush to hospital

Referral Criteria

Refer to infectious disease specialist or higher centre if suspected severe/complicated malaria, mixed species infection, treatment failure after 72 hours, or underlying conditions (sickle cell, immunodeficiency)

Hospitalisation Criteria

Admit if cerebral malaria (altered consciousness, seizures), severe anaemia (Hb <5 g/dL), respiratory distress, hypoglycaemia, renal failure, jaundice, parasitaemia >5%, or inability to tolerate oral medication

Notes

Monitor for delayed haemolysis after IV artesunate.

Guideline Changes

IV artesunate is now unambiguous first-line for severe malaria, replacing IV quinine. Artesunate-pyronaridine added as new ACT.

B51 — Plasmodium vivax malaria

SNOMED: 186792002 | Duration: 14 days | Source: WHO 2025, CDC

First-Line

- **Chloroquine** — 10 mg/kg PO day 1-2, then 5 mg/kg day 3 (total 25 mg/kg base); if CQ-sensitive
- **Primaquine (radical cure)** — 0.25-0.5 mg/kg/day PO for 14 days (max 30 mg/day); after G6PD testing

Second-Line

- **Artemether-lumefantrine** — Weight-based PO bid for 3 days; if CQ-resistant P. vivax area
- **Tafenoquine** — Single-dose radical cure; approved >=16yr only, requires G6PD normal **[NEW 2024-25]**

Investigations

- Peripheral blood smear — Confirm P. vivax (enlarged RBCs, Schuffner dots) [STAT]
- G6PD assay — Mandatory before primaquine [URGENT]
- CBC with reticulocyte count — Anemia assessment [URGENT]
- RDT (pLDH-based) — Rapid pan-Plasmodium detection [STAT]

Counselling

- P. vivax malaria requires 14 days of primaquine to prevent relapse — this is critical
- G6PD test must be done before starting primaquine to avoid a serious side effect
- Give primaquine with food to reduce stomach upset
- Watch for dark urine or sudden pallor during primaquine course — stop and see doctor immediately
- Use bed nets and repellents consistently to prevent reinfection

Referral Criteria

Refer if severe vivax malaria (rare), chloroquine-resistant vivax area, G6PD deficiency needing alternative radical cure, or splenic complications

Hospitalisation Criteria

Admit for severe anemia, splenic rupture, respiratory distress, or severe vivax malaria with organ dysfunction

Notes

Radical cure with primaquine (14 days) is essential to prevent relapse. Always test G6PD before primaquine; weekly dosing is alternative for G6PD-deficient patients.

Guideline Changes

WHO 2025 supports single low-dose primaquine for gametocyte clearance where G6PD test unavailable

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B54 — Malaria, unspecified

SNOMED: 61462000 | Duration: 3 days | Source: WHO 2025, CDC

First-Line

- **Artemether-lumefantrine** — Weight-based PO bid for 3 days (5-14 kg: 1 tab; 15-24 kg: 2 tabs per dose)
- **Artesunate-amodiaquine** — Weight-based PO daily for 3 days; WHO-recommended ACT

Second-Line

- **Artesunate IV** — 2.4 mg/kg IV at 0, 12, 24h, then daily; severe malaria, then switch to oral ACT
- **Quinine** — 10 mg/kg PO q8h for 7 days + clindamycin 7 mg/kg bid; if ACTs unavailable

Investigations

- Peripheral blood smear (thick and thin) — Gold standard for species identification and parasitemia [STAT]
- Rapid diagnostic test (RDT) — Point-of-care HRP2/pLDH detection [STAT]
- CBC — Anemia, thrombocytopenia [URGENT]
- Blood glucose — Hypoglycemia in severe malaria [STAT]

Counselling

- Complete the full 3-day antimalarial course even if fever resolves on day 1
- Ensure the child eats before taking the medicine — antimalarials should be taken with food
- Use mosquito nets (treated with insecticide) every night
- Watch for return of fever within 28 days — this may indicate treatment failure or relapse
- If P. vivax diagnosed, an additional 14-day course of primaquine will be needed to prevent relapse

Referral Criteria

Refer urgently if severe malaria: altered consciousness, seizures, severe anemia (Hb <5), respiratory distress, or jaundice

Hospitalisation Criteria

Admit for severe malaria (any danger sign), parasitemia >5%, age <6 months, inability to tolerate oral meds, or vomiting all doses

Notes

ACTs are first-line for all uncomplicated malaria. IV artesunate is mandatory for severe malaria; switch to oral ACT once patient can tolerate PO.

Guideline Changes

WHO 2025 guidelines reinforce ACTs as first-line; triple ACT (artemether-lumefantrine-amodiaquine) under evaluation

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B76 — Hookworm diseases

SNOMED: 187533003 | Duration: 3 days | Source: WHO 2024, CDC STH Guidelines

First-Line

- **Albendazole** — 400 mg PO single dose (200 mg if <2yr); 78-92% cure rate
- **Mebendazole** — 100 mg PO bid for 3 days OR 500 mg PO single dose

Second-Line

- **Pyrantel pamoate** — 11 mg/kg PO daily (max 1g) for 3 days
- **Iron supplementation** — 3-6 mg/kg/day elemental iron PO divided bid-tid; for anemia

Investigations

- Stool examination for ova — Confirm hookworm eggs
- CBC with peripheral smear — Microcytic anemia, eosinophilia [URGENT]
- Serum ferritin and iron studies — Assess iron deficiency from chronic blood loss

Counselling

- Give albendazole as prescribed; a 3-day course may be needed for heavy infection
- Iron supplement will also be given if the child is anemic — complete the full iron course
- Always wear footwear outdoors — hookworm enters through bare feet
- Wash hands and feet after playing in soil or mud
- Repeat deworming every 6 months in areas where hookworm is common

Referral Criteria

Refer to hematologist if severe anemia (Hb <5 g/dL) not responding to iron therapy, or if diagnosis is uncertain

Hospitalisation Criteria

Admit if severe anemia with cardiac symptoms, need for blood transfusion, or protein-losing enteropathy

Notes

Hookworm causes iron-deficiency anemia; check hemoglobin and supplement iron as needed. Albendazole 3-day course improves cure rate for heavy infections.

B77 — Ascariasis

SNOMED: 2435008 | Duration: 1 days | Source: WHO 2024, CDC STH Guidelines

First-Line

- **Albendazole** — 400 mg PO single dose (200 mg if <2yr); ~100% cure rate
- **Mebendazole** — 100 mg PO bid for 3 days OR 500 mg PO single dose

Second-Line

- **Ivermectin** — 150-200 mcg/kg PO single dose (max 12 mg); age >=5yr and >=15 kg
- **Pyrantel pamoate** — 11 mg/kg PO single dose (max 1g)

Investigations

- Stool examination for ova — Confirm Ascaris eggs
- Abdominal X-ray — If bowel obstruction suspected (worm bolus) [URGENT]
- CBC with differential — Eosinophilia supports parasitic infection

Counselling

- Single dose of albendazole is usually enough — no need for repeat unless instructed
- The child may pass worms in stool after treatment — this means the medicine is working
- Wash hands before eating and after toilet; trim nails short
- Wash all vegetables and fruits thoroughly before eating
- Deworm all family members together; repeat deworming every 6 months in endemic areas

Referral Criteria

Refer to pediatric surgeon if biliary or intestinal obstruction suspected from heavy worm burden

Hospitalisation Criteria

Admit if bowel obstruction, biliary ascariasis, pancreatitis from worm migration, or heavy burden with respiratory symptoms (Loeffler syndrome)

Notes

Single-dose albendazole is highly effective. Safe for children >=1 year per WHO MDA guidelines; same dose for adults and children.

B80 — Enterobiasis

SNOMED: 266162007 | Duration: 14 days | Source: WHO 2024, CDC

First-Line

- **Albendazole** — 400 mg PO single dose, repeat in 2 weeks (200 mg if <2yr)
- **Mebendazole** — 100 mg PO single dose, repeat in 2 weeks

Second-Line

- **Pyrantel pamoate** — 11 mg/kg PO single dose (max 1g), repeat in 2 weeks

Investigations

- Scotch tape test (perianal) — Detect eggs on perianal skin (early morning)
- Stool examination — Low sensitivity for pinworm (eggs rarely in stool)

Counselling

- Give the medicine now and repeat the same dose in exactly 2 weeks
- Treat all family members at the same time to prevent reinfection
- Wash all bed sheets, towels, and undergarments in hot water on the day of treatment
- Have the child wear tight-fitting underwear at night to prevent scratching
- Trim fingernails short and encourage handwashing before meals and after toilet

Referral Criteria

Referral rarely needed; consider pediatric GI if persistent despite repeated treatment of entire household

Hospitalisation Criteria

Hospitalisation not required for pinworm infection

Notes

Repeat dose in 2 weeks is essential as drugs do not kill eggs. Treat all household members simultaneously to prevent reinfection.

B81 — Other intestinal helminthiases

SNOMED: 187170004 | Duration: 3 days | Source: WHO 2024, CDC

First-Line

- **Albendazole** — 400 mg PO daily for 3 days (200 mg if <2yr); covers most helminths
- **Mebendazole** — 100 mg PO bid for 3 days

Second-Line

- **Ivermectin** — 200 mcg/kg PO single dose; Strongyloides, age >=5yr and >=15 kg
- **Praziquantel** — 25 mg/kg PO tid for 1-2 days; for tapeworms and flukes

Investigations

- Stool for ova and parasites — Species identification
- CBC with differential — Eosinophilia
- Strongyloides serology — If Strongyloides suspected

Counselling

- Complete the full 3-day deworming course for best results
- Different types of worms need different medicines — the stool test helps identify the right one
- Wash hands before eating and after toilet; do not eat raw/undercooked meat
- Keep the living area clean and ensure safe disposal of human waste
- Repeat deworming every 6 months in endemic areas as reinfection is common

Referral Criteria

Refer to pediatric GI if Strongyloides suspected (immunocompromised risk), or if refractory to standard treatment

Hospitalisation Criteria

Admit for Strongyloides hyperinfection syndrome, bowel obstruction, or severe malnutrition

Notes

Species-specific therapy is preferred once stool microscopy identifies the parasite. Extended 3-day courses improve efficacy for heavy infections.

B82.9 — Intestinal parasitism, unspecified

SNOMED: 240344008 | Duration: 3 days | Source: WHO 2024, CDC

First-Line

- **Albendazole** — 400 mg PO single dose (200 mg if <2yr); broad-spectrum antihelminthic
- **Mebendazole** — 100 mg PO bid for 3 days; broad-spectrum antihelminthic

Second-Line

- **Metronidazole** — 35-50 mg/kg/day PO divided tid (max 2.25g/day) for 7-10 days; if protozoal suspected
- **Ivermectin** — 200 mcg/kg PO single dose; if Strongyloides suspected, age >=5yr

Investigations

- Stool for ova and parasites (3 samples) — Identify specific parasite
- CBC with differential — Eosinophilia suggests parasitic infection
- Stool culture — If bacterial co-infection suspected

Counselling

- Give the deworming medicine as directed — usually for 1-3 days
- If worms are seen in stool after treatment, this is expected and normal
- Ensure clean drinking water and thoroughly wash all fruits and vegetables
- Practice good hand hygiene before eating and after using the toilet
- Bring a stool sample for repeat testing 2 weeks after completing treatment

Referral Criteria

Refer to pediatric GI if parasitic infection persists despite two rounds of treatment, or if Strongyloides hyperinfection suspected

Hospitalisation Criteria

Admit if severe anemia, bowel obstruction, Strongyloides hyperinfection, or significant malnutrition from chronic parasitism

Notes

Obtain stool ova and parasites to guide targeted therapy. Empiric albendazole covers most helminth infections pending diagnosis.

B85 — Pediculosis and phthiriasis

SNOMED: 81000006 | Duration: 14 days | Source: AAP 2024, CDC

First-Line

- **Permethrin 1% lotion** — Apply to damp hair for 10 min, rinse; repeat in 7-10 days
- **Dimethicone 4% lotion** — Apply to dry hair, leave 8h or overnight, wash out; repeat in 7 days

Second-Line

- **Ivermectin 0.5% lotion** — Apply to dry hair for 10 min, rinse; single application, age >=6mo **[NEW 2024-25]**
- **Oral ivermectin** — 200 mcg/kg PO, repeat in 7-10 days; >=15 kg, refractory cases
- **Spinosad 0.9% topical** — Apply to dry hair for 10 min, rinse; single treatment, age >=6mo

Investigations

- Clinical examination — Identify live lice and nits on hair shafts
- Dermoscopy — If diagnosis uncertain — visualize lice and nits

Counselling

- Apply the lotion exactly as directed — leave for the correct time then wash out
- Repeat treatment in 7-10 days to kill newly hatched lice from surviving eggs
- Comb hair with a fine-tooth nit comb every 3-4 days for 2 weeks after treatment
- Wash bed sheets, pillowcases, and towels in hot water on treatment day
- Check all family members and treat everyone who is affected at the same time

Referral Criteria

Referral rarely needed; consider dermatology if refractory to two rounds of permethrin + manual removal

Hospitalisation Criteria

Hospitalisation not needed for head lice

Notes

Permethrin resistance is increasing. Wet combing with nit comb every 3-4 days for 2 weeks is adjunctive. Treat all affected household contacts.

B86 — Scabies

SNOMED: 128869009 | Duration: 14 days | Source: CDC 2024, AAP 2024

First-Line

- **Permethrin 5% cream** — Apply neck to toes (include scalp in infants), wash off after 8-14h; repeat in 7 days
- **Oral ivermectin** — 200 mcg/kg PO, repeat in 7-14 days; >=15 kg (emerging data for <15 kg)

Second-Line

- **Precipitated sulfur 5-10%** — Apply in petrolatum nightly for 3 nights; infants <2mo
- **Crotamiton 10% cream** — Apply daily for 2 consecutive days, wash off 48h after last application

Investigations

- Skin scraping with KOH mount — Visualize mites, eggs, or fecal pellets
- Dermoscopy — Delta-wing jet sign (mite visualization)
- CBC — Eosinophilia may be present

Counselling

- Apply permethrin cream from neck to toes (include scalp in infants) and wash off after 8-14 hours
- Repeat the application in 7 days — one application is not enough
- Treat ALL family members and close contacts at the same time even if they have no symptoms
- Wash all clothes, bed sheets, and towels in hot water (>60C) on treatment day
- Itching may continue for 2-4 weeks after treatment — this does not mean treatment failed

Referral Criteria

Refer to dermatologist if crusted (Norwegian) scabies, refractory to two rounds of treatment, or immunocompromised child

Hospitalisation Criteria

Admit for crusted (Norwegian) scabies requiring isolation and intensive topical + oral treatment, or secondary bacterial infection with sepsis

Notes

Treat all household members and close contacts simultaneously. Launder clothing/bedding at >=60C. Itching may persist 2-4 weeks post-treatment (not treatment failure).

Guideline Changes

Oral ivermectin safety data expanding for children <15 kg per Lancet 2024 trial

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J10 — Influenza due to identified influenza virus

Duration: 5 days | Source: AAP 2025-2026, CDC 2024

First-Line

- **Oseltamivir** — <1yr: 3 mg/kg PO bid x5 days; 1-12yr weight-based PO bid x5 days (max 75 mg bid)
- **Baloxavir marboxil** — Single PO dose: 20-<80 kg 40 mg; >=80 kg 80 mg; age >=5yr, within 48h **[NEW 2024-25]**

Second-Line

- **Zanamivir** — 10 mg (2 inhalations) bid for 5 days; age >=7yr
- **Peramivir** — 12 mg/kg IV single dose (max 600 mg); age >=6mo, unable to take PO

Investigations

- Influenza rapid antigen test — Point-of-care diagnosis [URGENT]
- Influenza PCR (nasopharyngeal) — Confirm if rapid test negative but clinical suspicion high
- Chest X-ray — If pneumonia suspected
- CBC — Leukopenia supports viral etiology

Counselling

- Give oseltamivir (Tamiflu) within 48 hours of symptom onset for best results
- The child should rest at home and drink plenty of fluids
- Keep the child away from school for at least 24 hours after fever resolves
- Wash hands frequently and cover coughs/sneezes with elbow
- Annual influenza vaccination is the best prevention — get it every year before monsoon/winter season

Referral Criteria

Refer to hospital if respiratory distress, dehydration, altered consciousness, or high-risk child (asthma, immunocompromised, cardiac disease)

Hospitalisation Criteria

Admit for respiratory distress, hypoxia, encephalopathy, myocarditis, or severe dehydration

Notes

Start antivirals within 48h of symptom onset; still consider beyond 48h in high-risk children. Oseltamivir is preferred due to widest age range and formulation options.

J11 — Influenza, virus not identified

SNOMED: 6142004 | Duration: 5 days | Source: AAP 2025-2026, CDC 2024

First-Line

- **Oseltamivir** — <1yr: 3 mg/kg PO bid x5 days; 1-12yr weight-based PO bid x5 days (max 75 mg bid)
- **Baloxavir marboxil** — Single PO dose: 20-<80 kg 40 mg; >=80 kg 80 mg; age >=5yr, within 48h **[NEW 2024-25]**

Second-Line

- **Zanamivir** — 10 mg (2 inhalations) bid for 5 days; age >=7yr
- **Peramivir** — 12 mg/kg IV single dose (max 600 mg); age >=6mo, unable to take PO

Investigations

- Influenza rapid antigen test — Point-of-care screening [URGENT]
- Respiratory viral panel PCR — If influenza test negative but viral illness suspected
- Chest X-ray — If lower respiratory symptoms present

Counselling

- Start antiviral medicine (Tamiflu) as early as possible — even before test results return
- Keep the child home, hydrated, and resting
- Paracetamol or ibuprofen for fever and body aches
- Do not give aspirin to children with viral illness (risk of Reye syndrome)
- Annual flu vaccination recommended for all children above 6 months

Referral Criteria

Refer to hospital if respiratory distress, high-risk child, or no improvement after 48 hours of antiviral therapy

Hospitalisation Criteria

Admit for respiratory distress, hypoxia (SpO2 <92%), dehydration, seizures, or immunocompromised status

Notes

Treat clinically suspected influenza empirically when testing is pending or unavailable. Same antiviral approach regardless of virus identification status.

L03 — Cellulitis

SNOMED: 128045006 | Duration: 10 days | Source: IDSA 2014, AAP 2024

First-Line

- **Cephalexin** — 50-100 mg/kg/day PO divided tid (max 1.5g/dose); outpatient, non-purulent
- **Clindamycin** — 30-40 mg/kg/day PO divided tid (max 1.8g/day); MRSA suspected or PCN allergy

Second-Line

- **TMP-SMX** — 8-12 mg/kg/day TMP PO divided bid (max 320 mg TMP bid); MRSA, purulent
- **Cefazolin** — 100 mg/kg/day IV divided q8h (max 6g/day); moderate-severe, inpatient
- **Doxycycline** — 2-4 mg/kg/day PO divided bid (max 200 mg/day); age >=8yr, MRSA

Investigations

- CBC with differential — Leukocytosis in moderate-severe cases [URGENT]
- CRP — Monitor inflammatory response [URGENT]
- Blood culture — If systemic signs present [URGENT]
- Wound culture — If purulent drainage present
- Ultrasound — If abscess suspected

Counselling

- Give antibiotics exactly as prescribed — complete the full course
- Mark the border of the redness with a pen and watch if it spreads beyond the mark
- Keep the affected area elevated to reduce swelling
- Apply warm compresses for comfort if advised by the doctor
- Return if redness spreads rapidly, fever develops, or streaking appears — this needs urgent review

Referral Criteria

Refer to surgeon if abscess formation needing incision and drainage, or if necrotizing fasciitis suspected

Hospitalisation Criteria

Admit if rapid spread, systemic toxicity (fever, tachycardia), facial cellulitis, periorbital/orbital cellulitis, or failed outpatient antibiotics

Notes

Mark and monitor erythema borders. Elevate affected extremity. Hospitalize if systemic toxicity, rapid spread, or failed outpatient therapy.

M86 — Osteomyelitis

SNOMED: 60168000 | Duration: 28 days | Source: PIDS/IDSA 2021

First-Line

- **Cefazolin** — 150-200 mg/kg/day IV divided q8h (max 6g/day) for 3-4 weeks
- **Clindamycin** — 40 mg/kg/day IV divided q8h (max 2.7g/day); use if MRSA prevalence >=10%
- **Nafcillin/Oxacillin** — 150-200 mg/kg/day IV divided q6h (max 12g/day); low MRSA areas

Second-Line

- **Vancomycin** — 60 mg/kg/day IV divided q6h (max 4g/day); for confirmed MRSA
- **Cephalexin (step-down)** — 100-150 mg/kg/day PO divided q8h (max 4g/day); oral switch

Investigations

- CRP and ESR — Inflammatory markers — monitor response [STAT]
- Blood culture — Identify causative organism [STAT]
- MRI of affected bone — Gold standard for extent of infection [URGENT]
- X-ray of affected bone — Baseline (may be normal early) [URGENT]
- CBC — Leukocytosis supports diagnosis [URGENT]

Counselling

- Your child has a bone infection that needs several weeks of antibiotics to clear completely
- Do not stop antibiotics early even if the child feels better — incomplete treatment causes relapse
- Keep the affected limb rested and elevated initially
- Return if fever returns, swelling worsens, or the child refuses to use the limb
- Follow-up blood tests (CRP) will be done to confirm the infection is clearing

Referral Criteria

Refer to pediatric orthopedic surgeon if abscess needs drainage, if septic arthritis coexists, or if chronic osteomyelitis with sequestrum

Hospitalisation Criteria

Admit for IV antibiotics (initial 3-5 days minimum), surgical drainage if abscess present, sepsis, or inability to tolerate oral antibiotics

Notes

S. aureus is most common pathogen; Kingella kingae predominates in children <3yr. Transition to oral therapy guided by clinical response, typically after 3-5 days IV.

Guideline Changes

4-month shorter regimens now considered for non-severe pediatric cases

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R50.9 — Pyrexia Under Investigation

SNOMED: 386661006 | Duration: 5 days | Source: IAP Fever Guidelines, AIIMS Febrile Child Protocol, NICE Fever in Under 5s

First-Line

- **Paracetamol** — 10-15 mg/kg/dose PO 6-8 hourly SOS for fever >100°F
- **Mefenamic Acid** — As per weight, TDS PO for 3-5 days, add-on for bodyache/high fever

Second-Line

- **Cefpodoxime** — 8-10 mg/kg/day div BD PO for 5-7 days, only if persistent fever >3 days or lab evidence of bacterial infection

Investigations

- Complete blood count — Infection pattern and severity assessment [STAT]
- CRP — Assess degree of inflammation [STAT]
- Procalcitonin — Differentiate bacterial from viral infection [URGENT]
- Urine routine and microscopy — Rule out urinary tract infection [STAT]
- Blood culture — Severe or prolonged fever to identify organism [URGENT]
- Malaria antigen / Dengue NS1 + IgM — If endemic area or clinical suspicion [STAT]

Counselling

- Give paracetamol only when fever is above 100°F, do not give routinely
- Offer frequent fluids - ORS, water, coconut water - to prevent dehydration
- Use tepid sponging with lukewarm water for high fever, do not use cold water
- If fever persists beyond 2-3 days, bring child for review with reports
- Watch for warning signs: rash, bleeding, vomiting, altered sensorium, breathing difficulty - come to hospital immediately

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Persistent fever for more than 3 days
- Vomiting or signs of dehydration
- Rash or any bleeding from skin or gums
- Altered sensorium or confusion

Referral Criteria

Refer if fever persists >5 days without identifiable focus, suspected enteric fever or tropical infection requiring specialist evaluation, or immunocompromised child

Hospitalisation Criteria

Admit if toxic appearance, severe dehydration, altered sensorium, bleeding manifestations, persistent high-grade fever with hemodynamic instability, or age <3 months with fever

Notes

Symptomatic treatment first. Avoid unnecessary antibiotics unless persistent fever >3 days, toxic child, raised CRP/Procalcitonin, or localizing signs. Supportive: adequate fluids (ORS, water, coconut water), light diet, tepid sponging, rest.

Guideline Changes

IAP 2024: Procalcitonin recommended as key biomarker to differentiate bacterial vs viral fever; emphasis on avoiding empirical antibiotics in first 72 hours unless toxic appearance or positive cultures; AIIMS Protocol: Structured fever workup with CBC, CRP, urine R/M, blood culture for fever >3 days; NICE 2024: Age-stratified approach - infants <3 months with fever require urgent evaluation and empiric antibiotics

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U07.1 — COVID-19, virus identified

SNOMED: 840539006 | Duration: 5 days | Source: NIH 2024, IDSA 2024, AAP

First-Line

- **Nirmatrelvir/Ritonavir (Paxlovid)** — 300/100 mg PO bid for 5 days; EUA age >=12yr and >=40 kg, within 5 days of onset **[NEW 2024-25]**

- **Remdesivir** — 5 mg/kg IV day 1 (max 200 mg), then 2.5 mg/kg/day (max 100 mg) for 3 days; >=3 kg **[NEW 2024-25]**

Second-Line

- **Dexamethasone** — 0.15 mg/kg IV/PO daily (max 6 mg) for up to 10 days; hospitalized on O2/ventilator
- **Baricitinib** — 2-4 mg PO daily based on age; hospitalized, EUA >=2yr **[NEW 2024-25]**
- **Tocilizumab** — Weight-based IV single dose; MIS-C or rapid deterioration despite steroids **[NEW 2024-25]**

Investigations

- SARS-CoV-2 RT-PCR or rapid antigen — Confirm diagnosis [STAT]
- CBC, CRP, D-dimer, ferritin — Assess severity and MIS-C screening [URGENT]
- Chest X-ray — If hypoxia or respiratory symptoms [URGENT]
- Troponin, BNP, echocardiogram — If MIS-C suspected [URGENT]

Counselling

- Most children recover from COVID-19 with mild symptoms in 5-7 days
- Give paracetamol for fever and ensure plenty of fluids
- Watch for breathing difficulty, persistent chest pain, or inability to stay awake — seek emergency care
- Monitor for MIS-C symptoms 2-6 weeks after infection: high fever, rash, red eyes, abdominal pain
- Isolate the child for 5 days from symptom onset; masks for all household contacts

Referral Criteria

Refer urgently if MIS-C suspected, oxygen saturation <94%, respiratory distress, or high-risk child (immunocompromised, cardiac disease)

Hospitalisation Criteria

Admit for hypoxia (SpO2 <94%), respiratory distress, MIS-C, severe dehydration, or high-risk comorbidities

Notes

Paxlovid is preferred for outpatient high-risk children >=12yr. Remdesivir 3-day outpatient course is second-line; most beneficial within 7 days of symptom onset.

Guideline Changes

Remdesivir FDA-approved for children >=28 days and >=3 kg; Paxlovid EUA for >=12yr

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Neonatal

P07.3 — Prematurity Follow-up

SNOMED: 15750041000119101 | Duration: 90 days | Source: IAP 2024 Neonatal Guidelines, WHO 2022 Preterm Care Recommendations, AAP 2023

First-Line

- **Multivitamin Drops** — 0.5-1 ml (10 drops) once daily PO, contains Vitamins A, B-complex, C, D. Continue till 1 year of age
- **Ferrous Ascorbate + Folic Acid** — 2-3 mg/kg/day elemental iron PO once daily, start 2-4 weeks after birth, continue till 6-12 months corrected age
- **Cholecalciferol** — 400 IU/day PO once daily, continue at least 1 year
- **Calcium Supplement** — 50-75 mg/kg/day elemental calcium PO once daily, for bone mineralization and prevention of osteopenia of prematurity
- **MCT Oil** — 1-5 ml/day divided with feeds PO, for increased caloric intake and weight gain
- **Dicyclomine** — 5-6 drops SOS PO, only if symptomatic infantile colic. Avoid routine use
- **Preterm Formula / Fortified Feeds** — Pediasure or preterm formula if poor weight gain, 1/7 scoop per extra feed. Prefer breastfeeding first

Investigations

- Hemoglobin — Screen for anemia at 3-4 months
- ROP Screening — Retinopathy of prematurity screening as per schedule
- Hearing screening (OAE/BERA) — All preterm infants, once
- Weight monitoring — Weekly weight check for growth assessment
- Length and Head Circumference — Monthly anthropometry

Counselling

- Continue breastfeeding as first choice, feed every 2-3 hours
- Give all supplements regularly as prescribed - do not skip
- Target weight gain is 15-20 g/kg/day
- Report immediately if poor feeding or poor weight gain
- Report immediately if lethargy or repeated vomiting
- Report immediately if seizures or abnormal movements
- Report immediately if breathing difficulty
- Follow-up every 2 weeks initially, then monthly
- Keep all immunization appointments on schedule (use chronological age)
- Normal development is possible with proper follow-up

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Poor feeding or poor weight gain
- Repeated vomiting
- Seizures or abnormal movements
- Breathing difficulty

Referral Criteria

Refer to neonatologist if poor weight gain despite adequate feeding, developmental delay, suspected ROP (retinopathy of prematurity), or recurrent infections

Hospitalisation Criteria

Admit if failure to thrive, severe anemia, apnea episodes, suspected sepsis, or respiratory distress

Notes

Goals: adequate weight gain (15-20 g/kg/day), prevent nutritional deficiencies (iron, calcium, Vitamin D), monitor neurodevelopment, prevent anemia/rickets/infections. Exclusive breastfeeding till 6 months corrected age, feeds every 2-3 hours. Fortified feeds if poor weight gain. Use CORRECTED age for growth/development, CHRONOLOGICAL age for vaccinations. Catch-up growth expected by 1-2 years.

Guideline Changes

IAP 2025: New consensus on developmentally supportive follow-up with color-coded risk stratification and validated screening tools; WHO 2022: 11 new recommendations for preterm/LBW care including home visits by trained health workers; AAP 2023: Primary care framework for neurodevelopmental monitoring of preterm infants

Neonatal OPD Evaluation

P92.0 — Vomiting in a Neonate

SNOMED: 249497008 | Duration: 3 days | Source: NNF Guidelines 2024, Nelson Textbook of Pediatrics 22e

Investigations

- CBC, CRP, blood culture
- Serum electrolytes, blood gas
- Abdominal X-ray / ultrasound
- Upper GI contrast study
- CSF analysis (if sepsis suspected)
- Metabolic screening

Counselling

- Spitting up is common and improves by 4–6 months
- Avoid overfeeding and home remedies
- Maintain proper feeding position and burping
- Return immediately if green vomiting, fever, poor feeding, or weight loss
- Monitor baby's growth and activity

Warning Signs

- Bilious (green/yellow) vomiting — हरी या पीली उल्टी
- Projectile vomiting — तेज़ ज़ोर से उल्टी आना
- Blood in vomit — उल्टी में खून आना
- Abdominal distension — पेट का फूलना
- Failure to pass meconium within 24 hours — 24 घंटे में पहला मल न आना
- Lethargy or poor feeding — सुस्ती या दूध न पीना
- Dehydration (dry mouth, sunken eyes, no tears) — पानी की कमी
- Fever or hypothermia — बुखार या शरीर का ठंडा पड़ना

Referral Criteria

Refer to pediatric surgeon if bilious vomiting, abdominal distension, or suspected intestinal obstruction. Refer to pediatric GI if persistent non-bilious vomiting not responding to conservative management.

Hospitalisation Criteria

Admit if bilious vomiting, projectile vomiting with dehydration, suspected surgical cause, sepsis, or failure to thrive.

Notes

Bilious vomiting in a neonate is malrotation with volvulus until proven otherwise. Requires urgent upper GI contrast study.

Neonatology

A33 — Neonatal tetanus

SNOMED: 43424001 | Duration: 21 days | Source: WHO, MSF Guidelines

First-Line

- **Human TIG** — 500 IU IM as single dose (2 separate sites); neutralizes unbound toxin
- **Metronidazole** — 15 mg/kg IV load, then 7.5 mg/kg IV q12h x7-10 days; preferred over penicillin
- **Diazepam** — 0.1-0.3 mg/kg IV q1-4h PRN for spasms; or infusion 5-15 mg/kg/day

Second-Line

- **Phenobarbital** — 10-15 mg/kg/day PO/IV divided q6h as adjunct for spasm control
- **Magnesium sulfate** — 50 mg/kg IV loading then 30-50 mg/kg/day infusion for autonomic dysfunction

Investigations

- Clinical diagnosis — Tetanus is diagnosed clinically; no specific lab test [STAT]
- Wound/umbilical culture — Identify Clostridium tetani
- Blood gas — Assess respiratory status [URGENT]
- Electrolytes and glucose — Monitor metabolic status during illness [URGENT]

Counselling

- Baby has tetanus infection from the umbilical cord; this is a serious illness
- Treatment involves antitoxin, antibiotics, and medicines to control muscle spasms
- Baby will need to be in a quiet, dark room as noise and light trigger spasms
- Baby may need help with breathing during the illness
- This is preventable with maternal tetanus vaccination; ensure all future pregnancies are covered

Referral Criteria

Refer to NICU/PICU immediately; may need mechanical ventilation for severe spasms

Hospitalisation Criteria

All cases require ICU admission; life-threatening condition

Notes

Neonatal tetanus is preventable with maternal TT vaccination. Monitor respiratory function closely; intubation may be needed for severe spasms.

B37.0 — Oral Thrush in Neonate

SNOMED: 79740000

Investigations

- Clinical diagnosis
- KOH preparation / fungal culture — Severe, recurrent or treatment-resistant cases

Counselling

- Oral thrush is common and easily treatable. — मुंह का छाला आम है और आसानी से ठीक हो जाता है।
- It is not due to poor care. — यह खराब देखभाल की वजह से नहीं होता।
- Maintain proper hygiene of feeding equipment. — दूध पिलाने के बर्तन साफ़ रखें।
- Complete full course of medication even if symptoms improve early. — लक्षण कम होने पर भी दवाई पूरी करें।
- Consult doctor if baby develops fever, feeding refusal or spreading lesions. — बुखार, दूध न पीना, या छाले बढ़ना दिखे तो डॉक्टर को दिखाएं।

Warning Signs

- Feeding refusal or excessive crying during feeds — दूध पीने से मना करना या दूध पीते समय बहुत रोना
- Spreading lesions beyond mouth — छाले मुंह से बाहर फैलना
- Fever — बुखार
- Poor weight gain — वज़न न बढ़ना
- Persistent or recurring infection despite treatment — इलाज के बावजूद बार-बार होना
- Rash in diaper area — डायपर वाली जगह पर दाने

H35.10 — Retinopathy of Prematurity (ROP) Screening

SNOMED: 415297005

Counselling

- Screening is essential even if baby appears normal — बच्चा सामान्य दिखे तब भी आंखों की जांच ज़रूरी है।
- Multiple follow-up visits are required — कई बार जांच करानी पड़ेगी।
- Procedure is safe — जांच सुरक्षित है।
- Early treatment can prevent blindness — जल्दी इलाज से अंधापन रोका जा सकता है।
- Do not miss any scheduled eye examination — कोई भी आंख की जांच की तारीख़ न छोड़ें।
- Premature babies need long-term eye follow-up — समय से पहले जन्मे बच्चों को लंबे समय तक आंखों की जांच चाहिए।

Warning Signs

- White reflex in pupil — आंख की पुतली में सफ़ेद चमक
- Abnormal eye movements — आंखों की असामान्य हरकत
- Baby not following objects with eyes — बच्चा आंखों से चीज़ों को न देखना
- Squinting — तिरछा देखना
- Missed screening appointment — जांच की तारीख़ छूटना

K40.90 — Hernia and Hydrocele in Neonate

SNOMED: 396232000

Investigations

- Clinical diagnosis
- Ultrasound — Unclear diagnosis or atypical swelling
- Color Doppler — Assess testicular perfusion in suspected incarceration

Counselling

- Hernias do not resolve spontaneously and require surgery — हर्निया अपने आप ठीक नहीं होता, ऑपरेशन ज़रूरी है
- Hydroceles often resolve without treatment — हाइड्रोसील अक्सर बिना इलाज के ठीक हो जाता है
- Seek urgent care if swelling becomes painful or irreducible — अगर सूजन दर्दनाक हो या अंदर न जाए तो तुरंत डॉक्टर को दिखाएं
- Surgery is safe with excellent outcomes — ऑपरेशन सुरक्षित है और परिणाम बहुत अच्छे होते हैं
- Monitor swelling size and behavior — सूजन के आकार और बदलाव पर नज़र रखें

Warning Signs

- Irreducible swelling (does not go back in) — सूजन जो अंदर न जाए
- Sudden increase in size of swelling — सूजन का अचानक बढ़ जाना
- Tender, tense or red swelling — दर्दनाक, तनी हुई या लाल सूजन
- Vomiting or abdominal distension — उल्टी या पेट फूलना

- Refusal to feed or excessive crying — दूध न पीना या बहुत रोना
- Fever — बुखार

P05.1 — Small for gestational age

SNOMED: 722525006 | Duration: 5 days | Source: AAP, WHO 2022

First-Line

- **Glucose monitoring** — BG at 2h, then q3-6h for 24h; target >45 mg/dL after 4h of life
- **Early feeding** — Breastfeed within 1h; supplement with expressed milk or formula if needed
- **Thermoregulation** — Skin-to-skin or servo-controlled warmer; SGA infants are hypothermia-prone

Second-Line

- **IV dextrose** — D10W 2 mL/kg bolus then GIR 4-6 mg/kg/min if BG persistently <45
- **Polycythemia screening** — Venous Hct at 4-6h; PET if symptomatic with Hct >65%

Investigations

- Blood glucose monitoring — Screen at 2h, then q3-6h x24h [STAT]
- Venous hematocrit — Polycythemia screening at 4-6h [URGENT]
- Serum calcium — Hypocalcemia risk in SGA infants
- TORCH screening — If intrauterine infection suspected as cause of SGA

Counselling

- Baby is smaller than expected; frequent feeding is very important to prevent low sugar
- Keep baby warm with skin-to-skin contact and proper wrapping
- Blood sugar and temperature will be checked frequently in the first 1-2 days
- Baby may take some time to catch up in growth; regular follow-up with weight checks is important
- Breastfeed on demand, at least every 2-3 hours

Referral Criteria

Refer to neonatologist if refractory hypoglycemia, polycythemia requiring PET, or poor growth

Hospitalisation Criteria

Admit if symptomatic hypoglycemia, weight loss >10%, hypothermia, or polycythemia with Hct >65% symptomatic

Notes

SGA neonates at risk for hypoglycemia, polycythemia, and hypothermia. Monitor feeding adequacy and weight gain closely.

P05.9 — Intrauterine growth restriction

SNOMED: 22033007 | Duration: 5 days | Source: RCOG 2024, AAP

First-Line

- **Glucose monitoring** — Check blood glucose within 2h of birth, then q3-6h for 24h; feed early
- **Early feeding** — Initiate breastfeeding within 1h of birth; supplement if BG <45 mg/dL
- **Thermoregulation** — Skin-to-skin contact; servo-controlled warmer; target axillary 36.5-37.4C

Second-Line

- **IV dextrose** — D10W 2 mL/kg bolus then 4-6 mg/kg/min GIR if BG <25 mg/dL or symptomatic
- **Polycythemia screening** — Check venous Hct at 4-6h; PET if Hct >65% with symptoms

Investigations

- Blood glucose monitoring — High risk for hypoglycemia; check at 2h then q3-6h x 24h [STAT]

- Venous hematocrit — Screen for polycythemia at 4-6 hours of life [URGENT]
- Serum calcium — Risk of hypocalcemia in SGA/IUGR infants
- Thyroid function test — Screen if intrauterine infection or syndromic features suspected

Counselling

- Baby is smaller than expected for their age in the womb; close monitoring is needed
- Feed baby frequently - every 2-3 hours - to prevent low blood sugar
- Keep baby warm with skin-to-skin contact and warm clothing
- Blood sugar will be checked frequently in the first 24-48 hours
- Regular weight checks and follow-up are important to ensure baby is catching up in growth

Referral Criteria

Refer to neonatologist if persistent hypoglycemia, poor growth, or feeding difficulties

Hospitalisation Criteria

Admit if symptomatic hypoglycemia, weight loss >10%, hypothermia, or polycythemia requiring intervention

Notes

High risk for hypoglycemia, polycythemia, hypothermia, and feeding intolerance. Monitor closely for first 24-48 hours.

P07 — Prematurity

SNOMED: 44247006 | Duration: 30 days | Source: European RDS Guidelines 2022, AAP

First-Line

- **CPAP** — Nasal CPAP 5-8 cmH2O from delivery room; avoid intubation if possible
- **Caffeine citrate** — Loading 20 mg/kg IV/PO, maintenance 5-10 mg/kg/day for apnea of prematurity
- **Surfactant** — Poractant alfa 200 mg/kg if FiO2 >0.3 on CPAP; early rescue preferred

Second-Line

- **TPN** — Early parenteral nutrition; protein 2-3 g/kg/day, lipids 1-3 g/kg/day from Day 0-1
- **Iron supplementation** — 2-4 mg/kg/day from 2-6 weeks of age; continue through first year
- **Vitamin D** — 400 IU/day once on full enteral feeds; for bone health

Investigations

- Cranial ultrasound — IVH screening at day 3-7 and at 36 weeks PMA [URGENT]
- ROP screening — From 31 weeks PMA or 4 weeks postnatal age
- Hearing screen (ABR) — Before discharge; risk factor for sensorineural hearing loss
- Blood gas — Respiratory monitoring [STAT]
- CBC, CRP — Infection screening [URGENT]

Counselling

- Baby is born early and needs extra support for breathing, feeding, and warmth
- Mother's milk is the best medicine for premature babies; express milk every 2-3 hours
- Kangaroo mother care should be done daily; it helps baby grow and bond
- Baby will need vitamin D and iron supplements; continue as advised after discharge
- Use corrected age (not birth age) for developmental milestones; follow up regularly with pediatrician

Referral Criteria

Refer to NICU at birth; ROP screening, audiology, and developmental follow-up after discharge

Hospitalisation Criteria

All preterm infants require monitoring; admit to NICU if <34 weeks GA or with respiratory distress

Notes

Delayed cord clamping >60 seconds recommended. Minimize handling, maintain thermoregulation, and promote human milk feeding.

Guideline Changes

LISA increasingly adopted; delayed cord clamping (>60s) now standard

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P07.0 — Very low birth weight

SNOMED: 276611006 | Duration: 45 days | Source: European RDS Guidelines 2022, AAP

First-Line

- **TPN** — Start Day 0: amino acids 2-3 g/kg/day, lipids Day 1 1-3 g/kg/day, D10W
- **Trophic feeds** — Mother's own milk 10-20 mL/kg/day from Day 1; advance 20-30 mL/kg/day
- **Caffeine citrate** — Loading 20 mg/kg, maintenance 5-10 mg/kg/day for apnea prevention

Second-Line

- **Human milk fortifier** — Add at 100 mL/kg/day enteral volume; human milk-based preferred for VLBW
- **Erythropoietin** — 250-400 IU/kg SC 3x/week with iron; limited benefit, ROP risk debated
- **Iron supplementation** — 2-4 mg/kg/day elemental iron from 2 weeks of age

Investigations

- Cranial ultrasound (serial) — IVH screening at day 3, 7, 14, and 36 weeks PMA [URGENT]
- ROP screening — From 31 weeks PMA
- CBC (serial) — Monitor hemoglobin; guide transfusion decisions
- Hearing screen (ABR) — Before discharge
- Metabolic screening — Newborn metabolic screen

Counselling

- Baby is very small and will need NICU care for several weeks
- Mother's breastmilk is the most important medicine; express milk every 2-3 hours around the clock
- Blood transfusions may be needed as small babies become anemic; this is expected
- Baby will need supplements (iron, vitamin D, calcium) after discharge
- Developmental follow-up and eye screening are essential; attend all appointments

Referral Criteria

Born at tertiary NICU; refer for ROP, audiology, and neurodevelopmental follow-up

Hospitalisation Criteria

All VLBW infants require NICU admission until stable feeding, thermoregulation, and breathing (typically 34-36 weeks PMA)

Notes

Prioritize human milk to reduce NEC. Minimize blood draws, use delayed cord clamping, and target restrictive transfusion thresholds.

Guideline Changes

Human milk-based fortifier shown to reduce NEC vs bovine-based in VLBW

P07.1 — Low birth weight

SNOMED: 276610007 | Duration: 14 days | Source: WHO 2022, AAP

First-Line

- **Kangaroo mother care** — Continuous skin-to-skin contact; improves thermoregulation and BF

- **Early enteral feeding** — Mother's own milk preferred; start trophic feeds 10-20 mL/kg/day
- **Thermoregulation** — Maintain 36.5-37.4C; use warmer/incubator for infants <1500g

Second-Line

- **TPN** — Start if enteral feeds not tolerated; amino acids 2-3 g/kg/day from Day 0
- **Iron supplementation** — 2-4 mg/kg/day elemental iron from 2-6 weeks of age

Investigations

- Blood glucose — Hypoglycemia screening in first 24-48 hours [STAT]
- CBC — Baseline hemoglobin; screen for anemia
- Thyroid screening — Newborn thyroid screen if not done
- Hearing screen (OAE) — Before discharge; risk factor for hearing loss

Counselling

- Baby is small and needs extra care with warmth and feeding
- Kangaroo mother care (skin-to-skin) is the best way to keep baby warm and help them grow
- Breastfeed frequently; mother's milk is the best nutrition for low birth weight babies
- Baby will need iron and vitamin D supplements from 2 weeks of age
- Regular weight checks are important; follow up every week until weight gain is satisfactory

Referral Criteria

Refer to NICU if birth weight <1500g, unable to maintain temperature, or feeding intolerance

Hospitalisation Criteria

Admit if BW <1800g, unable to maintain temperature >36.5C, or feeding difficulties

Notes

WHO recommends immediate KMC for all LBW infants regardless of stability. Monitor glucose, temperature, and weight gain closely.

Guideline Changes

WHO 2022 updated guidelines emphasizing kangaroo mother care immediately after birth

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P07.2 — Extreme prematurity

SNOMED: 15887011000119107 | Duration: 60 days | Source: European RDS Guidelines 2022, NRP 2025

First-Line

- **Surfactant** — Poractant alfa 200 mg/kg via LISA or ETT within 2h of birth for RDS
- **Caffeine citrate** — Loading 20 mg/kg IV/PO, then 5-10 mg/kg/day maintenance for apnea prevention
- **CPAP** — Start CPAP 6-8 cmH2O from birth; avoid intubation if possible
- **Antenatal steroids** — Betamethasone 12 mg IM x2 doses 24h apart to mother before delivery

Second-Line

- **TPN** — Start amino acids Day 0 at 2-3 g/kg/day; lipids Day 1 at 1-3 g/kg/day
- **Indomethacin prophylaxis** — 0.1 mg/kg IV q24h x3 for IVH prevention in <28 wk; debated

Investigations

- Cranial ultrasound — Screen for IVH at day 3-7, then weekly [URGENT]
- ROP screening — Retinopathy of prematurity screening from 31 weeks PMA
- Blood gas — Respiratory status monitoring [STAT]
- CBC, CRP — Infection screening [URGENT]
- Hearing screen (OAE/ABR) — Before discharge; high risk for hearing loss

Counselling

- Baby is born very early and needs special care in NICU for many weeks
- Mother's breastmilk is extremely important; express milk every 2-3 hours
- Kangaroo care (skin-to-skin) is beneficial and parents should do it daily
- Baby may have breathing, feeding, and growth challenges that will improve with time
- Follow-up with developmental specialist is essential after discharge; use corrected age for milestones

Referral Criteria

Deliver at tertiary NICU with capacity for extreme preterm care; refer for ROP screening, audiology, and developmental follow-up

Hospitalisation Criteria

All extremely preterm infants require NICU admission until at least 34-36 weeks corrected gestational age

Notes

Active resuscitation now considered from 22 weeks at proactive centers. Thermoregulation with polyethylene wrap critical at delivery.

Guideline Changes

LISA (less invasive surfactant) emerging as preferred delivery method for surfactant

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P08 — Post-term newborn

Duration: 3 days | Source: AAP, ACOG

First-Line

- **Glucose monitoring** — Screen BG at 2h of birth; q3-6h for 24h if at risk for hypoglycemia
- **Early feeding** — Initiate breastfeeding within 1h; monitor for adequate intake
- **Clinical monitoring** — Observe for meconium aspiration, polycythemia, hypoglycemia signs

Second-Line

- **IV dextrose** — D10W infusion if symptomatic hypoglycemia; GIR 4-6 mg/kg/min
- **Polycythemia screening** — Venous Hct at 4-6h of life; treat if >65% with symptoms

Investigations

- Blood glucose — Screen for hypoglycemia at 2h then q3-6h x24h [STAT]
- Venous hematocrit — Screen for polycythemia at 4-6h of life [URGENT]
- Serum bilirubin — Monitor for jaundice if clinically indicated
- Chest X-ray — If respiratory distress; rule out meconium aspiration [URGENT]

Counselling

- Baby was born after the due date; this requires monitoring for a few days
- Feed baby early and frequently to prevent low blood sugar
- Watch for difficulty breathing, yellow skin color, or poor feeding
- Dry, peeling skin is normal for post-term babies and will improve on its own
- Follow up within 3-5 days of discharge for weight and jaundice check

Referral Criteria

Refer to neonatologist if meconium aspiration, persistent hypoglycemia, or polycythemia

Hospitalisation Criteria

Admit if respiratory distress, meconium aspiration, symptomatic polycythemia, or persistent hypoglycemia

Notes

Post-term neonates at higher risk for meconium aspiration, hypoglycemia, and polycythemia. Close monitoring first 24-48h.

P08.1 — Large for gestational age

SNOMED: 15635491000119102 | Duration: 3 days | Source: AAP 2022

First-Line

- **Glucose monitoring** — Screen BG at 2h of age; if <25 mg/dL refeed and recheck in 1h
- **Early feeding** — Breastfeed or formula within 1h of birth; q2-3h feeds for first 24h

Second-Line

- **IV dextrose** — D10W 2 mL/kg bolus then GIR 4-6 mg/kg/min for persistent hypoglycemia
- **Dextrose gel 40%** — 0.5 mL/kg buccal for mild asymptomatic hypoglycemia; massage into mucosa **[NEW 2024-25]**

Investigations

- Blood glucose monitoring — High risk for hypoglycemia; screen at 2h then q3-6h x12h [STAT]
- Serum calcium and magnesium — Risk of hypocalcemia in IDM
- CBC with differential — Polycythemia screening
- Echocardiography — If IDM; screen for hypertrophic cardiomyopathy

Counselling

- Baby is larger than expected; blood sugar will be checked frequently in first 12-24 hours
- Feed baby within the first hour of birth and then every 2-3 hours
- If mother has diabetes, baby needs closer monitoring
- Watch for excessive sleepiness, tremors, or poor feeding and report to doctor
- Baby may have bruising or swelling from birth which will resolve on its own

Referral Criteria

Refer to neonatologist if persistent hypoglycemia requiring IV dextrose or birth injuries suspected

Hospitalisation Criteria

Admit if symptomatic hypoglycemia, birth injuries (clavicle fracture, brachial plexus injury), or BG persistently <45 mg/dL

Notes

LGA infants of diabetic mothers have highest hypoglycemia risk. Screen glucose for 12h; monitor for birth injuries from macrosomia.

P12.0 — Cephalohematoma due to birth injury

SNOMED: 206200000 | Duration: 7 days | Source: AAP

First-Line

- **Observation** — Monitor size, watch for jaundice; resolves spontaneously in 2 weeks to 3 months
- **Bilirubin monitoring** — Serial TSB if jaundice develops; phototherapy per AAP thresholds if indicated

Second-Line

- **Phototherapy** — If hyperbilirubinemia develops from hematoma reabsorption; per hour-specific nomogram
- **CBC monitoring** — Monitor hemoglobin if large cephalohematoma; transfuse only if symptomatic anemia

Investigations

- Serum bilirubin (serial) — Monitor for hyperbilirubinemia from hematoma reabsorption [URGENT]
- CBC — Large hematomas may cause drop in hemoglobin
- Skull X-ray — Only if underlying skull fracture suspected

Counselling

- Baby has a blood collection under the scalp bone from birth; this is common and not dangerous
- Do NOT squeeze or try to drain the swelling; it will absorb on its own in weeks to months

- The swelling may feel hard at the edges as it heals; this is normal
- Watch for increasing yellowness of skin (jaundice) and bring baby for bilirubin check if concerned
- No special care is needed; handle baby normally

Referral Criteria

Refer if rapidly enlarging, associated skull fracture, or coagulopathy suspected

Hospitalisation Criteria

Admit if associated significant jaundice requiring phototherapy or symptomatic anemia from large hematoma

Notes

Do NOT aspirate cephalohematoma — risk of infection. Resolves spontaneously. Calcified rim may persist for months but is benign.

P15.9 — Birth Injuries in a Neonate

SNOMED: 56110009

Counselling

- Most birth injuries are mild and heal without long-term problems — ज़्यादातर जन्म-चोटें हल्की होती हैं और बिना किसी लंबी समस्या के ठीक हो जाती हैं
- Minor scalp swelling or bruising often settles on its own — सिर की हल्की सूजन या नील अपने आप ठीक हो जाती है
- Handle the baby gently — बच्चे को नरमी से पकड़ें
- Do not pull or lift forcefully under the arms — बच्चे को बाजुओं के नीचे से ज़ोर से न उठाएं
- Follow-up is important if there is decreased limb movement or persistent swelling — अगर हाथ-पैर की हलचल कम हो या सूजन बनी रहे तो जांच ज़रूरी है
- Seek urgent care if baby becomes pale, lethargic, poorly feeding, or swelling increases — अगर बच्चा पीला, सुस्त, कम दूध पीने वाला हो जाए या सूजन बढ़ जाए तो तुरंत डॉक्टर को दिखाएं

Warning Signs

- Expanding or tense swelling on scalp — सिर पर बढ़ती या तनी हुई सूजन
- Baby not moving one arm or leg — बच्चे का एक हाथ या पैर न हिलाना
- Facial asymmetry during crying — रोते समय चेहरे का एक तरफ़ लटकना
- Pale or blue color — बच्चे का पीला या नीला पड़ जाना
- Seizures or excessive irritability — दौरे या बहुत ज़्यादा चिड़चिड़ापन
- Poor feeding or vomiting — ठीक से दूध न पीना या उल्टी
- Increasing jaundice — बढ़ता हुआ पीलिया

Referral Criteria

Suspected subgaleal hemorrhage,Shock or poor perfusion,Persistent seizures,Organ injury suspected,Depressed skull fracture,Persistent brachial plexus weakness,Major fracture requiring splinting/orthopedic care

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P21.9 — Birth asphyxia, unspecified

SNOMED: 28314004 | Duration: 7 days | Source: NRP 2025, AHA/AAP

First-Line

- **Positive pressure ventilation** — 21-30% FiO2 for term, increase to 100% if no response; initiate within 60 sec
- **Chest compressions** — 3:1 ratio with PPV; 90 compressions + 30 breaths/min if HR <60 despite PPV
- **Epinephrine** — 0.01-0.03 mg/kg IV (1:10,000) via UVC if HR <60 after 60s of CC + PPV

Second-Line

- **Endotracheal epinephrine** — 0.05-0.1 mg/kg via ETT if IV access delayed; less reliable absorption

- **Volume expansion** — NS or O-neg blood 10 mL/kg IV over 5-10 min if hypovolemia suspected

Investigations

- Blood gas (cord or early arterial) — pH <7.0 and base deficit >12 indicate significant asphyxia [STAT]
- Blood glucose — Hypoglycemia common after asphyxia [STAT]
- CBC and coagulation profile — DIC screening [URGENT]
- Brain MRI — At 4-7 days for prognosis
- aEEG monitoring — Continuous seizure monitoring; guides cooling decision [URGENT]

Counselling

- Baby did not breathe well at birth and needed help; this can affect the brain
- Baby will be closely monitored in the NICU for at least a few days
- If brain injury is moderate-severe, cooling therapy may be offered within 6 hours of birth
- Long-term follow-up with developmental specialist is very important
- Most babies with mild birth asphyxia recover completely; outcome depends on severity

Referral Criteria

Refer to tertiary NICU with cooling capability if moderate-severe encephalopathy; neurology follow-up

Hospitalisation Criteria

All cases require NICU admission; cooling facility needed for moderate-severe HIE

Notes

Effective ventilation is the single most important step. Evaluate for therapeutic hypothermia if moderate-severe encephalopathy develops.

Guideline Changes

NRP 2025 updates: establish alternate airway before compressions; epinephrine after 60s of CC

P22.0 — Respiratory distress syndrome of newborn

SNOMED: 46775006 | Duration: 10 days | Source: European RDS Guidelines 2022, RDS-NExT 2023

First-Line

- **Surfactant (LISA)** — Poractant alfa 200 mg/kg via thin catheter on CPAP; within 2h of birth preferred **[NEW 2024-25]**
- **CPAP** — Start 6-8 cmH2O immediately after birth; primary respiratory support
- **Antenatal steroids** — Betamethasone 12 mg IM x2 to mother 24h apart before preterm delivery

Second-Line

- **Surfactant (INSURE)** — Intubate-Surfactant-Extubate technique if LISA not available
- **Mechanical ventilation** — Volume-targeted; TV 4-6 mL/kg if intubation required; minimize duration
- **Caffeine citrate** — 20 mg/kg loading then 5-10 mg/kg/day; facilitates extubation and prevents apnea

Investigations

- Chest X-ray — Ground glass appearance with air bronchograms confirms RDS [STAT]
- Blood gas (arterial/capillary) — Assess oxygenation and guide FiO2/surfactant decision [STAT]
- CBC, CRP — Rule out concurrent pneumonia/sepsis [URGENT]
- Blood glucose — Prematurity-related hypoglycemia screening [STAT]

Counselling

- Baby's lungs lack a substance called surfactant needed for breathing; this is common in premature babies
- Surfactant medicine will be given directly into the lungs through a thin tube
- CPAP or ventilator support will help baby breathe until lungs mature
- Antenatal steroid injection (if given before birth) has already helped lung development

- Most babies improve significantly after surfactant treatment and gradually need less breathing support

Referral Criteria

Deliver at center with NICU and surfactant availability; refer for antenatal steroids before preterm delivery

Hospitalisation Criteria

All RDS cases require NICU admission until weaned off respiratory support

Notes

LISA reduces need for mechanical ventilation and BPD compared to INSURE. Poractant alfa 200 mg/kg superior to lower doses and beractant.

Guideline Changes

LISA (less invasive surfactant administration) recommended over INSURE where expertise available

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P22.1 — Transient tachypnea of newborn

SNOMED: 7550008 | Duration: 3 days | Source: AAP

First-Line

- **Supplemental oxygen** — FiO2 to maintain SpO2 92-97%; usually <40% FiO2 sufficient
- **CPAP** — 5-7 cmH2O nasal CPAP if moderate respiratory distress; aids fluid clearance
- **IV fluids** — D10W + electrolytes at maintenance; NPO until RR <60-70 to reduce aspiration risk

Second-Line

- **High-flow nasal cannula** — 2-8 L/min heated humidified; alternative to CPAP for mild TTN
- **Gavage feeding** — OG/NG feeds if RR 60-80 and unable to safely PO feed

Investigations

- Chest X-ray — Perihilar streaking, fluid in fissures; hyperinflation [URGENT]
- Blood gas — Mild respiratory alkalosis typical [URGENT]
- CBC, CRP — Rule out pneumonia if not improving
- Blood glucose — Hypoglycemia screening [STAT]

Counselling

- Baby is breathing fast because of fluid left in the lungs from birth; this is temporary
- This condition resolves on its own within 1-3 days; no specific medicine is needed
- Baby may need oxygen support and IV fluids until breathing normalizes
- Once breathing rate comes down, breastfeeding can start
- This is common after cesarean delivery and is not dangerous

Referral Criteria

Refer to NICU if requiring >40% FiO2 or not improving within 72 hours

Hospitalisation Criteria

Admit if tachypnea (RR >60) with increased work of breathing or requiring supplemental oxygen

Notes

TTN is self-limiting; resolves within 24-72h. No proven pharmacotherapy (diuretics, salbutamol, steroids not recommended). Rule out other causes.

P22.9 — Neonatal respiratory distress, unspecified

SNOMED: 1179634008 | Duration: 7 days | Source: European RDS Guidelines 2022

First-Line

- **CPAP** — Nasal CPAP 5-8 cmH2O from birth; first-line respiratory support for preterm
- **Supplemental oxygen** — Titrate to SpO2 90-95% (preterm) or 92-97% (term); avoid hyperoxia
- **Surfactant** — Poractant alfa 200 mg/kg if FiO2 >0.3 on CPAP for RDS; via LISA if available

Second-Line

- **Mechanical ventilation** — Intubate if FiO2 >0.5-0.6 on CPAP, recurrent apnea, or respiratory failure
- **High-flow nasal cannula** — 1-8 L/min for mild-moderate respiratory distress; heated humidified

Investigations

- Chest X-ray — Identify cause: RDS, TTN, pneumonia, pneumothorax, CDH [STAT]
- Blood gas — Assess oxygenation and ventilation [STAT]
- CBC with differential — Rule out sepsis/pneumonia [URGENT]
- Blood glucose — Hypoglycemia screening [STAT]
- Pre and post-ductal SpO2 — Screen for cyanotic heart disease or PPHN [STAT]

Counselling

- Baby is having difficulty breathing which is common in newborns, especially premature babies
- Oxygen or CPAP (gentle air pressure through the nose) will help baby breathe
- If needed, surfactant medicine can be given directly into the lungs
- Most cases of breathing difficulty improve within 2-3 days
- Baby will remain in NICU until breathing comfortably on room air

Referral Criteria

Refer to NICU; tertiary center if requiring advanced respiratory support

Hospitalisation Criteria

All neonates with respiratory distress require admission for monitoring and support

Notes

Identify underlying cause: RDS, TTN, pneumonia, pneumothorax, or CDH. Chest X-ray and blood gas guide management. Most TTN resolves by 72h.

P23 — Congenital pneumonia

SNOMED: 78895009 | Duration: 10 days | Source: WHO 2024, AAP

First-Line

- **Ampicillin** — 50-100 mg/kg IV q8-12h; covers GBS, Listeria, and common pathogens
- **Gentamicin** — 5 mg/kg IV q24-48h based on GA; synergistic with ampicillin for gram-negatives
- **Respiratory support** — Supplemental O2, CPAP, or mechanical ventilation based on severity

Second-Line

- **Cefotaxime** — 50 mg/kg IV q8-12h; alternative to gentamicin if renal concerns or CSF coverage needed
- **Surfactant** — Consider if concurrent RDS; poractant alfa 200 mg/kg via ETT

Investigations

- Blood culture — Identify organism before starting antibiotics [STAT]
- Chest X-ray — Infiltrates, consolidation confirming pneumonia [STAT]
- CBC with differential — WBC count, neutrophil ratio for infection [STAT]
- CRP — Inflammatory marker; monitor treatment response [URGENT]
- Blood gas — Assess respiratory status and oxygen needs [URGENT]

Counselling

- Baby has a lung infection that started before or during birth
- IV antibiotics will be given for 7-10 days; full course is important

- Baby may need oxygen or breathing support during treatment
- Blood cultures will guide the specific antibiotic choice
- Baby will recover fully in most cases; follow up if breathing problems persist after discharge

Referral Criteria

Refer to NICU if requiring mechanical ventilation or not responding to first-line antibiotics

Hospitalisation Criteria

All cases require admission for IV antibiotics; NICU if respiratory support needed

Notes

Treat for 7-10 days; shorter courses (<=5 days) may suffice for culture-negative cases. Obtain blood culture before antibiotics.

P24.0 — Meconium aspiration syndrome

SNOMED: 86649001 | Duration: 7 days | Source: NRP 2025, AAP

First-Line

- **Supportive ventilation** — Supplemental O2, CPAP, or MV; PEEP 6-8 cmH2O; rate 40-60; Ti 0.5-0.6s
- **Surfactant** — 100-200 mg/kg via ETT for severe MAS with high FiO2 requirement; may reduce ECMO need
- **Inhaled nitric oxide** — 20 ppm for PPHN complicating MAS (OI >15-25); wean as oxygenation improves

Second-Line

- **HFOV** — High-frequency oscillatory ventilation for refractory hypoxemia or air trapping
- **Antibiotics** — Ampicillin + gentamicin empirically; MAS mimics pneumonia and culture often sent
- **ECMO** — For severe refractory respiratory failure; last resort after maximal medical therapy

Investigations

- Chest X-ray — Patchy infiltrates, hyperinflation, air trapping [STAT]
- Blood gas (arterial) — Assess oxygenation and calculate oxygenation index [STAT]
- CBC with differential — Rule out concurrent infection [URGENT]
- Blood culture — MAS mimics pneumonia; rule out infection [URGENT]

Counselling

- Baby breathed in meconium (first stool) during birth, causing lung irritation
- Baby needs oxygen and breathing support; most cases resolve in 3-5 days
- In severe cases, special gas (nitric oxide) may be used to help the lungs
- Baby will be in NICU until breathing normalizes
- Full recovery is expected in most cases; follow up if coughing or fast breathing continues

Referral Criteria

Refer to tertiary NICU if requiring iNO, HFOV, or consideration for ECMO

Hospitalisation Criteria

All MAS cases require admission; NICU if supplemental O2 needed

Notes

Supportive care is the mainstay; no specific therapy. Do not routinely suction non-vigorous infants. Most resolve in 3-5 days.

Guideline Changes

Routine endotracheal suctioning of non-vigorous meconium-stained infants no longer recommended (since 2015 NRP)

P25 — Neonatal pneumothorax

Duration: 7 days | Source: Neonatal Consensus Guidelines

First-Line

- **Needle aspiration** — 23-25G needle or 18-24G angiocath; 2nd-3rd ICS MCL; for tension pneumothorax emergently
- **Chest tube (thoracostomy)** — Pigtail catheter (6-8 Fr) or trocar; connect to underwater seal -10 to -20 cmH2O suction

Second-Line

- **Observation** — For small asymptomatic pneumothorax; high-flow or 100% O2 may hasten reabsorption
- **High-frequency ventilation** — Transition to HFOV if pneumothorax recurs on conventional ventilation

Investigations

- Chest X-ray — Confirm pneumothorax; assess size and tension [STAT]
- Transillumination — Rapid bedside diagnosis in NICU [STAT]
- Blood gas — Assess respiratory compromise [STAT]
- Pre and post-ductal SpO2 — Assess for associated PPHN [URGENT]

Counselling

- Air has leaked from baby's lung into the chest space, causing breathing difficulty
- Needle or tube drainage may be needed to remove the trapped air
- Baby will be on continuous monitoring and may need oxygen support
- Most cases resolve within a few days after drainage
- In rare cases, the air leak may recur; doctor will monitor closely

Referral Criteria

Refer to NICU; pediatric surgeon if recurrent or complicated pneumothorax

Hospitalisation Criteria

All symptomatic pneumothorax require NICU admission; tension pneumothorax is a life-threatening emergency

Notes

Tension pneumothorax is a life-threatening emergency requiring immediate needle decompression. Transillumination is rapid bedside diagnostic tool.

P26 — Pulmonary hemorrhage originating in perinatal period

SNOMED: 206306009 | Duration: 10 days | Source: Expert Consensus 2022

First-Line

- **Endotracheal epinephrine** — 0.5 mL of 1:10,000 via ETT for vasoconstriction; may repeat x1
- **Increase PEEP/MAP** — Raise PEEP to 8-10 cmH2O or switch to HFOV to tamponade bleeding
- **Packed RBC transfusion** — 10-15 mL/kg to replace blood loss; maintain Hb >12 during acute phase

Second-Line

- **Surfactant** — 100-200 mg/kg after bleeding stops; restores surfactant inactivated by blood/protein
- **FFP/Cryoprecipitate** — 10-15 mL/kg FFP for coagulopathy; vitamin K 1 mg IV if not already given
- **Recombinant Factor VIIa** — 50-100 mcg/kg IV for refractory hemorrhage; case reports only

Investigations

- CBC (serial) — Monitor hemoglobin; guide transfusion [STAT]
- Coagulation profile (PT, aPTT, fibrinogen) — DIC assessment and correction [STAT]
- Blood gas — Assess severity of respiratory compromise [STAT]
- Chest X-ray — White-out pattern indicating bilateral hemorrhage [STAT]

- Blood type and crossmatch — For packed RBC transfusion [STAT]

Counselling

- Baby has bleeding from the lungs which is a serious emergency
- Baby will need a breathing tube and ventilator to support breathing
- Blood transfusion may be needed to replace lost blood
- This is more common in very premature and sick babies
- Treatment is aggressive but many babies recover; doctor will keep you updated

Referral Criteria

Immediate NICU with ventilator support; may need tertiary transfer

Hospitalisation Criteria

Life-threatening emergency requiring immediate NICU admission and mechanical ventilation

Notes

Life-threatening emergency. Increase MAP to tamponade bleeding, correct coagulopathy, and transfuse aggressively. Surfactant after bleeding stops.

P28.4 — Neonatal apnea

SNOMED: 70073005 | Duration: 14 days | Source: AAP, CAP Trial

First-Line

- **Caffeine citrate** — Loading 20 mg/kg IV/PO, maintenance 5-10 mg/kg/day; may increase to 10 if persistent
- **CPAP** — Nasal CPAP 5-7 cmH2O to splint airway and reduce apnea frequency

Second-Line

- **NIPPV** — Nasal intermittent PPV; may be more effective than CPAP for severe apnea
- **Doxapram** — 0.5-2 mg/kg/h IV infusion; for caffeine-refractory apnea; limited use

Investigations

- CBC, CRP — Rule out sepsis as trigger for new-onset apnea [STAT]
- Blood glucose — Hypoglycemia can trigger apnea [STAT]
- Blood gas — Assess for respiratory acidosis [URGENT]
- Cranial ultrasound — Rule out IVH or CNS cause in preterm
- EEG — If seizures suspected as cause of apnea events

Counselling

- Baby stops breathing for short periods; this is common in premature babies
- Caffeine medicine will help baby remember to breathe; it is safe
- Baby will be on a monitor that alerts nurses when breathing pauses occur
- Baby can go home when there have been no apnea episodes for 5-7 days
- If baby turns blue or stops breathing at home, stimulate gently and call emergency

Referral Criteria

Refer to NICU; investigate for underlying causes if apnea is new-onset in a previously well neonate

Hospitalisation Criteria

All significant apnea episodes require hospital monitoring until event-free for 5-7 days

Notes

Caffeine citrate (CAP trial) reduces apnea, BPD, and improves neurodevelopment. Rule out sepsis, anemia, and metabolic causes for new-onset apnea.

P29.3 — Persistent pulmonary hypertension of newborn

SNOMED: 233815004 | Duration: 10 days | Source: AHA/ATS 2015, Expert Consensus 2024

First-Line

- **Inhaled nitric oxide** — Start 20 ppm when OI >15-25; wean by 5 ppm increments; avoid abrupt stop
- **Oxygen therapy** — Maintain preductal SpO2 >95%; target PaO2 60-80 mmHg; avoid hyperoxia
- **Mechanical ventilation** — Optimize lung recruitment; target pH 7.35-7.45; avoid overventilation

Second-Line

- **Sildenafil** — 1-2 mg/kg PO q6h or 0.4 mg/kg IV over 3h; for iNO-refractory PPHN
- **Milrinone** — 0.33-0.99 mcg/kg/min IV; inodilator for PPHN with cardiac dysfunction
- **Epoprostenol** — 1-20 ng/kg/min IV; prostacyclin analog for refractory PPHN
- **ECMO** — For severe refractory PPHN (OI >40) failing maximal medical therapy

Investigations

- Echocardiography — Confirm PPHN; estimate PA pressure; rule out structural heart disease [STAT]
- Pre and post-ductal SpO2 — >5% difference suggests right-to-left shunting [STAT]
- Blood gas (arterial) — Calculate oxygenation index to guide therapy [STAT]
- Chest X-ray — Identify underlying lung pathology [STAT]
- CBC, blood culture — Rule out sepsis as precipitant [URGENT]

Counselling

- Baby's lung blood vessels are not opening properly, making it hard to get oxygen
- Special gas (nitric oxide) breathed through a tube helps open these blood vessels
- This is a serious condition requiring intensive care
- Most babies improve within 5-7 days with treatment
- Follow-up after discharge for lung and heart health is important

Referral Criteria

Refer to tertiary NICU with iNO capability; refer for ECMO if not responding to maximal therapy

Hospitalisation Criteria

All cases require NICU admission with mechanical ventilation and hemodynamic monitoring

Notes

iNO is the only FDA-approved pulmonary vasodilator for neonatal PPHN. 30% are iNO non-responders; sildenafil is best alternative.

Guideline Changes

Sildenafil IV formulation and milrinone increasingly used for iNO-refractory PPHN

P35.0 — Congenital rubella syndrome

SNOMED: 1857005 | Duration: 30 days | Source: AAP Red Book 2024, CDC

First-Line

- **Supportive care** — Multidisciplinary: cardiology, ophthalmology, audiology, neurology assessments
- **Cardiac surgery** — Repair of PDA or other CHD when indicated; timing based on cardiac status

Second-Line

- **Hearing aids/CI** — Early audiology referral; hearing aids or cochlear implant for SNHL
- **Cataract surgery** — Ophthalmology referral; lens aspiration if visually significant cataract

Investigations

- Rubella IgM antibody — Confirm congenital infection [STAT]

- Echocardiography — Screen for PDA, pulmonary stenosis, and other CHD [URGENT]
- Ophthalmologic examination — Cataract, glaucoma, pigmentary retinopathy screening [URGENT]
- ABR hearing test — Screen for sensorineural hearing loss [URGENT]
- CBC with platelet count — Thrombocytopenia and blueberry muffin rash evaluation [URGENT]

Counselling

- Baby was infected with rubella virus before birth; there is no specific antiviral treatment
- Baby may have heart, eye, and hearing problems that need specialist care
- Baby can spread the virus for several months; strict hand hygiene and isolation from pregnant women
- Regular follow-up with multiple specialists (heart, eye, ear) is essential
- Early hearing aids or cochlear implant can help if hearing loss is found

Referral Criteria

Immediate referral to cardiology, ophthalmology, and audiology; genetics consultation

Hospitalisation Criteria

Admit for stabilization, cardiac evaluation, and multidisciplinary assessment at birth

Notes

No specific antiviral treatment. Infants shed virus for months; isolate and enforce hand hygiene. Screen glucose and thyroid function.

P35.1 — Congenital cytomegalovirus infection

SNOMED: 1857005 | Duration: 180 days | Source: ESPID 2024, AAP

First-Line

- **Valganciclovir** — 16 mg/kg PO q12h for 6 months; start within first month of life
- **Ganciclovir IV** — 6 mg/kg IV q12h; for life-threatening disease, switch to oral when stable

Second-Line

- **Monitoring only** — For asymptomatic cCMV or mild symptoms; serial hearing tests recommended

Investigations

- CMV PCR (urine/saliva) — Confirm diagnosis within first 3 weeks of life [STAT]
- CBC with differential (weekly) — Monitor for neutropenia from valganciclovir [URGENT]
- Auditory brainstem response (ABR) — Baseline hearing assessment; repeat q6 months [URGENT]
- Brain MRI — Evaluate for intracranial calcifications, ventriculomegaly
- Ophthalmologic examination — Screen for chorioretinitis [URGENT]

Counselling

- Baby has an infection (CMV virus) caught from mother during pregnancy
- Antiviral medicine will be given for 6 months to protect baby's hearing
- Regular blood tests are needed to monitor for side effects of medication
- Hearing tests will be done every 6 months as hearing loss can develop later
- Good hand hygiene is important; CMV spreads through body fluids

Referral Criteria

Refer to infectious disease specialist, audiology, and ophthalmology at diagnosis

Hospitalisation Criteria

Admit for IV ganciclovir if severe/life-threatening disease; outpatient oral valganciclovir for stable cases

Notes

Monitor CBC weekly (neutropenia common side effect). Treatment improves hearing outcomes if started within 1 month of life.

Guideline Changes

6 months valganciclovir now standard for symptomatic cCMV; oral preferred over IV for non-critical

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P36.9 — Neonatal septic shock

SNOMED: 713854001 | Duration: 7 days | Source: Surviving Sepsis Campaign 2020, NNF 2024

First-Line

- **Normal saline bolus** — 10-20 mL/kg IV over 15-30 min; repeat up to 40-60 mL/kg total if responsive
- **Epinephrine** — 0.05-0.3 mcg/kg/min IV continuous infusion for fluid-refractory shock
- **Ampicillin + Gentamicin** — Empiric antibiotics within 1h of recognition; meningitis dosing if suspected

Second-Line

- **Dopamine** — 5-20 mcg/kg/min IV; alternative first-line vasopressor to epinephrine
- **Hydrocortisone** — 1-2 mg/kg IV q8h for catecholamine-resistant shock; stress dose steroids
- **Norepinephrine** — 0.05-0.5 mcg/kg/min IV for warm shock with low SVR

Investigations

- Blood culture (before antibiotics) — Identify organism [STAT]
- Blood gas with lactate — Lactate elevation indicates tissue hypoperfusion [STAT]
- CBC with differential — WBC, platelet count, I:T ratio [STAT]
- Coagulation profile — DIC screening [STAT]
- Echocardiography — Assess cardiac function and guide vasopressor choice [URGENT]

Counselling

- Baby is critically ill with a severe blood infection causing shock; intensive care is needed
- Multiple medicines through IV are needed to support blood pressure and fight infection
- Baby may need a breathing machine and close monitoring of heart and blood pressure
- Blood transfusions and other treatments may be needed
- Recovery depends on how quickly treatment started; the NICU team is doing everything possible

Referral Criteria

Immediate transfer to tertiary NICU with vasopressor and ventilator capability

Hospitalisation Criteria

All cases require immediate NICU/PICU admission; this is a life-threatening emergency

Notes

Start vasopressors if still shocked after 40-60 mL/kg fluid. Monitor lactate, UOP, and perfusion. Hydrocortisone for refractory shock.

Guideline Changes

Epinephrine now favored equally with dopamine as first-line vasopressor per recent RCTs

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P37.1 — Congenital toxoplasmosis

SNOMED: 73893000 | Duration: 365 days | Source: AAP 2017, CDC 2024

First-Line

- **Pyrimethamine** — 2 mg/kg/day PO x2 days, then 1 mg/kg/day x2-6 mo, then 1 mg/kg 3x/wk to 12 mo
- **Sulfadiazine** — 100 mg/kg/day PO divided q12h for 12 months
- **Leucovorin (folinic acid)** — 10 mg PO 3x/week throughout pyrimethamine therapy; protects bone marrow

Second-Line

- **Corticosteroids** — Prednisone 1 mg/kg/day if CSF protein >1 g/dL or severe chorioretinitis

Investigations

- Toxoplasma IgM and IgA — Confirm congenital infection [STAT]
- CBC with differential (weekly) — Monitor for pyrimethamine-induced bone marrow suppression [URGENT]
- Brain MRI or CT — Intracranial calcifications, hydrocephalus assessment [URGENT]
- Ophthalmologic examination — Screen for chorioretinitis [URGENT]
- CSF analysis — Elevated protein indicates CNS involvement [URGENT]

Counselling

- Baby has a parasitic infection caught from mother during pregnancy; treatment is for 12 months
- Give all three medicines as prescribed; do not stop without doctor's advice
- Blood tests every 1-2 weeks are essential to check for medicine side effects on blood cells
- Eye checkups are needed every 3-6 months as eye problems can appear later
- The infection is not contagious from baby to others

Referral Criteria

Refer to pediatric infectious disease specialist, ophthalmology, and neurology

Hospitalisation Criteria

Admit if severe symptomatic disease (seizures, hydrocephalus) or CNS involvement

Notes

Treat for 12 months minimum. Monitor CBC weekly while on daily pyrimethamine due to bone marrow suppression.

P37.9 — Neonatal meningitis

SNOMED: 82353009 | Duration: 21 days | Source: WHO 2024, AAP

First-Line

- **Ampicillin** — 200-300 mg/kg/day IV divided q6-8h for meningitis dosing; 3 weeks minimum
- **Gentamicin** — 5 mg/kg IV q24h (term) or q36-48h (preterm); monitor troughs <2 mcg/mL
- **Cefotaxime** — 200 mg/kg/day IV divided q6-8h; better CSF penetration than gentamicin

Second-Line

- **Vancomycin** — 15 mg/kg IV q8-12h for suspected MRSA or late-onset; target trough 10-15
- **Meropenem** — 40 mg/kg IV q8h for resistant gram-negative meningitis

Investigations

- Lumbar puncture with CSF analysis — Cell count, glucose, protein, gram stain, culture [STAT]
- Blood culture — Identify causative organism [STAT]
- CBC with differential — WBC count and markers of infection [STAT]
- CRP — Inflammatory marker for monitoring treatment response [URGENT]
- Cranial ultrasound — Screen for ventriculitis, abscess, hydrocephalus [URGENT]

Counselling

- Baby has a serious brain infection that needs strong IV antibiotics for at least 3 weeks
- Baby will stay in NICU during the entire treatment
- Repeat spinal tap (lumbar puncture) is needed to check if infection is clearing
- Some babies may develop complications like hydrocephalus; head size will be monitored
- Long-term follow-up for hearing and development is essential after discharge

Referral Criteria

Refer to pediatric neurology and infectious disease; neurosurgery if hydrocephalus develops

Hospitalisation Criteria

All cases require NICU admission for minimum 21 days of IV antibiotics

Notes

Treat minimum 21 days or 14 days after CSF sterilization, whichever is longer. Repeat LP at 48-72h to document sterilization.

P38 — Umbilical Sepsis (Omphalitis)

SNOMED: 239095007

Second-Line

- **Vancomycin** — 15 mg/kg IV q8-12h if MRSA suspected; adjust based on culture results
- **Clindamycin** — 5-7.5 mg/kg IV q8h if anaerobic or necrotizing fasciitis concern
- **Surgical debridement** — Rarely needed; only for necrotizing fasciitis or abscess formation

Investigations

- Swab for culture (if discharge purulent) — localized no systemic signs
- CBC and CRP if not improving — localized no systemic signs
- CBC — systemic signs present
- CRP — systemic signs present
- Blood culture — systemic signs present
- Urine culture — systemic signs present
- Lumbar puncture if indicated — systemic signs present
- Admit for full sepsis work-up — systemic signs present

Counselling

- Keep umbilical cord clean and dry. — नाल को साफ़ और सूखा रखें।
- Avoid tight clothing over cord. — नाल पर कसे कपड़े न पहनाएं।
- Do not apply oils, powders or traditional substances. — तेल, पाउडर या घरेलू चीज़ें न लगाएं।
- Watch for warning signs: redness, swelling, foul smell, pus, fever. — लालिमा, सूजन, बदबू, पीप या बुखार पर ध्यान दें।
- Normal cord separation occurs in 7–14 days. — नाल आमतौर पर 7-14 दिन में गिर जाती है।
- Seek medical help if discharge or redness increases. — रिसाव या लालिमा बढ़े तो तुरंत डॉक्टर को दिखाएं।

Warning Signs

- Purulent or foul-smelling umbilical discharge — नाल से पीप या बदबू आना
- Spreading redness beyond 1 cm from stump — नाल से 1 सेमी से ज़्यादा लालिमा फैलना
- Fever or hypothermia — बुखार या शरीर ठंडा पड़ना
- Lethargy or poor feeding — सुस्ती या दूध न पीना
- Abdominal distension — पेट का फूलना
- Skin discoloration around umbilicus — नाभि के आस-पास त्वचा का रंग बदलना

P39.1 — Neonatal conjunctivitis

SNOMED: 276675009 | Duration: 14 days | Source: CDC STI Guidelines 2021, AAP

First-Line

- **Erythromycin ophthalmic** — 0.5% ointment prophylaxis to both eyes within 1h of birth
- **Ceftriaxone** — 25-50 mg/kg IM/IV x1 dose (max 125 mg) for gonococcal conjunctivitis
- **Erythromycin oral** — 50 mg/kg/day PO divided q6h x14 days for chlamydial conjunctivitis

Second-Line

- **Saline irrigation** — Frequent saline eye lavage for gonococcal conjunctivitis as adjunct
- **Azithromycin oral** — 20 mg/kg/day PO x3 days for chlamydial; if erythromycin fails

Investigations

- Conjunctival swab for gram stain and culture — Identify organism; gram-negative diplococci for gonococcal [STAT]
- Chlamydia PCR (conjunctival swab) — Rule out chlamydial conjunctivitis [URGENT]
- Blood culture — If systemic signs present [URGENT]

Counselling

- Baby has an eye infection that needs treatment; clean eyes gently with warm water
- Complete the full course of antibiotic medicine even if the eye looks better
- If pus increases, eye becomes very swollen, or baby seems unwell, come back immediately
- Gonococcal eye infection is serious and needs injection medicine
- Good hand hygiene before and after touching baby's eyes is essential

Referral Criteria

Refer to ophthalmology if gonococcal conjunctivitis, corneal involvement, or not improving with treatment

Hospitalisation Criteria

Admit for gonococcal ophthalmia (medical emergency); systemic antibiotics required

Notes

Gonococcal ophthalmia is a medical emergency requiring systemic antibiotics. Topical treatment alone insufficient for chlamydial infection.

P52 — Intracranial nontraumatic hemorrhage of fetus and newborn

SNOMED: 206417009 | Duration: 21 days | Source: AAP, Papile Classification

First-Line

- **Supportive care** — Maintain hemodynamic stability, correct coagulopathy, minimize handling
- **Serial cranial ultrasound** — Head US at Day 3-7, repeat at 14d and 36 wk PMA; monitor for hydrocephalus

Second-Line

- **RBC transfusion** — 10-15 mL/kg for symptomatic anemia from hemorrhage; maintain Hb >10 in acute phase
- **VP shunt/reservoir** — Neurosurgery consult for progressive post-hemorrhagic hydrocephalus
- **Acetazolamide** — 25 mg/kg/day PO for temporizing slowly progressive hydrocephalus; limited evidence

Investigations

- Cranial ultrasound (serial) — Grade IVH; repeat at day 3, 7, 14, and 36 weeks PMA [STAT]
- CBC — Monitor hemoglobin for blood loss [STAT]
- Coagulation profile — Rule out coagulopathy contributing to hemorrhage [URGENT]
- Brain MRI — At term-equivalent age for detailed assessment

Counselling

- Baby has bleeding inside the brain; this is more common in premature babies
- Handle baby gently; minimize disturbances and loud noises
- Head size will be measured regularly to check for fluid buildup (hydrocephalus)
- Mild bleeding (grade 1-2) usually resolves on its own
- Regular follow-up with brain scans and developmental assessments is important after discharge

Referral Criteria

Refer to neurosurgery if progressive hydrocephalus; neurology for developmental follow-up

Hospitalisation Criteria

All cases require NICU monitoring; neurosurgery consultation for grade III-IV IVH

Notes

Prevention key: antenatal steroids, delayed cord clamping, minimal handling. Most grade I-II IVH resolve; grade III-IV carry higher neurodevelopmental risk.

P53 — Hemorrhagic disease of newborn

SNOMED: 24149006 | Duration: 7 days | Source: AAP 2022, CPS

First-Line

- **Vitamin K1 (treatment)** — 1 mg IV/SC for active bleeding; 0.3-0.5 mg/kg if <1500g birth weight
- **Vitamin K1 (prophylaxis)** — 0.5-1 mg IM at birth for all neonates; single dose prevents VKDB

Second-Line

- **Fresh frozen plasma** — 10-15 mL/kg IV for moderate-severe active bleeding with coagulopathy
- **Oral Vitamin K** — 2 mg PO at birth, 2-4 wks, and 6-8 wks if parents decline IM; less reliable

Investigations

- PT/INR — Markedly prolonged in VKDB; normalizes after vitamin K [STAT]
- aPTT — Prolonged in severe cases [STAT]
- CBC — Assess hemoglobin drop from bleeding [STAT]
- Cranial ultrasound — If intracranial hemorrhage suspected (irritability, bulging fontanelle) [URGENT]

Counselling

- Baby has a bleeding problem due to low vitamin K; injection will correct this
- Vitamin K injection at birth prevents this condition; it is safe and strongly recommended
- If baby is actively bleeding, treatment with vitamin K and blood products will be given
- Watch for unusual bruising, blood in stool, or excessive bleeding from umbilical cord
- Follow up as advised; most babies recover completely with treatment

Referral Criteria

Refer to pediatric hematologist if bleeding is severe or does not respond to vitamin K

Hospitalisation Criteria

Admit all active bleeding cases; ICU admission for intracranial hemorrhage

Notes

IM vitamin K at birth is the standard of care for VKDB prevention. Late VKDB (2-12 weeks) occurs with missed or oral-only prophylaxis.

P54.2 — Blood in Stool in a Neonate

SNOMED: 12063002

Investigations

- undefined — Infection suspected
- undefined — CMPA suspicion
- undefined — Sepsis / NEC
- undefined — Bleeding disorder
- undefined — NEC / obstruction / perforation
- undefined — Differentiate swallowed maternal blood

Counselling

- Small streaks of blood are often benign (anal fissure or allergy) — खून की हल्की लकीरें अक्सर सामान्य होती हैं (गुदा में दरार या एलर्जी)
- Reassure if baby is active and feeding well — अगर बच्चा सक्रिय है और अच्छे से दूध पी रहा है तो चिंता न करें
- Avoid frequent rectal temperature checks — बार-बार गुदा से तापमान न लें
- Continue breastfeeding unless advised otherwise — जब तक डॉक्टर न कहे, माँ का दूध पिलाना जारी रखें
- Watch for warning signs: fever, distension, vomiting, poor feeding — खतरे के संकेत देखें: बुखार, पेट फूलना, उल्टी, कम दूध पीना
- Seek urgent care if symptoms worsen — अगर लक्षण बिगड़ें तो तुरंत डॉक्टर को दिखाएं

Warning Signs

- Large amount of blood in stool — पखाने में ज़्यादा खून आना
- Abdominal distension or hardness — पेट का फूलना या सख्त होना
- Bilious (green) vomiting — हरे रंग की उल्टी
- Fever or hypothermia — बुखार या शरीर ठंडा पड़ना
- Baby becoming lethargic or pale — बच्चे का सुस्त या पीला पड़ जाना
- Refusal to feed — दूध न पीना
- Black tarry stools — काला चिपचिपा पखाना

Referral Criteria

Suspected NEC,Sepsis or shock,Persistent or large bleeding,Abdominal distension or obstruction,Failure to thrive,Coagulopathy suspected

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P55 — Hemolytic disease of newborn

SNOMED: 206431009 | Duration: 7 days | Source: AAP 2022 Hyperbilirubinemia Guidelines

First-Line

- **Intensive phototherapy** — Irradiance >=30 mcW/cm2/nm; start when TSB at or above threshold for GA/age
- **IVIG** — 0.5-1 g/kg IV over 2h for isoimmune hemolytic disease if TSB approaching exchange level

Second-Line

- **Exchange transfusion** — Double-volume (160 mL/kg); when TSB exceeds exchange threshold despite phototherapy
- **IV fluids** — D10W + electrolytes at 1.5x maintenance as supplement to feeds during phototherapy

Investigations

- Total serum bilirubin (serial) — Monitor q4-8h during phototherapy [STAT]
- Direct Coombs test (DAT) — Confirm immune-mediated hemolysis [STAT]
- CBC with reticulocyte count — Assess severity of hemolysis and anemia [URGENT]
- Blood group and Rh (mother and baby) — Identify incompatibility [STAT]
- G6PD screen — Rule out enzyme deficiency contributing to hemolysis

Counselling

- Baby's blood cells are being destroyed by mother's antibodies; phototherapy (blue light) helps
- Breastfeeding should continue during phototherapy; it does not need to be stopped
- Baby's eyes will be covered during light therapy to protect them
- Bilirubin levels will be checked frequently with blood tests
- If light therapy is not enough, blood exchange may be needed; doctor will explain if required

Referral Criteria

Refer to neonatologist if TSB approaching exchange transfusion level, rapidly rising bilirubin (>0.5 mg/dL/h), or hemoglobin dropping

Hospitalisation Criteria

Admit for phototherapy if TSB at or above treatment threshold; NICU for exchange transfusion or IVIG

Notes

Rh and ABO incompatibility most common causes. Check DAT, reticulocyte count, and serial bilirubin. IVIG may reduce need for exchange.

Guideline Changes

AAP 2022 raised phototherapy thresholds; IVIG role more selective

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P59.8 — Neonatal Cholestasis

SNOMED: 276555003

Counselling

- Not all jaundice is benign after 2 weeks of age. — 2 हफ्ते बाद भी पीलिया सामान्य नहीं माना जाना चाहिए।
- Pale stools and dark urine are danger signs and need urgent evaluation. — सफ़ेद पोटी और गहरे रंग का पेशाब खतरे के निशान हैं, तुरंत जांच कराएं।
- Early surgery in biliary atresia can preserve native liver function. — पित्त नली की बीमारी में जल्दी ऑपरेशन से लीवर बचाया जा सकता है।
- Good nutrition and vitamin supplementation are important. — अच्छा पोषण और विटामिन की खुराक ज़रूरी है।
- Regular follow-up and monitoring are essential. — नियमित जांच बहुत ज़रूरी है।

Warning Signs

- Pale or white stools — सफ़ेद या मिट्टी जैसी पोटी
- Dark yellow or brown urine — गहरे पीले या भूरे रंग का पेशाब
- Worsening jaundice — पीलिया बढ़ना
- Bleeding from any site — कहीं से भी खून आना
- Poor feeding or weight loss — दूध न पीना या वज़न कम होना
- Swollen abdomen — पेट फूलना

P59.9 — Unconjugated Neonatal Jaundice

SNOMED: 387712008

Second-Line

- **Exchange transfusion** — Double-volume 160 mL/kg; if TSB exceeds exchange threshold despite phototherapy
- **IVIG** — 0.5-1 g/kg IV over 2h for isoimmune hemolytic disease approaching exchange level

Counselling

- Mild jaundice is common but timing and progression are important — हल्का पीलापन आम है लेकिन समय और बढ़ोतरी पर ध्यान दें
- Ensure adequate feeding (8–12 feeds/day) — दिन में 8-12 बार दूध पिलाएं
- Return if jaundice worsens or feeding decreases — पीलापन बढ़े या दूध कम पिए तो तुरंत आएं
- Educate about danger signs — खतरे के लक्षण पहचानें: सुस्ती, तेज़ रोना, पीठ तनना, बुखार

Warning Signs

- Jaundice within first 24 hours of life — जन्म के 24 घंटे में पीलापन (हमेशा खतरनाक)
- Poor feeding or lethargy — दूध न पीना या सुस्ती
- High-pitched cry — तेज़ पतली आवाज़ में रोना
- Back arching (opisthotonus) — पीठ का तनना

- Fever or pallor — बुखार या पीलापन
- Rapidly worsening jaundice — तेज़ी से बढ़ता पीलापन
- Dark urine or pale stools — गहरे रंग का पेशाब या सफ़ेद मल

Guideline Changes

AAP 2022 raised phototherapy thresholds by ~2 mg/dL for healthy term infants

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P61 — Neonatal anemia

Duration: 14 days | Source: JAMA 2024, Cochrane

First-Line

- **Packed RBC transfusion** — 10-20 mL/kg over 2-4h; use restrictive thresholds (Hb 7-10 based on support)
- **Iron supplementation** — 2-4 mg/kg/day elemental iron from 2-6 weeks in preterm infants

Second-Line

- **Erythropoietin** — 250-400 IU/kg SC 3x/week; modest reduction in late transfusions; ROP concern
- **Delayed cord clamping** — 60-120 seconds at birth; reduces anemia and transfusion need in preterms

Investigations

- CBC with reticulocyte count — Severity of anemia and bone marrow response [STAT]
- Direct Coombs test — Rule out immune-mediated hemolysis [URGENT]
- Peripheral blood smear — RBC morphology to identify cause
- Serum ferritin and iron studies — Assess iron stores

Counselling

- Baby has low blood count and may need a blood transfusion
- Blood tests (heel pricks) are needed frequently; each test takes only a small amount
- Baby will receive iron supplements to help build blood
- Premature babies commonly need transfusions; this is expected and safe
- Breast milk and iron-rich supplements will help after discharge

Referral Criteria

Refer to hematologist if anemia is persistent, unusual, or not responding to standard treatment

Hospitalisation Criteria

Admit if symptomatic anemia (poor feeding, tachycardia, apnea) or hemoglobin below transfusion threshold

Notes

Minimize phlebotomy losses. Restrictive transfusion thresholds are safe per ETTNO/TOP trials. Delayed cord clamping is standard.

Guideline Changes

Restrictive transfusion thresholds (Hb 7-10 g/dL based on age/support) validated by ETTNO/TOP trials

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P61.0 — Transient neonatal thrombocytopenia

SNOMED: 206512000 | Duration: 10 days | Source: PlaNeT-2 Trial 2019, AAP

First-Line

- **Platelet transfusion** — 10-15 mL/kg; threshold <25,000/mcL (restrictive) per PlaNeT-2 trial **[NEW 2024-25]**
- **Treat underlying cause** — Antibiotics for sepsis, maternal ITP workup; most cases self-limiting

Second-Line

- **IVIG** — 1 g/kg IV prior to platelet transfusion for alloimmune thrombocytopenia (NAIT)

- **Platelet transfusion (liberal)** — 10-15 mL/kg at <50,000/mcL only for active bleeding or pre-procedure

Investigations

- CBC with platelet count (serial) — Monitor trend q12-24h [STAT]
- Blood culture — Rule out sepsis as cause [STAT]
- Maternal platelet count and anti-platelet antibodies — Evaluate for NAIT or maternal ITP [URGENT]
- Coagulation profile (PT/aPTT) — Assess overall coagulation status [URGENT]

Counselling

- Baby has low platelet count which means blood clotting is reduced
- Blood tests will be repeated frequently to monitor the platelet count
- Platelet transfusion may be needed if count is very low or baby is bleeding
- Most cases in newborns resolve on their own within 1-2 weeks
- Handle baby gently; avoid unnecessary procedures to reduce bleeding risk

Referral Criteria

Refer to pediatric hematologist if thrombocytopenia persists >2 weeks or suspected NAIT

Hospitalisation Criteria

Admit if platelets <50,000/mcL with bleeding or <25,000/mcL regardless of symptoms

Notes

Restrictive threshold (25K) is safer than liberal (50K) per PlaNeT-2 trial. For NAIT, use HPA-matched or washed maternal platelets.

Guideline Changes

PlaNeT-2 trial established restrictive platelet transfusion threshold of 25,000/mcL

P61.1 — Neonatal polycythemia

SNOMED: 737210007 | Duration: 5 days | Source: UCSF Consensus Guidelines

First-Line

- **IV normal saline hydration** — Increase fluid intake; NS bolus 10-20 mL/kg for mild symptomatic polycythemia
- **Partial exchange transfusion** — PET with NS; Vol = (Hct observed - 50)/Hct observed x 90 mL/kg x wt(kg)

Second-Line

- **Observation** — Monitor Hct q6-12h; many cases resolve with hydration alone

Investigations

- Venous hematocrit — Confirm diagnosis; must be venous not capillary [STAT]
- Blood glucose — Polycythemia causes hypoglycemia [STAT]
- Serum bilirubin — Risk of hyperbilirubinemia [URGENT]
- Serum calcium — Screen for associated hypocalcemia

Counselling

- Baby's blood is thicker than normal which can cause problems with blood flow
- Extra fluids and close monitoring will be given
- If the blood is very thick or baby has symptoms, a partial blood exchange may be needed
- This condition usually resolves within a few days with treatment
- Follow up for a blood count check after discharge as advised

Referral Criteria

Refer to neonatologist if symptomatic polycythemia requiring partial exchange transfusion

Hospitalisation Criteria

Admit if venous Hct >70% (asymptomatic) or >65% with symptoms (jitteriness, poor feeding, cyanosis)

Notes

PET for Hct >70% asymptomatic or >65% symptomatic. Use venous (not capillary) Hct for decision-making. Long-term benefit of PET unproven.

P70.4 — Neonatal hypoglycemia

SNOMED: 52767006 | Duration: 5 days | Source: AAP 2022, PES 2023

First-Line

- **IV dextrose bolus** — D10W 2 mL/kg (200 mg/kg) IV push for BG <25 or symptomatic; then infusion
- **Dextrose infusion** — D10W at GIR 4-6 mg/kg/min; increase by 2 mg/kg/min q30min to max 12-15
- **Early feeding** — Breastfeed or formula within 1h of birth for at-risk infants; q2-3h

Second-Line

- **Dextrose gel 40%** — 0.5 mL/kg buccal massage for mild asymptomatic hypoglycemia; recheck in 30 min **[NEW 2024-25]**
- **Glucagon** — 0.02-0.03 mg/kg IM/IV (max 1 mg) for refractory hypoglycemia; temporary bridge
- **Diazoxide** — 5-15 mg/kg/day PO divided q8h for persistent hyperinsulinemic hypoglycemia

Investigations

- Blood glucose (serial) — Bedside and lab glucose monitoring q30min-1h during treatment [STAT]
- Insulin level (critical sample) — Collect during hypoglycemia episode to evaluate for hyperinsulinism [URGENT]
- Cortisol and growth hormone — Collect during hypoglycemia if persistent; rule out hormonal causes [URGENT]
- Urine ketones — Absence of ketones during hypoglycemia suggests hyperinsulinism [URGENT]
- Blood gas — Metabolic acidosis if associated organic acidemia [URGENT]

Counselling

- Baby has low blood sugar which needs to be corrected quickly
- Feed baby frequently - every 2-3 hours - breastmilk is preferred
- Blood sugar will be checked frequently by heel prick; this is necessary
- If blood sugar stays low despite feeding, IV sugar water (dextrose) will be given
- Watch for shakiness, poor feeding, or excessive sleepiness at home after discharge

Referral Criteria

Refer to endocrinologist if hypoglycemia persists beyond 48-72 hours or requires GIR >12 mg/kg/min

Hospitalisation Criteria

Admit if BG <25 mg/dL, symptomatic, or requiring IV dextrose infusion

Notes

Screen at-risk groups: IDM, SGA, LGA, preterm. Target BG >45 mg/dL after 4h of life. Max peripheral dextrose D12.5%; higher needs central line.

Guideline Changes

Dextrose gel 40% increasingly adopted as first intervention for mild asymptomatic hypoglycemia

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P71 — Transitory neonatal disorders of calcium and magnesium metabolism

SNOMED: 206488006 | Duration: 7 days | Source: AAP, ANMF 2023

First-Line

- **Calcium gluconate 10%** — 1-2 mL/kg (10-20 mg/kg elemental Ca) IV slow push over 10-30 min with cardiac monitoring
- **Calcium gluconate maintenance** — 50-75 mg/kg/day elemental Ca IV continuous or 80 mg/kg/day PO divided q6h

Second-Line

- **Magnesium sulfate** — 25-50 mg/kg IV/IM q8-12h for hypomagnesemia; monitor for bradycardia
- **Calcitriol** — 0.05-0.1 mcg/kg/day PO for persistent hypocalcemia from hypoparathyroidism

Investigations

- Serum calcium (total and ionized) — Confirm hypocalcemia; ionized Ca more accurate [STAT]
- Serum magnesium — Must correct hypomagnesemia first [STAT]
- Serum phosphorus — Elevated in hypoparathyroidism [URGENT]
- PTH level — Low in hypoparathyroidism, high in vitamin D deficiency
- 25-OH Vitamin D — Rule out vitamin D deficiency

Counselling

- Baby has low calcium or magnesium in the blood causing jitteriness or seizures
- Calcium medicine will be given through IV slowly with heart monitoring
- Oral calcium supplement will continue after IV treatment
- Breastmilk or low-phosphorus formula helps maintain calcium levels
- Follow up for repeat blood tests to ensure levels remain normal

Referral Criteria

Refer to endocrinologist if hypocalcemia persists >1 week or hypoparathyroidism suspected

Hospitalisation Criteria

Admit if symptomatic (seizures, tetany) or ionized calcium <4 mg/dL

Notes

IV calcium must be given slowly under cardiac monitoring to prevent bradycardia. Correct hypomagnesemia first as it impairs Ca correction.

P74.1 — Neonatal dehydration

SNOMED: 78812008 | Duration: 5 days | Source: AAP, NICE 2020

First-Line

- **IV normal saline bolus** — NS 10-20 mL/kg over 15-30 min for shock; repeat as needed up to 40-60 mL/kg
- **IV maintenance fluids** — D10W + 0.45% NaCl at maintenance rate; adjust Na+ by <=10-12 mEq/L/day
- **Breastfeeding support** — Assess latch, frequency, and adequacy; supplement if weight loss >10%

Second-Line

- **Oral rehydration** — Small frequent breastmilk/formula feeds if mild and tolerated orally
- **Electrolyte monitoring** — Check Na, K, BUN, Cr q8-12h during correction; slow correction for hypernatremia

Investigations

- Serum electrolytes (Na, K) — Hypernatremic dehydration common in breastfed neonates [STAT]
- BUN and creatinine — Assess renal function; pre-renal azotemia [URGENT]
- Blood glucose — Hypoglycemia risk with poor intake [STAT]
- Urine specific gravity — Assess degree of dehydration [URGENT]
- Serum bilirubin — Dehydration worsens jaundice

Counselling

- Baby is dehydrated, usually from inadequate feeding; fluids will be given through IV

- Breastfeeding support will be provided to improve feeding technique
- Monitor baby's urine output - at least 6-8 wet diapers per day means adequate feeding
- Weight will be checked daily; expect steady weight gain after rehydration
- If weight loss >10% from birth weight, bring baby to hospital immediately

Referral Criteria

Refer to neonatologist if hypernatremic dehydration (Na >150), seizures, or AKI

Hospitalisation Criteria

Admit if moderate-severe dehydration, weight loss >10%, hypernatremic, or hemodynamically unstable

Notes

Hypernatremic dehydration common in breastfed neonates with poor intake. Correct sodium slowly (<=10-12 mEq/L/day) to prevent cerebral edema.

P76.9 — Intestinal obstruction of newborn

SNOMED: 276521007 | Duration: 14 days | Source: Pediatric Surgery Guidelines

First-Line

- **NGT decompression** — Large-bore NGT to continuous or intermittent suction; strict NPO
- **IV fluid resuscitation** — NS 10-20 mL/kg bolus; then maintenance + deficit replacement
- **Surgical exploration** — Laparotomy for mechanical obstruction; urgency depends on etiology

Second-Line

- **Contrast enema** — Gastrografin enema; diagnostic and therapeutic for meconium ileus
- **N-acetylcysteine** — Mucolytic via NGT or enema for meconium ileus; 2-4% solution
- **Antibiotics** — Broad-spectrum: ampicillin + gentamicin if perforation suspected

Investigations

- Abdominal X-ray (erect and supine) — Air-fluid levels, dilated bowel loops, absence of distal gas [STAT]
- Contrast enema — Diagnostic and potentially therapeutic for meconium ileus [URGENT]
- CBC with differential — Evaluate for infection/sepsis [URGENT]
- Serum electrolytes — Assess fluid and electrolyte status [URGENT]

Counselling

- Baby has a blockage in the intestine; the green vomit is a warning sign that needs urgent treatment
- Baby cannot be fed by mouth until the blockage is resolved
- Surgery may be needed depending on the cause of the blockage
- A tube will be placed through the nose to drain the stomach and reduce pressure
- After treatment, feeds will be started slowly once the intestine recovers

Referral Criteria

Immediate pediatric surgery referral for all neonatal intestinal obstruction

Hospitalisation Criteria

All cases require emergency admission; bilious vomiting in a neonate is obstruction until proven otherwise

Notes

Bilious vomiting in a neonate is obstruction until proven otherwise. Obtain upright abdominal XR and surgical consult urgently.

P77 — Necrotizing enterocolitis of newborn

SNOMED: 788989006 | Duration: 14 days | Source: Bell Staging, AAP 2024

First-Line

- **Ampicillin + Gentamicin** — Ampicillin 50 mg/kg IV q8-12h + Gentamicin 5 mg/kg IV q24-48h
- **Metronidazole** — 15 mg/kg IV load, then 7.5 mg/kg IV q12h for anaerobic coverage
- **NPO + NGT** — Bowel rest 7-14 days; NGT decompression; TPN for nutrition

Second-Line

- **Surgical intervention** — Laparotomy or peritoneal drain for perforation (pneumoperitoneum)
- **Vancomycin** — 15 mg/kg IV q8-12h if MRSA or late-onset concern; adjust per culture
- **Probiotics (prevention)** — Lactobacillus/Bifidobacterium for VLBW NEC prevention **[NEW 2024-25]**

Investigations

- Abdominal X-ray (serial) — Pneumatosis intestinalis, portal venous gas, pneumoperitoneum [STAT]
- CBC with differential — Thrombocytopenia, left shift, neutropenia markers [STAT]
- Blood culture — Identify causative organism before antibiotics [STAT]
- CRP / Procalcitonin — Inflammatory markers for monitoring [URGENT]
- Blood gas (arterial/capillary) — Metabolic acidosis assessment [URGENT]

Counselling

- Baby has a serious bowel infection that needs strong antibiotics and bowel rest
- Baby will not be fed by mouth for 7-14 days; nutrition will be given through IV
- Mother should continue to express and store breastmilk as it will be used when feeds restart
- Surgery may be needed if the bowel becomes perforated (develops a hole)
- Recovery takes time; baby will be closely monitored with X-rays and blood tests

Referral Criteria

Immediate pediatric surgical consultation for all NEC cases; transfer to center with pediatric surgery if not available

Hospitalisation Criteria

All cases require NICU admission; Bell stage II and above require intensive monitoring

Notes

Stage with Bell criteria. Pneumoperitoneum is absolute surgical indication. Human milk feeding significantly reduces NEC incidence.

Guideline Changes

Probiotics increasingly recommended for NEC prevention in VLBW infants; lactoferrin under study

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P81 — Disturbance of temperature regulation of newborn

SNOMED: 68550008 | Duration: 5 days | Source: WHO Thermal Protection, AAP

First-Line

- **Radiant warmer/incubator** — Servo-controlled to maintain axillary temp 36.5-37.4C; for hypothermia rewarming
- **Skin-to-skin (KMC)** — Kangaroo mother care for stable infants; effective thermoregulation method
- **Polyethylene wrap** — Apply at birth for preterms <32 wk without drying; reduces heat loss

Second-Line

- **Warm IV fluids** — Warmed NS/D10W for severely hypothermic infants; slow rewarming 0.5C/h
- **Sepsis evaluation** — Obtain cultures and start empiric antibiotics if hypothermia unexplained

Investigations

- Blood glucose — Hypothermia causes hypoglycemia [STAT]
- Blood culture and CBC — Sepsis screening if temperature instability is unexplained [URGENT]

- Blood gas — Assess for metabolic acidosis in severe hypothermia [URGENT]
- Thyroid function test — Rule out congenital hypothyroidism if persistent hypothermia

Counselling

- Baby is unable to maintain normal body temperature; needs help staying warm
- Kangaroo mother care (skin-to-skin) is one of the best ways to warm baby
- Keep baby well wrapped; avoid bathing until temperature is stable
- Hypothermia (low temperature) can be a sign of infection; doctor will investigate
- At home, ensure room temperature is warm and baby is well dressed, especially at night

Referral Criteria

Refer to NICU if severe hypothermia (<32C), not responding to rewarming, or suspected sepsis

Hospitalisation Criteria

Admit if temperature <36C despite rewarming measures or if associated with other danger signs

Notes

Hypothermia (<36.5C) increases mortality in neonates. Rewarm slowly at 0.5C/h. Unexplained temperature instability warrants sepsis workup.

P81.9 / R50.9 — Fever in Neonate

SNOMED: 60025004

Investigations

- CBC with differential
- CRP / Procalcitonin
- Blood culture
- Urine analysis + culture
- Lumbar puncture
- Chest X-ray (if respiratory signs)
- HSV PCR (if suspected)

Counselling

- Neonatal fever is serious and requires admission — नवजात शिशु में बुखार गंभीर है और भर्ती होना ज़रूरी है।
- Early treatment improves outcome — जल्दी इलाज शुरू करने से बच्चा जल्दी ठीक होता है।
- Avoid overheating or over-wrapping — बच्चे को ज़्यादा कपड़े या गर्मी में न रखें।
- Watch for danger signs — खतरे के संकेतों पर ध्यान रखें।

Warning Signs

- Poor feeding or refusal to feed
- Lethargy or excessive sleepiness
- Seizures or abnormal movements
- Respiratory distress or fast breathing
- Cyanosis (bluish discoloration)
- Bulging fontanelle
- Vesicular skin rash (suspect HSV)
- Jaundice worsening rapidly
- Hypothermia (temperature <36.5 C)

P83.1 — Benign Neonatal Rash

SNOMED: 240302002

Counselling

- Most newborn rashes are harmless and self-resolving — ज़्यादातर नवजात के दाने हानिरहित होते हैं और अपने आप ठीक हो जाते हैं
- No need for creams, antibiotics, or medications — किसी क्रीम, एंटीबायोटिक या दवा की ज़रूरत नहीं है
- Avoid talcum powder and scented products — टैल्कम पाउडर और खुशबूदार चीज़ों से बचें
- Maintain hygiene with plain lukewarm water bathing — गुनगुने पानी से साफ़ सफ़ाई रखें
- Rash will gradually resolve over days to weeks — दाने धीरे-धीरे कुछ दिनों से हफ़्तों में ठीक हो जाएंगे
- Seek medical care if red flag symptoms appear — अगर खतरे के संकेत दिखें तो तुरंत डॉक्टर को दिखाएं

Warning Signs

- Fever with rash — दाने के साथ बुखार
- Blisters or pus-filled bumps — छाले या मवाद वाले दाने
- Rash spreading rapidly — दाने तेज़ी से फैलना
- Baby not feeding well — बच्चे का ठीक से दूध न पीना
- Baby becoming lethargic or irritable — बच्चे का सुस्त या चिड़चिड़ा हो जाना
- Purple or dark red spots on skin — त्वचा पर बैंगनी या गहरे लाल धब्बे

P83.4 — Breast Masses, Vaginal Bleeding and Other Genital Anomalies in Neonates

SNOMED: 281610003

Investigations

- undefined
- undefined
- undefined
- undefined
- undefined

Counselling

- These findings are common and usually hormonal / physiological — ये लक्षण आम हैं और आमतौर पर हार्मोन के कारण होते हैं
- Breast swelling or milky discharge often resolves spontaneously — स्तन की सूजन या दूध जैसा रिसाव अपने आप ठीक हो जाता है
- Do not squeeze or massage the breasts — स्तनों को दबाएं या मालिश न करें
- First-week vaginal bleeding is usually self-limiting — पहले हफ्ते में योनि से खून आना आमतौर पर अपने आप रुक जाता है
- Maintain gentle hygiene only; no internal cleaning — सिर्फ़ हल्की सफ़ाई रखें; अंदर की सफ़ाई न करें
- Seek review if bleeding persists, discharge is foul-smelling, breast becomes red/tender, or genitalia appear unclear — अगर खून बंद न हो, बदबूदार रिसाव हो, स्तन लाल/दर्दनाक हो, या जननांग स्पष्ट न दिखें तो डॉक्टर को दिखाएं

Warning Signs

- Persistent vaginal bleeding beyond first week — पहले हफ्ते के बाद भी योनि से खून आना
- Breast redness, warmth or pus — स्तन पर लालिमा, गर्मी या मवाद
- Ambiguous genitalia — अस्पष्ट जननांग
- Fever or poor feeding — बुखार या कम दूध पीना
- Inguinal swelling that does not reduce — जांघ की सूजन जो कम न हो

• Signs of salt loss: vomiting, poor weight gain, dehydration — नमक की कमी के संकेत: उल्टी, वज़न न बढ़ना, पानी की कमी

Referral Criteria

Ambiguous genitalia,Persistent vaginal bleeding,Breast abscess / mastitis,Hard or non-reducible inguinal / labial mass,Suspected imperforate hymen / cloacal anomaly,Salt-loss symptoms or suspicion of CAH

P90 — Neonatal convulsions

SNOMED: 87476004 | Duration: 10 days | Source: ILAE 2023, WHO

First-Line

• **Phenobarbital** — 20 mg/kg IV loading over 10-15 min; additional 10 mg/kg if seizures persist (max 40)
• **Correct metabolic causes** — Check and correct glucose, calcium, magnesium, sodium before/with anticonvulsants

Second-Line

• **Levetiracetam** — 40-60 mg/kg IV loading; maintenance 10 mg/kg IV q8h; fewer sedation side effects
• **Phenytoin/Fosphenytoin** — 20 mg/kg IV PE loading; second-line per ILAE if phenobarbital fails
• **Midazolam** — 0.05-0.15 mg/kg IV bolus then 0.1-0.4 mg/kg/h infusion for refractory seizures

Investigations

• Blood glucose — Hypoglycemia is most common treatable cause [STAT]
• Serum calcium and magnesium — Metabolic causes of seizures [STAT]
• EEG or aEEG — Confirm electrical seizure activity; guide treatment [URGENT]
• Lumbar puncture — Rule out meningitis/encephalitis [URGENT]
• Brain MRI — Identify structural or ischemic cause

Counselling

• Baby is having seizures which need immediate treatment to protect the brain
• Seizures in newborns can look different from older children - subtle movements, lip smacking, or cycling movements
• Blood tests will identify if low sugar, calcium, or infection is causing the seizures
• Medicine (phenobarbital) will be given to stop and prevent seizures
• Follow-up with a brain specialist (neurologist) and EEG testing is important

Referral Criteria

Refer to neonatologist/neurologist; tertiary NICU for continuous EEG monitoring

Hospitalisation Criteria

All neonatal seizures require NICU admission for monitoring and treatment

Notes

Phenobarbital is first-line regardless of etiology (ILAE 2023). Confirm seizures with aEEG/EEG when possible. Treat underlying cause concurrently.

Guideline Changes

ILAE 2023 guidelines confirm phenobarbital as first-line; levetiracetam has lower efficacy (28% vs 80%)

P91.6 — Hypoxic ischemic encephalopathy

SNOMED: 126945001 | Duration: 14 days | Source: AAP 2026, AHA

First-Line

• **Therapeutic hypothermia** — Cool to 33.5-34.5C within 6h of birth; maintain x72h then rewarm 0.5C/h

- **Supportive NICU care** — Correct acidosis, maintain euglycemia, avoid hyperthermia, seizure monitoring

Second-Line

- **Phenobarbital** — 20 mg/kg IV loading for seizures; additional 10 mg/kg if seizures persist
- **Erythropoietin (investigational)** — 1000 IU/kg IV q24h x5d; neuroprotective adjunct under study; not yet standard **[NEW 2024-25]**
- **Late hypothermia** — Cooling at 6-24h may provide small benefit if missed <6h window **[NEW 2024-25]**

Investigations

- aEEG/EEG (continuous) — Seizure monitoring and encephalopathy grading [STAT]
- Brain MRI — At day 4-7 for prognostication [URGENT]
- Blood gas — pH, lactate, base deficit for severity assessment [STAT]
- Renal and hepatic function — Multi-organ injury assessment [URGENT]
- Coagulation profile — DIC screening in severe asphyxia [URGENT]

Counselling

- Baby's brain was injured due to lack of oxygen at birth; cooling therapy protects the brain
- Baby will be cooled for 72 hours then slowly rewarmed; this is a proven treatment
- Baby may have seizures which will be treated with medicines
- Brain MRI at 4-7 days will help understand how much the brain was affected
- Long-term developmental follow-up is essential; early intervention can greatly help

Referral Criteria

Refer to tertiary NICU with therapeutic hypothermia capability within 6 hours of birth; neurology and developmental follow-up

Hospitalisation Criteria

All moderate-severe HIE require NICU admission; therapeutic hypothermia must start within 6 hours

Notes

Eligible: >=36 wk GA with moderate-severe HIE. Requires continuous neuromonitoring (aEEG) and MRI at 4-7 days. Mild HIE not cooled outside trials.

Guideline Changes

AAP 2026 clinical report: hypothermia may benefit if started 6-24h post-birth; mild HIE still not indicated outside trials

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P92.9 — Feeding problems of newborn

SNOMED: 276717003 | Duration: 14 days | Source: AAP, ESPGHAN 2023

First-Line

- **Gavage feeding** — OG/NG tube feeding for preterm <32-34 wks unable to coordinate suck-swallow
- **Breast milk fortifier** — Add HMF when volume reaches 100 mL/kg/day in preterm infants
- **IV fluids/TPN** — D10W + electrolytes; TPN if enteral feeds not tolerated >48-72h

Second-Line

- **Occupational therapy** — Feeding therapy and oral stimulation program for dysphagia
- **Preterm formula** — 24 kcal/oz preterm formula if mother's milk unavailable

Investigations

- Blood glucose — Rule out hypoglycemia contributing to poor feeding [STAT]
- Serum electrolytes — Assess hydration status and electrolyte balance [URGENT]
- Thyroid function test — Rule out congenital hypothyroidism causing poor feeding
- Modified barium swallow — Evaluate swallowing mechanism if aspiration suspected

Counselling

- Premature babies often cannot suck and swallow well; this improves with time
- Mother's own milk is the best food; keep expressing milk regularly even if baby is tube-fed
- Baby will gradually move from tube feeds to breast/bottle as they grow stronger
- Kangaroo care (skin-to-skin) helps baby learn to feed
- Weight gain will be monitored closely; do not worry if it is slow initially

Referral Criteria

Refer to feeding therapist/OT if poor feeding persists beyond expected corrected age; refer to gastroenterology if GERD or anatomical cause suspected

Hospitalisation Criteria

Admit if weight loss >10% birth weight, signs of dehydration, or inability to tolerate any oral feeds

Notes

Prioritize mother's own milk; human milk reduces NEC risk. Advance feeds by 10-30 mL/kg/day as tolerated.

P96.1 — Neonatal withdrawal symptoms from maternal drug use

SNOMED: 414819007 | Duration: 21 days | Source: AAP 2024, Finnegan Score

First-Line

- **Morphine sulfate** — 0.04-0.08 mg/kg PO q3-4h; titrate by 0.04 mg/kg per Finnegan score; wean 10-20%/day
- **Non-pharmacologic care** — Eat-Sleep-Console model: swaddling, low-stimulation, rooming-in, breastfeeding **[NEW 2024-25]**

Second-Line

- **Methadone** — 0.05-0.1 mg/kg PO q6h; longer acting; may reduce treatment duration
- **Clonidine** — 0.5-1 mcg/kg PO q4-6h as adjunct for refractory symptoms
- **Phenobarbital** — Loading 20 mg/kg, maintenance 5 mg/kg/day; for polysubstance or non-opioid withdrawal

Investigations

- Urine drug screen (baby) — Confirm substance exposure [STAT]
- Meconium drug screen — Detects exposure over longer window than urine [URGENT]
- Blood glucose — Hypoglycemia monitoring in withdrawal [STAT]
- Electrolytes — Dehydration from poor feeding and excessive losses

Counselling

- Baby is showing withdrawal symptoms from medicines/substances mother took during pregnancy
- Keeping baby with mother (rooming-in) and breastfeeding help reduce withdrawal symptoms
- Swaddling, quiet environment, and gentle handling comfort the baby
- Medicine may be needed if baby cannot eat or sleep well despite comfort measures
- This is not the baby's fault; the goal is to make baby comfortable while symptoms resolve

Referral Criteria

Refer to neonatologist if high Finnegan scores requiring pharmacotherapy; social work involvement for family support

Hospitalisation Criteria

Admit for observation minimum 3-5 days; longer if pharmacotherapy needed (ESC model)

Notes

Eat-Sleep-Console approach reduces pharmacotherapy use by 50%. Prioritize rooming-in and breastfeeding (if not contraindicated).

Guideline Changes

Eat-Sleep-Console (ESC) approach replacing Finnegan scoring at many centers; reduces pharmacotherapy need

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P96.8 — Shock following birth

SNOMED: 17849001 | Duration: 7 days | Source: NRP 2025, NNF India

First-Line

- **Normal saline bolus** — NS 10-20 mL/kg IV over 15-30 min via UVC; repeat up to 40-60 mL/kg total
- **Epinephrine** — 0.01-0.03 mg/kg IV (0.1-0.3 mL/kg of 1:10,000) via UVC; repeat q3-5 min
- **Positive pressure ventilation** — Effective PPV is the most critical step; initiate within 60 seconds of birth

Second-Line

- **Dopamine** — 5-20 mcg/kg/min IV infusion for fluid-refractory hypotension
- **Packed RBCs** — O-negative 10-20 mL/kg for hemorrhagic shock; uncrossmatched if emergent
- **Hydrocortisone** — 1-2 mg/kg IV for vasopressor-resistant shock; adrenal insufficiency

Investigations

- Blood gas (arterial) — Lactate, pH, and base deficit to assess shock severity [STAT]
- CBC with differential — Hemoglobin for hemorrhagic shock; WBC for sepsis [STAT]
- Blood glucose — Hypoglycemia common in shock [STAT]
- Blood culture — Rule out sepsis as cause [STAT]
- Echocardiography — Evaluate cardiac function and etiology of shock [URGENT]

Counselling

- Baby was very sick at birth and needed emergency resuscitation
- Baby will be monitored closely in the NICU for several days
- Multiple medicines and fluids through IV are needed to support blood pressure and circulation
- Cause of the shock will be investigated and treated specifically
- Recovery depends on the underlying cause; doctor will update you regularly

Referral Criteria

Immediate NICU/PICU care; cardiology if cardiogenic shock suspected

Hospitalisation Criteria

All cases require immediate NICU admission; life-threatening emergency

Notes

Identify shock etiology: hypovolemic, cardiogenic, or distributive. Effective ventilation is the single most important intervention in neonatal resuscitation.

Q25.0 — Patent ductus arteriosus

SNOMED: 83330001 | Duration: 7 days | Source: AAP 2025, Cochrane 2020

First-Line

- **Ibuprofen IV** — 10 mg/kg IV x1, then 5 mg/kg IV q24h x2 doses
- **Acetaminophen IV** — 15 mg/kg IV q6h for 3-7 days; alternative to NSAIDs **[NEW 2024-25]**
- **Indomethacin IV** — 0.2 mg/kg IV q12h x3 doses; adjust by postnatal age

Second-Line

- **Transcatheter closure** — Device closure for hsPDA in preterm >700g; growing evidence **[NEW 2024-25]**
- **Surgical ligation** — Open thoracotomy; reserved for failed medical/catheter therapy
- **Conservative management** — Fluid restriction, diuretics, monitor; many close spontaneously

Investigations

- Echocardiography — Confirm PDA, assess hemodynamic significance [URGENT]
- Chest X-ray — Cardiomegaly, pulmonary congestion
- CBC, RFT — Baseline before pharmacological closure
- BNP/NT-proBNP — Biomarker of hemodynamic significance

Counselling

- PDA is a heart condition where a blood vessel that should close after birth stays open
- Medicine (ibuprofen or paracetamol) is tried first to close the PDA
- If medicine does not work, a small procedure or surgery may be needed
- Watch for fast breathing, poor feeding, or excessive sweating during feeds — these suggest the PDA is significant
- Many small PDAs close on their own and may just need monitoring

Referral Criteria

Refer to pediatric cardiologist for echocardiography-confirmed hemodynamically significant PDA; interventional cardiology if catheter closure considered

Hospitalisation Criteria

Admit for hemodynamically significant PDA with heart failure, initiation of pharmacological closure, or transcatheter/surgical closure

Notes

Treat only hemodynamically significant PDA; echo-guided assessment essential. Early closure <14 days not shown to improve outcomes per AAP 2025.

Guideline Changes

AAP 2025 report: prophylactic treatment no longer recommended; acetaminophen now recognized option

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Q30.0 — Choanal atresia

SNOMED: 890228005 | Duration: 14 days | Source: ENT Expert Consensus 2023

First-Line

- **Surgical repair** — Transnasal endoscopic choanoplasty; gold standard approach
- **Oral airway placement** — McGovern nipple or oral airway as emergency bridge to surgery

Second-Line

- **Nasal stenting** — Controversial; if used, stents left 2-6 weeks post-repair
- **Topical steroid drops** — Combined antibiotic-steroid drops x2 weeks post-op to reduce granulation

Investigations

- CT scan of nose and sinuses — Confirm atresia and plan surgical approach [URGENT]
- Flexible nasoendoscopy — Direct visualization of atretic plate [URGENT]
- Echocardiography — Screen for CHARGE syndrome associated anomalies
- Hearing test (ABR) — CHARGE syndrome screening

Counselling

- Bilateral choanal atresia is a breathing emergency in newborns — the baby cannot breathe through the nose
- An oral airway will be used temporarily until surgery can be done
- After surgery, nasal drops and suctioning will be needed to keep the airway open
- Watch for noisy breathing, feeding difficulty, or blue spells and report immediately
- Follow-up endoscopy will be needed to check for re-narrowing of the nasal passages

Referral Criteria

Refer urgently to pediatric ENT surgeon; emergency if bilateral atresia in neonate (obligate nasal breather)

Hospitalisation Criteria

Admit all bilateral cases immediately (neonatal emergency); unilateral cases may be managed electively but admit if feeding/breathing difficulty

Notes

Bilateral atresia is a neonatal emergency as newborns are obligate nasal breathers. Unilateral cases may be repaired electively.

Guideline Changes

Evidence against routine stenting post-repair growing; stented repairs show higher restenosis

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Q38.1 — Tongue Tie (Ankyloglossia)

SNOMED: 67787004

Counselling

- Tongue tie is a common cause of breastfeeding difficulty in newborns — जीभ का बंधन (टंग टाई) स्तनपान में कठिनाई का आम कारण है
- Mild tongue tie without feeding issues may only need observation — हल्के टंग टाई में सिर्फ़ निगरानी काफ़ी हो सकती है
- Frenotomy is generally safe, quick, and effective when indicated — फ्रेनोटॉमी सुरक्षित, जल्दी और असरदार प्रक्रिया है
- Most babies feed better soon after release procedure — अधिकतर बच्चे प्रक्रिया के बाद बेहतर दूध पीते हैं
- Simple anterior tongue tie usually does not require general anesthesia — साधारण टंग टाई में बेहोशी की ज़रूरत नहीं होती
- Seek review if feeding remains poor or bleeding occurs after procedure — अगर दूध पीने में सुधार न हो या खून आए तो दोबारा दिखाएं

Warning Signs

- Persistent poor latch despite correct positioning — सही स्थिति में भी दूध सही से न पकड़ पाना
- Inadequate weight gain — वज़न पर्याप्त न बढ़ना
- Maternal nipple trauma or bleeding — माँ के निप्पल में दर्द या खून
- Heart-shaped or notched tongue tip — जीभ का दिल जैसा आकार
- Clicking sounds during feeds — दूध पीते समय क्लिक की आवाज़

Q39.0 — Esophageal atresia without fistula

SNOMED: 26179002 | Duration: 21 days | Source: ESPGHAN-NASPGHAN 2023

First-Line

- **Surgical repair** — Primary anastomosis via thoracotomy if gap allows; staged if long-gap EA
- **Replogle tube** — Continuous low suction of upper pouch to prevent aspiration
- **IV fluids/TPN** — NPO with TPN support until surgical repair and feeding tolerance

Second-Line

- **Gastrostomy tube** — G-tube for feeding if delayed primary repair (long-gap EA)
- **Foker process** — Serial traction sutures for long-gap EA to achieve primary anastomosis

Investigations

- Failure to pass NGT (coiling in upper pouch) — Initial diagnostic sign [STAT]
- Chest and abdomen X-ray — NGT coiled in pouch; gas in stomach suggests fistula [STAT]
- Echocardiography — VACTERL cardiac screening [URGENT]

- Renal ultrasound — VACTERL renal anomaly screening [URGENT]
- Spinal X-ray — VACTERL vertebral anomaly screening

Counselling

- Esophageal atresia means the food pipe is not connected — surgery is needed to repair it
- A Replogle tube in the baby's throat continuously suctions saliva to prevent choking
- The baby will be fed through IV or a tube initially; oral feeding begins after surgical repair heals
- After repair, the baby may have reflux — positioning and medications help manage this
- Long-term follow-up with a pediatric surgeon and GI specialist is needed for years

Referral Criteria

Refer immediately to neonatal surgical center; screen for VACTERL association

Hospitalisation Criteria

Admit immediately at birth; NICU care with Replogle tube, NPO, IV fluids, and surgical planning

Notes

Pure EA (no fistula) often has long gap requiring staged repair. Screen for VACTERL; start acid suppression post-repair.

Guideline Changes

Thoracoscopic repair feasibility increasing but with higher complication rates in neonates

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Q39.1 — Tracheoesophageal fistula

SNOMED: 788869006 | Duration: 21 days | Source: ESPGHAN-NASPGHAN 2023

First-Line

- **Surgical repair** — Fistula ligation + esophageal anastomosis via right thoracotomy
- **Replogle tube** — Continuous suction of upper pouch; head-up position to prevent aspiration
- **IV fluids/TPN** — NPO with IV fluids and TPN until postoperative feeding

Second-Line

- **Acid suppression** — PPI or H2 blocker post-repair; GER very common after TEF repair
- **Thoracoscopic repair** — Feasible but technically challenging; higher leak rates in neonates

Investigations

- Echocardiography — VACTERL screening; cardiac anomalies in 50% [URGENT]
- Renal ultrasound — VACTERL screening for renal anomalies
- Spinal X-ray — Screen for vertebral anomalies (VACTERL)
- Contrast esophagogram — Postoperative to check anastomosis before feeding

Counselling

- Baby has a connection problem between the food pipe and windpipe that needs surgery
- Baby cannot be fed by mouth until after surgical repair
- After surgery, baby may have reflux (milk coming back up) which is common and treatable
- Baby will need follow-up for breathing and swallowing problems
- VACTERL association means doctor will check heart, kidneys, spine, and limbs

Referral Criteria

Immediate referral to pediatric surgeon upon diagnosis; prenatal referral to tertiary center if suspected

Hospitalisation Criteria

All cases require NICU admission; surgery within first few days of life

Notes

Urgent but not emergent; stabilize first. 50% have VACTERL association; obtain cardiac echo, renal US, spine imaging.

Q40.0 — Congenital pyloric stenosis

SNOMED: 204671009 | Duration: 7 days | Source: AAP, RCH Guidelines 2024

First-Line

- **Ramstedt pyloromyotomy** — Open or laparoscopic; definitive cure after fluid resuscitation
- **IV fluid resuscitation** — D5 0.45% NaCl at 1.5x maintenance; correct alkalosis and K+ before surgery

Second-Line

- **Atropine sulfate** — 0.01 mg/kg IV/PO before feeds; rarely used, for non-surgical candidates

Investigations

- Abdominal ultrasound (pyloric) — Pyloric muscle thickness >3 mm, channel length >15 mm [URGENT]
- Serum electrolytes and blood gas — Hypochloremic hypokalemic metabolic alkalosis [STAT]
- BUN and creatinine — Assess dehydration severity [URGENT]

Counselling

- Pyloric stenosis means the stomach outlet is too narrow — surgery (pyloromyotomy) is the cure
- Surgery is safe with excellent outcomes — feeding resumes within hours after surgery
- Before surgery, IV fluids will correct the baby's dehydration and electrolyte imbalance
- After surgery, start with small frequent feeds and gradually increase
- Watch for vomiting after discharge — if persistent projectile vomiting returns, contact the surgeon

Referral Criteria

Refer to pediatric surgeon once diagnosed on ultrasound; stabilize with IV fluids and electrolyte correction before transfer

Hospitalisation Criteria

Admit for IV fluid resuscitation, electrolyte correction (chloride, potassium), and pyloromyotomy

Notes

Urgent resuscitation, not emergency surgery. Correct hypochloremic metabolic alkalosis and dehydration before OR.

Q41.0 — Duodenal atresia

SNOMED: 51118003 | Duration: 21 days | Source: Pediatric Surgery Guidelines

First-Line

- **Surgical repair** — Duodeno-duodenostomy (diamond anastomosis); definitive treatment
- **NGT decompression** — Nasogastric tube to low intermittent suction; prevent aspiration
- **IV fluids/TPN** — Stabilize with IV fluids and TPN preoperatively and postoperatively

Second-Line

- **Duodeno-jejunostomy** — Alternative bypass if duodeno-duodenostomy not feasible

Investigations

- Abdominal X-ray — Classic double-bubble sign [STAT]
- Karyotype/microarray — 30% association with Down syndrome [URGENT]
- Echocardiography — Associated cardiac anomalies [URGENT]
- Renal ultrasound — Associated renal anomalies

Counselling

- Duodenal atresia means the baby's intestine is blocked — surgery is the only treatment
- The baby will have a nasogastric tube to drain the stomach before surgery
- After surgery, feeding starts slowly through a tube, then orally as the baby tolerates
- Watch for vomiting, abdominal distension, or feeding intolerance after surgery and report
- Screen for Down syndrome (30% association) and cardiac anomalies is recommended

Referral Criteria

Refer immediately to neonatal surgical center; screen for Down syndrome and associated anomalies

Hospitalisation Criteria

Admit immediately after birth for surgical planning; NPO with NGT decompression and IV fluids/TPN

Notes

Classic double-bubble sign on X-ray. Screen for Down syndrome (30% association) and cardiac anomalies.

Q41.1 — Jejunal atresia

SNOMED: 721650001 | Duration: 21 days | Source: Pediatric Surgery Guidelines

First-Line

- **Surgical repair** — Resection of atretic segment with primary anastomosis; taper if needed
- **NGT decompression** — Nasogastric tube for proximal decompression preoperatively
- **IV fluids/TPN** — Parenteral nutrition until bowel function returns postoperatively

Second-Line

- **Temporary jejunostomy** — Diverting stoma for multiple atresias or compromised bowel

Investigations

- Abdominal X-ray — Dilated proximal bowel with absence of distal gas [URGENT]
- Upper GI contrast study — Identify level and type of atresia [URGENT]
- Echocardiography — Screen for associated cardiac anomalies
- Sweat chloride test — Cystic fibrosis screening if indicated

Counselling

- Baby has a blockage in the small intestine that needs surgery to fix
- After surgery, baby will need IV nutrition for days to weeks until the bowel starts working
- Breastmilk can be started slowly once the doctor gives permission
- Some babies need a longer time to tolerate full feeds; this is normal
- Follow up regularly to monitor growth and nutrition after discharge

Referral Criteria

Immediate referral to pediatric surgeon upon diagnosis

Hospitalisation Criteria

All cases require NICU admission for surgical management

Notes

Dilated proximal bowel may require tapering enteroplasty to improve peristalsis. Prolonged TPN often needed.

Q41.2 — Ileal atresia

SNOMED: 204697003 | Duration: 21 days | Source: Pediatric Surgery Guidelines

First-Line

- **Surgical repair** — Resection of atretic segment with primary end-to-end anastomosis

- **NGT decompression** — Nasogastric decompression; NPO preoperatively
- **IV fluids/TPN** — IV fluid resuscitation and TPN until enteral feeds tolerated

Second-Line

- **Temporary stoma** — Ileostomy for complicated cases; reversed at 6-8 weeks

Investigations

- Abdominal X-ray (erect) — Triple bubble sign / dilated bowel loops [URGENT]
- Upper GI contrast study — Confirm level and type of atresia [URGENT]
- Echocardiography — Screen for associated congenital heart defects
- Sweat chloride test — Rule out cystic fibrosis if meconium ileus suspected

Counselling

- Surgery is necessary and cannot be delayed; the blocked portion of intestine must be removed
- Baby will be on IV nutrition (drip feeding) until the intestine heals and milk feeds can restart
- Recovery takes 2-4 weeks; baby will stay in NICU during this time
- Watch for vomiting, abdominal distension, or green-colored vomit after discharge and report immediately
- Long-term growth and feeding are usually normal after successful surgery

Referral Criteria

Immediate referral to pediatric surgeon at birth upon diagnosis of intestinal atresia

Hospitalisation Criteria

All cases require hospitalization; this is a surgical emergency requiring NICU admission

Notes

Multiple atresias possible; preserve bowel length to prevent short bowel syndrome. Screen for cystic fibrosis if meconium ileus coexists.

Q42.2 — Congenital absence, atresia and stenosis of anus with fistula

SNOMED: 204721004 | Duration: 14 days | Source: Pena Classification, APSA Guidelines

First-Line

- **Surgical repair** — PSARP or primary anoplasty based on fistula type and severity
- **Colostomy** — Diverting colostomy for complex malformations; definitive repair at 3-6 months

Second-Line

- **IV fluids/NPO** — Preop stabilization with IV fluids, NPO, NGT decompression
- **Antibiotics** — Perioperative broad-spectrum antibiotics; ampicillin + gentamicin

Investigations

- Invertogram/cross-table lateral X-ray — Determine level of atresia (high vs low) [URGENT]
- Echocardiography — VACTERL cardiac screening [URGENT]
- Renal ultrasound — VACTERL renal anomaly screening [URGENT]
- Spinal X-ray/MRI — VACTERL vertebral anomalies, tethered cord
- Voiding cystourethrogram — If fistula location needs definition

Counselling

- Your baby has an anorectal malformation — surgery is needed to create a normal anal opening
- A colostomy (temporary stoma) may be needed first, with definitive surgery at 3-6 months
- Learning to care for the colostomy will be taught before discharge
- After definitive repair, regular anal dilatations may be needed for several months
- Long-term follow-up for bowel function is important — constipation and soiling may need management

Referral Criteria

Refer immediately to pediatric surgeon at birth; screen for VACTERL association (vertebral, cardiac, renal, limb anomalies)

Hospitalisation Criteria

Admit at birth for surgical evaluation and management; emergency colostomy if high malformation without perineal fistula

Notes

Perineal fistula can undergo primary repair without colostomy. Screen for VACTERL association with echo, renal US, spine imaging.

Q53 — Undescended Testes (Cryptorchidism)

SNOMED: 204878001

Second-Line

- **hCG** — 250-1500 IU IM 2x/wk for 4 wk; limited role; not recommended by AUA for unilateral UDT
- **GnRH (Buserelin)** — Nasal spray 1.2 mg/day for 4 wk; limited efficacy; not widely recommended

Investigations

- undefined — Bilateral non-palpable testes
- undefined — Suspected anorchia
- undefined — Imaging

Counselling

- Condition is common in newborn males — नवजात लड़कों में यह आम समस्या है
- Most cases resolve by 6 months — अधिकतर 6 महीने तक अपने आप ठीक हो जाती है
- Surgery is simple and safe if required — ज़रूरत हो तो ऑपरेशन सरल और सुरक्षित है
- Early surgery improves fertility — जल्दी ऑपरेशन से प्रजनन क्षमता बेहतर रहती है
- Cancer risk is low but higher than normal — कैंसर का खतरा कम है पर सामान्य से ज़्यादा
- Regular follow-up till puberty is essential — यौवन तक नियमित जांच ज़रूरी है

Warning Signs

- Bilateral non-palpable testes — दोनों तरफ़ अंडकोष न मिलना
- Empty scrotum with ambiguous genitalia — खाली अंडकोष थैली और अस्पष्ट जननांग
- Associated inguinal hernia — साथ में हर्निया होना
- Ascending testis (previously normal) — पहले सामान्य था पर अब ऊपर चला गया
- Testis not descended by 6 months — 6 महीने तक अंडकोष नीचे न आना

Guideline Changes

AUA 2025 guideline update reinforces early orchidopexy by 12 months

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Q62.0 — Antenatally Detected Hydronephrosis

SNOMED: 111631000119106

Investigations

- Ultrasound (USG KUB)
- VCUG (Voiding Cystourethrogram) — Bilateral dilation, ureteral dilation, abnormal bladder, or UTI
- DTPA / MAG3 scan — Obstruction or worsening hydronephrosis
- Serum creatinine — Bilateral or high-grade disease

Counselling

- Most cases resolve without surgery — ज़्यादातर मामले बिना ऑपरेशन के ठीक हो जाते हैं
- Regular follow-up and scans are essential — नियमित जांच और अल्ट्रासाउंड ज़रूरी है
- Watch for signs of UTI such as fever, irritability, or poor feeding — पेशाब के संक्रमण के संकेत देखें जैसे बुखार, चिड़चिड़ापन, या कम दूध पीना
- Serial imaging helps guide treatment decisions — बार-बार की जाने वाली तस्वीरें इलाज का फैसला करने में मदद करती हैं
- Early detection prevents kidney damage — जल्दी पता लगने से गुर्दे की क्षति रुकती है

Warning Signs

- Fever or irritability suggesting UTI — बुखार या चिड़चिड़ापन जो पेशाब के संक्रमण का संकेत हो
- Poor feeding or vomiting — कम दूध पीना या उल्टी
- Palpable abdominal mass — पेट में गांठ महसूस होना
- Poor urine stream or dribbling — पेशाब की कमज़ोर धार या टपकना
- Failure to thrive — वज़न और विकास न बढ़ना
- Decreased urine output — पेशाब कम आना

Q67.3 — Deformational Plagiocephaly

SNOMED: 21880004

Investigations

- Not required in typical cases — routine
- Cranial ultrasound (if fontanelle open) — if doubt
- 3D CT scan (only if craniosynostosis suspected) — if doubt

Counselling

- Condition is common and usually benign — यह आम है और आमतौर पर गंभीर नहीं है।
- Does not affect brain development or intelligence — इससे दिमाग के विकास या बुद्धि पर कोई असर नहीं पड़ता।
- Early repositioning gives excellent outcomes — जल्दी करवट बदलने से बहुत अच्छे नतीजे मिलते हैं।
- Head shape improves gradually with growth — सिर की शक्ल बढ़ने के साथ धीरे-धीरे ठीक होती है।
- Helmet therapy rarely needed and only in selected cases — हेलमेट की ज़रूरत बहुत कम होती है।
- Follow advice consistently for best results — सलाह का नियमित रूप से पालन करें।
- Increase tummy time during awake hours — जागते समय बच्चे को पेट के बल ज़्यादा लिटाएं।

Warning Signs

- Cranial ridging or hard ridge along sutures — सिर पर कड़ी लकीर या उभार
- Progressive worsening despite repositioning — करवट बदलने के बावजूद बिगड़ना
- Developmental delay — विकास में देरी
- Abnormal head circumference growth — सिर की गोलाई असामान्य बढ़ना या न बढ़ना
- Associated syndromic features — अन्य असामान्य लक्षण दिखना

Q79.0 — Congenital diaphragmatic hernia

SNOMED: 17190001 | Duration: 30 days | Source: CDH EURO Consortium 2024, AHA

First-Line

- **Surgical repair** — Primary closure or GORE-TEX patch via thoracotomy after stabilization
- **Gentle ventilation** — Peak pressure <25 cmH2O, permissive hypercapnia, avoid barotrauma

- **Inhaled nitric oxide** — 20 ppm iNO for associated PPHN; wean as oxygenation improves

Second-Line

- **ECMO** — For refractory hypoxemia; 25-30% of CDH neonates require ECMO
- **Surfactant** — Consider if concurrent RDS; not routine for CDH
- **Milrinone** — 0.33-0.99 mcg/kg/min IV for pulmonary hypertension support

Investigations

- Chest X-ray — Bowel loops in chest, mediastinal shift [STAT]
- Echocardiography — Assess pulmonary hypertension and cardiac function [URGENT]
- Blood gas — Assess oxygenation and ventilation [STAT]
- Prenatal ultrasound (if available) — Lung-to-head ratio for prognosis
- Karyotype — Rule out chromosomal abnormalities

Counselling

- CDH is a serious condition where the baby's organs are in the wrong position affecting breathing
- Surgery will be done once the baby is stabilized — typically 2-3 days after birth
- The baby will be on a ventilator initially — this is expected and necessary
- After surgery, the baby may need continued respiratory support for days to weeks
- Long-term follow-up for lung function, feeding, growth, and development is essential

Referral Criteria

Refer immediately to tertiary NICU with ECMO capability and pediatric surgical team

Hospitalisation Criteria

Admit all CDH cases to NICU immediately at birth; transfer to surgical NICU for definitive management

Notes

Stabilize before surgery; avoid bag-mask ventilation. Repair timing when hemodynamically stable, typically 48-72h post-birth.

Guideline Changes

LISA/gentle ventilation strategies increasingly preferred; minimally invasive repair discouraged in neonates

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Q79.2 — Omphalocele

SNOMED: 18735004 | Duration: 28 days | Source: APSA Guidelines

First-Line

- **Surgical closure** — Primary fascial closure if defect small (<4 cm) and stable infant
- **Staged silo closure** — Silo/mesh for giant omphalocele; gradual reduction over days-weeks
- **IV fluids/TPN** — IV fluid resuscitation; TPN until enteral feeding established

Second-Line

- **Paint and wait** — Topical silver sulfadiazine/povidone-iodine; epithelialization for giant defects
- **Antibiotics** — Perioperative broad-spectrum coverage; ampicillin + gentamicin

Investigations

- Echocardiography — 50% have associated cardiac defects; Beckwith-Wiedemann screening [URGENT]
- Renal ultrasound — Screen for genitourinary anomalies
- Blood glucose monitoring — Beckwith-Wiedemann syndrome causes hypoglycemia [STAT]
- Karyotype / Microarray — High association with chromosomal abnormalities

Counselling

- Baby is born with abdominal organs outside the belly covered by a thin sac; surgery is needed

- Large defects may need multiple surgeries over weeks
- Baby will be in NICU for several weeks; parents can visit and participate in care
- Doctor will check for associated heart or kidney problems
- Long-term outcome is usually good once the defect is closed

Referral Criteria

Prenatal diagnosis should trigger referral to tertiary center with pediatric surgery and NICU for planned delivery

Hospitalisation Criteria

All cases require immediate NICU admission at birth

Notes

Giant omphalocele (>5 cm) often requires staged approach. Screen for Beckwith-Wiedemann syndrome and cardiac anomalies.

Q79.3 — Gastroschisis

SNOMED: 72951007 | Duration: 30 days | Source: APSA Guidelines 2024

First-Line

- **Silo placement** — Preformed spring-loaded silo; bedside placement, gradual reduction over 3-7 days
- **Primary closure** — Direct fascial closure if bowel edema minimal; in OR under GA
- **IV fluids/TPN** — Aggressive fluid resuscitation (up to 2-3x maintenance); TPN until feeds

Second-Line

- **Sutureless closure** — Umbilical cord or ward closure techniques; emerging evidence **[NEW 2024-25]**
- **Antibiotics** — Broad-spectrum prophylaxis; ampicillin + gentamicin perioperatively

Investigations

- Prenatal ultrasound — Diagnosis usually made antenatally
- CBC, blood gas, electrolytes — Assess fluid status and third-spacing [STAT]
- Blood culture — If sepsis suspected [URGENT]
- Abdominal X-ray (post-reduction) — Assess bowel gas pattern and complications

Counselling

- Gastroschisis means the baby's intestines are outside the body through a hole near the belly button
- Surgery will place the intestines back and close the abdomen — this may take several days if done gradually
- The baby will need IV nutrition (TPN) until the intestines start working, which can take weeks
- Watch for feeding tolerance milestones — this determines when the baby can go home
- Long-term outcomes are generally good — most children grow and develop normally

Referral Criteria

Refer immediately to neonatal surgical center prenatally or at birth; planned delivery at surgical center is ideal

Hospitalisation Criteria

Admit immediately at birth; NICU with aggressive IV fluid resuscitation, bowel protection (warm saline wrap), and surgical planning

Notes

Cover exposed bowel with warm saline-soaked gauze and plastic wrap immediately. High fluid losses require aggressive IV resuscitation.

Guideline Changes

Bedside spring-loaded silo increasingly preferred over OR primary closure

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R01.1 — Murmur in a Neonate

SNOMED: 88610006

Investigations

- Pulse oximetry screening
- Chest X-ray
- ECG
- Echocardiography
- Pediatric cardiology referral

Counselling

- Not all murmurs are dangerous; some resolve spontaneously. — सभी दिल की आवाज़ें खतरनाक नहीं होतीं; कुछ अपने आप ठीक हो जाती हैं।
- If echocardiography is advised, it is to ensure the baby's heart is normal. — अगर इको टेस्ट बोला गया है तो यह बच्चे के दिल की जांच के लिए है।
- Explain warning signs: cyanosis, breathing difficulty, poor feeding. — खतरे के निशान: नीलापन, सांस में तकलीफ, दूध न पीना।
- Encourage regular follow-up even after a normal echocardiogram. — सामान्य रिपोर्ट आने पर भी नियमित जांच कराएं।

Warning Signs

- Blue discoloration of lips or tongue — होंठ या जीभ का नीला पड़ना
- Fast or difficulty breathing — तेज़ या मुश्किल सांस लेना
- Poor feeding or excessive sweating during feeds — दूध न पीना या दूध पीते समय बहुत पसीना आना
- Lethargy or reduced activity — सुस्ती या कम हिलना-डुलना
- Cold hands and feet — हाथ-पैर ठंडे पड़ना

R06.1 — Noisy Breathing in a Neonate

SNOMED: 267036007

Investigations

- Pulse oximetry
- Nasal catheter test
- Flexible laryngoscopy
- Chest X-ray / Neck X-ray
- Barium swallow
- Polysomnography

Counselling

- Many neonates have immature, soft airways causing noise. — कई नवजात शिशुओं की सांस की नली कोमल होती है जिससे आवाज़ आती है।
- Laryngomalacia peaks at 6 months and resolves by 12-18 months. — लैरिंगोमलेशिया 6 महीने में सबसे ज़्यादा होता है और 12-18 महीने तक ठीक हो जाता है।
- Most cases are self-limiting. — ज़्यादातर मामले अपने आप ठीक हो जाते हैं।
- Watch for warning signs like bluish lips, chest indrawing, choking. — नीले होंठ, छाती धंसना, या दम घुटना दिखे तो तुरंत डॉक्टर को दिखाएं।
- Feed baby in upright position to reduce noisy breathing. — बच्चे को ऊपर उठाकर (बैठी स्थिति में) दूध पिलाएं।

Warning Signs

- Blue discoloration of lips — होंठों का नीला पड़ना

- Chest retractions or indrawing — छाती का धंसना
- Apnea or breathing pauses — सांस रुकना
- Choking or aspiration during feeds — दूध पीते समय दम घुटना
- Poor weight gain — वज़न न बढ़ना
- Stridor at rest or worsening — आराम में भी सांस की आवाज़ या बढ़ना

R10.4 — Infantile Colic

SNOMED: 48340000

Counselling

- Colic peaks at 6-8 weeks and usually resolves by 3-4 months. — कोलिक 6-8 हफ़्ते में सबसे ज़्यादा होता है और 3-4 महीने तक अपने आप ठीक हो जाता है।
- The crying is not due to poor parenting, pain or hunger in most cases. — ज़्यादातर मामलों में रोना खराब परवरिश, दर्द या भूख की वजह से नहीं होता।
- There are no long-term consequences for baby's health. — बच्चे की सेहत पर इसका कोई लंबे समय का असर नहीं होता।
- Parents should practice self-care, share caregiving and seek support. — माता-पिता अपना ख़्याल रखें, बारी-बारी से बच्चे को संभालें और मदद लें।
- Never shake the baby. — बच्चे को कभी झटके से न हिलाएं।

Warning Signs

- Persistent vomiting
- Blood in stool
- Fever or lethargy
- Poor weight gain or failure to thrive
- Excessive sleepiness or unresponsiveness
- Abdominal distension
- Refusal to feed

R25.8 — Seizure Mimics in a Neonate

SNOMED: 60238002

Counselling

- Many newborn movements are normal or benign. — नवजात शिशु की कई हलचलें सामान्य होती हैं।
- Seizure mimics do not usually cause brain damage. — ये दौरे जैसी हलचलें आमतौर पर दिमाग को नुकसान नहीं पहुँचाती।
- Avoid unnecessary panic or medication without confirmation. — बिना पुष्टि के घबराएं नहीं और दवा शुरू न करें।
- Observe for red flags such as apnea, cyanosis, feeding difficulty, eye deviation, or persistent rhythmic movements. — सांस रुकना, नीलापन, दूध न पीना, आंखों का टेढ़ा होना जैसे लक्षणों पर ध्यान दें।
- Most mimics resolve with time. — ये अधिकतर अपने आप ठीक हो जाती हैं।

Warning Signs

- Apnea with bradycardia or cyanosis — सांस रुकना, धड़कन धीमी होना या नीलापन
- Eye deviation or lip smacking — आंखों का टेढ़ा होना या होंठ चटकाना
- Persistent tonic or clonic movements — लगातार अकड़न या झटके
- Non-suppressible movements — पकड़ने पर भी न रुकने वाली हलचल
- Poor feeding or lethargy — दूध न पीना या सुस्ती

- Abnormal neurological examination — असामान्य तंत्रिका जांच

Z00.1 — Umbilical Cord Care

SNOMED: 268525005

Second-Line

- **Dicyclomine** — 5-7 drops TDS PO only if symptomatic colic, short duration only

Counselling

- Cord will dry and fall off on its own; do not pull it — नाल सूखकर खुद गिर जाएगी; खींचें नहीं
- Keep the area clean and dry at all times — जगह को हमेशा साफ़ और सूखा रखें
- Fold diaper below the stump to avoid moisture — डायपर को नाल के नीचे मोड़कर रखें
- Avoid any home remedies or powders — कोई घरेलू उपचार या पाउडर न लगाएं
- Mild discharge or bleeding after fall is normal — नाल गिरने के बाद हल्का रिसाव या खून सामान्य है
- Seek medical help if redness, pus, foul smell, fever, or excessive crying occurs — लालिमा, पीप, बदबू, बुखार या ज़्यादा रोना हो तो डॉक्टर को दिखाएं

Warning Signs

- Redness extending beyond base of stump — नाल के आस-पास लालिमा फैलना
- Purulent or foul-smelling discharge — पीप या बदबूदार रिसाव
- Fever, lethargy, poor feeding — बुखार, सुस्ती, दूध न पीना
- Persistent bleeding — लगातार खून आना
- Cord not falling off by 3-4 weeks — 3-4 हफ़्ते में नाल न गिरना

Guideline Changes

IAP 2024: Vitamin D 400 IU/day universally recommended from birth; iron supplementation from 6 weeks for LBW/preterm, 6 months for term; emphasis on developmental milestone screening at every well-baby visit using validated tools. WHO 2024 growth standards reaffirmed.

Z00.110 — Red Flag Signs in the Neonatal OPD

SNOMED: 267038008

Investigations

- CBC
- CRP
- Blood culture
- Blood sugar
- Serum bilirubin
- Sepsis screen
- Chest X-ray
- Abdominal X-ray
- ABG
- Neurosonography / imaging as clinically indicated

Counselling

- Teach parents to recognize red flag signs — माता-पिता को खतरे के निशान पहचानना सिखाएं।
- Explain urgency of seeking care immediately if warning signs appear — खतरे के निशान दिखें तो तुरंत डॉक्टर के पास जाएं।
- Provide printed or digital list of warning signs — खतरे के निशानों की सूची लिखकर या फ़ोन में दें।

- Reassure that early visits are encouraged even if unsure — अगर शक हो तो भी बच्चे को जल्दी दिखाना सही है।
- Keep hospital emergency numbers saved in phone — अस्पताल के इमरजेंसी नंबर फ़ोन में सेव रखें।

Warning Signs

- Poor feeding or refusal to feed — दूध न पीना या मना करना
- Fever (>=38 degrees C) or hypothermia (<36.5 degrees C) — बुखार या शरीर ठंडा पड़ना
- Fast breathing or chest retractions — तेज़ सांस या छाती धंसना
- Blue discoloration of lips or body — होंठ या शरीर का नीला पड़ना
- Seizures or abnormal movements — दौरे या असामान्य हरकत
- Lethargy or excessive sleepiness — सुस्ती या बहुत ज़्यादा सोना
- Bilious (green) vomiting — हरी उल्टी
- Abdominal distension — पेट फूलना
- Blood in stools — पोटी में खून
- Bulging fontanel — तालू का उभरना

Z13.228 — Newborn Screening (NBS) Repeat and Follow-up

SNOMED: 60151004

Counselling

- Repeat testing ensures accurate diagnosis — दोबारा जांच से सही पहचान होती है।
- Repeat does not always indicate disease — दोबारा जांच का मतलब बीमारी नहीं है।
- Early detection improves outcomes — जल्दी पहचान से इलाज बेहतर होता है।
- Strict follow-up is essential — समय पर जांच कराना बहुत ज़रूरी है।
- Keep all reports safely for every visit — सभी रिपोर्ट संभालकर रखें और हर बार लाएं।

Warning Signs

- Poor feeding or vomiting — दूध न पीना या उल्टी
- Lethargy or excessive sleepiness — सुस्ती या बहुत ज़्यादा सोना
- Seizures — दौरे पड़ना
- Jaundice worsening — पीलिया बढ़ना
- Failure to gain weight — वज़न न बढ़ना
- Unusual body odor — शरीर से अजीब गंध

Z13.5 — Newborn Hearing Screening and Follow-up

SNOMED: 417491009

Counselling

- Screening is safe, painless, and quick — जांच सुरक्षित, दर्दरहित और जल्दी होती है।
- Failure does not always mean permanent hearing loss — फेल होने का मतलब हमेशा बहरापन नहीं है।
- Follow-up is essential — फॉलो-अप ज़रूरी है।
- Early detection improves speech and language development — जल्दी पहचान से बोलने और भाषा के विकास में सुधार होता है।
- The 1-2-3 rule: Screen by 1 month, diagnose by 2 months, intervene by 3 months — 1-2-3 नियम: 1 महीने तक जांच, 2 महीने तक पहचान, 3 महीने तक इलाज शुरू।

Warning Signs

- No startle to loud sound by 1 month — 1 महीने तक तेज़ आवाज़ पर न चौंकना

- No babbling by 6 months — 6 महीने तक बोलने की कोशिश न करना
- No response to name by 9 months — 9 महीने तक नाम पर प्रतिक्रिया न देना
- No words by 12 months — 12 महीने तक कोई शब्द न बोलना
- Parent concern about hearing — माता-पिता को सुनने में संदेह

Z39.1 — Breastfeeding Support (Positioning, Latching and Techniques)

SNOMED: 243094000

Counselling

- Breastfeeding is a learned skill that improves with time — स्तनपान एक सीखने वाली कला है जो समय के साथ बेहतर होती है
- Feed on demand and watch baby cues — बच्चे की ज़रूरत के अनुसार दूध पिलाएं और उसके संकेत देखें
- Avoid pacifiers in first 4-6 weeks — पहले 4-6 हफ़्ते चुसनी न दें
- Seek help early if pain or poor feeding — अगर दर्द हो या बच्चा ठीक से दूध न पिए तो जल्दी मदद लें
- Exclusive breastfeeding recommended for first 6 months — पहले 6 महीने केवल माँ का दूध पिलाएं

Warning Signs

- Baby not waking for feeds for more than 4 hours — बच्चा 4 घंटे से ज़्यादा दूध के लिए न जागे
- Less than 6 wet diapers in 24 hours after day 4 — चौथे दिन के बाद 24 घंटे में 6 से कम गीले डायपर
- Poor weight gain or weight loss after day 5 — पांचवें दिन के बाद वज़न न बढ़ना या घटना
- Persistent clicking sounds during feeding — दूध पीते समय लगातार क्लिक की आवाज़
- Cracked or bleeding nipples — फटे या खून निकलते निप्पल
- Baby appears jaundiced or lethargic — बच्चा पीला या सुस्त दिखे

Z41.2 — Circumcision in Neonates

SNOMED: 72310004

Counselling

- Mild redness, swelling, and yellow film can be part of normal healing — हल्की लालिमा, सूजन और पीली परत सामान्य ठीक होने का हिस्सा है
- Bleeding should stop with gentle pressure — हल्का दबाव देने से खून रुक जाना चाहिए
- Seek urgent review if baby does not pass urine, develops fever, or has continuous bleeding — अगर बच्चा पेशाब न करे, बुखार हो, या लगातार खून बहे तो तुरंत डॉक्टर को दिखाएं
- Do not apply home remedies, powders, or antiseptics unless advised — बिना डॉक्टर की सलाह के घरेलू उपचार, पाउडर या एंटीसेप्टिक न लगाएं
- Apply petroleum jelly to prevent sticking to diaper — डायपर से चिपकने से बचाने के लिए वैसलीन लगाएं
- Follow-up if healing appears delayed or abnormal — अगर ठीक होने में देरी लगे या असामान्य लगे तो दोबारा दिखाएं

Warning Signs

- Continuous bleeding that does not stop with pressure — लगातार खून बहना जो दबाव से न रुके
- No urine output for more than 6-8 hours — 6-8 घंटे से ज़्यादा पेशाब न आना
- Fever — बुखार
- Pus or foul-smelling discharge — मवाद या बदबूदार रिसाव
- Excessive swelling or redness — बहुत ज़्यादा सूजन या लालिमा
- Baby refusing feeds or becoming lethargic — बच्चे का दूध न पीना या सुस्त हो जाना

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null — Sacral Dimple (Neonate)

SNOMED: 311897005

Investigations

- High-risk sacral dimple — indications
- Deep or atypical dimple — indications
- Associated skin or neurological findings — indications
- Spinal Ultrasound (<3 months) — recommended tests
- MRI Spine (>3 months or if US unavailable) — recommended tests
- Pediatric Neurology / Neurosurgery referral if abnormal — recommended tests

Counselling

- Most sacral dimples are benign and harmless — अधिकतर सैक्रल डिम्पल सामान्य और हानिरहित होते हैं
- Small, midline dimples with visible base usually do not need tests — छोटे, बीच में स्थित गड्ढे जिनका तल दिखता है, आमतौर पर जांच की ज़रूरत नहीं
- Certain features may require imaging to rule out spinal problems — कुछ विशेषताओं में रीढ़ की समस्या जांचने के लिए स्कैन ज़रूरी हो सकता है
- Observe for warning signs like limb weakness or urinary issues — पैरों की कमज़ोरी या पेशाब की समस्या जैसे लक्षणों पर ध्यान दें
- Early detection ensures better outcomes — जल्दी पहचान से बेहतर इलाज होता है

Warning Signs

- Limb weakness or asymmetry — पैरों में कमज़ोरी या असमानता
- Bladder or bowel dysfunction — पेशाब या मल त्याग में समस्या
- Foot deformity — पैर की विकृति
- Growing lesion or discharge from dimple — गड्ढे का बढ़ना या उसमें से रिसाव
- Hair tuft or hemangioma near dimple — गड्ढे के पास बालों का गुच्छा या लाल धब्बा

Neurological

A87 — Viral meningitis

SNOMED: 58170007 | Duration: 10 days | Source: IDSA, AAP, WHO 2025

First-Line

- **Ibuprofen** — 10 mg/kg PO q6-8h for headache and fever; max 40 mg/kg/day
- **Acetaminophen** — 15 mg/kg PO q4-6h for analgesia/antipyresis; max 75 mg/kg/day
- **IV fluids** — Maintenance fluids; monitor for SIADH; restrict if serum Na <135 mEq/L

Second-Line

- **Ondansetron** — 0.15 mg/kg IV/PO q8h; max 4 mg/dose; for nausea/vomiting
- **Acyclovir** — 10 mg/kg IV q8h if HSV-2 meningitis confirmed or immunocompromised patient

Investigations

- Lumbar puncture with CSF analysis — Lymphocytic pleocytosis, normal glucose, mildly elevated protein [STAT]
- CSF enterovirus PCR — Most common cause of viral meningitis in children [URGENT]
- CBC, CRP — Normal/mildly elevated CRP helps distinguish from bacterial [URGENT]
- Blood glucose (paired with CSF) — CSF/blood glucose ratio >0.6 suggests viral [STAT]

Counselling

- Child has a viral infection of the brain covering; this is usually milder than bacterial meningitis
- Treatment is mainly supportive: rest, fluids, and pain medicine
- Most children recover completely within 1-2 weeks
- Keep child comfortable; give pain medicine regularly for headache
- Return to hospital if child becomes more drowsy, has seizures, or symptoms worsen

Referral Criteria

Refer to pediatric neurologist if altered consciousness, seizures, or not improving within expected timeframe

Hospitalisation Criteria

Admit neonates, immunocompromised, altered mental status, or if unable to differentiate from bacterial meningitis

Notes

Most cases are self-limited and resolve in 1-2 weeks with supportive care. Admit neonates, immunocompromised, or those with altered mental status or seizures for observation.

Guideline Changes

WHO launched first-ever meningitis diagnosis/treatment/care guidelines April 2025

F80 — Specific developmental disorders of speech and language

SNOMED: 229721007 | Duration: 365 days | Source: ASHA 2024 CPG, German Interdisciplinary CPG 2024

First-Line

- **Parent-based language intervention** — Structured parent training (e.g. Heidelberg model) for expressive language delay; start by age 2-3
- **Speech-language therapy** — Direct child-centered therapy by SLP; 1-3 sessions/wk; phonological and language-focused
- **Phonological therapy** — Targeted phonological or integrated phonological treatment for speech sound disorders

Second-Line

- **Augmentative communication (AAC)** — Picture exchange, speech-generating devices for severe expressive impairment
- **Oral motor therapy** — Motor approach for isolated phonetic speech sound disorders

Investigations

- Hearing assessment (OAE/BERA) — Mandatory for all children with speech delay; hearing loss is leading treatable cause [URGENT]
- Developmental assessment — Evaluate for global developmental delay vs isolated language delay
- Autism screening (M-CHAT) — Rule out ASD which commonly presents with speech delay
- Thyroid function test — Rule out hypothyroidism

Counselling

- Child has delayed speech or language development; early therapy gives the best results
- Talk to your child frequently; narrate daily activities in simple language
- Reduce screen time; face-to-face interaction is best for language learning
- Read picture books together daily; let child point and name objects
- Speech therapy should be regular (1-3 times per week); practice exercises at home too

Referral Criteria

Refer to speech-language pathologist by age 2 if not using words; audiology for all speech-delayed children

Hospitalisation Criteria

Hospitalization not required for speech/language disorders

Notes

Begin intervention by age 2-3 as only ~1/3 of late talkers catch up spontaneously. For multilingual children, provide therapy in all languages when possible.

Guideline Changes

2024 interdisciplinary CPG published with updated intervention recommendations for DLD

F82 — Developmental disorder of motor function

SNOMED: 268674003 | Duration: 365 days | Source: APTA Pediatric CPG 2020, EACD 2019

First-Line

- **Task-oriented PT** — Physical therapy with task-specific motor training; evidence-based per APTA CPG; 1-3x/wk
- **Occupational therapy** — Activity-based intervention targeting daily living skills and fine motor tasks
- **Parent education/home program** — Structured home exercises and environmental modifications per therapist guidance

Second-Line

- **CO-OP approach** — Cognitive Orientation to Occupational Performance; child-directed problem solving for motor tasks
- **Methylphenidate** — 0.3-1 mg/kg/day PO for comorbid ADHD if present; start low and titrate

Investigations

- Movement ABC-2 — Standardized motor assessment for DCD diagnosis
- Vision screening — Rule out visual problems affecting coordination
- ADHD screening (Conners/Vanderbilt) — 50% comorbidity with ADHD
- Cognitive assessment — Rule out intellectual disability; identify learning needs

Counselling

- Child has difficulty with coordination and motor skills; this is called DCD (developmental coordination disorder)
- Occupational therapy and physical therapy are the main treatments
- Practice motor skills through fun activities and play at home
- This is not laziness; child is trying hard but their brain processes movement differently
- Many children improve significantly with consistent therapy; be patient and encouraging

Referral Criteria

Refer to occupational therapist and physiotherapist; screen for ADHD and learning disabilities

Hospitalisation Criteria

Hospitalization not typically required

Notes

Screen for comorbid ADHD, ASD, and language disorders which are common. No pharmacotherapy for DCD itself; intervention focuses on task-oriented motor skill training.

F90.9 — ADHD

SNOMED: 406506008 | Duration: 180 days | Source: AAP 2019

First-Line

- **Methylphenidate short-acting** — Start 5mg BID for age 6yr+, titrate by 5mg/week, max 60mg/day
- **Methylphenidate LA** — 18-54mg once daily AM, Concerta or Ritalin LA

Second-Line

- **Atomoxetine** — 0.5 mg/kg/day for 4 weeks then 1.2 mg/kg/day (max 100mg), non-stimulant
- **Guanfacine ER** — 1mg/day HS, titrate to 1-4mg/day, non-stimulant alpha-2 agonist

Investigations

- Vanderbilt Assessment Scale (parent and teacher) — Standardised ADHD rating for diagnosis and monitoring
- Vision and hearing screening — Rule out sensory deficits mimicking inattention
- Thyroid function tests — If clinical suspicion of thyroid disorder
- ECG — Before starting stimulants if family history of cardiac disease or sudden death

Counselling

- ADHD is a medical condition, not bad behaviour - the child needs support, not punishment
- Medicine works best when combined with behavioural strategies and structured routines
- Give methylphenidate in the morning; monitor appetite and sleep
- Regular follow-up every 3 months to check weight, height, blood pressure and heart rate
- Inform school teachers about the condition; structured classroom seating and shorter task durations help

Referral Criteria

Refer to child psychiatrist/developmental pediatrician for initial diagnosis, comorbid conditions (anxiety, learning disability, autism), medication non-response, or significant side effects

Hospitalisation Criteria

Hospitalisation rarely needed; consider if acute suicidal ideation, severe aggression with safety concerns, or need for inpatient behavioural assessment

Notes

Behaviour therapy first-line for age 4-5yr. Stimulants first-line for age 6yr+. Monitor growth, BP, HR.

G00 — Bacterial meningitis, not elsewhere classified

SNOMED: 95883001 | Duration: 14 days | Source: IDSA 2004, NICE 2024 (NG240), AAP

First-Line

- **Ceftriaxone** — 100 mg/kg IV day 1 then 50 mg/kg IV q12h; max 4 g/day; 10-14 days
- **Vancomycin** — 15 mg/kg IV q6h; target trough 15-20 mcg/mL; add empirically for DRSP coverage
- **Dexamethasone** — 0.15 mg/kg IV q6h x 2-4 days; give 10-20 min before or with first antibiotic dose

Second-Line

- **Cefotaxime** — 50 mg/kg IV q6h; max 12 g/day; alternative to ceftriaxone
- **Meropenem** — 40 mg/kg IV q8h; max 6 g/day; for resistant organisms or penicillin allergy

Investigations

- Lumbar puncture with CSF analysis — Cell count, protein, glucose, gram stain, culture; do not delay antibiotics if LP delayed [STAT]
- Blood culture — Positive in 50-80% of bacterial meningitis [STAT]
- CBC, CRP, procalcitonin — Inflammatory markers [STAT]
- CSF PCR panel — Rapid pathogen identification [URGENT]
- Brain CT/MRI — If focal signs, papilledema, or clinical deterioration [URGENT]

Counselling

- Child has a serious bacterial infection of the brain lining; strong IV antibiotics are needed immediately
- Treatment requires at least 10-14 days of IV antibiotics in hospital
- Steroid medicine is given to reduce brain swelling and protect hearing
- Hearing test must be done before discharge as hearing loss can occur
- Vaccinations (pneumococcal, meningococcal, Hib) are the best prevention for close contacts and future protection

Referral Criteria

Immediate referral to pediatric ICU; neurology consultation; audiology before discharge

Hospitalisation Criteria

All cases require emergency admission; ICU if altered consciousness, shock, or seizures

Notes

Do not delay antibiotics for LP; start empiric therapy immediately if meningitis suspected. Adjunctive dexamethasone reduces hearing loss especially in H. influenzae meningitis.

Guideline Changes

NICE updated meningitis guidelines March 2024 with revised empiric antibiotic recommendations

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G03 — Meningitis due to other and unspecified causes

SNOMED: 7180009 | Duration: 14 days | Source: IDSA 2004, AAP, NICE 2024

First-Line

- **Ceftriaxone** — 100 mg/kg IV day 1 then 50 mg/kg IV q12h; empiric until etiology clarified; max 4 g/day
- **Vancomycin** — 15 mg/kg IV q6h empirically; de-escalate once organism identified
- **Acyclovir** — 10 mg/kg IV q8h empirically if viral (HSV) etiology possible; pending PCR

Second-Line

- **Dexamethasone** — 0.15 mg/kg IV q6h x 2-4 days if bacterial etiology suspected
- **Ampicillin** — 50 mg/kg IV q6h; add for Listeria coverage in neonates and immunocompromised

Investigations

- Lumbar puncture with CSF analysis — Cell count, protein, glucose, gram stain, culture, viral PCR [STAT]
- Blood culture — Identify bacteremia [STAT]

- CBC, CRP — Distinguish bacterial from viral [STAT]
- CSF PCR (multiplex) — Rapid pathogen identification panel [URGENT]
- Brain MRI — If atypical course or focal signs

Counselling

- Child has an infection of the brain covering; cause is being investigated
- Antibiotics and antiviral medicines are started until the exact cause is known
- Blood and spinal fluid tests will identify whether bacteria, virus, or other cause is responsible
- If it is viral meningitis, it usually gets better on its own in 1-2 weeks
- Watch for worsening headache, neck stiffness, drowsiness, or seizures and come immediately

Referral Criteria

Refer to pediatric neurologist and infectious disease; ICU if clinical deterioration

Hospitalisation Criteria

All suspected meningitis cases require admission until etiology is established

Notes

Start empiric antibiotics plus acyclovir until CSF culture, PCR, and gram stain clarify etiology. Up to 92% of aseptic meningitis cases receive empirical antibiotics pending workup.

Guideline Changes

NICE 2024 guidelines updated empiric treatment recommendations

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G04 — Encephalitis, myelitis and encephalomyelitis

SNOMED: 45170000 | Duration: 21 days | Source: IDSA 2008, AAP

First-Line

- **Acyclovir** — Neonates: 20 mg/kg IV q8h x 21 days; Children >3mo: 10 mg/kg IV q8h x 14-21 days
- **Ceftriaxone** — 100 mg/kg IV day 1 then 50 mg/kg IV q12h empirically until bacterial meningitis excluded

Second-Line

- **IV methylprednisolone** — 20-30 mg/kg/day IV x 3-5 days for autoimmune/ADEM; max 1 g/day
- **IVIG** — 2 g/kg IV over 2-5 days for autoimmune encephalitis or ADEM refractory to steroids
- **Ganciclovir** — 5 mg/kg IV q12h for CMV encephalitis in immunocompromised

Investigations

- Lumbar puncture with CSF analysis — Lymphocytic pleocytosis; HSV PCR, enterovirus PCR [STAT]
- Brain MRI — Temporal lobe involvement suggests HSV; diffuse ADEM pattern [URGENT]
- EEG — Seizure detection; periodic lateralized discharges in HSV [URGENT]
- Blood culture and CBC — Rule out bacterial cause [STAT]
- Anti-NMDA receptor antibodies — Autoimmune encephalitis panel if viral causes excluded

Counselling

- Child has inflammation of the brain which can be caused by virus or immune system problems
- IV antiviral medicine (acyclovir) is started immediately and continued until tests confirm the cause
- Child may be confused, have seizures, or behave differently during the illness
- Recovery can take weeks to months; rehabilitation may be needed
- Follow-up with neurologist is important for long-term cognitive and developmental monitoring

Referral Criteria

Immediate referral to pediatric neurology and infectious disease; ICU if altered consciousness

Hospitalisation Criteria

All cases require admission; ICU for seizures, altered consciousness, or need for intubation

Notes

Start acyclovir empirically in all suspected encephalitis; do not wait for HSV PCR results. Autoimmune encephalitis (anti-NMDAR) increasingly recognized and requires immunotherapy.

G06 — Intracranial abscess and granuloma

SNOMED: 441806004 | Duration: 42 days | Source: ESCMID 2023, IDSA

First-Line

- **Ceftriaxone** — 100 mg/kg/day IV divided q12h; max 4 g/day; for aerobic gram-negative and streptococcal coverage
- **Metronidazole** — 30 mg/kg/day IV divided q8h; max 4 g/day; excellent anaerobic coverage and CNS penetration
- **Vancomycin** — 60 mg/kg/day IV divided q6h; target trough 15-20 mcg/mL; for MRSA/resistant strep coverage

Second-Line

- **Meropenem** — 120 mg/kg/day IV divided q8h; max 6 g/day; for resistant organisms or Pseudomonas coverage
- **Surgical aspiration/excision** — Stereotactic aspiration for abscess >2.5 cm or causing mass effect; culture-directed therapy
- **Dexamethasone** — 0.5-1 mg/kg/day IV divided q6h short-term for significant perilesional edema and mass effect

Investigations

- MRI brain with contrast — Confirm abscess, assess size, ring enhancement, edema [STAT]
- Blood cultures — Identify causative organism [STAT]
- CT sinuses/mastoids — Identify ENT source of infection [URGENT]
- CRP and ESR — Monitor treatment response [URGENT]
- Stereotactic aspiration culture — Culture-directed antibiotic therapy if surgical drainage done [URGENT]

Counselling

- Your child has a serious infection inside the brain that needs long-term antibiotics through IV
- Treatment usually takes 4-6 weeks or longer - completing the full course is critical
- Surgery may be needed to drain the infection if medicines alone are not enough
- Regular CT or MRI scans will be done to check if the infection is shrinking
- After treatment, your child will need follow-up to watch for seizures or other complications

Referral Criteria

Refer to pediatric neurosurgeon for all brain abscesses; infectious disease specialist for antibiotic guidance; ENT if sinogenic or otogenic source

Hospitalisation Criteria

All intracranial abscesses require hospitalization for IV antibiotics; urgent neurosurgical intervention for abscess >2.5 cm, mass effect, or clinical deterioration

Notes

Empiric triple therapy (cephalosporin + metronidazole + vancomycin) covers most pathogens. Duration is 4-6 weeks post-surgery or 6-8 weeks for medically managed abscesses.

Guideline Changes

ESCMID 2023 published first comprehensive brain abscess guidelines for children and adults

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G24 — Dystonia

SNOMED: 15802004 | Duration: 90 days | Source: Dystonia Coalition, Frontiers in Dystonia 2022

First-Line

- **Trihexyphenidyl** — Start 1 mg/day PO; increase 1 mg q3-5 days to 2 mg TID; max 30 mg TID; 51% response rate
- **Levodopa/carbidopa** — 1-5 mg/kg/day levodopa PO divided TID; diagnostic trial for dopa-responsive dystonia first
- **Botulinum toxin A** — Focal injection into dystonic muscles; dose varies by muscle group; repeat q3 months

Second-Line

- **Baclofen** — Start 5 mg/day PO; increase 5 mg q3-5 days; max 40-80 mg/day divided TID; 10% response rate
- **Deep brain stimulation** — GPi DBS for severe generalized dystonia refractory to medical therapy; best for DYT1+
- **Intrathecal baclofen** — ITB pump for severe dystonia-spasticity; test dose 25-100 mcg intrathecal
- **Clonazepam** — 0.01-0.03 mg/kg/day PO divided BID-TID; adjunctive for dystonia with myoclonus

Investigations

- MRI brain — Structural cause of dystonia
- Genetic testing (DYT gene panel) — Identify genetic dystonia subtypes
- Levodopa trial — Diagnostic trial for dopa-responsive dystonia
- Copper studies (ceruloplasmin, 24h urine copper) — Rule out Wilson disease

Counselling

- Dystonia causes involuntary muscle contractions - your child is not doing this on purpose
- Medicines take several weeks to show full effect; do not stop them suddenly
- A trial of levodopa is important because some types of dystonia respond dramatically to it
- Stress and fatigue can worsen symptoms; ensure adequate rest and a calm environment
- Regular follow-up is needed to adjust medicine doses as your child grows

Referral Criteria

Refer to pediatric movement disorder specialist for classification and genetic testing; neurosurgery referral for DBS evaluation if refractory to medications

Hospitalisation Criteria

Admit for status dystonicus (life-threatening), acute dystonic storm with respiratory compromise, or DBS implantation

Notes

Always trial levodopa first to rule out dopa-responsive dystonia (Segawa disease). Anticholinergics (trihexyphenidyl) are most effective first-line oral agents in children.

G25 — Other extrapyramidal and movement disorders

SNOMED: 267580003 | Duration: 14 days | Source: AAN, AAP, Sanger 2003 review

First-Line

- **Benztropine** — 0.02-0.05 mg/kg/dose IM/IV for acute dystonic reaction; then 1-2 mg PO BID maintenance
- **Diphenhydramine** — 1.25 mg/kg IV/IM for acute EPS; then 5 mg/kg/day PO divided QID; max 300 mg/day
- **Trihexyphenidyl** — Start 1 mg PO daily; titrate by 1 mg q3-5 days; for chronic movement disorders

Second-Line

- **Valbenazine** — VMAT2 inhibitor for tardive dyskinesia; limited pediatric data; FDA-approved adults only **[NEW 2024-25]**

- **Tetrabenazine** — 12.5 mg PO daily; titrate to 12.5 mg TID; for chorea/hyperkinetic disorders; monitor for depression
- **Propranolol** — 1-4 mg/kg/day PO divided BID-TID for essential/action tremor

Investigations

- Drug history review — Identify offending medication causing EPS [STAT]
- Serum ceruloplasmin and copper — Rule out Wilson disease
- Thyroid function tests — Rule out thyroid-related movement disorder
- MRI brain — If structural cause suspected

Counselling

- If your child developed these movements after starting a new medicine, inform the doctor immediately
- Do not stop any prescribed medicine without consulting the doctor first
- These movements are often treatable once the cause is identified
- Keep a diary of when movements are worst and what triggers them
- Ensure adequate sleep as tiredness can worsen movement problems

Referral Criteria

Refer to pediatric neurologist if movement disorder persists beyond 2 weeks, etiology unclear, or not responding to initial treatment

Hospitalisation Criteria

Admit for severe acute dystonic reaction with airway compromise, tardive dyskinesia with functional impairment, or diagnostic workup requiring prolonged monitoring

Notes

Identify and discontinue offending medication for drug-induced movement disorders first. Treatment depends on specific movement disorder type; anticholinergics for acute EPS, VMAT2 inhibitors for tardive syndromes.

Guideline Changes

VMAT2 inhibitors emerging for tardive dyskinesia; novel gene therapies under investigation

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G40 — Epilepsy

SNOMED: 84757009 | Duration: 730 days | Source: ILAE 2022, NICE 2022 (NG217)

First-Line

- **Valproate** — 15-30 mg/kg/day PO divided BID-TID; first-line for generalized epilepsy; monitor LFTs; avoid in pregnancy
- **Levetiracetam** — 10-30 mg/kg/day PO divided BID; titrate over 2-4 wks; first-line for focal and generalized seizures
- **Carbamazepine** — 10-20 mg/kg/day PO divided BID-TID; first-line for focal seizures; check HLA-B*1502 in at-risk groups
- **Ethosuximide** — 20 mg/kg/day PO divided BID; first-line for childhood absence epilepsy; max 40 mg/kg/day

Second-Line

- **Lamotrigine** — 0.3-1 mg/kg/day PO; titrate slowly over 8 wks to 5-15 mg/kg/day; first-line for focal/generalized
- **Oxcarbazepine** — 8-10 mg/kg/day PO divided BID; titrate to 30-46 mg/kg/day; for focal seizures
- **Topiramate** — 1-3 mg/kg/day PO; titrate to 5-9 mg/kg/day; adjunctive or monotherapy for focal/generalized
- **Cenobamate** — Starting dose 12.5 mg/day PO; titrate slowly; for drug-resistant focal epilepsy >=18 yrs; peds trials ongoing **[NEW 2024-25]**

Investigations

- EEG (awake and sleep) — Classify epilepsy syndrome and guide medication choice

- MRI brain (epilepsy protocol) — Identify structural cause of epilepsy
- CBC, LFT, renal function — Baseline before starting antiseizure medicines
- Drug levels — Therapeutic drug monitoring for valproate, carbamazepine, phenytoin

Counselling

- Epilepsy medicine must be taken every day at the same time - never skip doses
- Do not stop the medicine suddenly even if seizures have stopped - this is dangerous
- Keep a seizure diary noting date, time, duration, and description of each episode
- Your child can attend school normally but inform the teacher about the condition and first aid
- Avoid swimming alone and heights; ensure adequate sleep as lack of sleep triggers seizures

Referral Criteria

Refer to pediatric neurologist if seizures not controlled with first medication, diagnostic uncertainty, or consideration of epilepsy surgery

Hospitalisation Criteria

Admit for status epilepticus, cluster seizures, new-onset seizures with altered consciousness, or for EEG monitoring

Notes

Choice of ASM depends on seizure type and epilepsy syndrome; monotherapy is preferred. Avoid valproate in females of childbearing potential due to teratogenicity per ILAE/NICE guidance.

Guideline Changes

ILAE updated initial monotherapy recommendations 2022; cenobamate emerging for refractory epilepsy

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G41 — Status epilepticus

SNOMED: 230456007 | Duration: 3 days | Source: AES 2016, ESETT Trial 2019, NICE 2022

First-Line

- **Lorazepam** — 0.1 mg/kg IV over 2 min; max 4 mg; repeat x1 after 5 min if seizure persists; first-line
- **Midazolam** — 0.2 mg/kg IM/IN (max 10 mg) if no IV access; or 0.1 mg/kg IV; faster onset via IM than IV diazepam
- **Diazepam** — 0.2-0.3 mg/kg IV (max 10 mg) or 0.5 mg/kg PR (max 20 mg); repeat x1 after 5 min

Second-Line

- **Fosphenytoin** — 20 mg PE/kg IV at 3 mg PE/kg/min; max 1500 mg PE; 2nd-line if benzodiazepine fails
- **Levetiracetam** — 60 mg/kg IV over 15 min; max 4500 mg; 2nd-line per ESETT trial; equivalent efficacy to fosphenytoin
- **Valproate** — 20-40 mg/kg IV over 5-10 min; max 3000 mg; 2nd-line; avoid in <2 yrs or liver disease
- **Midazolam infusion** — Load 0.2 mg/kg IV then 0.05-2 mg/kg/hr for refractory SE; titrate to EEG burst suppression

Investigations

- Blood glucose (bedside) — Rule out hypoglycemia as cause/complication [STAT]
- Electrolytes (Na, Ca, Mg) — Rule out metabolic causes [STAT]
- Continuous EEG — Monitor for ongoing subclinical seizures in refractory SE [STAT]
- CT brain — If first presentation or focal signs suggesting structural cause [URGENT]
- Antiepileptic drug levels — Check therapeutic levels in known epilepsy patients [URGENT]

Counselling

- Status epilepticus is a medical emergency - your child needed emergency treatment for prolonged seizures
- If seizure lasts more than 5 minutes at home, give the emergency medicine and call an ambulance
- Keep rescue medicine (midazolam nasal spray or diazepam rectal) with you at all times

- After discharge, regular follow-up with the neurologist is essential to prevent recurrence
- Ensure your child takes daily seizure medicines on time and never misses a dose

Referral Criteria

All status epilepticus cases should be managed by or in consultation with pediatric neurologist; refer for refractory SE or if underlying cause unclear

Hospitalisation Criteria

All status epilepticus requires emergency hospitalization; ICU admission for refractory status epilepticus, respiratory compromise, or need for continuous infusions

Notes

Benzodiazepines must be given within 5 minutes of seizure onset; underdosing is common and increases morbidity. ESETT trial showed levetiracetam, fosphenytoin, and valproate equally effective as second-line agents.

Guideline Changes

ESETT trial confirmed equivalent efficacy of levetiracetam, fosphenytoin, and valproate as 2nd-line

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G43 — Migraine

SNOMED: 37796009 | Duration: 3 days | Source: AAN 2019 (reaffirmed 2025), AHS

First-Line

- **Ibuprofen** — 10 mg/kg PO at onset; max 400 mg; first-line acute treatment
- **Rizatriptan** — 5 mg PO if <40 kg, 10 mg if >=40 kg; FDA-approved for pediatric migraine age >=6
- **Sumatriptan nasal spray** — 5-20 mg intranasal; FDA-approved >=12 yrs for acute migraine
- **Topiramate** — 1-3 mg/kg/day PO divided BID for prophylaxis; only FDA-approved preventive in children 12-17

Second-Line

- **Amitriptyline** — 0.25-1 mg/kg PO qHS for prophylaxis; max 1 mg/kg/day; limited evidence vs placebo
- **Propranolol** — 1-4 mg/kg/day PO divided BID-TID for prophylaxis; monitor HR and BP
- **Almotriptan** — 6.25-12.5 mg PO once; FDA-approved >=12 yrs for acute migraine
- **CBT + amitriptyline** — Combined cognitive behavioral therapy with amitriptyline shown beneficial in CHAMP trial

Investigations

- Clinical diagnosis (ICHD-3 criteria) — Migraine is a clinical diagnosis; no specific test needed
- Brain MRI — Only if atypical features, aura prolonged >60 min, or focal deficits
- Vision screening — Rule out refractive error contributing to headaches

Counselling

- Migraine is a recurring headache condition; it can be managed but not cured
- Take pain medicine within 30 minutes of headache onset for best effect
- Identify and avoid triggers: skipped meals, poor sleep, stress, bright lights, certain foods
- If headaches are frequent (>4/month), preventive medicine may be needed
- Relaxation techniques and regular exercise help reduce migraine frequency

Referral Criteria

Refer to pediatric neurologist if >4 headaches/month, not responding to acute treatment, or atypical features

Hospitalisation Criteria

Admit for status migrainosus (>72h), associated stroke-like symptoms, or intractable vomiting

Notes

Acute treatment should be given early at headache onset for best effect. Evidence for preventive medications is limited; CBT combined with medication may be superior to medication alone.

Guideline Changes

AAN acute and preventive guidelines reaffirmed Oct 2025

G47.9 — Sleep disorder, unspecified

SNOMED: 39898005 | Duration: 30 days | Source: AASM 2022, IPSA Expert Consensus 2025

First-Line

- **Sleep hygiene/behavioral therapy** — Consistent bedtime routine, limit screens 1h before bed; graduated extinction; first-line always
- **Melatonin** — Start 0.5 mg PO 30-60 min before bedtime; titrate weekly to max 5 mg; for sleep-onset insomnia

Second-Line

- **Clonidine** — 0.05-0.1 mg PO qHS for sleep-onset difficulty with comorbid ADHD; max 0.4 mg/day
- **Iron supplementation** — 3-6 mg/kg/day elemental iron PO if ferritin <50 ng/mL and restless legs/PLMD suspected
- **Trazodone** — 1-2 mg/kg PO qHS for sleep-maintenance insomnia; limited pediatric data; off-label use

Investigations

- Sleep diary (2 weeks) — Document sleep-wake patterns before intervention
- Serum ferritin — Low ferritin (<50 ng/mL) associated with restless legs/PLMD
- Polysomnography — If sleep-disordered breathing or periodic limb movements suspected

Counselling

- Good sleep habits are the most important treatment - set a fixed bedtime and wake time every day
- Remove all screens (TV, tablet, phone) from the bedroom and stop screen use 1 hour before bed
- Create a calm bedtime routine: bath, story, dim lights, then sleep
- Avoid caffeinated drinks (cola, tea, coffee, chocolate) especially after lunch
- If using melatonin, give it 30-60 minutes before the desired bedtime

Referral Criteria

Refer to pediatric sleep specialist if behavioral measures and melatonin fail after 4-6 weeks, suspected sleep-disordered breathing, or narcolepsy

Hospitalisation Criteria

Hospitalization rarely needed; consider admission for polysomnography if severe obstructive symptoms or for safety if parasomnia causing injury

Notes

Behavioral interventions should always precede and accompany any pharmacotherapy for pediatric insomnia. Melatonin not recommended under age 2; avoid in typically developing children without clear clinical indication.

Guideline Changes

2025 IPSA expert consensus on melatonin use in typically developing children published

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G61.0 — Guillain-Barre syndrome

SNOMED: 40956001 | Duration: 28 days | Source: AAN 2003, AANS/CNS

First-Line

- **IVIG** — 0.4 g/kg/day IV x 5 days (total 2 g/kg); preferred in children; can also give 1 g/kg/day x 2 days

- **Supportive care** — Monitor respiratory function q4h; intubate if FVC <20 mL/kg; DVT prophylaxis; pain management

Second-Line

- **Plasmapheresis** — 1-1.5 plasma volumes x 5-6 sessions over 10-14 days; for children >10-15 kg; requires central line
- **Gabapentin** — 5-10 mg/kg/day PO divided TID for neuropathic pain; titrate to effect

Investigations

- Nerve conduction study/EMG — Confirm diagnosis and classify subtype (AIDP vs AMAN) [URGENT]
- Lumbar puncture (CSF) — Albuminocytological dissociation (high protein, normal cells) [URGENT]
- Forced vital capacity (FVC) serial — Monitor respiratory function q4-6h [STAT]
- Anti-ganglioside antibodies — Help classify GBS subtype

Counselling

- GBS causes weakness that usually starts in the legs and moves upward - this is expected
- Most children recover fully but recovery can take weeks to months
- If your child has difficulty breathing or swallowing, rush to the hospital immediately
- Physiotherapy during and after recovery is very important to regain strength
- Pain is common in GBS and should be reported to the doctor for treatment

Referral Criteria

All GBS cases should be managed by or in consultation with pediatric neurologist; ICU referral if respiratory function declining

Hospitalisation Criteria

All suspected GBS cases require hospitalization for monitoring; ICU admission if FVC <20 mL/kg, bulbar weakness, rapid progression, or autonomic instability

Notes

IVIG preferred over plasmapheresis in children due to easier administration and fewer side effects. Monitor for respiratory failure and autonomic instability closely; ICU admission for severe cases.

G70 — Myasthenia gravis and other myoneural disorders

SNOMED: 91637004 | Duration: 365 days | Source: MGFA, AAN 2016, Consensus Guidelines 2025

First-Line

- **Pyridostigmine** — 1 mg/kg PO q4-6h; start 30 mg TID then titrate to max 120 mg QID based on symptoms
- **Prednisolone** — 1-2 mg/kg/day PO; start low and increase slowly to minimize transient worsening; taper once stable

Second-Line

- **Azathioprine** — 2-3 mg/kg/day PO as steroid-sparing agent; check TPMT before starting; onset 3-6 months
- **Mycophenolate mofetil** — 600 mg/m2 PO BID; max 2 g/day; steroid-sparing; onset 2-6 months
- **IVIG** — 2 g/kg IV over 2-5 days for exacerbations or pre-thymectomy
- **Eculizumab** — Weight-based IV q2wk for AChR-Ab+ generalized MG refractory to standard therapy; FDA-approved peds **[NEW 2024-25]**

Investigations

- Anti-AChR antibodies — Confirm autoimmune MG diagnosis [URGENT]
- Anti-MuSK antibodies — If AChR-negative and MG suspected
- CT chest — Evaluate for thymoma or thymic hyperplasia
- Repetitive nerve stimulation (RNS) — Electrophysiological confirmation of neuromuscular junction disorder
- Pulmonary function tests — Baseline and serial monitoring of respiratory muscle strength

Counselling

- Myasthenia gravis causes muscle weakness that worsens with activity and improves with rest
- Give medicines exactly on time as even small delays can increase weakness
- Learn the emergency signs: difficulty breathing, unable to swallow, or very weak voice
- Avoid very hot weather and overexertion as these worsen symptoms
- Inform any doctor or dentist about myasthenia before any procedure or new medicine

Referral Criteria

Refer to pediatric neurologist for all new diagnoses; thoracic surgery referral for thymectomy evaluation in generalized MG

Hospitalisation Criteria

Admit for myasthenic crisis (respiratory failure, severe dysphagia), cholinergic crisis, or pre-thymectomy preparation

Notes

Children with ocular MG may achieve spontaneous remission; start with pyridostigmine alone initially. Thymectomy should be considered for generalized MG in children; vaccinate against meningococcus before eculizumab.

Guideline Changes

FDA approved eculizumab expanded indication for pediatric generalized MG

G71 — Primary disorders of muscles

SNOMED: 73297009 | Duration: 365 days | Source: AAN 2016 (DMD corticosteroids), French Soc Ped Neuro 2024

First-Line

- **Prednisone** — 0.75 mg/kg/day PO for Duchenne MD; reduces strength loss; delays cardiomyopathy; long-term use
- **Deflazacort** — 0.9 mg/kg/day PO for DMD; FDA-approved; possibly less weight gain than prednisone
- **Physical therapy** — Regular stretching, strengthening, and cardiopulmonary rehab; prevent contractures; ongoing

Second-Line

- **Ataluren** — 40 mg/kg/day PO divided TID for DMD with nonsense mutations; EMA-approved
- **Eteplirsen** — 30 mg/kg IV weekly for DMD amenable to exon 51 skipping; FDA-approved
- **Cardiac medications** — ACE inhibitor (enalapril 0.1 mg/kg/day) or beta-blocker started by age 10 for cardiomyopathy prevention
- **Delandistrogene moxeparvovec** — Gene therapy for DMD age 4-5; single IV dose 1.33x10^14 vg/kg; FDA-approved 2023 **[NEW 2024-25]**

Investigations

- CK (creatine kinase) — Markedly elevated in DMD/BMD [URGENT]
- Genetic testing (dystrophin gene) — Confirm specific mutation for treatment eligibility
- Echocardiography — Baseline cardiac function; annual from age 10
- Pulmonary function tests — Annual monitoring of respiratory decline
- DEXA bone density — Monitor for steroid-induced osteoporosis

Counselling

- Muscular dystrophy is a genetic condition - steroid medicine slows the progression significantly
- Daily steroid medicine must not be stopped suddenly; your doctor will adjust the dose
- Regular physiotherapy and stretching prevents stiff joints and maintains mobility longer
- Watch for signs of heart or breathing problems: getting tired easily, cough at night, swollen feet

- Genetic counseling is available for family planning and testing other family members

Referral Criteria

Refer to pediatric neurologist and geneticist at diagnosis; cardiology referral by age 10 or at diagnosis for cardiac monitoring

Hospitalisation Criteria

Admit for acute respiratory failure, pneumonia, cardiac decompensation, fractures in osteoporotic bone, or post-surgical care

Notes

Corticosteroids remain the cornerstone of DMD treatment and should be initiated when motor function plateaus. Gene therapies are emerging; specific genetic testing guides exon-skipping and gene therapy eligibility.

Guideline Changes

2024 French guidelines on corticosteroid compound choice and side effect management in DMD

G72 — Myopathy

Duration: 180 days | Source: BSR 2022, SHARE/CARRA 2025

First-Line

- **Prednisolone** — 1-2 mg/kg/day PO (max 60 mg); taper over 6-12 months once response achieved
- **Methotrexate** — 15 mg/m2/wk PO or SC (max 25 mg/wk); preferred steroid-sparing agent; give with folic acid

Second-Line

- **IVIG** — 2 g/kg IV monthly for refractory disease or severe dysphagia; as adjunct to immunosuppression
- **Rituximab** — 375 mg/m2 IV x 4 weekly doses for treatment-resistant IIM
- **Mycophenolate mofetil** — 600 mg/m2 PO BID; max 2 g/day; for skin-predominant or steroid-intolerant disease
- **Cyclophosphamide** — 500 mg/m2 IV monthly x 6 for severe refractory IIM with lung involvement

Investigations

- CK (creatine kinase) — Marker of muscle damage and disease activity [URGENT]
- MRI muscles (thighs) — Assess extent of muscle inflammation and guide biopsy
- Myositis-specific antibodies panel — Classify myositis subtype and prognosis
- Muscle biopsy — Histological confirmation if diagnosis uncertain
- CT chest (HRCT) — Screen for interstitial lung disease

Counselling

- Your child has muscle inflammation that needs long-term treatment to control
- Steroid medicines are essential but have side effects - do not stop them without doctor advice
- Protect your child from sun exposure with sunscreen and clothing as skin is very sensitive
- Regular gentle exercise and physiotherapy help maintain muscle strength
- Blood tests are needed regularly to monitor disease activity and medicine side effects

Referral Criteria

Refer to pediatric rheumatologist for all inflammatory myopathies; dermatology for skin-predominant disease

Hospitalisation Criteria

Admit for severe muscle weakness affecting swallowing or breathing, pulse IV steroid therapy, or disease flare not responding to outpatient treatment

Notes

Juvenile IIM should be managed by pediatric rheumatology specialist. Exercise and physiotherapy across all disease activity levels improves outcomes and quality of life.

Guideline Changes

2025 PRES/CARRA treat-to-target recommendations published for juvenile dermatomyositis

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G80 — Cerebral palsy

SNOMED: 128188000 | Duration: 365 days | Source: AAN 2010, NICE 2017, China CPG 2024

First-Line

- **Baclofen (oral)** — 0.3 mg/kg/day PO divided TID; titrate weekly to 0.75-2 mg/kg/day; max 40 mg/day <8 yrs
- **Botulinum toxin A** — OnabotulinumtoxinA 10-15 U/kg per session for focal spasticity; max 340 U total per session
- **Physical therapy** — Ongoing PT/OT; goal-directed task-specific training; minimum 2-3 sessions/wk
- **Diazepam** — 0.12-0.8 mg/kg/day PO divided TID-QID for generalized spasticity; short-term use preferred

Second-Line

- **Intrathecal baclofen** — ITB pump: test dose 25-100 mcg intrathecal; for severe generalized spasticity refractory to oral meds
- **Tizanidine** — 0.05 mg/kg PO qHS; titrate to 0.05 mg/kg TID; max 0.1 mg/kg TID
- **Dantrolene** — 0.5 mg/kg PO BID; titrate to max 3 mg/kg QID or 400 mg/day; monitor LFTs
- **Selective dorsal rhizotomy** — Surgical option for children with spastic diplegia age 3-8 yrs with good selective motor control

Investigations

- MRI brain — Determine etiology and extent of brain injury
- Hip X-ray (AP pelvis) — Hip surveillance for subluxation/dislocation
- EEG — If seizures suspected
- Swallowing assessment (VFSS) — If feeding difficulties or recurrent pneumonia

Counselling

- Cerebral palsy is a lifelong condition but therapy can greatly improve your child's abilities
- Regular physiotherapy and occupational therapy at home is as important as hospital sessions
- Medicines for stiffness need to be given on time every day and dose changes should only be done by the doctor
- Watch for difficulty swallowing or frequent choking during meals and report to doctor
- Your child can attend school with appropriate support - ask about inclusive education options

Referral Criteria

Refer to pediatric neurologist for classification, botulinum toxin planning, or if seizures develop; orthopedic referral for progressive contractures or hip surveillance

Hospitalisation Criteria

Admit for severe aspiration pneumonia, status epilepticus, post-surgical care (SDR, ITB pump placement), or acute hip dislocation

Notes

Multimodal approach combining physical therapy, orthotics, and pharmacology is standard. BoNT-A is preferred for focal/segmental spasticity; oral agents for generalized spasticity.

Guideline Changes

2024 China neurorestorative therapy CPG published with updated rehabilitation recommendations

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G81 — Hemiplegia

SNOMED: 50582007 | Duration: 365 days | Source: AAN 2010, NICE 2017, China CPG 2024

First-Line

- **Botulinum toxin A** — OnabotulinumtoxinA 10-15 U/kg per session into affected limb muscles; repeat q3-6 months
- **Constraint-induced movement therapy** — CIMT: restrain unaffected limb 2-6 hrs/day for 2-8 wks to promote affected side use
- **Physical therapy** — Task-specific upper and lower limb rehabilitation; 3-5 sessions/wk during intensive programs

Second-Line

- **Baclofen (oral)** — 0.3 mg/kg/day PO divided TID; titrate to 0.75-2 mg/kg/day for associated spasticity
- **Serial casting** — Short leg cast changed weekly x 4-6 wks for ankle equinus contracture after BoNT-A
- **Intrathecal baclofen** — ITB pump for severe refractory spasticity; test dose 25-100 mcg intrathecal

Investigations

- MRI brain — Identify cause of hemiplegia (stroke, malformation)
- EEG — If seizures present or suspected
- Coagulation profile and thrombophilia screen — If stroke suspected as cause
- Echocardiography — If cardioembolic stroke suspected

Counselling

- Encourage your child to use the affected hand/leg during daily activities like eating and dressing
- Home exercises prescribed by the therapist should be done daily for best results
- Botulinum toxin injections help reduce stiffness and need to be repeated every 3-6 months
- Use proper footwear and ankle braces as recommended to prevent falls
- Your child can participate in most activities with appropriate adaptations

Referral Criteria

Refer to pediatric neurologist for etiology workup; orthopedic surgeon for progressive contractures or gait abnormalities

Hospitalisation Criteria

Admit for intensive rehabilitation programs, post-surgical recovery (tendon transfers, SDR), or acute neurological deterioration

Notes

BoNT-A combined with intensive PT or CIMT gives best functional outcomes for hemiplegic CP. Address both upper and lower limb involvement for optimal rehabilitation.

G91 — Hydrocephalus

SNOMED: 267687006 | Duration: 365 days | Source: AANS/CNS 2014 (updated 2021)

First-Line

- **VP shunt** — Ventriculoperitoneal shunt; use antibiotic-impregnated catheter (Level I); programmable valve preferred
- **ETV** — Endoscopic third ventriculostomy for obstructive hydrocephalus; lower infection/reoperation rates

Second-Line

- **ETV + CPC** — ETV with choroid plexus cauterization; reduces CSF production; emerging evidence in infants
- **Acetazolamide** — 25-100 mg/kg/day PO divided TID as temporizing measure; max 2 g/day; monitor electrolytes
- **Neuroendoscopic lavage** — For post-hemorrhagic hydrocephalus; removes intraventricular clots; may reduce shunt need (Level III) **[NEW 2024-25]**

Investigations

- CT brain — Rapid assessment of ventricular size [STAT]
- MRI brain — Detailed assessment of hydrocephalus etiology [URGENT]
- Shunt series X-ray — Assess shunt continuity if malfunction suspected [URGENT]
- Fundoscopy — Screen for papilledema

Counselling

- Learn the danger signs of shunt blockage: sudden headache, vomiting, drowsiness, vision changes
- Bring the child immediately to hospital if these signs appear at any time
- Regular follow-up with neurosurgeon is essential even if the child seems well
- Keep the shunt identification card with you at all times
- Inform any doctor treating your child about the VP shunt before procedures or medications

Referral Criteria

Refer to pediatric neurosurgeon for all new diagnoses; ophthalmology for papilledema monitoring

Hospitalisation Criteria

Admit for acute raised ICP, shunt malfunction or infection, new-onset hydrocephalus, or pre-surgical workup

Notes

Both CSF shunts and ETV are acceptable options (Level II evidence). Shunt infection prevention with antibiotic catheters and strict protocols is a priority.

Guideline Changes

2021 CNS systematic review update reinforced antibiotic-impregnated shunt use

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G93.1 — Anoxic brain damage, not elsewhere classified

SNOMED: 389098007 | Duration: 14 days | Source: AAP 2014 (updated 2026), AHA NRP

First-Line

- **Therapeutic hypothermia** — Cool to 33.5-34.5°C within 6h of birth x 72h then rewarm over 6-12h; for mod-severe HIE >=36wk GA
- **Phenobarbital** — 20 mg/kg IV loading dose for seizures; then 3-5 mg/kg/day maintenance; first-line neonatal ASM
- **Supportive ICU care** — Maintain normoglycemia, normocarbia, avoid hypotension; target MAP for gestational age

Second-Line

- **Levetiracetam** — 40-60 mg/kg IV loading then 20-30 mg/kg/day divided BID for neonatal seizures; preferred if cardiac
- **Erythropoietin (EPO)** — 1000 U/kg IV x 3-5 doses adjunct to cooling; investigational neuroprotectant; mixed trial results **[NEW 2024-25]**

Investigations

- MRI brain — Assess extent of hypoxic-ischemic injury (day 3-5) [URGENT]
- aEEG/continuous EEG — Monitor for seizures during cooling [STAT]
- Blood gas (cord or postnatal) — Assess severity of acidosis [STAT]
- Serum lactate — Marker of tissue hypoxia severity [STAT]

Counselling

- Your baby's brain was affected by low oxygen - cooling treatment helps protect the brain
- Your baby will be closely monitored in the NICU for several days
- Seizures are common after brain injury and will be treated with medicines
- Long-term developmental follow-up is very important even if your baby seems to recover well
- Early intervention therapy (physiotherapy, occupational therapy) should start as soon as recommended

Referral Criteria

Refer to pediatric neurologist for all moderate-severe HIE; developmental pediatrician for long-term follow-up

Hospitalisation Criteria

All moderate-severe HIE requires NICU admission for therapeutic hypothermia, seizure management, and multiorgan support

Notes

Therapeutic hypothermia is standard of care for moderate-severe neonatal HIE within 6 hours of birth. Late cooling (6-24h) may still provide modest benefit per updated 2026 AAP guidance.

Guideline Changes

AAP released updated clinical report on therapeutic hypothermia for neonatal HIE Feb 2026

G93.2 — Benign intracranial hypertension

SNOMED: 68267002 | Duration: 90 days | Source: Alder Hey IIH Guideline 2024, IIH Consensus 2018

First-Line

- **Acetazolamide** — 15-25 mg/kg/day PO divided TID; titrate up to max 100 mg/kg/day (max 2 g/day); monitor electrolytes
- **Weight management** — Target 5-10% weight loss if overweight/obese; dietary counseling; primary intervention in obese teens

Second-Line

- **Topiramate** — 1-3 mg/kg/day PO divided BID; has carbonic anhydrase activity; also aids weight loss
- **Furosemide** — 1-2 mg/kg/day PO divided BID; adjunct to acetazolamide if insufficient response
- **Optic nerve sheath fenestration** — Surgical; for progressive visual loss refractory to medical therapy
- **CSF shunting (LP shunt)** — Lumboperitoneal or VP shunt for refractory headache with visual loss despite medical therapy

Investigations

- MRI brain with MRV — Rule out cerebral venous sinus thrombosis and mass lesion [URGENT]
- Lumbar puncture with opening pressure — Confirm elevated CSF pressure (>28 cmH2O in children) [URGENT]
- Visual field perimetry — Baseline and serial monitoring of visual field loss [URGENT]
- Fundoscopy — Grade papilledema severity [URGENT]

Counselling

- Your child has increased pressure inside the head that can affect vision if not treated
- Take the medicine regularly even if headaches improve - stopping too soon causes relapse
- Regular eye checkups are very important to protect your child's vision
- If your child is overweight, losing weight is one of the most important treatments
- Report any worsening headache, vision changes, or double vision immediately

Referral Criteria

Refer to ophthalmologist urgently for visual field and fundus assessment; neurosurgeon if vision deteriorating despite medical therapy

Hospitalisation Criteria

Admit for acute vision loss, severe papilledema with visual field loss, need for therapeutic lumbar puncture, or surgical intervention

Notes

Acetazolamide is first-line pharmacotherapy; monitor for metabolic acidosis and electrolyte abnormalities. Ophthalmology follow-up is essential to monitor papilledema and visual fields.

Guideline Changes

2024 Alder Hey hospital published updated pediatric IIH clinical guideline

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G93.4 — Encephalopathy, unspecified

SNOMED: 81308009 | Duration: 14 days | Source: AAP, Japanese Encephalopathy Guidelines 2020

First-Line

- **Supportive ICU care** — Maintain airway, normoglycemia, normothermia; treat raised ICP; target MAP for age; GCS monitoring
- **Lorazepam** — 0.1 mg/kg IV (max 4 mg) for seizures; may repeat x1 in 5 min; first-line anticonvulsant
- **Acyclovir** — 10 mg/kg IV q8h empirically pending HSV PCR if infectious etiology suspected

Second-Line

- **IV methylprednisolone** — 20-30 mg/kg/day IV x 3-5 days for suspected autoimmune/inflammatory encephalopathy; max 1 g/day
- **IVIG** — 2 g/kg IV over 2-5 days for autoimmune encephalopathy refractory to steroids
- **Mannitol** — 0.25-1 g/kg IV over 20 min for raised ICP; can repeat q4-6h; monitor serum osmolality

Investigations

- CT brain (urgent) — Rapid screen for mass lesion, hemorrhage, edema [STAT]
- MRI brain — Detailed assessment of brain parenchyma [URGENT]
- Lumbar puncture with CSF analysis — Rule out meningitis/encephalitis (cell count, protein, glucose, HSV PCR) [STAT]
- Metabolic screen (ammonia, lactate, blood gas) — Rule out metabolic encephalopathy [STAT]
- EEG — Assess for subclinical seizures and encephalopathy severity [URGENT]

Counselling

- Your child's brain is not functioning normally - the cause needs to be found and treated
- Your child will need close monitoring in the hospital, possibly in the ICU
- Many causes of encephalopathy are treatable and children can recover well
- Report any seizures, change in alertness, or breathing difficulty to the nursing staff immediately
- After recovery, developmental follow-up may be needed to check for any lasting effects

Referral Criteria

Refer to pediatric neurologist urgently for all encephalopathy cases; infectious disease if infection suspected; metabolic genetics if metabolic cause suspected

Hospitalisation Criteria

All children with encephalopathy require hospitalization; ICU admission for GCS <8, seizures, hemodynamic instability, or need for airway protection

Notes

Identify and treat the underlying cause (metabolic, infectious, autoimmune, toxic). PICU admission required for GCS <8, raised ICP, or hemodynamic instability.

Q02 — Microcephaly

SNOMED: 127551000119100 | Duration: 365 days | Source: AAP 2022, WHO 2016

First-Line

- **Physical therapy** — Early intervention PT/OT/Speech starting from infancy; individualized goals
- **Levetiracetam** — 10-30 mg/kg/day PO divided BID for associated seizures; titrate over 2-4 wks
- **Occupational therapy** — Ongoing developmental support; fine motor and adaptive skill training

Second-Line

- **Phenobarbital** — 3-5 mg/kg/day PO divided BID for seizures in infants
- **Gastrostomy feeding** — Surgical placement if oral feeding unsafe; for severe cases with dysphagia

Investigations

- Brain MRI — Identify structural brain anomaly
- Genetic testing (microarray, WES) — Identify genetic cause
- TORCH screening — Rule out congenital infection (CMV, Zika, toxoplasmosis)
- EEG — If seizures present or suspected
- Thyroid function test — Rule out congenital hypothyroidism

Counselling

- Child's head is smaller than normal which may affect development
- Early intervention (physiotherapy, occupational therapy, speech therapy) is the most important treatment
- There is no medicine to grow the head, but therapy helps the child reach their best potential
- Seizures may develop; if child has jerking movements or staring spells, report immediately
- Join a support group for families of children with developmental conditions; many resources are available

Referral Criteria

Refer to pediatric neurologist and genetics at diagnosis; early intervention program

Hospitalisation Criteria

Admit for seizure control, feeding difficulties requiring tube placement, or acute illness in compromised child

Notes

No treatment reverses microcephaly; management is supportive with early intervention services. Seizure control and nutritional support are key priorities.

Guideline Changes

CDC/AAP updated developmental milestone surveillance tools in 2022

Q03 — Congenital hydrocephalus

SNOMED: 156894003 | Duration: 365 days | Source: AANS/CNS Task Force 2014 (updated 2021)

First-Line

- **VP shunt** — Ventriculoperitoneal shunt with programmable valve; antibiotic-impregnated catheter recommended (Level I)
- **ETV** — Endoscopic third ventriculostomy; preferred in obstructive hydrocephalus >6 months age

Second-Line

- **ETV + CPC** — ETV with choroid plexus cauterization; lower shunt dependency rates in selected patients
- **Acetazolamide** — 25-100 mg/kg/day PO divided TID; temporary CSF reduction; NOT recommended for post-hemorrhagic

Investigations

- MRI brain with CSF flow study — Determine type and severity of hydrocephalus [URGENT]
- Head circumference serial measurement — Monitor for progressive hydrocephalus
- CT brain — Rapid assessment if acute decompensation [STAT]
- Ophthalmoscopy/fundoscopy — Screen for papilledema

Counselling

- Watch for signs of increased head pressure: vomiting, excessive sleepiness, bulging fontanelle, sunset eyes
- Keep regular follow-up appointments for head circumference measurement
- If a shunt is placed, learn the signs of shunt blockage: headache, vomiting, drowsiness

- Avoid rough play or contact sports that could damage the shunt
- Development may be delayed; early intervention therapy helps your child reach their potential

Referral Criteria

Refer to pediatric neurosurgeon at diagnosis for surgical planning; ophthalmology for papilledema screening

Hospitalisation Criteria

Admit for acute raised ICP symptoms, shunt malfunction or infection, pre-operative preparation for VP shunt or ETV

Notes

Antibiotic-impregnated shunt tubing reduces infection risk (Level I evidence). ETV success depends on age and etiology; higher failure rates in premature and very young infants.

Guideline Changes

2024 narrative review supports increasing ETV+CPC use over VP shunt in select patients

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Q04 — Other congenital malformations of brain

SNOMED: 253153000 | Duration: 365 days | Source: AANS/CNS Pediatric Section 2022, Johns Hopkins

First-Line

- **Neurosurgical decompression** — Individualized surgical approach based on malformation type; e.g. Chiari decompression with duraplasty
- **Levetiracetam** — 10-30 mg/kg/day PO divided BID for associated seizures
- **Physical therapy** — Early intervention PT/OT/Speech to maximize cognitive and motor function

Second-Line

- **VP shunt** — For associated hydrocephalus; programmable valve preferred
- **Valproate** — 15-30 mg/kg/day PO divided BID-TID for seizures; monitor LFTs

Investigations

- Brain MRI — Detailed structural assessment of the malformation [URGENT]
- EEG — Seizure evaluation
- Genetic testing — Identify underlying genetic cause; guide counseling
- Ophthalmologic examination — Associated visual pathway abnormalities
- Developmental assessment — Baseline and serial developmental evaluation

Counselling

- Child has a brain structure abnormality present from birth
- Management depends on the specific type; the neurologist will explain the treatment plan
- Early intervention therapies are essential to maximize development
- Some children may need surgery (shunt for fluid, decompression)
- Long-term follow-up with multiple specialists is needed; keep all appointments

Referral Criteria

Refer to pediatric neurology and neurosurgery at diagnosis; genetics for syndromic evaluation

Hospitalisation Criteria

Admit for raised intracranial pressure, hydrocephalus, intractable seizures, or surgical intervention

Notes

Management is highly individualized based on specific malformation and severity. Multidisciplinary team including neurosurgery, neurology, genetics, and rehabilitation is essential.

R27.0 — Ataxia

SNOMED: 20262006 | Duration: 30 days | Source: AAN 2018, National Ataxia Foundation

First-Line

- **Physical therapy** — Balance and coordination training; gait rehab; 4-wk inpatient rehab improves ataxia scores
- **4-Aminopyridine** — 5 mg PO BID for episodic ataxia type 2; reduces attack frequency (Level B AAN)
- **Riluzole** — 50 mg PO BID for mixed/Friedreich/SCA ataxias; probably improves ataxia signs

Second-Line

- **Omaveloxolone (Skyclarys)** — 150 mg PO daily for Friedreich ataxia age >=16 yrs; FDA-approved 2023; NRF2 activator **[NEW 2024-25]**
- **Levacetylleucine (Aqneursa)** — PO for Niemann-Pick type C; FDA-approved 2024; dose per weight-based schedule **[NEW 2024-25]**
- **IVIG** — 2 g/kg IV over 2-5 days for acute post-infectious cerebellar ataxia; for refractory cases
- **Acetazolamide** — 5-10 mg/kg/day PO divided BID for episodic ataxia type 2 if 4-AP not tolerated

Investigations

- MRI brain with posterior fossa views — Rule out posterior fossa tumor, stroke, demyelination [URGENT]
- Urine toxicology screen — Rule out drug ingestion in acute ataxia [STAT]
- Genetic testing panel — For chronic/hereditary ataxia (Friedreich, SCA)
- Serum anti-neuronal antibodies — If autoimmune cerebellar ataxia suspected

Counselling

- Ataxia means your child has difficulty with balance and coordination - this has many possible causes
- Keep the home environment safe: remove loose rugs, install grab bars, pad sharp furniture corners
- Some causes of ataxia are treatable, so completing all recommended tests is very important
- If ataxia came on suddenly, bring the child to the hospital immediately
- Physical therapy exercises at home help improve balance and should be practiced daily

Referral Criteria

Refer to pediatric neurologist for all new-onset ataxia for etiology workup; urgent referral if acute onset suggesting stroke or posterior fossa lesion

Hospitalisation Criteria

Admit for acute-onset ataxia (rule out posterior fossa pathology, stroke, intoxication), progressive ataxia with raised ICP signs, or inability to walk/stand safely

Notes

Treatment is etiology-specific; identify underlying cause before selecting therapy. New disease-modifying agents for Friedreich ataxia and NPC represent major recent advances.

Guideline Changes

FDA approved omaveloxolone (Skyclarys) 2023 for Friedreich ataxia; levacetylleucine (Aqneursa) 2024 for NPC

R51 — Headache

SNOMED: 25064002 | Duration: 3 days | Source: AAN 2019 (reaffirmed 2025), AAP endorsed

First-Line

- **Ibuprofen** — 10 mg/kg PO once at headache onset; max 400 mg/dose; 68% response rate in trials
- **Acetaminophen** — 15 mg/kg PO once at headache onset; max 1000 mg/dose; 54% response rate

Second-Line

- **Rizatriptan** — 5 mg PO if <40 kg, 10 mg PO if >=40 kg; for moderate-severe headache

- **Sumatriptan nasal spray** — 5-20 mg intranasal once; FDA-approved for adolescents >=12 yrs

Investigations

- Clinical examination with fundoscopy — Papilledema suggests raised ICP [URGENT]
- Brain MRI — Only if red flags present (progressive, focal signs, early morning vomiting)
- Vision screening — Refractive errors contribute to headaches in children

Counselling

- Give pain medicine (ibuprofen/paracetamol) early at the start of headache for best effect
- Maintain regular sleep schedule, adequate hydration, and regular meals
- Limit screen time; take regular breaks from computer/phone
- Keep a headache diary noting triggers, duration, and what helped
- Stress and anxiety are common triggers in school-going children; address them

Referral Criteria

Refer to pediatric neurologist if headaches are progressive, waking child from sleep, associated with vomiting, vision changes, or focal neurological signs

Hospitalisation Criteria

Admit for thunderclap headache, signs of raised ICP, meningism, or new focal neurological deficits

Notes

Rule out secondary causes with red flag screening before treating as primary headache. NSAIDs are first-line; triptans reserved for moderate-severe or NSAID-refractory cases.

Guideline Changes

AAN guideline reaffirmed Oct 2025; IHS global acute migraine recs published 2024

R56.00 — Febrile seizures

SNOMED: 41497008 | Duration: 1 days | Source: AAP 2011 reaffirmed

First-Line

- **Midazolam buccal/intranasal** — 0.2-0.3 mg/kg (max 10mg), preferred pre-hospital non-IV route
- **Diazepam rectal** — 0.5 mg/kg (max 20mg), seizure over 5 minutes, pre-hospital

Second-Line

- **Lorazepam IV** — 0.1 mg/kg IV (max 4mg), hospital management
- **Phenobarbital IV** — 15-20 mg/kg IV loading, status epilepticus

Investigations

- Blood glucose — Rule out hypoglycaemia as seizure cause [STAT]
- Serum electrolytes — If diarrhoea, vomiting, or abnormal fluid intake [URGENT]
- Lumbar puncture — Mandatory if <12 months or meningeal signs present; consider in 12-18 months if unimmunised [STAT]
- EEG — Not indicated for simple febrile seizure; only if atypical features or developmental concerns

Counselling

- Febrile seizures look scary but are generally harmless and do not cause brain damage
- During a seizure: lay child on side, do not put anything in mouth, time the seizure
- If seizure lasts more than 5 minutes, call ambulance or rush to hospital
- Paracetamol and ibuprofen do NOT prevent febrile seizures but help with fever comfort
- Most children outgrow febrile seizures by age 5-6 years; long-term medicine is usually not needed

Referral Criteria

Refer to pediatric neurologist if complex febrile seizure (focal, >15 minutes, recurrent within 24 hours), seizure in child <6 months or >5 years, or developmental concerns

Hospitalisation Criteria

Admit if seizure duration >15 minutes (status epilepticus), post-ictal altered consciousness >1 hour, suspected meningitis (neck stiffness, bulging fontanelle), first seizure in infant <12 months, or complex febrile seizure for observation

Notes

Antipyretics do NOT prevent recurrence. Long-term anticonvulsants not recommended for simple febrile seizures.

R62.0 — Delayed milestones in childhood

SNOMED: 401169009 | Duration: 365 days | Source: AAP 2022, CDC Learn the Signs Act Early

First-Line

- **Early intervention program** — Multidisciplinary PT/OT/Speech therapy; federally mandated for ages 0-3 yrs
- **Speech-language therapy** — 1-3 sessions/wk targeting expressive/receptive language goals
- **Occupational therapy** — Fine motor, sensory integration, adaptive skills training; 1-3 sessions/wk

Second-Line

- **Iron supplementation** — 3-6 mg/kg/day elemental iron PO if iron-deficiency anemia contributing to delay
- **Thyroid hormone replacement** — Levothyroxine 10-15 mcg/kg/day PO if hypothyroidism identified

Investigations

- Developmental screening (ASQ/DASII) — Formal developmental assessment to quantify delay
- Thyroid function test — Rule out hypothyroidism as treatable cause
- Hearing assessment (OAE/BERA) — Hearing loss is common treatable cause of speech delay
- Vision screening — Visual impairment contributes to global delay
- Genetic testing (microarray) — If etiology unclear or dysmorphic features

Counselling

- Child is not reaching milestones at the expected age; early therapy makes a big difference
- Start physiotherapy, speech therapy, and occupational therapy as recommended
- Practice exercises at home daily as taught by the therapist
- Screen hearing and vision as they can contribute to developmental delay
- Track milestones using a milestone diary; share progress with the doctor at each visit

Referral Criteria

Refer to developmental pediatrician, early intervention program; audiology and ophthalmology screening

Hospitalisation Criteria

Admission usually not required unless associated with acute seizures, severe malnutrition, or metabolic crisis

Notes

Earlier identification leads to better outcomes; screen at 9, 18, and 30 months per AAP. Treat underlying causes (metabolic, genetic, nutritional) while providing therapy.

Guideline Changes

AAP/CDC updated milestone checklists in 2022 raising threshold from 50% to 75%

Renal

N00 — Acute nephritic syndrome

SNOMED: 61503006 | Duration: 14 days | Source: KDIGO 2021 GN, CDC APSGN 2024

First-Line

- **Furosemide** — 1-2 mg/kg IV/PO q6-12h for edema and fluid overload; avoid high doses if oliguric
- **Amlodipine** — 0.06-0.3 mg/kg/day PO once daily for hypertension; max 5-10 mg/day based on age
- **Fluid restriction** — Restrict fluid to insensible losses + urine output during acute phase; low-salt diet

Second-Line

- **Nifedipine** — 0.25-0.5 mg/kg PO q6-8h for hypertensive urgency; short-acting for acute BP control
- **Penicillin V** — 250-500 mg PO BID-TID for 10 days if active strep infection; eradication only, does not alter GN course
- **Methylprednisolone** — IV pulse 30 mg/kg (max 1g) x3 days for rapidly progressive/crescentic disease only

Investigations

- Urinalysis with microscopy — Dysmorphic RBCs, RBC casts confirm glomerulonephritis [STAT]
- Serum complement C3 and C4 — Low C3 with normal C4 typical of APSGN [URGENT]
- Serum creatinine and BUN — Assess renal function [URGENT]
- ASO titer / Anti-DNase B — Evidence of recent streptococcal infection
- Urine protein/creatinine ratio — Quantify proteinuria

Counselling

- Child's kidneys are inflamed, usually after a throat or skin infection
- Restrict salt and water intake during the swelling phase
- Blood pressure medicine and water pills will help control swelling and BP
- This condition usually heals completely in 2-4 weeks
- Follow up with urine and blood tests; protein in urine should disappear within months

Referral Criteria

Refer to pediatric nephrologist if rapidly progressive course, persistent low complement >8 weeks, or nephrotic-range proteinuria

Hospitalisation Criteria

Admit if hypertensive emergency, significant fluid overload, oliguria, or rapidly deteriorating renal function

Notes

APSGN is self-limiting; supportive care with diuretics and antihypertensives is mainstay. Immunosuppression reserved for crescentic disease on biopsy.

Guideline Changes

Updated CDC clinical guidelines for post-streptococcal GN published

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N01 — Rapidly progressive nephritic syndrome

SNOMED: 236392004 | Duration: 30 days | Source: KDIGO 2021 GN, Japan RPGN 2014

First-Line

- **Methylprednisolone** — IV pulse 30 mg/kg/day (max 1 g) for 3 consecutive days; then oral pred 1-2 mg/kg/day taper
- **Cyclophosphamide** — IV pulse 500-750 mg/m2 monthly x6 or PO 2 mg/kg/day x8-12 wk; for induction

Second-Line

- **Rituximab** — 375 mg/m2 IV weekly x4 for ANCA-associated RPGN; non-inferior to cyclophosphamide **[NEW 2024-25]**
- **Plasmapheresis** — 5-7 exchanges over 7-14 days for anti-GBM disease or severe ANCA vasculitis with DAH
- **Mycophenolate mofetil** — 1200 mg/m2/day PO divided BID for maintenance after induction; alternative to azathioprine
- **Azathioprine** — 1-2 mg/kg/day PO for maintenance therapy after induction remission achieved

Investigations

- Renal biopsy — Urgent; crescentic GN classification guides therapy [STAT]
- ANCA (c-ANCA, p-ANCA) — ANCA-associated vasculitis [STAT]
- Anti-GBM antibodies — Anti-GBM disease (Goodpasture) [STAT]
- Serum complement C3, C4 — Low in immune complex GN [URGENT]
- Serum creatinine (serial) — Monitor rate of renal function decline [STAT]

Counselling

- Child has rapidly worsening kidney disease that needs urgent treatment
- Strong medicines to suppress the immune system are needed to stop kidney damage
- Kidney biopsy (taking a small piece of kidney with a needle) is needed to guide treatment
- Treatment requires close monitoring with frequent blood tests
- With early and aggressive treatment, kidney function can be preserved in many cases

Referral Criteria

Urgent referral to pediatric nephrologist; renal biopsy within 48 hours

Hospitalisation Criteria

All cases require admission for urgent renal biopsy, pulse immunosuppression, and monitoring

Notes

Urgent renal biopsy is essential to guide therapy. Aggressive induction immunosuppression has markedly improved outcomes; dialysis-dependent AKI predicts poor prognosis.

Guideline Changes

Rituximab increasingly used in pediatric ANCA-associated RPGN

N02 — Recurrent and persistent hematuria

SNOMED: 197698000 | Duration: 180 days | Source: KDIGO 2021 GN, Kidney360 2020

First-Line

- **Observation** — Serial urinalysis q3-6mo; monitor BP, proteinuria, serum creatinine; reassurance if isolated
- **ACE inhibitor** — Enalapril 0.08-0.6 mg/kg/day PO if proteinuria >0.5 g/day develops; renoprotective

Second-Line

- **ARB** — Losartan 0.7-1.4 mg/kg/day PO if ACEi not tolerated; for proteinuric hematuria
- **Fish oil (omega-3)** — Adjunctive therapy for IgA nephropathy with proteinuria >1 g/day; per KDIGO GN guidelines

Investigations

- Urinalysis with microscopy — Dysmorphic RBCs suggest glomerular origin
- Urine calcium/creatinine ratio — Hypercalciuria is common cause of isolated hematuria
- Serum creatinine, C3, C4 — Assess renal function and complement levels
- Renal ultrasound — Structural causes, stones, masses
- Family urine screening — Thin basement membrane disease and Alport syndrome

Counselling

- Blood in urine is being investigated; most causes in children are not dangerous
- Regular check-ups with urine tests every 3-6 months are needed
- If protein starts appearing in urine or blood pressure rises, medicine may be needed
- Vigorous exercise may temporarily worsen visible blood in urine; this is usually harmless
- Report any swelling of face/legs, tea-colored urine, or high BP immediately

Referral Criteria

Refer to pediatric nephrologist if proteinuria develops, hypertension, family history of renal disease, or hearing loss (Alport syndrome)

Hospitalisation Criteria

Admit if gross hematuria with clots causing retention, or associated nephritic syndrome features

Notes

Isolated microscopic hematuria without proteinuria has good prognosis. Refer to nephrology if proteinuria, hypertension, or family history of renal disease.

N04 — Nephrotic syndrome

SNOMED: 52254009 | Duration: 56 days | Source: KDIGO 2025, IPNA 2022 SSNS

First-Line

- **Prednisolone** — 2 mg/kg/day PO (max 60 mg) daily x4 wk, then 1.5 mg/kg (max 50 mg) alt-day x4 wk
- **Albumin + Furosemide** — Albumin 1 g/kg IV over 2-4h with furosemide 1 mg/kg IV for severe edema/anasarca

Second-Line

- **Cyclophosphamide** — 2 mg/kg/day PO for 8-12 wk for frequent relapsers; cumulative max 168 mg/kg
- **Cyclosporine** — 4-6 mg/kg/day PO divided BID; for steroid-dependent or steroid-resistant NS; monitor trough levels
- **Mycophenolate mofetil** — 1200 mg/m2/day PO divided BID (max 2 g/day); for SDNS/FRNS; continue >=12 months
- **Rituximab** — 375 mg/m2 IV x1-2 doses; for SDNS refractory to standard steroid-sparing agents **[NEW 2024-25]**

Investigations

- Urine protein/creatinine ratio — Quantify proteinuria; nephrotic range >2 mg/mg [STAT]
- Serum albumin — Low (<2.5 g/dL) confirms nephrotic syndrome [STAT]
- Serum cholesterol — Hyperlipidemia supports diagnosis
- Serum creatinine — Assess renal function [URGENT]
- Complement C3, C4 — Normal in MCNS; low suggests other glomerulopathy

Counselling

- Child's kidneys are leaking protein into urine causing body swelling
- Steroid medicine (prednisolone) is the main treatment; follow the dose schedule carefully
- Check urine with dipstick daily at home; record results in a diary
- Restrict salt in diet during active disease; no need to restrict protein
- Relapses are common (60-70% of children relapse); come back if swelling returns or urine shows protein

Referral Criteria

Refer to pediatric nephrologist if steroid-resistant (no remission by 4 weeks), frequent relapses (>2/6 months), or steroid-dependent

Hospitalisation Criteria

Admit for severe edema/anasarca, respiratory distress from pleural effusion, peritonitis, or thrombosis

Notes

85-90% of children achieve remission with steroids. Steroid-sparing agents selected based on toxicity profile, adherence, and relapse pattern.

Guideline Changes

KDIGO 2025 first dedicated guideline for nephrotic syndrome in children; rituximab now recommended

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N05 — Unspecified nephritic syndrome

SNOMED: 52845002 | Duration: 14 days | Source: KDIGO 2021 GN, AAP

First-Line

- **Furosemide** — 1-2 mg/kg IV/PO q6-12h for edema and volume overload; monitor electrolytes
- **Amlodipine** — 0.06-0.3 mg/kg/day PO once daily for hypertension; max 5-10 mg/day
- **Supportive care** — Fluid and sodium restriction; monitor renal function, electrolytes, complement levels

Second-Line

- **Enalapril** — 0.08-0.6 mg/kg/day PO for persistent proteinuria or hypertension; renoprotective
- **Prednisolone** — 1-2 mg/kg/day PO if biopsy shows crescentic or proliferative GN requiring immunosuppression

Investigations

- Urinalysis with microscopy — RBC casts, dysmorphic RBCs confirm glomerular disease [STAT]
- Serum complement C3, C4 — Guide differential diagnosis [URGENT]
- Serum creatinine and BUN — Assess renal function [URGENT]
- ANA, anti-dsDNA — Screen for lupus nephritis
- Renal biopsy — If atypical features or not improving as expected

Counselling

- Child's kidneys are inflamed; the exact cause is being investigated
- Restrict salt and water during the swelling phase
- Blood pressure and kidney function will be monitored closely
- Additional tests including kidney biopsy may be needed to find the exact cause
- Most cases in children respond well to treatment once the cause is identified

Referral Criteria

Refer to pediatric nephrologist for full evaluation; renal biopsy if not typical APSGN

Hospitalisation Criteria

Admit if hypertensive emergency, significant fluid overload, or rapidly deteriorating renal function

Notes

Establish specific diagnosis via serology and biopsy when possible. Treatment is primarily supportive unless biopsy reveals disease requiring immunosuppression.

N10 — Acute pyelonephritis

SNOMED: 36689008 | Duration: 14 days | Source: AAP 2024, NICE, EAU/ESPU 2024

First-Line

- **Ceftriaxone** — 50-75 mg/kg IV/IM daily (max 2 g) for initial empiric therapy; switch to oral when afebrile x24-48h
- **Cefixime** — 8 mg/kg PO daily (max 400 mg) for outpatient oral therapy; 10-14 day total course
- **Amoxicillin-clavulanate** — 25-45 mg/kg/day PO divided BID-TID for 10-14 days; oral alternative for outpatient treatment

Second-Line

- **Ampicillin + Gentamicin** — Ampicillin 100 mg/kg/day IV divided q6h + gentamicin 5-7.5 mg/kg/day IV; for neonates/young infants
- **Ciprofloxacin** — 10-20 mg/kg/day PO divided BID (max 750 mg/dose); reserve for resistant organisms in older children

Investigations

- Urine culture (before antibiotics) — Identify organism and sensitivity [STAT]
- CBC, CRP — Elevated WBC and CRP support pyelonephritis [URGENT]
- Renal ultrasound — Rule out obstruction, abscess; assess kidney size [URGENT]
- VCUG — After first febrile UTI in child <2 years per AAP
- DMSA scan — Detect renal scarring 4-6 months after acute infection

Counselling

- Child has a kidney infection with fever that needs antibiotic treatment for 10-14 days
- Complete the full antibiotic course even if child feels better
- Ensure child drinks plenty of fluids
- If child has recurrent infections, further tests will be done to check for kidney problems
- Toilet training and regular voiding habits are important to prevent future infections

Referral Criteria

Refer to pediatric nephrologist/urologist for first febrile UTI in child <2 years; recurrent pyelonephritis; or abnormal imaging

Hospitalisation Criteria

Admit if age <2 months, toxic-appearing, unable to tolerate oral medication, or immunocompromised

Notes

Oral therapy is equivalent to IV for non-toxic children >2 months. Obtain urine culture before antibiotics; VCUG if febrile UTI in child <2 years.

Guideline Changes

AAP 2024 Redbook supports shorter 5-10 day courses for uncomplicated cases

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N12 — Tubulo-interstitial nephritis, not specified as acute or chronic

SNOMED: 1092831000119109 | Duration: 30 days | Source: KDIGO 2021, Pediatric Nephrology Consensus

First-Line

- **Remove offending agent** — Identify and discontinue causative drug (NSAIDs, antibiotics, PPIs); treat underlying infection if present
- **Prednisolone** — 1 mg/kg/day PO (max 60 mg) for 4-6 wk then gradual taper; for drug-induced or immune-mediated ATIN

Second-Line

- **Mycophenolate mofetil** — 600-1200 mg/m2/day PO divided BID for steroid-refractory TIN; emerging evidence **[NEW 2024-25]**
- **Supportive care** — BP control, fluid/electrolyte management; monitor renal function for recovery; low-salt diet

Investigations

- Serum creatinine (serial) — Monitor renal function recovery [URGENT]
- Urinalysis — White cell casts, eosinophiluria (Hansel stain) [URGENT]
- CBC with differential — Peripheral eosinophilia in drug-induced TIN [URGENT]
- Renal biopsy — If diagnosis uncertain or not improving with drug withdrawal

- Renal ultrasound — Assess kidney size; enlarged in acute TIN

Counselling

- Child's kidney is inflamed, most likely from a medicine; the causative medicine must be stopped
- Kidney function usually recovers once the medicine is stopped
- Steroid medicine may be given to help the kidney heal faster
- Blood tests will be repeated regularly to ensure kidney function is improving
- Always inform doctors about this drug reaction to avoid the medicine in future

Referral Criteria

Refer to pediatric nephrologist if renal function not recovering after stopping offending agent, or renal biopsy needed

Hospitalisation Criteria

Admit if AKI, significant electrolyte disturbance, or need for renal biopsy

Notes

Removing the offending agent is most important step; most cases recover spontaneously. Steroids may accelerate recovery but evidence from RCTs is limited.

Guideline Changes

Role of mycophenolate mofetil emerging for steroid-refractory TIN

N13 — Obstructive and reflux uropathy

SNOMED: 197764002 | Duration: 365 days | Source: AUA VUR 2010, EAU/ESPU 2024 VUR Update

First-Line

- **Antibiotic prophylaxis** — TMP-SMX 2 mg/kg PO daily or nitrofurantoin 1-2 mg/kg PO daily for VUR grade III-V or post-UTI
- **BBD treatment** — Treat bladder-bowel dysfunction with timed voiding, constipation management before surgical intervention

Second-Line

- **Endoscopic injection** — Deflux (dextranomer/HA) subureteral injection for VUR grade III-IV; 70-80% success per injection
- **Ureteral reimplantation** — Open or robotic-assisted surgical correction for high-grade VUR refractory to conservative management
- **Oxybutynin** — 0.1-0.2 mg/kg PO BID-TID if associated overactive bladder; treat BBD concomitantly

Investigations

- VCUG — Grade VUR and assess bladder
- Renal ultrasound — Assess for hydronephrosis and renal size
- DMSA scan — Detect renal scarring
- Urine culture — Before starting and during prophylaxis
- Uroflowmetry — Evaluate for bladder-bowel dysfunction

Counselling

- Child's urine flows backward from the bladder to the kidneys which can cause infections
- Daily low-dose antibiotic prevents urinary infections while waiting for reflux to resolve
- Treat constipation and ensure regular toilet habits as these affect reflux
- Many children outgrow reflux by age 5-6 years; surgery is only for severe cases
- Bring child for urine test if there is fever without obvious source

Referral Criteria

Refer to pediatric urologist for high-grade VUR (IV-V), recurrent febrile UTIs despite prophylaxis, or renal scarring

Hospitalisation Criteria

Admit for febrile UTI, post-surgical management, or acute pyelonephritis

Notes

Risk-based approach: low-grade VUR often resolves spontaneously. Treat BBD before considering surgery; antibiotic prophylaxis for recurrent febrile UTIs.

Guideline Changes

2024 EAU/ESPU VUR chapter updated with risk-based management approach

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N13.3 — Hydronephrosis

SNOMED: 43064006 | Duration: 180 days | Source: SFU Grading, EAU/ESPU 2024

First-Line

- **Observation** — Serial renal US q3-6mo; ~50% resolve spontaneously; SFU grade 1-2 typically managed conservatively
- **Prophylactic antibiotics** — Amoxicillin 10-15 mg/kg PO daily or TMP-SMX 2 mg/kg PO daily for high-grade HN to prevent UTI

Second-Line

- **Pyeloplasty** — Open or laparoscopic repair for UPJ obstruction with declining function or SFU grade 3-4
- **MAG3 renogram** — Diuretic renography to assess drainage and differential renal function; guides surgical decision

Investigations

- Renal ultrasound (serial) — Monitor APD and SFU grade q3-6 months
- MAG3 diuretic renogram — Assess drainage and differential renal function for SFU 3-4
- VCUG — If UTI occurs to rule out vesicoureteral reflux
- Urine culture — If febrile illness to detect UTI early [URGENT]

Counselling

- Baby's kidney is swollen with urine because of slow drainage; this was found on ultrasound
- Many cases improve on their own without surgery; regular ultrasound monitoring is needed
- Antibiotic medicine may be given daily to prevent urinary infections
- Surgery is only needed if the kidney function is getting worse
- Keep all follow-up ultrasound appointments every 3-6 months

Referral Criteria

Refer to pediatric urologist if SFU grade 3-4, declining function on renogram, or febrile UTI

Hospitalisation Criteria

Admit for febrile UTI with hydronephrosis or postoperatively after pyeloplasty

Notes

About half of hydronephrosis cases resolve spontaneously. Surgery indicated for febrile UTIs, deteriorating hydronephrosis, or declining renal function on renogram.

N17 — Acute kidney failure

SNOMED: 14669001 | Duration: 14 days | Source: KDIGO 2012 AKI, pRIFLE Criteria

First-Line

- **Fluid management** — Assess volume status; restrict fluids to insensible losses + urine output if oliguric; avoid fluid overload
- **Furosemide** — 1-2 mg/kg IV bolus or 0.1-0.4 mg/kg/h infusion; convert oliguric to non-oliguric AKI if euvolemic
- **Electrolyte correction** — Calcium gluconate 100 mg/kg IV, insulin-glucose, kayexalate for hyperkalemia; bicarb for acidosis

Second-Line

- **CRRT** — Continuous renal replacement therapy for fluid overload >10%, refractory electrolyte/acid-base derangement
- **Peritoneal dialysis** — Preferred RRT modality in small children; for AKI refractory to conservative measures
- **Rasburicase** — 0.2 mg/kg IV once for tumor lysis syndrome-associated AKI; prevents uric acid nephropathy

Investigations

- Serum creatinine (serial) — Monitor trend; staging per pRIFLE criteria [STAT]
- Serum electrolytes (K, Na, Ca, PO4) — Hyperkalemia is life-threatening; frequent monitoring needed [STAT]
- Blood gas — Metabolic acidosis assessment [STAT]
- Renal ultrasound — Assess kidney size and rule out obstruction [URGENT]
- Urinalysis — Casts and proteinuria guide etiology [URGENT]

Counselling

- Child's kidneys have suddenly stopped working properly; this needs urgent treatment
- Fluid intake and urine output will be carefully measured
- Blood tests will be checked frequently; some may show high potassium which needs immediate treatment
- Dialysis (blood cleaning machine) may be needed if kidneys do not recover quickly
- Most children recover kidney function if the cause is treated early

Referral Criteria

Immediate referral to pediatric nephrologist; ICU if hemodynamically unstable or needing dialysis

Hospitalisation Criteria

All AKI cases require admission; ICU for dialysis, hyperkalemia, fluid overload >10%, or metabolic acidosis

Notes

Identify and treat underlying cause promptly. Avoid nephrotoxins; adjust all renally-cleared drug doses; fluid overload >10% associated with poor outcomes.

Guideline Changes

Pediatric-specific AKI biomarkers (NGAL, KIM-1) increasingly used for early detection

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N18 — Chronic kidney disease

SNOMED: 431855005 | Duration: 365 days | Source: KDIGO 2024 CKD, KDOQI Pediatric

First-Line

- **Enalapril** — 0.08-0.6 mg/kg/day PO daily; ACEi for proteinuria and BP control; target BP <50th percentile
- **Erythropoietin** — 50 U/kg SC/IV 3x/wk for CKD anemia; target Hb 11-12 g/dL; supplement iron first
- **Calcium carbonate** — Phosphate binder; dose titrated to normalize serum phosphorus; give with meals

Second-Line

- **Growth hormone** — 0.05 mg/kg/day SC for CKD-related growth failure; per KDOQI; continue until target height or transplant
- **Cholecalciferol** — Vitamin D 1000-2000 IU/day PO; calcitriol 0.01-0.05 mcg/kg/day if secondary hyperparathyroidism

- **Sevelamer** — Non-calcium phosphate binder; 400-800 mg PO TID with meals if calcium-based binders insufficient
- **Darbepoetin** — 0.45 mcg/kg SC/IV q2-4wk; longer-acting ESA alternative to erythropoietin

Investigations

- Serum creatinine and eGFR — Monitor renal function q3-6 months
- Urine albumin/creatinine ratio — Proteinuria monitoring; adjust ACEi accordingly
- CBC — Anemia of CKD; guide EPO therapy
- Serum calcium, phosphorus, PTH — CKD-MBD (mineral bone disease) monitoring
- Renal ultrasound — Kidney size and structural assessment

Counselling

- Child's kidneys are not working fully; this is a long-term condition needing regular care
- Blood pressure medicine (ACE inhibitor) protects the remaining kidney function
- Follow the dietary advice: reduce salt, ensure adequate protein (not too much or too little)
- Growth may be slow; growth hormone injection may help
- Keep all follow-up appointments; blood and urine tests are needed every 3-6 months

Referral Criteria

Refer to pediatric nephrologist at diagnosis; dietitian, and transplant team for CKD stage 4-5

Hospitalisation Criteria

Admit for hyperkalemia, severe fluid overload, uremic symptoms, or dialysis initiation

Notes

Multidisciplinary CKD clinic follow-up essential. Preterm and SGA infants at increased CKD risk; early RAS blockade for proteinuria slows progression.

Guideline Changes

KDIGO 2024 major update with dedicated pediatric chapter including growth and development focus

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N19 — Unspecified kidney failure

SNOMED: 42399005 | Duration: 14 days | Source: KDIGO 2012 AKI, KDIGO 2024 CKD

First-Line

- **Identify and treat cause** — Urgent workup: renal US, electrolytes, urinalysis, serology; classify as AKI vs CKD to guide therapy
- **Furosemide** — 1-2 mg/kg IV/PO for fluid overload; dose adjust based on renal function response
- **Electrolyte management** — Correct hyperkalemia, acidosis, hyperphosphatemia; restrict potassium and phosphorus intake

Second-Line

- **Dialysis** — PD or HD if conservative measures fail; PD preferred in young children; preemptive transplant evaluation
- **Enalapril** — 0.08-0.6 mg/kg/day PO if CKD with proteinuria/HTN confirmed; avoid in AKI with hemodynamic instability

Investigations

- Serum creatinine and BUN — Baseline renal function assessment [STAT]
- Renal ultrasound — Kidney size differentiates AKI (normal/enlarged) from CKD (small) [STAT]
- Serum electrolytes — Hyperkalemia, hyperphosphatemia assessment [STAT]
- CBC — Normocytic anemia suggests CKD [URGENT]
- Urinalysis — Casts, protein, and blood guide etiology [URGENT]

Counselling

- Child's kidneys are not working well; the doctor is investigating whether this is sudden or long-standing
- Blood tests and urine tests will help determine the cause and best treatment
- Fluid and diet may need to be restricted temporarily
- If kidney function does not improve, dialysis or transplant may be discussed
- Regular follow-up with the kidney specialist is essential

Referral Criteria

Urgent referral to pediatric nephrologist for workup and classification

Hospitalisation Criteria

Admit for electrolyte derangements, fluid overload, uremic symptoms, or need for dialysis

Notes

Priority is distinguishing AKI from CKD via renal US (kidney size) and history. Multidisciplinary team approach with nephrology, dietitian, and social work.

N20 — Calculus of kidney and ureter

SNOMED: 95570007 | Duration: 14 days | Source: AUA 2025 Stones, EAU 2025 Urolithiasis

First-Line

- **Hydration + analgesics** — IV NS bolus then maintenance fluids; ketorolac 0.5 mg/kg IV (max 30 mg) for acute colic
- **ESWL** — Extracorporeal shock wave lithotripsy for stones <2 cm; preferred first-line for most pediatric stones

Second-Line

- **Ureteroscopy** — Flexible URS with laser lithotripsy for ureteral or lower pole stones; holmium laser preferred
- **Tamsulosin** — 0.4 mg PO daily (off-label) for medical expulsive therapy of distal ureteral stones 5-10 mm
- **Potassium citrate** — 1-2 mEq/kg/day PO divided BID-TID for hypocitraturia prevention; adjust per urine citrate
- **Hydrochlorothiazide** — 0.5-1 mg/kg/day PO for hypercalciuria prevention; monitor electrolytes

Investigations

- Non-contrast CT abdomen (low-dose) — Gold standard for stone detection; size and location [URGENT]
- Renal and bladder ultrasound — First-line in children; avoid radiation when possible [URGENT]
- 24-hour urine metabolic profile — Calcium, oxalate, citrate, uric acid, cystine
- Stone analysis — Composition guides preventive therapy
- Serum electrolytes, calcium, uric acid, creatinine — Metabolic evaluation [URGENT]

Counselling

- Child has a kidney or ureter stone causing pain
- Increase water intake to at least 2 liters per day to prevent new stones
- Pain medicine will be given; most small stones pass on their own
- Stone should be collected and sent for analysis to find the cause
- Diet changes (less salt, less animal protein) help prevent recurrence

Referral Criteria

Refer to pediatric urologist if stone >5mm, obstruction, infection, or not passing spontaneously

Hospitalisation Criteria

Admit if infected obstructed stone (surgical emergency), intractable pain, or AKI

Notes

Metabolic evaluation with stone analysis is mandatory in all pediatric stone formers. Tailor prevention to stone composition; dietary modifications are first-line prevention.

Guideline Changes

AUA 2025 updated surgical stones guideline with pediatric-specific sections

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N31 — Neuromuscular dysfunction of bladder

SNOMED: 397732007 | Duration: 365 days | Source: AUA 2024 NLUTD, EAU/ESPU 2024

First-Line

- **Oxybutynin** — 0.1-0.2 mg/kg PO BID-TID; anticholinergic for detrusor overactivity; max 5 mg TID
- **Clean Intermittent Catheterization** — CIC 4-6x daily from birth; cornerstone of neurogenic bladder management
- **Tolterodine** — 0.05-0.1 mg/kg PO BID; antimuscarinic alternative if oxybutynin not tolerated

Second-Line

- **Mirabegron** — 0.4-0.8 mg/kg PO daily; beta-3 agonist for refractory detrusor overactivity **[NEW 2024-25]**
- **Botulinum toxin A** — 6-12 units/kg intradetrusor injection; max 200 units; for refractory cases
- **Augmentation cystoplasty** — Surgical bladder augmentation for patients refractory to medical therapy

Investigations

- Urodynamic study — Characterize bladder function; guide treatment
- Renal ultrasound — Screen for hydronephrosis and renal damage [URGENT]
- VCUG — Evaluate for vesicoureteral reflux
- Serum creatinine — Monitor renal function
- Urine culture — Baseline and if symptomatic UTI suspected

Counselling

- Child's bladder does not empty properly due to nerve problems; regular catheterization is needed
- Clean intermittent catheterization (CIC) is the most important treatment; learn the proper technique
- Medicine will help the bladder relax and hold more urine
- Regular kidney ultrasounds are needed to protect kidneys from damage
- Proper bowel management is also important; constipation worsens bladder problems

Referral Criteria

Refer to pediatric urologist and nephrologist at diagnosis; urodynamics needed

Hospitalisation Criteria

Admit for febrile UTI, obstructive uropathy, or acute kidney injury

Notes

Proactive CIC and antimuscarinics from early life preserve upper tract function. Urodynamic studies guide therapy; monitor renal function and bladder compliance regularly.

Guideline Changes

Beta-3 agonists (mirabegron) now approved for pediatric use

N32 — Other disorders of bladder

SNOMED: 197858004 | Duration: 90 days | Source: EAU/ESPU 2024, AAP

First-Line

- **Behavioral therapy** — Timed voiding, bladder training, pelvic floor exercises; first-line for all pediatric bladder dysfunction

- **Oxybutynin** — 0.1-0.2 mg/kg PO BID-TID; anticholinergic for overactive bladder; max 5 mg TID

Second-Line

- **Mirabegron** — 25-50 mg PO daily (>5 yr); beta-3 agonist for refractory OAB **[NEW 2024-25]**
- **Botulinum toxin A** — Intradetrusor injection 50-200 units for refractory detrusor overactivity
- **Desmopressin** — 0.2-0.6 mg PO at bedtime for associated nocturnal enuresis; restrict fluids

Investigations

- Urinalysis and urine culture — Rule out UTI
- Renal and bladder ultrasound — Post-void residual; bladder wall thickness
- Uroflowmetry with EMG — Non-invasive assessment of voiding pattern
- Voiding diary — Record frequency, volume, and incontinence episodes

Counselling

- Child has bladder problems that can be improved with regular toilet habits
- Timed voiding (going to the toilet every 2-3 hours) is the most important step
- Treat constipation as it makes bladder problems worse
- Medicine helps the bladder relax; give it regularly as prescribed
- Be patient; improvement takes 3-6 months of consistent effort

Referral Criteria

Refer to pediatric urologist if not responding to behavioral therapy and antimuscarinics after 3 months

Hospitalisation Criteria

Admit for febrile UTI or acute urinary retention

Notes

Behavioral modifications are first-line before pharmacotherapy. Screen for bladder-bowel dysfunction as combined management improves outcomes.

Guideline Changes

FDA approval of beta-3 agonists for pediatric bladder dysfunction

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N39.0 — Urinary tract infection, site not specified

SNOMED: 68566005 | Duration: 10 days | Source: AAP 2024 Redbook, JAMA WikiGuidelines 2024

First-Line

- **Cephalexin** — 25-50 mg/kg/day PO divided TID-QID for 7-10 days; first-line for uncomplicated lower UTI
- **Amoxicillin-clavulanate** — 25-45 mg/kg/day PO divided BID-TID for 7-10 days; alternative oral first-line
- **Cefixime** — 8 mg/kg PO daily (max 400 mg) for 7-10 days; once-daily dosing improves adherence

Second-Line

- **TMP-SMX** — 6-12 mg TMP/kg/day PO divided BID for 7-10 days; avoid if local resistance >20%
- **Nitrofurantoin** — 5-7 mg/kg/day PO divided QID for 7 days; for lower UTI only; not for pyelonephritis
- **Ceftriaxone** — 50-75 mg/kg IV/IM daily (max 2 g) for toxic-appearing children or infants <2 months

Investigations

- Urine culture (clean catch or catheter) — Gold standard for diagnosis; must be collected before antibiotics [STAT]
- Urinalysis — Pyuria (WBC >5/hpf) and nitrites support UTI [STAT]
- Renal ultrasound — First febrile UTI; structural evaluation
- VCUG — Recurrent febrile UTI or abnormal renal US

Counselling

- Child has a urine infection that needs antibiotic treatment

- Complete the full antibiotic course; do not stop early even if symptoms improve
- Encourage child to drink plenty of water throughout the day
- Toilet habits are important: do not hold urine, empty bladder completely, wipe front to back
- Treat constipation as it contributes to urinary infections

Referral Criteria

Refer to pediatric nephrologist/urologist for first febrile UTI <2 years, recurrent UTIs (>3/year), or abnormal imaging

Hospitalisation Criteria

Admit if age <2 months, toxic-appearing, unable to tolerate oral medication, or complicated UTI with obstruction

Notes

Obtain urine culture before starting empiric antibiotics. Screen toilet-trained children for bladder-bowel dysfunction; shorter courses may suffice for afebrile UTI.

Guideline Changes

AAP 2024 supports shorter antibiotic courses (5-10 days) for uncomplicated UTI

N43 — Hydrocele and spermatocele

Duration: 7 days | Source: EAU/ESPU 2024, AAP

First-Line

- **Observation** — Watch and wait for noncommunicating hydrocele in infants; most resolve by 12-24 months of age
- **Surgical repair** — Inguinal approach hydrocelectomy if persists >12-24 mo, communicating type, or symptomatic

Second-Line

- **Laparoscopic repair** — Laparoscopic extraperitoneal closure; safe and effective in children; shorter recovery **[NEW 2024-25]**
- **Ibuprofen** — 10 mg/kg PO q6h post-op for pain control; acetaminophen 15 mg/kg q6h alternative; avoid narcotics in infants

Investigations

- Clinical examination with transillumination — Transillumination positive for hydrocele; confirms fluid
- Scrotal ultrasound — If diagnosis uncertain; differentiate from inguinal hernia or testicular pathology

Counselling

- Baby has fluid around the testicle causing swelling; this is usually harmless
- In infants, most hydroceles resolve on their own by 12-24 months
- Surgery is only needed if the swelling persists beyond 1-2 years or increases in size
- Surgery is a minor procedure; baby usually goes home the same day
- If swelling suddenly becomes painful or red, come to hospital immediately (may be hernia)

Referral Criteria

Refer to pediatric surgeon if hydrocele persists beyond 12-24 months, communicating type, or rapidly enlarging

Hospitalisation Criteria

Day-care surgery; overnight admission rarely needed unless complications

Notes

Noncommunicating hydroceles in infants usually resolve spontaneously. Spermatoceles are benign, asymptomatic, and rarely require intervention in children.

Guideline Changes

Laparoscopic hydrocele repair increasingly adopted as safe alternative

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N44 — Torsion of testis

SNOMED: 69282001 | Duration: 5 days | Source: EAU/ESPU 2024, AUA

First-Line

- **Emergency surgical exploration** — Scrotal exploration and detorsion ASAP; >90% salvage if within 6h; do not delay for imaging
- **Bilateral orchidopexy** — Fix both testes to dartos fascia at time of surgery; prevents recurrence and contralateral torsion

Second-Line

- **Manual detorsion** — Outward rotation (open book) at bedside as temporizing measure; does not replace surgery
- **Orchiectomy** — Removal of non-viable testis if necrotic on exploration; consider prosthesis discussion later

Investigations

- Clinical diagnosis (no delay for imaging) — Surgery should not be delayed for ultrasound if clinical suspicion is high [STAT]
- Scrotal ultrasound with Doppler — Only if diagnosis uncertain and will not delay surgery; absent blood flow [STAT]
- CBC, coagulation profile — Pre-operative workup [STAT]

Counselling

- Child's testicle has twisted, cutting off blood supply; this is a surgical emergency
- Surgery must happen within 6 hours for the best chance of saving the testicle
- Both testicles will be fixed during surgery to prevent future torsion
- After surgery, some swelling and bruising is normal; ice packs and pain medicine help
- If the testicle could not be saved, discuss prosthesis options when child is older

Referral Criteria

IMMEDIATE surgical referral; do not delay for imaging when clinical suspicion is high

Hospitalisation Criteria

Emergency admission for immediate scrotal exploration; no delay for imaging

Notes

Testicular torsion is a surgical emergency; salvage rate drops rapidly after 6 hours. Clinical diagnosis should not be delayed by imaging when suspicion is high.

N45 — Orchitis and epididymitis

SNOMED: 274718005 | Duration: 14 days | Source: AAP, EAU/ESPU 2024

First-Line

- **Supportive care** — Bed rest, scrotal elevation, ice packs, ibuprofen 10 mg/kg PO q6-8h or acetaminophen 15 mg/kg q6h
- **Ceftriaxone + Doxycycline** — Ceftriaxone 250 mg IM x1 + doxycycline 2 mg/kg PO BID x10 days; for sexually active adolescents only

Second-Line

- **TMP-SMX** — 4-6 mg TMP/kg/day PO divided BID for 10-14 days if UTI-associated epididymitis with positive culture
- **Cephalexin** — 25-50 mg/kg/day PO divided QID for 10-14 days for culture-positive bacterial epididymitis

Investigations

- Scrotal ultrasound with Doppler — Differentiate from testicular torsion; assess blood flow [URGENT]
- Urinalysis and urine culture — Document UTI if present [URGENT]
- CBC, CRP — Assess for systemic infection
- VCUG — Consider if UTI-associated; rule out structural abnormality

Counselling

- Child has swelling and pain in the testicle area; this is usually due to infection
- Rest, scrotal support, and pain medicine will help
- In prepubertal boys, antibiotics are only needed if urine test shows infection
- It is important to rule out testicular torsion (twisted testicle) which is an emergency
- Complete the antibiotic course if prescribed; follow up if not improving in 3-5 days

Referral Criteria

Refer to pediatric urologist to evaluate for anatomic abnormalities; emergency referral if torsion cannot be ruled out

Hospitalisation Criteria

Admit if testicular torsion cannot be excluded clinically or if systemically unwell

Notes

In prepubertal boys, antibiotics reserved for those with pyuria or positive urine culture. Refer to urology to evaluate for anatomic abnormalities.

Guideline Changes

Evidence supports withholding antibiotics in prepubertal boys without pyuria

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N47 — Phimosis

SNOMED: 449826002 | Duration: 56 days | Source: EAU/ESPU 2024, AAP

First-Line

- **Betamethasone 0.05%** — Apply to foreskin tip BID with gentle retraction for 4-8 wk; ~65% success rate
- **Mometasone 0.1%** — Apply to foreskin BID for 4-8 wk; alternative topical steroid with good efficacy

Second-Line

- **Clobetasol 0.05%** — Apply to foreskin BID for 4-6 wk; potent steroid for resistant physiologic phimosis
- **Preputioplasty** — Dorsal slit or limited preputioplasty for failed steroid therapy; foreskin-sparing alternative
- **Circumcision** — Definitive surgical option for pathologic phimosis, BXO, recurrent UTIs, or failed steroid therapy

Investigations

- Urinalysis — Screen for UTI if history of recurrent infections
- Clinical examination — Differentiate physiologic from pathologic (BXO) phimosis

Counselling

- Tight foreskin is normal in young boys and usually loosens by age 3-4 years
- Steroid cream applied to the foreskin tip twice daily for 4-8 weeks works in most cases
- Gently retract the foreskin as far as it goes without force while applying cream
- Surgery (circumcision) is only needed if cream does not work or skin disease (BXO) is present
- Good hygiene with gentle retraction during bathing is important once foreskin is retractable

Referral Criteria

Refer to pediatric urologist if steroid cream fails after 2 courses or pathologic phimosis suspected

Hospitalisation Criteria

Admit for paraphimosis (emergency) or circumcision under general anesthesia

Notes

Topical steroids are first-line; treatment can be repeated if phimosis recurs. BXO is an absolute indication for circumcision at any age.

N48.2 — Other inflammatory disorders of penis

SNOMED: 44882003 | Duration: 10 days | Source: EBM Guidelines, Perth Children's Hospital

First-Line

- **Warm sitz baths** — Warm water soaks 2-3x daily; gentle retraction and hygiene; supportive care for mild balanitis
- **Betamethasone 0.05%** — Apply to foreskin BID for 4-6 wk; for associated phimosis in children >3 yr

Second-Line

- **Clotrimazole 1%** — Apply topically BID for 7-14 days if candidal balanitis suspected
- **Bacitracin ointment** — Apply topically TID for 5-7 days for suspected bacterial infection; avoid neosporin in children
- **Oral antibiotics** — Cephalexin 25-50 mg/kg/day PO divided TID-QID for 7 days if refractory or strep-positive

Investigations

- Urine culture — Rule out associated UTI
- Wound swab culture — If purulent discharge present
- Blood glucose — Screen for diabetes in recurrent balanitis in older children

Counselling

- Child has an infection/inflammation of the foreskin/penis tip
- Warm water soaks 2-3 times daily will help reduce swelling and pain
- Do not forcefully retract the foreskin; gentle cleaning is sufficient
- Apply the prescribed cream/ointment as directed
- If symptoms return frequently, the doctor may recommend steroid cream or circumcision

Referral Criteria

Refer to pediatric urologist if recurrent episodes or suspected balanitis xerotica obliterans

Hospitalisation Criteria

Admit if unable to pass urine (acute retention), spreading cellulitis, or systemic signs of infection

Notes

Most cases resolve with hygiene and topical therapy. Refer to urology if recurrent or refractory to exclude balanitis xerotica obliterans.

Q54 — Hypospadias

SNOMED: 416010008 | Duration: 14 days | Source: EAU/ESPU 2024, AAP

First-Line

- **Surgical repair** — Urethroplasty at 6-18 months of age; technique based on severity (TIP, Mathieu, or staged)
- **Testosterone** — Topical DHT cream or IM testosterone 2 mg/kg q4wk x2-3 doses preop for small glans/proximal hypospadias

Second-Line

- **hCG** — 250-1000 IU IM 1-2x/wk for 5-6 wk preop; stimulates local testosterone for penile growth
- **Caudal block** — Bupivacaine 0.25% 1 mL/kg caudal block periop for analgesia; reduces opioid need

Investigations

- Urinalysis — Baseline before surgery
- Karyotype — Only if severe hypospadias with undescended testes (DSD evaluation)
- Renal ultrasound — Screen for associated urinary tract anomalies in severe cases
- Endocrine workup — 17-OHP, testosterone, DHT if DSD suspected

Counselling

- Child's urinary opening is not at the normal position; surgery will correct this
- Surgery is usually done between 6-18 months of age when the baby is healthy
- Do NOT get the child circumcised before surgery; the foreskin may be used for repair
- After surgery, a catheter (tube) will drain urine for a few days
- Follow-up visits are important to check for complications like narrowing of the new opening

Referral Criteria

Refer to pediatric urologist by 6 months of age for surgical planning

Hospitalisation Criteria

Admit for day-care or short-stay surgery; overnight observation for proximal hypospadias repair

Notes

Optimal surgical age is 6-18 months. Long-term follow-up is essential to detect stricture, voiding dysfunction, or recurrent curvature.

Guideline Changes

2024 EAU/ESPU Part II update on hypospadias surgical outcomes and follow-up

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Q60 — Renal agenesis and other reduction defects of kidney

SNOMED: 204938007 | Duration: 365 days | Source: Italian SPN Consensus 2022, CAKUT Guidelines

First-Line

- **Monitoring** — Yearly BP, urine albumin-creatinine ratio; serial renal US to confirm compensatory hypertrophy
- **ACE inhibitor** — Enalapril 0.08-0.6 mg/kg/day PO if hypertension or proteinuria develops; target BP <75th percentile

Second-Line

- **ARB** — Losartan 0.7-1.4 mg/kg/day PO once daily if ACEi not tolerated; renoprotective
- **Dietary counseling** — Moderate protein intake, adequate hydration, avoid nephrotoxins; no routine sports restriction per AAP

Investigations

- Renal ultrasound — Confirm solitary kidney; assess compensatory hypertrophy
- Blood pressure (ambulatory) — Annual ABPM for early hypertension detection
- Urine albumin/creatinine ratio — Annual screening for proteinuria
- Serum creatinine and eGFR — Annual renal function monitoring
- VCUG — Screen for VUR in contralateral kidney if UTI history

Counselling

- Child has only one kidney (or a very small kidney); most people live normal lives with one kidney
- Regular blood pressure checks and urine tests are important to protect the remaining kidney
- Encourage adequate water intake; avoid dehydration
- Contact sports with risk of abdominal injury should be discussed with the doctor
- Medicine may be needed if blood pressure rises or protein appears in urine

Referral Criteria

Refer to pediatric nephrologist for long-term monitoring; genetics if bilateral or syndromic

Hospitalisation Criteria

Admit for hypertensive emergency, acute kidney injury, or severe UTI

Notes

Solitary kidney predisposes to hypertension and proteinuria over time. Ambulatory BP monitoring preferred; multidisciplinary follow-up for complex CAKUT.

Guideline Changes

2022 Italian consensus on congenital solitary kidney management published

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Q61 — Cystic kidney disease

SNOMED: 236439005 | Duration: 365 days | Source: KDIGO 2025 ADPKD, International Consensus 2019

First-Line

- **BP control** — Target BP <50th-75th percentile; ACEi (enalapril 0.08-0.6 mg/kg/day PO) or ARB preferred
- **Monitoring** — Serial renal US, eGFR, urine albumin-creatinine ratio q6-12mo; Mayo Imaging Classification for risk

Second-Line

- **Tolvaptan** — V2-receptor antagonist; slows cyst growth in ADPKD; pediatric trials ongoing; adult dose 45-90 mg/d **[NEW 2024-25]**
- **Pravastatin** — Shown to slow total kidney volume growth in children with ADPKD; dose per age-based statin protocols **[NEW 2024-25]**
- **Metformin** — Emerging cyst-modifying agent; IMPEDE-PKD trial ongoing; not yet standard pediatric use **[NEW 2024-25]**

Investigations

- Renal ultrasound — Monitor cyst burden and kidney size
- Blood pressure monitoring (24h ABPM) — Detect hypertension early
- Serum creatinine and eGFR — Monitor renal function over time
- Urine albumin/creatinine ratio — Early proteinuria detection
- Genetic testing — Confirm PKD1/PKD2 mutation; family counseling

Counselling

- Child has cysts (fluid-filled sacs) in the kidneys; this is a lifelong condition
- Blood pressure control is the most important treatment to protect kidneys
- Regular check-ups with blood tests and ultrasound are essential
- Adequate water intake is important; encourage child to drink plenty of fluids
- Genetic counseling is recommended as this condition can be inherited

Referral Criteria

Refer to pediatric nephrologist at diagnosis; genetics counseling for family

Hospitalisation Criteria

Admit for hypertensive emergency, acute kidney injury, or infected cyst

Notes

BP control with RAS blockade is cornerstone of renoprotection. Novel disease-modifying agents (tolvaptan, metformin) are under active investigation in pediatric ADPKD.

Guideline Changes

KDIGO 2025 first guideline on ADPKD; tolvaptan and metformin trials ongoing in children

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Q62 — Congenital obstructive defects of renal pelvis and congenital malformations of ureter

SNOMED: 236807008 | Duration: 180 days | Source: SFU Consensus, EAU/ESPU 2024

First-Line

- **Observation** — Serial renal US q3-6mo for mild-moderate hydronephrosis; majority resolve spontaneously
- **Pyeloplasty** — Open or laparoscopic surgical repair of UPJ obstruction if worsening function or SFU grade 3-4

Second-Line

- **Prophylactic antibiotics** — Amoxicillin 10-15 mg/kg PO daily or TMP-SMX 2 mg/kg PO daily for high-grade HN
- **Robotic pyeloplasty** — Robotic-assisted laparoscopic approach; emerging in older children with UPJ obstruction **[NEW 2024-25]**
- **Ureteral reimplantation** — Surgical correction for obstructive megaureter or ureteral malposition if symptomatic

Investigations

- Renal ultrasound (serial) — Monitor hydronephrosis severity q3-6 months
- MAG3 diuretic renogram — Assess drainage and differential renal function
- VCUG — Evaluate for associated vesicoureteral reflux
- Serum creatinine — Monitor renal function in bilateral disease

Counselling

- Baby has a blockage or narrowing in the kidney drainage system found on ultrasound
- Many cases resolve on their own with time; regular ultrasound monitoring is needed
- Surgery is only needed if the kidney function is declining or infections occur
- Antibiotic prophylaxis may be given to prevent urinary infections
- Keep all follow-up appointments for ultrasound scans every 3-6 months

Referral Criteria

Refer to pediatric urologist at diagnosis; nephrology if renal function declining

Hospitalisation Criteria

Admit for febrile UTI, worsening hydronephrosis with declining function, or surgical intervention

Notes

Many cases resolve spontaneously; surgery reserved for declining renal function or recurrent infections. Early diagnosis via prenatal US enables timely monitoring.

Guideline Changes

Robotic-assisted laparoscopic pyeloplasty gaining wider pediatric use

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Q64.2 — Posterior urethral valves

SNOMED: 297163001 | Duration: 14 days | Source: French NDCP 2025, EAU/ESPU 2024

First-Line

- **Endoscopic valve ablation** — Cystoscopic incision of valves within 48h of birth once metabolically stable; gold standard
- **Bladder drainage** — Urethral catheter or suprapubic catheter as initial decompression in unstable neonates

Second-Line

- **Vesicostomy** — Cutaneous vesicostomy if infant too small or unstable for primary ablation; temporary diversion
- **Anticholinergics** — Oxybutynin 0.1-0.2 mg/kg PO BID for secondary bladder dysfunction post-ablation
- **CIC** — Clean intermittent catheterization for persistent bladder dysfunction post valve ablation

Investigations

- Renal ultrasound — Bilateral hydronephrosis, thick bladder wall, dilated posterior urethra [STAT]
- VCUG — Confirms posterior urethral valves; shows dilated posterior urethra [URGENT]
- Serum creatinine (nadir) — Best predictor of long-term renal outcome; check after drainage [URGENT]
- Serum electrolytes — Hyperkalemia and acidosis common [STAT]
- MAG3 renogram — Assess differential renal function after stabilization

Counselling

- Baby boy has valve-like flaps blocking urine flow; this needs urgent treatment
- A catheter will be placed to drain the bladder immediately
- Surgery (endoscopic) to remove the valves will be done once baby is stable
- Kidney function needs lifelong monitoring as some damage may already have occurred
- Child may need regular catheterization and medicines for bladder function long-term

Referral Criteria

Immediate referral to pediatric urologist; nephrology co-management for renal function

Hospitalisation Criteria

All cases require admission at diagnosis for catheter drainage and stabilization before surgery

Notes

Prenatal diagnosis enables early intervention. Multidisciplinary follow-up (urology + nephrology) is essential to monitor bladder and renal function long-term.

Guideline Changes

Miniaturized endoscopes (<2mm) now enable valve ablation in neonates >2 kg

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R31 — Hematuria, unspecified

SNOMED: 53298000 | Duration: 30 days | Source: AAP, KDIGO 2021 GN

First-Line

- **Evaluation workup** — Urinalysis with microscopy, urine Ca/Cr ratio, serum creatinine, C3/C4, renal US; repeat UA x3 to confirm
- **Observation** — Reassurance and serial monitoring q3-6mo if isolated microscopic hematuria without proteinuria or HTN

Second-Line

- **ACE inhibitor** — Enalapril 0.08-0.6 mg/kg/day PO if concurrent proteinuria >0.5 g/day develops
- **Antibiotics** — Treat underlying UTI if identified; cephalexin 25-50 mg/kg/day PO divided TID-QID for 7-10 days

Investigations

- Urinalysis with microscopy (x3) — Confirm persistence; dysmorphic RBCs suggest glomerular origin
- Urine calcium/creatinine ratio — Hypercalciuria screening
- Serum creatinine, C3, C4 — Renal function and complement assessment
- Renal ultrasound — Structural causes, stones, tumors
- Urine culture — Rule out UTI as cause

Counselling

- Blood has been found in child's urine; most causes in children are not serious
- Several tests are needed to find the cause; this may take a few visits
- If urine is tea-colored, face is puffy, or blood pressure is high, come immediately
- Avoid dehydration; encourage child to drink adequate water
- Follow up regularly for urine tests until the cause is identified and resolved

Referral Criteria

Refer to pediatric nephrologist if proteinuria present, hypertension, low complement, family history of renal disease, or persistent gross hematuria

Hospitalisation Criteria

Admit if gross hematuria with clots, associated nephritic syndrome, or significant proteinuria

Notes

Most isolated hematuria is benign; confirm persistence with 3 samples 2 weeks apart. Refer to nephrology if proteinuria, HTN, decreased C3, or family history of renal disease.

R80 — Proteinuria, unspecified

SNOMED: 29738008 | Duration: 90 days | Source: KDIGO 2024 CKD, AAP

First-Line

- **Confirm and quantify** — First morning urine protein/creatinine ratio; repeat x3 to exclude orthostatic proteinuria; renal function panel
- **ACE inhibitor** — Enalapril 0.08-0.6 mg/kg/day PO for persistent proteinuria >0.5 g/day; uptitrate q4-8 wk to effect

Second-Line

- **ARB** — Losartan 0.7-1.4 mg/kg/day PO once daily if ACEi not tolerated; antiproteinuric
- **Low-salt diet** — Sodium restriction <2 g/day enhances ACEi/ARB antiproteinuric effect; dietary counseling

Investigations

- First morning urine protein/creatinine ratio — Distinguish orthostatic from persistent proteinuria
- Serum albumin — Low albumin suggests nephrotic syndrome
- Serum creatinine — Assess renal function
- Complement C3, C4 — Screen for glomerulonephritis
- Renal ultrasound — Structural assessment

Counselling

- Protein is leaking into child's urine; this needs investigation to find the cause
- First morning urine sample is important for accurate testing
- Orthostatic proteinuria (only when standing) is harmless and common in adolescents
- Low-salt diet helps reduce proteinuria along with medicine if needed
- Regular follow-up urine tests are needed every 1-3 months

Referral Criteria

Refer to pediatric nephrologist if persistent non-orthostatic proteinuria >0.5 g/day, associated hematuria, or declining renal function

Hospitalisation Criteria

Admit if nephrotic-range proteinuria with edema or associated renal dysfunction

Notes

Exclude orthostatic proteinuria (benign) with first morning sample. Persistent non-orthostatic proteinuria warrants nephrology referral and consideration of biopsy.

Respiratory

A15 — Respiratory tuberculosis, bacteriologically and histologically confirmed

SNOMED: 186192006 | Duration: 120 days | Source: WHO 2022, ATS/CDC/IDSA 2025 Update

First-Line

- **Isoniazid (H)** — 10-15 mg/kg/day PO daily (max 300 mg); give with pyridoxine 1-2 mg/kg/day
- **Rifampicin (R)** — 10-20 mg/kg/day PO daily (max 600 mg); take on empty stomach
- **Pyrazinamide (Z)** — 30-40 mg/kg/day PO daily (max 2 g); used in intensive phase (first 2 months)
- **Ethambutol (E)** — 15-25 mg/kg/day PO daily (max 2.5 g); used in intensive phase; monitor visual acuity

Second-Line

- **4-month HPZM regimen (>=12 yr)** — Isoniazid + rifapentine + pyrazinamide + moxifloxacin daily x4 months; for adolescents >=12 yr **[NEW 2024-25]**
- **Levofloxacin** — 15-20 mg/kg/day PO daily (max 750 mg); for drug-resistant TB or ethambutol intolerance
- **Amikacin** — 15-30 mg/kg/day IV/IM daily (max 1 g); for MDR-TB regimens; monitor audiology

Investigations

- Sputum/gastric aspirate for AFB and GeneXpert — Confirm TB and detect rifampicin resistance [URGENT]
- Chest X-ray — Assess extent of pulmonary disease [URGENT]
- Mantoux test (TST) — Support diagnosis in children
- LFT baseline — Before starting hepatotoxic drugs (INH, RIF, PZA)
- HIV test — TB-HIV coinfection screening per NACO guidelines

Counselling

- Tuberculosis is curable but treatment takes 4-6 months - completing all medicines is critical
- All medicines must be taken daily without missing any dose; directly observed therapy is recommended
- Common side effects include orange urine (rifampicin) - this is normal and harmless
- Contact screening: all family members especially children should be tested for TB
- Your child can return to school after 2 weeks of treatment if feeling well and cough improving

Referral Criteria

Refer to pediatric TB specialist if drug-resistant TB, extrapulmonary TB, HIV co-infection, or treatment failure

Hospitalisation Criteria

Admit for severe or miliary TB, TB meningitis, respiratory distress, or when directly observed therapy initiation is needed

Notes

Standard regimen: 2 months HRZE then 2-4 months HR (4-month total for non-severe, 6-month for severe/cavitary). Directly Observed Therapy (DOT) is recommended.

Guideline Changes

ATS/CDC/ERS/IDSA 2025 update: 4-month regimen (2HRZE/2HR) now standard for non-severe pediatric TB

A16 — Respiratory tuberculosis, not confirmed bacteriologically or histologically

SNOMED: 154283005 | Duration: 120 days | Source: WHO 2022, ATS/CDC/IDSA 2025 Update

First-Line

- **Isoniazid (H)** — 10-15 mg/kg/day PO daily (max 300 mg); give with pyridoxine 1-2 mg/kg/day
- **Rifampicin (R)** — 10-20 mg/kg/day PO daily (max 600 mg); take on empty stomach
- **Pyrazinamide (Z)** — 30-40 mg/kg/day PO daily (max 2 g); intensive phase (first 2 months)

Second-Line

- **Ethambutol (E)** — 15-25 mg/kg/day PO daily (max 2.5 g); add to intensive phase if drug resistance suspected
- **4-month HPZM regimen (>=12 yr)** — Isoniazid + rifapentine + pyrazinamide + moxifloxacin daily x4 months; adolescents >=12 yr **[NEW 2024-25]**
- **Levofloxacin** — 15-20 mg/kg/day PO daily (max 750 mg); for drug-resistant TB or intolerance

Investigations

- Chest X-ray — Assess pulmonary involvement [URGENT]
- Mantoux test (TST) or IGRA — Support clinical diagnosis
- Gastric aspirate or induced sputum for GeneXpert — Attempt microbiological confirmation [URGENT]
- LFT baseline — Before starting ATT drugs

Counselling

- Your child is being treated for TB even though the test did not grow the bacteria - this is common in children
- Treatment is the same and just as important; complete all 4-6 months of medicines
- Give medicines on empty stomach in the morning as directed
- Watch for side effects: jaundice (yellow eyes), persistent vomiting, severe itching - report immediately
- All close contacts should be screened for TB; keep follow-up appointments

Referral Criteria

Refer to pediatric TB specialist if not improving after 2 months of treatment, drug-resistant contacts, or extrapulmonary involvement

Hospitalisation Criteria

Admit for severe disease, miliary/meningeal TB, respiratory distress, or for treatment initiation in unreliable home setting

Notes

Clinically diagnosed TB (smear/culture negative) in children is treated identically to confirmed TB. The 4-month regimen is non-inferior for non-severe disease per SHINE trial.

Guideline Changes

4-month regimen (2HRZ(E)/2HR) validated for clinically diagnosed non-severe pediatric TB per SHINE trial

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G47.3 — Obstructive sleep apnea

SNOMED: 78275009 | Duration: 90 days | Source: AAP 2012 (reaffirmed 2017), ATS 2024

First-Line

- **Adenotonsillectomy** — First-line surgical treatment for pediatric OSA; refer to ENT for evaluation
- **CPAP therapy** — Titrated via polysomnography; for children not candidates for surgery or with persistent OSA post-surgery
- **Intranasal corticosteroids** — Fluticasone 1 spray/nostril daily or mometasone 1 spray/nostril daily for mild OSA

Second-Line

- **Montelukast** — 1-5 yr: 4 mg PO daily; 6-14 yr: 5 mg PO daily; for mild residual OSA
- **Weight management** — Dietary and lifestyle interventions for obese children with OSA
- **Rapid maxillary expansion** — Orthodontic device for children with maxillary constriction and OSA

Investigations

- Polysomnography (sleep study) — Gold standard for diagnosing and grading OSA severity
- Lateral neck X-ray — Assess adenoid size if PSG not available
- Echocardiography — If severe OSA to rule out cor pulmonale

Counselling

- Your child stops breathing briefly during sleep which reduces oxygen to the brain and body
- Enlarged tonsils and adenoids are the most common cause in children and surgery usually cures it
- If your child snores loudly, breathes through the mouth, or is restless during sleep, these are important signs
- Overweight children should be encouraged to eat healthy and exercise to help reduce symptoms
- After surgery, a follow-up sleep study may be needed to confirm the problem is resolved

Referral Criteria

Refer to ENT for adenotonsillectomy evaluation; sleep medicine if OSA persists post-surgery or child not a surgical candidate

Hospitalisation Criteria

Admit for adenotonsillectomy (day surgery or overnight for high-risk); severe OSA in young children (<3 yrs) or those with comorbidities may need overnight observation post-op

Notes

Up to 40% of children have persistent OSA after adenotonsillectomy. Polysomnography is recommended for diagnosis and post-surgical reassessment.

Guideline Changes

ATS 2024 guideline on persistent post-adenotonsillectomy OSA management

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J00 — Common Cold (Acute Viral Upper Respiratory Infection) *[mild]*

SNOMED: 82272006 | Duration: 5 days | Source: IAP 2024,WHO 2023,AAP Guidelines

First-Line

- **PARACETAMOL** — 10–15 mg/kg/dose PO every 6–8 hours SOS for fever >100°F, maximum 60 mg/kg/day, duration 3–5 days
- **NASAL SALINE (0.65% NORMAL SALINE)** — 2–3 drops in each nostril 4–6 times/day for nasal blockage relief and feeding improvement

Second-Line

- **PHENYLEPHRINE + CHLORPHENIRAMINE** — Oral drops as per weight (typically 2–6 drops TDS depending on age), use only if significant nasal discharge/congestion and child is symptomatic; avoid use >5 days and use cautiously in infants

Investigations

- CBC — Persistent fever >3 days, toxic appearance, suspicion of bacterial infection
- CRP — To rule out bacterial infection if symptoms persist or worsen

Counselling

- This is a common viral illness and usually resolves on its own within few days — यह एक सामान्य वायरल बीमारी है, कुछ दिनों में अपने आप ठीक हो जाती है।
- Continue breastfeeding and give adequate fluids; small frequent feeds are better — माँ का दूध जारी रखें और पर्याप्त तरल पदार्थ दें; थोड़ा-थोड़ा बार-बार पिलाएं।
- Use medicines only as prescribed; avoid over-the-counter cough syrups in infants — केवल डॉक्टर की बताई दवा दें; छोटे बच्चों को बाज़ार की खांसी की दवा न दें।
- Keep child warm, avoid cold exposure, and maintain hygiene — बच्चे को गर्म रखें, ठंड से बचाएं और साफ़-सफ़ाई रखें।
- Follow-up after 2–3 days or earlier if symptoms worsen — 2-3 दिन बाद या लक्षण बिगड़ें तो पहले डॉक्टर को दिखाएं।

Warning Signs

- Fast breathing or chest indrawing
- Bluish lips or tongue

- Convulsions or fits
- Excessive sleepiness or unresponsiveness
- Persistent vomiting
- Poor feeding or refusal to feed
- Fever lasting more than 3 days
- Drop in oxygen saturation

Referral Criteria

Refer to pediatrician or higher center if no improvement after 3 days, recurrent episodes, or development of warning signs

Hospitalisation Criteria

Admit if SpO2 <92%, severe respiratory distress, poor feeding, dehydration, lethargy, or age <2 months with symptoms

Notes

Viral upper respiratory infection; self-limiting illness lasting 5–7 days; antibiotics are not indicated routinely

Guideline Changes

IAP 2024: Continued emphasis on no routine antibiotics or OTC cough/cold medicines in children; ACCP: Recommends against OTC cough suppressants in pediatric URI; No major guideline changes in 2024-2025 for common cold management

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J01 — Acute sinusitis

SNOMED: 15805002 | Duration: 14 days | Source: AAP 2013

First-Line

- **Amoxicillin** — 45-90 mg/kg/day PO divided BID (max 2 g/dose) for 10-14 days; standard first-line
- **Amoxicillin-clavulanate** — 90 mg/kg/day (amoxicillin) PO divided BID (max 2 g/dose) for 10-14d; severe or risk factors

Second-Line

- **Ceftriaxone** — 50 mg/kg IM/IV x1 dose (max 1 g); for vomiting or unable to tolerate oral; then switch to PO
- **Clindamycin** — 30-40 mg/kg/day PO divided TID (max 1.8 g/day) for penicillin-allergic patients
- **Intranasal corticosteroids** — Fluticasone 1 spray/nostril daily as adjunct to antibiotics; age >=2 yr

Investigations

- Clinical diagnosis (no routine imaging) — Diagnosis based on clinical criteria per AAP
- CT sinuses — Only if orbital or intracranial complications suspected [STAT]
- CBC — If severe or complicated presentation [URGENT]

Counselling

- Sinusitis means infection of the sinuses (air spaces in the face bones)
- Antibiotics need to be taken for the full 10-14 days even if your child feels better earlier
- Saline nose drops or spray help clear the sinuses - use before meals and bedtime
- Avoid smoke exposure and keep the house dust-free
- Return if symptoms worsen after 48-72 hours of antibiotics or if high fever develops

Referral Criteria

Refer to ENT if complications (orbital/intracranial), recurrent acute sinusitis (>4 episodes/year), or progression to chronic sinusitis

Hospitalisation Criteria

Admit for orbital complications (periorbital/orbital cellulitis), intracranial complications, or toxic appearance

Notes

Diagnose bacterial sinusitis by clinical criteria (persistent >10d, worsening, or severe onset). Watchful waiting for 3 days may be appropriate for mild uncomplicated cases.

J02 — Acute pharyngitis

SNOMED: 363746003 | Duration: 10 days | Source: IDSA 2012, AAP, CDC GAS Guidelines

First-Line

- **Amoxicillin** — 50 mg/kg PO once daily (max 1000 mg) or 25 mg/kg BID (max 500 mg/dose) for 10 days
- **Penicillin V** — 250 mg PO BID-TID (<27 kg) or 500 mg PO BID (>=27 kg) for 10 days for confirmed GAS
- **Benzathine penicillin G** — <27 kg: 600,000 U IM x1; >=27 kg: 1.2 MU IM x1; for adherence concerns

Second-Line

- **Cephalexin** — 20 mg/kg/dose PO BID (max 500 mg/dose) for 10 days; non-anaphylactic penicillin allergy
- **Azithromycin** — 12 mg/kg PO day 1 (max 500 mg) then 6 mg/kg/day x4 days; for anaphylactic PCN allergy
- **Clindamycin** — 7 mg/kg/dose PO TID (max 300 mg/dose) for 10 days; for type 1 PCN allergy

Investigations

- Rapid strep test (throat swab) — Detect Group A Streptococcus before prescribing antibiotics [URGENT]
- Throat culture — If rapid test negative but clinical suspicion high
- ASO titer — If post-streptococcal complications suspected

Counselling

- Sore throat can be caused by virus or bacteria; a throat swab test will tell which one
- If it is strep throat (bacterial), antibiotics for full 10 days are needed to prevent rheumatic fever
- Give soft, warm foods and plenty of fluids; cold items like ice cream also help soothe the throat
- Paracetamol or ibuprofen can be given for pain and fever
- Your child can return to school 24 hours after starting antibiotics if feeling better

Referral Criteria

Refer to ENT if recurrent strep pharyngitis (>=7/year or >=5/year for 2 years) for tonsillectomy consideration; rheumatology if rheumatic fever develops

Hospitalisation Criteria

Admit for peritonsillar abscess, inability to swallow fluids, dehydration, or airway compromise

Notes

Rapid strep test or throat culture should confirm GAS before starting antibiotics. Treatment prevents rheumatic fever and reduces symptom duration by 1-2 days.

J03 — Acute tonsillitis

SNOMED: 90176007 | Duration: 10 days | Source: IDSA 2012, AAP/CDC GAS Guidelines

First-Line

- **Amoxicillin** — 50 mg/kg PO once daily (max 1000 mg) or 25 mg/kg BID (max 500 mg/dose) for 10 days
- **Penicillin V** — 250 mg PO BID-TID (<27 kg) or 500 mg PO BID (>=27 kg) for 10 days for confirmed GAS
- **Benzathine penicillin G** — <27 kg: 600,000 U IM x1; >=27 kg: 1.2 MU IM x1; for adherence concerns

Second-Line

- **Cephalexin** — 20 mg/kg/dose PO BID (max 500 mg/dose) for 10 days; for non-anaphylactic PCN allergy
- **Azithromycin** — 12 mg/kg PO day 1 (max 500 mg) then 6 mg/kg/day x4 days; for type 1 PCN allergy
- **Clindamycin** — 7 mg/kg/dose PO TID (max 300 mg/dose) for 10 days; penicillin allergy alternative

- **Acetaminophen/Ibuprofen** — Acetaminophen 15 mg/kg q4-6h or ibuprofen 10 mg/kg q6-8h PRN for pain/fever

Investigations

- Rapid strep test — Detect Group A Streptococcus [URGENT]
- Throat culture — If rapid test negative but GAS clinically suspected
- Monospot test — If EBV mononucleosis suspected (bilateral tonsillar enlargement, lymphadenopathy)

Counselling

- Tonsillitis is usually caused by a virus but strep bacteria also cause it
- If strep test is positive, complete 10 days of antibiotics to prevent rheumatic fever and kidney problems
- Warm salt water gargling (for children >6 years) and cold fluids help with pain
- Your child can go back to school 24 hours after starting antibiotics
- If tonsillitis keeps recurring (>7 times a year), surgery to remove tonsils may be recommended

Referral Criteria

Refer to ENT for recurrent tonsillitis (>=7/yr, >=5/yr x2 yrs, or >=3/yr x3 yrs) or peritonsillar abscess

Hospitalisation Criteria

Admit for peritonsillar abscess drainage, airway compromise from severe tonsillar enlargement, inability to swallow, or dehydration

Notes

Confirm GAS with rapid strep test or culture before prescribing antibiotics. Tonsillectomy is considered for recurrent GAS tonsillitis (>=7 episodes/yr).

J04 — Acute laryngitis and tracheitis

SNOMED: 195679003 | Duration: 5 days | Source: AAP, CPS 2018, Cochrane Reviews

First-Line

- **Dexamethasone** — 0.6 mg/kg PO/IM x1 dose (max 16 mg); onset 2-3h, lasts 24-48h; effective for all severity
- **Nebulized epinephrine** — L-epi 1:1000: 0.5 mL/kg (max 5 mL) or racemic epi 2.25%: 0.05 mL/kg (max 0.5 mL) neb
- **Cool mist/humidified air** — Cool mist therapy; limited evidence but commonly used for comfort; avoid steam (burn risk)

Second-Line

- **Prednisolone** — 1-2 mg/kg/day PO (max 60 mg) for 2-3 days; if dexamethasone unavailable
- **Budesonide nebulized** — 2 mg neb x1 dose; for patients unable to take PO dexamethasone or with persistent symptoms
- **Heliox** — 70:30 or 80:20 helium-oxygen mixture via face mask; for severe croup with stridor at rest

Investigations

- Clinical diagnosis (no routine investigations) — Croup is diagnosed clinically by barking cough and stridor
- Neck X-ray (AP view) — Steeple sign if diagnosis uncertain (not routine)
- Pulse oximetry — If significant respiratory distress [STAT]

Counselling

- Croup causes a barking cough and noisy breathing which sounds worse at night
- A single dose of dexamethasone steroid medicine usually helps a lot within a few hours
- Cool night air or cool mist can help ease the breathing - take the child near an open window
- Most croup gets better in 3-5 days; the barking cough may last a bit longer
- Rush to hospital if your child has noisy breathing at rest, drooling, or turns blue

Referral Criteria

Refer to ENT if recurrent croup (>3 episodes) to rule out subglottic stenosis or other airway abnormality

Hospitalisation Criteria

Admit if stridor at rest persists after nebulized epinephrine, severe respiratory distress, age <6 months, or toxic appearance (suspect bacterial tracheitis)

Notes

A single dose of oral dexamethasone is effective for all croup severities including mild. Observe for 2-4 hours after nebulized epinephrine due to rebound potential.

J05.0 — Viral Croup Syndrome

SNOMED: 71186008 | Duration: 3 days | Source: IAP 2024, AAP Croup Guidelines, WHO

First-Line

- **Dexamethasone** — 0.15-0.6 mg/kg single dose PO/IM/IV, max 16 mg; gold standard, reduces airway edema
- **Epinephrine (nebulisation)** — 0.5 ml/kg of 1:1000 solution (max 5 ml) nebulised, repeat after 20-30 min if needed; for moderate-severe cases with stridor at rest
- **Oxygen Therapy** — If SpO2 < 92%, via nasal prongs or mask
- **Levosalbutamol + Ambroxol** — Age/weight appropriate dose nebulised or PO, optional bronchial/cough support, not primary treatment for croup
- **Fexofenadine** — Age-appropriate dose PO, only if allergic component present

Investigations

- Pulse oximetry — All cases to assess oxygenation [STAT]
- Neck X-ray AP view — Steeple sign if diagnosis uncertain or atypical [URGENT]

Counselling

- Croup causes a barking cough and noisy breathing - keep child calm as crying worsens symptoms
- Use humidified air or steam in the room
- Give adequate fluids and small frequent meals
- Improvement expected within 24 hours after steroid dose
- Full recovery usually in 3-5 days
- If stridor persists at rest, child has severe breathing difficulty, SpO2 drops, or child becomes lethargic - come to hospital immediately

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Stridor at rest that does not improve
- Severe chest retractions
- SpO2 dropping below 92 percent
- Poor oral intake or refusal to drink

Referral Criteria

Refer to pediatric pulmonologist or ENT if recurrent croup (>2 episodes), atypical presentation, suspected foreign body, or subglottic stenosis

Hospitalisation Criteria

Admit if stridor at rest persists after dexamethasone and adrenaline, severe retractions, SpO2 <92%, lethargy/exhaustion, poor oral intake, or need for repeated adrenaline nebulisation

Notes

Dexamethasone single dose is the single most effective treatment. Adrenaline nebulisation for emergency relief in moderate-severe cases - observe 2-4 hours after dose for recurrence. Keep child calm (crying worsens stridor). Humidified air, adequate fluids. Avoid unnecessary antibiotics. Severity: Mild = barking cough no stridor at rest; Moderate = stridor at rest with mild retractions; Severe = marked distress with hypoxia.

Guideline Changes

2024 cohort data: single-dose dexamethasone (0.6 mg/kg) remains gold standard; repeat dosing in hospitalised children showed longer LOS without benefit on 30-day ED return; 0.15 mg/kg may suffice for mild croup; nebulised epinephrine observation period confirmed at 2-4 hr.

J06.9 — Acute upper respiratory infection, unspecified

SNOMED: 54150009 | Duration: 7 days | Source: AAP Choosing Wisely, AAFP 2022

First-Line

- **Acetaminophen** — 15 mg/kg/dose PO q4-6h PRN (max 75 mg/kg/day or 4 g/day) for fever and discomfort
- **Ibuprofen** — 10 mg/kg/dose PO q6-8h PRN (max 40 mg/kg/day); age >=6 months for fever/pain
- **Nasal saline irrigation** — Saline drops (infants) or spray/rinse (older children) PRN for nasal congestion

Second-Line

- **Honey (cough relief)** — 2.5-10 mL PO at bedtime for nocturnal cough; age >1 yr only (botulism risk in infants)
- **Cool-mist humidifier** — Bedside humidifier at night for cough/congestion relief; keep device clean
- **Dexamethasone (if croup)** — 0.6 mg/kg PO x1 (max 16 mg) if laryngotracheitis/croup component identified

Investigations

- None routinely needed — Diagnosis is clinical
- CBC — Only if fever prolonged >5 days to rule out bacterial infection

Counselling

- Upper respiratory infection (URI) is very common in children and caused by viruses
- Antibiotics are NOT helpful and should not be used for viral URI
- Keep your child hydrated with warm fluids; honey at bedtime for cough (>1 year only)
- Do not give OTC cough and cold medicines to children under 4 years
- See a doctor if fever lasts >5 days, breathing becomes fast, or ear pain develops

Referral Criteria

Referral not usually needed; refer to ENT if recurrent URI with complications or suspected anatomical abnormality

Hospitalisation Criteria

Hospitalization rarely needed; admit if febrile infant <3 months, severe dehydration, or complications develop

Notes

Antibiotics are NOT indicated for viral URI. Avoid OTC cough/cold medications in children <4 years per AAP recommendation due to lack of efficacy and safety concerns.

J12 — Viral pneumonia

SNOMED: 75570004 | Duration: 7 days | Source: IDSA 2018 Influenza, AAP, WHO 2023

First-Line

- **Supportive care** — O2 if SpO2 <92%; IV fluids if dehydrated; antipyretics PRN; humidified air
- **Oseltamivir (if influenza)** — <1 yr: 3 mg/kg PO BID; 1-12 yr: weight-based (30-75 mg BID); start within 48h x5 days

Second-Line

- **Baloxavir marboxil (if influenza)** — 5-<12 yr, >=20 kg: weight-based single dose PO; >=12 yr & 40-80 kg: 40 mg x1; >80 kg: 80 mg **[NEW 2024-25]**
- **Remdesivir (if COVID-19)** — >=3.5 kg: 5 mg/kg IV day 1, then 2.5 mg/kg IV daily for 5 days; for hospitalized pts **[NEW 2024-25]**
- **Ribavirin (rare)** — Aerosolized: 6 g reconstituted in 300 mL via SPAG-2 x12-18 hr/day; severe immunocompromised RSV

Investigations

- Pulse oximetry — Assess oxygenation [STAT]
- Chest X-ray — If moderate-severe or clinical uncertainty [URGENT]
- Respiratory viral panel (NP swab) — Identify specific virus (influenza, RSV, adenovirus) [URGENT]

Counselling

- Your child has a viral lung infection - antibiotics will NOT help as it is caused by a virus
- Most viral pneumonia gets better on its own in 1-2 weeks with rest and fluids
- Give paracetamol or ibuprofen for fever; ensure your child drinks plenty of fluids
- If your child has influenza, antiviral medicine (oseltamivir) works best if started within 48 hours
- Return if breathing worsens, fever persists beyond 5 days, or child stops eating/drinking

Referral Criteria

Refer to pediatric pulmonologist if severe course, not improving as expected, or underlying lung/immune condition

Hospitalisation Criteria

Admit if SpO2 <92%, severe respiratory distress, dehydration, or high-risk child (immunocompromised, chronic lung disease)

Notes

Most viral pneumonia in children is self-limiting and requires only supportive care. Oseltamivir should be started within 48 hours for confirmed or suspected influenza.

J13 — Pneumococcal pneumonia

SNOMED: 266350000 | Duration: 7 days | Source: IDSA/PIDS 2011, WHO 2023

First-Line

- **Amoxicillin (outpatient)** — 90 mg/kg/day PO divided BID (max 4 g/day) for 5-7 days; first-line oral therapy
- **Ampicillin IV (inpatient)** — 150-200 mg/kg/day IV divided q6h (max 12 g/day) for hospitalized uncomplicated cases
- **Penicillin G IV** — 200,000-250,000 U/kg/day IV divided q4-6h (max 24 MU/day) if penicillin-susceptible

Second-Line

- **Ceftriaxone** — 50-100 mg/kg/day IV/IM once daily (max 4 g) for penicillin-resistant strains (MIC >=2)
- **Clindamycin** — 30-40 mg/kg/day IV divided q6-8h (max 2.7 g/day) for penicillin-allergic patients
- **Levofloxacin** — >=5 yr: 8-10 mg/kg/day PO/IV daily (max 750 mg) for beta-lactam allergy
- **Vancomycin** — 40-60 mg/kg/day IV q6-8h (max 4 g/day) add if meningitis suspected or highly resistant

Investigations

- Chest X-ray — Confirm pneumonia and check for complications [URGENT]
- Blood culture — Confirm S. pneumoniae and susceptibility [URGENT]
- CBC — Assess severity of infection [URGENT]
- CRP — Guide treatment duration [URGENT]

Counselling

- Your child has pneumonia caused by pneumococcus bacteria which responds well to antibiotics
- Complete the full course of antibiotics even after your child starts feeling better
- Ensure your child has received all pneumococcal vaccine doses (PCV) as per schedule
- Give adequate fluids and rest; paracetamol for fever
- Return if fever persists after 48 hours of antibiotics or breathing worsens

Referral Criteria

Refer to pediatric pulmonologist if complicated pneumonia (empyema, necrotizing) or recurrent pneumococcal infections suggesting immunodeficiency

Hospitalisation Criteria

Admit if SpO2 <92%, unable to take oral medicines, moderate-severe respiratory distress, complicated pneumonia, or age <6 months

Notes

High-dose amoxicillin (90 mg/kg/day) overcomes intermediate penicillin resistance in S. pneumoniae. Ensure PCV13/15/20 vaccination is up to date.

J14 — Pneumonia due to Haemophilus influenzae

SNOMED: 10625231000119106 | Duration: 7 days | Source: IDSA/PIDS 2011, WHO 2023

First-Line

- **Amoxicillin-clavulanate** — 90 mg/kg/day (amoxicillin) PO divided BID (max 4 g/day) for 7 days; covers beta-lactamase
- **Ceftriaxone IV** — 50-100 mg/kg/day IV/IM once daily (max 4 g) for hospitalized patients

Second-Line

- **Cefotaxime** — 150-200 mg/kg/day IV divided q6-8h (max 12 g/day) alternative to ceftriaxone
- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg) then 5 mg/kg/day x4 days for beta-lactam allergy
- **Ampicillin IV** — 150-200 mg/kg/day IV divided q6h; only if beta-lactamase negative confirmed
- **TMP-SMX** — 8-10 mg/kg/day (TMP) PO divided BID for 7-10 days as oral step-down alternative

Investigations

- Chest X-ray — Confirm and assess pneumonia [URGENT]
- Blood culture — Confirm H. influenzae and susceptibility testing [URGENT]
- CBC with differential — Assess infection severity [URGENT]
- CRP — Monitor treatment response [URGENT]

Counselling

- Your child has pneumonia caused by Haemophilus bacteria - antibiotics will treat it
- Complete the full antibiotic course as prescribed even if your child improves early
- Ensure your child has received Hib vaccine doses as per schedule
- Give plenty of fluids and paracetamol for fever
- Return if fever does not improve in 48 hours or breathing worsens

Referral Criteria

Refer to pediatric pulmonologist if complicated or recurrent pneumonia; immunology if recurrent H. influenzae infections

Hospitalisation Criteria

Admit if SpO2 <92%, severe respiratory distress, unable to take oral medicines, or complicated pneumonia

Notes

25-50% of H. influenzae produce beta-lactamase; do not use ampicillin empirically. Use amoxicillin-clavulanate or a cephalosporin for empiric coverage.

J15.9 — Bacterial pneumonia, unspecified

SNOMED: 53084003 | Duration: 7 days | Source: IDSA/PIDS 2011 (update 2026), WHO 2023

First-Line

- **Amoxicillin** — 90 mg/kg/day PO divided BID-TID (max 4 g/day) for 5-7 days; first-line outpatient
- **Ampicillin IV** — 150-200 mg/kg/day IV divided q6h (max 12 g/day) for hospitalized uncomplicated CAP
- **Ceftriaxone** — 50-100 mg/kg/day IV/IM once daily (max 4 g) if not fully immunized or resistant organism

Second-Line

- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg) then 5 mg/kg/day x4 days for atypical coverage
- **Clindamycin** — 30-40 mg/kg/day IV divided q6-8h (max 2.7 g/day) if MRSA or anaerobe suspected
- **Levofloxacin** — 6 mo-4 yr: 16-20 mg/kg/day IV/PO BID; >=5 yr: 8-10 mg/kg/day daily (max 750 mg)
- **Cefotaxime** — 150-200 mg/kg/day IV divided q6-8h (max 12 g/day) for hospitalized patients

Investigations

- Chest X-ray — Confirm pneumonia and assess for complications [URGENT]
- CBC with differential — Assess infection severity [URGENT]
- Blood culture — Identify pathogen; obtain before antibiotics if possible [URGENT]
- CRP/Procalcitonin — Guide antibiotic therapy and monitor response [URGENT]

Counselling

- Your child has a bacterial lung infection (pneumonia) that needs antibiotics
- Complete the full antibiotic course even if your child feels better after a few days
- Give plenty of fluids and rest; keep your child in a well-ventilated room
- Return to hospital if fever does not come down after 48 hours of antibiotics or breathing worsens
- Make sure your child is up to date on pneumococcal and flu vaccines to prevent future pneumonia

Referral Criteria

Refer to pediatric pulmonologist if complicated pneumonia (effusion, empyema, necrotizing), recurrent pneumonia, or immunocompromised host

Hospitalisation Criteria

Admit if SpO2 <92%, moderate-severe respiratory distress, unable to tolerate oral fluids/medicines, age <6 months, or complicated pneumonia

Notes

Amoxicillin is first-line for outpatient CAP in immunized children. Viral pathogens cause most pneumonia in preschoolers; antibiotics not always needed.

Guideline Changes

IDSA/PIDS 2026 updated CAP guidelines released Feb 2026 replacing 2011 version

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J18.9 — Pneumonia, unspecified organism

SNOMED: 233604007 | Duration: 7 days | Source: IDSA/PIDS 2011 (update 2026), WHO 2023

First-Line

- **Amoxicillin** — 90 mg/kg/day PO divided BID (max 4 g/day) for 5-7 days; outpatient empiric first-line
- **Ampicillin IV** — 150-200 mg/kg/day IV divided q6h (max 12 g/day) for hospitalized children
- **Supportive care** — O2 if SpO2 <92%; IV fluids if dehydrated; antipyretics (acetaminophen 15 mg/kg q4-6h)

Second-Line

- **Ceftriaxone** — 50-100 mg/kg/day IV/IM once daily (max 4 g) for under-immunized or severe illness
- **Azithromycin** — 10 mg/kg PO day 1 then 5 mg/kg/day x4 days for school-age if atypical suspected
- **Oseltamivir** — <=15 kg: 30 mg BID; 15-23 kg: 45 mg; 23-40 kg: 60 mg; >40 kg: 75 mg BID x5d if influenza
- **Clindamycin** — 30-40 mg/kg/day IV q6-8h (max 2.7 g/day) add if necrotizing/MRSA suspected

Investigations

- Chest X-ray — Confirm pneumonia diagnosis [URGENT]
- CBC with differential — Help differentiate viral vs bacterial [URGENT]
- CRP or Procalcitonin — Guide decision on antibiotic therapy [URGENT]
- Blood culture — If hospitalized or toxic-appearing [URGENT]

Counselling

- Your child has a lung infection (pneumonia) - most cases in young children are caused by viruses
- If antibiotics are prescribed, complete the full course; do not stop early
- Give plenty of warm fluids, rest, and paracetamol/ibuprofen for fever
- Watch for danger signs: very fast breathing, chest indrawing, refusing to eat, becoming very sleepy
- Pneumococcal and influenza vaccines help prevent many types of pneumonia

Referral Criteria

Refer to pediatric pulmonologist if complicated pneumonia, recurrent pneumonia (>2 episodes in 1 year), or underlying risk factors

Hospitalisation Criteria

Admit if SpO2 <92%, severe respiratory distress, unable to feed/drink, toxic appearance, or age <6 months with suspected bacterial pneumonia

Notes

Viral causes predominate in children <5 years; antibiotics are not always required. Obtain CXR and labs to help differentiate bacterial vs viral etiology.

Guideline Changes

IDSA/PIDS 2026 updated empiric CAP therapy recommendations

J20 — Acute bronchitis

SNOMED: 10509002 | Duration: 10 days | Source: AAP, AAFP 2016, CDC Antibiotic Stewardship

First-Line

- **Acetaminophen** — 15 mg/kg/dose PO q4-6h PRN (max 75 mg/kg/day or 4 g/day) for fever and discomfort
- **Ibuprofen** — 10 mg/kg/dose PO q6-8h PRN (max 40 mg/kg/day); age >=6 months for fever/pain
- **Honey (cough)** — 2.5-10 mL PO at bedtime for cough in children >1 yr; NOT for infants <1 yr (botulism risk)

Second-Line

- **Cool-mist humidifier** — Bedside use at night for cough relief; keep clean to prevent mold
- **Dextromethorphan** — 1-2 mg/kg/day divided q6-8h PO; age >4 yr only; limited evidence of efficacy
- **Amoxicillin (if PBB suspected)** — 40-50 mg/kg/day PO divided BID-TID for 2-4 wk if chronic wet cough >4 wk (PBB)

Investigations

- None routinely needed — Diagnosis is clinical
- Chest X-ray — Only if pneumonia or complications suspected

Counselling

- Acute bronchitis is usually caused by a virus and antibiotics are NOT needed

- Cough may last 2-3 weeks even after the child is better - this is normal
- Give plenty of warm fluids; honey at bedtime helps cough in children over 1 year
- Avoid cough syrups in children under 4 years as they do not work and can be harmful
- Return if fever persists beyond 5 days, or if your child develops difficulty breathing

Referral Criteria

Refer to pediatric pulmonologist if wet cough persists >4 weeks (suspected protracted bacterial bronchitis) or recurrent bronchitis

Hospitalisation Criteria

Hospitalization rarely needed; admit only if respiratory distress, inability to maintain hydration, or suspected complications (pneumonia)

Notes

Antibiotics should NOT be prescribed for acute bronchitis as etiology is almost always viral. Consider protracted bacterial bronchitis only if wet cough persists >4 weeks.

J21 — Acute bronchiolitis

SNOMED: 4120002 | Duration: 7 days | Source: AAP 2014 (reaffirmed 2022)

First-Line

- **Supplemental oxygen** — Via nasal cannula if SpO2 persistently <90%; titrate to maintain SpO2 >=90%
- **Nasal suctioning** — Gentle superficial suctioning of nares PRN; avoid deep suctioning
- **IV/NG fluids** — If unable to maintain oral hydration; NS or D5-NS at maintenance rate

Second-Line

- **HFNC (high-flow nasal cannula)** — 2 L/kg/min (max 8-10 L/min in infants) if worsening on standard O2; FiO2 titrated
- **Nebulized hypertonic saline** — 3% NaCl 4 mL neb q4-8h; may reduce hospital stay in admitted patients; not for ED
- **Trial of albuterol** — 0.15 mg/kg neb x1 dose; consider in >12 mo with prior wheeze/atopy; stop if no response

Investigations

- Pulse oximetry — Assess need for supplemental oxygen [STAT]
- Nasopharyngeal aspirate for RSV/viral panel — Identify causative virus for cohorting
- Chest X-ray — Only if diagnostic uncertainty or clinical deterioration (not routine)

Counselling

- Bronchiolitis is a viral chest infection very common in babies during winter months
- Antibiotics do not help as this is a viral infection - supportive care is the main treatment
- Keep your baby's nose clear by gentle suctioning before feeds
- Feed small amounts frequently as babies tire easily while breathing fast
- Come back to hospital if your baby stops feeding, breathes very fast, or turns blue

Referral Criteria

Refer to pediatric pulmonologist if atypical course, recurrent bronchiolitis, or underlying lung/heart disease

Hospitalisation Criteria

Admit if SpO2 persistently <90%, unable to maintain oral feeding, marked respiratory distress, age <3 months, or significant comorbidity (CHD, prematurity, immunodeficiency)

Notes

Bronchodilators, corticosteroids, and antibiotics are NOT routinely recommended. Supportive care with oxygen and hydration is the mainstay of treatment.

J21.0 — Acute bronchiolitis due to RSV

SNOMED: 57089007 | Duration: 7 days | Source: AAP 2014, CDC/AAP 2023 RSV Immunization

First-Line

- **Supportive care** — O2 if SpO2 <90%, nasal suctioning PRN, IV/NG fluids if poor oral intake; no routine meds
- **Nirsevimab (prophylaxis)** — <5 kg: 50 mg IM x1; >=5 kg: 100 mg IM x1; before first RSV season; high-risk 2nd season: 200 mg **[NEW 2024-25]**
- **HFNC** — 2 L/kg/min for infants with worsening respiratory distress despite standard O2

Second-Line

- **Palivizumab (prophylaxis)** — 15 mg/kg IM monthly x5 doses during RSV season; being phased out by Dec 2025
- **Nebulized hypertonic saline** — 3% NaCl 4 mL neb q4-8h; for hospitalized patients; may reduce LOS
- **Ribavirin (rare use)** — Aerosolized ribavirin reserved for severely immunocompromised with life-threatening RSV

Investigations

- RSV rapid antigen test or PCR — Confirm RSV for cohorting and epidemiology [URGENT]
- Pulse oximetry — Determine need for supplemental oxygen [STAT]
- Chest X-ray — Only if clinical deterioration or suspected complication

Counselling

- Your baby has an RSV infection which is the most common cause of bronchiolitis
- Treatment is mainly supportive - oxygen and fluids if needed; medicines do not help much
- Keep your baby away from other sick children and crowded places during illness
- Handwashing is the most important way to prevent spreading the infection
- Ask your doctor about nirsevimab injection to protect your baby before the next RSV season

Referral Criteria

Refer to pediatric pulmonologist if prolonged course, underlying chronic lung disease, or recurrent RSV infections

Hospitalisation Criteria

Admit if SpO2 <90%, unable to feed, marked respiratory distress, age <6 weeks, or high-risk (preterm, CHD, immunodeficiency)

Notes

Nirsevimab is now the preferred single-dose prophylaxis for RSV in infants entering their first RSV season. Treatment remains supportive; no antiviral has proven benefit in routine RSV bronchiolitis.

Guideline Changes

Nirsevimab (2023) approved as single-dose RSV prophylaxis replacing monthly palivizumab for most infants

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J22 — Acute lower respiratory infection, unspecified

SNOMED: 195742007 | Duration: 7 days | Source: WHO 2023, IDSA/PIDS 2011

First-Line

- **Amoxicillin** — 90 mg/kg/day PO divided BID (max 4 g/day) for 5 days if bacterial cause suspected
- **Supportive care** — Supplemental O2 if SpO2 <92%; oral rehydration; antipyretics PRN

Second-Line

- **Amoxicillin-clavulanate** — 90 mg/kg/day (amoxicillin) PO divided BID (max 4 g/day) if resistant organisms suspected
- **Azithromycin** — 10 mg/kg PO day 1 (max 500 mg) then 5 mg/kg/day days 2-5 if atypical pathogen suspected

- **Ceftriaxone** — 50-100 mg/kg/day IV/IM once daily (max 2 g) if hospitalized or unable to take PO

Investigations

- Chest X-ray — Assess for consolidation or complications [URGENT]
- CBC with differential — Help differentiate bacterial vs viral [URGENT]
- CRP — Inflammatory marker to guide antibiotic use [URGENT]
- Blood culture — If high fever or toxic-appearing [URGENT]

Counselling

- Your child has a lower respiratory infection - it may be viral or bacterial
- If antibiotics are prescribed, complete the full course even if your child feels better
- Give plenty of fluids and rest; paracetamol or ibuprofen for fever
- Watch for danger signs: fast breathing, chest indrawing, not feeding, becoming drowsy
- Vaccination (pneumococcal, Hib, influenza) helps prevent many chest infections

Referral Criteria

Refer to pediatric pulmonologist if recurrent LRTI, poor response to antibiotics, or underlying chronic condition

Hospitalisation Criteria

Admit if SpO2 <92%, severe respiratory distress, unable to feed, danger signs per WHO criteria, or failed outpatient treatment

Notes

Most pediatric LRTIs are viral and require only supportive care. Reserve antibiotics for suspected bacterial infection based on clinical features and inflammatory markers.

J32 — Chronic sinusitis

SNOMED: 40055000 | Duration: 90 days | Source: AAP 2013, AAO-HNS 2015

First-Line

- **Intranasal corticosteroids** — Fluticasone 1-2 sprays/nostril daily or mometasone 1 spray/nostril daily; age >=2 yr
- **Nasal saline irrigation** — Isotonic or hypertonic saline rinse 1-2 times daily via squeeze bottle or neti pot
- **Amoxicillin-clavulanate** — 45 mg/kg/day (amoxicillin) PO divided BID for acute exacerbations; max 875 mg/dose

Second-Line

- **Adenoidectomy** — Surgical removal of adenoids; first-line surgical option when medical therapy fails
- **Endoscopic sinus surgery** — Reserved for refractory CRS after failed medical therapy and adenoidectomy
- **Clindamycin** — 30-40 mg/kg/day PO divided TID for 21 days if anaerobes suspected; max 1.8 g/day

Investigations

- CT sinuses (non-contrast) — Confirm chronic sinusitis and assess extent
- Nasal endoscopy — Visualize middle meatus, polyps, adenoid hypertrophy
- Allergy testing (skin prick or specific IgE) — Identify allergic component contributing to CRS

Counselling

- Chronic sinusitis means long-standing sinus infection lasting more than 12 weeks
- Daily nasal saline rinse is very helpful - teach your child to do it morning and evening
- Use the nasal spray medicine regularly as prescribed even when symptoms improve
- Avoid exposure to cigarette smoke and dust as these worsen sinus problems
- If your child has nasal polyps or does not improve with medicines, surgery may be needed

Referral Criteria

Refer to ENT if symptoms persist after 3 months of medical therapy, nasal polyps identified, or recurrent acute exacerbations

Hospitalisation Criteria

Admit for orbital or intracranial complications of sinusitis (orbital cellulitis, epidural abscess), or for endoscopic sinus surgery

Notes

Chronic sinusitis requires symptoms >12 weeks without resolution. Routine systemic antibiotics are not recommended for uncomplicated chronic sinusitis in children.

Guideline Changes

Emerging biologic therapies under investigation for pediatric CRS with nasal polyps

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J45 — Asthma Inhaler-Based Management (OPD) *[mild]*

SNOMED: 195967001 | Duration: 90 days | Source: GINA 2024/2025, IAP Asthma Guidelines 2024

First-Line

- **Salbutamol Inhaler** — MDI + Spacer + Mask, 100 mcg/puff, 2 puffs QID (5-6 breaths per puff via spacer). Reliever - continue till symptoms controlled, stop 5 days after complete relief
- **Budesonide Inhaler** — 100-200 mcg/puff MDI, 1 puff BD (q12h) via spacer. Controller - minimum 3 months, then step-down. Do NOT stop abruptly
- **Levosalbutamol** — NEB: <10 kg 0.63 mg, >=10 kg 1.25 mg, every 20 min x3 doses for acute relief, then q6-8h
- **Prednisolone** — 1-2 mg/kg/day PO for 3-5 days, only for moderate episodes
- **Fexofenadine** — Age-appropriate dose PO BD, if allergic component

Investigations

- Pulse oximetry — Assess oxygenation during acute exacerbation [STAT]
- Peak expiratory flow rate — Children >5 years for severity and treatment response monitoring
- Chest X-ray — If first episode or suspected complication

Counselling

- Inhaler is always better than syrup - faster, safer, fewer side effects
- Use spacer with mask for proper drug delivery
- Inhaler technique: shake, attach spacer, place mask tightly, press 1 puff, allow 5-6 breaths, repeat if needed
- Controller inhaler (Budesonide) must be taken daily for minimum 3 months - do NOT stop suddenly
- Reliever inhaler (Salbutamol) is for rescue use during symptoms only
- Report immediately if severe chest retractions or SpO2 <92%
- Report immediately if no response to inhaler
- Report immediately if feeding difficulty
- Follow up in 3-5 days, then regular follow-up for step-up/step-down assessment

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Severe chest retractions
- Drop in oxygen saturation below 92%
- No response to inhaler treatment
- Feeding difficulty or unable to eat

Referral Criteria

Refer to pediatric pulmonologist if requiring Step 3 or above, frequent exacerbations despite ICS, or diagnostic uncertainty

Hospitalisation Criteria

Admit if severe respiratory distress, SpO2 <92%, no response to inhaler therapy, or inability to feed

Notes

Inhaler therapy is the GOLD STANDARD - faster action, fewer side effects than oral drugs. Spacer + Mask mandatory for proper drug delivery in infants and young children. GINA Stepwise: Step 1 SABA PRN, Step 2 low-dose ICS, Step 3 ICS+LABA, Step 4 specialist referral. ICS is preventive (not immediate relief). SABA is rescue drug. Early inhaler use reduces hospital admissions.

Guideline Changes

GINA 2025: New diagnostic criteria for asthma in children <5 years; expanded role of T2 biomarkers (blood eosinophils, FeNO) in assessment; GINA 2024: MDI+Spacer remains preferred delivery for all ages; Step-down only after 3 months symptom-free on low-dose ICS

J45.901 — Asthma - acute exacerbation

SNOMED: 304527002 | Duration: 5 days | Source: GINA 2024

First-Line

- **Albuterol/Salbutamol** — 0.15 mg/kg nebulised q20min x3, min 2.5mg max 5mg
- **ICS-Formoterol MART** — Budesonide-formoterol as both maintenance AND reliever at Steps 3-5 **[NEW 2024-25]**

Second-Line

- **Prednisolone oral** — 1-2 mg/kg/day (max 40mg) x3-5 days, moderate-severe
- **Dexamethasone** — 0.3-0.6 mg/kg single dose, alternative to prednisolone course

Investigations

- Pulse oximetry — Assess severity of acute exacerbation [STAT]
- Peak expiratory flow (PEF) — Objective assessment in children >6 years [URGENT]
- Spirometry (FEV1/FVC) — Baseline assessment and monitoring in children >6 years
- Chest X-ray — If first episode, atypical presentation, or suspected complication

Counselling

- Always carry the reliever inhaler (blue puffer) wherever the child goes
- Use spacer device with mask for children under 5 years for better medicine delivery
- Identify and avoid triggers: dust, smoke, pets, cold air, strong smells
- Do not stop controller inhaler even when child is well - it prevents attacks
- Seek emergency care if child cannot speak in full sentences, lips turn blue, or reliever not helping

Referral Criteria

Refer to pediatric pulmonologist if uncontrolled asthma despite Step 3 therapy, life-threatening exacerbation, diagnostic uncertainty, or suspected occupational/allergic triggers needing immunotherapy

Hospitalisation Criteria

Admit if severe exacerbation not responding to 3 nebulizations in 1 hour, SpO2 <92% on room air, altered consciousness, silent chest, or need for IV magnesium/aminophylline

Notes

GINA 2024: SABA alone without ICS not recommended for age 6yr+.

Guideline Changes

ICS-formoterol preferred reliever at Steps 3-5. Tiotropium add-on for age 6yr+. No fixed FEV1/FVC cutoff.

J46 — Status asthmaticus

SNOMED: 10675551000119104 | Duration: 5 days | Source: GINA 2024, NAEPP EPR-3, CHOP PICU Pathway

First-Line

- **Continuous albuterol nebulization** — 5-10 kg: 7.5 mg/hr; 10-20 kg: 11.25 mg/hr; >20 kg: 15 mg/hr via continuous neb
- **Ipratropium bromide** — Neb: 250-500 mcg q4-6h; give with albuterol in first hour q20min x3
- **Methylprednisolone IV** — 1 mg/kg IV q6h (max 30 mg/dose) or 2 mg/kg load then 0.5 mg/kg q6h
- **Magnesium sulfate IV** — 50 mg/kg IV over 20-30 min (max 2 g); single dose for severe/refractory cases

Second-Line

- **Terbutaline IV** — Load: 10 mcg/kg (max 400 mcg); infusion: 0.1-10 mcg/kg/min titrated to effect
- **Epinephrine** — 0.01 mg/kg IM (1:1000, max 0.5 mg) q20min x3; or IV infusion 0.1-1 mcg/kg/min
- **Ketamine** — 0.5-2 mg/kg IV bolus then 0.5-2 mg/kg/hr infusion for bronchodilation and sedation
- **Aminophylline IV** — Load: 6 mg/kg IV over 20 min; maintenance: 0.5-1 mg/kg/hr; monitor levels

Investigations

- Arterial blood gas — Assess severity of respiratory failure and CO2 retention [STAT]
- Chest X-ray — Rule out pneumothorax, pneumonia, foreign body [STAT]
- Serum electrolytes — Monitor for hypokalemia from beta-agonist use [URGENT]
- Peak flow (if cooperative) — Objective measure of airflow obstruction and response to treatment [URGENT]

Counselling

- Your child had a very severe asthma attack that needed intensive hospital treatment
- After discharge, take all prescribed medicines regularly especially the preventer inhaler daily
- Get an asthma action plan from your doctor and keep it accessible at home and school
- Avoid known triggers: dust, smoke, strong perfumes, cold air, and pets if allergic
- If the reliever inhaler is not working or your child is struggling to breathe, come to hospital immediately

Referral Criteria

Refer to pediatric pulmonologist/allergist for all cases; ICU consultation for refractory cases

Hospitalisation Criteria

All status asthmaticus requires hospitalization, typically in PICU; admit if not responding to repeated bronchodilators, SpO2 <90%, altered consciousness, or silent chest

Notes

Status asthmaticus is refractory to standard bronchodilator therapy and requires ICU admission. Monitor for cardiac arrhythmias with continuous beta-agonist use.

J80 — Adult respiratory distress syndrome / ARDS

SNOMED: 67782005 | Duration: 14 days | Source: PALICC-2 2023

First-Line

- **Lung-protective ventilation** — Tidal volume 6-8 mL/kg IBW; PEEP >=5 cmH2O per FiO2 table; Pplat <28 cmH2O
- **Supplemental oxygen** — Target SpO2 92-97% (mild-moderate); SpO2 >=88% acceptable in severe PARDS after PEEP optimization
- **Conservative fluid management** — Target negative or neutral fluid balance after initial resuscitation; furosemide 1 mg/kg IV PRN

Second-Line

- **Prone positioning** — >=12 hr/day for moderate-severe PARDS; improves V/Q matching and oxygenation
- **Inhaled nitric oxide (iNO)** — 5-20 ppm for refractory hypoxemia; discontinue if no response in 30-60 min
- **Neuromuscular blockade** — Cisatracurium 1-2 mcg/kg/min or vecuronium 1-2 mcg/kg/min for severe PARDS; <=48h
- **ECMO** — Consider VV-ECMO for refractory hypoxemia despite optimized ventilation; OI >40

Investigations

- Arterial blood gas — Calculate P/F ratio for PARDS severity classification [STAT]
- Chest X-ray — Bilateral opacities consistent with ARDS [STAT]
- Echocardiography — Rule out cardiogenic pulmonary edema [URGENT]
- Blood culture and respiratory cultures — Identify infectious etiology [STAT]

Counselling

- Your child has a serious lung condition called ARDS that needs intensive care treatment
- A breathing machine (ventilator) is necessary to help the lungs heal
- Recovery takes time - lungs need days to weeks to recover
- The medical team will keep you informed about your child's progress daily
- After recovery, some children may need follow-up lung function tests

Referral Criteria

Manage in PICU; pediatric intensivist leads care; consult pulmonology and infectious disease as needed

Hospitalisation Criteria

All PARDS cases require PICU admission for mechanical ventilation, close monitoring, and potentially ECMO for refractory cases

Notes

No specific pharmacotherapy for PARDS; management is supportive with lung-protective ventilation. Non-adherence to PALICC recommendations is associated with higher mortality.

Guideline Changes

PALICC-2 replaced PALICC-1 (2015) with updated severity stratification and ventilation targets

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J85 — Lung abscess and necrosis of lung

Duration: 42 days | Source: IDSA/PIDS 2011, Expert Consensus

First-Line

- **Clindamycin IV** — 30-40 mg/kg/day IV divided q6-8h (max 2.7 g/day); covers S. aureus and anaerobes
- **Ceftriaxone** — 50-100 mg/kg/day IV once daily (max 4 g); combined with clindamycin for broad coverage
- **Ampicillin-sulbactam IV** — 200-300 mg/kg/day (ampicillin) IV divided q6h (max 12 g/day); alternative empiric

Second-Line

- **Piperacillin-tazobactam** — 300 mg/kg/day (piperacillin) IV divided q6-8h (max 16 g/day) for polymicrobial
- **Vancomycin** — 40-60 mg/kg/day IV divided q6-8h (max 4 g/day) if MRSA confirmed; trough 15-20
- **CT-guided percutaneous drainage** — For large (>4 cm) peripheral abscess not responding to 7-10 days of antibiotics
- **Linezolid** — 10 mg/kg IV/PO q8h (max 600 mg/dose); for MRSA if vancomycin not tolerated

Investigations

- CT chest with contrast — Confirm abscess, assess size and guide intervention [URGENT]
- Blood cultures — Identify causative organism [STAT]
- CBC with differential — Monitor infection markers [URGENT]
- CRP/ESR — Track treatment response [URGENT]

Counselling

- Your child has a serious lung infection with abscess formation that needs prolonged antibiotics
- IV antibiotics will be given for 2-4 weeks followed by oral antibiotics to complete 4-6 weeks total
- Your child will need to stay in hospital for several weeks
- Regular CT scans will monitor if the abscess is shrinking
- Most children recover completely but complete the full antibiotic course to prevent relapse

Referral Criteria

Refer to pediatric pulmonologist and infectious disease; interventional radiology or surgery if abscess >4 cm not responding to 7-10 days of antibiotics

Hospitalisation Criteria

All lung abscesses require hospitalization for prolonged IV antibiotics; ICU admission if sepsis or respiratory failure

Notes

Prolonged IV antibiotics (2-4 weeks) followed by oral step-down (total 4-6 weeks) is standard. Surgical drainage is rarely needed if antibiotics started early.

J86 — Empyema

SNOMED: 312682007 | Duration: 28 days | Source: BTS 2005, AATS 2017, IDSA/PIDS 2011

First-Line

- **Ampicillin-sulbactam IV** — 200-300 mg/kg/day (ampicillin) IV divided q6h (max 12 g amp/day); empiric first-line
- **Ceftriaxone + Clindamycin** — Ceftriaxone 50-100 mg/kg/day IV daily + clindamycin 30-40 mg/kg/day IV q6-8h
- **Chest tube drainage + fibrinolytics** — 10-14 Fr tube; alteplase 0.1 mg/kg (max 6 mg) q12h x4-6 doses with 1h dwell time

Second-Line

- **VATS (video-assisted thoracoscopic surgery)** — For multiloculated empyema or failure of chest tube + fibrinolytics after 48-72h
- **Vancomycin** — 40-60 mg/kg/day IV divided q6-8h (max 4 g/day) if MRSA suspected; trough 15-20 mcg/mL
- **Meropenem** — 60 mg/kg/day IV divided q8h (max 6 g/day) for polymicrobial or resistant infection

Investigations

- Chest X-ray — Initial assessment of effusion size [STAT]
- Chest ultrasound — Characterize effusion (simple vs loculated) and guide drainage [URGENT]
- CT chest with contrast — Assess for loculations, lung parenchyma, and guide intervention [URGENT]
- Pleural fluid analysis (cell count, pH, glucose, LDH, culture) — Confirm empyema and identify organism [STAT]

Counselling

- Your child has pus collection around the lung which needs long-term antibiotic treatment
- A chest tube may be placed to drain the fluid - this is necessary for recovery
- IV antibiotics will be given for 1-2 weeks followed by oral antibiotics for 2-4 more weeks
- Your child may need to stay in hospital for 1-2 weeks or longer
- Follow-up chest X-ray will be done to ensure the infection has cleared completely

Referral Criteria

Refer to pediatric surgeon/thoracic surgeon for chest tube placement and VATS if needed; infectious disease for antibiotic guidance

Hospitalisation Criteria

All empyema cases require hospitalization for IV antibiotics and drainage; surgical referral for multiloculated collections

Notes

Total antibiotic duration (IV + oral) is typically 3-4 weeks. Intrapleural fibrinolytics with chest tube drainage are superior to drainage alone.

J90 — Pleural effusion, not elsewhere classified

SNOMED: 60046008 | Duration: 14 days | Source: BTS 2005, IDSA/PIDS 2011

First-Line

- **Diagnostic thoracentesis** — 18-20G needle; send fluid for cell count, glucose, LDH, pH, Gram stain, culture
- **IV antibiotics (if parapneumonic)** — Ceftriaxone 50-100 mg/kg/day IV daily + clindamycin 30-40 mg/kg/day IV q6-8h
- **Chest tube drainage** — 10-14 Fr pigtail catheter for complicated effusion (pH <7.0, glucose <40, LDH >1000, pus)

Second-Line

- **Intrapleural urokinase** — 40,000 IU in 40 mL NS (>=10 kg) BID x3 days via chest tube; dwell 4h then drain
- **VATS** — Video-assisted thoracoscopic surgery for multiloculated effusion or fibrinolytic failure
- **Diuretics** — Furosemide 1-2 mg/kg IV/PO q6-12h for transudative effusion (e.g., nephrotic, CHF)

Investigations

- Chest X-ray — Confirm effusion and estimate size [URGENT]
- Chest ultrasound — Guide thoracentesis and characterize effusion [URGENT]
- Pleural fluid analysis (Light criteria) — Differentiate transudative from exudative [URGENT]
- Serum albumin and LDH — Compare with pleural fluid for Light criteria [URGENT]

Counselling

- There is fluid collection around your child's lung that needs to be investigated and treated
- A needle may be used to take a sample of the fluid to find the cause
- If the fluid is infected, antibiotics and a drainage tube may be needed
- Some effusions resolve with treating the underlying cause without needing drainage
- Regular follow-up and repeat imaging will ensure the fluid is clearing

Referral Criteria

Refer to pediatric pulmonologist for all significant effusions; surgeon if drainage needed; nephrology/cardiology if transudative

Hospitalisation Criteria

Admit for symptomatic effusion causing respiratory distress, complicated parapneumonic effusion, or need for thoracentesis/chest tube

Notes

Determine if effusion is transudative or exudative using Light criteria to guide management. Parapneumonic effusion requires antibiotics and may need drainage if complicated.

J93 — Pneumothorax

SNOMED: 36118008 | Duration: 7 days | Source: BTS 2010, Medscape/ACEP Guidelines

First-Line

- **Observation + O2** — 100% O2 via NRB mask for small (<20%) stable primary spontaneous PTX; monitor q4-6h

- **Needle decompression** — 14-16G needle at 4th ICS mid-axillary line (preferred in peds); for tension PTX or large PTX
- **Chest tube (tube thoracostomy)** — 10-14 Fr pigtail catheter; connect to water seal or -20 cmH2O suction; 4th-5th ICS MAL

Second-Line

- **Needle aspiration** — 16-18G over-needle catheter with 3-way stopcock; for stable patients with large 1st episode PSP
- **Heimlich valve** — One-way flutter valve attached to pigtail catheter for ambulatory management of stable PTX
- **VATS with pleurodesis** — For recurrent PTX (>=2 episodes); bleb resection + mechanical or chemical pleurodesis

Investigations

- Chest X-ray (upright, PA) — Confirm pneumothorax and estimate size [STAT]
- CT chest — If recurrent PTX or to plan surgical intervention
- Pulse oximetry — Assess oxygenation [STAT]
- ABG — If respiratory distress or large PTX [URGENT]

Counselling

- Your child has air leaking from the lung into the space around it
- A small pneumothorax may heal on its own with rest and oxygen; larger ones need a chest tube
- Your child should avoid vigorous exercise and air travel until fully healed
- If sudden chest pain or breathlessness occurs again, come to hospital immediately
- Recurrence is possible; discuss with doctor about preventive surgery if it happens again

Referral Criteria

Refer to pediatric surgeon/thoracic surgeon for chest tube placement; VATS referral for recurrent pneumothorax

Hospitalisation Criteria

Admit all symptomatic pneumothorax; immediate intervention for tension pneumothorax; observation for small (<20%) asymptomatic primary spontaneous PTX

Notes

Tension pneumothorax requires immediate needle decompression followed by chest tube placement. The 4th ICS at the anterior axillary line is the preferred site in pediatric patients.

J96.0 — Acute respiratory failure

SNOMED: 65710008 | Duration: 10 days | Source: PALICC-2 2023, PCCM Guidelines

First-Line

- **Supplemental oxygen** — Titrate to SpO2 92-97%; via nasal cannula, face mask, or high-flow nasal cannula (HFNC)
- **Non-invasive ventilation (NIV)** — CPAP 5-10 cmH2O or BiPAP via nasal/face mask; trial before intubation if able
- **Mechanical ventilation** — Tidal volume 6-8 mL/kg IBW; PEEP per ARDSNet table; plateau pressure <28 cmH2O

Second-Line

- **HFNC (high-flow nasal cannula)** — Flow rate 2 L/kg/min (max ~60 L/min); FiO2 titrated to SpO2 target
- **Surfactant** — Calfactant 3 mL/kg or beractant 4 mL/kg via ETT for severe ARDS in young children
- **Prone positioning** — 12-16 hr/day in mechanically ventilated patients with severe hypoxemia (P/F <150)
- **Inhaled nitric oxide (iNO)** — Start 20 ppm; titrate down; for refractory hypoxemia; assess response at 30 min

Investigations

- Arterial blood gas — Confirm type of respiratory failure (hypoxemic vs hypercapnic) [STAT]
- Chest X-ray — Identify underlying cause (pneumonia, ARDS, effusion) [STAT]
- CBC with differential — Assess for infection [STAT]
- Blood culture — If sepsis or pneumonia suspected [STAT]

Counselling

- Your child is having difficulty breathing and needs oxygen and close monitoring in the hospital
- The medical team may need to use a breathing machine if your child is not improving
- This is a serious condition but most children recover well with proper treatment
- After discharge, follow-up with the doctor to ensure lungs have recovered fully
- If your child was on a ventilator, some cough and tiredness is normal for a few weeks after

Referral Criteria

Manage in PICU; consult pediatric pulmonologist and intensivist for all cases

Hospitalisation Criteria

All acute respiratory failure requires hospitalization; PICU admission for need for mechanical ventilation, severe hypoxemia (P/F <200), or hemodynamic instability

Notes

Treat underlying cause aggressively. Lung-protective ventilation with low tidal volumes is the cornerstone of management in pediatric acute respiratory failure.

Guideline Changes

PALICC-2 (2023) updated ventilation and oxygenation targets for pediatric respiratory failure

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J98.0 — Bronchospasm

SNOMED: 4386001 | Duration: 5 days | Source: GINA 2024, NAEPP EPR-3

First-Line

- **Albuterol (salbutamol)** — Neb: 0.15 mg/kg (min 2.5 mg) q20min x3 doses; MDI: 4-8 puffs via spacer q20min
- **Ipratropium bromide** — Neb: 250 mcg (<20 kg) or 500 mcg (>=20 kg) q20min x3 doses with albuterol
- **Prednisolone** — 1-2 mg/kg/day PO (max 60 mg) for 3-5 days for moderate-severe episodes

Second-Line

- **Dexamethasone** — 0.6 mg/kg PO/IM x1 dose (max 16 mg); may repeat x1 in 24-48h as alternative to pred
- **Epinephrine** — 0.01 mg/kg IM (1:1000, max 0.5 mg) q15-20min x3 for severe bronchospasm
- **Magnesium sulfate** — 25-50 mg/kg IV over 20 min (max 2 g) for refractory bronchospasm

Investigations

- Pulse oximetry — Assess oxygenation status [STAT]
- Chest X-ray — Rule out pneumonia, foreign body, or pneumothorax [URGENT]
- Peak flow meter (if >6 yrs) — Objective measure of airway obstruction [URGENT]

Counselling

- Bronchospasm means the airways in the lungs have tightened making it hard to breathe
- Give the inhaler medicine with a spacer as shown by the doctor - this is the most effective way
- If your child is not improving after 3 puffs of the blue inhaler, come to the hospital immediately
- Keep the rescue inhaler available at all times including at school
- Identify and avoid triggers like dust, smoke, cold air, and strong smells

Referral Criteria

Refer to pediatric pulmonologist if recurrent bronchospasm suggesting asthma, poor response to standard bronchodilators, or diagnostic uncertainty

Hospitalisation Criteria

Admit for severe bronchospasm not responding to initial bronchodilator therapy, oxygen saturation <92% despite treatment, or respiratory distress

Notes

Inhaled SABA via MDI with spacer is preferred over nebulizer in cooperative children. Assess response after each treatment and escalate as needed.

R06.2 — Wheezing

SNOMED: 56018004 | Duration: 7 days | Source: GINA 2024, ERS Task Force 2024

First-Line

- **Albuterol (salbutamol)** — MDI: 2-4 puffs via spacer q4-6h PRN; Neb: 0.15 mg/kg (min 2.5 mg) q4-6h PRN
- **Prednisolone** — 1-2 mg/kg/day PO (max 40 mg) for 3-5 days for acute wheezing exacerbation
- **Inhaled corticosteroids (ICS)** — Fluticasone 44-88 mcg 2 puffs BID or budesonide neb 0.25-0.5 mg BID for MTW

Second-Line

- **Montelukast** — 6 mo-5 yr: 4 mg PO daily; 6-14 yr: 5 mg PO daily; for recurrent viral wheeze
- **Intermittent high-dose ICS** — Budesonide 1 mg neb BID or fluticasone 250 mcg BID at onset of viral illness x7-10d
- **Ipratropium bromide** — Neb: 250-500 mcg q6-8h as add-on bronchodilator for acute episodes

Investigations

- Pulse oximetry — Assess oxygenation [STAT]
- Chest X-ray — If first episode, atypical presentation, or not responding to treatment [URGENT]
- Viral respiratory panel (nasopharyngeal swab) — Identify viral trigger

Counselling

- Wheezing means there is a whistling sound when your child breathes out, caused by narrowed airways
- Many young children wheeze only with viral infections and outgrow it by age 5-6
- Give the inhaler with spacer and mask as demonstrated; this is more effective than nebulizer
- If wheezing happens frequently (more than once a month), a daily preventer medicine may be needed
- Keep your home smoke-free and reduce dust exposure to minimize episodes

Referral Criteria

Refer to pediatric pulmonologist if wheezing is recurrent (>3 episodes), persistent between episodes, or not responding to standard treatment

Hospitalisation Criteria

Admit if SpO2 <92%, severe respiratory distress, poor feeding in infants, or not improving with bronchodilator therapy

Notes

Classify as episodic viral wheeze or multiple-trigger wheeze to guide maintenance therapy. Daily ICS is most effective for multiple-trigger wheeze phenotype.

Guideline Changes

ERS updated phenotype-based approach distinguishing episodic viral vs multiple-trigger wheeze

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T17 — Foreign body in respiratory tract

SNOMED: 262599003 | Duration: 3 days | Source: AHA/AAP BLS Guidelines, Medscape/StatPearls

First-Line

- **Rigid bronchoscopy** — Definitive removal under general anesthesia; procedure of choice in pediatric FB aspiration
- **Back blows/chest thrusts** — Infant <1 yr: 5 back blows + 5 chest thrusts; for acute choking with complete obstruction
- **Abdominal thrusts (Heimlich)** — Child >1 yr: subdiaphragmatic thrusts until object expelled or child becomes unresponsive

Second-Line

- **Flexible bronchoscopy** — Diagnostic or therapeutic in select cases; smaller caliber but limited extraction capability
- **Dexamethasone** — 0.5-0.6 mg/kg IV/PO (max 10 mg) pre/post bronchoscopy to reduce airway edema
- **Antibiotics** — Amoxicillin-clavulanate 45 mg/kg/day PO BID if secondary infection from prolonged impaction

Investigations

- Chest X-ray (inspiratory and expiratory) — Look for air trapping, radio-opaque foreign body [STAT]
- CT chest — If X-ray negative but clinical suspicion high [URGENT]
- Pulse oximetry — Assess oxygenation status [STAT]

Counselling

- Your child had a foreign body in the airway which has been removed
- Keep small objects (nuts, coins, button batteries, small toys) away from children under 3 years
- Cut food into small pieces for young children - avoid whole grapes, popcorn, raw carrots
- Teach older children to chew food properly and not to talk or laugh while eating
- If your child suddenly starts coughing, choking, or turns blue, learn basic choking first aid

Referral Criteria

Urgent ENT/pediatric surgery referral for bronchoscopic removal; immediate referral for complete airway obstruction

Hospitalisation Criteria

All confirmed or suspected airway foreign bodies require hospital admission for bronchoscopy; observe 24h post-removal for airway edema

Notes

Complications increase significantly after 24-48 hours of impaction. Urgent rigid bronchoscopy should not be delayed for imaging if clinical suspicion is high.

Rheumatology

D80 — Immunodeficiency with predominantly antibody defects

Duration: 365 days | Source: ESID 2024, AAAAI/ACAAI 2022

First-Line

- **IVIG** — 400-600 mg/kg IV q3-4wks; individualize trough target based on infection frequency
- **SCIG** — 100-200 mg/kg SC weekly; preferred for home therapy; fewer systemic side effects
- **Amoxicillin prophylaxis** — 20 mg/kg/day PO for sinopulmonary infections pending Ig replacement initiation

Second-Line

- **Azithromycin prophylaxis** — 10 mg/kg PO weekly (max 500mg) for recurrent lower respiratory infections
- **Hyaluronidase-facilitated SCIG** — HyQvia: rHuPH20 + IgG 10% SC monthly; allows higher volumes per site **[NEW 2024-25]**
- **Chest physiotherapy** — Airway clearance techniques for established bronchiectasis; non-pharmacologic

Investigations

- Immunoglobulin levels (IgG, IgA, IgM, IgE) — Characterize antibody deficiency pattern [URGENT]
- Vaccine response titers (tetanus, pneumococcal) — Assess functional antibody response
- Lymphocyte subsets — Rule out combined immunodeficiency
- HRCT chest — Screen for bronchiectasis

Counselling

- Your child's body does not make enough antibodies so regular immunoglobulin replacement is needed
- IVIG is given monthly at hospital or SCIG can be given weekly at home
- Your child can attend school and play normally; inform school about the condition
- Report all infections promptly for early treatment
- Annual lung function tests help detect any lung damage early

Referral Criteria

All cases managed by pediatric immunologist; pulmonology referral if recurrent respiratory infections or bronchiectasis

Hospitalisation Criteria

Admit for severe or unusual infections, initial IVIG therapy, or complications such as pneumonia or sepsis

Notes

Antibody deficiency is the most common primary immunodeficiency group. Regular monitoring of lung function and CT chest for bronchiectasis is essential.

Guideline Changes

2024 review supports individualized trough targets rather than fixed 500mg/dL threshold

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D81 — Combined immunodeficiencies

SNOMED: 31323000 | Duration: 365 days | Source: EBMT/ESID 2022, IUIS/WHO 2024

First-Line

- **HSCT** — Definitive cure; best outcomes if <3.5mo age; HLA-matched sibling preferred; ~97% survival
- **IVIG** — 400-600 mg/kg IV q3-4wks; maintain trough IgG >500mg/dL; bridge to HSCT
- **TMP-SMX prophylaxis** — 5 mg/kg/day TMP PO divided BID for PJP prophylaxis until immune reconstitution

Second-Line

- **Gene therapy** — Autologous CD34+ cell modification; for ADA-SCID and SCID-X1 without matched donor **[NEW 2024-25]**

- **Fluconazole** — 3-6 mg/kg/day PO daily for fungal prophylaxis pre-transplant
- **Acyclovir** — 20 mg/kg/dose IV q8h or 20 mg/kg/dose PO QID for HSV/VZV prophylaxis

Investigations

- Lymphocyte subsets (CD3, CD4, CD8, CD19, CD16/56) — Confirm T-cell, B-cell, NK-cell deficiency pattern [STAT]
- TREC assay (newborn screen) — Detect SCID at birth [STAT]
- Genetic testing (SCID gene panel) — Identify specific gene defect for treatment planning [URGENT]
- Immunoglobulin levels (IgG, IgA, IgM) — Assess antibody production [URGENT]
- CMV PCR — Screen for CMV viremia before HSCT [URGENT]

Counselling

- Your baby has a serious immune deficiency that needs urgent specialized treatment
- Keep your baby away from sick people, crowded places, and avoid live vaccines
- Hand hygiene is critical for everyone who touches your baby
- HSCT (bone marrow transplant) is the cure and best done as early as possible
- Your baby will need to take preventive antibiotics and antifungals until the immune system is restored

Referral Criteria

Urgent referral to pediatric immunologist and HSCT center at diagnosis; this is a pediatric emergency

Hospitalisation Criteria

Most SCID patients require hospitalization from diagnosis through HSCT for protective isolation and infection management

Notes

SCID is a pediatric emergency; early HSCT before 3.5 months yields best survival. Avoid live vaccines and use irradiated, CMV-negative blood products.

Guideline Changes

Gene therapy trials showing >90% survival for SCID-X1 and ADA-SCID

D82 — Immunodeficiency associated with other major defects

SNOMED: 191029005 | Duration: 365 days | Source: ESID/EBMT 2022, Expert consensus 2024

First-Line

- **IVIG** — 400-600 mg/kg IV q3-4wks for antibody deficiency component; target trough >500mg/dL
- **HSCT** — Definitive cure for WAS and severe DiGeorge; HLA-matched donor preferred; early referral
- **TMP-SMX prophylaxis** — 5 mg/kg/day TMP PO divided BID for PJP and bacterial infection prophylaxis

Second-Line

- **Thymus transplant** — For complete DiGeorge (T-cell absent); cultured thymus tissue allograft
- **Gene therapy** — Lentiviral CD34+ cell transduction for WAS without matched donor; investigational **[NEW 2024-25]**
- **Platelet transfusion** — For WAS with severe thrombocytopenia and bleeding; irradiated products only
- **Eltrombopag** — Weight-based dosing PO daily for WAS thrombocytopenia; off-label in children

Investigations

- Lymphocyte subsets with T-cell markers — Characterize immune defect pattern [URGENT]
- Immunoglobulin levels — Assess antibody deficiency [URGENT]
- Genetic testing (targeted or WES) — Confirm specific syndrome
- Echocardiography — Cardiac defects in DiGeorge and other syndromes
- Serum calcium — Hypoparathyroidism screening in DiGeorge [URGENT]

Counselling

- Your child has an immune deficiency combined with other medical problems that need coordinated care
- Multiple specialists will be involved in your child's care - keep all appointments
- Preventive antibiotics and immunoglobulin may be needed to prevent infections
- Avoid live vaccines unless specifically cleared by the immunologist
- Bone marrow transplant may be recommended for certain conditions and should be discussed early

Referral Criteria

Refer to pediatric immunologist at diagnosis; genetics for syndrome-specific evaluation; cardiology for DiGeorge

Hospitalisation Criteria

Admit for serious infections, pre-HSCT conditioning, thymus transplant, or acute complications of the underlying syndrome

Notes

Syndrome-specific management is essential; DiGeorge requires cardiac and endocrine care alongside immunology. WAS patients should receive early curative therapy.

Guideline Changes

Gene therapy for Wiskott-Aldrich showing promising long-term outcomes in trials

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D83 — Common variable immunodeficiency

SNOMED: 191010004 | Duration: 365 days | Source: ESID 2024, AAAAI 2022

First-Line

- **IVIG** — 400-600 mg/kg IV q3-4wks; target trough IgG >500mg/dL (>700 if lung disease)
- **SCIG** — 100-200 mg/kg SC weekly; equivalent efficacy to IVIG; fewer systemic reactions
- **Amoxicillin prophylaxis** — 20 mg/kg/day PO daily for breakthrough sinopulmonary infections if needed

Second-Line

- **Azithromycin prophylaxis** — 10 mg/kg PO weekly (max 500mg) for recurrent respiratory infections
- **Budesonide** — Enteric-coated 3-9mg PO daily for CVID-associated enteropathy
- **Rituximab** — 375 mg/m2 IV weekly x4 for CVID-associated autoimmune cytopenias

Investigations

- Immunoglobulin levels (IgG, IgA, IgM) — Confirm hypogammaglobulinemia [URGENT]
- Vaccine response titers — Assess functional antibody production
- Lymphocyte subsets — Evaluate T-cell and B-cell compartments
- HRCT chest — Screen for bronchiectasis and granulomatous disease

Counselling

- CVID means your child's body cannot make enough antibodies to fight infections
- Monthly immunoglobulin infusions (IVIG or weekly SCIG) are needed lifelong
- Your child can attend school and lead a normal life with regular treatment
- Report any new infections, diarrhea, enlarged lymph nodes, or unexplained weight loss promptly
- Annual blood tests and periodic lung scans are needed to monitor for complications

Referral Criteria

All cases should be managed by pediatric immunologist; pulmonology for lung complications; gastroenterology for CVID enteropathy

Hospitalisation Criteria

Admit for severe infections despite replacement therapy, CVID-related autoimmune complications, or for initial IVIG loading doses

Notes

Lifelong immunoglobulin replacement is standard of care. Monitor for granulomatous disease, autoimmune cytopenias, and lymphoid malignancy.

Guideline Changes

2024 review supports SCIG as preferred over IVIG for replacement therapy in stable patients

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D84 — Other immunodeficiencies

SNOMED: 234532001 | Duration: 180 days | Source: AAAAI 2022, IUIS 2024

First-Line

- **IVIG** — 400-600 mg/kg IV q3-4wks if hypogammaglobulinemia confirmed; monitor trough levels
- **TMP-SMX prophylaxis** — 5 mg/kg/day TMP PO divided BID for patients at risk of opportunistic infections
- **Treat underlying cause** — Discontinue/reduce offending drug if secondary; nutritional repletion if malnourished

Second-Line

- **Azithromycin prophylaxis** — 10 mg/kg PO weekly (max 500mg) for recurrent sinopulmonary infections
- **Fluconazole** — 3-6 mg/kg/day PO daily for fungal prophylaxis in combined defects
- **G-CSF (Filgrastim)** — 5-10 mcg/kg/day SC for congenital or severe neutropenia component

Investigations

- CBC with differential — Assess for neutropenia, lymphopenia [URGENT]
- Immunoglobulin levels — Screen for antibody deficiency [URGENT]
- Lymphocyte subsets — Assess cellular immunity
- Complement levels (CH50, C3, C4) — Screen for complement deficiency

Counselling

- Your child has an immune problem that makes infections more common or severe
- Finding and treating the underlying cause is the first step
- Preventive antibiotics or immunoglobulin may be needed depending on the specific deficiency
- Good nutrition is important for immune function; ensure a balanced diet
- Report any new or unusual infections to your doctor promptly

Referral Criteria

Refer to pediatric immunologist for evaluation and management plan; address underlying cause

Hospitalisation Criteria

Admit for serious or opportunistic infections, or for diagnostic workup of complex immunodeficiency

Notes

Identify and treat reversible secondary causes first. Monitor immune function q6-12mo and initiate IgG replacement if recurrent infections persist.

L50 — Urticaria

SNOMED: 126485001 | Duration: 14 days | Source: EAACI/GA2LEN/EuroGuiDerm 2022, BSACI 2024

First-Line

- **Cetirizine** — 0.25 mg/kg/dose PO daily; can updose to 2-4x standard dose if uncontrolled
- **Loratadine** — 2-5yr: 5mg PO daily; >=6yr: 10mg daily; updose to 2x if needed
- **Fexofenadine** — 6-11yr: 30mg PO BID; >=12yr: 180mg PO daily; updose up to 4x if needed

Second-Line

- **Omalizumab** — 150-300mg SC q4wks for chronic spontaneous urticaria; age >=12yr
- **Montelukast** — 6mo-5yr: 4mg PO daily; 6-14yr: 5mg; add-on to antihistamine
- **Cyclosporine** — 3-5 mg/kg/day PO divided BID; for refractory CSU, short-term use
- **Prednisolone** — 0.5-1 mg/kg/day PO short course 3-7 days for acute flares only

Investigations

- CBC with differential — Screen for eosinophilia or other abnormalities
- ESR and CRP — Screen for underlying inflammatory condition in chronic urticaria
- Thyroid function and anti-thyroid antibodies — Autoimmune thyroid disease associated with chronic urticaria

Counselling

- Hives (urticaria) are itchy raised welts that can appear anywhere and usually resolve within 24 hours per lesion
- Antihistamine medicine (cetirizine or loratadine) should be taken daily, not just when itchy
- If hives last more than 6 weeks, it is called chronic urticaria and may need longer treatment
- Cool compresses and loose cotton clothing help reduce itching
- If hives come with swelling of lips/tongue or difficulty breathing, use epinephrine and go to hospital

Referral Criteria

Refer to pediatric allergist/dermatologist for chronic urticaria (>6 weeks), angioedema, or if standard antihistamines at 4x dose are ineffective

Hospitalisation Criteria

Admit if associated angioedema with airway compromise, anaphylaxis, or systemic vasculitis

Notes

Updose second-gen antihistamines up to 4x before adding biologics. Avoid first-generation antihistamines per international guidelines.

Guideline Changes

2024 global update recommends same algorithm for children as adults with weight-based dosing

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M04.1 — Periodic fever syndromes

SNOMED: 234538002 | Duration: 365 days | Source: EULAR/PRINTO 2024, Expert consensus 2023

First-Line

- **Colchicine** — FMF: <5yr 0.5mg/day, 5-10yr 1mg/day, >10yr 1.5mg/day PO; PFAPA: same dosing
- **Prednisolone** — PFAPA: 1-2 mg/kg PO single dose at flare onset; aborts episode rapidly
- **Anakinra** — 1-2 mg/kg/day SC (max 100mg); for colchicine-resistant FMF or CAPS; age >=8mo

Second-Line

- **Canakinumab** — 2-4 mg/kg SC q4-8wks (max 300mg); for CAPS age >=2yr and refractory FMF
- **Rilonacept** — Loading: 4.4 mg/kg SC then 2.2 mg/kg SC weekly (max 320mg); CAPS age >=2yr
- **Tonsillectomy** — Surgical; definitive for PFAPA if medical therapy fails; high cure rate

Investigations

- CBC, ESR, CRP, SAA during fever — Markedly elevated during attacks; SAA for FMF monitoring [URGENT]
- Genetic testing (FMF/CAPS/TRAPS panel) — Confirm specific autoinflammatory syndrome
- 24-hour urine protein — Screen for AA amyloidosis in FMF
- Ferritin — Screen for MAS if very high during flare [URGENT]

Counselling

- Periodic fever syndromes cause regular episodes of high fever lasting a few days each time
- Colchicine must be taken daily to prevent attacks and protect the kidneys (in FMF)

- For PFAPA, a single dose of steroid can stop a fever episode within hours
- Keep a fever diary documenting frequency, duration, and associated symptoms
- Genetic testing may help confirm the exact type of periodic fever syndrome

Referral Criteria

Refer to pediatric rheumatologist for all suspected autoinflammatory syndromes; genetics for confirmation

Hospitalisation Criteria

Admit for severe disease flares with serositis, very high inflammatory markers, suspected MAS, or diagnostic workup requiring serial sampling

Notes

Colchicine is lifelong for FMF to prevent amyloidosis. IL-1 blockade is highly effective for CAPS and colchicine-resistant autoinflammatory syndromes.

M08 — Juvenile arthritis

SNOMED: 410795001 | Duration: 365 days | Source: ACR 2022 (2025 update draft), CARRA 2024

First-Line

- **Methotrexate** — 10-15 mg/m2/wk PO (preferred) or SC, max 25mg/wk; with folic acid 1mg/day
- **Naproxen** — 5-10 mg/kg/dose PO BID, max 500mg BID; for mild oligo-JIA
- **Adalimumab** — 20mg SC q2wks (<30kg) or 40mg SC q2wks (>=30kg); first-line biologic
- **Triamcinolone hexacetonide** — 1 mg/kg intra-articular (max 40mg large joint, 20mg small); for oligoarthritis

Second-Line

- **Etanercept** — 0.8 mg/kg SC weekly (max 50mg) or 0.4 mg/kg SC 2x/wk
- **Tocilizumab** — sJIA: 8-12 mg/kg IV q2wks (<30kg) or 8mg/kg q2wks (>=30kg)
- **Abatacept** — 10 mg/kg IV q4wks (max 1g) for polyarticular JIA
- **Emapalumab** — 6 mg/kg IV load then 3 mg/kg q3days; for MAS in sJIA refractory to steroids **[NEW 2024-25]**

Investigations

- CBC, ESR, CRP — Assess inflammatory activity [URGENT]
- ANA — Predict uveitis risk in JIA
- RF and anti-CCP antibodies — Classify JIA subtype and predict erosive disease
- Slit-lamp eye examination — Screen for asymptomatic anterior uveitis
- Joint X-ray or ultrasound — Assess joint damage and synovitis

Counselling

- Juvenile arthritis is a condition where your child's immune system causes joint inflammation
- Medicines are needed for a long time (years) to control the disease and prevent joint damage
- Regular exercise and physiotherapy are very important to keep joints flexible and muscles strong
- Eye checkups every 3-6 months are essential as arthritis can silently affect the eyes (uveitis)
- Your child can and should attend school and participate in activities with appropriate support

Referral Criteria

All JIA cases should be managed by pediatric rheumatologist; ophthalmology screening for uveitis; orthopedics for joint damage

Hospitalisation Criteria

Admit for macrophage activation syndrome (MAS), severe systemic JIA flare with serositis, or for IV biologic infusions if no outpatient infusion center

Notes

Early DMARD/biologic initiation improves inactive disease rates significantly (83% if biologic started within 6mo). Screen all sJIA patients for lung disease.

Guideline Changes

Draft 2025 ACR update: prefer oral MTX, earlier biologics, less NSAID/steroid reliance; emapalumab approved 2025 for MAS

M13 — Other arthritis

SNOMED: 3723001 | Duration: 30 days | Source: ACR 2022, PIDS/IDSA 2023

First-Line

- **Naproxen** — 5-10 mg/kg/dose PO BID (max 500mg BID) for symptomatic relief
- **Ibuprofen** — 10 mg/kg/dose PO q6-8h (max 40mg/kg/day or 2400mg/day)
- **Prednisolone** — 0.5-1 mg/kg/day PO short course for moderate inflammation, taper over 2-4 wks

Second-Line

- **Sulfasalazine** — 30-50 mg/kg/day PO divided BID (max 2g/day); for reactive arthritis
- **Methotrexate** — 10-15 mg/m2/wk PO/SC if persistent >3mo; with folic acid supplementation
- **Intra-articular triamcinolone** — 1 mg/kg per joint (max 40mg large, 20mg small) for monoarticular disease

Investigations

- CBC, ESR, CRP — Assess for inflammation and infection [URGENT]
- Joint aspiration (synovial fluid analysis) — Rule out septic arthritis if acute monoarthritis with fever [STAT]
- Blood culture — If septic arthritis suspected [STAT]
- X-ray of affected joint — Baseline assessment for joint damage

Counselling

- Your child's joint pain and swelling need investigation to find the exact cause
- Anti-inflammatory medicines (NSAIDs) will help with pain but the underlying cause must be treated
- If the joint is hot, red, and very painful, infection must be ruled out urgently
- Rest the affected joint but encourage gentle movement once pain improves
- Most reactive arthritis in children resolves completely within a few months

Referral Criteria

Refer to pediatric rheumatologist if arthritis persists >6 weeks; urgent orthopedic referral if septic arthritis suspected

Hospitalisation Criteria

Admit if septic arthritis suspected (hot, swollen joint with fever), for joint aspiration, or if unable to bear weight

Notes

Rule out septic arthritis before immunosuppressive therapy. Most reactive arthritis in children resolves within 3-12 months with NSAIDs alone.

M32 — Systemic lupus erythematosus

SNOMED: 55464009 | Duration: 365 days | Source: ACR 2025, EULAR 2023, EULAR jSLE 2025

First-Line

- **Hydroxychloroquine** — 5 mg/kg/day PO (max 400mg/day); recommended for ALL SLE patients long-term
- **Prednisolone** — 0.5-2 mg/kg/day PO (max 60mg); taper rapidly once controlled; minimize cumulative dose
- **Mycophenolate mofetil** — 600 mg/m2/dose PO BID (max 1.5g BID); preferred for lupus nephritis induction
- **Belimumab** — 10 mg/kg IV q4wks or 200mg SC weekly; age >=5yr; add-on to standard therapy

Second-Line

- **Azathioprine** — 1-3 mg/kg/day PO for maintenance after nephritis induction; check TPMT first
- **Cyclophosphamide** — 500-1000 mg/m2 IV monthly x6 for severe nephritis or CNS lupus
- **Rituximab** — 375 mg/m2 IV weekly x4 or 750 mg/m2 x2 doses; for refractory disease
- **Voclosporin** — 23.7mg PO BID (adult dose); approved for lupus nephritis; limited peds data **[NEW 2024-25]**

Investigations

- ANA and anti-dsDNA antibodies — Confirm diagnosis and monitor disease activity [URGENT]
- Complement C3, C4 — Low levels indicate active lupus [URGENT]
- CBC with differential — Screen for cytopenias [URGENT]
- Urinalysis and urine protein/creatinine ratio — Screen for lupus nephritis [URGENT]
- Renal biopsy — If proteinuria or active urine sediment (nephritis) [URGENT]

Counselling

- Lupus is a chronic autoimmune condition that needs lifelong treatment and monitoring
- Hydroxychloroquine is a very important medicine that should be continued long-term
- Strict sun protection is essential: sunscreen daily, avoid midday sun, wear protective clothing
- Report any new symptoms: face rash, joint pain, mouth sores, swelling, blood in urine
- Regular blood and urine tests are needed to detect organ involvement early

Referral Criteria

All pediatric SLE should be managed by pediatric rheumatologist; nephrology for lupus nephritis; hematology for severe cytopenias

Hospitalisation Criteria

Admit for lupus nephritis flare, severe cytopenias, CNS lupus, serositis, pulmonary hemorrhage, or MAS

Notes

Hydroxychloroquine is cornerstone therapy for all pediatric SLE. Monitor growth velocity and pubertal development given cumulative steroid exposure risk.

Guideline Changes

ACR 2025: uniform HCQ use, limit glucocorticoid duration, early biologic/DMARD introduction

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M33 — Dermatopolymyositis

SNOMED: 396230008 | Duration: 365 days | Source: BSR 2022, CARRA 2024, SHARE/EULAR 2023

First-Line

- **Prednisolone** — 1-2 mg/kg/day PO (max 60mg) or IV methylprednisolone 30mg/kg (max 1g) x3 days
- **Methotrexate** — 15-20 mg/m2/wk PO or SC (max 40mg/wk) with folic acid 1mg/day
- **IVIG** — 2 g/kg IV divided over 2-5 days, monthly for moderate-severe disease

Second-Line

- **Rituximab** — 375 mg/m2 IV weekly x4 or 750 mg/m2 (max 1g) x2 doses 2wks apart
- **Mycophenolate mofetil** — 600 mg/m2/dose PO BID (max 1.5g BID) for steroid-sparing
- **Abatacept** — 10 mg/kg IV q4wks (max 1g); for refractory JDM
- **Tofacitinib** — 5mg PO BID or weight-based; emerging for refractory cases **[NEW 2024-25]**

Investigations

- CK, LDH, AST, aldolase — Muscle enzyme levels to assess disease activity [URGENT]
- MRI muscles (thighs) — Assess muscle inflammation pattern and guide biopsy
- Myositis-specific antibodies (MSA panel) — Classify subtype and predict organ involvement
- Nailfold capillaroscopy — Assess microvasculopathy

- HRCT chest — Screen for interstitial lung disease

Counselling

- Dermatomyositis is an autoimmune condition where the immune system attacks muscles and skin
- Medicines to suppress the immune system are needed for months to years - do not stop without doctor advice
- Strict sun protection is essential: sunscreen SPF 50+, full sleeves, hat, and avoid midday sun
- Regular gentle exercise and physiotherapy help maintain muscle strength
- Blood tests are needed every few weeks to monitor disease activity and medicine side effects

Referral Criteria

All cases should be managed by pediatric rheumatologist; dermatology co-management for skin disease; pulmonology if ILD suspected

Hospitalisation Criteria

Admit for severe muscle weakness (unable to sit/stand), dysphagia, respiratory muscle weakness, pulmonary involvement, or for IV pulse steroid therapy

Notes

Combine corticosteroids with MTX from onset to improve outcomes and reduce steroid dependence. Sun protection and physical therapy are essential adjuncts.

Guideline Changes

JAK inhibitors emerging for refractory JDM; treat-to-target recommendations published 2025

M35 — Other systemic involvement of connective tissue

SNOMED: 194464009 | Duration: 365 days | Source: Expert consensus 2022, ACR/EULAR 2023

First-Line

- **Ibuprofen** — 10 mg/kg/dose PO q6-8h (max 40mg/kg/day); up to 1/3 of MCTD controlled with NSAIDs
- **Hydroxychloroquine** — 5 mg/kg/day PO (max 400mg/day); for arthritis, skin, and constitutional symptoms
- **Prednisolone** — 0.5-1 mg/kg/day PO for moderate disease; taper as tolerated over weeks

Second-Line

- **Methotrexate** — 10-15 mg/m2/wk PO/SC for arthritis or myositis component; with folic acid
- **Mycophenolate mofetil** — 600 mg/m2/dose PO BID (max 1.5g BID) for organ-threatening features
- **Nifedipine** — 0.25-0.5 mg/kg/dose PO TID (max 3mg/kg/day) for Raynaud phenomenon
- **Azathioprine** — 1-3 mg/kg/day PO daily for steroid-sparing maintenance

Investigations

- ANA and anti-U1 RNP antibodies — Confirm MCTD diagnosis
- CBC, ESR, CRP, complement levels — Assess disease activity
- Echocardiography — Screen for pulmonary hypertension
- HRCT chest — Screen for interstitial lung disease
- Renal function and urinalysis — Monitor for renal involvement

Counselling

- Mixed connective tissue disease affects multiple body systems and needs long-term treatment
- Take all medicines regularly as prescribed; do not stop without doctor advice
- Cold weather can cause fingers to turn white/blue (Raynaud phenomenon) - keep hands warm with gloves
- Regular blood tests and checkups are needed to monitor for organ involvement
- Report any new symptoms like breathlessness, chest pain, or skin changes to your doctor

Referral Criteria

All cases should be managed by pediatric rheumatologist; pulmonology for ILD or pulmonary hypertension screening; cardiology if cardiac involvement

Hospitalisation Criteria

Admit for organ-threatening disease (renal crisis, pulmonary hypertension, severe myositis), or disease flares not responding to outpatient therapy

Notes

Treatment is tailored to the dominant clinical phenotype (lupus-like, scleroderma-like, myositis-like). Monitor for pulmonary hypertension and ILD as major complications.

M60 — Myositis

SNOMED: 26889001 | Duration: 180 days | Source: BSR 2022, CARRA 2024

First-Line

- **Prednisolone** — 1-2 mg/kg/day PO (max 60mg) x4-6wks then slow taper; or IV pulse 30mg/kg (max 1g) x3d
- **Methotrexate** — 15-20 mg/m2/wk SC (max 40mg/wk) with folic acid 1mg/day; start at disease onset
- **IVIG** — 2 g/kg IV divided over 2-5 days monthly for moderate-severe or refractory disease

Second-Line

- **Mycophenolate mofetil** — 600 mg/m2/dose PO BID (max 1.5g BID) as steroid-sparing agent
- **Rituximab** — 375 mg/m2 IV weekly x4; for refractory myositis with autoantibodies
- **Azathioprine** — 1-3 mg/kg/day PO daily for maintenance; check TPMT before starting
- **Tofacitinib** — 5mg PO BID (or weight-based); emerging for refractory inflammatory myopathy **[NEW 2024-25]**

Investigations

- CK, LDH, AST, aldolase — Muscle enzyme levels to confirm myositis and track activity [URGENT]
- MRI muscles — Assess inflammation pattern and guide biopsy site
- Myositis-specific antibody panel — Classify subtype and predict complications
- EMG/NCS — Confirm myopathic pattern and exclude neuropathy
- Muscle biopsy — Histological confirmation if diagnosis uncertain

Counselling

- Myositis means inflammation of muscles; treatment needs to start early and continue for a long time
- Steroid medicine is essential initially - do not stop without doctor advice as the disease can flare
- Regular blood tests are needed to monitor muscle enzymes and medicine side effects
- Gentle physiotherapy helps maintain strength; avoid very intense exercise during active disease
- Sun protection is important if skin rash is present (dermatomyositis)

Referral Criteria

All inflammatory myositis should be managed by pediatric rheumatologist; pulmonology if respiratory muscle weakness or ILD

Hospitalisation Criteria

Admit for severe weakness (unable to stand/sit), dysphagia, respiratory muscle involvement, or for IV pulse steroid therapy

Notes

Combine steroids with methotrexate from onset to achieve steroid-free remission. Steroid taper is slow (1-2 years) to prevent relapse; monitor CK and muscle strength.

Guideline Changes

JAK inhibitors emerging as option for refractory inflammatory myositis

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M79.1 — Myalgia

SNOMED: 68962001 | Duration: 7 days | Source: AAP 2023, Expert consensus 2024

First-Line

- **Ibuprofen** — 10 mg/kg/dose PO q6-8h (max 40mg/kg/day or 2400mg/day); age >=3mo
- **Acetaminophen** — 15 mg/kg/dose PO q4-6h (max 75mg/kg/day or 4g/day); if NSAIDs contraindicated
- **Naproxen** — 5-7 mg/kg/dose PO BID (max 500mg BID); for older children >=2yr

Second-Line

- **Diclofenac** — 1 mg/kg/day PO divided TID (max 150mg/day); age >=1yr
- **Physical therapy** — Stretching, gentle strengthening, heat/ice; non-pharmacologic first-line adjunct

Investigations

- CK (creatine kinase) — Screen for muscle damage/myositis [URGENT]
- CBC, ESR, CRP — Assess for underlying inflammation or infection
- Urinalysis — Check for myoglobinuria if rhabdomyolysis suspected [URGENT]
- Thyroid function tests — Hypothyroidism can cause myalgia

Counselling

- Muscle pain in children usually gets better with rest and pain medicine within a few days
- Give ibuprofen or paracetamol as directed for pain relief
- If muscle pain follows a viral illness, it is usually benign and self-limiting
- Encourage gentle stretching and movement once pain improves
- Return if pain is severe, not improving in a week, or if weakness, rash, or dark urine develops

Referral Criteria

Refer to pediatric rheumatologist if myalgia persists >2 weeks, associated with weakness or elevated CK, or suspected inflammatory myopathy

Hospitalisation Criteria

Admit if severe myalgia with elevated CK suggesting rhabdomyolysis, dark/cola-colored urine, or suspected inflammatory myositis with systemic features

Notes

Investigate underlying cause before chronic NSAID use; limit OTC NSAID to 10 days without physician review. NSAIDs reduce opioid requirements significantly in pediatric pain.

T78.2 — Anaphylactic shock, unspecified

SNOMED: 39579001 | Duration: 2 days | Source: AAAAI/ACAAI 2023, WAO 2024

First-Line

- **Epinephrine IM** — 0.01 mg/kg IM anterolateral thigh (1:1000), max 0.5mg; repeat q5-15min PRN
- **Normal saline bolus** — 20 mL/kg IV bolus, repeat PRN for hypotension; max 1L/dose
- **Diphenhydramine** — 1-2 mg/kg IV/IM/PO q6h, max 50mg/dose; adjunct only, not substitute for epi
- **Methylprednisolone** — 1-2 mg/kg IV, max 125mg; for biphasic reaction prevention

Second-Line

- **Epinephrine infusion** — 0.1-1 mcg/kg/min IV continuous for refractory anaphylaxis; max 10 mcg/min
- **Glucagon** — 20-30 mcg/kg IV bolus, max 1mg; for beta-blocker-associated refractory hypotension
- **Neffy nasal spray** — 2mg intranasal single spray; for patients >=30kg as alternative to IM epi **[NEW 2024-25]**

Investigations

- Serum tryptase (within 1-4 hours) — Confirm mast cell degranulation in anaphylaxis [STAT]
- Specific IgE panel — Identify trigger allergen (after acute episode resolves)

- Skin prick testing — Comprehensive allergen identification (4-6 weeks after episode)

Counselling

- Anaphylaxis is a life-threatening allergic reaction - your child must always carry emergency epinephrine
- Learn how to use the epinephrine auto-injector and practice with the trainer device
- Give epinephrine FIRST if severe reaction occurs, then call ambulance - do not delay
- Get a medical alert bracelet for your child listing their allergy
- Strictly avoid the identified trigger; inform school, relatives, and caretakers about the allergy

Referral Criteria

Urgent referral to pediatric allergist for all anaphylaxis cases for trigger identification and management plan

Hospitalisation Criteria

All anaphylaxis requires emergency department evaluation; observe minimum 4-6 hours for biphasic reaction; admit if severe, required multiple epinephrine doses, or has asthma

Notes

Epinephrine IM is the ONLY first-line treatment; delays increase mortality. Observe 4-6h minimum post-reaction for biphasic anaphylaxis risk.

Guideline Changes

Neffy (epinephrine nasal spray) approved Aug 2024 for patients >=30kg

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T78.4 — Allergy, unspecified

SNOMED: 419076005 | Duration: 14 days | Source: ARIA-EAACI 2024, AAP 2024

First-Line

- **Cetirizine** — 0.25 mg/kg/dose PO daily; 6-23mo: 2.5mg, 2-5yr: 5mg, >=6yr: 10mg daily
- **Loratadine** — 2-5yr: 5mg PO daily; >=6yr: 10mg PO daily
- **Prednisolone** — 1-2 mg/kg/day PO for acute allergic reactions, max 60mg, short course 3-5 days

Second-Line

- **Omalizumab** — 75-375mg SC q2-4wks based on IgE level and weight; age >=1yr for food allergy **[NEW 2024-25]**
- **Montelukast** — 6mo-5yr: 4mg PO daily; 6-14yr: 5mg PO daily; >=15yr: 10mg PO daily
- **Fexofenadine** — 2-11yr: 30mg PO BID; >=12yr: 60mg PO BID or 180mg daily

Investigations

- Specific IgE panel (blood) — Identify specific allergens
- Skin prick testing — Gold standard for IgE-mediated allergy identification
- CBC with eosinophil count — Elevated eosinophils support allergic etiology

Counselling

- Identify and strictly avoid the allergen that caused the reaction
- Always carry the allergy card listing your child's allergens
- Second-generation antihistamines (cetirizine, loratadine) are preferred over older ones that cause drowsiness
- If your child has severe food allergy, read all food labels carefully
- Ask your doctor if your child needs to carry an emergency epinephrine auto-injector

Referral Criteria

Refer to pediatric allergist for skin prick testing, specific IgE testing, and allergen identification; refer urgently if anaphylaxis history

Hospitalisation Criteria

Admit for severe allergic reactions with systemic involvement, anaphylaxis, or for supervised oral food challenge

Notes

Second-generation antihistamines are first-line; avoid first-gen due to sedation and CNS effects. Identify and eliminate allergen triggers before escalating pharmacotherapy.

Guideline Changes

Omalizumab approved Feb 2024 for IgE-mediated food allergy age >=1yr; Neffy nasal epinephrine approved Aug 2024

Skin

B35 — Dermatophytosis

SNOMED: 47382004 | Duration: 28 days | Source: AAD 2024, AAP 2024

First-Line

- **Terbinafine cream 1%** — Apply to affected area daily-BID for 1-2 weeks; for tinea corporis/cruris/pedis
- **Griseofulvin microsize oral** — 20-25 mg/kg/day PO divided BID (max 1 g/day) for 6-8 weeks; for tinea capitis
- **Clotrimazole 1% cream** — Apply to affected area BID for 2-4 weeks; OTC topical for tinea corporis/pedis

Second-Line

- **Terbinafine oral** — <20 kg: 62.5 mg/day; 20-40 kg: 125 mg/day; >40 kg: 250 mg/day PO for 4-6 weeks
- **Fluconazole oral** — 6 mg/kg/day PO for 2-4 weeks (tinea corporis) or 6-8 weeks (tinea capitis)
- **Itraconazole oral** — 3-5 mg/kg/day PO for 2-4 weeks; for resistant dermatophytosis or onychomycosis

Investigations

- KOH preparation — Confirm dermatophyte hyphae before starting treatment
- Fungal culture — Species identification; important if treatment failure
- Wood lamp examination — For Microsporum species (green fluorescence)

Counselling

- Apply antifungal cream beyond the visible rash edges by 2 cm; continue for 1 week after rash clears
- For tinea capitis (scalp ringworm), oral medicine is essential - cream alone will not work
- Do not share combs, towels, caps, or pillows to prevent spread
- Wash clothing and bedding in hot water during treatment
- If family members or pets have similar patches, they need treatment too

Referral Criteria

Refer to dermatologist if tinea capitis not responding to 8 weeks of griseofulvin, suspected T. indotineae (resistant strain), extensive tinea corporis, or onychomycosis

Hospitalisation Criteria

Hospitalisation not needed; consider if kerion (inflammatory tinea capitis) with secondary bacterial infection requiring IV antibiotics

Notes

Tinea capitis always requires systemic antifungal therapy; topical agents alone are insufficient. Monitor for T. indotineae resistance if poor response to terbinafine.

Guideline Changes

Rising T. indotineae antifungal resistance reported; CDC alert 2024

B36 — Other superficial mycoses

SNOMED: 56454009 | Duration: 14 days | Source: Medscape 2024, StatPearls 2024

First-Line

- **Ketoconazole 2% shampoo** — Apply to body/scalp as wash, leave 5 min, rinse daily for 2 weeks; for pityriasis versicolor
- **Selenium sulfide 2.5% lotion** — Apply to affected area, leave 10 min, rinse daily for 1-2 weeks; OTC antifungal
- **Clotrimazole 1% cream** — Apply to affected area BID for 2-4 weeks; topical azole for localized disease

Second-Line

- **Fluconazole oral** — 6 mg/kg PO (max 400 mg) single dose or weekly x2-3 weeks; for extensive disease
- **Itraconazole oral** — 3-5 mg/kg/day PO for 5-7 days; for recurrent or widespread pityriasis versicolor

- **Zinc pyrithione 1% shampoo** — Apply as body wash, leave 5 min, rinse; use 2-3x/week; OTC maintenance option

Investigations

- KOH preparation — Spaghetti and meatballs appearance confirms Malassezia
- Wood lamp examination — Yellow-green fluorescence of affected patches

Counselling

- Pityriasis versicolor is caused by a normal skin yeast - it is not contagious
- Apply antifungal shampoo as body wash; leave on for 5-10 minutes before rinsing
- The white/brown patches may take months to match normal skin colour even after treatment
- Use antifungal shampoo once monthly as maintenance to prevent recurrence
- The condition commonly recurs in hot humid weather - preventive treatment before summer helps

Referral Criteria

Refer to dermatologist if widespread recurrent disease not responding to topical treatment, or diagnostic uncertainty

Hospitalisation Criteria

Hospitalisation not needed for superficial mycoses

Notes

Pityriasis versicolor has high recurrence; discuss monthly prophylactic antifungal wash. Dyspigmentation may persist for months after successful treatment.

B37.2 — Candidiasis of skin and nail

SNOMED: 78048006 | Duration: 14 days | Source: IDSA 2016, AAP 2024

First-Line

- **Nystatin cream 100,000 U/g** — Apply to affected skin area TID-QID for 14 days; first-line for cutaneous candidiasis
- **Clotrimazole 1% cream** — Apply to affected area BID-TID for 7-14 days; broad-spectrum topical antifungal
- **Miconazole 2% cream** — Apply to affected area BID for 14 days; effective for skin and intertriginous candidiasis

Second-Line

- **Fluconazole oral** — 6 mg/kg PO day 1 then 3 mg/kg/day for 14-21 days; for refractory or nail candidiasis
- **Ciclopirox 8% nail lacquer** — Apply to affected nails daily for up to 48 weeks; for onychomycosis in older children
- **Ketoconazole 2% cream** — Apply to affected area daily for 2-4 weeks; alternative topical antifungal

Investigations

- KOH preparation — Confirm fungal elements if diagnosis uncertain
- Fungal culture — Species identification if resistant to treatment
- Blood glucose — Screen for diabetes in recurrent candidiasis

Counselling

- Keep the affected skin clean and completely dry; pat dry after washing
- Apply antifungal cream to the rash and 2 cm beyond the edges twice daily
- Avoid tight clothing and synthetic fabrics over the affected area
- Complete the full 14-day course even if the rash looks better
- If fungal infections keep recurring, see doctor - may need blood sugar check

Referral Criteria

Refer to dermatologist if chronic or recurrent mucocutaneous candidiasis; refer to immunologist if suspected underlying immunodeficiency (chronic mucocutaneous candidiasis)

Hospitalisation Criteria

Rarely needed for cutaneous candidiasis; admit if disseminated candidiasis suspected (immunocompromised child), or extensive erosive lesions in neonate

Notes

Keep affected skin dry and well-ventilated. For recurrent candidiasis, evaluate for predisposing factors (diabetes, immunodeficiency, prolonged antibiotic use).

L01.0 — Impetigo

SNOMED: 48277006 | Duration: 7 days | Source: IDSA 2014

First-Line

- **Mupirocin 2% ointment** — TID x5-7 days, first-line for localised non-bullous impetigo

Second-Line

- **Cephalexin** — 25-50 mg/kg/day divided q6-8h x7 days, extensive or bullous impetigo
- **TMP-SMX** — 8-10 mg/kg/day TMP component q12h x7 days, if MRSA suspected

Investigations

- Wound swab culture and sensitivity — Recurrent impetigo or failure of first-line treatment
- Urinalysis — If GAS impetigo - screen for post-streptococcal glomerulonephritis 2-3 weeks later

Counselling

- Keep affected area clean; wash gently with soap and water before applying ointment
- Child should not share towels, clothes, or bedding to prevent spread to siblings
- Keep fingernails short and discourage scratching of the sores
- Child can return to school after 24 hours of antibiotic treatment if lesions are covered
- Complete the full course of antibiotics; do not stop early even if skin looks better

Referral Criteria

Refer to dermatologist if recurrent impetigo despite adequate treatment, suspected MRSA (culture-guided), or deep tissue involvement (ecthyma)

Hospitalisation Criteria

Admit if extensive bullous impetigo with systemic toxicity, suspected staphylococcal scalded skin syndrome, or post-streptococcal complications (glomerulonephritis signs)

Notes

Systemic antibiotics for extensive disease, bullous impetigo, or suspected MRSA.

L21 — Seborrheic dermatitis

SNOMED: 200764003 | Duration: 14 days | Source: PMC/IDOJ 2024, StatPearls 2024

First-Line

- **Ketoconazole 2% shampoo** — Apply to scalp, lather, leave 5 min then rinse; daily for 2 weeks then 1-2x/week
- **Hydrocortisone 1% cream** — Apply to affected area BID for 7-14 days; for erythema in skin folds
- **Emollients** — Apply after bathing to restore epidermal barrier; use fragrance-free products

Second-Line

- **Ciclopirox 1% cream/shampoo** — Apply to affected area BID for 2-4 weeks or shampoo 2x/week; antifungal alternative

- **Selenium sulfide 2.5% shampoo** — Apply to scalp, leave 5-10 min, rinse; 2-3x/week for 2-4 weeks
- **Zinc pyrithione 1% shampoo** — Apply to scalp, lather, rinse; 2-3x/week; mild OTC antifungal option
- **Tacrolimus 0.03% ointment** — Apply BID to facial SD; steroid-sparing for children >=2 yr

Investigations

- KOH preparation — Confirm Malassezia if diagnosis uncertain
- Skin biopsy — If atypical presentation or suspected Langerhans cell histiocytosis

Counselling

- Cradle cap in babies is very common and harmless - it usually resolves by 6-12 months
- Apply mineral oil or coconut oil to scalp, leave 15 minutes, then gently comb off scales
- Use antifungal shampoo 2-3 times per week on scalp; leave on for 5 minutes before rinsing
- Mild steroid cream can be used on red, inflamed areas for 1-2 weeks only
- Condition may recur; maintenance antifungal shampoo once weekly can help prevent flares

Referral Criteria

Refer to dermatologist if severe seborrheic dermatitis not responding to antifungal + topical steroid, diagnostic uncertainty (consider psoriasis, Langerhans cell histiocytosis in infants)

Hospitalisation Criteria

Hospitalisation not needed for typical SD; consider if erythrodermic presentation or suspected Leiner disease in infants

Notes

Infantile seborrheic dermatitis (cradle cap) is usually self-limiting and responds to gentle scale removal with mineral oil. Recurrence is common; maintenance therapy may be needed.

Guideline Changes

Tapinarof 1% and ruxolitinib 1.5% cream in pipeline for adolescent facial SD

L22 — Diaper dermatitis

SNOMED: 91487003 | Duration: 7 days | Source: AAP 2024, StatPearls 2024

First-Line

- **Zinc oxide ointment** — Apply thick layer to affected area with every diaper change; first-line barrier protection
- **Nystatin cream 100,000 U/g** — Apply to affected area 3-4x daily after diaper change for 7-14 days if candidal
- **Petrolatum barrier ointment** — Apply generously at each diaper change as barrier; use with frequent diaper changes

Second-Line

- **Hydrocortisone 1% cream** — Apply BID for 3-7 days for moderate inflammation; do not use under occlusion long-term
- **Miconazole 2% cream** — Apply to affected area BID for 7-14 days; for candidal diaper dermatitis
- **Clotrimazole 1% cream** — Apply to affected area BID-TID for 7-14 days; alternative antifungal for candidal rash

Investigations

- KOH preparation — If candidal superinfection suspected (satellite lesions, >3 days)
- Skin swab culture — If bacterial superinfection suspected (purulent discharge)

Counselling

- Change diaper frequently - at least every 2-3 hours and immediately after soiling
- Apply thick layer of zinc oxide cream as barrier at every diaper change

- Allow diaper-free time for 10-15 minutes several times a day for air exposure
- Use plain water or mild wipes for cleaning; avoid scented wipes
- If rash has satellite lesions (small red dots around main rash), it may be fungal - tell doctor

Referral Criteria

Refer to dermatologist if not responding to standard treatment in 7 days, suspected underlying condition (psoriasis, Langerhans cell histiocytosis), or recurrent severe episodes

Hospitalisation Criteria

Hospitalisation not typically needed; consider if extensive erosive dermatitis with secondary infection or failure to thrive

Notes

Frequent diaper changes and air exposure are the cornerstone of management. If rash persists >3 days despite barrier care, consider candidal superinfection.

L23 — Allergic contact dermatitis

SNOMED: 238575004 | Duration: 14 days | Source: AAP 2025, AAD 2024

First-Line

- **Hydrocortisone 1% cream** — Apply thin layer to affected area BID for 10-14 days; use lowest potency effective
- **Triamcinolone 0.1% cream** — Apply to affected area BID for 10-14 days; avoid face/intertriginous areas
- **Emollients (ceramide-based)** — Apply liberally after bathing and PRN; use fragrance-free products (e.g. CeraVe)

Second-Line

- **Prednisolone oral** — 0.5-1 mg/kg/day PO for 7 days then taper over 7 days; for severe/widespread disease
- **Tacrolimus 0.03% ointment** — Apply BID to affected area for up to 3 weeks; for face/eyelids in children >=2 yr
- **Hydroxyzine** — 2 mg/kg/day PO divided Q6-8h (max 25 mg/dose) for pruritus relief

Investigations

- Patch testing — Identify contact allergen in recurrent cases
- Skin swab culture — If secondary infection suspected

Counselling

- Identify and strictly avoid the substance causing the allergic reaction
- Apply steroid cream as directed - thin layer only; do not use on face for more than 1 week
- Use fragrance-free soap, detergent, and moisturisers
- Keep child nails short to prevent scratching and secondary infection
- If rash recurs, patch testing by a dermatologist can identify the exact allergen

Referral Criteria

Refer to dermatologist/allergist if recurrent contact dermatitis, allergen identification needed (patch testing), or not responding to topical treatment

Hospitalisation Criteria

Rarely needed; admit if severe widespread reaction with systemic symptoms, secondary bacterial infection requiring IV antibiotics, or erythroderma

Notes

Identify and avoid the offending allergen via patch testing if recurrent. Pre-treatment with barrier cream or topical steroid may help if adhesive/dressing exposure is unavoidable.

Guideline Changes

AAP 2025 clinical report emphasizes daily bathing with fragrance-free cleansers

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L24 — Irritant contact dermatitis

SNOMED: 110979008 | Duration: 14 days | Source: AAP 2024, Medscape 2024

First-Line

- **Emollients (ceramide-based)** — Apply liberally BID and after hand washing; restore epidermal barrier (e.g. CeraVe, Cetaphil)
- **Hydrocortisone 1% cream** — Apply to affected area BID for 10-14 days; low potency suitable for children
- **Triamcinolone 0.1% ointment** — Apply BID for 10-14 days; for moderate lesions on trunk/extremities

Second-Line

- **Prednisolone oral** — 0.5-1 mg/kg/day PO for 7 days then taper; for severe widespread ICD only
- **Tacrolimus 0.03% ointment** — Apply BID for up to 3 weeks; steroid-sparing for face/flexures in children >=2 yr
- **Hydroxyzine** — 2 mg/kg/day PO divided Q6-8h (max 25 mg/dose) for pruritus

Investigations

- Patch testing — To differentiate from allergic contact dermatitis if diagnosis uncertain
- Skin biopsy — If diagnosis uncertain after clinical assessment and patch testing

Counselling

- Identify and remove the irritant causing the rash - this is the key to healing
- Apply barrier cream (zinc oxide or petroleum jelly) before exposure to known irritants
- Use fragrance-free soap and moisturisers; avoid harsh detergents
- Moisturise skin regularly - at least twice daily and after hand washing
- Cotton gloves at night can prevent scratching during sleep

Referral Criteria

Refer to dermatologist if chronic ICD not responding to irritant avoidance and topical treatment, or occupational exposure in adolescents

Hospitalisation Criteria

Rarely needed; admit if severe widespread ICD with secondary infection or significant skin barrier breakdown

Notes

Identify and remove the causative irritant. Barrier creams (zinc oxide, dimethicone) can prevent recurrence in occupational or habitual exposure.

L25.9 — Unspecified contact dermatitis

SNOMED: 40275004 | Duration: 14 days | Source: AAP 2024, Medscape 2024

First-Line

- **Hydrocortisone 1% cream** — Apply to affected area BID for 10-14 days; low potency for pediatric use
- **Emollients (ceramide-based)** — Apply liberally BID and after bathing; restore skin barrier
- **Cetirizine** — 6-23 mo: 2.5 mg PO daily; 2-5 yr: 5 mg daily; >=6 yr: 10 mg daily for itch

Second-Line

- **Triamcinolone 0.1% ointment** — Apply BID for 10-14 days; for moderate dermatitis on trunk/extremities
- **Prednisolone oral** — 0.5-1 mg/kg/day PO for 7 days then taper over 7 days; severe cases only
- **Tacrolimus 0.03% ointment** — Apply BID for face/intertriginous areas; steroid-sparing for children >=2 yr

Investigations

- Patch testing — Identify specific contact allergen if recurrent
- Skin biopsy — If diagnosis uncertain; differentiate from psoriasis, dermatophytosis

Counselling

- Try to identify what is causing the rash - common culprits are detergent, nickel jewellery, plants
- Apply moisturiser regularly and steroid cream on inflamed areas only
- Use fragrance-free products for skin care and laundry
- Avoid scratching; keep nails short and apply cool compresses for itch relief
- If rash keeps coming back, allergy testing (patch test) can identify the specific cause

Referral Criteria

Refer to dermatologist/allergist for patch testing if recurrent or chronic contact dermatitis, or diagnosis uncertain

Hospitalisation Criteria

Rarely needed; consider if severe generalised reaction or secondary infection requiring IV treatment

Notes

Attempt to classify as allergic vs irritant via history and patch testing. Avoid irritants; use mild cleansers and fragrance-free products.

L27 — Dermatitis due to ingested substances

Duration: 7 days | Source: AAP 2025, AAAAI/ACAAI 2023

First-Line

- **Cetirizine** — 6-23 mo: 2.5 mg PO daily; 2-5 yr: 2.5-5 mg daily; >=6 yr: 5-10 mg daily
- **Hydrocortisone 1% cream** — Apply thin layer to affected area BID for 7-14 days; for symptomatic skin lesions
- **Emollients** — Apply liberally after bathing and PRN to restore skin barrier

Second-Line

- **Prednisolone oral** — 1-2 mg/kg/day PO (max 60 mg) for 3-5 days for severe systemic reactions
- **Diphenhydramine** — 1.25 mg/kg/dose PO Q6h (max 50 mg/dose); useful for acute pruritus/urticaria
- **Hydroxyzine** — 2 mg/kg/day PO divided Q6-8h (max 25 mg/dose); for persistent itch

Investigations

- Serum specific IgE — Identify food allergen triggers
- Skin prick testing — If food allergy suspected; done by allergist
- Complete blood count with eosinophils — Eosinophilia in drug hypersensitivity

Counselling

- Stop the suspected food or medicine immediately - this is the most important step
- Keep a food diary to help identify which food is causing the skin reaction
- Give antihistamine as prescribed for itching; apply moisturiser for skin dryness
- If swelling of lips, tongue, or breathing difficulty occurs, use epinephrine and go to hospital
- See an allergist for proper allergy testing if food allergy is suspected

Referral Criteria

Refer to allergist if food allergy suspected for IgE testing and oral food challenge; refer to dermatologist if drug reaction with skin involvement

Hospitalisation Criteria

Admit if severe drug reaction (SJS/TEN), anaphylaxis, widespread blistering, mucosal involvement, or systemic symptoms (fever, organ involvement)

Notes

Discontinue offending ingested substance (food, drug) immediately. Referral to allergist for IgE testing if food allergy suspected.

L28 — Lichen simplex chronicus and prurigo

SNOMED: 201020007 | Duration: 28 days | Source: Medscape 2024, StatPearls 2024

First-Line

- **Clobetasol 0.05% ointment** — Apply to lichenified plaques BID for 2-4 weeks; may use under occlusion overnight
- **Hydroxyzine** — 2 mg/kg/day PO divided Q6-8h (max 25 mg/dose); break itch-scratch cycle at bedtime
- **Emollients** — Apply liberally TID and after bathing to reduce dryness-triggered pruritus

Second-Line

- **Tacrolimus 0.1% ointment** — Apply BID to affected area; steroid-sparing for face/genital LSC in children >=2 yr
- **Dupilumab** — 6-17 yr weight-based: 15-<30 kg 300 mg Q4W; >=60 kg 600 mg load then 300 mg Q2W SC **[NEW 2024-25]**
- **Gabapentin** — 5-10 mg/kg/day PO divided TID; for neuropathic itch component; off-label in children

Investigations

- Skin biopsy — If diagnosis uncertain; shows epidermal thickening and dermal fibrosis
- Complete blood count with eosinophils — Exclude underlying atopic or parasitic conditions

Counselling

- The main goal is to break the itch-scratch cycle - keep nails very short
- Apply thick moisturiser and steroid cream as directed; cover area with bandage at night
- Hydroxyzine at bedtime helps reduce scratching during sleep
- Stress and anxiety can worsen the condition - relaxation techniques may help
- Treatment takes weeks to months; be patient and consistent with cream application

Referral Criteria

Refer to dermatologist if not responding to potent topical steroids, need for systemic therapy, or suspicion of underlying neuropathic itch

Hospitalisation Criteria

Rarely needed; consider if severe secondary infection or need for inpatient behavioural intervention

Notes

No FDA-approved therapy exists for LSC; breaking the itch-scratch cycle is the primary goal. Behavioral intervention and covering affected areas at night can help in children.

Guideline Changes

Dupilumab and nemolizumab emerging for refractory prurigo nodularis

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L29 — Pruritus

SNOMED: 418290006 | Duration: 14 days | Source: AAP 2024, AAAAI 2023

First-Line

- **Cetirizine** — 6-23 mo: 2.5 mg PO daily; 2-5 yr: 2.5-5 mg daily; >=6 yr: 5-10 mg daily
- **Hydroxyzine** — 2 mg/kg/day PO divided Q6-8h (max 25 mg/dose); sedating, useful for nocturnal itch
- **Emollients** — Apply liberally after bathing to reduce xerosis-related pruritus

Second-Line

- **Fexofenadine** — 2-11 yr: 30 mg PO BID; >=12 yr: 60 mg BID or 180 mg daily; non-sedating
- **Loratadine** — 2-5 yr: 5 mg PO daily; >=6 yr: 10 mg daily; non-sedating option
- **Diphenhydramine** — 1.25 mg/kg/dose PO Q6h (max 50 mg/dose); for acute severe itch; avoid in <2 yr

Investigations

- Complete blood count — Screen for eosinophilia, iron deficiency, lymphoma
- Liver and renal function tests — Exclude cholestatic or uraemic pruritus
- Thyroid function tests — Hyperthyroidism can cause generalised pruritus
- Skin scraping for scabies — If interdigital or nocturnal itch pattern

Counselling

- Moisturise skin liberally; dry skin is the most common cause of itching in children
- Use lukewarm baths and mild soap; avoid hot water which worsens itching
- Give antihistamine at bedtime if itching disturbs sleep
- Keep fingernails short; cotton gloves at night can prevent scratching
- If itching persists despite moisturising, see doctor for further investigation

Referral Criteria

Refer to dermatologist if chronic pruritus >6 weeks not responding to antihistamines and moisturisers, or underlying systemic cause suspected

Hospitalisation Criteria

Rarely needed; consider if severe excoriations with secondary infection, or pruritus as part of systemic illness (liver, kidney, lymphoma)

Notes

Investigate underlying cause (xerosis, scabies, systemic disease) before chronic antihistamine use. Up-dose second-gen antihistamines up to 4x before switching agents.

Guideline Changes

Up-dosing of second-gen antihistamines (2-4x) now recommended before escalation

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L30 — Other and unspecified dermatitis

SNOMED: 43116000 | Duration: 14 days | Source: AAP 2025, AAAAI/ACAAI 2023

First-Line

- **Hydrocortisone 1% cream** — Apply to affected area BID for 7-14 days; first-line for mild unspecified dermatitis
- **Emollients** — Apply liberally BID-TID and after bathing; fragrance-free ceramide-based preferred
- **Cetirizine** — 6-23 mo: 2.5 mg PO daily; 2-5 yr: 5 mg daily; >=6 yr: 10 mg daily for pruritus

Second-Line

- **Triamcinolone 0.1% ointment** — Apply BID for 14 days; for moderate dermatitis on trunk/extremities
- **Tacrolimus 0.03% ointment** — Apply BID for face/eyelids/flexures; for children >=2 yr; steroid-sparing
- **Prednisolone oral** — 1 mg/kg/day PO for 5-7 days then taper; for severe flares only

Investigations

- Skin biopsy — If diagnosis uncertain or chronic unresponsive dermatitis
- Patch testing — If contact allergen suspected
- Complete blood count — Eosinophilia may suggest allergic or parasitic cause

Counselling

- Apply moisturiser generously twice daily and after bathing

- Use mild steroid cream only on the red, itchy areas - not on normal skin
- Use fragrance-free soap and detergent; dress in soft cotton clothing
- If rash does not improve in 2 weeks, come back for further evaluation
- Keep a diary of potential triggers (food, detergent, stress) to identify the cause

Referral Criteria

Refer to dermatologist if persistent dermatitis not responding to topical treatment, diagnostic uncertainty, or need for patch testing/skin biopsy

Hospitalisation Criteria

Rarely needed; admit if erythroderma, severe secondary infection, or suspected systemic cause requiring inpatient workup

Notes

Attempt to identify specific etiology before treating empirically. If persistent, consider patch testing, skin biopsy, or dermatology referral.

Guideline Changes

AAP 2025 update on skin-directed management for atopic/unspecified dermatitis

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L40 — Psoriasis

SNOMED: 9014002 | Duration: 84 days | Source: AAD-NPF 2024, AAP 2024

First-Line

- **Calcipotriene 0.005% cream** — Apply to plaques daily-BID; for <30% BSA; can combine with topical steroid; >=12 yr
- **Betamethasone dipropionate 0.05%** — Apply to plaques BID for 2-4 weeks; medium-high potency for trunk/extremities
- **Hydrocortisone 2.5% cream** — Apply BID to face/flexural psoriasis for 2-4 weeks; lower potency for sensitive areas

Second-Line

- **Methotrexate** — 0.2-0.7 mg/kg PO/SC once weekly (max 25 mg/wk); for moderate-severe; monitor CBC/LFTs
- **Apremilast** — 6-17 yr: 35-70 kg 20 mg PO BID; >70 kg 30 mg BID; titrate over 5 days **[NEW 2024-25]**
- **Ustekinumab** — >=6 yr: <60 kg 0.75 mg/kg SC wk 0,4 then Q12W; 60-100 kg 45 mg; >100 kg 90 mg
- **Adalimumab** — >=4 yr: 0.8 mg/kg SC Q2W (max 40 mg); for moderate-severe plaque psoriasis

Investigations

- Skin biopsy — If diagnosis uncertain; classic acanthosis and Munro microabscesses
- Throat swab culture — If guttate psoriasis triggered by streptococcal infection
- ASO titre — Evidence of recent streptococcal infection in guttate psoriasis
- Liver and renal function tests — Baseline before starting methotrexate or systemic therapy

Counselling

- Psoriasis is a chronic condition with flare-ups and remissions - it is manageable but not curable
- Apply steroid cream or calcipotriene as directed; do not use strong steroids on face or groin
- Keep skin well moisturised - this reduces scaling and itch
- Stress and infections (especially strep throat) can trigger flares
- Emotional support is important; psoriasis is not contagious - educate school and friends

Referral Criteria

Refer to dermatologist for moderate-severe psoriasis (>10% BSA), psoriatic arthritis symptoms, failure of topical therapy, or need for systemic/biologic therapy

Hospitalisation Criteria

Admit if erythrodermic or pustular psoriasis, psoriasis with severe secondary infection, or metabolic derangement from extensive disease

Notes

Topical therapy is first-line for mild-moderate psoriasis; screen for psoriatic arthritis, metabolic syndrome, and psychosocial impact. Biologics require pre-treatment TB and hepatitis screening.

Guideline Changes

Apremilast approved for pediatric psoriasis ages 6-17 in Aug 2024

L42 — Pityriasis rosea

SNOMED: 77252004 | Duration: 42 days | Source: AAP 2024, Nationwide Children's 2024

First-Line

- **Emollients/calamine lotion** — Apply to affected area TID-QID PRN for itch relief; lukewarm oatmeal baths may help
- **Cetirizine** — 2-5 yr: 2.5-5 mg PO daily; >=6 yr: 5-10 mg daily; for symptomatic pruritus
- **Hydrocortisone 1% cream** — Apply to itchy lesions BID for 1-2 weeks; for moderate pruritus

Second-Line

- **Acyclovir** — 20 mg/kg/dose PO QID (max 800 mg/dose) x 7 days; some evidence for faster resolution
- **NB-UVB phototherapy** — 3x/week for 2-4 weeks; for severe/widespread pruritic cases; accelerates clearance
- **Erythromycin** — 25-40 mg/kg/day PO divided QID for 14 days; limited evidence for faster resolution

Investigations

- Clinical diagnosis (usually sufficient) — Diagnosis is clinical; testing only if atypical
- RPR/VDRL — If sexually active adolescent to rule out secondary syphilis
- KOH preparation — If fungal infection (tinea) needs to be excluded

Counselling

- Pityriasis rosea is harmless and heals on its own in 6-8 weeks
- It is not contagious - child can attend school normally
- Avoid hot baths and vigorous exercise which may make the rash worse
- Use calamine lotion or mild steroid cream for itch relief
- The rash follows a Christmas tree pattern on the back - this is characteristic and not worrying

Referral Criteria

Refer to dermatologist if atypical presentation (no herald patch, face involvement), persisting >12 weeks, or diagnostic uncertainty (rule out secondary syphilis in adolescents)

Hospitalisation Criteria

Hospitalisation not needed for pityriasis rosea

Notes

Pityriasis rosea is self-limited (4-8 weeks); treatment is symptomatic. Avoid hot water and strenuous exercise which may exacerbate the rash.

L51 — Erythema multiforme

SNOMED: 36715001 | Duration: 14 days | Source: PMC/Ped Derm 2023, AAFP 2019, StatPearls 2024

First-Line

- **Ibuprofen** — 5-10 mg/kg/dose PO Q6-8h (max 40 mg/kg/day); for pain and inflammation in EM minor
- **Diphenhydramine elixir** — 1.25 mg/kg/dose PO Q6h (max 50 mg/dose); for pruritus; can also use as oral rinse

- **Triamcinolone 0.1% cream** — Apply to skin lesions BID for 7-14 days; for symptomatic relief of EM minor

Second-Line

- **Acyclovir (prophylactic)** — 20 mg/kg/dose PO QID (max 400 mg/dose) for 5 days; if HSV-triggered EM confirmed
- **Valacyclovir (prophylactic)** — >=12 yr: 500 mg PO BID x 5 days; for HSV-associated recurrent EM; suppressive dosing
- **Viscous lidocaine 2%** — Apply to oral lesions Q3h PRN (max 4 doses/day); for painful oral mucosal involvement

Investigations

- Skin biopsy — If diagnosis uncertain; interface dermatitis with lymphocytic infiltrate
- HSV PCR or serology — If HSV-triggered EM suspected
- Mycoplasma IgM — If preceded by respiratory symptoms; Mycoplasma-induced rash

Counselling

- Erythema multiforme usually heals on its own in 1-4 weeks
- Give pain medicine as needed; soft cool foods if mouth sores are present
- Avoid hot or spicy foods that may irritate mouth ulcers
- If triggered by cold sores (herpes), antiviral medicine can prevent recurrence
- Watch for spreading rash, blistering of skin, or difficulty eating/drinking - see doctor urgently

Referral Criteria

Refer to dermatologist if recurrent EM (>3 episodes/year), EM major with mucosal involvement, suspected SJS (>1 mucosal site, extensive skin detachment), or need for HSV suppressive therapy

Hospitalisation Criteria

Admit if EM major with severe oral involvement preventing eating, suspected SJS/TEN (detachment >10% BSA), dehydration, or systemic toxicity

Notes

EM is usually self-limited (1-4 weeks); supportive care is the mainstay. If HSV-associated and recurrent, consider long-term antiviral suppression.

L52 — Erythema nodosum

SNOMED: 32861005 | Duration: 21 days | Source: PMC 2022, StatPearls 2024

First-Line

- **Ibuprofen** — 5-10 mg/kg/dose PO Q6-8h (max 40 mg/kg/day); first-line NSAID for pain/inflammation
- **Naproxen** — 5-7 mg/kg/dose PO Q12h (max 500 mg/dose); alternative NSAID for older children
- **Rest and elevation** — Bed rest with leg elevation; compressive bandaging; reduces inflammation and pain

Second-Line

- **Potassium iodide** — 150-300 mg PO TID for 3-4 weeks; for refractory EN; monitor thyroid function
- **Prednisolone oral** — 0.5-1 mg/kg/day PO for 1-2 weeks then taper; only after infection excluded
- **Colchicine** — 0.5 mg PO daily-BID (children >4 yr); for recurrent EN; off-label in pediatrics

Investigations

- ASO titre — Streptococcal infection is commonest trigger in children
- Chest X-ray — Screen for sarcoidosis, TB, lymphoma
- ESR and CRP — Inflammatory markers; usually elevated
- Tuberculin skin test (Mantoux) — Rule out tuberculosis as trigger, especially in India
- Stool examination — If GI symptoms present; screen for IBD

Counselling

- Erythema nodosum usually resolves on its own in 2-6 weeks
- Rest legs as much as possible; elevate legs when sitting
- Take ibuprofen regularly with food for pain and inflammation
- The bumps may leave bruise-like marks that fade gradually - this is normal
- It is important to find the underlying cause - complete all prescribed blood tests

Referral Criteria

Refer to rheumatologist or dermatologist if recurrent EN, suspected underlying systemic disease (IBD, sarcoidosis, TB), or not responding to NSAIDs

Hospitalisation Criteria

Rarely needed; admit if unable to walk due to pain, suspected underlying serious condition (TB, lymphoma), or systemic toxicity

Notes

EN is usually self-limited (2-6 weeks); investigate underlying cause (streptococcal infection, IBD, sarcoidosis, TB). Treat the underlying condition when identified.

L53 — Other erythematous conditions

SNOMED: 247441003 | Duration: 7 days | Source: StatPearls 2024, AAP 2024

First-Line

- **Cetirizine** — 6-23 mo: 2.5 mg PO daily; 2-5 yr: 5 mg daily; >=6 yr: 10 mg daily; for itch/urticaria
- **Hydrocortisone 1% cream** — Apply to affected area BID for 7-14 days; for localized erythematous rash
- **Emollients** — Apply liberally BID-TID; soothe and protect inflamed skin

Second-Line

- **Prednisolone oral** — 1 mg/kg/day PO for 3-5 days; for severe drug-induced or toxic erythema
- **Diphenhydramine** — 1.25 mg/kg/dose PO Q6h (max 50 mg/dose); for acute symptomatic relief
- **Hydroxyzine** — 2 mg/kg/day PO divided Q6-8h (max 25 mg/dose); for persistent pruritus

Investigations

- Complete blood count with differential — Eosinophilia in drug reactions; atypical lymphocytes in viral
- Skin biopsy — If diagnosis uncertain or chronic erythema

Counselling

- Most erythematous rashes in children are self-limiting and resolve within 1-2 weeks
- Apply moisturiser and mild steroid cream as directed for itch and redness
- If caused by a medicine, stop the medicine and inform your doctor
- Take photos of the rash progression to show the doctor at follow-up
- Seek urgent care if rash spreads rapidly, blisters form, or child develops fever

Referral Criteria

Refer to dermatologist if diagnosis uncertain, rash persisting >2 weeks, suspected drug reaction, or mucosal involvement

Hospitalisation Criteria

Admit if suspected severe drug eruption (SJS/TEN), widespread blistering, mucosal involvement, or systemic toxicity

Notes

Identify and discontinue causative drug or irritant if toxic erythema is suspected. Most cases resolve spontaneously once the trigger is removed.

L70 — Acne

SNOMED: 11381005 | Duration: 84 days | Source: AAD 2024

First-Line

- **Benzoyl peroxide 2.5-5%** — Apply thin layer to affected area daily; start low concentration and titrate; reduces resistance
- **Adapalene 0.1% gel** — Apply pea-sized amount to face at bedtime; first-line retinoid; OTC available
- **Clindamycin 1% + BP 5% gel** — Apply to affected area daily; combination preferred over antibiotic monotherapy

Second-Line

- **Doxycycline** — Children >8 yr & >45 kg: 50-100 mg PO daily-BID; <=45 kg: 2.2 mg/kg BID; 8-12 wk course
- **Sarecycline** — 33-54 kg: 60 mg PO daily; 55-84 kg: 100 mg daily; 85-136 kg: 150 mg daily **[NEW 2024-25]**
- **Isotretinoin** — 0.5-1 mg/kg/day PO divided BID for 15-20 weeks; for severe/refractory acne; iPLEDGE req
- **Spironolactone** — Start 50 mg PO daily, titrate to 100-200 mg daily; for hormonal acne in adolescent females

Investigations

- Serum lipid profile — Before and during isotretinoin therapy
- Liver function tests — Before and monthly during isotretinoin therapy
- Pregnancy test (urine hCG) — Mandatory before isotretinoin in females of childbearing potential [STAT]
- Total and free testosterone — If suspected hormonal cause (hirsutism, irregular periods) in adolescent females

Counselling

- Wash face gently twice daily with mild cleanser; do not scrub or pick at pimples
- Apply benzoyl peroxide and retinoid at bedtime on dry skin; start every other day and increase
- Skin may get slightly dry and irritated initially - this improves with regular moisturiser use
- Results take 6-8 weeks to show; continue treatment patiently
- Isotretinoin (if prescribed) requires strict pregnancy prevention in females and regular blood tests

Referral Criteria

Refer to dermatologist if moderate-severe acne not responding to topical therapy + oral antibiotic, severe nodular/cystic acne, acne scarring, or isotretinoin candidacy

Hospitalisation Criteria

Rarely needed; consider if acne fulminans (explosive onset with fever, joint pain, ulcerative acne)

Notes

Multimodal topical therapy is first-line per AAD 2024; always pair topical antibiotics with benzoyl peroxide to prevent resistance. Isotretinoin requires pregnancy prevention (iPLEDGE).

Guideline Changes

AAD 2024 now conditionally recommends sarecycline; emphasizes multimodal topical therapy

L73 — Follicular disorders

SNOMED: 13600006 | Duration: 10 days | Source: StatPearls 2024, AAP 2024

First-Line

- **Mupirocin 2% ointment** — Apply to affected follicles TID for 7-10 days; for superficial bacterial folliculitis
- **Clindamycin 1% lotion** — Apply to affected area BID for 7-14 days; for localized bacterial folliculitis
- **Warm compresses** — Apply for 15-20 min TID-QID to promote drainage; supportive measure for mild cases

Second-Line

- **Cephalexin** — 25-50 mg/kg/day PO divided Q6-8h (max 500 mg/dose) for 7-10 days; deep folliculitis
- **Doxycycline** — Children >8 yr: 2.2 mg/kg/dose PO BID (max 100 mg/dose) for 7-10 days
- **Ketoconazole 2% cream** — Apply BID for 2-4 weeks; for Malassezia (pityrosporum) folliculitis

Investigations

- Wound swab culture and sensitivity — Recurrent or treatment-resistant folliculitis
- KOH preparation — If pityrosporum (Malassezia) folliculitis suspected
- Blood glucose — If recurrent bacterial skin infections (screen for diabetes)

Counselling

- Keep the affected area clean and dry; wash with mild antibacterial soap
- Do not squeeze or pick at the bumps as this can spread infection
- Wear loose-fitting cotton clothing to reduce friction and sweating
- Use separate towels from other family members until infection clears
- If bumps keep coming back, see doctor for culture to check if MRSA is the cause

Referral Criteria

Refer to dermatologist if recurrent folliculitis despite treatment, suspected pityrosporum folliculitis, or deep folliculitis/furuncles needing drainage

Hospitalisation Criteria

Rarely needed; consider if carbuncle/large abscess requiring surgical drainage, systemic toxicity, or immunocompromised child

Notes

Most mild folliculitis resolves spontaneously with hygiene measures. Culture swab if MRSA suspected; TMP-SMX or clindamycin may be needed.

L74 — Miliaria

SNOMED: 63951004 | Duration: 3 days | Source: AAP/HealthyChildren 2024, Mayo Clinic 2024

First-Line

- **Cooling measures** — Move child to cool environment; remove excess clothing; allow skin to air dry
- **Calamine lotion** — Apply to affected area 3-4x daily (avoid face/genitals); for children >2 yr; soothes itch
- **Lukewarm baths** — Bathe every 2-3 hours without soap; allow air drying; reduces skin temperature

Second-Line

- **Hydrocortisone 0.5-1% cream** — Apply to affected area BID for 3-5 days if significant inflammation/itch persists
- **Anhydrous lanolin** — Apply thin layer to affected area to prevent ductal obstruction in recurrent miliaria

Counselling

- Move child to a cool, well-ventilated room - this is the most important treatment
- Dress in loose, light cotton clothing; avoid overdressing the baby
- Keep room temperature comfortable; use fan or air conditioner if available
- Bathe with lukewarm water 2-3 times daily without soap; pat dry gently
- Calamine lotion can be applied for comfort; no other medicines are usually needed

Referral Criteria

Referral usually not needed; refer to dermatologist if miliaria profunda (deep, persistent variant) or recurrent severe episodes

Hospitalisation Criteria

Hospitalisation not needed for miliaria; consider only if secondary bacterial infection (miliaria pustulosa) with systemic symptoms

Notes

Miliaria resolves in 2-3 days with cooling alone; pharmacotherapy rarely needed. Ensure lightweight cotton clothing and adequate ventilation in warm environments.

L81 — Other disorders of pigmentation

SNOMED: 190302002 | Duration: 180 days | Source: JAMA Derm 2024, AAD 2024

First-Line

- **Tacrolimus 0.1% ointment** — Apply to depigmented areas BID for >=6 months; first-line for facial vitiligo; >=2 yr
- **Pimecrolimus 1% cream** — Apply to affected area BID; first-line alternative TCI for vitiligo; >=2 yr
- **Mometasone 0.1% cream** — Apply to depigmented patches daily for 3-6 months; medium-potency TCS for trunk/limbs

Second-Line

- **Ruxolitinib 1.5% cream** — Apply to affected area BID (max 10% BSA); >=12 yr; topical JAK inhibitor for vitiligo **[NEW 2024-25]**
- **NB-UVB phototherapy** — 2-3x/week for 6-12 months; combine with topical TCI/TCS for enhanced repigmentation
- **Hydroquinone 2-4% cream** — Apply to hyperpigmented area BID for 8-12 weeks; for post-inflammatory hyperpigmentation
- **Azelaic acid 15-20% cream** — Apply BID for 3-6 months; for melasma/PIH; well tolerated in children

Investigations

- Thyroid function tests — Vitiligo associated with autoimmune thyroiditis
- CBC with differential — Screen for pernicious anaemia (vitamin B12 deficiency)
- Wood lamp examination — Enhances depigmented patches for diagnosis and extent mapping
- Anti-TPO antibodies — Autoimmune thyroid screening in vitiligo patients

Counselling

- Vitiligo treatment takes 6-12 months to show results - be patient and consistent
- Apply topical cream as directed; combine with sun exposure (10-15 min) for better results
- Use sunscreen on depigmented areas to prevent sunburn and protect from further damage
- Vitiligo is not contagious and does not cause physical harm
- Emotional and psychological support for the child is very important; counselling may help

Referral Criteria

Refer to dermatologist for rapidly spreading vitiligo, facial/acral involvement, or consideration for phototherapy/systemic therapy; refer to ophthalmologist for associated eye involvement

Hospitalisation Criteria

Hospitalisation not needed for pigmentation disorders

Notes

Early aggressive treatment of vitiligo in children (<12 yr onset) yields better outcomes. Topical calcineurin inhibitors are preferred over steroids for facial lesions due to lower atrophy risk.

Guideline Changes

Ruxolitinib 1.5% cream (Opzelura) approved for vitiligo >=12 yr in 2022

Q84 — Congenital malformations of skin appendages

SNOMED: 205845004 | Duration: 30 days | Source: StatPearls 2024, AAP 2024

First-Line

- **Topical antibacterial ointment** — Apply to small aplasia cutis lesions BID until healed; mupirocin 2% or bacitracin preferred
- **Wound care/dressings** — Daily gentle cleansing with saline, petroleum jelly, and non-adherent dressing for ACC

Second-Line

- **Surgical excision** — For accessory skin appendages, dermoid cysts, or large ACC lesions (>4 cm); refer surgery
- **Minoxidil 2% solution** — Apply to scalp BID for localized congenital alopecia; off-label; limited pediatric evidence

Investigations

- Ultrasound of lesion — Assess depth and underlying structures
- MRI (scalp lesions) — Rule out connection to sagittal sinus in large scalp aplasia cutis [URGENT]
- Genetic evaluation — If suspected syndromic association (e.g., Adams-Oliver syndrome)

Counselling

- Most congenital skin appendage anomalies are cosmetic and managed conservatively
- Keep the area clean and watch for signs of infection (redness, warmth, pus)
- For aplasia cutis: keep wound clean and covered; most small defects heal without surgery
- Some skin anomalies may be associated with underlying conditions - follow-up blood tests may be needed
- Surgical removal of accessory appendages can be done electively when child is older

Referral Criteria

Refer to pediatric dermatologist/plastic surgeon if large aplasia cutis (>4 cm), aplasia cutis with exposed deep structures, cosmetically significant lesions, or suspected underlying syndrome

Hospitalisation Criteria

Admit if large scalp aplasia cutis with sagittal sinus exposure, significant bleeding, secondary infection requiring IV antibiotics, or associated systemic anomalies

Notes

Most congenital skin appendage malformations are managed conservatively or surgically. Large aplasia cutis defects with exposed dura require urgent neurosurgical consultation.

Surgical

K40 — Inguinal hernia

SNOMED: 396232000 | Duration: 1 days | Source: AAP 2023 (Pediatrics), EUPSA Evidence & Guideline Committee 2022

First-Line

- **Open herniotomy** — High ligation of processus vaginalis; day surgery; inguinal incision; gold standard technique
- **Laparoscopic hernia repair** — Equivalent outcomes to open; allows contralateral ring evaluation; shorter bilateral OR time
- **Urgent surgical repair** — Incarcerated hernia: attempt gentle reduction then semi-urgent repair within 24-48 hrs

Second-Line

- **Contralateral exploration** — Case-by-case decision; laparoscopic visualization of contralateral ring preferred over blind exploration
- **Manual reduction + delayed repair** — Sedation + Trendelenburg for incarcerated hernia; schedule repair within 2-5 days after reduction

Investigations

- Clinical diagnosis (no routine investigations) — Diagnosis is clinical by examination
- Ultrasound (inguinal) — If diagnosis uncertain (hydrocele vs hernia)
- Pre-operative blood work — CBC before surgery

Counselling

- Inguinal hernia in children always needs surgery as it will not close on its own
- Surgery is safe and usually done as day care; your child goes home the same day
- If the lump becomes hard, painful, and does not go back in, rush to hospital immediately as it could be stuck
- After surgery, avoid rough play for 1-2 weeks; normal activities can resume gradually
- Slight swelling around the wound is normal; fever or redness around the wound needs medical attention

Referral Criteria

Refer to pediatric surgeon at diagnosis for elective repair; urgent referral for irreducible hernia

Hospitalisation Criteria

Day surgery for elective repair; admit for incarcerated hernia (urgent reduction and repair within 24-48h), or if premature infant needing observation post-anesthesia

Notes

Pediatric inguinal hernias do not close spontaneously and always require surgical repair. Premature infants have higher incidence and should be repaired before NICU discharge or shortly after.

Guideline Changes

AAP 2023 guideline: laparoscopic approach equivalent to open; allows contralateral ring evaluation

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K42 — Umbilical hernia

SNOMED: 396347007 | Duration: 1 days | Source: AAFP/AAP Choosing Wisely, ACS Pediatric Surgery Guidelines

First-Line

- **Watchful waiting** — Observe until age 4-5 yrs; 95% of defects <1cm close spontaneously; no strapping or coins
- **Umbilical hernia repair** — Suture repair (Mayo technique) if persistent after age 5 or symptomatic; day surgery; infraumbilical incision

Second-Line

- **Urgent surgical repair** — Immediate repair for strangulated hernia; resuscitate, broad-spectrum antibiotics, emergent OR
- **Early repair consideration** — May consider repair before age 5 for large defects >2cm, proboscoid hernias, or symptomatic cases

Investigations

- Clinical diagnosis — No investigations needed; diagnosis is clinical
- Ultrasound (abdomen) — Only if associated anomalies suspected

Counselling

- Umbilical hernia is very common in babies and almost always closes by itself by age 4-5 years
- Do NOT tape coins or apply any binding on the hernia - this does not help and can cause skin problems
- If the hernia becomes hard, painful, or changes color, rush to hospital immediately
- Surgery is only needed if it has not closed by age 5 or if it is very large
- Avoid lifting heavy objects for 2 weeks after surgery if done

Referral Criteria

Refer to pediatric surgeon if hernia persists after age 4-5 years, defect >2 cm, symptomatic, or proboscoid hernia

Hospitalisation Criteria

Day surgery for elective repair; emergency admission for strangulated hernia (very rare)

Notes

Umbilical hernias are extremely common in infants, especially in African American and preterm infants. Avoid referral to surgery before age 4-5 unless symptomatic or incarcerated.

Guideline Changes

Choosing Wisely recommends deferring surgical referral until age 4-5 years for asymptomatic cases

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K60 — Anal fissure and fistula

SNOMED: 197150008 | Duration: 42 days | Source: ASCRS 2025, APSA/Pediatric Surgery NaT

First-Line

- **Conservative management** — Stool softeners, high-fiber diet, sitz baths after each BM; topical glyceryl trinitrate 0.2% ointment BID x6-8 wks
- **Fistulotomy** — Simple fistulotomy for persistent fistula-in-ano; sphincter-sparing in children; heals by secondary intention
- **Incision & drainage** — For perianal abscess: bedside or OR I&D under GA; pack wound loosely; sitz baths post-op

Second-Line

- **Topical diltiazem 2%** — Applied BID to anal margin for chronic fissure unresponsive to GTN; fewer headaches than GTN
- **Botulinum toxin injection** — Injection into internal anal sphincter for refractory chronic fissure; used off-label in older children
- **Fistulectomy** — Complete excision of fistula tract for complex or recurrent fistulae; rarely needed in children

Investigations

- Clinical examination — Diagnosis is usually clinical
- Examination under anesthesia — If complex perianal disease or suspected Crohn
- MRI pelvis — If fistula-in-ano or suspected Crohn perianal disease

Counselling

- Anal fissures are common in children and usually heal with stool softeners and proper diet

- Give high-fiber foods (fruits, vegetables, whole grains) and plenty of water to prevent hard stools
- Apply the prescribed ointment gently around the anal area as directed
- Sitz baths (sitting in warm water for 10-15 minutes) after each bowel movement help healing
- Do not delay bowel movements - encourage the child to go when they feel the urge

Referral Criteria

Refer to pediatric surgeon if fissure not healing after 6-8 weeks of conservative treatment, recurrent fissures, or if fistula present

Hospitalisation Criteria

Admit for perianal abscess requiring I&D under anesthesia, or if Crohn disease suspected with complex perianal disease

Notes

Most pediatric perianal fistulae are superficial and may resolve spontaneously without sphincter involvement. Exclude Crohn disease in any child with recurrent or complex perianal disease.

Guideline Changes

2025 systematic review supports initial surgical intervention for recurrent pediatric perianal fistula

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Q35 — Cleft palate

SNOMED: 87979003 | Duration: 365 days | Source: ACPA 2024 Parameters of Care, NEJM 2023 Timing Study

First-Line

- **Palatoplasty** — Primary palate repair at 9-12 months; Furlow or von Langenbeck technique; before speech development
- **Feeding support** — Palatal obturator if needed; specialized bottles; upright positioning; dietitian monitoring growth
- **Myringotomy with tubes** — PE tubes at time of palate repair or earlier for chronic OME; audiometry follow-up q6mo

Second-Line

- **Speech therapy** — Begin by age 12-18 mo post-repair; target velopharyngeal competence; nasometry assessment at age 3-4
- **Pharyngeal flap/sphincter pharyngoplasty** — Secondary surgery for velopharyngeal insufficiency if persistent after age 4; improves speech
- **Alveolar bone graft** — Mixed dentition stage (age 7-11); iliac crest bone graft to alveolar cleft for canine eruption
- **Palatal fistula repair** — For symptomatic oronasal fistulae post-palatoplasty; local flaps or obturator

Investigations

- Audiological assessment — Baseline hearing before palate repair
- Pre-operative blood work — CBC, coagulation before palatoplasty
- Tympanometry — Assess middle ear effusion
- Nasometry/nasal emission test — Post-surgery speech assessment at age 3-4

Counselling

- Cleft palate repair surgery is done around 9-12 months to help with speech development
- Use special feeding bottles and hold baby upright during feeds to prevent milk going into the nose
- Ear infections are common; your child will need regular hearing tests
- Speech therapy after surgery is very important for clear speech
- Multiple specialists will see your child as part of the cleft team over many years

Referral Criteria

Refer to multidisciplinary cleft team at birth: plastic surgery, ENT, audiology, speech-language pathology, orthodontics

Hospitalisation Criteria

Admit for palatoplasty, PE tube insertion, and any secondary surgeries (pharyngeal flap, bone graft)

Notes

Timing of palate repair balances speech outcomes (earlier is better) against maxillary growth (later preserves growth). Interdisciplinary cleft team follow-up is required through adolescence.

Guideline Changes

NEJM 2023 RCT showed 6-month palate repair noninferior to 12-month repair for speech outcomes

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Q36 — Cleft lip

SNOMED: 80281008 | Duration: 365 days | Source: ACPA 2024 Parameters of Care, WHO Smile Train Guidelines

First-Line

- **Primary lip repair** — Millard or Fisher technique at 3-6 months; rule of 10s (10 wks, 10 lb, Hb 10g/dL); under GA
- **Nasoalveolar molding (NAM)** — Pre-surgical orthodontic device from 1-2 wks of age; narrows cleft, molds nose; improves surgical result
- **Feeding support** — Specialized bottles (Haberman, Dr. Brown's); lactation consultant; upright feeding position

Second-Line

- **Lip adhesion** — Preliminary procedure at 2-4 wks for wide clefts to approximate lip segments before definitive repair
- **Secondary rhinoplasty** — Nasal revision at age 4-6 or after skeletal maturity; addresses residual nasal asymmetry
- **Lip revision surgery** — Scar revision and vermilion adjustment if needed; typically deferred until school age or later

Investigations

- Feeding assessment — Ensure adequate nutrition before surgery [URGENT]
- Pre-operative blood work (CBC, coagulation) — Before surgical repair
- Echocardiography — Rule out associated cardiac defects if syndromic

Counselling

- Cleft lip is a common birth defect that is fully correctable with surgery
- Surgery to repair the lip is usually done around 3-6 months of age
- Use specialized feeding bottles as shown; your baby can feed well with the right technique
- Nasoalveolar molding (NAM) device before surgery gives better results if started early
- Your child will need follow-up with the cleft team for several years but the results are excellent

Referral Criteria

Refer to multidisciplinary cleft team at birth: plastic surgery, ENT, speech therapy, orthodontics, audiology

Hospitalisation Criteria

Admit for lip repair surgery (day surgery or 1-2 nights), and for any revision surgeries

Notes

Multidisciplinary cleft team management from birth through adulthood is the standard of care. NAM if started within first 2 weeks significantly improves primary surgical outcomes.

Guideline Changes

2024 ACPA update emphasizes nasoalveolar molding (NAM) pre-surgical orthodontics

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Q37 — Cleft palate with cleft lip

SNOMED: 66948001 | Duration: 365 days | Source: ACPA 2024 Parameters of Care, WHO/Smile Train Protocol

First-Line

- **Staged surgical repair** — Lip repair at 3-6 mo, palate repair at 9-12 mo; NAM from birth if available; PE tubes at palatoplasty
- **Nasoalveolar molding** — Custom appliance from 1-2 wks; molds cleft lip, alveolus, and nose pre-surgically; improves symmetry
- **Feeding management** — Specialized bottles; upright positioning; weight monitoring; supplemental NG feeds if growth faltering
- **Audiological care** — ABR at birth; tympanometry regularly; PE tubes for chronic OME; hearing aids if sensorineural loss

Second-Line

- **Alveolar bone graft** — At mixed dentition (age 7-11); iliac crest autograft; enables canine eruption and stabilizes arch
- **Secondary speech surgery** — Pharyngeal flap or sphincter pharyngoplasty for VPI; after age 4 if speech therapy insufficient
- **Orthognathic surgery** — Le Fort I advancement at skeletal maturity (age 16-18) for midface hypoplasia; orthodontics pre/post
- **Definitive rhinoplasty** — Final nasal reconstruction after skeletal maturity; septorhinoplasty for residual deformity

Investigations

- Pre-operative blood work (for each surgery) — CBC, coagulation screen
- Audiological assessment — Baseline and regular monitoring
- Echocardiography — If associated syndrome suspected (Pierre Robin, velocardiofacial)
- Genetic evaluation — If dysmorphic features suggest a syndrome

Counselling

- Your baby has both cleft lip and palate which will need staged surgeries over several years
- Lip repair is done around 3-6 months and palate repair around 9-12 months
- Feeding support with special bottles is needed from birth - your cleft nurse will teach you
- Your child will need speech therapy, dental care, and hearing checks throughout childhood
- With proper care, children with cleft lip and palate grow up to live completely normal lives

Referral Criteria

Refer to multidisciplinary cleft team at birth; ENT, audiology, speech therapy, orthodontics, and psychosocial support

Hospitalisation Criteria

Admit for staged surgical repairs (lip, palate, bone graft, secondary procedures), PE tube insertion, and any revision surgeries

Notes

Combined cleft lip and palate requires the most complex staged surgical protocol with 5+ procedures over 18 years. Multidisciplinary team follow-up from birth through adulthood is mandatory.

Guideline Changes

2024 ACPA emphasizes integrated NAM + staged surgical protocol; earlier palate repair gaining evidence

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Q65 — Congenital deformities of hip

SNOMED: 52781008 | Duration: 180 days | Source: AAP 2024, AAOS DDH Quality Measure, BSCOS 2023 Consensus

First-Line

- **Pavlik harness** — First-line for DDH diagnosed 0-6 months; 95% success in dysplasia/subluxation, 80% in dislocation
- **Ultrasound screening** — Graf method US at 4-6 wks for risk factors or clinical concern; universal screening under debate
- **Closed reduction + spica cast** — Under GA with arthrogram for failed Pavlik or late diagnosis (6-18 mo); hip spica 12 wks

Second-Line

- **Open reduction** — For failed closed reduction or age >18 mo; medial or anterior approach; followed by spica cast
- **Femoral/pelvic osteotomy** — Salter innominate or femoral shortening osteotomy for residual dysplasia in older children
- **Abduction brace/splint** — Rhino or Tubingen splint alternative to Pavlik for mild dysplasia; ensures hip in flexion/abduction

Investigations

- Hip ultrasound (Graf method) — Primary screening and monitoring tool for infants <6 months [URGENT]
- Hip X-ray (AP pelvis) — Assessment after age 4-6 months when ossification visible
- MRI hip — If closed reduction performed to confirm hip position in cast

Counselling

- DDH means the hip joint is not properly formed; early treatment gives the best results
- The Pavlik harness must be worn full-time as directed - do not remove it for bathing unless told
- Keep all follow-up ultrasound appointments to check if the hip is developing properly
- Double or triple diapering does NOT treat DDH and should not be relied upon
- If started early (before 6 months), most babies recover completely with just the harness

Referral Criteria

Refer to pediatric orthopedic surgeon for all confirmed DDH; urgent referral for dislocated hip

Hospitalisation Criteria

Admit for closed reduction under anesthesia and spica casting, open reduction surgery, or osteotomy procedures

Notes

Early detection and treatment is critical: Pavlik harness before 6 months yields best outcomes. Avascular necrosis of the femoral head is the most serious complication of treatment.

Guideline Changes

2024 evidence increasingly supports universal ultrasound screening over selective screening

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Q66 — Congenital deformities of feet

SNOMED: 397932003 | Duration: 365 days | Source: AAP 2022 Clinical Report (Pediatrics), POSNA/Ponseti International

First-Line

- **Ponseti serial casting** — Weekly long-leg casts from birth; correct cavus, adductus, varus, then equinus; typically 5-7 casts
- **Percutaneous Achilles tenotomy** — Performed after casting if equinus persists (>85% of cases); outpatient under local anesthesia

- **Foot abduction brace** — Boots-and-bar orthosis 23 hrs/day x3 mo, then nighttime until age 5; critical for preventing relapse

Second-Line

- **French physiotherapy method** — Daily stretching, taping, continuous passive motion; alternative to casting in select centers
- **Tibialis anterior tendon transfer** — For relapsed clubfoot with dynamic supination; transfer to lateral cuneiform; after age 2.5-3
- **Posteromedial release** — Extensive surgical release reserved for severe resistant cases; higher complication rate than Ponseti
- **Metatarsus adductus stretching** — Gentle serial stretching and reverse-last shoes for non-clubfoot forefoot adduction; resolves in most

Investigations

- Clinical diagnosis — Clubfoot is diagnosed at birth by physical exam
- X-ray foot (weight-bearing, after correction) — Assess alignment after treatment completion
- Ultrasound hip — Screen for associated DDH

Counselling

- Clubfoot is very treatable with the Ponseti method of weekly plaster casts starting from birth
- Most babies need 5-7 weekly casts followed by a small procedure to lengthen the Achilles tendon
- The brace (boots and bar) must be worn 23 hours daily for 3 months then at night until age 5
- Brace compliance is the single most important factor in preventing recurrence
- Your child will walk and run normally if treatment is completed properly

Referral Criteria

Refer to pediatric orthopedic surgeon experienced in Ponseti method at birth; re-refer if relapse occurs

Hospitalisation Criteria

Admit only for Achilles tenotomy (day procedure) or for major reconstructive surgery if needed for relapsed/resistant clubfoot

Notes

The Ponseti method is the worldwide standard for clubfoot with 95% success rate when bracing compliance is maintained. Bracing noncompliance is the leading cause of recurrence.

Guideline Changes

2022 AAP report reinforces Ponseti method as definitive first-line with 95% success rate

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Q67 — Congenital musculoskeletal deformities of skull, face, spine and chest

SNOMED: 205117007 | Duration: 180 days | Source: APTA Pediatrics 2024 CPG (Torticollis), AAP 2023

First-Line

- **Physical therapy** — Manual stretching, positioning, tummy time from birth; 2024 APTA CPG: start PT within 1-3 months for best outcome **[NEW 2024-25]**
- **Caregiver education** — Home stretching program, environmental modifications, positional strategies to prevent plagiocephaly
- **Helmet therapy** — Custom cranial orthosis for positional plagiocephaly; start 4-6 mo, wear 23 hrs/day for 3-6 months

Second-Line

- **Botulinum toxin injection** — OnabotulinumtoxinA into SCM for refractory CMT; used as adjunct to stretching in older infants

- **SCM surgical release** — Unipolar or bipolar SCM release for persistent torticollis after >1 yr of PT; improves ROM
- **Bracing for pectus/scoliosis** — TLSO brace for congenital scoliosis monitoring; vacuum bell or Nuss bar for pectus excavatum
- **Spinal surgery** — Hemivertebra excision or growing rods for progressive congenital scoliosis; individualized timing

Investigations

- Neck ultrasound — Assess SCM muscle for fibromatosis in torticollis
- Cervical spine X-ray — Rule out skeletal cause of torticollis (hemivertebra)
- Spinal X-ray (full spine PA) — Baseline for congenital scoliosis; serial monitoring

Counselling

- Congenital torticollis (head tilt) responds very well to physiotherapy started early
- Do daily stretching exercises at home as shown by the physiotherapist
- Encourage tummy time and turning head to both sides during play
- A molding helmet may be needed if the head shape is significantly flattened on one side
- Most cases resolve completely with consistent physiotherapy within the first year

Referral Criteria

Refer to pediatric physiotherapist immediately; orthopedic/neurosurgery referral for congenital scoliosis or if torticollis not improving after 6 months of PT

Hospitalisation Criteria

Admit for SCM surgical release if torticollis persistent after >1 year of PT, spinal surgery for progressive congenital scoliosis, or pectus repair

Notes

Congenital muscular torticollis resolves in >90% with early PT initiated before age 3 months. Congenital vertebral anomalies require serial imaging to monitor curve progression.

Guideline Changes

2024 APTA published first evidence-based CPG for congenital muscular torticollis physical therapy

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Q69 — Polydactyly

SNOMED: 367506006 | Duration: 1 days | Source: ASSH/POSNA Guidelines, Current Opinion Pediatrics 2023

First-Line

- **Surgical excision** — Optimal age 6 mo-3 yrs; reconstruct collateral ligaments and tendons; preserve best-functioning digit
- **Suture ligation (type B postaxial)** — Office suture ligation within first 3 months for pedunculated postaxial polydactyly; simple and safe

Second-Line

- **Reconstructive surgery** — Bilhaut-Cloquet procedure for thumb (preaxial) polydactyly; combines two digits into one functional thumb
- **Delayed surgery** — Defer to age 1+ for complex preaxial polydactyly to allow ossification for accurate anatomic assessment
- **Foot polydactyly excision** — Surgery at ~1 yr before walking; excise border toe preserving metatarsal alignment; minimize gait impact

Investigations

- X-ray of affected hand/foot — Assess bony anatomy and joint involvement
- Clinical photography — Document pre-operative appearance

Counselling

- Extra finger or toe is a common birth variation that is corrected with surgery
- For a skin-tag type extra finger, simple removal can be done very early
- For bony extra digits, surgery is planned around 6-12 months for best results
- After surgery, hand therapy may be needed to ensure good function
- The cosmetic and functional results are usually excellent

Referral Criteria

Refer to pediatric hand surgeon for preaxial polydactyly; pediatric orthopedics for postaxial; genetics if other anomalies present

Hospitalisation Criteria

Day surgery for excision; observation may be needed for neonatal suture ligation

Notes

Surgical timing should be individualized based on polydactyly type, location, and complexity. Earlier surgery (before age 3) yields better postoperative finger shaping and functional recovery.

Guideline Changes

Recent evidence favors surgery at 6 mo-3 yrs for better functional and cosmetic outcomes vs. age 4-6

Q70 — Syndactyly

SNOMED: 373413006 | Duration: 1 days | Source: ASSH Guidelines, Emedicine/Medscape 2024 Update

First-Line

- **Surgical web space release** — Release at 6-24 mo; dorsal flap commissuroplasty with full-thickness skin grafts; staged if adjacent webs
- **Early release for border digits** — Prioritize thumb-index and ring-small web spaces first (by 6 mo) to prevent growth deviation

Second-Line

- **Graftless release techniques** — Local flap designs avoiding skin graft donor site; suitable for simple incomplete syndactyly **[NEW 2024-25]**
- **Staged bilateral release** — Never release both sides of same digit simultaneously; 3+ months between stages to protect vascularity
- **Complex syndactyly reconstruction** — Bony separation, tendon/nerve reconstruction for osseous syndactyly; higher complication rate
- **Foot syndactyly observation** — Toe syndactyly often does not require surgery unless functional impairment or shoe-fitting problems

Investigations

- X-ray of affected hand/foot — Determine if simple (skin) or complex (bony) syndactyly
- Genetic evaluation — If bilateral, multiple limbs, or associated anomalies (Apert, Poland)

Counselling

- Syndactyly (joined fingers/toes) is a common birth variation corrected with surgery
- Surgery is planned between 6-24 months of age for hand syndactyly
- If adjacent web spaces are affected, surgery is done in stages to protect blood supply
- Hand therapy after surgery helps achieve the best function
- Toe syndactyly often does not need surgery unless it causes problems with shoes or walking

Referral Criteria

Refer to pediatric hand surgeon at birth for hand syndactyly; orthopedics for foot syndactyly; genetics if syndromic

Hospitalisation Criteria

Day surgery or 1-night stay for web space release; staged admissions if multiple web spaces involved

Notes

Release before age 2 prevents angular deformity in digits of unequal length. Staged surgery is mandatory when adjacent web spaces are involved to protect digit blood supply.

Guideline Changes

Graftless syndactyly release techniques gaining adoption, reducing donor site morbidity

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Q71 — Reduction defects of upper limb

SNOMED: 205955001 | Duration: 365 days | Source: AAPM&R Pediatric Limb Deficiency, JPOSNA Prosthetics Essentials

First-Line

- **Passive prosthesis** — Fit at sitting age (~6 mo); maintains arm length symmetry; supports bimanual tasks and balance
- **Body-powered prosthesis** — Introduce at age 2-3 with prehensor; harness-operated; durable for active children
- **Occupational therapy** — Bimanual training, adaptive techniques, fine motor skills; ongoing with prosthetic transitions

Second-Line

- **Myoelectric prosthesis** — Sensor-controlled powered hand; fit from age 2-3; multi-articulated digits available (i-Limb, Michelangelo) **[NEW 2024-25]**
- **Surgical pollicization** — Index finger transfer to thumb position for radial club hand or thumb aplasia; age 6-18 months
- **Limb lengthening** — Distraction osteogenesis for partial defects (e.g., short forearm); case-by-case basis
- **No prosthesis approach** — Some children function well without prosthesis using adaptive techniques; reassess every 2-3 years

Investigations

- X-ray of affected limb — Characterize skeletal anatomy
- Genetic evaluation — If bilateral or syndromic features
- Renal ultrasound — Screen for associated renal anomalies (radial deficiency)

Counselling

- Your child has an upper limb difference - many children do remarkably well with or without a prosthesis
- A prosthetic hand can be fitted as early as 6 months of age
- Occupational therapy helps your child develop skills for daily activities
- Let your child choose whether to use a prosthesis as they grow older
- Children with upper limb differences often develop excellent compensatory skills

Referral Criteria

Refer to pediatric orthopedic/hand surgeon, prosthetics team, and occupational therapist at birth

Hospitalisation Criteria

Admit for surgical reconstruction (pollicization, limb lengthening), or prosthetic fitting under sedation in young children

Notes

Many children with unilateral upper limb deficiency adapt remarkably well with or without prostheses. Prosthetic choice should prioritize function and the child's own preferences as they develop.

Guideline Changes

Multi-articulated myoelectric prostheses (i-Limb) now available for children with gesture control software

Q72 — Reduction defects of lower limb

SNOMED: 205955001 | Duration: 365 days | Source: AAPM&R Pediatric Limb Deficiency, JPOSNA 2024 Prosthetics Review

First-Line

- **Prosthetic fitting** — Fit when child pulls to stand (~9-12 mo); update annually during growth; multidisciplinary team approach
- **Physical therapy** — Gait training, strength, balance; adaptive equipment; ongoing as prosthetic needs change with growth
- **Surgical reconstruction** — Limb lengthening (Ilizarov or IM nail) for partial defects; individualized based on defect severity

Second-Line

- **Intramedullary nail lengthening** — Newer motorized IM nails (PRECICE) for older children; less external hardware; standard procedure now **[NEW 2024-25]**
- **Syme or Boyd amputation** — Ankle disarticulation for severe fibular hemimelia; enables prosthetic fitting with end-bearing stump
- **Rotationplasty (Van Nes)** — For proximal femoral focal deficiency; rotates foot 180 degrees to serve as biological knee joint
- **Orthotic management** — AFO or KAFO for mild defects; shoe lifts for leg length discrepancy <2cm

Investigations

- X-ray of affected limb — Characterize skeletal deficiency
- MRI of affected limb — Assess soft tissue and joint anatomy for surgical planning
- Genetic evaluation — If bilateral or syndromic presentation

Counselling

- Your child has a limb difference that can be managed very well with modern prosthetics and therapy
- Prosthetic fitting starts early (around 9-12 months) when the child begins to stand
- Regular prosthetic updates are needed as your child grows
- Children with limb differences adapt remarkably well and can participate in sports and activities
- Connect with other families through limb difference support groups for shared experience

Referral Criteria

Refer to pediatric orthopedic surgeon and rehabilitation medicine at birth; prosthetics and orthotics team

Hospitalisation Criteria

Admit for limb lengthening surgery, amputation/revision surgery, or reconstructive procedures

Notes

Decision between limb reconstruction and amputation with prosthetic fitting must involve family and multidisciplinary team. Prosthetic technology advances rapidly, requiring frequent reassessment.

Guideline Changes

2024 advances in intramedullary nail lengthening and extramedullary implantable lengthening devices

Q75 — Congenital malformations of skull and face bones

SNOMED: 205845004 | Duration: 365 days | Source: AAP/ACPA Craniosynostosis Guidelines 2020 (updated), Children's National

First-Line

- **Cranial vault remodeling** — Open surgery at 6-12 months for single-suture craniosynostosis; fronto-orbital advancement for coronal

- **Endoscopic strip craniectomy** — Minimally invasive option for infants <3-4 months; followed by helmet therapy for 6-12 months
- **Tranexamic acid** — IV TXA 10-15 mg/kg bolus then infusion intra-op; reduces blood loss during cranial vault surgery

Second-Line

- **Helmet therapy** — Post-endoscopic or for positional plagiocephaly; custom molding helmet worn 23 hrs/day for 6-12 mo
- **Staged midface advancement** — Le Fort III or monobloc distraction for syndromic craniosynostosis (Apert, Crouzon); age 4-8
- **ICP monitoring** — For suspected raised ICP; papilledema screening; VP shunt if hydrocephalus develops
- **Distraction osteogenesis** — Mandibular or midface distraction for severe micrognathia/midface hypoplasia; external devices **[NEW 2024-25]**

Investigations

- CT skull with 3D reconstruction — Confirm craniosynostosis and plan surgery [URGENT]
- Ophthalmology exam (fundoscopy) — Screen for papilledema indicating raised ICP [URGENT]
- MRI brain — Assess brain if developmental concerns or syndromic
- Genetic testing — If syndromic craniosynostosis suspected (FGFR mutations)

Counselling

- Craniosynostosis means bones of the skull fused too early; surgery corrects the head shape and protects the brain
- The best time for surgery is usually before 6-12 months of age
- If your baby had endoscopic surgery, the helmet must be worn 23 hours a day for best results
- After surgery, head shape improves significantly over several months
- Some complex cases need more than one surgery as your child grows

Referral Criteria

Refer to pediatric neurosurgeon and craniofacial surgeon at diagnosis; ophthalmology for raised ICP screening

Hospitalisation Criteria

Admit for cranial vault surgery, ICU post-operatively, and for any staged craniofacial procedures

Notes

Best surgical outcomes for single-suture craniosynostosis occur before 6 months of age. Syndromic cases (Apert, Crouzon, Pfeiffer) require lifelong multidisciplinary craniofacial team care.

Guideline Changes

Minimally invasive endoscopic suturectomy gaining wider adoption for early-presenting cases

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Q78.0 — Osteogenesis imperfecta

SNOMED: 78314001 | Duration: 365 days | Source: OIF Guidelines, Calcified Tissue International 2024 Review

First-Line

- **IV pamidronate** — 0.5-1 mg/kg/day x3 days q2-4 mo; start in infancy for moderate-severe OI; improves BMD, reduces pain
- **IV zoledronic acid** — 0.025-0.05 mg/kg q6mo; alternative to pamidronate; fewer infusion days; monitor calcium post-infusion
- **Fracture management** — Gentle handling; lightweight splints/casts; minimize immobilization to prevent further bone loss
- **Physical therapy** — Low-impact PT/OT to build muscle strength and mobility; aquatic therapy; avoid contact sports

Second-Line

- **Oral risedronate** — 0.5-1 mg/kg/wk oral; less effective than IV bisphosphonates but avoids infusions; for milder types
- **Telescoping IM rods** — Fassier-Duval rods for recurrent long bone fractures; grow with child; reduce fracture frequency
- **Denosumab** — Anti-RANKL antibody; emerging use in children unresponsive to bisphosphonates; rebound risk on stopping **[NEW 2024-25]**
- **Dental care** — Screen for dentinogenesis imperfecta; preventive dental care; crowns for fragile teeth

Investigations

- DEXA bone density scan — Baseline and annual BMD monitoring
- Genetic testing (COL1A1/COL1A2) — Confirm OI type and guide prognosis
- Skeletal survey X-ray — Baseline assessment of fractures and deformities
- Serum calcium, phosphate, vitamin D, ALP — Assess bone metabolism
- Audiometry — Screen for hearing loss (type I OI)

Counselling

- Osteogenesis imperfecta (brittle bone disease) causes bones to break easily
- Handle your child gently but do not restrict all activity - supervised safe play builds strength
- Bisphosphonate infusions (given every few months) strengthen bones and reduce fractures
- Swimming and water therapy are excellent exercises that build strength without fracture risk
- Dental care is important as teeth may also be affected; see a dentist regularly

Referral Criteria

Managed by multidisciplinary team: pediatric endocrinologist, orthopedic surgeon, physiotherapist, geneticist, dentist

Hospitalisation Criteria

Admit for fracture management, bisphosphonate infusions (day admission), intramedullary rod surgery, or spinal surgery

Notes

Start bisphosphonate therapy as early as possible since effect on skeleton is greatest during growth. Multidisciplinary care including orthopedics, endocrine, PT, and genetics is essential.

Guideline Changes

2024 meta-analysis confirms IV bisphosphonates improve BMD; comparative trials still needed for optimal regimen

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Q87 — Other congenital malformation syndromes affecting multiple systems

SNOMED: 192956008 | Duration: 365 days | Source: ACMG 2023, BWS Consensus 2024, AAP PWS Health Supervision, Marfan/GeneReviews

First-Line

- **Genetic diagnosis and counseling** — CMA, targeted gene panels, or exome sequencing; confirm specific syndrome; family recurrence counseling
- **Multisystem screening** — Cardiac echo, renal US, ophthalmology, audiology, skeletal survey tailored to specific syndrome
- **Growth monitoring** — Plot on syndrome-specific growth charts if available; GH evaluation for short stature syndromes (e.g., PWS)
- **Developmental services** — Early intervention PT/OT/speech; behavioral support; educational planning and IEP from age 3

Second-Line

- **Tumor surveillance (BWS)** — Abdominal US q3mo until age 7 for Wilms tumor/hepatoblastoma; AFP levels q3mo until age 4
- **GH therapy (PWS)** — Somatropin 0.035 mg/kg/day; improves body composition, growth, cognition; exclude OSA before starting
- **Cardiac monitoring (Marfan)** — Annual echo for aortic root dilation; beta-blocker (atenolol/losartan) if root Z-score >3
- **Dietary management (PWS)** — Strict caloric control and supervision for hyperphagia; behavioral strategies; dietitian involvement

Investigations

- Genetic testing (CMA, gene panel, or exome) — Confirm specific syndrome diagnosis
- Echocardiography — Screen for cardiac defects [URGENT]
- Abdominal ultrasound — Screen for tumors (BWS) or organ anomalies
- Skeletal survey — If skeletal dysplasia or connective tissue disorder suspected
- Developmental assessment — Guide early intervention services

Counselling

- Your child has a genetic syndrome affecting multiple body systems that needs coordinated care
- Accurate genetic diagnosis is the first step to plan proper monitoring and treatment
- Multiple specialist appointments will be needed - keep a health folder to track everything
- Growth and development may be different but early intervention helps maximize potential
- Connect with syndrome-specific support groups for practical advice and emotional support

Referral Criteria

Refer to clinical geneticist for diagnosis; multidisciplinary team: cardiology, endocrinology, neurology, developmental pediatrics as indicated by specific syndrome

Hospitalisation Criteria

Admit based on syndrome-specific complications: cardiac surgery, tumor surveillance findings, metabolic crises, or acute illness

Notes

Q87 encompasses diverse syndromes requiring syndrome-specific management protocols. Accurate genetic diagnosis is the critical first step to guide appropriate surveillance and treatment.

Guideline Changes

2024 Italian BWS update expanded tumor surveillance; 2024 PWS guidance for transition to adult care published

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Q90 — Down syndrome

SNOMED: 41040004 | Duration: 365 days | Source: AAP 2022 (Pediatrics), Global Down Syndrome Foundation 2024

First-Line

- **Echocardiogram** — Screen all neonates for congenital heart defects (AVSD in ~40%); surgical repair as indicated
- **Thyroid screening** — TSH at birth, 6 mo, 12 mo, then annually; treat hypothyroidism with levothyroxine 10-15 mcg/kg/day
- **Early intervention** — PT/OT/speech from infancy; developmental assessment by 12 mo; IEP by age 3
- **Ophthalmology & audiology** — Eye exam by 6 mo then annually; audiology at birth, 6 mo, annually; ABR if needed

Second-Line

- **Atlantoaxial instability screening** — Cervical spine X-ray if symptomatic; flexion/extension views before sports participation
- **Sleep study** — Polysomnography by age 4 for obstructive sleep apnea; adenotonsillectomy or CPAP if positive

- **Celiac screening** — Tissue transglutaminase IgA at age 2 then as indicated; prevalence 5-16% in DS
- **Behavioral health support** — Screen for ASD, ADHD, anxiety, depression; psychiatric meds used cautiously per AAP 2022 guidance **[NEW 2024-25]**

Investigations

- Echocardiography — Screen for congenital heart defects (within first month) [URGENT]
- Thyroid function test (TSH) — Screen at birth, 6mo, 12mo, then annually
- CBC — Screen for transient myeloproliferative disorder/leukemia in neonates [URGENT]
- Audiological assessment — Screen at birth and every 6 months
- Ophthalmology exam — Screen by 6 months

Counselling

- Down syndrome is a genetic condition - your child will need regular checkups with multiple specialists
- Early intervention (physiotherapy, speech therapy) from infancy greatly helps development
- Heart checkups are essential as many children with Down syndrome have heart defects
- Thyroid testing is done every year; hearing and eye tests are also important
- Your child can attend school and achieve milestones with appropriate support and encouragement

Referral Criteria

Multidisciplinary team from birth: cardiology, ENT, ophthalmology, endocrinology, developmental pediatrics; genetics for family counseling

Hospitalisation Criteria

Admit for cardiac surgery, acute leukemia, severe infections, airway concerns, or atlantoaxial subluxation with neurological signs

Notes

Comprehensive health supervision requires lifelong multisystem screening per AAP schedule. The 2024 collaboration ensures guideline translations are freely available worldwide.

Guideline Changes

2022 AAP update added autism screening (7-19% prevalence), expanded psychiatric medication guidance

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Q91 — Edwards syndrome and Patau syndrome

SNOMED: 51500006 | Duration: 365 days | Source: AAP 2025 Clinical Report (Pediatrics), NRP Guidelines

First-Line

- **Supportive neonatal care** — Full resuscitation per NRP; thermoregulation, respiratory support, NG/G-tube feeding as needed **[NEW 2024-25]**
- **Cardiac management** — Echo at birth; diuretics + digoxin for CHF; palliative or corrective cardiac surgery if appropriate
- **Feeding support** — NG tube or gastrostomy for poor feeding; treat GER medically first, fundoplication if refractory

Second-Line

- **Palliative care pathway** — Comfort-focused care with family-centered decision-making when prognosis is very poor
- **Infection management** — Standard antibiotics for pneumonia, UTI, otitis media; these are leading causes of morbidity
- **Genetic counseling** — Karyotype confirmation; recurrence risk counseling; prenatal diagnosis options for future pregnancies

Investigations

- Karyotype — Confirm chromosomal abnormality [URGENT]

- Echocardiography — Screen for cardiac defects (present in >80%) [URGENT]
- Renal ultrasound — Screen for renal malformations
- Cranial ultrasound — Screen for CNS malformations

Counselling

- Trisomy 13/18 are serious genetic conditions with significant medical challenges
- Your medical team will discuss all treatment options with you and support your decisions
- Many families find comfort in connecting with support groups for these conditions
- Each child is different - some children live longer than expected with active medical care
- Palliative care services are available to help manage comfort and quality of life

Referral Criteria

Multidisciplinary approach from birth: neonatology, cardiology, genetics, palliative care; surgical specialties as needed based on shared decision-making

Hospitalisation Criteria

Most require NICU admission at birth; readmit for acute infections, cardiac decompensation, or surgical interventions based on family-directed goals of care

Notes

Median survival is 5-15 days for trisomy 13 and 5-15 days for trisomy 18, but long-term survivors exist with active intervention. Shared decision-making with families is essential.

Guideline Changes

AAP 2025 clinical report formally supports active management including resuscitation; no longer comfort-only default

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Q92 — Other trisomies and partial trisomies of autosomes

Duration: 365 days | Source: AAP 2025, ACMG Cytogenetics Guidelines

First-Line

- **Supportive neonatal care** — Full resuscitation; respiratory support; NG/G-tube feeding; thermoregulation per NRP guidelines **[NEW 2024-25]**
- **Cardiac screening** — Echo at birth; CHD present in majority; medical management of CHF; surgical repair when appropriate
- **Genetic confirmation** — Karyotype + CMA to define exact trisomy/partial trisomy; guides prognosis and surveillance plan

Second-Line

- **Palliative care** — Family-centered palliative approach when prognosis is very poor; shared decision-making essential
- **Developmental services** — Early intervention PT/OT/speech; special education referral; developmental pediatrician follow-up
- **Infection prophylaxis** — Prompt treatment of respiratory and urinary infections; influenza and pneumococcal vaccines on schedule

Investigations

- Chromosomal microarray — Define exact trisomy/partial trisomy and breakpoints [URGENT]
- Echocardiography — Screen for cardiac defects [URGENT]
- Cranial ultrasound or MRI — Screen for brain anomalies
- Renal ultrasound — Screen for renal anomalies

Counselling

- Your child has an extra piece of chromosome material which affects development

- The outlook depends on which chromosome is involved and how much extra material there is
- Your medical team will create a personalized monitoring plan for your child
- Early intervention services help maximize your child's potential
- Genetic counseling is important for understanding the condition and planning

Referral Criteria

Refer to clinical geneticist at diagnosis; multidisciplinary team based on specific findings: cardiology, neurology, developmental pediatrics

Hospitalisation Criteria

Admit based on specific complications: cardiac surgery, respiratory support, or acute illness management

Notes

Prognosis varies widely depending on specific chromosomal abnormality and degree of mosaicism. Shared decision-making with families regarding intensity of intervention is critical.

Guideline Changes

AAP 2025 guidance supports active management for trisomies previously considered lethal

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Q96 — Turner syndrome

SNOMED: 38804009 | Duration: 365 days | Source: International TS Consensus 2024 (Eur J Endocrinol), PES/ESE

First-Line

- **Growth hormone therapy** — Somatropin 0.045-0.050 mg/kg/day SC; start from age 2 if growth failure; continue until bone age >=14
- **Estrogen replacement** — Low-dose transdermal estradiol at age 11-12; titrate over 2-3 yrs; add progesterone for cycling
- **Cardiac surveillance** — Echo + cardiac MRI at diagnosis; aortic monitoring q3-5 yrs; screen for bicuspid aortic valve, coarctation
- **Renal and thyroid screening** — Renal US at diagnosis; TSH annually; autoimmune thyroiditis in ~25%; levothyroxine if needed

Second-Line

- **Oxandrolone** — Low-dose oxandrolone 0.03-0.05 mg/kg/day as GH adjunct for severe short stature; monitor virilization
- **Fertility preservation** — Oocyte/ovarian tissue cryopreservation if ovarian function present; counsel in adolescence **[NEW 2024-25]**
- **Hearing assessment** — Audiometry every 1-3 yrs; sensorineural and conductive loss common; ENT referral as needed
- **Psychosocial support** — Screen for nonverbal learning difficulties, social anxiety; neuropsych evaluation recommended

Investigations

- Karyotype — Confirm 45,X or mosaic variant
- Echocardiography and cardiac MRI — Baseline cardiac assessment and aortic monitoring [URGENT]
- Renal ultrasound — Screen for horseshoe kidney and other anomalies
- Thyroid function tests — Annual screening for autoimmune thyroiditis
- Audiometry — Screen for sensorineural and conductive hearing loss

Counselling

- Turner syndrome (45,X) affects girls and requires lifelong medical care
- Growth hormone injections from early childhood help achieve better adult height
- Estrogen treatment at puberty helps develop normal secondary sexual characteristics
- Heart monitoring is very important throughout life as aorta problems can occur

- Your daughter can live a full, productive life with appropriate medical support

Referral Criteria

All cases managed by pediatric endocrinologist; cardiology for aortic surveillance; ENT for hearing; reproductive endocrinology in adolescence

Hospitalisation Criteria

Admit for cardiac surgery (coarctation repair), acute aortic dissection (rare but emergency), or lymphedema management in neonates

Notes

The 2024 international consensus is the most comprehensive TS guideline to date with 87 recommendations. Cardiovascular surveillance is the leading cause of excess mortality and requires lifelong monitoring.

Guideline Changes

2024 Aarhus guidelines updated GH dosing, cardiovascular surveillance, and fertility preservation protocols

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Q97 — Other sex chromosome abnormalities, female phenotype, not elsewhere classified

Duration: 365 days | Source: AAP Genetics 2023, Endocrine Society Guidelines

First-Line

- **Karyotype and genetic testing** — Full karyotype; CMA for mosaicism detection; includes Triple X (47,XXX) and other variants
- **Developmental support** — Early intervention for speech/language delays; educational accommodations; IEP as needed
- **Endocrine monitoring** — Puberty assessment; FSH/LH/estradiol if delayed puberty; bone age annually from age 8

Second-Line

- **Estrogen replacement** — Low-dose estradiol if ovarian insufficiency or delayed puberty; start age 11-12, titrate gradually
- **Psychological support** — Screen for anxiety, social difficulties, ADHD; CBT or behavioral therapy as needed
- **Fertility counseling** — Assess ovarian function; discuss fertility preservation options; refer to reproductive endocrinology

Investigations

- Karyotype — Confirm specific sex chromosome abnormality
- Developmental assessment — Screen for speech-language and learning delays
- FSH, LH, estradiol — If delayed puberty or ovarian insufficiency suspected

Counselling

- Your daughter has an extra X chromosome which is usually a mild condition
- Most girls with Triple X (47,XXX) have normal appearance and can lead normal lives
- Some girls may need help with speech, language, or learning - early support is key
- Puberty and fertility are usually normal but should be monitored
- Regular developmental checkups help identify and address any concerns early

Referral Criteria

Refer to clinical geneticist for confirmation; developmental pediatrician; speech-language pathologist if delays noted; endocrinology if pubertal concerns

Hospitalisation Criteria

Hospitalization rarely needed for the chromosomal condition itself; admit only for associated complications

Notes

Triple X syndrome (47,XXX) is the most common condition in this category with generally mild phenotype. Individualized screening and support based on specific karyotype findings is key.

Guideline Changes

Expanded use of chromosomal microarray for precise characterization of sex chromosome variants

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Q98 — Klinefelter syndrome and other male chromosome abnormalities

SNOMED: 405769009 | Duration: 365 days | Source: European Academy of Andrology/ESE 2021, JCEM 2025 Update

First-Line

- **Testosterone replacement** — Begin at puberty (age 11-14) if levels low; IM testosterone enanthate 50mg monthly, titrate upward
- **Developmental support** — Speech-language therapy, educational support; screen for ADHD and learning disabilities by school age
- **Endocrine monitoring** — Annual LH/FSH/testosterone from age 10; bone density (DXA) at puberty onset and periodically

Second-Line

- **Transdermal testosterone gel 1%** — Alternative to IM injections; daily application; titrate to maintain physiologic levels
- **Fertility preservation** — Micro-TESE for sperm retrieval in adolescents/young adults; counsel early about fertility options
- **Psychological support** — CBT or counseling for anxiety, depression, social difficulties; multidisciplinary team approach **[NEW 2024-25]**
- **Gynecomastia surgery** — Reduction mammoplasty if persistent gynecomastia causes distress; typically after puberty

Investigations

- Karyotype — Confirm 47,XXY or variant
- FSH, LH, testosterone — Baseline hormonal assessment from age 10
- DEXA bone density — Screen for osteoporosis at puberty
- Developmental/neuropsychological assessment — Identify learning needs and guide intervention

Counselling

- Klinefelter syndrome is a common chromosome condition (47,XXY) affecting boys
- Most boys with Klinefelter have normal intelligence but may need help with speech and learning
- Testosterone treatment will be started at puberty to support normal male development
- Early speech therapy and educational support make a big difference
- Discuss fertility options with your doctor when your child reaches adolescence

Referral Criteria

Refer to pediatric endocrinologist for pubertal management; speech-language pathologist for early intervention; psychology for behavioral support

Hospitalisation Criteria

Hospitalization rarely needed; admit only for surgical procedures (gynecomastia correction) or acute complications

Notes

Early diagnosis (ideally prenatal or infancy) allows proactive developmental and endocrine management. Multidisciplinary care including endocrinology, psychology, and andrology is essential.

Guideline Changes

2025 JCEM review expanded psychological and quality-of-life management recommendations

Q99 — Other chromosome abnormalities, not elsewhere classified

SNOMED: 409709004 | Duration: 365 days | Source: ACMG 2023, AAP Genetics Guidelines

First-Line

- **Genetic evaluation** — Chromosomal microarray + karyotype; consider exome/genome sequencing if CMA non-diagnostic
- **Developmental monitoring** — Early intervention referral; PT/OT/speech as indicated; developmental pediatrician follow-up
- **Multisystem screening** — Cardiac echo, renal US, ophthalmology, audiology based on phenotype; tailor to specific abnormality

Second-Line

- **Genetic counseling** — Recurrence risk assessment; family karyotyping; prenatal diagnosis options for future pregnancies
- **Behavioral/psychiatric support** — Screen for ASD, ADHD, anxiety; therapy and medication as indicated by symptoms
- **Growth monitoring** — Serial growth parameters; consider GH evaluation if short stature; endocrine referral as needed

Investigations

- Chromosomal microarray — Precise characterization of chromosomal abnormality
- Karyotype — Confirm and characterize chromosome abnormality
- Echocardiography — Screen for cardiac defects
- Renal ultrasound — Screen for renal anomalies

Counselling

- Your child has a chromosome difference that needs ongoing medical monitoring
- The specific effects depend on the exact chromosome involved - your doctor will explain
- Regular checkups with developmental assessment help identify and address any issues early
- Genetic counseling can help you understand recurrence risk for future pregnancies
- Connect with support groups for families with similar chromosome conditions

Referral Criteria

Refer to clinical geneticist for diagnosis and syndrome-specific surveillance; developmental pediatrician; other specialists based on phenotype

Hospitalisation Criteria

Admit based on specific complications: cardiac surgery, acute infections, seizure management, or surgical correction of associated anomalies

Notes

Management is symptom-directed as phenotypes vary widely by specific chromosomal abnormality. Regular multidisciplinary follow-up with tailored screening protocols is essential.

Guideline Changes

Chromosomal microarray and exome sequencing now standard first-tier diagnostic tests

Urology

N39.44 — Enuresis (Nocturnal Bedwetting)

SNOMED: 236660009 | Duration: 56 days | Source: ICCS 2020 Standardization Document, IAP Enuresis Protocol, J Pediatr Urol 2024

First-Line

- **Reassurance (non-pharmacological)** — Explain to parents: common condition, improves with age, not child's fault; mandatory first step
- **Behavioral Therapy (non-pharmacological)** — Fluid management (avoid excess after 6-7 PM), timed voiding before bed, bladder training (hold urine for increasing intervals), positive reinforcement with bladder diary
- **Enuresis Alarm (non-pharmacological)** — Most effective long-term therapy, use for 3-4 months if available
- **Desmopressin** — 0.2 mg (200 mcg) PO at bedtime, can increase to 0.4 mg if needed; duration 4-12 weeks then taper; restrict fluids after dose

Second-Line

- **Imipramine** — 0.9-1.5 mg/kg/dose PO at bedtime (max 50 mg for <12y, 75 mg for >=12y); use only if desmopressin fails; monitor for cardiac side effects; avoid in overdose-risk families
- **Oxybutynin** — 0.2-0.3 mg/kg/dose BD-TDS PO (max 5 mg TDS); useful when overactive bladder component present; may combine with desmopressin for refractory cases
- **Tolterodine** — 1-2 mg PO at bedtime; alternative anticholinergic if oxybutynin not tolerated; fewer side effects than oxybutynin

Investigations

- Urine routine and microscopy — Rule out UTI in all cases
- Ultrasound KUB — Suspected structural abnormality or secondary enuresis
- Blood sugar — Rule out diabetes if polyuria or secondary enuresis
- Serum electrolytes — Monitor if on desmopressin therapy

Counselling

- Bedwetting is common and not the child's fault - never punish
- Restrict fluids after 6-7 PM, encourage more fluids during daytime
- Child should pass urine before bedtime every night
- Reward dry nights with a star chart - positive reinforcement works
- Most children outgrow bedwetting naturally; improvement expected in 4-8 weeks
- If on desmopressin, restrict fluids after taking the dose to avoid low sodium

Warning Signs

- Fast breathing or chest indrawing
- Bluish discoloration of lips or tongue
- Convulsions or fits
- Excessive sleepiness or unresponsive
- Daytime incontinence in addition to bedwetting
- Weak urine stream or straining to urinate
- Recurrent urinary tract infections
- Secondary enuresis after being dry for more than 6 months

Referral Criteria

Refer to pediatric nephrologist/urologist if daytime incontinence, weak urinary stream, recurrent UTI, secondary enuresis with red flags, or no response to 3 months of treatment

Hospitalisation Criteria

Hospitalisation not typically required for primary enuresis. Admit only if severe hyponatremia on desmopressin or if secondary enuresis requires workup for underlying neurological/renal cause

Notes

Behavioral therapy is first-line: fluid restriction after 6-7 PM, timed voiding before bed, wake child once at night, bladder training with increasing daytime holding intervals, positive reinforcement with reward calendar. Enuresis alarm is the most effective long-term therapy (use 3-4 months). Medication only if moderate-severe or significant social impact. Monitor for hyponatremia on desmopressin. Never punish the child.

Guideline Changes

ICCS 2020 updated standardization: active therapy from age 6 years; enuresis alarm remains most effective long-term treatment (3-4 months); combination desmopressin + imipramine shown more effective than desmopressin + oxybutynin for refractory cases (2024 study). No major 2024-2025 ICCS guideline revision.